

REVISED & UPDATED**GATE PREP SERIES**

GATE

Graduate Aptitude Test in Engineering

GENERAL APTITUDE

Highlights:

- Sections Included: General Aptitude.
- 1300+ MCQs & NTQs.
- Complete GATE 2016-2021 solved papers.
- 3 full length mock tests.

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 **GATE PREP SERIES** 

GATE

Graduate Aptitude Test in Engineering

2022

General Aptitude



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GATE

Graduate Aptitude Test in Engineering

IIT Institutes



GATE 2021 conducted by
Indian Institute of Technology, Mumbai



GATE 2022 will be conducted by
Indian Institute of Technology, Kharagpur



Indian Institute of Technology, Delhi



Indian Institute of Technology, Chennai



Indian Institute of Technology, Guwahati



Indian Institute of Technology, Roorkee



Indian Institute of Technology, Kanpur

Preface

The Graduate Aptitude Test in Engineering (GATE) is an online exam conducted by the IITs for admissions to PG courses in IITs, IISc Bangalore, NITs and many state run universities as well as private universities. Also there are more than 37 PSUs that use GATE score for recruitments. A large number of corporates are also using GATE score as a tool to screen students for placements.

GK Publications is well known as the “publisher of choice” to students preparing for GATE and other technical test prep examinations in the country. We published first set of books in 1994 when GATE exam, both objective and conventional, was conducted in the paper and pencil environment, and used as a check point for entry to post graduate courses in IITs and IISCs. At that time, students had little access to technology and relied mainly on instructor led learning followed by practice with books available for these examinations.

A lot has changed since then!

Today, GATE is conducted in an online only mode with Multiple Choice Questions, Multiple Select Questions and Numerical Based Questions. The score is valid for three years and is used not only for post graduate courses but is also used by major PSUs for recruitment. Today's students have easy access to technology and the concept of a monologue within the classroom has changed to dialogue where students come prepared with concepts and then discuss topics. They learn a lot of things on the go with their mobile devices and practice for mock tests online.

We, as a leading publisher of GATE books, have also embraced change. Today, our books are no more guides only but come with a fully supported mobile app and a web portal. The mobile App provides access to video lectures, short tests and regular updates about the exam. The web portal in addition to what is available on the App provides full length mock tests to mimic the actual exam and help you gauge your level of preparedness. The combination of practice content in print, video lectures, and short and full length tests on mobile and web makes this product a complete courseware for GATE preparation.

We also know that improvement is a never ending process and hence we welcome your suggestions and feedback or spelling and technical errors if any. Please write to us at gkp@gkpublications.com

We hope that our small effort will help you prepare well for the examination.

We wish you all the best!

GK Publications Pvt. Ltd.

About GATE

The Graduate Aptitude Test in Engineering (GATE) conducted by IISc and IITs has emerged as one of the bench mark tests for engineering and science aptitude in facilitating admissions for higher education (M.Tech./Ph.D.) in IITs, IISc and various other Institutes/Universities/Laboratories in India. With the standard and high quality of the GATE examination in 27 disciplines of engineering and science, Humanities and Social Sciences subjects, it identifies the candidate's understanding of a subject and aptitude and eligibility for higher studies. During the last few years, GATE score is also being used as one of the criteria for recruitment in Government Organizations such as Cabinet Secretariat, and National/State Public Sector Undertakings in India. Because of the importance of the GATE examination, the number of candidates taking up GATE exams has increased tremendously. GATE exams are conducted by the IITs and IISc as a computer based test having multiple choice questions and numerical answer type questions. The questions are mostly fundamental, concept based and thought provoking. From 2017 onwards GATE Exam is being held in Bangladesh, Ethiopia, Nepal, Singapore, Sri Lanka and United Arab Emirates. An Institute with various nationalities in its campus widens the horizons of an academic environment. A foreign student brings with him/her a great diversity, culture and wisdom to share. Many GATE qualified candidates are paid scholarships/assistantship, especially funded by Ministry of Human Resources Development, Government of India and by other Ministries. IIT, Kharagpur is the Organizing Institute for GATE 2022.

Why GATE?

Admission to Post Graduate and Doctoral Programmes

Admission to postgraduate programmes with MHRD and some other government scholarships/assistantships in engineering colleges/institutes is open to those who qualify through GATE. GATE qualified candidates with Bachelor's degree in Engineering/Technology/Architecture or Master's degree in any branch of Science/Mathematics/Statistics/Computer Applications are eligible for admission to Master/Doctoral programmes in Engineering/Technology/Architecture as well as for Doctoral programmes in relevant branches of Science with MHRD or other government scholarships/assistantships. Candidates with Master's degree in Engineering/Technology/Architecture may seek admission to relevant Ph.D programmes with scholarship/assistantship without appearing in the GATE examination.

Financial Assistance

A valid GATE score is essential for obtaining financial assistance during Master's programs and direct Doctoral programs in Engineering/Technology/Architecture, and Doctoral programs in relevant branches of Science in Institutes supported by the MHRD or other Government agencies. As per the directives of the MHRD, the following procedure is to be adopted for admission to the post-graduate programs (Master's and Doctoral) with MHRD scholarship/assistantship. Depending upon the norms adopted by a specific institute or department of the Institute, a candidate may be admitted directly into a course based on his/her performance in GATE only **or** based on his/her performance in GATE **and** an admission test/interview conducted by the department to which he/she has applied **and/or** the

candidate's academic record. If the candidate is to be selected through test/interview for post-graduate programs, a minimum of 70% weightage will be given to the performance in GATE and the remaining 30% weightage will be given to the candidate's performance in test/interview and/or academic record, as per MHRD guidelines. The admitting institutes could however prescribe a minimum passing percentage of marks in the test/interview. Some colleges/institutes specify GATE qualification as the mandatory requirement even for admission without MHRD scholarship/assistantship.

To avail of the financial assistance (scholarship), the candidate must first secure admission to a program in these Institutes, by a procedure that could vary from institute to institute. Qualification in GATE is also a minimum requirement to apply for various fellowships awarded by many Government organizations. Candidates are advised to seek complete details of admission procedures and availability of MHRD scholarship/assistantship from the concerned admitting institution. The criteria for postgraduate admission with scholarship/assistantship could be different for different institutions. The management of the post-graduate scholarship/assistantship is also the responsibility of the admitting institution. Similarly, reservation of seats under different categories is as per the policies and norms prevailing at the admitting institution and Government of India rules. *GATE offices will usually not entertain any enquiry about admission, reservation of seats and /or award of scholarship / assistantship.*

PSU Recruitments

As many as 37 PSUs are using GATE score for recruitment. It is likely that more number of PSUs may start doing so by next year. Below is the list of PSUs:

MDL, BPCL, GAIL, NLC LTD, CEL, Indian Oil, HPCL, NBPC, NECC, BHEL, WBSEDCL, NTPC, ONGC, Oil India, Power Grid, Cabinet Secretariat, Govt. of India, BAARC, NFL, IPR, PSPCL, PSTCL, DRDO, OPGC Ltd., THDC India Ltd., BBNL, RITES, IRCON, GHECL, NHAI, KRIBHCO, Mumbai Railway Vikas Corporation Ltd. (MRVC Ltd.), National Textile Corporation, Coal India Ltd., BNPM, AAI, NALCO, EdCIL India.

Important :

1. Admissions in IITs/IISc or other Institutes for M.Tech./Ph.D. through GATE scores shall be advertised separately by the Institutes and GATE does not take the responsibility of admissions.
2. Cabinet Secretariat has decided to recruit officers for the post of Senior Field Officer (Tele) (From GATE papers of EC, CS, PH), Senior Research Officer (Crypto) (From GATE papers of EC, CS, MA), Senior Research Officer (S&T) (From GATE papers EC, CS, CY, PH, AE, BT) in the Telecommunication Cadre, Cryptographic Cadre and Science & Technology Unit respectively of Cabinet Secretariat. The details of the scheme of recruitment shall be published in National Newspaper/Employment News by the concerned authority.
3. Some PSUs in India have expressed their interest to utilize GATE scores for their recruitment purpose. The Organizations who intend to utilize GATE scores shall make separate advertisement for this purpose in Newspapers etc.

Who Can Appear for GATE?

Eligibility for GATE

Before starting the application process, the candidate must ensure that he/she meets the eligibility criteria of GATE given in Table.

Eligibility Criteria for GATE 2022

Degree/Program	Qualifying Degree/Examination	Description of Eligible Candidates	Expected Year of Completion
B.E. / B.Tech. / B.Pharm.	Bachelor's degree in Engineering / Technology (4 years after 10+2 or 3 years after B.Sc. / Diploma in Engineering / Technology)	Currently in the 3 rd year or higher grade or already completed	2022
B. Arch.	Bachelor's degree of Architecture (5- year course) / Naval Architecture (4- year course) / Planning (4- year course)	Currently in the 3 rd year or higher grade or already completed	2023 (for 5-year program), 2022 (for 4-year program)
B.Sc. (Research) / B.S.	Bachelor's degree in Science (Post-Diploma/4 years after 10+2)	Currently in the 3 rd year or higher grade or already completed	2022
Pharm. D. (after 10+2)	6 years degree program, consisting of internship or residency training, during third year onwards	Currently in the 3 rd /4 th /5 th /6 th year or already completed	2024
M.B.B.S.	Degree holders of M.B.B.S. and those who are in the 5 th /6 th /7 th semester or higher semester of such programme.	5 th , 6 th , 7 th or higher semester or already completed	2022
M. Sc. / M.A. / MCA or equivalent	Master's degree in any branch of Arts/Science/Mathematics/Statistics/ Computer Applications or equivalent	Currently in the first year or higher or already Completed	2022
Int. M.E./ M.Tech. (Post-B.Sc.)	Post-B.Sc Integrated Master's degree programs in Engineering/ Technology (4-year program)	Currently in the 1 st / 2 nd /3 rd /4 th year or already completed	2024
Int. M.E./ M.Tech. or Dual Degree (after Diploma or 10+2)	Integrated Master's degree program or Dual Degree program in Engineering/Technology (5-year program)	Currently in the 3 rd /4 th /5 th year or already completed	2023
B.Sc. / B.A. / B.Com.	Bachelor degree in any branch of Science / Arts / Commerce (3 years program)	Currently in the 3 rd year or already completed	2021
Int. M.Sc. / Int. B.S. / M.S.	Integrated M.Sc. or 5-year integrated B.S.-M.S. program	Currently in the 3 rd year or higher or already completed	2022
Professional Society Examinations (equivalent to B.E. / B.Tech. / B.Arch.)	B.E./B.Tech./B.Arch. equivalent examinations of Professional Societies, recognized by MoE/UPSC/AICTE (e.g. AMIE by Institution of Engineers-India, AMICE by the Institute of Civil Engineers-India and so on)	Completed Section A or equivalent of such professional courses	NA

In case a candidate has passed one of the qualifying examinations as mentioned above in 2020 or earlier, the candidate has to submit the degree certificate / provisional certificate / course completion certificate / professional certificate / membership certificate issued by the society or institute. In case, the candidate is expected to complete one of the qualifying criteria in 2022 or later as mentioned above, he/she has to submit a certificate from Principal or a copy of marks card for section A of AMIE.

Certificate From Principal

Candidates who have to submit a certificate from their college Principal have to obtain one from his/her institution beforehand and upload the same during the online submission of the application form.

Candidates With Backlogs

Candidates, who have appeared in the final semester/year exam in 2022, but with a backlog (arrears/failed subjects) in any of the papers in their qualifying degree should upload a copy of any one of the mark sheets of the final year,

OR

obtain a declaration from their Principal along with the signature and seal beforehand and upload the same during the online submission of the application form.

GATE Structure

Structure of GATE

GATE 2022 will be conducted on 27 subjects (papers). Table below shows the list of papers and paper codes for GATE 2022. A candidate is allowed to appear in ANY ONE or UP TO TWO papers of the GATE examination. However, note that the combination of TWO papers in which a candidate can appear MUST be from the pre-defined list as given in Table. Also note that for a paper running in multiple sessions, a candidate will be mapped to appear for the examination in one of the sessions ONLY.

List of GATE Papers and Corresponding Codes

GATE Paper	Code	GATE Paper	Code
Aerospace Engineering	AE	Instrumentation Engineering	IN
Agricultural Engineering	AG	Mathematics	MA
Architecture and Planning	AR	Mechanical Engineering	ME
Bio-medical Engineering	BM	Mining Engineering	MN
Biotechnology	BT	Metallurgical Engineering	MT
Civil Engineering	CE	Petroleum Engineering	PE
Chemical Engineering	CH	Physics	PH
Computer Science and Information Technology	CS	Production and Industrial Engineering	PI
Chemistry	CY	Statistics	ST
Electronics and Communication Engineering	EC	Textile Engineering and Fibre Science	TF
Electrical Engineering	EE	Engineering Sciences	XE*
Environmental Science & Engineering	ES	Humanities & Social Sciences	XH**
Ecology and Evolution	EY	Life Sciences	XL***
Geology and Geophysics	GG		

Note: *Environmental Science and Engineering (ES) and Humanities and Social Sciences (XH) are two new papers introduced in GATE 2022.*

*XE Paper Sections	Code	**XH Paper Sections	Code	***XL Paper Sections	Code
Engineering Mathematics (Compulsory) (15 marks)	A	Reasoning and Comprehension (Compulsory) (25 marks)	B1	Chemistry (Compulsory) (25 marks)	P
Any TWO optional Sections (2x35 = 70 marks)		Any ONE optional Section (60 marks)		Any TWO optional Sections (2x30 = 60 marks)	
Fluid Mechanics	B	Economics	C1	Biochemistry	Q
Materials Science	C	English	C2	Botany	R
Solid Mechanics	D	Linguistics	C3	Microbiology	S
Thermodynamics	E	Philosophy	C4	Zoology	T
Polymer Science and Engineering	F	Psychology	C5	Food Technology	U
Food Technology	G	Sociology	C6		
Atmospheric and Oceanic Sciences	H				

*XE (Engineering Sciences), **XH (Humanities & Social Sciences), ***XL (Life Sciences), papers are of general nature and will be comprised of Sections listed in the above table

Note: Each subject / paper is of total 100 marks. General Aptitude (GA) section of 15 marks is common for all papers. Hence remaining 85 marks are for the respective subject / paper code.

**Combination of Two Papers Allowed to Appear in GATE 2022
(subject to availability of infrastructure and schedule)**

Code of The First (Primary) Paper	Codes of Papers Allowed as The Second Paper
AE	XE
AG	ES
AR	CE
BM	BT / XL
BT	BM / XL
CE	AR / ES
CH	CY / PE / XE
CS	MA
CY	CH / XL
EC	IN / PH
EE	IN
ES	AG / CE
EY	XL
GG	MN / PE / PH
IN	EC / EE / PH
MA	CS / PH / ST
ME	XE
MN	GG / XE
MT	PH / XE
PE	CH / GG / XE
PH	EC / GG / IN / MA / MT / ST
PI	XE
ST	MA / PH
TF	XE
XE	AE / CH / ME / MN / MT / PE / PI / TF
XH	----
XL	BM / BT / CY / EY

General Aptitude Questions

All the papers will have a few questions that test the General Aptitude (Language and Analytical Skills), apart from the core subject of the paper.

Duration and Examination Type

All the papers of the GATE 2022 examination will be for 3 hours duration and they consist of 65 questions for a total of 100 marks. Since the examination is an ONLINE computer based test (CBT), at the end of the stipulated time (3-hours), the computer screen will automatically close the examination inhibiting any further action.

Candidates will be permitted to occupy their allotted seats 40 minutes before the scheduled start of the examination. Candidates can login and start reading the instructions 20 minutes before the start of examination. The late login time (if any) recorded by the computer system **MUST NOT** be beyond 30 minutes from the actual starting time of the examination. Under **NO** circumstances, will a candidate be permitted to login after 30 minutes from the actual examination starting time. Candidates will **NOT** be permitted to leave the examination hall before the end of the examination.

Pattern of Question Papers

*GATE 2022 may contain questions of **THREE** different types in all the papers:*

- (i) **Multiple Choice Questions (MCQ)** carrying 1 or 2 marks each, in all the papers and sections. These questions are objective in nature, and each will have choice of four answers, out of which **ONLY ONE** choice is correct.

Negative Marking for Wrong Answers: For a wrong answer chosen in a MCQ, there will be negative marking. For 1-mark MCQ, 1/3 mark will be deducted for a wrong answer. Likewise, for 2-mark MCQ, 2/3 mark will be deducted for a wrong answer.

- (ii) **Multiple Select Questions (MSQ)** carrying 1 or 2 marks each in all the papers and sections. These questions are objective in nature, and each will have choice of four answers, out of which **ONE or MORE** than **ONE** choice(s) is / are correct.

Note: There is **NO** negative marking for a wrong answer in MSQ questions. However, there is **NO** partial credit for choosing partially correct combinations of choices or any single wrong choice.

- (iii) **Numerical Answer Type (NAT)** Questions carrying 1 or 2 marks each in most of the papers and sections. For these questions, the answer is a signed real number, which needs to be entered by the candidate using the virtual numeric keypad on the monitor (keyboard of the computer will be disabled). No choices will be shown for these types of questions. The answer can be a number such as 10 or –10 (an integer only). The answer may be in decimals as well, for example, 10.1 (one decimal) or 10.01 (two decimals) or –10.001 (three decimals). These questions will be mentioned with, up to which decimal places, the candidates need to present the answer. Also, for some NAT type problems an appropriate range will be considered while evaluating these questions so that the candidate is not unduly penalized due to the usual round-off errors. Candidates are advised to do the rounding off at the end of the calculation (not in between steps). Wherever required and possible, it is better to give NAT answer up to a maximum of three decimal places.

Note: *There is **NO** negative marking for a wrong answer in NAT questions.*

*Also, there is **NO** partial credit in NAT questions.*

Marking Scheme – Distribution of Marks and Questions

General Aptitude (GA) Questions

In all papers, GA questions carry a total of 15 marks. The GA section includes 5 questions carrying 1-mark each (sub-total 5 marks) and 5 questions carrying 2-marks each (sub-total 10 marks).

Question Papers other than GG, XE, XH and XL

These papers would contain 25 questions carrying 1-mark each (sub-total 25 marks) and 30 questions carrying 2-marks each (sub-total 60 marks) consisting of some MCQ type questions, while the remaining may be MSQ and / or NAT questions.

GG (Geology and Geophysics) Paper

Apart from the General Aptitude (GA) section, the GG question paper consists of two parts: Part A and Part B. Part A is compulsory for all the candidates. Part B contains two sections: Section 1 (Geology) and Section 2 (Geophysics). Candidates will have to attempt questions in Part A and questions in either Section 1 or Section 2 of Part B. The choice of Section 1 OR Section 2 of Part B has to be made at the time of online registration in GOAPS. At the examination hall, candidate cannot request for change of section.

Part A consists of 25 questions carrying 1-mark each (sub-total 25 marks and some of these will be MSQ and/or numerical answer type questions while remaining questions will be MCQ type). Either section of Part B (Section 1 and Section 2) consists of 30 questions carrying 2-marks each (sub-total 60 marks and some of these will be MSQ and/or numerical answer type questions while remaining questions will be MCQ type).

XE Paper (Engineering Sciences)

A candidate appearing in the XE paper has to answer the following:

- GA – General Aptitude carrying a total of 15 marks.
- Section A – Engineering Mathematics (Compulsory): This section contains 11 questions carrying a total of 15 marks: 7 questions carrying 1-mark each (sub-total 7 marks), and 4 questions carrying 2-marks each (sub-total 8 marks). Some questions will be of numerical answer type while remaining questions will be MCQ type.
- Any two of XE Sections B to H: The choice of two sections from B to H can be made during the examination after viewing the questions. Only TWO optional sections can be answered at a time. A candidate wishing to change midway of the examination to another optional section must first choose to deselect one of the previously chosen optional sections (B to H). Each of the optional sections of the XE paper (Sections B through H) contains 22 questions carrying a total of 35 marks: 9 questions carrying 1-mark each (sub-total 9 marks) and 13 questions carrying 2-marks each (sub-total 26 marks). Some questions will be of MSQ and/or numerical answer type while remaining questions will be MCQ type.

XH Paper (Humanities and Social Sciences)

A candidate appearing in the XH paper has to answer the following:

- **GA – General Aptitude** carrying a total of 15 marks.
- **Section B1 – Reasoning and Comprehension (Compulsory):** This section contains 15 questions carrying a total of 25 marks: 5 questions carrying 1-mark each (sub-total 5 marks) and 10 questions carrying 2-marks each (sub-total 20 marks). Some questions will be of MSQ and/or numerical answer type while remaining questions will be MCQ type.
- **Any ONE of XH Sections C1 to C6:** The ONE choice of section from C1 to C6 has to be made at the time of online registration in GOAPS. At the examination hall, candidate cannot request for change of section. Each of the optional sections of the XH paper (Sections C1 through C6) contains 40 questions carrying a total of 60 marks: 20 questions carrying 1-mark each (sub-total 20 marks) and 20 questions carrying 2-marks each (sub-total 40 marks). Some questions will be of MSQ and/or numerical answer type while remaining questions will be MCQ type.

XL Paper (Life Sciences)

A candidate appearing in the XL paper has to answer the following:

- **GA – General Aptitude** carrying a total of 15 marks.
- **Section P–Chemistry (Compulsory):** This section contains 15 questions carrying a total of 25 marks: 5 questions carrying 1-mark each (sub-total 5 marks) and 10 questions carrying 2-marks each (sub-total 20 marks). Some questions will be of MSQ and/or numerical answer type while remaining questions will be MCQ type.
- **Any two of XL Sections Q to U:** The choice of two sections from Q to U can be made during the examination after viewing the questions. Only TWO optional sections can be answered at a time. A candidate wishing to change midway of the examination to another optional section must first choose to deselect one of the previously chosen optional sections (Q to U). Each of the optional sections of the XL paper (Sections Q through U) contains 20 questions carrying a total of 30 marks: 10 questions carrying 1-mark each (sub-total 10 marks) and 10 questions carrying 2-marks each (sub-total 20 marks). Some questions will be of MSQ and/or numerical answer type while remaining questions will be MCQ type.

GATE Score

After the evaluation of the answers, the actual (raw) marks obtained by a candidate will be considered for computing the GATE Score. For multi-session papers (subjects), raw marks obtained by the candidates in different sessions will be converted to Normalized marks for that particular subject. Thus, raw marks (for single session papers) or normalized marks (for multi-session papers) will be used for computing the GATE Score, based on the qualifying marks.

Calculation of Normalized Marks for Multi-Session Papers

In GATE 2022 examination, some papers may be conducted in multi-sessions. Hence, for these papers, a suitable normalization is applied to take into account any variation in the difficulty levels of the question papers across different sessions. The normalization is done based on the fundamental assumption that "in all multi-session GATE papers, the distribution of abilities of candidates is the same across all the sessions". This assumption is justified since the number of candidates appearing in multi-session papers in GATE 2022 is large and the procedure for allocation of session to candidates is random. Further, it is also ensured that for the same multi-session paper, the number of candidates allotted in each session is of the same order of magnitude.

Based on the above, and considering various normalization methods, the committee arrived at the following formula for calculating the normalized marks for the multi-session papers.

Normalization mark of j^{th} candidate in the i^{th} session $\widehat{M}_{ij}$ is given by

$$\widehat{M}_{ij} = \frac{\bar{M}_t^g - M_q^g}{\bar{M}_{ti} - M_{iq}} (M_{ij} - M_{iq}) + M_q^g$$

where

M_{ij} : is the actual marks obtained by the j^{th} candidate in i^{th} session

$\bar{M}_t^g$: is the average marks of the top 0.1% of the candidates considering all sessions

M_q^g : is the sum of mean and standard deviation marks of the candidates in the paper considering all sessions

$\bar{M}_{ti}$: is the average marks of the top 0.1% of the candidates in the i^{th} session

M_{iq} : is the sum of the mean marks and standard deviation of the i^{th} session

Calculation of GATE Score for All Papers

For all papers for which there is only one session, actual marks obtained by the candidates will be used for calculating the GATE 2022 Score. For papers in multi-sessions, normalized marks will be calculated corresponding to the raw marks obtained by a candidate and the GATE 2022 Score will be calculated based on the normalized marks.

The GATE 2022 score will be computed using the formula given below.

$$\text{GATE Score} = S_q + (S_t - S_q) \frac{(M - M_q)}{(\bar{M}_t - M_q)}$$

where

M : marks obtained by the candidate (actual marks for single session papers and normalized marks for multi-session papers)

M_q : is the qualifying marks for general category candidate in the paper

$\bar{M}_t$: is the mean of marks of top 0.1% or top 10 (whichever is larger) of the candidates who appeared in the paper (in case of multi-session papers including all sessions)

S_q : 350, is the score assigned to M_q

S_t : 900, is the score assigned to $\bar{M}_t$

In the GATE 2022 the qualifying marks (M_q) for general category student in each subject will be 25 marks (out of 100) or $\mu + \sigma$, whichever is larger. Here μ is the mean and σ is the standard deviation of marks of all the candidates who appeared in the paper.

After the declaration of results, GATE Scorecards can be downloaded by the GATE qualified candidates ONLY.

The GATE 2022 Committee has the authority to decide the qualifying mark/score for each GATE paper. In case of any claim or dispute with respect to GATE 2022 examination or score, the Courts and Tribunals in Mumbai alone will have the exclusive jurisdiction to entertain and settle them.

GATE Syllabus

Verbal Aptitude

Basic English Grammar: tenses, articles, adjectives, prepositions, conjunctions, verb-noun agreement, and other parts of speech.

Basic Vocabulary: words, idioms, and phrases in context, Reading and comprehension, Narrative sequencing.

Quantitative Aptitude

Data Interpretation: data graphs (bar graphs, pie charts, and other graphs representing data), 2-and 3-dimensional plots, maps, and tables.

Numerical Computation and Estimation: ratios, percentages, powers, exponents and logarithms, permutations and combinations, and series, Mensuration and geometry, Elementary statistics and probability.

Analytical Aptitude

Logic: Deduction and induction, Analogy, Numerical relations and reasoning.

Spatial Aptitude

Transformation of shapes: translation, rotation, scaling, mirroring, assembling, and grouping Paper folding, cutting, and patterns in 2 and 3 dimensions.

Verbal Aptitude

1

CHAPTER

English Grammar

ERRORS IN USE OF ARTICLES

ARTICLES

Indefinite (A, An)

Use of 'An'

- Before words beginning with vowel sounds [a, e, i, o, u are called vowels, others are consonants].
e.g. an apple, an egg, an owl.
- Before words beginning with silent 'h' but sounds as vowel.
e.g. an hour, an honourable man, an heir, an honest man.
- F, H, L, M, N, R, S, X are letters that are not vowels but begin with vowel sound 'M' has the sound of 'em'. So, 'an' is used before abbreviations beginning with vowels of these letters.
e.g. an M.L.A., an R.A.F., an N.C.C. officer, an F.I.R., an X-ray, an H.E. school, an S.P.

Use of 'A'

- In the sense of one.
e.g. He couldn't speak a word to save himself.
With 'one' (since 'one' begins with sound of 'w')
e.g. a one-man show, a one-rupee note.
- Before words beginning with consonant sound
e.g. a boy, a box, a dog.
- With Vowel letters having consonant value.
e.g. a university, a unique article, a euphenism, a unit, a European language
- With units and rate (per).
e.g. He earns rupees five hundred a month.
- In exclamatory expressions before singular countable nouns.
e.g. What a pretty girl !
- When two subjects or articles are thought of as a single unit.
e.g. He was ready with a cup and saucer.
- With certain expressions of quantity.
e.g. a lot of, a dozen, a great deal of, a couple.
- With a person's name to indicate that the person is perhaps unknown to the person addressed.
e.g. A Mr. roy is at the door.
- With a special meal (to celebrate something or in someone's honour).
e.g. I called my friends to a lunch to celebrate my success.
- To make a common noun of a proper noun.
*e.g. This man is 'a second Newton'.
(This phrase means 'a philosopher as great as Newton')*

Definite (The)

Use of 'The'

- When we speak of a particular person or thing already referred to.
e.g. I dislike the follow.
- When a singular noun represents a whole class.
e.g. The mango is considered the king among fruits.
- With name of
 - gulfs, rivers, oceans, islands and mountains e.g. the Himalayas, the Indian ocean, the Persian Gulf, the Red sea, the Andaman islands, the Brahmaputra river.
 - Certain books
e.g. the Vedas, the Puranas, the Bible, the Ramacharitmanas.
 - Musical instruments
e.g. the flute, the violin, the tabla, the trumpet.
 - The inventions
e.g. I hate the telephone for its constant ringing.
 - Parts of body
e.g. He was wounded in the leg.
 - Religious groups
e.g. the Sikhs, the Hindus, the Parsees.
 - Names enforcing law
e.g. the Police, the Navy, the Air Force.
 - Political parties
e.g. the Congress, the Janata Pary, the B.J.P.
 - Aeroplanes, ships, trains etc
e.g. the Makalu (aeroplane), the Vikrant (Ship), the Rajdhani express (train).
 - Before names of an empire, dynasty or historical event
e.g. the Gupta dynasty, the Old Stone Age, the First World War, the American Revolution.
 - Clubs, foundations etc.
e.g. the Lion's Club, the Ford Foundation.
 - Before common nouns denoting unique things
e.g. the sun, the sky, the earth, the world.
 - With superlatives
e.g. He is the best boy in the class.
 - With ordinals
e.g. He took the first taxi that came his way.
 - Before the comparative degree
e.g. The more they get, the more they want.
 - Before an adjective when the noun is understood
e.g. The poor would favour him.

ERRORS IN USE OF NOUNS

1. Nouns which are used in singular form

- (i) Scenery, information, furniture, advice, machinery, stationery, news, poetry, business, mischief, fuel, issue, repair, bedding.
*e.g. The **scenery** of this place is worth seeing.*
- (ii) Physics, Mathematics, Economics, Classics, Ethics, Athletics, innings, gallows.
*e.g. **Economics** is a very interesting subject.*
- (iii) Brick, bread, fruit, word (as 'promise')
*e.g. Let me buy some **fruit**.*
- (iv) Words like dozen, score, hundred, thousand, million when preceded by a numeral.
*e.g. He bought ten **dozen** oranges.*
- (v) Expressions as a ten-rupee note, a two-hour journey, a four-mile walk, a Five-year plan, a six-man committee etc.
e.g. A ten-rupee note is lying there.

2. Nouns used only in plural form

- (i) Cattle, police, poultry, people, gentry, peasantry, artillery.
*e.g. The **police** have caught the thief.*
- (ii) Scissors, trousers, stockings, spectacles, shorts, alms, remains, riches, goods, measles.
e.g. My scissors are very sharp.

3. Nouns used both as singular and plural in the same form

- (i) Dear, sheep, fish, apparatus, wages.
*e.g. The **wages** of sin **is** death.*
*The **wages** of the workers **have** been raised.*

*I saw **a sheep** grazing in the field.*

***Sheep are** sold cheaper than goat.*

- (ii) Collective nouns as jury, public, team, audience, committee, government, audience, congregation, orchestra.

*e.g. The **team are** looking quite fit.*

*The **team has** not turned up yet.*

4. Use of Collective Nouns

- **crowd** of people
- **herd** of cattle/cows
- **team** of players
- **flight** of birds
- **bouquet** of flowers
- **Shoal/school** of fish
- **bundle** of sticks
- **army** of soldiers
- **flock** of sheep
- **crew** of sailors
- **swarm** of bees
- **garland** of beads
- **gang** of thieves
- **library** of books
- **mob** of angry people
- **fleet** of ships/cruisers
- **pack** of wolves
- **pack** of cards
- **loaf** of bread
- **heap** of corn
- **stock** of grain
- **class** of pupils
- **bunch** of grapes
- **herd** of lions
- **pile** of books
- **hoarde** of nomads
- **block** of flats
- **file** of papers

5. One of or any of is followed by plural words.

*e.g. I want **one of the books** kept on the table.*

***Any of these tools** may serve the purpose.*

6. Plural nouns are used with fractions and decimal over 1.

*e.g. It took us one and a half **hours**.*

ERRORS IN USE OF PRONOUNS

Pronouns are words used to replace nouns or noun groups already mentioned.

Nominative/ Vocative case (comes before verb)	Accurative/ Objective case (comes after verb)	Possessive case	Reflexive Pronoun
I	Me	My	Myself
We	Us	Our	Ourselves
He	Him	His	Himself
She	Her	Her	Herself
They	Them	Their	Themselves
Who	Whom	Whose	—

- The pronoun 'One' must be followed by 'one's'.
e.g. One must do one's duty to one's country.
- When 'one' means 'one in number', the pronoun for it is third person singular pronoun (he, she, it).
e.g. One of them has given up one's studies. (×)
One of them has given up his studies. (✓)
- 'Everyone' or 'Everybody' must be followed by 'his'.
e.g. Everyone should love his country.
- Each, every, anyone, anybody must be followed by the singular pronoun of their person.
e.g. Anyone can do this if he tries.

- 'Let' is followed by pronoun in the objective case.
e.g. Let him go.

- 'But' and 'except' are followed by pronoun in the objective case.

e.g. Everyone attended the party except him.

'such as' is followed by pronoun in the subjective case.

e.g. I have no liking for such a man as he.

- Verbs like enjoy, avail, pride, resign, apply, acquit, assert, absent are followed by reflexive pronouns.
e.g. He absented himself from the class.

We enjoyed ourselves at the party.

8. Reflexive pronouns are never used with verbs keep, conceal, quality, spread, rest, stay.
e.g. *I stayed away from my class.*
He qualified in the test.
9. When first, second and third person singular pronouns (I, you and He) are used together, they are placed in the order : You, he and I.
e.g. *You he and I are neighbours.*
In case of plural pronouns, 'we' comes first, then 'you' and then 'they'.
e.g. *We, you and they must work together.*
But if we have only two persons including first, then first person pronoun is written first.
e.g. *I and Sanjeev have done this job.*
10. 'Who' denotes subject and 'whom' denotes object.
e.g. *Who do you think did the job ?*
11. 'Whose' is used for persons and 'which' for lifeless objects.
e.g. *This is the table which I was talking about.*
12. 'Which' conveys additional information and 'that' explains a certain thing.
e.g. *I will tell you the first thing which I remember.*
13. The following expressions usually take 'that' in place of 'who' or 'which'.
Only, Any, It is, All, Superlatives.
e.g. *He is the only man that can do it.*
Any man that listens to you is a fool.
14. 'Each other' is used for two; 'one another' for more than two.
e.g. *Rahul and Renu love each other.*
15. The complement of the verb *to be*, when it is expressed by a pronoun, should be in Nominative case.
e.g. *It was he who did it.*
16. When the same person is the subject and object, it is necessary to use reflexive pronouns.
e.g. *I cut me shaving this morning. (×)*
I cut myself shaving this morning. (✓)
17. When a pronoun is the object of a verb or preposition it should be in objective case.
e.g. *These books are for you and I. (×)*
These books are for you and me. (✓)
Between him and me there is an understanding.
18. The Relative Pronoun should be placed as near as possible to the antecedent.
e.g. *I have read Shakespeare's works who was a great dramatist. (×)*
I have read the works of Shakespeare who was a great dramatist. (✓)
19. The case of the pronoun following 'than' and 'as' is decided by mentally supplying the verb and completing the sentence.
e.g. *She is taller than I (am)*
I love more than (I love) him.

PARTS OF SPEECH

The **parts of speech** explain how a word is used in a sentence.

There are eight main parts of speech (also known as word classes): **nouns, pronouns, adjectives, verbs, adverbs, prepositions, conjunctions and interjections.**

Most **parts of speech** can be divided into sub-classes. Prepositions can be divided into prepositions of time, prepositions of place etc. Nouns can be divided into proper nouns, common nouns, concrete nouns etc.

It is important to know that a word can sometimes be in more than one part of speech. For example with the word *increase*.

Increase can be a verb e.g. Prices *increased*

and *increase* can also be a noun e.g. There was an *increase* in the number of followers.

The eight main parts of speech in English are:

1. NOUN

A noun is the name of a person, place, thing, or idea.

man... Butte College... house... happiness

A noun is a word for a person, place, thing, or idea. Nouns are often used with an article (*the, a, an*), but not always. Proper nouns always start with a capital letter; common nouns do not. Nouns can be singular or plural, concrete or abstract. Nouns show possession by adding 's. Nouns can function in different roles within a sentence; for example, a noun can be a subject, direct object, indirect object, subject complement, or object of a preposition.

Examples of nouns: Daniel, London, table, dog, teacher, pen, city, happiness, hope

Example sentences:

- **Steve** lives in **Sydney**. **Mary** uses **pen** and **paper** to write **letters**.
- The young **girl** brought me a very long **letter** from the **teacher**, and then she quickly disappeared. Oh my!

2. PRONOUN

A pronoun is a word used in place of a noun.

She... we... they... it

A pronoun is a word used in place of a noun. A pronoun is usually substituted for a specific noun, which is called its antecedent. In the sentence above, the antecedent for the pronoun *she* is the girl. Pronouns are further defined by type: personal pronouns refer to specific persons or things; possessive pronouns indicate ownership; reflexive pronouns are used to

emphasize another noun or pronoun; relative pronouns introduce a subordinate clause; and demonstrative pronouns identify, point to, or refer to nouns.

Examples of pronouns: I, you, we, they, he, she, it, me, us, them, him, her, this, those

Example sentences:

- Mary is tired. **She** wants to sleep. I want **her** to dance with **me**.
- The young girl brought **me** a very long letter from the teacher, and then **she** quickly disappeared. Oh my!

3. VERB

A verb expresses action or being.

jump... is... write... become

The verb in a sentence expresses action or being. There is a main verb and sometimes one or more helping verbs. ("She can sing." *Sing* is the main verb; *can* is the helping verb.) A verb must agree with its subject in number (both are singular or both are plural). Verbs also take different forms to express tense.

Examples: go, speak, run, eat, play, live, walk, have, like, are, is

Example sentences:

- I **like** Woodward English. I **study** their charts and **play** their games.
- The young girl **brought** me a very long letter from the teacher, and then she quickly **disappeared**. Oh my!

4. ADJECTIVE

An adjective modifies or describes a noun or pronoun.

pretty... old... blue... smart

An adjective is a word used to modify or describe a noun or a pronoun. It usually answers the question of which one, what kind, or how many. (Articles [a, an, the] are usually classified as adjectives.)

Examples: big, happy, green, young, fun, crazy, three

Example sentences:

- The **little** girl had a **pink** hat.
- The **young** girl brought me a very **long** letter from the teacher, and then she quickly disappeared. Oh my!

5. ADVERB

An adverb modifies or describes a verb, an adjective, or another adverb.

gently... extremely... carefully... well

An adverb describes or modifies a verb, an adjective, or another adverb, but never a noun. It usually

answers the questions of when, where, how, why, under what conditions, or to what degree. Adverbs often end in -ly.

Examples: slowly, quietly, very, always, never, too, well, tomorrow, here

Example sentences:

- I am **usually** busy. **Yesterday**, I ate my lunch **quickly**.
- The young girl brought me a **very** long letter from the teacher, and **then** she **quickly** disappeared. Oh my!

6. PREPOSITION

A preposition is a word placed before a noun or pronoun to form a phrase modifying another word in the sentence.

by... with.... about... until

(by the tree, with our friends, about the book, until tomorrow)

A preposition is a word placed before a noun or pronoun to form a phrase modifying another word in the sentence. Therefore a preposition is always part of a prepositional phrase. The prepositional phrase almost always functions as an adjective or as an adverb. The following list includes the most common prepositions:

Examples: at, on, in, from, with, near, between, about, under

Example sentences:

- I left my keys **on** the table **for** you.
- The young girl brought me a **very** long letter from the teacher, and **then** she **quickly** disappeared. Oh my!

7. CONJUNCTION

A conjunction joins words, phrases, or clauses.

and... but... or... while... because

A conjunction joins words, phrases, or clauses, and indicates the relationship between the elements joined. Coordinating conjunctions connect grammatically equal elements: and, but, or, nor, for, so, yet. Subordinating conjunctions connect clauses that are not equal: because, although, while, since, etc. There are other types of conjunctions as well.

Examples: and, or, but, because, so, yet, unless, since, if.

Example sentences:

- I was hot **and** exhausted **but** I still finished the marathon.
- The young girl brought me a very long letter from the teacher, **and** then she quickly disappeared. Oh my!

8. INTERJECTION

An interjection is a word used to express emotion.

Oh!... Wow!... Oops!

An interjection is a word used to express emotion. It is often followed by an exclamation point.

Examples: Ouch! Wow! Great! Help! Oh! Hey! Hi!

Example sentences:

- Wow! I passed my English test. Great! – Ouch! That hurt.
- The young girl brought me a very long letter from the teacher, and then she quickly disappeared. **Oh my!**

ERRORS IN USE OF PREPOSITIONS

A preposition is a word used with a noun or pronoun to show its relation to some other word in a sentence.

PREPOSITIONS OF TIME

1. **At** is used
 - (a) with a definite point of time
e.g. I usually get up at 5 o'clock.
 - (b) with festivals
e.g. He will come at Holi.
2. **In** is used :
 - (a) with the parts of the day, (with **noon**, use **at**), months, seasons and years
e.g. He takes a walk in the afternoon.
 - (b) with the future tense referring to the period in which action may take place
e.g. You must be careful in future.
3. **On** is used : with days and dates
e.g. My brother will arrive on Monday.
4. **By** refers to the latest time at which an action will be over
e.g. The examination will be over by 5 p.m.
5. **For** is used with perfect continuous tense showing the duration of action
e.g. I have been here for three years.
6. **Since** is used with the point of time when action begins and continues :
e.g. He has been ill since last Tuesday.
7. **From** refers to the starting point of action.
e.g. This water came from a spring.

PREPOSITIONS OF POSITION

1. **At** refers to an exact point.
e.g. The tourist stayed at the Tourist Hotel.
He studied at Oxford.
2. **In** refers to larger areas.
e.g. He lives in Bombay.
3. **Between** is used for two persons or two things.
e.g. Share these sweets between him and me.
4. **Among** is used with more than two persons or

things but before the word which starts with a consonant letter.

e.g. Divide the sweets among the three boys.

5. **Amongst** is also used with more than two persons or things but before the word which starts with a vowel letter.
e.g. Divide the sweets amongst us.
6. **Above** is used for **higher than**.
e.g. The sun rose above the horizon.
7. **Under** is used for **vertically below**.
e.g. It is shady under the trees.
8. **Below** is used for **lower than**.
e.g. When the sun sets it goes below the horizon.
9. **Over** is used for **vertically above**.
e.g. There is an aircraft coming over.
10. **Beneath** means a lower position.
e.g. the ground was slippery beneath her.

PREPOSITIONS OF DIRECTION

1. **To** is used to express motion from one place to another.
e.g. We walked to the river and back.
2. **Towards** refers to direction.
e.g. He saw me running towards him.
3. **Into** denotes motion towards the inside of something.
e.g. She fell into a ditch.
4. **At** refers to aim.
e.g. He aimed at the bird.
5. **For** denotes direction.
e.g. I will leave for Pune today.
6. **Against** shows pressure.
e.g. She cleaned the edge of her knife against the plate.
7. **Off** refers to separation.
e.g. He was wiping sweat off his face.
8. **From** refers to the point of departure.
e.g. The man parted from his friends.

OTHER USES OF PREPOSITION

1. **About** shows nearness.
e.g. His father is about to retire.
2. **Along** stands for **in the same line**.
e.g. She led them along the corridor.
3. **After** refers to sequence.
e.g. She came after me.
4. **Across** means from one side.
e.g. He blew a cloud of smoke across the table.
5. **Before** stands for **in front of**.
e.g. A thief was brought before the judge.
6. **Behind** means **at the back of**.
e.g. She sat down behind the hedge.
7. **Beyond** means **on the farther side of**.
e.g. This is beyond his power.
8. **Beside** means **by the side of**.
e.g. I sat down beside my wife.
9. **Besides** means **in addition to**.
e.g. He is guilty of five killings and more besides.

WORDS FOLLOWED BY PREPOSITIONS

A

- abound with
- abstain from
- according to
- accordance with
- accuse of
- accustomed to
- acquaint with
- acquit of
- adverse to
- affiliate to/with
- afflict with
- afraid of
- agree to/on
- aim at
- allegation
- about
- allude to
- alternate with
- alternative to
- angry with
- apply to (body)
- appreciation of
- apprise of
- approximate of
- arraign against
- arrest in
- arrive at/in
- ashamed of
- aspire to/after/at
- associate with
- assure of
- attribute to

B, C

- beset with
- beware of
- blink at
- boast of
- break off
- bungle over
- careful of
- cause of
- centre on
- charge with
- clamour for
- clamp on
- coerce into
- collide with
- comment on
- compatible with
- complain of
- comply with
- compliment on
- composed of
- concur in
- condole with

- conducive to
- confer on
- confirmation of
- conform to
- congratulate on
- conscious of
- contiguity with
- converge on
- convict of
- co-opt to
- cope with
- correspond with
- converge on
- convict of
- co-opt to
- cope with
- correspond with
- cost of
- credit with
- criticism of
- cure of

D

- decide on
- debar from
- delight in
- deliberate on
- denude of
- depart to
- departure for
- depend on/upon
- deprive of
- derive from
- derogate from
- derogatory to
- desist from
- despair of
- detrimental to
- devoid of
- difficulty in
- diffident of
- disabuse of
- disagree with
- disgusted with
- dispose of
- dissociate from
- divest of
- divide into
- dressed in

E

- effort in (—ing)
- election to
- embroil in
- emphasis on
- employ in
- enamoured of
- encroach on

- endow with
- enlarge on
- engage in
- essential to
- expostulate with
- extend to

F, G, I

- fed up with
- focus on
- forbid to
- fraught with
- fritter in
- full of
- give to
- grapple with
- good at
- identical with
- impose on
- incur on
- indict for
- indifferent to
- indulge in
- infatuate with
- infect with
- infest with
- information on
- insensible to
- insist on
- interest in
- intimate to
- intoxicate with
- intrigue with
- intrude on
- investigation of
- involve in

J, K, L

- jealous of
- jeer at
- jump at
- just to
- knock at
- lash at
- lay siege to
- leave for
- live on (income)
- long for

O, P

- oblivious of
- operate on
- opinion on
- overcome with
- pay a call on
- persevere in
- persist in
- plan for
- plead for

- pleased with
- prefer to
- preference for
- present with
- prevent from
- priority to
- probe into
- prohibit from
- proud of

R, S

- recoil from
- reconcile to
- remand to
- remonstrate with
- repent of
- replete with
- reply to
- research on
- resolve on/to
- sail for
- satiate with
- satisfied with
- sentence to
- sick of
- side with
- sparing
- square with
- stickler for
- subversive of
- succeed in
- sue for
- superior to
- supply with
- sure of
- surprised at
- susceptible to
- suspect of
- sympathy for

T, U, V, W

- testify to
- thank for
- throw at
- tinker at
- tire of
- trace to
- trample on
- translate into
- try to (do)
- unequal to (the task)
- vie with
- vote for (a candidate)
- wait for (or await)
- ways of
- wish for

SOME SPECIAL CASES

- absolve from (blame) / of (sin)
- admit to (hospital, place, class)
- appear for (examination)/at
- begin on (not from Sunday)
- blow down (the roof)/off (steam)/over (pass off)
- buy in (market, store)/at (shop)
- come/go by (bus, train)
- come into (conflict with)
- compare with (one man height with another's)/to (one man's height to a building)
- consist of (to denote the substance)/in (define the subject)
- deal with (somebody)/ in (goods)
- die of (disease)/from (other causes)
- differ from (to be different from)/with (to disagree)
- due to (used after a noun, not after a verb)
- entrust to (somebody a job)
- entrust (somebody) with a job)
- fill with (enthusiasm)/ in (a form)
- fire at (when not hit)/on (when injured to killed)
- glad of (the news)/at (having arrived safe)
- hostility to/in hand (being attended to)/ on hand (in possession of)
- ingratiate (oneself) with
- inquire into (matter) / of (person)/about. often (thing)
- invest with (power/in (blank)
- key of (the door)/to (a problem)
- level (charge) against.
- look at (consider evidence)/to (somebody for aid)
- meeting (called) for (Sunday)
- motive in (doing)/for (an act)
- neglect of (duty)/in (doing a thing)
- nominated to (board)
- noted for (good things)
- owing to (used after a verb, not after a noun)
- part from (people)/with (things)
- preventive for priority
- put out (light)/off (trip)
- Shoot at (when injured)/in (leg, head)
- take up (job)/to (a hobby)
- taste of (sensation)/for (liking)
- tear off (remove)/up (to pieces)
- vest with (The P.M. is vested with power to ...)/ in (Powers are vested in the P.M. to ...)

ERRORS IN USE OF CONJUNCTIONS

1. **Scarcely or hardly** is followed by **when**.
e.g. Hardly had I slept when the telephone rang.
2. **Though** is followed by **yet**.
e.g. Though he worked hard yet he failed.

3. **No sooner** is followed by **than**.
e.g. No sooner did we reach there than it began to rain.
4. **Not only** is followed by **but also**.
e.g. Not only did he help her, but also dropped her home safely.
5. **Lest** is negative and so should not be followed by **not**; it is followed by **should**.
e.g. Work hard lest you should fail.
If **would** or **may** is used in place of **should**, then **else** should be used in place of **lest**.
e.g. Give him water, else he may die.
6. **Both** is complemented by **and**, not by **as well as**.
e.g. Both Amit and Satish are good at Science.
7. **So.....as** is used in negative sentences, whereas **asas** is used in affirmative sentences.
e.g. He is not so tall as his brother.
8. **Other** is followed by **than**.
e.g. He has no other claim than his wealth.
9. The word **reason** is not followed by **because**, but by **that**.
e.g. The reason why he didn't go was that his mother was ill.
10. **Because** denotes reason.
in order that denotes purpose.
e.g. He went to the doctor because he was ill.
11. Words such as regard, describe, define, treat, mention, depict, portray are followed by **as**.
e.g. I regard her as my sister.
12. **As** and **since** are also used to express reason.
e.g. As he was not there, I spoke to his brother.
13. **Neither** is followed by **nor** and both are followed by same auxiliary verb.
e.g. Neither Ravi nor Raju was seen.
14. **Either** is followed by **or**.
e.g. Either you are or he is to blame.
15. **Unless, until, if not, so that** should not be followed by **not**.
e.g. Wait here until I come.
16. **If** is used in conditional series
Whether is used in uncertainty.
e.g. I don't know whether he was present.
17. To express **time before** use **until** or **till** and to express **how long** use **as long as**.
e.g. Work as long as you live.
18. **Such** is followed by **as**.
e.g. We talked about such subjects as the weather.
Such is followed by **that** if we emphasize degree of something by mentioning the result.
e.g. The extent of the disaster was such that not a single man could survive.
19. **Like** is followed by pronoun; **as** is followed by a clause.
e.g. He runs like me.

ERRORS IN SUBJECT-VERB AGREEMENT

1. Singular subject must have singular verb.
e.g. He writes; I write.
2. Plural subject must have plural verb.
e.g. They write; We write.
3. Two subjects joined by **and** will always take a plural verb.
e.g. The doctor and nurse work together.
The doctor and nurses work together.
The doctors and nurse work together.
4. Two singular subjects joined by **or** or **nor** will take a singular verb.
e.g. A doctor or a nurse is working in the hospital.
5. A singular subject and a plural subject joined by **or** or **nor** will take a singular or plural verb depending on which subject is nearer the verb.
e.g. Neither Deepak nor his friends are joining the tour.
Neither his friends nor Deepak is joining the tour.
6. If the subject is singular and the predicate is plural, the verb must agree with its subject and not its predicate.
e.g. Physical conditioning and mental attitude are winning combination.
The winning combination is physical conditioning and mental attitude.
7. Indefinite pronouns such as someone, somebody, each, nobody, anyone, anybody, one, no one, everyone, everybody, either, neither, etc. always take a singular verb.
e.g. Each of my friends calls me once a month.
8. Indefinite pronouns which indicate more than one (several, few, both, many) always take plural verbs
e.g. Both of the books required careful reading.
9. Collective nouns (fleet, army, committee, crowd) are singular when the group works together as a unit and hence take singular verbs.
e.g. The jury has reached its verdict.
10. Collective nouns are plural when the members of the group are acting individually and hence take plural verbs.
e.g. The jury have argued for five hours.
11. Some words (such as news, measles, mumps etc.) end in **—s** but represent a single thing. These words need singular verbs.
e.g. The 7 o'clock news is about to begin.
12. Some words (such as scissors, trousers, spectacles, shorts etc.) end in **—s** and seem to represent a single thing, but they are two parts to that single thing. These words take plural verbs.
e.g. The scissors are on the table.
13. Words (such as politics, ethics, athletics etc.) that end in **—ies** are usually singular and hence take singular verbs.
e.g. Mathematics is his favourite subject.

But these words are singular when they refer to a study, science or practice. If these words have modifiers with them, they become plural and hence take plural verbs.

e.g. His politics are somewhat divided.

14. Title of the books need singular verbs.
e.g. Great Expectations is a good book.
15. Some nouns in the plural form represent an amount, a fraction, or an element of time. These nouns are considered singular and hence take singular verbs.
e.g. Sixty minutes is enough to finish this work.
16. If two subjects are joined together by **as well as** the verb will act according to the first subject.
e.g. Students as well as the teacher are playing.
17. The subject **Many a** is always followed by the singular verb.
e.g. Many a man was drowned in the sea.
18. If two subjects are joined together by **with** the verb will act according to the subject.
e.g. The principal together with his students was seeing the final match.
19. If subject is **The number of ...** use a singular verb.
e.g. The number of books is very small.
20. If the subject begins with **A number of** (= many), use a plural verb.
e.g. A number of books are missing.
21. If two subjects express one idea, use a singular verb.
e.g. Bread and butter is wholesome food.
22. When adjectives such as **much, less, little** and **more** are used as nouns, they must have a singular verb.
e.g. A little of good habits makes our life happy.

ERRORS IN THE USE OF TENSES

The changed forms of a verb that indicate time of the action are called **tenses** of the verb.

1. When the verb in the Principal Clause is in the Past tense, the verbs of the Subordinate Clauses should be in the Past tense.
e.g. He said that he had finished his work.
2. But a past tense in the Principal Clause may or may not be followed by the Past tense in the subordinate clause if the latter expresses universal or habitual truth.
e.g. The teacher said that the earth revolves round the sun.
3. Any tense may be used in the sub-ordinate clause if it gives a comparison by using the word **than**.
e.g. The teacher liked Anil better than he liked me.
4. Any tense can be used when the subordinate clause is in a quotation.
e.g. I said, "I am going to Delhi today".
5. The Present Perfect Tense (subject + has/have + V₃) cannot be used when an expression of Past time (yesterday, last night, ago etc.) is used.
e.g. We wrote to you yesterday about his mother.

6. In conditional sentences the subordinate clause, beginning with **if**, is generally in Present or Past tense but not in future tense.
e.g. If I go to Delhi, I shall see the Red Fort.
7. With the phrases **as if** and **as though**, Past tense and plural form of the verb should be used.
e.g. He behaves as if he were a millionaire.
8. Past Perfect tense (subject + had + V₃) is used when a sentence refers to two past actions and one of them occurs earlier than the other.
e.g. When my uncle came to India I had passed M.A.
9. Words like **usually, generally, often, whenever** etc. are used in Present Indefinite tense.
e.g. I usually go to play cricket.
10. If the action began in the past and is still continuing in the present, use Present Perfect tense (Subject + has/have + been V₁ + ing).
e.g. I have been reading in this class for two months.
11. Do not use the Future or the Present tense after such expressions as **suppose that, it is high time, it is time, as if** etc.
e.g. It is high time that you went home.

GERUND & INFINITIVE

Use of Gerund (V₁ + ing)

1. When an action is being considered in general sense, gerund is used as subject.
e.g. Swimming is his favourite pastime.
2. Gerund is used as subject in short prohibitions.
e.g. Smoking is prohibited.
3. Verbs such as help, stop, detest, avoid, finish, dread, mind, prevent, dislike, risk, deny, recollect, no good, no use, resent, delay, postpone, defer, enjoy, forgive, pardon, excuse, suggest etc. are followed by the gerund.
e.g. It is no use saying.
4. A gerund is placed after a preposition :
e.g. She is tired of walking.
But the prepositions **except** and **but** are followed by an infinitive.
5. Agerund, showing physical activity, is put after **come** or **go**.
e.g. He is to go shooting.
6. Nouns, followed by a gerund, are put in the possessive case.
e.g. He hates his brother's drinking everyday.
7. If there is a sense of dislike, hesitation, risk etc. in a sentence, use gerund.
e.g. I dislike reading cheap novels.
8. Adjective before gerund are put in the possessive case.
e.g. I do not mind your going away.
9. **With a view to** is always followed by a gerund and not an adjective.
e.g. We go to school with a view to studying.

INFINITIVE (TO + V₁)

1. Verbs such as learn, remember, promise, swear, consent, neglect, refuse, purpose, try, endeavour, attempt, fail, care, hope, decide, hesitate, prepare, determine, undertake, manage, arrange, seem, forget, agree, regret etc. are followed by infinitive:
e.g. We endeavour to invite such persons.
We promise to do this work soon.
2. Verbs such as order, tell, invite, oblige, allow, permit, compel, teach, instruct, advise, tempt, encourage, urge, request, show, remind, forbid, warm etc. are followed by object and infinitive
e.g. The teacher instructed the students to go.
3. Verbs or expressions like—will, can, do, must, may, let are followed by infinitive without **to**
e.g. Let him do this work.
4. Expressions like would rather, would sooner, rather than, sooner than, had better are followed by infinitive without **to**
e.g. You had better tell him.
5. **See** can be used with infinitive or without it
e.g. Go to see him.
6. The infinitive is used after adjectives like delight, angry, glad, astonished
e.g. I was astonished to find him.
7. The verb know is never directly followed by the infinitive. It is followed by a conjunction and then the infinitive
e.g. Do you know to play the harmonium ? (×)
Do you know how to play the harmonium ? (✓)
8. **...had better** is always followed by the infinitive without **to**
e.g. You had better to stop taking the medicine which has harmful side-effects. (×)
You had better stop taking the medicine which has harmful side-effects. (✓)
9. Infinitive is not used with **to** after verbs of perception like-to see, to hear, to feel, to watch etc.
e.g. I saw him cross the street.
10. Avoid using split-infinitive :
e.g. He planned to not go on a vacation this year. (×)
He planned not to go on a vacation this year. (✓)

AUXILIARY VERBS

These are helping verbs.

1. **May** implies permission, doubt or possibility.
e.g. It may rain tonight.
2. **Might** is the past form of **may** but it does not necessarily represent past time.
Often it implies more doubt than **may**.
e.g. If the clouds are salted, the rains might come.
3. **Can** is used to express ability.
e.g. He can do this work.
4. **Could** is the past form of **can** but it does not necessarily represent past time. Often it implies a more uncertain condition than **can**.
e.g. She could refuse, but she never does.

5. **Should** is the past form of **shall**. It is used in subordinate clauses after **in case** and sometimes after **if**.

e.g. I shall get some money in case brother comes.

It is also used in past sentences with **so that** and **in order that** :

e.g. He turned the stereo down very low so that he should not disturb him.

- **Should** can also be used in sub-ordinate clauses when we are expressing the idea that something must be done or is important. The fact is that this happens after verbs like command, order, request, insist, suggest, advise etc. & after adjectives like important, vital, essential, necessary, eager, anxious, concerned etc.

e.g. I am anxious that nobody should be hurt.

- **Should** is also used in sub-ordinate clauses in sentences where we express personal reactions to events. We express our reactions with words like—amazing, interesting, shocked, sorry, normal, natural, it's a shame etc.

e.g. I am sorry you should think I did it on purpose.

6. **Must** is followed by the infinitive without **to**.

e.g. I must get up at five tomorrow.

Must is used to give strong advice or orders :

e.g. I really must stop drinking.

Must is used to say that we are sure about something :

e.g. I am in love—that must be nice.

Must is used after a past reporting verb.

e.g. I felt there must be something wrong.

7. **Should** and **Ought** have similar meanings, but **ought** is followed by **to**. **Ought to** has a more objective force and is used when we are talking about laws, duties and regulations.

e.g. We ought to see her tomorrow.

Should and **ought to** can also be used to talk about strong probability.

e.g. He has bought thirty pints of whisky—that ought to be enough.

To talk about things which did not happen, although they were supposed to, we use

should and **ought to** with the perfect infinitive :

e.g. The taxi should have arrived at 8.30.

8. **Do** is used to make question and negative forms of ordinary verbs.

e.g. Do you know Rajat ?

Do is used in question-tags and short answers

e.g. Does he know I am here ? —Yes, he does.

Do is used before an imperative to make the request more persuasive

e.g. Do accompany us.

Do is not used in questions which have **who**, **what** or **which** as their subject

e.g. Who said that ?

What happened ?

Which one's that ?

But if **who**, **what** or **which** is the object of the sentence, **do** is used

e.g. Who did you see ?

Which department do you want ?

Do is also used with an **—ing** form when we want to talk of activity that takes a certain time or that is repeated. In this case we find determiners like—the, my, some, much, etc.

e.g. I usually do most of my washing on Sundays.

ERRORS IN USE OF ADJECTIVES

Word qualifying a noun or pronoun is called an **adjective**.

1. The adjectives ending in **—ior** (prior, junior, senior, superior, inferior, posterior) take 'to' and not 'than' after them.

e.g. He is senior to me.

2. Some adjectives like unique, ideal, perfect, extreme, complete, universal, infinite, perpetual, chief, entire, round, impossible are not compared.

e.g. It is the most unique book. (×)

It is a unique book (✓)

3. Comparative degree is used in comparing two things or persons.

e.g. It is the better of the two books.

Superlative degree is used in comparing more than two things or persons.

e.g. He is the best of the three boys.

4. Double comparatives and double superlatives must not be used.

e.g. He is more wiser than his brother. (×)

He is wiser than his brother. (✓)

5. When we compare two qualities in the same person or thing, the comparative ending **—er** is not used.

e.g. You are wiser than old. (×)

You are more wiser than old. (✓)

6. When two adjectives in superlative or comparative degree are used together, the one formed by adding 'more' or 'most' must follow the other adjective.

e.g. He is more intelligent and wiser than his brother. (×)

He is wiser and more intelligent than his brother. (✓)

7. When two adjectives with differing degrees of comparison are used they should be complete in themselves.

e.g. He is as wise, if not wiser than his brother. (×)

He is as wise as, if not wiser than his brother. (✓)

8. When two changes happen together, comparative degree is used in both.

e.g. The higher you go, the cooler you feel.

9. When comparative degree is used in superlative sense, it is followed by 'any other'.
e.g. *Kapil is better than any bowler.* (×)
Kapil is better than any other bowler. (✓)
10. Compound adjective formed by adding 'worth' is placed after the noun it qualifies.
e.g. *This is a worth seeing sight.* (×)
This is a sight worth seeing. (✓)
11. When two or more comparatives are joined by 'and', they must be in the same degree.
e.g. *Russel was one of the wisest and most learned men of the world.*
12. When there are two objects of comparison, then to avoid repetition of noun, 'that' is used for singular noun and 'those' for plural noun.
e.g. *The climate of Ranchi is better than Gaya.* (×)
The climate of Ranchi is better than that of Gaya. (✓)
13. If comparison is made by using 'other', 'than' is used instead of 'but'.
e.g. *He turned out to be no other than my old friend.*
14. **Likely, certain** and **sure** are followed by 'to'.
e.g. *He is likely to win.*

CONFUSED ADJECTIVES

1. **Beautiful** is used for woman; **handsome** for man.
e.g. *He is a handsome youth.*
She is a beautiful girl.
2. **Less** refers to quantity, **fewer** denotes number.
e.g. *He takes no less than a litre of milk.*
They have fewer books than I have.
3. **Last** is the final one; **Latest** is last upto the present.
e.g. *Z is the last letter of the alphabet.*
This is the latest edition of the book.
4. **Each** is used for one of two or more things; **every** is used for more than two things, taken as a group.
e.g. *Each of the two boys was wrong.*
He read every book I gave him.
5. **Older** refers to persons or things.
e.g. *This tree is older than that.*
Elder refers to persons only.
e.g. *He is my elder brother.*
6. **Little** means 'not much'.
A little means 'at least some'.
e.g. *He slept little.*
He slept a little.
7. **Farther** means 'more distant'.
Further means 'additional'.
e.g. *Mumbai is farther than Delhi.*
I shall get further information.

8. **Latter** means the second of two things.
e.g. *Keats and Byron are romantic poets, but I prefer the latter.*

Later refers to time.

e.g. *She came to school later than I.*

ERRORS IN USE OF ADVERBS

Adverbs are words that add information about the verb.

1. Adverbs of manner, place and time are usually placed after the verb or object.
e.g. *He was running slowly.*
Reena does her work carefully.
2. Adverbs of frequency (e.g. never, often, usually, always, rarely, etc.) and other adverbs (like already, almost, just, quite, nearly, hardly) are normally put between subject and verb. If there is more than one word in the verb, they are put after first word.
e.g. *He often goes to Delhi.*
I quite agree with you.
But if verb is 'am', 'is' and 'are', adverb is placed after the verb.
e.g. *I am never late for school.*
3. The adverb **enough** is placed after the adjective.
e.g. *She is cunning enough to tackle him.*
4. 'Ever' is sometimes incorrectly used for 'never'.
e.g. *He seldom or ever tells a lie.* (×)
He seldom or never tells a lie. (✓)
5. Adverb 'not' shouldn't be used with words having negative meaning.
e.g. *The teacher forbade me not to go.* (×)
The teacher forbade me to go. (✓)
6. The word 'only' should be placed immediately before the word it modifies.
e.g. *Hari answered only two questions.*
7. An adverb should not be used before an infinitive.
e.g. *He quickly did the job.* (×)
He did the job quickly. (✓)
8. The auxiliaries *have to* and *used to* come after the adverb.
e.g. *He often used to go to cinema.*

SOME MORE TIPS OF USAGE

1. Avoid the use of **lots, a lot** and **a whole lot** in the sense of **much** or **a great deal**.
e.g. *He expects to earn a lot of money on his sale of farm products.* (×)
He expects to earn a great deal of money on his sale of farm products. (✓)
2. **Plenty** is a noun and is always followed by **of**.
e.g. *He has plenty of room in his old house.*

3. **Due to** always modifies a noun and not a verb. Hence, no sentence should begin with **due to**, it must be used only after some form of the verb **to be**.
e.g. *His death was due to natural causes.*
4. **Alright** is incorrect, use **all right**.
e.g. *I think it is a quite alright if you stay.* (×)
I think it is quite all right if you stay. (✓)
5. If the gender is not determined, use the pronoun of the masculine gender.
e.g. *If anybody has got the book let her return it* (×)
If anybody has got the book let him return it (✓)
6. When two nouns are closely connected, apostrophe —'s will be written after second only.
e.g. *Ravi's and Shikha's mother is ill.* (×)
Ravi and Shikha's mother is ill. (✓)
7. **Both** and **as well as** cannot be used together in the same sentence because both convey the same sense.
e.g. *Both Pinki as well as Pooja are beautiful* (×)
Both Pinki and Pooja are beautiful. (✓)
8. **Supposing** and **if** cannot be used together in the same sentence.
e.g. *Supposing if he fails, what will he do ?* (×)
If he fails, what will he do ? (✓)
9. **Endure** suggests **suffering**, usually in silence and **tolerate**, means to allow it with some degree of approval.
e.g. *He endured the pain without complaint.*
He will not tolerate laziness.
10. Avoid the use of **had** with **ought**.
e.g. *You had not ought to do it.* (×)
You ought not do it. (✓)
11. **As** is used when one compares things or persons of equal or about equal size or quality, **so** is used when one compares things or persons which are unequal.
e.g. *He weighs as much as his father.* (×)
He does not weigh so much as his father. (✓)
12. Avoid using the double comparative and double superlative.
e.g. *This pen is the most costliest in this store.* (×)
This pen is the costliest in this store. (✓)
13. **Cent per cent** and **word by word** are wrong uses. The real terms are **hundred percent** and **word for word**.
14. When two comparatives are used in a sentence for proportion, **the** is used before both of them.
e.g. *The higher we go, the cooler it is.*
15. **Do the needful** is incorrect, write **do what is necessary**.
16. Never say **family members**, say **members of the family**.
e.g. *His family members have gone to Mumbai.* (×)
The members of his family have gone to Mumbai (✓)
17. **Else** is always followed by **but** and never by **than**.
e.g. *It is nothing else but your pride which makes you say such a thing.*
18. **All of** is used in conversation, not in standard written English.
e.g. *She gave all of her antiques to the museum* (×)
She gave all her antiques to the museum (✓)
19. **Talking terms** is wrong, use **speaking terms**.
e.g. *He is not on talking terms with his brother* (×)
He is not on speaking terms with his brother (✓)
20. Verbs like resemble, recommend, comprise, order, accompany, reach, join, shirk, attack, emphasise, discuss, fear, succeed, resist, request, assist, benefit, afford etc. are not followed by any preposition when used in the active voice.
e.g. *He emphasised on the need for discipline.* (×)
He emphasised the need for discipline. (✓)
She will accompany with you to Chennai. (×)
She will accompany you to Chennai. (✓)
21. The phrase **type of**, **sort of** and **kind of** do not follow **a** or **an**.
e.g. *What kind of a friend are you ?* (×)
What kind of friend are you ? (✓)
22. A full hour number follows **o'clock**, but in fraction it does not follow.
e.g. *He left this place by 10.40 o'clock.* (×)
He left this place by 10.40. (✓)
23. **Yet** means up to the present time something that has not happened.
e.g. *His brothers are not married even yet.*

EXERCISE

MCQ TYPE QUESTIONS

Directions (Q. 1 – 20)

In this section you are required to spot errors in sentences. Read each sentence to find out whether there is any error in any of the parts. Errors, if any, are only in one of the parts. No sentence has more than one error. When you detect an error, in any of the parts of the sentence, choose the corresponding alphabet.

1. (a) The scenery here is not good
(b) I have lost my furnitures
(c) We have received no information
(d) He told his mother this news
(e) No error.
2. (a) Please excuse the trouble
(b) He took pains over his work
(c) In India there are many poor
(d) Scouts wear shorts
(e) No error.
3. (a) I spent the holidays with my family
(b) Gone him some blotting paper
(c) Write this in your note book
(d) Gone my kind regards to all
(e) No error.
4. (a) We had a good play of football
(b) We saw a play
(c) We like acting
(d) He is a tall man
(e) No error.
5. (a) The boy was wearing a new suit
(b) He took offence at this
(c) Please put your sign here
(d) Please put your signature here
(e) No error.
6. (a) He is my cousin brother
(b) We are all brothers
(c) I spent the middle of the day working
(d) He had a large amount of money
(e) No error.
7. (a) He got into bad company
(b) One of my servant tells me
(c) Some of my servants tell me
(d) The front of the house
(e) No error.
8. (a) I had a bathe in the sea
(b) I have hurt a toe
(c) Each of these days play games
(d) None of us went
(e) No error.
9. (a) A man should work hard
(b) A man should not waste his time
(c) A boy should not waste his time
(d) Here is my cup; please fill it
(e) No error.
10. (a) Have you a pencil ? I have not got one
(b) Is he coming ? Yes I think so
(c) He enjoyed during the holidays
(d) I asked for my pencil but he did not give it to me
(e) No error.
11. (a) The boy who does best he will get a prize
(b) Whoever does best will get a prize
(c) Who did this ? I
(d) He and I are brothers
(e) No error.
12. (a) I went for a walk with some friends
(b) He is wiser than I
(c) The master tested the boy if he could read English
(d) I shall see whether the brakes work well
(e) No error.
13. (a) Everyone is frightened when he sees a tiger
(b) None of us has seen him
(c) People strave when they have no money
(d) The size of the shoe should be the same as this shoe
(e) No error.
14. (a) Everyone knows this
(b) Every man knows this
(c) These all mangoes are ripe
(d) He hold the book in both hands
(e) No error.
15. (a) I have no any friends
(b) Neither man has come
(c) All idle man should do some work or other
(d) Shakespeare is greater than any other poet
(e) No error.
16. (a) Open your book at page six
(b) He is in class ninth
(c) This article costs ten rupees
(d) He came a second time
(e) No error.
17. (a) King George VI
(b) He is older than I
(c) He is better than I
(d) Raipur is Hotter than Simla
(e) No error.
18. (a) He is worse than I
(b) A horse is more useful than a car
(c) He gets a small salary
(d) I our library the number of books is less
(e) No error.
19. (a) He is the more clever of the two
(b) He is the cleverer of the two
(c) From the three he is more clever
(d) Of the two plans this is the better
(e) No error.

20. (a) He is becoming strong
 (b) There is a very good teacher in that class
 (c) He will spend the rest of his life here
 (d) This is a sight worth seeing
 (e) No error.

Directions (Q. 21 – 30)

Each question below has a sentence, from the choices provided, identify the one which best restates the given sentence and mark its number as the answer.

21. Uneasy lies the head that wears a crown.
 (a) Uneasily lies the head that wears a crown.
 (b) Uneasy is the head wearing a crown.
 (c) The head which wears the crown lies uneasy.
 (d) The head which is wearing the crown lies uneasy.
22. Had I been a little early, I would not have missed the train.
 (a) Had I been a little early, I would not have missed the train.
 (b) Had I been a little early than now, I would not have missed the train.
 (c) Had I been a little earlier, I would not have missed the train.
 (d) Had I only been a little early than now, I would not have missed the train.
23. She is as intelligent as her sister if not more.
 (a) She is more intelligent than her sister.
 (b) She is as intelligent as her sister if not more intelligent.
 (c) She and her sister are equally intelligent.
 (d) She is less intelligent than her sister.
24. You will not succeed unless you are optimistic.
 (a) You will not be successful unless you are optimistic.
 (b) You will not succeed until you are an optimist.
 (c) You will not succeed unless you don't be optimistic.
 (d) You will not succeed unless you don't be an optimistic.
25. The students were advised to follow the instructions of the examiner.
 (a) The instructions of the examiner were followed by the students.
 (b) The students followed the examiner's instructions.
 (c) The examiner's instructions were advised to be followed by the students..
 (d) It has been advised to the students that they should follow the examiner's instructions.
26. Despite being ill, he attended the classes.
 (a) Despite of being ill, he attended the classes.
 (b) Despite his illness, he attended the classes.
 (c) In spite of his being ill, he attended classes.
 (d) The classes were attended by him in spite of being ill.
27. That man is aggressive by nature is a hard fact of life and no one can deny it.
 (a) That man is aggressive by nature is a hard fact of life which none can deny.
 (b) That man is aggressive by nature is a hard fact of life and no one can deny it.
 (c) That man is aggressive by nature is a hard fact of life and not one can deny.
 (d) That man is aggressive by nature is hard for anyone to deny.
28. He is as old as I.
 (a) He is as old as me.
 (b) He and I are equally old.
 (c) He and I are equally older.
 (d) Both he and I are of the same age.
29. A man becomes wiser with age and experience.
 (a) A man as he is aged and experienced becomes wiser.
 (b) A man with age and experience becomes wiser.
 (c) A man as he is aged and experienced becomes the wiser.
 (d) With age and experience, a man becomes wiser.
30. Hazards of life cannot be negated but they can be quite effortlessly evaded.
 (a) Hazards in life can be quite effortlessly evaded and completely nullified.
 (b) Hazards of life cannot be avoided but they can be made ineffective.
 (c) To be made ineffective, hazards of life must be avoided.
 (d) It may not be possible to nullify the hazards of life but they can quite easily be avoided.

Directions (Q. 31 – 40)

Each sentence is broken into four parts a, b, c, d. Mark the part which has an error. Ignore errors of punctuation.

31. (a) Every man, woman
 (b) and child
 (c) in the house on fire
 (d) have been saved.
32. (a) One of the
 (b) best lawyers in town
 (c) have been
 (d) hired.
33. (a) I request you (b) kindly to
 (c) come to me (d) immediately.
34. (a) My friend's mother
 (b) is the principal
 (c) of a
 (d) girl's college.
35. (a) To succeed in these tests
 (b) it is absolutely necessary
 (c) for us
 (d) to aim for speed and accuracy.

36. (a) The aeroplane is a
(b) powerful instrument of war
(c) and their peacetime role
(d) is just as important.
37. (a) The depletion of ozone in the atmosphere
(b) is a reminder to all of us
(c) both in the developed and developing world
(d) that they cannot continue to use contraptions which give out noxious gases.
38. (a) Scotsmen are notorious
(b) for their thrift
(c) and he is the butt
(d) of many exaggerated jokes.
39. (a) A mixed economy
(b) is it in which
(c) the public sector
(d) and the private sector co - exist.
40. (a) Although the beaver's hind feet
(b) are webbed for swimming
(c) their front feet
(d) are small and handlike.

Directions(Q. 41 – 45)

Given below are five sentences, each of which may or may not have errors.

Mark (a), if there is only one error.

Mark (b), if there are two errors.

Mark(c), if there are more than two errors.

Mark (d), if there is no error.

41. "This feels like one big dream," says a weary American woman to her husband, as they sit in a plane flying home.
42. The worst mistake a non - profit organization can make is to take all the money it is given and be beholden to doing things on someone else's terms.

43. Although most hotels in Bhutan has a distinctly local flavour, some pander with the perceived tastes of Western tour groups - and can be rather bland.
44. The crustaceans find sanctuary in the jords, often clinging to the walls in large, gregarious groups.
45. In these tropical lowlands the Maya built an agrarian civilization that eventually supported one of the highest population densities in the pre - industrial America.

Directions (Q. 46–50)

Each of the following questions has two sentences A and B.

Mark (a), if you think sentence A has an error.

Mark (b), if you think sentence B has an error.

Mark (c), if you think both sentences A and B have errors.

Mark (d), if you think neither sentence has an error.

46. A. The boss himself or his secretary answer the phone on Saturday.
B. Neither the quality nor the prices have changed.
47. A. The members re-commended that all delinquents be fined.
B. A stone lying in one position for a long time may gather moss.
48. A. Until I received that letter, I was hoping to have had a visit from Krishnan.
B. Follow the main road for a mile; then you need to take the next road at the left.
49. A. The people to watch closely are the ones ruling behind the political scene.
B. Give the tools to whoever can use them best.
50. A. The magazine has been published continuous since 1951, and it has the funniest cartoons you can possibly imagine.
B. I feel glumly when I read his nightmarish tales.

ANSWERS

MCQ Type Questions

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (b) | 2. (a) | 3. (e) | 4. (a) | 5. (c) | 6. (a) | 7. (b) | 8. (c) | 9. (e) | 10. (c) |
| 11. (a) | 12. (c) | 13. (d) | 14. (c) | 15. (a) | 16. (b) | 17. (d) | 18. (d) | 19. (c) | 20. (a) |
| 21. (c) | 22. (a) | 23. (c) | 24. (a) | 25. (d) | 26. (b) | 27. (a) | 28. (d) | 29. (d) | 30. (d) |
| 31. (d) | 32. (c) | 33. (b) | 34. (d) | 35. (d) | 36. (c) | 37. (d) | 38. (c) | 39. (b) | 40. (c) |
| 41. (a) | 42. (d) | 43. (b) | 44. (d) | 45. (d) | 46. (a) | 47. (d) | 48. (c) | 49. (d) | 50. (c) |

EXPLANATIONS

MCQ TYPE QUESTIONS

21. Uneasily does not denote a state, which is required in the context. 'Wearing a crown' denotes that the head is doing it, 'is wearing' denotes a continuous action, which is impossible, as the head cannot do it. Hence choice (c) restates the given sentence correctly.
22. All the statements have the correct combination of tenses. So, we need not check it. Let us find the errors in other statements. 'Early than now' and 'earlier' suggest that I reached early; in that case, I did not miss the train. So, choices (b), (c), and (d) do not state the same idea as in the question statement. Hence correct choice is (a).
23. The given statement states that both she and her sister are equally intelligent. This is given in statement. Hence correct choice is (c).
24. 'Unless' denotes condition, 'until' denotes time. Choice (b) has adverb clause of time. With unless, another negative is not used. 'Unless don't' is wrong. So, Choices (c) and (d) are incorrect. Hence correct choice is (a).
25. Choices (a) and (b) state the same idea [(a) – passive voice; (b) Active voice] that the students followed the instructions, which differs from the question statement. Instructions cannot be advised. So, choice (c) is meaningless. Choice (d) states the context of the given statement aptly.
26. Despite means inspite of. Despite of is a wrong usage. 'His' need not be used in choice (c), as the sentence is about one person. Choice (d) means that the classes were ill. Hence correct choice is (b).
27. Choice (a) best restates the given sentence, though choice (b) is the repetition of the given statement. 'Not one can deny' in choice (c) and 'for any one to deny' in choice (d) make them incorrect. Hence correct choice is (a).
28. In this type of comparison, the pronouns are in subjective case 'Me', which in objective case is wrongly used in choice (a). Equally is not used with 'old'. Of the same age is the correct expression. Hence correct choice is (d).
29. Choice (a) and (b) mean that the prerequisite for wisdom is age and experience, which is incorrect. Just by growing old and working for a long period, a person cannot improve his wisdom. 'The wiser' is used only when we compare wisdom with some other quality. Choice (d) is the correct restatement of the given one. Hence correct choice is (d).
30. The given statement says that one cannot completely nullify the hazards of life but they can be evaded quite easily. Choice (a) which states that hazards cannot be avoided is wrong (b) states that hazards can be made ineffective and hence this is also wrong (c) does not convey the meaning as stated and hence it too is incorrect. (d) correctly restates the idea and is our answer.
31. Every man refers to a singular noun, therefore.... has been saved.
32. has
33. drop 'kindly'
34. Girl's implies possessive noun, whereas the college is meant for girls.
35. aim at, aim for is wrong usage.
36. delete their - add : its
37. delete they - add : we
38. delete he is - add : they are.
39. delete it - add : that
40. delete their - add : its.
41. One sits 'on' a plane.
42. There are no errors.
43. Hotels 'have' and some pander 'to....'
44. There are no errors in this sentence.
45. There are no errors.
48. (A) hoping to have a visit;
(B) then take the next road to
50. (A) continuously; (B) glum.

2

CHAPTER

Sentence Completion

A Sentence Completion question consists of an incomplete sentence, which has one or two blanks. From the answer choices given, you are asked to choose the appropriate word / words that can take the place of the blanks in order to best complete the meaning of the given sentence.

Directions:

Each of the sentences below has one or two blanks; each blank indicates omission of a word or set of words. Beneath the sentence are given four answer choices, each containing a word or a set of words. Choose the word or set of words for each blank that gives the best meaning to the sentence as a whole.

A thinking man resists the vulgar prosperity that _____ ever to _____ by preserving and communicating heroic sentiments.

- A. disintegrates ... banality
- B. transforms... neutrality . .
- C. decays... luxury
- D. retrogrades ... barbarism
- E. level ...hurtly

The theory underlying a sentence completion stimulus is that the meaning and the tone of a sentence can be gathered without the missing word/ words. If so, it is possible for you to supply the appropriate word(s) to complete the sentence.

Invariably, in every sentence completion question, there will be a clue or two that could be used as lead(s) toward identifying the correct word(s).

These clues could be structural (syntax/grammar clues) or semantic (meaning clues). The word(s) that you choose should agree and merge with the sentence structurally and functionally or semantically.

Every sentence completion question can be analyzed for the negative or positive nature or tone of words to be filled in.

What is measured by a Sentence completion question?

i. Skill of vocabulary

Sentence completion question is a test of your word power and diction. It is only if you know the meaning of word(s) that you will be able to choose confidently, the appropriate one(s) to fill the

incomplete sentence. Moreover the question sentence also may have difficult words in itself, knowing the meaning of which is essential for understanding the sentence holistically and thus to choose the appropriate word(s).

Look at the following question:

The department can use information received from dissatisfied employees and __ to launch an investigation against the company.

- A. pettifoggers
- B. buccaneers
- C. ambassadors
- D. whistleblowers

The knowledge of the meaning of the words in choices is essential for you to select the answer to this question with confidence. Let's analyze meaning of these words.

A Pettifogger is one who practices chicanery or unlawful business. Can this word fit the sentence? No.

A Buccaneer is a pirate; again not a word that would go with dissatisfied employees to supply information about a company.

A whistleblower is one who informs about corruption or wrongdoing. Yes, this fits the sentence.

An Ambassador is not one to do anything with a company. In fact this is the first choice you could eliminate.

ii. Skill of semantics and syntax

Every sentence has a semantic dimension (meaning) and a syntactical dimension (word arrangement). Knowledge of these two is a mandatory condition for you to be able to choose the appropriate word. You must understand the meaning of a sentence in order to identify the appropriate choice word that fills the blank, and completes the sentence coherently.

You should have a good understanding of the functions of structural words in a sentence such as unless, rather, yet, while unfortunately, in contrast, despite.

Look at the following question:

The meaning of words is destroyed if you change them, whereas the meaning of a thought _____ whatever word you think proper to use.

- A. alters
- B. transforms
- C. remains
- D. settles
- E. Ruins

When you read the sentence you must recognize the function of 'whereas' in the sentence. It signals a shift in the meaning of the sentence.

'One thing is destroyed whereas the other is not' the answer thus is (C) remains.

The semantic feature of a sentence refers to its meaning, and how the meaning of the given words in a sentence helps us to identify the word to complete the sentence.

Study the following sentence:

A cult brand _____ the imagination of a small group who spreads the word, makes converts, helps turn a fringe product into a _____ name.

- A. ceases... redoubtable
- B. elevates ... dynamic
- C. fires... dream
- D. seizes... mainstream

The first clue is the word Imagination; hence options (A) and (D) it, ceases and seizes can easily be eliminated. You have to work for fires and elevates. To fire One's Imagination is a standard idiomatic usage. Moreover the meaning of the sentence implies turning a fringe (peripheral) product into (the opposite of it) a dream product. Hence, answer is (C).

iii. Skill of identifying clues

A sentence completion question is not merely a test of your vocabulary. It also tests your ability to analyze the given sentence and to identify the structural and semantic clues that decide on the meaning and nature of the word(s) to be chosen.

You will find one or two clues in a sentence completion question. Common key words such as and or but are all more valuable for us in sentence completion. These are structural clues. (These are dealt with in detail later)

Sentence Completion Strategies

In order to answer sentence completion question correctly, you should first read the entire sentence as you see it, without trying to fill the word(s). Desist from working on each blank separately. The second part of the sentence may have a bearing on its first

part. And both words must fit into the meaning of the sentence; therefore read your choice into the sentence supplying and evaluating both words.

Each sentence contains a few crucial clues that determine the answers, and finding these clues will guide you to the correct answer. Look for what is directly stated. In sentence completion question, you may also find poetic and literary language, which tests your understanding of figurative usage. The correct answer is the one that is directly implied in the sentence.

- Determine where the sentence is heading, using structural clues.
- Anticipate answers in your words.
- Look for answers that are similar to yours.
- Fix the words in the sentence to see if they get with the whole sentence.

Sentence completion strategy in a nutshell.

➤ Strategy 1

Read the entire sentence.

In both one- and two-blank sentences read the entire sentence before you inspect the choices. An overall understanding of the sentence is essential.

➤ Strategy 2

Identify structural clues to determine the nature of the word(s) required.

You can determine the positive or negative tone of the expected word using structural clues. And identify the words whose synonym, antonym, parallel word or cause/effect you have to supply.

➤ Strategy 3

Substitute your own words.

Place your own words in the blanks assuming that answer choices are not given, going by the clues you have identified.

➤ Strategy 4

Identify the words similar to the ones you substituted.

If the word you have substituted goes with the requirements of the sentence, you are sure to find, a word(s) similar in meaning and tone to the ones you have substituted.

➤ Strategy 5

Read the sentence again with word(s) filled in.

Now read the entire sentence with the words filled in to see if the word(s) fit literally and figuratively.

A. THOUGHT CONTINUITY CLUES

In some sentences, the first part of the sentence and the second part are parallel ideas. There is no shift in meaning. The missing word in the sentence is parallel to, or has a meaning similar to a word already given in the sentence. One part of the sentence may be a support to another. Some other times, though, one part of the sentence may be complementary to the other in meaning. These clues continue the sentence in the same direction. The positive or negative tone of what follows is not changed by these clue words.

The words that indicate such continuity are:

- Similarly
- Because
- And
- Since
- Thus
- In fact
- Moreover
- Indeed
- In addition
- Consequently

Study the following sentences:

1. The film script showed depth and maturity; likewise the treatment showed profundity.
2. His argument was convincing and persuasive.
3. Shakespeare lives forever; in fact he is immortal.

B. PARALLEL IDEA CLUES

These clues are similar to the thought continuity clues. The word that completes the sentence is in the same tone, as in the word already given. The word(s) that complete the sentence complements a word or phrase already given.

- i. All that the young ask of the political class is compassion and concern/ sympathy/ care.
- ii. On question about political and electoral reforms, the respondents showed great forethought and vision/rationality.

In the above sentences you may observe that the words given and the words that have to be filled in are parallel/ complementary or in the same tone.

C. THOUGHT REVERSAL CLUES

These terms change the direction of the sentence. They make one part of the sentence contradictory to the other part. The word or phrase that completes the sentence is opposite in meaning to a word or phrase already given. The positive or negative charge

of an answer is changed by these clues.

Such sign – word include:

- But
- On the contrary
- In spite of
- Yet
- However
- Unless
- Rather
- Although
- While
- Unfortunately
- Nonetheless
- Ironically
- Paradoxically
- On the other hand
- Contradictorily
- Contrastively
- Despite
- Even though
- Even if

Illustrations:

His argument was flawed but persuasive.

There is no cynicism in the way the young deal with issues; on the contrary they are optimistic toward life.

The emphasis will not be on physical assets but on intangibles.

In the sentences above the word in italics are structural clues, which indicate that the words to be filled in are opposite in meaning to the words, that are already given in the sentences.

D. CAUSE EFFECT CLUES

While studying a sentence completion question, you should be able to gauge as to which direction the sentence is heading. If the sentence has an idea, it is possible to infer what would logically follow the causal phenomenon. In other words, we can anticipate the type of effect that the given cause would lead to. Similarly by analyzing the nature of an effect (sentences in which effect is given), we can also infer the plausible causal factor that would've contributed to the type of effect given. The easiest way is to

identify the tone of the cause or effect to be supplied into the sentence-negative or positive

Let's work on sentences with cause effect relationship.

Torture is so prevalent and _____ of the helpless so common that these helpless beings are certainly _____

When they pair out their story of agony.

- (1) indemnifications... unhinged
- (2) felicitation... believable
- (3) disparaging ...credulous
- (4) victimization ... credible

In the above sentence you observe that there are two clues: parallel and causal. Signaled by the word and the first blank needs a word parallel to torture i.e. victimization. The intensity of these two results in the possibility of the stories believable or credible.

Look at another example:

Higher education in India can no longer be allowed to remain as a/ an _____ instrument that crushes the spirit of the individual while goading him into a _____ and consumerist mode.

- (1) innervating... boisterous
- (2) emasculating... materialist
- (3) invigorating ... pragmatic
- (4) relegating ... humanistic

An instrument that crushes the spirit of the individual cannot be innervating or invigorating (both means energizing) nor can it be riveting; since rivet means to hold firmly and relegate means to lower in rank is closer but the latter word cannot be humanistic (3) The word that is parallel to consumerist is materialist.

In a similar mode identify the cause effect ideas in the incomplete sentences in the following drill and choose appropriate words to complete them.

E. WORD DEFINATION CLUES

Yet another common clue found in sentence completion stimuli is the meaning or definition of the word to be filled given as part of the incomplete sentence itself. It is similar to the thought continuity clue, except that in case the meaning of the word or definition of the word to be selected is already given.

Let us study an example :

The great irony of Apple Computers is that very posture that won Steve jobs' a cult following creating an exclusive group of _____ technicians who _____ refused to bow to IBM, hobbled his company.

iconoclastic.....stubbornly

conservative.....invariably

rebellious.....timidly

traditional.....outrightly

In the sentence above you would've already observed the words 'refused to bow', which signals the word to be placed in the second blank. This phrase clearly defines the word stubbornly.

Another example :

The crisis in modern education is not merely a constraint of _____ of funds but of _____ practices, unsuited for contemporary system.

scarcity.....doctrinaire

fulsomeness.....authoritative

restrictiveness.....autocratic

paucity.....anachronistic

In the above sentence the key to the letter word is 'unsuited for contemporary education', you can now easily eliminate unsuitable choices and arrive at choice D, i.e., anachronist.

EXERCISE

MCQ TYPE QUESTIONS

Directions for Q. 1 to 41 : In each of the following sentences, a part/parts of the sentence is left unfinished. Beneath each sentence, four different ways of completing the sentence are indicated. Choose the best alternative among the four.

1. For some of our ruling politicians, social justice has apparently come to mean that _____.
 (a) they have a right to plunder public money from the treasury
 (b) they have a right to the plunder of public money from the treasury
 (c) theirs is the right to plunder public money from the treasury
 (d) the right is theirs to plunder public money from the treasury
2. Some critics believe that Satyajit Ray never quite came back to the great beginning he made in his path-breaking film Pather Panchali. _____ have endured decades of well-travelled bad prints to become a signpost in cinematic history.
 (a) The bizarre history of its misty origins
 (b) Its haunting images
 (c) Its compelling munificence
 (d) The breathtaking awe it inspires
3. _____, the more they remain the same.
 (a) The less the dynamism
 (b) The more things change
 (c) The more pronounced the transformation
 (d) The more the merrier
4. Revenues are likely to register a dramatic increase if _____ on the hundred million people who are said to comprise the rising Indian middle class.
 (a) a flat tax is collected
 (b) a flat tax is placed
 (c) taxes at a flat rate are charged
 (d) tax is imposed at a flat rate
5. I am an entertainer. _____ I have to keep smiling because, deep in my heart, laughter and sorrow have an affinity.
 (a) Even if I have tears in me
 (b) Despite conditions of extreme adversity
 (c) In spite of misery around me
 (d) Although I have yet to make it big
6. The stock market is probably _____. And the way the markets are plunging says a lot about investor confidence.
 (a) an ideal indication of the health of public sentiment
 (b) the least imperfect mechanism for judging the quantity of the sentiment of the public.
 (c) the best indicator of public sentiment
 (d) the best barometer to assess the sentiment of the public
7. Home is _____ one has yet to improve.
 (a) a discovery over which
 (b) a certain over which
 (c) an invention on which
 (d) an institution upon which
8. The highest reward for a man's toil is not what he gets for it but what _____.
 (a) he makes out of it.
 (b) he gets for others.
 (c) he has overcome through it.
 (d) he becomes by it.
9. An act of justice closes the book on a misdeed; an act of vengeance _____.
 (a) reopens the first chapter
 (b) writes an epilogue
 (c) writes one of its own
 (d) opens new books
10. The trail is the thing, not the end of the trail. Travel too fast and you miss _____.
 (a) all you are travelling for.
 (b) all the sights you are supposed to see.
 (c) the very excitement of your travel.
 (d) all the enjoyments of travel.
11. _____ that in this apparent mess, two things not be interfered with.
 (a) It is important.
 (b) it is of cardinal importance.
 (c) It should be urgently understood.
 (d) It cannot be emphasised.
12. A country's wealth is its people. But instead of drawing out the strengths of the people, instead of drawing out their talent, this use of religion debases, degrades and depresses than _____.
 (a) in greater and greater measure
 (b) further
 (c) beyond reasonable limit
 (d) more and more
13. Wines that yielded a good commercial profit _____ in the same limited areas of France as of now.
 (a) seem to have been produced
 (b) appear to have a remarkable semblance
 (c) bear a significant similarity in terms of production to those grown
 (d) appear to have been similarly produced
14. This is about _____ a sociological analysis can penetrate.
 (a) the other limits that
 (b) just how far into the subject that
 (c) the relative distance that
 (d) as far as

15. The safest general characterisation of the European philosophical tradition as it has developed up to now, with all its diverse proponents, is that it consists of a _____ Plato.
 (a) series of footnotes to
 (b) set of prologues to
 (c) collection of chapters on
 (d) string of commentaries to
16. The ideas that these companies used seem so simple with _____ that their competitors will now _____ themselves for not thinking of them first.
 (a) advanced technology, hit
 (b) analysis, abuse
 (c) passage of time, curse
 (d) hindsight, kick
17. No doubt, it was our own government but it was being run on borrowed ideas, using _____ solutions.
 (a) worn out (b) second hand
 (c) impractical (d) appropriate
18. Overall, all the recent policy changes by the government only amount to a _____ in the sugar industry.
 (a) superficial attempt at liberalisation
 (b) drop in the ocean
 (c) large change
 (d) small regulating authority
19. In pursuance of their decision to resist what they saw as anti-labour policies, the company employees, union launched agitation to _____.
 (a) show their virility
 (b) reaffirm their commitment to the company.
 (c) bring down the government
 (d) demonstrate their strength
20. Scientists have found _____ evidence that Neanderthals, now an extinct species of humans _____ Europe, were cannibals.
 (a) grisly, living in
 (b) incontrovertible, a cross
 (c) chilling inhabiting
 (d) proper, in
21. The telephone symbolises that awkward _____ in all communication technologies ; while it _____ to bring us together, it keeps us apart.
 (a) paradox, needs (b) irony, intends
 (c) paradox, tries (d) irony wishes
22. The world is fast moving into a period of 'civilisational clash' in which the primary identification of people and countries will not be _____ as during the Cold War, but _____.
 (a) ideological, technological
 (b) cultural, ethnic
 (c) political, economic
 (d) ideological, cultural
23. In our country, the challenges are to raise _____ incomes to reduce poverty, and to _____ inefficient public sector enterprises.
 (a) farm, liberalise
 (b) middle-class, privatise
 (c) workers, take over
 (d) rural, restructure
24. The interest generated by the Soccer World Cup is _____ compared to the way cricket _____ the nation.
 (a) milder, fascinates
 (b) lukewarm, electrifies
 (c) tepid, inspires
 (d) unusual, grips
25. Though one eye is kept firmly on the _____, the company now also promotes _____ contemporary art.
 (a) present, experimental
 (b) future, popular
 (c) present, popular
 (d) market, popular
26. The law prohibits a person from felling a sandalwood tree, even if it grows on one's own land, without prior permission from the government. As poor people cannot deal with the government, this legal provision leads to a rip-roaring business for _____, who care neither for the _____, nor for the trees.
 (a) middlemen, rich (b) the government, poor
 (c) touts, rich (d) touts, poor
27. It will take some time for many South Koreans to _____ the conflicting images of North Korea, let alone to _____ what to make of their northern cousins.
 (a) reconcile, decide (b) understand, clarify
 (c) make out, decide (d) reconcile, understand
28. In these bleak and depressing times of _____ prices, non-performing governments and _____ crime rates, Sourav Ganguly has given us, Indians, a lot to cheer about.
 (a) escalating, increasing
 (b) spiralling, booming
 (c) spiralling, soaring
 (d) ascending, debilitating
29. The manners and _____ of the nouveau riche is a recurrent _____ in the literature.
 (a) style, motif
 (b) morals, story
 (c) wealth, theme
 (d) morals, theme

30. Many people suggest _____ and still others would like to convince people not to buy pirated cassettes.
- (a) to bring down audio-cassette prices to reduce the incidence of music piracy, others advocate strong legal action against the offenders,
 - (b) bringing down audio-cassette prices to reduce the incidents of music piracy, others are advocating strong legal action against offenders,
 - (c) bringing down audio-cassette prices to reduce the incidence of music piracy, others advocate strong legal action against offenders,
 - (d) audio-cassette prices to be brought down to reduce incidence of music piracy, others advocate that strong legal action must be taken against offenders,
31. The ancient Egyptians believed _____ so that when these objects were magically reanimated through the correct rituals, they would be able to function effectively.
- (a) that it was essential that things they portrayed must have every relevant feature shown as clearly as possible
 - (b) it was essential for things they portray to have had every relevant feature shown as clearly as possible,
 - (c) it was essential that the things they portrayed had every relevant feature shown as clearly as possible,
 - (d) that when they portrayed things, it should have every relevant feature shown as clearly as possible
32. Archaeologists believe that the pieces of red-ware pottery excavated recently near Bhavnagar and _____ shed light on a hitherto dark 600-year period in the Harappan history of Gujarat.
- (a) estimated with a reasonable certainty as being about 3400 years old,
 - (b) are estimated reasonably certain to be about 3400 years old
 - (c) estimated at about 3400 years old with reasonable certainty,
 - (d) estimated with reasonable certainty to be about 3400 years old,
33. The genocides in Bosnia and Rwanda, apart from being mis-described in the most sinister and _____ manner as 'ethnic cleansing', were also blamed, in further hand-washing rhetoric, on something dark and interior to _____ and perpetrators alike.
- (a) innovative; communicator
 - (b) enchanting; leaders
 - (c) disingenuous; victims
 - (d) exigent; exploiters
34. As navigators, calendar makers, and other _____ of the night sky accumulated evidence to the contrary, ancient astronomers were forced to _____ that certain bodies might move in circles about points, which in turn moved in circles about the earth.
- (a) scrutinizers; believe
 - (b) observers; agree
 - (c) scrutinizers; suggest
 - (d) observers; concede
35. Every human being, after the first few days of his life, is a product of two factors: on the one hand, there is his _____ endowment; and on the other hand, there is the effect of environment, including _____.
- (a) constitutional; weather
 - (b) congenital; education
 - (c) personal; climate
 - (d) genetic; pedagogy
36. Exhaustion of natural resources, destruction of individual initiative by governments, control over man's minds by central _____ of education and propaganda are some of the major evils which appear to be on the increase as a result of the impact of science upon minds suited by _____ to an earlier kind of world.
- (a) tenets; fixation
 - (b) aspects; inhibitions
 - (c) institutions; inhibitions
 - (d) organs; tradition
37. But _____ are now regularly written not just for tools, but well-established practices, organisations and institutions, not all of which seem to be _____ away.
- (a) reports, withering
 - (b) stories, trading
 - (c) books, dying
 - (d) obituaries, fading
38. The Darwin who _____ is most remarkable for the way in which he _____ the attributes of the world class thinker and head of the household.
- (a) comes, figures
 - (b) arises, adds
 - (c) emerges, combines
 - (d) appeared, combines
39. Since her face was free of _____ there was no way to _____ if she appreciated what had happened.
- (a) make-up, realise
 - (b) expression, ascertain
 - (c) emotion, diagnose
 - (d) scars, understand

40. In this context, the _____ of the British labour movement is particularly _____.
 (a) affiar, weird
 (b) activity, moving
 (c) experience, significant
 (d) atmosphere, gloomy
41. Indian intellectuals may boast, if they are so inclined, of being _____ to the most elitist among the intellectual _____ of the world.
 (a) subordinate, traditions
 (b) heirs, cliques
 (c) ancestors, societies
 (d) heir, traditions

Directions for Q. 42 to 47.

Fill the gaps in the passages below with the most appropriate word from the options given for each gap. The right words are the ones used by the author. Be guided by the author's overall style and meaning when you choose the answers.

Von Nuemann and Morgenstern assume a decision framework in which all options are thoroughly considered, each option being independent of others, with a numerical value derived for the utility of each possible outcome (these outcomes reflecting, in turn, all possible combinations of choices). The decision is then made to maximize the expected utility.

(42) _____, such a model reflects major simplifications of the way decisions are made in the real world. Humans are not able to process information as quickly and effectively as the model assumes; they tend not to think (43) _____ as easily as the model calls for; they often deal with a particular option without really assessing its (44) _____, and when they do assess alternatives, they may be extremely nebulous about their criteria of evaluation.

42. (a) Regrettably (b) Firstly
 (c) Obviously (d) Apparently
43. (a) quantitatively (b) systematically
 (c) scientifically (d) analytically
44. (a) implications (b) disadvantages
 (c) utility (d) alternatives

In a large company, (45) _____ people is about as common as using a gun or a switch-blade to (46) _____ an argument. As a result, most managers have little or no experience of firing people, and they find it emotionally traumatic; as result, they often delay the act interminably, much as an unhappy spouse will prolong a bad marriage. And when the firing is done, it's often done clumsily, with far worse side effects than are necessary.

Do the world-class softward organizations have a different way of firing people ? No, but they do the deed swiftly, humanely, and professionally.

The key point here is to view the fired employees as a "failed product" and to ask how the process (47) _____ such a phenomenon in the first place.

45. (a) dismissing (b) punishing
 (c) firing (d) admonishing
46. (a) resolve (b) thwart
 (c) defeat (d) close
47. (a) derived (b) engineered
 (c) produced (d) allowed

Directions for Q. 48 to 57 :

Fill up the blanks, numbered [48], [49]...up to [57], in the two passages below with the most appropriate word from the options given for each blank.

"Between the year 1946 and the year 1955, I did not file any income tax returns." With that [48] statement, Ramesh embarked on an account of his encounter with the Income Tax Department. "I originally owed Rs. 20,000 in unpaid taxes. With [49] and [50], the 20,000 became 60,000. The Income Tax Department then went into action, and I learned first hand just how much power the Tax Department wields. Royalties and trust funds can be [51]; automobiles may be [52], and auctioned off. Nothing belongs to the [53] until the case is settled."

48. (a) devious (b) blunt
 (c) tactful (d) pretentious
49. (a) interest (b) taxes
 (c) principal (d) returns
50. (a) sanctions (b) refunds
 (c) fees (d) fines
51. (a) closed (b) detached
 (c) attached (d) impounded
52. (a) smashed (b) seized
 (c) dismantled (d) frozen
53. (a) purchaser (b) victim
 (c) investor (d) offender

At that time the White House was as serene as a resort hotel out of season. The corridors were [54]. In the various offices, [55] gray men in waistcoats talked to one another in low-pitched voices. The only colour, or choler, curiously enough, was provided by President Eisenhower himself. Apparently, his [56] was easily set off; he scowled when he [57] the corridors.

54. (a) striking (b) hollow
 (c) empty (d) white
55. (a) quiet (b) faded
 (c) loud (d) stentorian
56. (a) laughter (b) curiosity
 (c) humour (d) temper
57. (a) paced (b) strolled
 (c) stormed (d) prowled

ANSWERS

MCQ Type Questions

1. (a)	2. (b)	3. (b)	4. (d)	5. (b)	6. (c)	7. (d)	8. (d)	9. (c)	10. (c)
11. (a)	12. (d)	13. (a)	14. (d)	15. (a)	16. (d)	17. (b)	18. (a)	19. (d)	20. (c)
21. (3)	22. (c)	23. (d)	24. (b)	25. (b)	26. (d)	27. (a)	28. (c)	29. (d)	30. (c)
31. (c)	32. (d)	33. (c)	34. (d)	35. (d)	36. (d)	37. (d)	38. (c)	39. (b)	40. (c)
41. (d)	42. (d)	43. (d)	44. (d)	45. (c)	46. (a)	47. (d)	48. (b)	49. (a)	50. (d)
51. (c)	52. (b)	53. (d)	54. (c)	55. (a)	56. (d)	57. (d)			

EXPLANATIONS

MCQ TYPE QUESTIONS

1. They in choice (a) means 'some of our ruling politicians' (the subject). Also grammatically 'they have a right to plunder' is correct whereas '..... right to the plunderer' is incorrect.
2. 'Bizarre' means 'weird'. Hence, choice (a) is ruled out. "Its haunting images have endured" fits syntactically. choice (c) can also be ruled out as 'munificence' means generosity and this is not intended in the sentence. Also, the word 'prints' in the given sentence goes with the word 'images' of choice (b).
3. 'They' in the part after the comma must refer to somebody or something. Hence, only choice (b) will fit as 'they' in that case refers to 'things'.
4. The right word to precede the proposition 'on' is imposed. Tax is not collected on people. Tax is not placed on people, Tax is not charged on people. Hence, choices (a), (b) and (c) are not the answers. Tax is imposed on
5. Choice (d) is ruled out as it does not logically fit. Choice (a) is also ruled out because of the expression 'tears in me'.
6. The key words are 'plunging' and 'investor confidence'. If markets are plunging, investor confidence is low. This aspect is brought out best by choice (c).
7. (a) 'Discovery' refers to something that existed previously and came to knowledge later. Home is not a discovery.
(b) Home is not a certain. It is a set up which comes into being as a result of a certain composition of various factors.
(c) Home is not a new invention of anyone at any stage of development.
So choice (d) is the best answer. Home is an institution just like 'marriage'. Home is a reality only when its members accept certain principles.
8. (a) A man's toil may help to make out wealth, but it is not the ultimate aim.

(b) A man's toil may secure a lot of things for his children or friends. That too is not something great.

(c) A man's toil may help him to overcome many difficulties, but that does not give him much pleasure.

So, choice (d) is the best answer. What one becomes by one's toil is greater than what one gets.

9. (a) 'Reopens the first chapter' means begins from the very start.

(b) 'Writes an epilogue' means ends the book.

(c) 'Opens new books' means creates new problems.

The above three are only partly correct. An act of justice pardons for ever an act of vengeance creates new problems. It is a saying. "An act of justice closes the book on a misdeed; an act of vengeance writes one of its own."

10. Here, the trail is the important thing and hence the excitement of the trail is what you will be looking for.

11. (b) The expression must be "it is of cardinal importance".

(c) There is no urgency in understanding, "It should be understood" is enough.

(d) Instead of "it cannot be," the expression should be "it must be".

So choice (a) is the best. It is important that the two things are not to be interfered with.

13. (b) It is not mentioned with what do wines bear semblance.

(c) The 'similarity' is only in terms of production. The similarity in production does not achieve commercial profit.

(d) For the same commercial profit wines need not be produced in the same of similar manner.

So, choice (a) is the best. It does not mention that the wines of similar profit are similarly produced or have semblance with one another.

14. (a) A sociological analysis does not stop at the outer limits of the subjects.
(b) No one can indicate how far into the subject can a sociological analysis go.
(c) 'Relative distance' refers to the distance of something else which is not mentioned.
So, choice (d) is the best. This indicates the probable distance to which the sociological analysis can go to.
30. Choices (a) and (d) are ruled out because 'suggest' does not go with 'to+ verb' form. Choice (2) cannot fit in the blank because of the error in parallelism. The appropriate structure is "Many people suggest.....and others advocate".
31. The tense in choices (a) and (b) is not consistent. Moreover, choice (a) uses both 'essential' and 'must', thus rendering either one of these words redundant. In choice (d) 'things' the plural form cannot take it'.
32. Choice (d) fits into the blank in choice (a) the usage of the indefinite article is erroneous. Choices (b) and (c) are erroneous due to the incorrect ordering of words.
33. 'Disingenuous' means 'insincere' and this best fits the context as the phrase 'ethnic cleansing' is marked by apostrophes indicating a misrepresentation. Also, the word 'victims' is best in contrast to 'perpetrators'. Hence, option (c) is correct.
36. 'Navigators, calendar makers' are best clubbed under the category of 'observers' rather than 'scrutinizers', as the latter term is more used to refer to specialists in a particular field. 'Students' can also fit to a certain extent and this leaves us with options (b), (d). 'Concede' has more to do with defeat or revision which is more apt in this context than 'agree' or 'conclude'. Hence, option (c) is correct.
42. Going by options, 'regrettably' doesn't fit as there is nothing to regret. 'Firstly' is no choice and nothing 'secondly' is mentioned. Paragraph has nothing aggressive tone so 'obviously' does not suit. 'Apparently' is only and most suitable option.
43. All options are suitable for the space but to choose one, the best, is required here 'Analytically' appears to be best. It expresses to analyse something using statistically proven methods to reach a decision.
44. It is a very obvious choice determined from the paragraph giving it an access.
45. 'Firing' comes as best suited option as paragraph explains it as a subject of discussion.
46. An argument is always 'resolved', it is never thwart, defeat or closed.
47. Going by options makes choice easier. Only "how the process allowed....." suits logically.
48. 'Blunt' means 'very direct' with respect to a remark. 'Devious' in choice (d) means 'dishonest' and is incorrect as the statement is a frank one. For the same reason 'tactful' in choice (c) is also not appropriate.
49. 'Interest' is the most appropriate work because the original amount he owed to the department accumulated to the said figure. 'Taxes' is redundant while 'principal' usually refers to a loan (in this case it isn't). 'Returns' is what one gets rather than pays.
50. 'Sanctions' is used for an official permission or order that limits trade and is rarely used for an amount. 'Fees' are paid for a service and 'refunds' are what one gets back. 'Fines' is the best choice because of the penalty component.
51. 'Attach' is the word used for officially taking something away. 'Impound' means 'confiscate' and is generally used in the context of illegal goods or contraband. Other choices are in no way comparable.
52. 'Smashed' in choice (a) and 'dismantled' in choice (c) are far from being logical. 'Frozen' is a wrong choice. 'Frozen' is used in the context of making bank accounts inoperable etc. Automobiles are seized.
53. Clearly the topic is about a person who has violated income tax regulations and he should be called an 'offender' of the law.
55. Choices (c) and (d) have similar meaning and are misnomers in the given context, as the keywords 'lowpitched voices' require a word that is synonymous with 'quiet' or 'calm'. The usage of the adjective 'faded' in the sentence to describe men is untenable.
56. The given sentence depicts the President as a very angry person as suggested by the keyword 'scowled'. Choices (a), (b) and (c) are either positive or neutral and cannot fit in the sentence.
57. 'Paced' in choice (a) is not consistent with the mood of the President as depicted in the first part of the sentence. It means 'to walk fast'. 'Strolled' in (b) means 'Walk casually' and hence is discarded. 'Stormed' in choice (c) refers to 'attack a place suddenly' or 'say something in a loud, angry way'. But here, neither of them fits in. 'Prowled' refers to moving in a restless manner and hence fits.

3

CHAPTER

Synonyms

INTRODUCTION

A Synonym may be defined as a word that has the same meaning or the same general meaning as a particular word in the same Language or is in some applications a more or less satisfactory application of it. Thus the synonym of a word can be an exact equivalent in meaning or should atleast convey its meaning in one way or the other or should show in what sense it can best be used.

SOME IMPORTANT WORDS WITH THEIR SYNONYMS

A

abandon	: desert, for sake, jilt, leave, evacuate, quilt, vacate, abdicate, renounce.	acquaint	: advice, announce, apprise, disclose, enlighten, notify, reveal, tell.
abduct	: kidnap, run away with, seize, snatch.	acid	: caustic, harsh, irritating, pungent, stinging, acrimonious, mordant, nasty, sarcastic, trenchant, vitriolic.
abet	: aid, assist, back, condone, help, promote, sanction, support, sustain, uphold.	acrimony	: bitterness, harshness, churlishness, peevishness, rancour, spleen, tartness, virulence.
abeyance	: adjournment, deferral, discontinuation, inactivity, intermission, postponement, recess.	adamant	: determined, firm, immovable, inexorable, intransigent, obdurate, resolute, stubborn, unrelenting, flinty, steely, stony.
abide	: accept, bear, brook, endure, dwell, linger, live, lodge, reside, sojourn, persist.	adept	: adroit, dexterous, masterful, proficient, versed, expert, genius.
ability	: adeptness, aptitude, capability, capacity, competence, dexterity, endowment, expertise.	adequate	: capable, commensurate, competent, requisite, tolerable, sufficient.
ablaze	: afire, aflame, alight, blazing, burning, fiery, flaming, aglow, brilliant, flashing, gleaming, illuminated, radiant, sparkling.	adhere	: cleave, cling, faster, obey, observe, respect.
abstain	: avoid, decline, deny, desist, renounce, refrain, withhold, forgo.	adieu	: conge, goodbye, valediction.
abundance	: affluence, ampleness, bounty, copiousness, exuberance, pro-fusion, fortune, wealth.	adjourne	: defer, delay, discontinue, prorogue.
abysmal	: bottomless, boundless, complete, deep, extreme, incalculable, profound, vast.	admirable	: commendable, excellent, exquisite, wonderful, worthy.
accede	: accept, admit, agree, comply, concede, concur, consent, endorse, assume, attain.	admirer	: bean, lover, wooer, devotee, enthusiast, votary.
access	: admission, admittance, approach, avenue, course, entering, outburst, paroxysm.	adrift	: drifting, unanchored, aimless, amiss, astray.
acclaim	: applaud, approve, celebrate, eulogize, exalt, hail, praise, salute, clapping.	adroit	: clever, cunning, deft, dexterous, ingenuous, nimble, proficient, quick-witted, able.
accompany	: attend, chaperon, conduct, convoy, squire, usher, supplement.	audacious	: courageous, daring, dauntless, intrepid, rash.
accord	: agree, assent, concur, conform, harmonize, bestow, concede, endow, render, vouchsafe.	austere	: exacting, forbidding, grave, grim, harsh, rigorous, severe, solemn, stern, stringent.
accuse	: allege, blame, cite, denounce, impench, incriminate, indict, recriminate.	averse	: antipathetic, disinclined, hostile, inimical, loath, reluctant, unwilling.
accustom	: acclimatize, acquaint, adapt, discipline, familiarize, habituate, inure.	avid	: ardent, eager, fanatical, ferrent, passionate, zealous, avaricious,
			B
		baptism	: immersion, purification, sprinkling, debut.

bastion : citadel, defence, fortress, mainstay.
 blaspheme : abuse, anathematize, curse, desecrate, execrate, profane, revile, swear.
 breach : break, aperture, charm, cleft, fissure, infraction, infringement, estrangement.
 broaden : augment, develop, fatten, spread, stretch.
 brood : agonize, dwell, mope, ponder, repine, ruminate.

C

catastrophe: affliction, adversity, cataclysm, devastation, disaster, fiasco, culmination.
 chaos : anarchy, bedlam, confusion, disorganization, entropy, tumult, lawlessness,
 chastity : celibacy, continence, innocence, modesty, purity.
 clan : band, brotherhood, fraternity, gens.
 clinch : assure, conclude, confirm, determine, secure.
 colloquial : conversational, demotic, everyday, familiar, idiomatic, informal, vernacular.
 commotion: ado, agitation, bustle, disorder, disturbance, excitement.
 complicate : confuse, entangle, involve, muddle.
 contour : curve, figure, lines, profile, relief.
 conviction : assurance, certainty, confidence, ferrous.
 creditable : admirable, commendable, deserving, estimable, honourable, loudable.
 crest : apex, zenith, pinnacle, summit, ridge.
 culminate : climax, close, conclude, finish.
 cynical : sarcastic, scornful, sneering, distrustful.

D

daunt : dismay, alarm, appal, frighten, intimidate, scare, subdue, terrify.
 deed : achievement, act, action, exploit, feat.
 deflate : collapse, constrict, empty, exhaust, shrink, chasten.
 deliberate : cogitate, consult, debate, mediate, ponder, reflect.
 diligent : active, attentive, busy, careful, constant.
 dire : alarming, appalling, awful, catastrophic, cruel, disastrous.
 discreet : careful, cautious, circumspect, diplomatic, judicious, politic, prudent.
 dissent : decline, differ, disagree, protest, refuse, disagreement, discord.
 ditch : channel, drain, furrow, trench, discord.
 dodge : dangle, duck, shift, sidestep, avoid, deceive.
 dubious : doubtful, hesitant, uncertain, unconvinced, undecided, unsure, ambiguous
 dwell : abide, establish, inhabit, settle, sojourn.

E

edgy : anxious, irritable, nervous, touchy.
 elusive : difficult to catch, shift, slippery, tricky, baffling, fleeting, puzzling, subtle.
 emblem : badge, gest, device, figure, image, representation.
 enchant : beguile, bewitch, captivate, charm, delight, enamour, enrapture, fascinate, hypnotize.
 enrage : aggravate, anger incense, incite, inflame, irritate, provoke.
 exalt : advance, dignify, elevate, honour, promote, acclaim, applaud, extol.
 exhilarating: breathtaking, cheering, exalting, exciting, vigorating, stimulating.
 expedite : accelerate, advance, assist, dispatch, facilitate, hasten, precipitate.

F

fame : celebrity, eminence, glory, honour, prominence, renown, repute, stardom.
 famine : scarcity, dearth, destitution, hunger, starvation.
 feeble : weak, delicate, etiolated, exhausted, infirm, languid, puny, shilpit.
 fierce : barbarous, brutal, cruel, dangerous, ferocious, murderous, passionate, savage.
 fishy : doubtful, dubious, funny, improbable, suspect, suspicious, deadpan, dull.
 flawless : faultless, impeccable, perfect, unblemished.
 foresee : anticipate, divine, envisage, forebode, portend, predict, prophesy, presage.
 fortress : castle, citadel, fastness, redoubt, stronghold.
 fraternity : brotherhood, association, company, comradeship.
 fretful : captious, complaining, factions, irritable, peevish, splenetic, touchy.

G

gaiety : animation, blitheness, cheerfulness, elation, exhilaration, glee, humour.
 gaudy : bright, brilliant, flashy, florid, garish, glaring, meretricious, showy, vulgar.
 gestation : development, evolution, incubation, pregnancy, repening.
 gracious : affable, amiable, beneficent, benevolent, cheritable, chivalrous, cordial.
 grimace : face, frown, mouth, scowl, sneer.
 grubby : dirty, filthy, frowzy, manky, shabby.

H

haggle : bargain, barter, beat down, dicker, higgler, dispute, quarrel, wrangle.
 haphazard : accidental, arbitrary, chance, fluky, random, aimless, careless.

Synonyms

3.3

hatred	: abomination, animosity, animus, antagonism, antipathy, enmity.
hectic	: animated, boisterous, chaotic, frantic, furious, turbulent, wild.
hilarious	: amusing, comical, convivial, exhilarated, humorous, jolly, jovial, joyful, joyous.
holocaust	: annihilation, carnage, destruction, devastation, massacre.
hurricane	: cyclone, gale, storm, tempest, tornado, twister.

I

idle	: dead, empty, gathering, dust, inactive, jobless, redundant.
immense	: enormous, extensive, giant, gigantic, huge, illimitable, : interminable, jumbo.
implicated	: incriminated, involved, suspected, under suspicion.
impudence	: assurance, audacity, boldness, bumptiousness, insolence.
incense	: anger, enrage, excite, aroma, balm, fragrance, redolence, scent.
indent	: order, request, dint, nick, notch.
inept	: awkward, bumbling, bungling, clumsy, gauche, inexperienced, maladroit, absurd.
insolent	: abusive, bold, contemptuous, impertinent, impudent, insulting, rude, saucy.
instigate	: actuate, encourage, forment, impel, incite, influence, provoke, persuade.

J, K, L, M

jaunty	: airy, breezy, buoyant, carefree, dapper, gay, lively, perky, showy, smart.
jurisdiction	: authority, command, control, influence, prerogative, sway.
keepsake	: emblem, favour, memento, relic, reminder, souvenir, symbol, token.
latch	: bar, bolt, catch, fastening, lock, sneck.
link	: component, constituent, division, association, attachment.
meander	: ramble, snake, straving, stroll, wander, bend, curve, loop, twist.
melancholy	: dejection, depression, despondency, gloom.

N, O, P, Q

nostalgic	: homesick, longing, regretful, sentimental, wistful.
obnoxious	: abhorrent, abominable, detestable, disagreeable, disgusting, hateable, horrid.
onerous	: burdensome, crushing, demanding, difficult, exciting, exhausting, exigent.

paw	: grab, handle, roughly, manhandle, maul, molest.
percolate	: drain, drip, exude, filter, leache, ooze, penetrate, permeate, pervade.
piety	: devotion, dutifulness, faith, grace, piousness, religion, sanctify, veneration.
pudding	: dessert, sweet.
quandary	: bewilderment, stick, delicate, situation, difficult, dilemma, embarrassment.
quiver	: agitate, oscillate, palpitate, pulsate, shudder, tremble, vibrate.

R, S, T

rapt	: absorbed, carried away, enthralled, entranced, fascinated, gripped, intent.
reveal	: announce, betray, broadcast, communicate, disclose, divulge, impart, proclaim.
rowdy	: bristerous, disorderly, loutish, noisy, unruly, uproarious, brawler.
sage	: acute, canny, intelligent, judicious, polite, prudent, sagacious.
scorn	: contempt, despite, disdain, disparagement, slight, sneer, curl, condemn.
shake	: fluctuate, bump, joggle, jounce, quake, shiver, shudder, totter, tremble.
sinner	: offender, reprobate, trespasser.
snigger	: giggle, laugh, smirk, sneer, titter.
stimulus	: encouragement, incentive, incitement, inducement, provocation.
tepid	: slightly warm, luke warm.
timid	: afraid, bashful, cowardly, diffident.

U, V, W, Y, Z

unmask	: disclose, discover, expose, reveal, unveil.
urbane	: civil, civilized, cosmopolitan, courteous, cultivated, debonair.
vehement	: ardent, eager, earnest, emphatic, enthusiastic, fervent, fervid, fierce, forcible.
weired	: bizarre, creepy, eerie, freakish, ghostly, mysterious, queer, spooky.
writhe	: contort, distant, jerk, struggle, thrash, wiggle, twist, wriggle.
yearn	: ache, covet, crave, desire, hanker, hunger, itch, languish.
zenith	: acme, apex, apogee, climax, height, meridian, peak, pinnacle, summit, top.

EXERCISE

MCQ TYPE QUESTIONS

Directions: Each of the questions below consists of a word followed by four lettered words or phrases. Choose the lettered word or phrase that is most nearly same in meaning to the word.

1. Abase

- | | |
|--------------|---------------|
| (a) incur | (b) tax |
| (c) estimate | (d) humiliate |

2. Aberration

- | | |
|---------------|----------------|
| (a) deviation | (b) abhorrence |
| (c) dislike | (d) absence |

3. Abet

- | | |
|---------------|-----------|
| (a) conceive | (b) wager |
| (c) encourage | (d) evade |

4. Aabeyance

- | | |
|----------------|----------------------|
| (a) obedience | (b) discussion |
| (c) excitement | (d) suspended action |

5. Abjure

- | | |
|----------------------|--------------|
| (a) discuss | (b) renounce |
| (c) run off secretly | (d) perjure |

6. Ablution

- | | |
|-------------|-----------------|
| (a) censure | (b) forgiveness |
| (c) mutiny | (d) washing |

7. Abnegation

- | | |
|-----------------|-----------------|
| (a) blackness | (b) self-denial |
| (c) selfishness | (d) cause |

8. Aborigine

- | |
|--------------------------|
| (a) first design |
| (b) absolution |
| (c) finale |
| (d) primitive inhabitant |

9. Abortive

- | |
|------------------|
| (a) unsuccessful |
| (b) consuming |
| (c) financing |
| (d) familiar |

10. Abstinence

- | | |
|----------------|--------------------|
| (a) restrained | (b) vulgar display |
| (c) deportment | (d) reluctance |

11. Abstruse

- | | |
|---------------|------------------|
| (a) profound | (b) irrespective |
| (c) suspended | (d) protesting |

12. Abut

- | | |
|---------------|------------|
| (a) stimulate | (b) grasp |
| (c) oppose | (d) adjoin |

13. Abysmal

- | | |
|----------------|--------------|
| (a) bottomless | (b) eternal |
| (c) meteric | (d) diabolic |

14. Accede

- | | |
|-------------|----------------|
| (a) fail | (b) compromise |
| (c) correct | (d) consent |

15. Acclivity

- | |
|-----------------------|
| (a) index |
| (b) report |
| (c) upslope of a hill |
| (d) negotiator |

16. Astute

- | | |
|------------|------------|
| (a) sheer | (b) noisy |
| (c) astral | (d) clever |

17. Atrocity

- | | |
|---------------|-----------------|
| (a) endurance | (b) fortitude |
| (c) session | (d) heinous act |

18. Atrophy

- | | |
|-------------|-----------------|
| (a) capture | (b) waste away |
| (c) govern | (d) award prize |

19. Attenuate

- | | |
|------------|---------------|
| (a) appear | (b) be absent |
| (c) weaken | (d) testify |

20. Atypical

- | | |
|-----------------|---------------|
| (a) superfluous | (b) fortitude |
| (c) unusual | (d) clashing |

21. Audacity

- | | |
|--------------|--------------|
| (a) boldness | (b) asperity |
| (c) strength | (d) stature |

22. Augment

- | | |
|----------------|-------------|
| (a) make noble | (b) anoint |
| (c) increase | (d) harvest |

23. Auxiliary

- | | |
|---------------|-----------------|
| (a) righteous | (b) prospective |
| (c) assistant | (d) archaic |

24. Avarice

- | | |
|---------------|-------------|
| (a) easiness | (b) greed |
| (c) statement | (d) invoice |

25. Avert

- (a) entertain (b) transform
(c) turn away (d) lead toward

26. Awry

- (a) recommended
(b) commiserating
(c) started
(d) crooked

27. Baleful

- (a) doubtful (b) virtual
(c) deadly (d) conventional

28. Balmy

- (a) venturesome (b) dedicated
(c) mild (d) fanatic

29. Banal

- (a) philosophical (b) trite
(c) dramatic (d) heedless

30. Baneful

- (a) intellectual (b) thankful
(c) decisive (d) poisonous

31. Boisterous

- (a) conflicting (b) noisy
(c) testimonial (d) grateful

32. Bombastic

- (a) sensitive (b) pompous
(c) rapid (d) sufficient

33. Bouillon

- (a) insight
(b) chowder
(c) gold
(d) clear soup

34. Brackish

- (a) careful (b) salty
(c) chosen (d) tough

35. Braggadocio

- (a) weaponry (b) boasting
(c) skirmish (d) encounter

36. Brazen

- (a) shameless (b) quick
(c) modest (d) pleasant

37. Brindled

- (a) equine (b) pathetic
(c) hasty (d) spotted

38. Brochure

- (a) opening (b) pamphlet
(c) censor (d) bureau

39. bucolic

- (a) diseased (b) repulsive
(c) rustic (d) twinkling

40. Buxom

- (a) voluminous (b) indecisive
(c) convincing (d) plump

41. Cache

- (a) lock
(b) hiding place
(c) tide
(d) automobile

42. Cacophony

- (a) discord
(b) dance
(c) applause
(d) type of telephone

43. Callow

- (a) youthful (b) holy
(c) mild (d) colored

44. Candid

- (a) vague (b) outspoken
(c) experienced (d) anxious

45. Chaste

- (a) loyal (b) timid
(c) curt (d) pure

46. Chide

- (a) unite (b) fear
(c) record (d) scold

47. Chimerical

- (a) developing (b) brief
(c) distant (d) fantastic

48. Choleric

- (a) musical (b) episodic
(c) hot headed (d) global

49. Churlish

- (a) marine (b) economical
(c) impolite (d) compact

50. Ciliated

- (a) foolish (b) swift
(c) early (d) hairy

51. Circuitous

- (a) indirect (b) complete
(c) obvious (d) aware

52. City

- (a) galvanize (b) visualize
(c) locate (d) quote

53. Clandestine

- (a) abortive (b) secret
(c) tangible (d) doomed

54. Claustrophobia

- (a) lack of confidence
(b) fear of spiders
(c) love of books
(d) fear of closed places

55. Cleft

- (a) split (b) waterfall
(c) assembly (d) adherence

56. Cliche

- (a) increase (b) vehicle
(c) morale (d) platitude

57. Coerce

- (a) recover (b) begin
(c) force (d) license

58. Cognizance

- (a) policy (b) knowledge
(c) advance (d) omission

59. Cohere

- (a) hold together
(b) occur simultaneously
(c) recollect
(d) materialize

60. Conjecture

- (a) magic (b) guess
(c) position (d) form

61. Connoisseur

- (a) gourmand (b) lover of art
(c) humorist (d) delinquent

62. Consanguinity

- (a) kinship (b) friendship
(c) bloodletting (d) relief

63. Consensus

- (a) general agreement (b) project
(c) insignificance (d) sheaf

64. Construe

- (a) explain (b) promote
(c) reserve (d) erect

65. Contaminate

- (a) arrest (b) prepare
(c) pollute (d) beam

66. Contentious

- (a) squealing (b) surprising
(c) quarrelsome (d) smug

67. Contenance

- (a) humanity (b) research
(c) embryology (d) self-restraint

68. Contraband

- (a) purpose (b) rogue
(c) rascality (d) difficulty

69. Contrite

- (a) smart
(b) penitent
(c) restful
(d) recognized

70. Controvert

- (a) turnover (b) contradict
(c) mind (d) expolain

71. Convene

- (a) propose (b) restore
(c) question (d) gather

72. Conversant

- (a) ignorant
(b) speaking
(c) incorporated
(d) familiar

73. Copious

- (a) plentiful (b) cheating
(c) dishonorable (d) adventurous

74. Corpulent

- (a) regenerate (b) obese
(c) different (d) hungry

75. Decimate

- (a) kill (b) disgrace
(c) search (d) collide

76. Declivite

- (a) trap (b) quadrangle
(c) quarter (d) downward slope

77. Decollete

- (a) flavored (b) demure
(c) flowery (d) low-necked

78. Decorous

- (a) momentary (b) emotional
(c) suppressed (d) proper

79. Decrepitude

- (a) feebleness (b) disease
(c) coolness (d) melee

80. Default

- (a) failure to act
(b) tendency to err
(c) desire to remedy
(d) debt

81. Defection

- (a) determination (b) desertion
(c) invitation (d) affection

82. Defile

- (a) maincure (b) ride
(c) pollute (d) assemble

83. Degraded

- (a) surprised (b) lowered
(c) ascended (d) learned

84. Deleterious

- (a) delaying
(b) experimental
(c) harmful
(d) graduating

85. Deluge

- (a) confusion (b) deception
(c) flood (d) mountain

86. Denigrate

- (a) refuse (b) blacken
(c) terrify (d) admit

87. Denouement

- (a) action (b) scenery
(c) resort (d) solution

88. Depravity

- (a) wickedness (b) sadness
(c) heaviness (d) tidiness

89. Deranged

- (a) insane (b) systematic
(c) neighborly (d) alphabetical

90. Dwindle

- (a) blow (b) inhabit
(c) spin (d) lessen

91. Ecstasy

- (a) joy (b) speed
(c) treasure (d) warmth

92. Edify

- (a) mystify (b) suffice
(c) improve (d) erect

93. Efface

- (a) countenance
(b) encourage
(c) recognize
(d) rub out

94. Effigy

- (a) requisition (b) organ
(c) charge (d) dummy

95. Egregious

- (a) pious (b) shocking
(c) anxious (d) sociable

96. Egress

- (a) entrance (b) bird
(c) exit (d) double

97. Elated

- (a) debased
(b) respectful
(c) drooping
(d) gay

98. Elusive

- (a) deadly (b) eloping
(c) evasive (d) simple

99. Emaciated

- (a) garrulous (b) primeval
(c) vigorous (d) thin

100. Emancipate

- (a) set free (b) take back
(c) make worse (d) embolden

101. Embellish

- (a) doff (b) don
(c) abscond (d) adon

102. Embroil

- (a) cherish (b) overheat
(c) entangle (d) assure

103. Emendation

- (a) correction
- (b) interpretation
- (c) exhumation
- (d) inquiry

104. Eminent

- (a) purposeful
- (b) high
- (c) delectable
- (d) currious

105. Flored

- (a) ruddy
- (b) rusty
- (c) ruined
- (d) patient

106. Foil

- (a) bury
- (b) frustrate
- (c) shield
- (d) desire

107. Foment

- (a) spoil
- (b) instigate
- (c) interrogate
- (d) spray

108. Foolhardy

- (a) strong
- (b) unwise
- (c) brave
- (d) futile

109. Foppish

- (a) scanty
- (b) radical
- (c) orthodox
- (d) dandyish

110. Foray

- (a) excursion
- (b) contest
- (c) ranger
- (d) intuition

111. Formidable

- (a) dangerous
- (b) outlandish
- (c) grandiloquent
- (d) impenetrable

112. Foster

- (a) accelerate
- (b) fondle
- (c) become infected
- (d) raise

113. Franchise

- (a) subway
- (b) discount
- (c) license
- (d) reason

114. Fritter

- (a) sour
- (b) chafe
- (c) dissipate
- (d) cancel

115. Frugality

- (a) foolishness
- (b) extremity
- (c) indifference
- (d) economy

116. Fulminate

- (a) fulfill
- (b) contemplate
- (c) talk nonsense
- (d) protest

117. Furor

- (a) excitement
- (b) worry
- (c) flux
- (d) anteroom

118. Furtive

- (a) underhanded
- (b) coy
- (c) brilliant
- (d) quick

119. Gadfly

- (a) humorist
- (b) nuisance
- (c) scholar
- (d) bum

120. Galleon

- (a) liquid measure
- (b) ship
- (c) armada
- (d) company

121. Garisth

- (a) sordid
- (b) flashy
- (c) prominent
- (d) lusty

122. Garner

- (a) prevent
- (b) assist
- (c) collect
- (d) compute

123. Garnish

- (a) paint
- (b) gamer
- (c) adorn
- (d) abuse

124. Garrulity

- (a) credulity
- (b) senility
- (c) loquaciousness
- (d) speciousness

125. Garrulous

- (a) arid
- (b) hasty
- (c) sociable
- (d) talkative

126. Gauche

- (a) rigid
- (b) swift
- (c) awkward
- (d) taciturn

127. Gaudy

- (a) holy
- (b) showy
- (c) sentimental
- (d) mild

128. Gaunt

- (a) victorious
- (b) tiny
- (c) stylish
- (d) haggard

129. Genuflect

- (a) falsify (b) trick
(c) project (d) bend the knee

130. Germane

- (a) bacteriological
(b) middle european
(c) prominent
(d) relevant

131. Germinal

- (a) creative (b) excused
(c) sterilized (d) primitive

132. Gist

- (a) chaff (b) summary
(c) expostulation (d) expiation

133. Glib

- (a) slippery (b) fashionable
(c) antiquated (d) articulate

134. Gnome

- (a) fury (b) giant
(c) dwarf (d) native

135. Inclement

- (a) unfavorable (b) abandoned
(c) kindly (d) selfish

136. Incompatible

- (a) capable
(b) reasonable
(c) faulty
(d) alienated

137. Inconsequential

- (a) disorderly (b) insignificant
(c) subsequent (d) insufficient

138. Incontinent

- (a) insular (b) complaisant
(c) crass (d) wanton

139. Incurable

- (a) narrow (b) straight
(c) inconceivable (d) unreformable

140. Incriminate

- (a) exacerbate (b) involve
(c) intimidate (d) lacerate

141. Inculcate

- (a) exculpate (b) educate
(c) exonerate (d) prepare

142. Indigent

- (a) lazy (b) pusillanimous
(c) penurious (d) affluent

143. Indignity

- (a) pomposity (b) bombast
(c) obeisance (d) insult

144. Indolence

- (a) sloth (b) poverty
(c) latitude (d) aptitude

145. Indubitably

- (a) flagrantly (b) doubtfully
(c) carefully (d) certainly

146. Inebriety

- (a) revelation (b) drunkenness
(c) felony (d) starvation

147. Inept

- (a) outward (b) spiritual
(c) foolish (d) clumsy

148. Infallible

- (a) final
(b) unbelievable
(c) perfect
(d) inaccurate

149. Infirmary

- (a) disability (b) age
(c) inoculation (d) hospital

150. Ingenuous

- (a) clever (b) stimulating
(c) naive (d) worried

151. Inimical

- (a) antagonistic
(b) anonymous
(c) fanciful
(d) accurate

152. Innocuous

- (a) not capable (b) not dangerous
(c) not eager (d) not frank

153. Insinuate

- (a) resist (b) suggest
(c) report (d) rectify

154. Insipid

- (a) witty (b) flat
(c) wily (d) talkative

155. Integrate

- (a) tolerate (b) unite
(c) flow (d) copy

156. Inter

- (a) bury (b) amuse
(c) relate (d) frequent

157. Interdict

- (a) acclaim (b) dispute
(c) prohibit (d) decide

158. Intermittent

- (a) heavy (b) fleet
(c) occasional (d) fearless

159. Intractable

- (a) culpable (b) flexible
(c) unruly (d) efficient

160. Intransigence

- (a) lack of training (b) stubbornness
(c) novelty (d) timidity

161. Intrepid

- (a) cold (b) hot
(c) understood (d) courageous

162. Intrinsic

- (a) extrinsic (b) abnormal
(c) above (d) basic

163. Inundate

- (a) abuse (b) deny
(c) swallow (d) flood

164. Inveigh

- (a) speak violently (b) orate
(c) disturb (d) apply

165. Lethargy

- (a) dynamism (b) orate
(c) disturb (d) apply

166. Loquacious

- (a) verbose (b) tacturn
(c) rational (d) alluring

167. Loath

- (a) tired (b) unwilling
(c) sickly (d) spirited

168. Mawkish

- (a) sentimental (b) true
(c) certain (d) devious

169. Mediocre

- (a) average (b) bitter
(c) medieval (d) industrial

170. Melee

- (a) heat (b) brawl
(c) attempt (d) weapon

171. Mellifluous

- (a) porous (b) honeycombed
(c) strong (d) smooth

172. Menial

- (a) intellectual (b) clairvoyant
(c) servile (d) arrogant

173. Mentor

- (a) guide (b) genius
(c) talker (d) philosopher

174. Mesmerize

- (a) remember (b) hypnotize
(c) delay (d) bore

175. Meticulous

- (a) steadfast (b) recent
(c) quaint (d) painstaking

176. Miasma

- (a) dream (b) noxious fumes
(c) scenario (d) quantity

177. Militant

- (a) combative (b) dramatic
(c) religious (d) quaintity

178. Minon

- (a) monster (b) quorum
(c) majority (d) dependent

179. Mirage

- (a) dessert (b) illusion
(c) water (d) mirror

180. Misanthrope

- (a) benefactor (b) philanderer
(c) man-hater (d) aesthete

181. Mischance

- (a) gamble (b) ordinance
(c) aperture (d) anecdote

182. Misdemeanor

- (a) felony (b) peccadillo
(c) indignity (d) flat

183. Modish

- (a) sentimental (b) stylish
(c) vacillating (d) contrary

184. Mollify

- (a) avenge (b) attenuate
(c) attribute (d) appease

185. Monetary

- (a) boring (b) fascinating
(c) fiscal (d) stationary

186. Moratorium

- (a) burial (b) gathering
(c) delay (d) refusal

187. Mordant

- (a) dying (b) trenchant
(c) fabricating (d) controlling

188. Moribund

- (a) dying (b) trenchant
(c) fabricating (d) controlling

189. Motley

- (a) active (b) disguised
(c) variegated (d) somber

190. Muggy

- (a) attacking (b) fascinating
(c) humid (d) characteristic

191. Mulct

- (a) swindle (b) hold
(c) record (d) pring

192. Multilingual

- (a) variegated (b) polyglot
(c) multilateral (d) polyandrous

193. Mundane

- (a) global (b) futile
(c) spatial (d) worldly

194. Munificent

- (a) grandiose (b) puny
(c) philanthropic (d) poor

195. Musty

- (a) flat (b) necessary
(c) indifferent (d) nonchalant

196. Myopic

- (a) visionary (b) nearsighted
(c) moral (d) glassy

197. Nascent

- (a) incipient (b) ignorant
(c) loyal (d) treacherous

198. Phlegmatic

- (a) calm (b) cryptic
(c) practical (d) salivary

199. Physiognomy

- (a) posture (b) head
(c) physique (d) face

200. Piebald

- (a) motley (b) coltish
(c) hairless (d) thoroughbred

201. Pillage

- (a) hoard (b) plunder
(c) versify (d) denigrate

202. Pinion

- (a) express (b) report
(c) reveal (d) restrain

203. Pinnacle

- (a) foothills (b) card game
(c) pass (d) peak

204. Pious

- (a) historic (b) devout
(c) multiple (d) fortunate

205. Pique

- (a) pyramid (b) revolt
(c) resentment (d) struggle

206. Placate

- (a) determine (b) transmit
(c) pacify (d) allow

207. Plagiarism

- (a) theft of funds (b) theft of ideas
(c) belief in god (d) arson

208. Plaintive

- (a) mournful (b) senseless
(c) persistent (d) rural

209. Platitude

- (a) fatness (b) bravery
(c) dimension (d) trite remark

210. Polemic

- (a) blackness (b) lighting
(c) magnetism (d) controversy

ANSWERS

MCQ Type Questions

1. (d)	2. (a)	3. (c)	4. (d)	5. (b)	6. (d)	7. (b)	8. (d)	9. (a)	10. (a)
11. (a)	12. (d)	13. (a)	14. (d)	15. (d)	16. (d)	17. (d)	18. (b)	19. (c)	20. (c)
21. (a)	22. (c)	23. (c)	24. (b)	25. (c)	26. (d)	27. (c)	28. (c)	29. (b)	30. (d)
31. (b)	32. (b)	33. (d)	34. (b)	35. (b)	36. (a)	37. (d)	38. (b)	39. (c)	40. (d)
41. (b)	42. (a)	43. (a)	44. (c)	45. (d)	46. (d)	47. (d)	48. (c)	49. (c)	50. (d)
51. (a)	52. (d)	53. (b)	54. (d)	55. (a)	56. (d)	57. (c)	58. (b)	59. (a)	60. (b)
61. (b)	62. (a)	63. (a)	64. (a)	65. (c)	66. (c)	67. (d)	68. (d)	69. (b)	70. (b)
71. (d)	72. (d)	73. (a)	74. (b)	75. (a)	76. (d)	77. (d)	78. (d)	79. (a)	80. (a)
81. (b)	82. (c)	83. (b)	84. (c)	85. (c)	86. (b)	87. (d)	88. (a)	89. (a)	90. (d)
91. (a)	92. (c)	93. (d)	94. (d)	95. (b)	96. (c)	97. (d)	98. (c)	99. (d)	100. (a)
101. (d)	102. (c)	103. (a)	104. (b)	105. (a)	106. (b)	107. (b)	108. (b)	109. (d)	110. (a)
111. (a)	112. (c)	113. (c)	114. (c)	115. (d)	116. (d)	117. (a)	118. (a)	119. (b)	120. (b)
121. (b)	122. (c)	123. (c)	124. (c)	125. (d)	126. (c)	127. (b)	128. (d)	129. (d)	130. (d)
131. (a)	132. (b)	133. (d)	134. (c)	135. (a)	136. (d)	137. (b)	138. (d)	139. (d)	140. (b)
141. (b)	142. (c)	143. (d)	144. (a)	145. (d)	146. (b)	147. (d)	148. (c)	149. (a)	150. (c)
151. (a)	152. (b)	153. (b)	154. (b)	155. (a)	156. (a)	157. (c)	158. (c)	159. (c)	160. (b)
161. (d)	162. (d)	163. (d)	164. (a)	165. (b)	166. (a)	167. (b)	168. (d)	169. (a)	170. (b)
171. (d)	172. (c)	173. (a)	174. (b)	175. (d)	176. (b)	177. (a)	178. (d)	179. (b)	180. (c)
181. (d)	182. (b)	183. (b)	184. (d)	185. (c)	186. (c)	187. (b)	188. (a)	189. (c)	190. (c)
191. (a)	192. (b)	193. (d)	194. (c)	195. (a)	196. (b)	197. (a)	198. (a)	199. (d)	200. (a)
201. (b)	202. (d)	203. (d)	204. (b)	205. (c)	206. (c)	207. (b)	208. (a)	209. (d)	210. (d)

4

CHAPTER

Antonyms

The basic idea of an antonym question is to find an opposite for a word. Antonym items are first and foremost a test of vocabulary.

If you have no idea of the meaning of the given word, there's not much you can do. When you don't know the meaning of an antonym, don't waste a lot of time trying to figure it out. In other words, once you recognize that you are out of ammunition, just do a random guess, confirm it, and move on to the next item.

Each antonym starts with a capitalized word, which is followed by four words or phrases in lowercase letters. The task is to find the word or phrase that is most nearly opposite in meaning to the bold word.

Plan to Answer Antonym Questions

1. Define the capitalized word.
2. Think of a meaning that is opposite to this word.
3. Read all the answer choices. Eliminate those that do not relate to the meaning you thought of. If only one choice remains, mark it and go on.
4. If more than one choice remains, go back and refine your thinking about the bold word.
5. Pick the answer choice that is most nearly opposite to the bold word.

Tedious

- (a) unlimited
- (b) confined
- (c) enthralling
- (d) appetizing
- (i) The capitalized word means "boring" or "tiresome"
- (ii) The opposite of "boring" would be something like *exciting* or *interesting*.
- (iii) Looking at all the answer choices, you can immediately eliminate (a), and (b) as having nothing to do with excitement. But that leaves (c) *enthralling* and (d) *appetizing*.
- (iv) Of the two remaining choices, *enthralling*, which means "captivating" or "fascinating," is more nearly opposite in meaning to *tedious* than *appetizing*, which means "savory," or "delicious".
- (v) Mark (c), *enthralling*, as the correct choice.

Note :

- (i) **Makes sure to read all the answers :** To answer antonym questions you'll sometimes have to distinguish fine shades of meaning. That's why you have to consider every answer choice.
- (ii) **Verb or Noun?** If the part of speech of the capitalized word is ambiguous, check the answer choices.

6. Sometimes the test writer selects a word you are likely to be familiar with, but sets up the question to test a meaning you do not ordinarily associate with the word. Here is an example:

e.g. precipitous

- (a) pleasantly sweet
- (b) overly ambitious
- (c) agreeably situated
- (d) gently sloping

The best choice is (d). We most often use the word *precipitous* to mean rash or foolhardy, but its central meaning is related to *precipice*, a sharp drop-off.

7. Eliminate non-answers

The simple way to eliminate answer choices is to toss out any words that don't have opposite meanings.

8. Concentrate on answer choices

Words can have different meanings based upon their part of speech. If you are uncertain about the part of speech of the bold word, just check the answer choices. If they're all verbs, for example, so is the bold word.

9. Play around with the part of speech of a word

You can alter the part of speech of the bold word and answer choices in your own mind. Sometimes an antonym will use a word you know but as a part of speech that is unfamiliar to you.

e.g. sublimity

- (a) erosion
- (b) baseness
- (c) conciseness
- (d) insistence

The best choice is (b). You may know the word *sublimity* better as the adjective *sublime*, meaning "lofty, high, or noble". So you may find it easier to think about the antonym by changing *sublimity* to the more familiar form *sublime*. As you think about each answer choice, you would then change it in your mind to an adjective *Baseness*, therefore, would become *base*; and *base* is an opposite of *sublime*.

10. Use word connotations to eliminate answer choices

Even if you don't know the exact meaning of a word, you may have a vague recollection of the context in which you first encountered it. So you may know whether the word has positive overtones or negative ones. This recollection may be sufficient to get a correct answer.

e.g. raffish

- (a) grotesque (b) delinquent
(c) uncaring (d) noble

The correct choice is (d). Let's assume that you do not know that it means "low, vulgar, the base." Let's further assume, however, that you have a vague knowledge of the word. You've seen it used to describe a character who is dishonest and not trustworthy. So even though you don't know the exact meaning of the word, you know that it has negative overtones. Since you are looking for the opposite of a word with negative overtones, you would eliminate every answer choice with negative overtones. As it turns out, this strategy worked perfectly with this antonym; only one word is left. *Noble* is the only word with positive overtones.

11. Unlock word meanings by taking the word apart

Even when you encounter a word for the first time, you may be able to ascertain its meaning from its parts.

e.g. cognoscitive

- (a) courageous (b) expensive
(c) unconscious (d) redundant

The best answer is (c). This is a very unusual word, but you can probably figure out its meaning by looking at its root *cog-*. This is the same root found in words such as *cognition* and *recognize*, and it has to do with knowledge. So we infer that *cognoscitive* has something to do with awareness, and (c) looks like a good opposite. A word of caution, however: Don't spend too much time trying to decipher the meaning of a word.

EXERCISE

MCQ TYPE QUESTIONS

Directions : Each question below consists of a word printed in capital letters, followed by four lettered words or phrases. Choose the lettered word or phrase that is most nearly opposite in meaning to the word in bold letters.

- 1. Abrogate**
(a) transgress (b) signify
(c) alleviate (d) ratify
- 2. Admonition**
(a) premonition (b) hallucination
(c) escape (d) commendation
- 3. Adulation**
(a) youth (b) purity
(c) brightness (d) criticism
- 4. Advocate**
(a) define (b) oppose
(c) remove (d) inspect
- 5. Affable**
(a) rude (b) ruddy
(c) needy (d) useless
- 6. Affected**
(a) weary (b) unfriendly
(c) divine (d) unfeigned
- 7. Affluence**
(a) poverty (b) fear
(c) persuasion (d) consideration
- 8. Agility**
(a) awkwardness (b) solidity
(c) temper (d) harmony
- 9. Alacrity**
(a) slowness (b) plenty
(c) fifth (d) courtesy
- 10. Alleviate**
(a) endure (b) worsen
(c) enlighten (d) maneuver
- 11. Allure**
(a) hinder (b) repel
(c) ignore (d) leave
- 12. Aloof**
(a) triangular (b) gregarious
(c) comparable (d) honorable
- 13. Amalgamate**
(a) equip (b) separate
(c) generate (d) materialize
- 14. Ambiguous**
(a) salvageable (b) corresponding
(c) responsible (d) clear
- 15. Amble**
(a) befriend (b) hasten
(c) steal (d) browse
- 16. Ambulatory**
(a) convalescent (b) valedictory
(c) bedridden (d) emergency
- 17. Ameliorate**
(a) make slow (b) make sure
(c) make young (d) make worse
- 18. Amicable**
(a) penetrating (b) compensating
(c) unfriendly (d) zigzag
- 19. Amoral**
(a) unusual (b) unfriendly
(c) ethical (d) suave
- 20. Amorphous**
(a) nauseous (b) obscene
(c) providential (d) definite
- 21. Amplify**
(a) distract (b) infer
(c) publicize (d) decrease
- 22. Analogous**
(a) not comparable (b) not capable
(c) not culpable (d) not corporeal

23. Anthematize

- (a) locate (b) deceive
(c) regulate (d) bless

24. Anemic

- (a) pailid (b) cruel
(c) red-blooded (d) ventilating

25. Animated

- (a) worthy (b) dull
(c) humorous (d) lengthy

26. Animus

- (a) pterodactyl (b) bastion
(c) giraffe (d) favor

27. Anomaly

- (a) desperation (b) requisition
(c) registry (d) regularity

28. Anonymous

- (a) desperate (b) signed
(c) defined (d) expert

29. Antediluvian

- (a) transported (b) subtle
(c) isolated (d) modern

30. Antipathy

- (a) profundity (b) objection
(c) willingness (d) foundness

31. Antithesis

- (a) velocity (b) maxim
(c) similarity (d) acceleration

32. Aphasia

- (a) volubility (b) necessity
(c) pain (d) crack

33. Baroque

- (a) polished (b) constant
(c) transformed (d) simple

34. Beatific

- (a) glorius (b) dreadful
(c) theatrical (d) crooked

35. Belittle

- (a) disobey (b) forget
(c) magnify (d) extol

36. Bellicose

- (a) peaceful (b) naval
(c) amusing (d) piecemeal

37. Benign

- (a) tenfold (b) peaceful
(c) blessed (d) malignant

38. Benison

- (a) curse (b) bachelor
(c) wedding (d) orgy

39. Berate

- (a) grant (b) praise
(c) refer (d) purchase

40. Bestial

- (a) animated (b) noble
(c) zoological (d) clear

41. Bigotry

- (a) arrogance (b) approval
(c) mourning (d) tolerance

42. Bizarre

- (a) roomy (b) veiled
(c) subdued (d) normal

43. Blanch

- (a) bleach (b) scatter
(c) darken (d) analyze

44. Bland

- (a) caustic (b) meager
(c) soft (d) uncooked

45. Blase

- (a) fiery (b) clever
(c) intriguing (d) ardent

46. Bleak

- (a) pale (b) sudden
(c) dry (d) cheerful

47. Blithe

- (a) spiritual (b) profuse
(c) cheerless (d) hybrid

48. Capacious

- (a) warlike (b) cordial
(c) curious (d) not spacious

49. Capricious

- (a) satisfied (b) insured
(c) photographic (d) steadfast

50. Captious

- (a) tolerant (b) capable
(c) frivolous (d) winning

51. Carnal

- (a) impressive (b) minute
(c) spiritual (d) actual

52. Carnivorous

- (a) gloomy (b) tangential
(c) productive (d) vegetarian

53. Carping

- (a) rapid (b) uncritical
(c) unintellectual (d) illegal

54. Castigation

- (a) commendation (b) patience
(c) generosity (d) understatement

55. Categorical

- (a) negative (b) ironic
(c) impartial (d) qualified

56. Catholic

- (a) religious (b) pacific
(c) narrow (d) weighty

57. Celerity

- (a) assurance (b) state
(c) acerbity (d) delay

58. Celibate

- (a) investing (b) married
(c) retired (d) commodious

59. Censure

- (a) process (b) enclose
(c) interest (d) praise

60. Centrifugal

- (a) centripetal (b) ephemeral
(c) lasting (d) barometric

61. Cessation

- (a) premium (b) gravity
(c) beginning (d) composition

62. Chaffing

- (a) achieving (b) serious
(c) caitalistic (d) sneezing

63. Coy

- (a) weak (b) airy
(c) brazen (d) old

64. Cozen

- (a) amuse (b) treat honestly
(c) prate (d) shackle

65. Craven

- (a) desirous (b) direct
(c) bold (d) civilized

66. Crux

- (a) affliction (b) spark
(c) events (d) trivial point

67. Cryptic

- (a) tomblike (b) futile
(c) famous (d) candid

68. Cupidity

- (a) anxiety (b) tragedy
(c) generosity (d) entertainment

69. Curtail

- (a) mutter (b) lengthen
(c) express (d) burden

70. Cynical

- (a) trusting (b) effortless
(c) conclusive (d) gallant

71. Dank

- (a) dry (b) guiltless
(c) warm (d) babbling

72. Dapper

- (a) unintelligent (b) untidy
(c) uncertain (d) ungrateful

73. Dauntless

- (a) stolid (b) cowardly
(c) irrelevant (d) peculiar

74. Dearth

- (a) life (b) abundance
(c) brightness (d) terror

75. Debacle

- (a) progress (b) refusal
(c) masque (d) cowardice

76. Debilitate

- (a) bedevil (b) repress
(c) strengthen (d) animate

77. Debonair

- (a) awkward (b) windy
(c) balmy (d) strong

78. Derogatory

- (a) roguish (b) immediate
(c) openionated (d) praising

79. Desecrate

- (a) desist (b) integrate
(c) confuse (d) consecrate

80. Despicable

- (a) streeing (b) worthy of esteem
(c) inevitable (d) featureless

81. Destitute

- (a) affluent (b) dazzling
(c) stationary (d) characteristic

82. Devoid

- (a) latent (b) eschewed
(c) full of (d) suspecting

83. Devout

- (a) quiet (b) dual
(c) impious (d) straightforward

84. Diabolical

- (a) mischievous (b) lavish
(c) seraphic (d) redolent

85. Diatribe

- (a) mass (b) range
(c) eulogy (d) elegy

86. Diffidence

- (a) sharapness (b) boldness
(c) malcontent (d) dialogue

87. Dilate

- (a) procrastinate (b) contract
(c) conclude (d) participate

88. Dilatory

- (a) narrowing (b) prompt
(c) enlarging (d) portentous

89. Diminution

- (a) expectation (b) context
(c) validity (d) appreciation

90. Din

- (a) lightness (b) safety
(c) silence (d) hunger

91. Disabuse

- (a) crash (b) violate
(c) renege (d) deceive

92. Disconsolate

- (a) unprejudiced (b) thankful
(c) theatrical (d) joyous

93. Enervate

- (a) strengthen (b) sputter
(c) arrange (d) scrutinize

94. Enhance

- (a) degrade (b) doubt
(c) scuff (d) gasp

- 95. Ennui**
 (a) hate (b) excitement
 (c) seriousness (d) humility
- 96. Enunciate**
 (a) pray (b) request
 (c) deliver (d) mumble
- 97. Ephemeral**
 (a) sensuous (b) passing
 (c) popular (d) eternal
- 98. Equable**
 (a) flat (b) decisive
 (c) stormy (d) dishonest
- 99. Equanimity**
 (a) agitation (b) stirring
 (c) volume (d) identity
- 100. Equilibrium**
 (a) imbalance (b) peace
 (c) inequity (d) directness
- 101. Equitable**
 (a) able to leave (b) able to learn
 (c) unfair (d) preferable
- 102. Equivocal**
 (a) mistaken (b) quaint
 (c) azure (d) clear
- 103. Erratic**
 (a) unromantic (b) free
 (c) popular (d) steady
- 104. Erroneous**
 (a) accurate (b) dignified
 (c) curious (d) abrupt
- 105. Erudite**
 (a) professorial (b) stately
 (c) short (d) ignorant
- 106. Ethereal**
 (a) long-lasting (b) earthy
 (c) ill (d) critical
- 107. Eulogistic**
 (a) pretty (b) critical
 (c) brief (d) stern
- 108. Euphonious**
 (a) strident (b) lethargic
 (c) literary (d) significant
- 109. Evasive**
 (a) frank (b) correct
 (c) empty (d) fertile
- 110. Exasperate**
 (a) confide (b) formalize
 (c) placate (d) betray
- 111. Excoriate**
 (a) scandalize (b) encourage
 (c) avoid (d) praise
- 112. Exculpate**
 (a) blame (b) prevail
 (c) acquire (d) ravish
- 113. Execrable**
 (a) innumerable (b) philosophic
 (c) physical (d) excellent
- 114. Excrate**
 (a) decipher (b) sadden
 (c) integrate (d) admit
- 115. Exhume**
 (a) decipher (b) sadden
 (c) integrate (d) inter
- 116. Exodus**
 (a) neglect (b) consent
 (c) entry (d) gain
- 117. Exonerate**
 (a) forge (b) accuse
 (c) record (d) doctor
- 118. Exorbitant**
 (a) moderate (b) partisan
 (c) military (d) barbaric
- 119. Extemporaneous**
 (a) rehearsed (b) hybrid
 (c) humiliating (d) satetesmanlike
- 120. Extraneous**
 (a) modern (b) decisive
 (c) essential (d) effective
- 121. Extrinsic**
 (a) reputable (b) inherent
 (c) swift (d) ambitious
- 122. Extrovert**
 (a) clown (b) hero
 (c) ectomorph (d) introvert
- 123. Grandiose**
 (a) false (b) ideal
 (c) proud (d) simple
- 124. Gratuitous**
 (a) warranted (b) frank
 (c) ingenuous (d) frugal
- 125. Gregarious**
 (a) antisocial (b) anticipatory
 (c) glorious (d) horrendous
- 126. Grisly**
 (a) suggestive (b) doubtful
 (c) untidy (d) pleasant
- 127. Gullible**
 (a) incredulous (b) fickle
 (c) tantamount (d) easy
- 128. Gusto**
 (a) noise (b) panic
 (c) atmosphere (d) distaste
- 129. Ghusty**
 (a) clam (b) noisy
 (c) fragrant (d) routine
- 130. Hackneyed**
 (a) carried (b) original
 (c) banned (d) timely

131. Haggard

- (a) shrewish (b) inspired
(c) plump (d) maidenly

132. Halcyon

- (a) wasteful (b) prior
(c) subsequent (d) martial

133. Haphazard

- (a) safe (b) indifferent
(c) deliberate (d) tense

134. Hapless

- (a) cheerful (b) consistent
(c) fortunate (d) considerate

135. Heed

- (a) ignore (b) hope
(c) overtake (d) nurture

136. Heretic

- (a) sophist (b) believer
(c) interpreter (d) pacifist

137. Heterogeneous

- (a) orthodox (b) pagan
(c) unlikely (d) similar

138. Hibernial

- (a) musical (b) summerlike
(c) local (d) seasonal

139. Hilarity

- (a) gloom (b) heartiness
(c) weakness (d) casualty

140. Hirsute

- (a) scaly (b) bald
(c) erudite (d) quiet

141. Hortatory

- (a) inquiring (b) denying
(c) killing (d) dissuading

142. Hovr

- (a) commence (b) soothe
(c) leave (d) transform

143. Hunbbub

- (a) calm (b) fury
(c) capital (d) axle

144. Hummock

- (a) ummusical (b) scorn
(c) wakefulness (d) vale

145. Husbandry

- (a) sportsmanship (b) dishonesty
(c) wastefulness (d) friction

146. Hybrid

- (a) productive (b) special
(c) purebred (d) oafish

147. Hyperbole

- (a) velocity (b) climax
(c) curve (d) understatement

148. Hypercritical

- (a) tolerant (b) false
(c) extreme (d) inarticulate

149. Hypocritical

- (a) sincere (b) narrow-minded
(c) shameful (d) amiable

150. Hypothetical

- (a) rational (b) fantastic
(c) wizened (d) axiomatic

151. Ignoble

- (a) produced by fire (b) worthy
(c) given to questioning (d) huge

152. Illusive

- (a) not deceptive (b) not certain
(c) not obvious (d) not coherent

153. Irsome

- (a) interesting (b) lazy
(c) tireless (d) devious

154. Irrelevant

- (a) lacking piety (b) fragile
(c) congruent (d) pertinent

155. Irreparable

- (a) legible (b) correctable
(c) proverbial (d) concise

156. Irtreverent

- (a) related (b) mischievous
(c) respective (d) pious

157. Jaded

- (a) upright (b) stimulated
(c) aspiring (d) applied

158. Jaundiced

- (a) whitened (b) inflamed
(c) quickened (d) unbiased

159. Jaunty

- (a) youthful (b) ruddy
(c) strong (d) sedate

160. Jeopardy

- (a) patience (b) courage
(c) safety (d) willingness

161. Jettison

- (a) salvage (b) submerge
(c) descend (d) decelerate

162. Jocular

- (a) arterial (b) bloodless
(c) verbose (d) serious

163. Judicious

- (a) punitive (b) unwise
(c) criminal (d) licit

164. Kindle

- (a) dislike (b) quench
(c) gather (d) sparkle

165. Lachrymose

- (a) cheering (b) smooth
(c) passionate (d) curt

166. Lackadasical

- (a) monthly (b) possessing time
(c) ambitious (d) pusillanimous

167. Laconic

- (a) milky (b) verbose
(c) wicked (d) flagrant

168. Lampoon

- (a) darken (b) praise
(c) abandon (d) sail

169. Languor

- (a) vitality (b) length
(c) embarrassment (d) wine

170. Latent

- (a) trim (b) forbidding
(c) execrable (d) obvious

171. Lavish

- (a) hostile (b) unwashed
(c) timely (d) frugal

172. Laudatory

- (a) dritying (b) disclaiming
(c) defamatory (d) inflammatory

173. Lax

- (a) salty (b) strick
(c) shrill (d) boring

174. Lechery

- (a) trust (b) compulsion
(c) zeal (d) purity

175. Lethargic

- (a) convalescent (b) beautiful
(c) enervating (d) invigorating

176. Levite

- (a) bridge (b) dam
(c) praise (d) solemnity

177. Lilliputian

- (a) destructive (b) proper
(c) gigantic (d) elegant

178. Limpid

- (a) erect (b) turbid
(c) tangential (d) timid

179. Lithe

- (a) stiff (b) limpid
(c) facetious (d) insipid

180. Livid

- (a) alive (b) mundane
(c) positive (d) undiscolored

181. Loath

- (a) loose (b) evident
(c) deliberate (d) eager

182. Loquacious

- (a) taciturn (b) sentimental
(c) soporific (d) soothing

183. Moist

- (a) crisp (b) hard
(c) dry (d) parched

184. Mask

- (a) hit (b) expose
(c) benign (d) pompous

185. Moisture

- (a) dryness (b) delegate
(c) dampness (d) fdefect

186. Magnify

- (a) produce (b) support
(c) reduce (d) destroy

187. Mendacious

- (a) broken (b) efficacious
(c) truthful (d) destructive

188. Manumit

- (a) throw (b) lock
(c) promise (d) enslave

189. Nebulous

- (a) starry (b) clear
(c) cold (d) fundamental

190. Nefarious

- (a) various (b) lacking
(c) benign (d) pompous

191. Negation

- (a) postulation (b) hypothecation
(c) affirmation (d) violation

192. Neophyte

- (a) veteran (b) satellite
(c) desperado (d) handwriting

193. Niggardly

- (a) protected (b) biased
(c) prodigal (d) bankrupt

194. Nocturnal

- (a) harsh (b) marauding
(c) patrolling (d) daily

195. Noisome

- (a) quiet (b) dismayed
(c) fragrant (d) sleepy

196. Notorious

- (a) fashionable (b) renowned
(c) inactive (d) intrepid

197. Obdurate

- (a) yielding (b) fleeting
(c) finite (d) fascinating

198. Obese

- (a) skillful (b) cadaverous
(c) clever (d) unpredictable

199. Objective

- (a) indecisive (b) apathetic
(c) markedly inferior (d) emotionally involved

200. Obligatory

- (a) demanding (b) optional
(c) facile (d) friendly

201. Obloquy

- (a) praise (b) rectangle
(c) circle (d) dialogue

202. Obsequious

- (a) successful (b) democratic
(c) supercilious (d) ambitious

203. Obsession

- | | |
|------------|--------------|
| (a) whim | (b) loss |
| (c) phobia | (d) delusion |

204. Obsolete

- | | |
|---------------|---------------|
| (a) heated | (b) desolate |
| (c) renovated | (d) frightful |

205. Obstreperous

- | | |
|------------|-----------|
| (a) turbid | (b) quiet |
| (c) remote | (d) lucid |

206. Obtuse

- | | |
|--------------|-----------------|
| (a) sheer | (b) transparent |
| (c) tranquil | (d) shrewd |

207. Odious

- | | |
|--------------|----------------|
| (a) fragrant | (b) redolent |
| (c) fetid | (d) delightful |

208. Odium

- | | |
|--------------|------------|
| (a) noise | (b) liking |
| (c) dominant | (d) hasty |

209. Omnipotent

- | | |
|--------------|----------------|
| (a) weak | (b) democratic |
| (c) despotic | (d) passionate |

210. Omniscient

- | | |
|-------------------|--------------|
| (a) sophisticated | (b) ignorant |
| (c) essential | (d) trivial |

211. Opiat

- | | |
|---------------|--------------|
| (a) distress | (b) sleep |
| (c) stimulant | (d) laziness |

212. Opportune

- | | |
|----------------|--------------|
| (a) occasional | (b) fragrant |
| (c) fragile | (d) awkward |

213. Opportunist

- | | |
|--------------------|----------------------|
| (a) man of destiny | (b) man of principle |
| (c) changeling | (d) adversary |

214. Opprobrium

- | | |
|-----------------|--------------|
| (a) delineation | (b) aptitude |
| (c) majesty | (d) praise |

215. Optimum

- | | |
|-----------------|-------------------|
| (a) pessimistic | (b) knowledgeable |
| (c) worst | (d) minimum |

216. Opulence

- | | |
|---------------|----------------|
| (a) pessimism | (b) patriotism |
| (c) potency | (d) poverty |

217. Ostentatious

- | | |
|--------------|----------------|
| (a) inactive | (b) unassuming |
| (c) impolite | (d) illicit |

218. Overweening

- | | |
|------------|-----------------|
| (a) humble | (b) impotent |
| (c) avid | (d) acrimonious |

219. Pejorative

- | | |
|-----------------|----------------|
| (a) positive | (b) legal |
| (c) determining | (d) delighting |

220. Pellucid

- | | |
|----------------------|------------------|
| (a) logistical | (b) philandering |
| (c) incomprehensible | (d) vagrant |

221. Penchant

- | | |
|--------------|---------------|
| (a) distance | (b) imminence |
| (c) dislike | (d) attitude |

222. Penurious

- | | |
|----------------|----------------|
| (a) imprisoned | (b) captivated |
| (c) generous | (d) vacant |

223. Perfunctory

- | | |
|--------------|---------------|
| (a) official | (b) through |
| (c) insipid | (d) vicarious |

224. Perigee

- | | |
|-------------|-------------|
| (a) eclipse | (b) planet |
| (c) apogee | (d) refugee |

225. Peripatetic

- | | |
|---------------|----------------|
| (a) wordly | (b) stationary |
| (c) disarming | (d) seeking |

226. Permeable

- | | |
|----------------|------------------|
| (a) perishable | (b) effective |
| (c) plodding | (d) impenetrable |

227. Pernicious

- | | |
|---------------|-----------------|
| (a) practical | (b) comparative |
| (c) harmless | (d) tangible |

228. Perpetual

- | | |
|---------------|----------------|
| (a) momentary | (b) standard |
| (c) serious | (d) industrial |

229. Perspicuity

- | | |
|------------|---------------|
| (a) grace | (b) feature |
| (c) review | (d) vagueness |

230. Pert

- | | |
|------------|----------------|
| (a) polite | (b) perishable |
| (c) moral | (d) deliberate |

231. Pertinacious

- | | |
|-----------------|-----------------|
| (a) vengeful | (b) consumptive |
| (c) superficial | (d) skilled |

232. Pertinent

- | | |
|------------------|----------------|
| (a) understood | (b) living |
| (c) discontented | (d) irrelevant |

233. Petulant

- | | |
|---------------|-------------------|
| (a) angry | (b) moral |
| (c) declining | (d) uncomplaining |

234. Precipitate

- | | |
|--------------|------------------|
| (a) dull | (b) anticipatory |
| (c) cautious | (d) considerate |

235. Prefatory

- | | |
|-----------------|-----------------|
| (a) outstanding | (b) magnificent |
| (c) conclusive | (d) intelligent |

236. Prelude

- | | |
|----------------|---------------|
| (a) intermezzo | (b) diva |
| (c) aria | (d) aftermath |

237. Presumption

- | | |
|----------------|----------------|
| (a) assertion | (b) activation |
| (c) motivation | (d) humility |

238. Pretentious

- | | |
|----------------|------------|
| (a) ominous | (b) calm |
| (c) unassuming | (d) futile |

239. Prim

- | | |
|--------------|-------------|
| (a) informal | (b) prior |
| (c) exterior | (d) private |

240. Pristine

- | | |
|-----------------|---------------|
| (a) cultivated | (b) condemned |
| (c) irreligious | (d) cautious |

241. Probity

- | | |
|--------------------|----------------|
| (a) aregert | (b) assumption |
| (c) corruptibility | (d) extent |

242. Prodigal

- | | |
|----------------|----------------|
| (a) large | (b) thrifty |
| (c) consistent | (d) compatible |

243. Prodigious

- | | |
|-------------------|----------------|
| (a) infinitesimal | (b) indignant |
| (c) indifferent | (d) indisposed |

244. Profane

- | | |
|--------------|---------------|
| (a) sanctify | (b) desecrate |
| (c) define | (d) manifest |

245. Prolific

- | | |
|----------------|-------------------|
| (a) unworkable | (b) backward |
| (c) barren | (d) controversial |

246. Prolix

- | | |
|---------------|-----------------|
| (a) stupid | (b) indifferent |
| (c) redundant | (d) pithy |

247. Prophylactic

- | | |
|--------------------|---------------------|
| (a) causing growth | (b) causing disease |
| (c) antagonistic | (d) brushing |

248. Propinquity

- | | |
|----------------|----------------|
| (a) remoteness | (b) uniqueness |
| (c) health | (d) virtue |

249. Propitious

- | | |
|--------------|-----------------|
| (a) rich | (b) induced |
| (c) promoted | (d) unfavorable |

250. Prosaic

- | | |
|--------------|-----------------|
| (a) pacified | (b) reprieved |
| (c) pensive | (d) imaginative |

251. Protean

- | | |
|----------------|--------------|
| (a) amateur | (b) catholic |
| (c) unchanging | (d) rapid |

252. Protract

- | | |
|-------------------|-------------|
| (a) make circular | (b) shorten |
| (c) further | (d) retrace |

253. Provident

- | | |
|---------------|--------------|
| (a) unholy | (b) rash |
| (c) miserable | (d) disabled |

254. Provincial

- | | |
|--------------|-------------------|
| (a) wealthy | (b) crass |
| (c) literary | (d) sophisticated |

255. Psychotic

- | | |
|---------------|-----------------|
| (a) dangerous | (b) clairvoyant |
| (c) criminal | (d) sane |

256. Puerile

- | | |
|--------------|--------------|
| (a) fragrant | (b) adult |
| (c) lonely | (d) feminine |

257. Pugnacious

- | | |
|-------------|-------------|
| (a) pacific | (b) feline |
| (c) mature | (d) angular |

258. Puissant

- | | |
|----------------|-----------------|
| (a) pouring | (b) fashionable |
| (c) articulate | (d) weak |

259. Pulchritude

- | | |
|----------------|-----------------|
| (a) ugliness | (b) notoriety |
| (c) bestiality | (d) masculinity |

260. Punctilious

- | | |
|-----------|--------------|
| (a) happy | (b) active |
| (c) vivid | (d) careless |

261. Punitive

- | | |
|---------------|--------------|
| (a) large | (b) humorous |
| (c) rewarding | (d) restive |

262. Pusillanimous

- | | |
|----------------------|----------------|
| (a) poverty-stricken | (b) chained |
| (c) posthumous | (d) courageous |

263. Putative

- | | |
|----------------|-----------------|
| (a) colonial | (b) quarrelsome |
| (c) undisputed | (d) powerful |

264. Resilient

- | | |
|-------------|----------------|
| (a) pungent | (b) foolish |
| (c) worthy | (d) unyielding |

265. Restive

- | | |
|----------------|---------------|
| (a) buoyant | (b) placid |
| (c) remorseful | (d) resistant |

266. Retentive

- | | |
|----------------|---------------|
| (a) forgetful | (b) accepting |
| (c) repetitive | (d) avoiding |

267. Reticence

- | | |
|---------------|--------------------|
| (a) fatigue | (b) fashion |
| (c) treachery | (d) loquaciousness |

268. Retrograde

- | | |
|-----------------|-------------------|
| (a) progressing | (b) inclining |
| (c) evaluating | (d) concentrating |

269. Revere

- | | |
|-------------|--------------|
| (a) advance | (b) dishonor |
| (c) age | (d) precede |

270. Rife

- | | |
|--------------|-----------|
| (a) direct | (b) scant |
| (c) peaceful | (d) grim |

271. Robust

- | | |
|-------------|----------------|
| (a) weak | (b) violent |
| (c) vicious | (d) villainous |

272. Rotundity

- | | |
|-------------------|--------------|
| (a) promenade | (b) nave |
| (c) grotesqueness | (d) slimness |

273. Rubble

- | | |
|-----------------------|--------------------|
| (a) artificial facade | (b) unbroken stone |
| (c) pale complexion | (d) strong defense |

274. Ruddy

- | | |
|------------|------------|
| (a) robust | (b) witty |
| (c) wan | (d) exotic |

275. Rudimentary

- | | |
|-------------|---------------|
| (a) pale | (b) polite |
| (c) asinine | (d) developed |

276. Rueful

- | | |
|-------------|-------------|
| (a) trite | (b) content |
| (c) capable | (d) capital |

277. Rustic

- | | |
|--------------|---------------|
| (a) urban | (b) slow |
| (c) corroded | (d) mercenary |

278. Ruthless

- | | |
|--------------|--------------|
| (a) merciful | (b) majestic |
| (c) mighty | (d) militant |

279. Sadistic

- | | |
|-----------------|------------|
| (a) happy | (b) quaint |
| (c) kindhearted | (d) vacant |

280. Sagacious

- | | |
|---------------|---------------|
| (a) foolish | (b) bitter |
| (c) voracious | (d) veracious |

281. Sallow

- | | |
|---------------|---------------|
| (a) salacious | (b) ruddy |
| (c) colorless | (d) permitted |

282. Salubrious

- | | |
|-------------|-------------|
| (a) salty | (b) bloody |
| (c) miasmic | (d) maudlin |

283. Salvage

- | | |
|------------|------------|
| (a) remove | (b) outfit |
| (c) burn | (d) lose |

284. Sanctimonious

- | | |
|------------------|------------|
| (a) hypothetical | (b) paltry |
| (c) mercenary | (d) pious |

285. Sanguine

- | | |
|---------------|-----------------|
| (a) choleric | (b) sickening |
| (c) warranted | (d) pessimistic |

286. Satiety

- | | |
|---------------|------------------|
| (a) emptiness | (b) warmth |
| (c) erectness | (d) straightness |

287. Scanty

- | | |
|---------------|---------------|
| (a) collected | (b) remote |
| (c) invisible | (d) plentiful |

288. Scurrilous

- | | |
|------------|--------------|
| (a) savage | (b) scabby |
| (c) decent | (d) volatile |

289. Secular

- | | |
|--------------|--------------|
| (a) vivid | (b) clerical |
| (c) punitive | (d) positive |

290. Sedentary

- | | |
|----------------|------------|
| (a) vicarious | (b) loyal |
| (c) accidental | (d) active |

291. Sedulous

- | | |
|----------------|---------------|
| (a) indolent | (b) guileless |
| (c) vindictive | (d) upright |

292. Senility

- | | |
|-------------------|--------------------|
| (a) verility | (b) loquaciousness |
| (c) forgetfulness | (d) youth |

293. Sententious

- | | |
|-----------------|--------------|
| (a) paragraphed | (b) positive |
| (c) posthumous | (d) wordy |

294. Tractable

- | | |
|------------------|-----------------|
| (a) unmanageable | (b) irreligious |
| (c) mortal | (d) incapable |

295. Traduce

- | | |
|--------------|--------------|
| (a) exhume | (b) increase |
| (c) purchase | (d) extol |

296. Tranquillity

- | | |
|-------------------|------------------|
| (a) lack of sleep | (b) lack of calm |
| (c) emptiness | (d) renewal |

297. Transient

- | | |
|---------------|-------------|
| (a) carried | (b) close |
| (c) permanent | (d) removed |

298. Tremulous

- | | |
|------------|-------------|
| (a) steady | (b) obese |
| (c) young | (d) healthy |

299. Trenchant

- | | |
|------------------|---------------|
| (a) lacking bite | (b) imperious |
| (c) inessential | (d) unafraid |

300. Trepidation

- | | |
|------------------|----------------|
| (a) slowness | (b) amputation |
| (c) fearlessness | (d) adroitness |

301. Trite

- | | |
|-------------------|---------------|
| (a) correct | (b) original |
| (c) distinguished | (d) premature |

302. Truculent

- | | |
|--------------|-----------------|
| (a) juicy | (b) overflowing |
| (c) peaceful | (d) determined |

303. Tumult

- | | |
|--------------|--------------|
| (a) scarcity | (b) defeat |
| (c) coolness | (d) serenity |

304. Turbid

- | | |
|----------------|----------------|
| (a) clear | (b) improbable |
| (c) invariable | (d) honest |

305. Turbulence

- | | |
|-----------------|--------------|
| (a) reaction | (b) approach |
| (c) impropriety | (d) calm |

306. Turgid

- | | |
|------------|---------------|
| (a) rancid | (b) shrunken |
| (c) cool | (d) explosive |

307. Turpitude

- | | |
|---------------|------------|
| (a) amplitude | (b) heat |
| (c) wealth | (d) virtue |

308. Tyro

- | | |
|---------------|------------|
| (a) infant | (b) rubber |
| (c) personnel | (d) expert |

309. Unearth

- | | |
|-------------|-----------|
| (a) conceal | (b) gnaw |
| (c) clean | (d) fling |

310. Unfeigned

- | | |
|---------------|-----------------|
| (a) pretended | (b) fashionable |
| (c) wary | (d) switched |

311. Ungainly

- (a) ignorant
- (b) graceful
- (c) detailed
- (d) dancing

312. Unimpeachable

- (a) fruitful
- (b) rampaging
- (c) faulty
- (d) pensive

313. Unkempt

- | | |
|------------|------------|
| (a) bombed | (b) washed |
| (c) neat | (d) tawdry |

314. Unruly

- | | |
|--------------|--------------|
| (a) chatting | (b) obedient |
| (c) definite | (d) lined |

315. Unseemly

- | | |
|----------------|-------------|
| (a) effortless | (b) proper |
| (c) conducive | (d) pointed |

316. Unsullied

- | | |
|---------------|---------------|
| (a) tranished | (b) countless |
| (c) soggy | (d) papered |

317. Untenable

- | | |
|-----------------|---------------|
| (a) supportable | (b) tender |
| (c) sheepish | (d) tremulous |

318. Unwitting

- | | |
|---------------|-----------------|
| (a) clever | (b) intense |
| (c) sensitive | (d) intentional |

319. Vacillation

- | | |
|-------------|-------------------|
| (a) remorse | (b) relief |
| (c) respect | (d) steadfastness |

320. Valedictory

- | | |
|----------------|----------------|
| (a) sad | (b) collegiate |
| (c) derivative | (d) salutatory |

321. Valor

- (a) admonition
- (b) injustice
- (c) cowardice
- (d) generosity

322. Vanguard

- (a) regiment
- (b) rear
- (c) echelon
- (d) protection

323. Vaunted

- (a) unvanquished
- (b) fell
- (c) belittled
- (d) believed

ANSWERS

MCQ Type Questions

1. (d)	2. (b)	3. (d)	4. (b)	5. (a)	6. (d)	7. (a)	8. (a)	9. (a)	10. (b)
11. (b)	12. (b)	13. (b)	14. (d)	15. (b)	16. (c)	17. (d)	18. (c)	19. (c)	20. (d)
21. (d)	22. (a)	23. (d)	24. (c)	25. (b)	26. (d)	27. (b)	28. (b)	29. (d)	30. (d)
31. (c)	32. (a)	33. (d)	34. (b)	35. (d)	36. (a)	37. (d)	38. (a)	39. (b)	40. (b)
41. (d)	42. (d)	43. (c)	44. (a)	45. (d)	46. (d)	47. (c)	48. (d)	49. (d)	50. (a)
51. (c)	52. (d)	53. (b)	54. (a)	55. (d)	56. (c)	57. (b)	58. (a)	59. (b)	60. (a)
61. (c)	62. (b)	63. (c)	64. (b)	65. (c)	66. (d)	67. (d)	68. (c)	69. (b)	70. (a)
71. (a)	72. (b)	73. (b)	74. (b)	75. (a)	76. (c)	77. (a)	78. (d)	79. (d)	80. (b)
81. (a)	82. (c)	83. (c)	84. (d)	85. (c)	86. (b)	87. (b)	88. (b)	89. (d)	90. (c)
91. (d)	92. (d)	93. (a)	94. (a)	95. (b)	96. (d)	97. (d)	98. (c)	99. (a)	100. (a)
101. (c)	102. (b)	103. (d)	104. (a)	105. (d)	106. (b)	107. (b)	108. (a)	109. (a)	110. (c)
111. (d)	112. (a)	113. (d)	114. (b)	115. (d)	116. (c)	117. (b)	118. (a)	119. (b)	120. (c)
121. (b)	122. (d)	123. (d)	124. (a)	125. (a)	126. (d)	127. (a)	128. (d)	129. (d)	130. (b)
131. (c)	132. (d)	133. (c)	134. (c)	135. (a)	136. (b)	137. (d)	138. (b)	139. (a)	140. (b)
141. (d)	142. (c)	143. (a)	144. (d)	145. (c)	146. (c)	147. (d)	148. (a)	149. (a)	150. (d)
151. (b)	152. (a)	153. (a)	154. (d)	155. (b)	156. (d)	157. (b)	158. (d)	159. (d)	160. (c)
161. (a)	162. (d)	163. (a)	164. (b)	165. (a)	166. (c)	167. (d)	168. (b)	169. (a)	170. (d)
171. (d)	172. (c)	173. (b)	174. (d)	175. (d)	176. (d)	177. (c)	178. (b)	179. (a)	180. (d)
181. (d)	182. (a)	183. (c)	184. (b)	185. (a)	186. (c)	187. (c)	188. (d)	189. (b)	190. (c)
191. (c)	192. (a)	193. (c)	194. (d)	195. (c)	196. (b)	197. (d)	198. (d)	199. (d)	200. (b)
201. (a)	202. (c)	203. (a)	204. (c)	205. (b)	206. (d)	207. (d)	208. (b)	209. (a)	210. (a)
211. (c)	212. (d)	213. (b)	214. (d)	215. (c)	216. (d)	217. (b)	218. (a)	219. (a)	220. (c)
221. (c)	222. (c)	223. (b)	224. (c)	225. (b)	226. (d)	227. (c)	228. (a)	229. (d)	230. (a)
231. (c)	232. (d)	233. (d)	234. (c)	235. (c)	236. (d)	237. (d)	238. (c)	239. (a)	240. (a)
241. (c)	242. (b)	243. (a)	244. (a)	245. (c)	246. (d)	247. (b)	248. (a)	249. (d)	250. (d)
251. (c)	252. (b)	253. (b)	254. (d)	255. (d)	256. (b)	257. (a)	258. (d)	259. (a)	260. (b)
261. (c)	262. (d)	263. (c)	264. (b)	265. (a)	266. (d)	267. (a)	268. (b)	269. (b)	270. (b)
271. (a)	272. (d)	273. (b)	274. (c)	275. (d)	276. (b)	277. (a)	278. (a)	279. (c)	280. (a)
281. (b)	282. (c)	283. (d)	284. (d)	285. (d)	286. (d)	287. (d)	288. (c)	289. (b)	290. (d)
291. (a)	292. (d)	293. (d)	294. (a)	295. (d)	296. (b)	297. (c)	298. (a)	299. (a)	300. (c)
301. (b)	302. (c)	303. (d)	304. (a)	305. (d)	306. (b)	307. (d)	308. (d)	309. (a)	310. (a)
311. (b)	312. (c)	313. (d)	314. (b)	315. (b)	316. (a)	317. (a)	318. (d)	319. (d)	320. (d)
321. (c)	322. (b)	323. (d)							

5

CHAPTER

VOCABULARY FOR READING COMPREHENSION

Not appreciating the subtle differences between words like - sarcastic and sardonic, ironical, cynical,... (the list goes on), many threaten your chances of a high score in R.C. (especially in inferential and true tone passages).

The beauty of English gets more enhanced if understood in the right perspective. On the other hand, one word misunderstood may stand you in for a comical interpretation of a serious passage. The 'Vocabulary for R.C.' section becomes a unique section, as it tries to encompass a kaleidoscope of tones, moods, opinions, characteristics, analogies which have statistically found a higher representation in the R.C. passages of management entrance exams.

Anarchist : A person who does not believe in any kind of system, law and order, governance and goes an extra mile to prove his view on political disorder.

Agnostic : A person who is unsure of the existence of God. Unlike atheists he is a little flexible with respect to his belief about God, but still unsure.

Atheism/atheist : A belief in the non-existence of God.

Authoritarian : A person following complete obedience to authority. Hitler is a good example of the kind.

Example : The authoritarian regime of Taliban finally met its nemesis.

Analytical : Characterized by logical and reasonable basis. Analytically sound managers - need of the hour.

Critic : A person who shows faults in something and usually expresses a base/low opinion.

Example : Saurav Ganguly's critics have suddenly found a new voice, after his litany of failures with shortpitch ball. A person who judges the quality of something especially works of art, literature, music, etc.

Example : He is a music critic, i.e. he judges

Callous : Having or showing no sympathy for other people's feelings or suffering.

Example : A callous disregard for the feeling of others.

Miscellaneous

Bacchanalian : Orgy; referred to a festival which was celebrated with dancing, song and revelry.

Blasphemy : The act of insulting or showing contempt or lack of reverence towards God.

Example : Blasphemy and independent thinking are very often confused.

Empiricism : The practice of relying on observation and experiment especially in the natural sciences.

A theory that all knowledge originates in experience.

Example : An empirical analysis of the situation yields different results.

Critical (for somebody/something) : It means indicating faults in someone.

The jihadis were quite critical of Pakistan supporting US against Afghanistan.

Critique : Critical analysis

Example : Presented in this column is a critique of the govt's economic policies.

Cynic : A person who believes that people do not do things for good, sincere or noble reasons, but only for their own advantage: concerned only with one's own interests.

Example : Stop giving these cynical smiles.

Example : Women in Afghanistan had over a period of time started viewing the cynical Taliban regime with much apprehension.

Example : His adventurism was very close to being termed as a cynical disregard for the safety of others.

Contemplative : Pondering over or considering thoughtfully.

Example : Contemplate changing careers.

Commendation : Praise or approval.

Castigating : To chastise, punish or to criticize severely.

Example : Hansie Cronje's castigation by the sports media.

Disparagement : The art or process of speaking in a belittling way; decrying, or reducing someone in rank or esteem.

Example : Disparaging someone's character or achievements.

Disinterested (noun) : Freedom from bias or self-interest, impartiality

Example : A disinterested act of kindness. Different from uninterested being without an interest.

Disillusion (ed) : To destroy the but mistaken beliefs or ideals.

Example : Disillusioned voters want an alternative to the two parties.

Disingenuous : Not straightforward; crafty.

Exploratory : To investigate systematically; to examine.

Example : Explore every suggestion given.

Ecclesiastical : Of or relating to a church, as an organized institution.

Example : The period witnessed strong anti-ecclesiastical view from the masses.

Eclectic : Selecting what appears to be the best form diverse sources systems, or styles. An eclectic approach.

Example : Different from the non-holistic: emphasizing the importance or the whole and the interdependence of its parts.

Frivolous : Something which demands lesser important or someone who is lacking in seriousness.

Example : The Zimbabweans are not frivolous when it comes to playing them in their own soil.

Indignation : Anger aroused by something unjust, unworthy, or mean.

Example : Kapil Dev's indignation is just.

Inquisitive : Someone who has the propensity to ask questions.

Example : My first class in the college saw many inquisitive students trying to unnerve me.

Ironic : Something which happens in the opposite way to what expected and often causes amusement because of this.

Example : It is ironic that he died of AIDS, though he was the one who discovered the first AIDS vaccine.

Myth : It is something untrue.

Mystify : To puzzle someone/something.

Morose : Unhappy and bad tempered; but not saying much.

Example: Morose mannerisms.

Moron : A very stupid person.

Misnomer : A wrong use of a name, word or description.

Example : To call CAT exam, a catwalk would be a misnomer.

Mystic : Something/someone which has a hidden meaning or spiritual power and arouses feeling of deep respect; and wonder.

Example : Mystical writings.

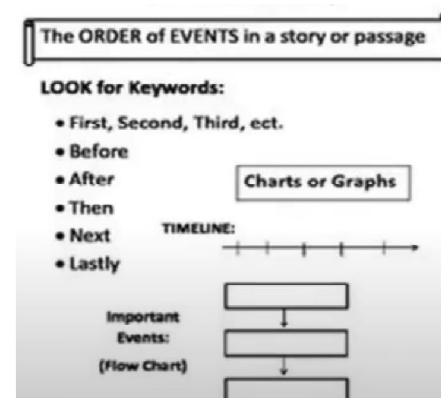
Mysticism : To believe that knowledge of God and a real truth may be reached by directing one's mind through spiritual insight.

NARRATIVE SEQUENCE

Narrative sequence is used to identify what happens at the beginning, middle and end of a sample paragraph and also identify the transition word in a given paragraph. Places of these identify words in given sentences used for to create a story.

When we read, it is important to be able to retell the events in a story. Those events need to be retold in the correct order for the story to make sense. Here, we will learn how to sequence events in a narrative.

Notes:



Sequence of Events

Have you ever seen a really good movie? Maybe it was an action-packed superhero movie or a fantasy adventure to a far-off land. When a director creates a new film, he or she has to make sure the scenes help the movie make sense from the opening scene until the credits start rolling.

Likewise, as a reader, it's important for you to identify the sequence of events in a story. The **sequence of events** is the order of events as they occur in a story. Just like when the heroes can't stop the world-ending doomsday machine before it's been built in a superhero movie, the sequence of events in story occurs in a precise order so that it makes sense to the person reading it.

So how do you identify the sequence of events? Get your popcorn because it's showtime!

Organize the Story

Example Sequence of Events Chart

The Boy Who Cried Wolf

Beginning	Middle	End
A wolf was eating the sheep of a town	The towns people came to help and found out the boy was lying.	A wolf really did approach the boy
The boy was bored of watching the sheep	The boy did this a couple more time.	The boy cried for help, but no one came because they thought he was lying again
He decided to pretend the wolf was attacking the sheep.	The town's people were upset.	The wolf attacked the boy and the sheep

Movie directors often use storyboards to help them visualize what will happen in each scene. Storyboards show what will happen in each camera shot just like the panels of a comic book tell a story. As a reader, you can also think of the sequence of events like a storyboard.

After you've read a story, you can organize the story into three sections and ask yourself the following questions:

- What happens at the beginning of the story?
- What happens in the middle of the story?
- What happens at the end of the story?

The beginning of the story usually introduces the main characters, setting, and problem faced by or purpose of the characters. The middle part of the story usually focuses on the characters as they try and solve the problem. The end of the story focuses on how the problem is resolved, or the end of the characters' journey.

It may help to create a chart, like this one, to help you organize the different events that make up the beginning, middle, and end of a story.

Sequence of Events:

Here we taught the importance of understanding the events of a story in order.

Structure of a Narrative: Easy Level

As this lesson explained, there are certain things that usually happen at the **beginning, middle, and end** of a story. Think of some stories that you are very familiar with. Nursery rhymes and fairy tales are a great place to start because they tend to be simple

and easy to remember. Choose at least three stories and divide them into their beginning, middle, and end sections. Check back on this lesson if you get stuck. When you feel confident that you've divided the stories properly, get a family member or teacher to check your work.

Story Scramble: Medium Level

There is more to a story than just beginning, middle, and end. Many events can happen in a single story, just like in the example of the tortoise and the hare in this story. Think of a book that you've recently read. On one piece of paper, write a clear list of events in the story. Now, cut them up into separate sentences and mix them all up. Can you put them back together again in the right order? Make sure that the end result of your activity makes logical sense.

Write Your Own: Hard Level

If you feel really confident in your understanding of narrative structure, try writing your own short story. Think carefully about the order of events as you write them. What does the audience have to be aware of for the events to make sense? What is your story's beginning, middle, and end? Consider these questions as you write.

Narrative Sequencers:

Narrative Sequencer is nothing but a word which tells you where in the paragraph transition words will take place.

Steps of sequencers:

- First
- Next
- Then
- After that
- Finally

Example 1:

Add sequencers in the correct spot in the following sentence:

I'll buy coffee and go to the movie.

- I'll buy first coffee and go then to the movie
- Then, I'll buy coffee and first go to the movie
- First, I'll buy coffee and then go to the movie
- I'll buy coffee and first go to the movie

Answer : (c)

Example 2:

Andrea loves to play soccer, but she broke her arm and cannot play for a while. Since breaking her arm, Andrea has been very sad. A friend of hers suggested that Andrea find another hobby to take her mind off of her injury. Andrea took her friend's advice and tried

knitting. She loved it! Now, Andrea spends her time knitting while waiting for her arm to heal, and she isn't sad anymore.

- (a) Andrea broke her arm and cannot play soccer for a while.
- (b) Andrea's friend suggested that she find another hobby, so she tried knitting
- (c) Andrea isn't sad anymore.
- (d) Andrea's arm healed and she started playing soccer again.

Answer (b)

SPEAKERS

Banal : A person whose speech is dull, flat, boring, unimaginative

Cogent : A person who can express his views brilliantly & convincingly

Garrulous : A person who goes on speaking a lot, all meaningless

Grandiloquent : A person who always uses heavy words

Inarticulate : A person who is very awkward while speaking

Laconic : A person saying few words full of deep meaning

Loquacious : A person who speaks just too much, quantity wise

Taciturn : An extremely quite person speaking very little
Syn : Reticent/ Reserved/Close-lipped

Verbose : A person who repeats the same idea again and again

Ventriloquist : A person who can throw his voice creating an illusion that it is coming from another source

Voluble : A very fluent and rapid speaker

Vociferous : A very noisy speaker

Magniloquent : A person who talks big

Eloquent : A person who uses great, artistic or emotional language

Soporific : A person whose speech puts the audience to sleep.

Egotist : A person who constantly boasts of his own achievements.

Soliloquist : A person talking to oneself.

JARGONS

These lists will help you remember groups of words and their interconnections. This will help you solve questions based on verbal analogies etc.

(1) Words Related to law:

admonition, evidence, prosecution, advocate, forgery, reprieve, allege, gibbet, scourage, attorney, incarceration, scoundrel, barrister, indictment, sentences, borstal, judicial, surety, circumstantial, culpable, perjury, vault, culprit

(2) Words related to death:

bier, decease, requiem, burial, epitaph, rites, cenotaph, obituary

(3) Words related to Human Nature:

arrogant, foppish, obsequious, beneficiaries, garrulous, peculiar, benefactor, haughty, propensity, benevolent, impudent, reckless, courteous, insolent, rogue, cringing, knave, sheer, dauntless, malevolent, sober, deficient, malicious, suave, enthusiastic.

Mendacious, taunt, finical, obstinate, tantalise

(4) Words related to Energy and Electricity:

aerial, crystal, galena, atmospheric, detector, insulate, battery, distortion, leads, bias, ebonite, neutralise, capacity, emitter, oscillate, phenomenon, circuit, energy, condenser, ether, radiating, current, frequency, radio, receiver. Telephone, sensitiveness,

(5) Words related to Automobile:

accelerator, differential, piston, armature, dynamo, radiator, axle, exhaust, silencer, sparking plug, bonnet, gear, bearings, generator, sump, carburetor, ignition, switch, clutch, induction, tyres, cylinder, magneto, valves, decarbonise, petrol, windscreen

(6) Words Related to Games and Entertainment:

aquatic, cycling, rehearsal, arena, hockey, sculling, athletics, opera, tennis, chess, pantomime, theatre, croquet, regatta, whistle

(7) Words Related to Geography:

agriculture, frigid, plantation, arable, frontier, sedimentary, atoll, furrow, solar, barrier, glade, survey, cascade, insular, swamp, climate, inundation, temperature, confluence, irrigation, tornado, continent, knoll, torrential, contour, mound, tillage, crater, meteor, typhoon, configuration, monsoon, vegetation, cyclone, oasis, verdure, eclipse, peninsular, zephyr, expanse, planetary, zone

(8) Words related to Aviation:

accessory, aviation, monoplane, aerodrome, aviator, propeller, aeroplane, hydroplane, pancake, rigid, skid, volplane, rudder, stunt

(9) Words related to Metallurgy:

amalgam, filament, platinum, brazen, flux, radium, copra, galvanise, shellac, concrete, kiln,

solder, ductile, mercury, tenacious, ebonite, nickel

(10) Words Related to Age and History :

antecedent, octogenarian, epoch, anterior, era, perpetual, anniversary, evanescent, prior, antique, jubilee, senile, centenary, simultaneous, matinee, chronometer, nocturnal, synchronise, contemporary, oblivion, ultimate, ephemeral

(11) Words Related to Shipping :

angler, freight, scupper, auxiliary, funnel, scuttle, berth, gear, sloop, bilge, keel, splice, bulkhead, mariner, starboard, bulwark, motor, stowaway, bunkers, naval, tackle, buoy, navigable, tarpaulin in, butt, nautical, tempest, canvas, pilot, towboat, chronometer, pirate, turbine, damage, quarantine, ventilator, derelict, rocket, veer, draught, saloon, wreckage, eddy, salvage, wrack, flare

(12) Words Related to Arms and Ammunition:

ammunition, knobkerrie, scimitar, bludgeon, knout, sheath, carbine, missile, sheathe, corslet, munitions, shrapnel, dynamite, poniard, visor, gully, quarrel

(13) Words related to Battle and Warfare:

adversary, dispatch, organise, armistice, enrolment, parley, assault, equipment, preliminary, belligerent, reinforcement, escort, casualty, reverberating, havoc, censor, hazard, revolution, cipher, indecisive, scheme, code, convoy, massacre, sniper, conscription, mutineer, strategic, decipher, mutual, strategic, declaration, neutral, violation, despatch, opponent

(14) Words Related to Navy:

admiralty, explosive, squadron, arsenal, magazine, submarine, artificer, monitor, submerge, boastswain, naval, submersion, camouflage, pinnacle, torpedo, cruiser, privateer, turret

(15) Words Related to Food and Cookery:

abattoir, essence, quail, albumen, farina, raisin, appetiser, frugal, relish, banquet, gammon, refrigerator, beverage, giblet, salad, caterer, gluttonous, sauce, cheddar, griskin, sausage, cocoa, intestine, spice, cinnamon, kidney, sponge, crumpet, knead, sumptuous, condensed, mace, steak, chutney, maize, slaughter, cutlet, mustard, tallow, chopper, nutriment, teal, cleaver, nitrogenous, tongue, cruet, offal, unleavened, culinary, palatable, vanilla, curry, prince, Vienna, dough, porridge, vegetables, dessert, pudding, viands, diet, preserve, victuals, digestible, provisions, vinegar

PHOBIAS, MANIAS AND COMPLEXES

FEAR	PHOBIA
Air	Aerophobia
Animals	Zoophobia
Beauty	Gallophobia
Birth	Genophobia
Blood	Hematophobia
Burglars	Scelerophobia
Burial alive	Taphephobia
Change	Neophobia
Childbirth	Maleusiophobia
Children	Pedophobia
Colours	Chromophobia
Crowd	Ochlophobia
Darkness	Nyctophobia
Death	Thanatophobia
Depth	Bathophobia
Disease	Pathophobia
Doctors	Latrophobia
Ghost	Phasmophobia
Hand	Chirophobia
Hair	Trichophobia
Heat	Thermophobia
Hell	Stygiophobia
Horses	Hippophobia
Knives	Aichmophobia
Large things	Megalophobia
Light	Photophobia
Lightning	Astrophobia
Males	Androphobia
Medicine	Pharmacophobia
Motherhood	Metrophobia
Motion	Kinesophobia
Pain	Agophobia; odynophobia
People	Demophobia
Poison	Toxicophobia
Poverty	Peniophobia
Red	Erythrophobia
Sexual intercourse	Coitophobia
Sinning	Peccatophobia
Sleep	Hypnophobia
Small things	Microphobia
Snake	Ophidiophobia

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Miscellaneous

Solitude	Autophobia; monophobia
Sound	Acousticophobia
Stairs	Climacophobia
Stealing	Kleptophobia
Study	Logophobia
Sunlight	Heliophobia
Thirteen (The number)	Triksiaidekaphobia
Water	Hydrophobia
Wealth	Plutophobia
Women	Gynaephobia
Work	Ergophobia
Writing	Graphophobia

Mania

Pyromaniac	Has a passion for fire
Dipsomaniac	Has an uncontrollable compulsion to drink
Monomaniac	Has an obsession with one thing
Kleptomaniac	Has a compulsion to steal
Manic depressive	Has alternating moods of extreme depression and uncontrollable exaltation
Schizophrenia	A split personality
Melancholic	Has a fixed condition of despondency
Paranoic	Has delusions of persecutions
Oedipus complex	A problem in "mama's boys" hostile to father and excessively devoted to mother
Electra complex	Female version of Oedipus complex, girl is hostile to mother and devoted to father

Plagued Pairs

These generally consist of Homonyms: words that have different meanings and different spellings, but sound the same (two, too)

Spellonyms: (a made - up term are words that have different meanings but are spelled so nearly the same (torturous and tortuous)

Study these words carefully

To abjure	To reject, Renounce
To adjure	To charge or command solemnly
To accept	Receive
Except (prep)	But
To except(v)	To exclude, Leave out (I will expect Peter from this exam)

To affect	To make a difference
To effect (v)	To bring about, cause to happen
Effect (n)	The result or outcome
To adapt	To modify
To adopt	To take as one's own
Averse	Strongly against
Adverse	Unfavourable, hostile
Capitol	Government building
Capital	First rate, a money stake, net worth
Censor (n)	Person judging offensive matter
To censor (v)	To delete questionable matter
Censure (n)	Hostile criticism
To Censure (v)	To criticize harshly
Complacent	Self-satisfied
Complaisant	Amiable, Docile
Complement (n)	Something that completes, goes well with
To Complement (v)	To supply a lack, to complete
Compliment	Approving remark
To compliment	To speak approvingly
Contemptible	Hateful, deserving of contempt or scorn
Contemptuous	Hateful, deserving of contempt or scorn
Contemptuous	Showing contempt
Credible	Plausible
Creditable	Worthy or credit
Credulous	Gullible, naive
To demur	To object
Demure	Well-behaved
To deprecate	To disapprove of (with regret)
To depreciate	To go down in value
Discrete	Separate, Distinct
Discreet	Careful
Disinterested	Without prejudice or bias
Uninterested	Not interested
Diverse	Different
Divers	Plural of diver
To elicit	To draw forth
Illicit	Illegal
To elude	To escape
To allude	To refer to indirectly
To emend	To correct by editing
To amend	Improve
Equable	Uniform, steady
Equitable	Fair, Just

Factious	Inclined to dispute
Factitious	Artificial, Sham
Gourmet	Lover of fine food
Gourmand	Indiscriminate lover of food
To hoard	To save for yourself
Horde	Swarm, Crowd
To immigrate	To move to another country
Emigrate	To leave one's country and move elsewhere
Ingenious	Inventive, clever
Ingenuous	Innocent, Naive
Interment	Burial
Internment	Imprisonment
To lie	To recline as on a bed
To lay	To put or place
Loath	Reluctant
Loathe	Detest
Marital	Of marriage
Martial	Warlike
Nauseated	Sick to the stomach
Nauseous	Revolting or Disgusting
Personable	Pleasant
Personal	Private
Perspicacious	Perceptive
Perspicuous	Clear
To prophesy	To foretell
Prophecy	That which is foretold
To prosecute	To bring legal action
To persecute	To harass
To regale	To entertain
Regal	Referring to royalty
Social	Referring to human society
Sociable	Friendly
Tortuous	Winding, twisted
Torturous	Cruelly painful
Urbane	Sophisticated
Urban	Referring to cities
Venal	Corrupt
Venial	Excusable
Vocation	Career
Avocation	Hobby

WORDS FROM MYTHS & LEGENDS

Achilles' heel	Weak spot; place of vulnerability
Aeolian	Referring to winds. Aeolus was God of the winds

Apollo and Adonis	A handsome desirable male Apollo was the Sun God of the Greeks and Romans. Adonis was a mortal man, lover of Venn
Atlantean, Atlantic	From Atlas; Indicates strength
Aurora	The dawn - Aurora was Goddess of the dawn
Bacchanal	An orgy; a bacchanal is a reveler or carouser at parties
Bacchanalian	Drunken revelry or wild party
Callope	Stream whistle or anything making a similar shrill sound
Cassandra	One who prophesies trouble.
Centaur	Having the head and arms of a man and body of a horse.
Chimera	Imaginative creature; wild and scary
Cupidity	Greed or avarice. Cupid was the young god who embodied sexual desire
Erotic	Of or pertaining to sexual love. Eros (cupid) was the God of passionate love
Furor (e)	Frenzy, excitement, Rage
Fury	An avenging spirit, Virago.
Gorgon	Ugly, frightening person repulsive creature
Halcyon	Peaceful, tranquil. Halcyon was a widow who was metamorphosed by Gods into a kingfisher
Harpy	Grasping, greedy, scary female. Harpies were part women, part bird
Hector	To bully or annoy
Helen (of Troy)	A woman of incomparable beauty (a face that launched a thousand ships)
Herculean	Extra ordinarily strong, Difficult
Hydra-headed	An evil with many sources; Difficult to control
Iridescent	Having a rainbow like play of colours
Jovial	Good-natured, Jolly
Juno-esque	Queenly, Statuesque (said of women) Juno wife of Jupiter was queenly
Lethargy	Sluggishness. The river Lethe ran through Hades (the underworld) and whoever drank the water forgot his past

Martial	Warlike, Suited for army or military life (from Mars, God of war)
Mentor	Respected guide, Teacher
Mercurial	Volatile in mood, Easily changeable
Mnemonics	The art of improving memory
Morphine	An addictive narcotic. Morpheus was God of dreams
Muse	Source of inspiration, a poet
Narcissism	Self-love; extreme egotism
Nemesis	Relentless pursuer of evil doers. Nemesis was Goddess of retribution
Neptune	Figuratively, the ocean. A Neptunist thinks the world emerged from water
Odyssey	Long wandering or voyage, the hero of Homer's Odyssey.
Olympian	Majestic, superior to others
Paeon	Hymn or song of praise. Paeon was the physician of Gods
Pandora's box	Source of evil. Malign influence needing to be kept under control
Parnassus	Having to do with poems; to climb Mount Parnassus is to write poetry
Pegasus	Extremely swift horse; Poetic inspiration
Phoenix	Symbol of immortality
Plutocracy	Government of the wealthy. Pluto was the god of the underworld
Promethean	Unusually original and creative, life-giving; Prometheus formed man from clay
Protean	Assuring many shapes or forms. Proteus, the sea god, kept changing shape
Psyche	The soul, Mind. Psyche was the mortal girl loved by Eros
Saturnalia	Wild party or orgy. Saturn, God of sowing
Sibyl	Prophetess of Ancient Greeks and Romans
Stentorian	Extremely loud(Stentor was the greek herald)
Stygian	Gloomy, Dark, Frightening
Tantalise	To torment or tease.

Terspsichorean	Referring to dance. Terpsichorean was the Muse of the Dance
Titanic	Gigantic in power
Zeus	Ruler, king Zeus was king of Greek gods. A Zeus has absolute power.
Cyclopean	Huge, Massive
Erotic	Pertainig to sexual desire
Gorgon	An ugly person, A horrifying face
Harpy	A grasping, ravenous person
Janus - Faced	Two faced, having two distinct faces
Jovial	Merry, joyous, Happy
Mentor	A teacher or a wise counselor/coach
Mercurial	Swift, Active
Narcissism	Self love & admiration, Narcissistic feelings
Nemesis	An agent of punishment, Upholder of moral code
Odyssey	Long dangerous journey
Olympian	Majestic, Awe-inspiring, detached and aloof
Paeon	A song of praise, Triumph or thanks giving
Palladium	A safeguard
Phoenix	A symbol of immortality
Protean	Changing in shape
Saturnine	Heavy, Dull, Gloomy, Serious, Morose
Stentorian	Very loud voiced; Bellowing

WORDS DEPICTING SUBJECTS OF STUDY

TERM	SUBJECT/ STUDY OF
Aetiology	Cause of diseases
Anthropology	Mankind
Astrology	Heavenly bodies
Cardiology	Heart function and diseases
Carpology	Fruits and seeds
Cetology	Whales
Cartography	Maps
Choreography	Dance
Chronology	Dates
Craniology	Skulls
Criminology	Crimes and criminals
Cryptology	Codes

Dactylology	Finger prints
Demography	Population statistics
Dermatology	Human skin
Endocrinology	Glands
Entomology	Insects
Epigraphy	Ancient inscriptions
Etymology	Word origins
Genealogy	Ancestry
Gerontology	Old age
Gynaecology	Women's disorders
Haematology	Blood
Harpetology	Reptiles and amphibians
Histology	Plant and animal tissue
Horology	Time, Timepieces
Hypnology	Sleep
Ichthyology	Marine life (fish)
Lexicography	Writing of dictionary
Lexicology	Dictionary
Lithology	Rocks
Metaphysics	Nature of existence, Truth and knowledge
Meteorology	Weather
Metrology	Measurement
Myology	Muscles
Obstetrics	Birth of children
Odontology	Teeth
Oenology	Wines
Oncology	Cancer, Tumors
Ontology	Nature of existence
Ophiology	Snakes
Ophthalmology	Eyes
Orology	Mountains
Osteology	Bones
Osteopathy	Manipulating bones and muscles
Otology	Ears
Paedology	Children
Palaeography	Old manuscripts
Palaeontology	Fossils
Pathology	Diseases
Pedagogy	Teaching
Pedology	Soil
Penology	Prisons and treatment of criminals
Pharmacology	Drugs
Philology	Languages
Physiology	Life process
Phytology	Plants
Podiatry	Feet
Polemology	Wars

Potamology	River
Psephology	Elections
Psychiatry	Treatment of mental illness
Psychology	Mind
Radiology	Radiation
Seismology	Earth quakes
Selenology	The moon
Speleology	Caves
Topography	Surface features of a region
Topology	Shapes and surfaces
Toxicology	Poison

WORDS FROM CHARACTERS OF LITERATURE

Babbit	An American business man, vastly pleased with himself; of Lewis' novel Babbit
Brobdingnagian	From Swift's Gulliver's Travels, synonym for gigantic
Bumble	Mr. Bumble was the pompous Orphanage official in Oliver Twist, anyone like him.
Cinderella	Like the fairy tale girl who goes from rags to riches.
DonJuan	An unsavoury lover, the original character killed men and seduced women, any fellow who is interesting to know.
Falstaffian	Shakespeare's bawdy, good humoured fat man in his historical plays, resemblance to this character.
Frankenstein	Name of the scientist in Mary Shelley's book by the same name, who created the monster; now the monster that destroys its creator.
Friday	Robinson Crusoe's original, a helper who is dependable (man Friday)
Galahad	Noble of all knights, both good looking and gallant
Jekyll&Hyde	One who alternates between good and evil; from Stevenson's story, Hyde ends with Jekyll trapped in the murderous Hyde personality
Lilliputian	Opposite of Brobdingnagians; anything miniature or tiny
Lothario	Well known lover, character in the play 'The Fair Penitent'

Malapropism	Mrs Malaprop from Sheridan's play 'the Rivals' gained immortality by her verbal mistakes, a hilarious misuse of words
Pickwickian	Dicken's character of 'Tickwick Papers': Pickwickian words are used in an esoteric special way (not in accordance with their usual meaning)
Pooh-bah	In Gilbert and Sullivan's opera 'The Mikado' was the Lord, any politician who holds several offices is a pooh-bah
Quixotic	Don Quixote was the lovable impractical old man in Cervante's novel with the same title. Quixotic schemes are unrealistic
Robot	Czech playwright Karel Čapek's play titled 'R.U.R. (Rossum's Universal Robots)'; a robot is anyone who works without thought
Scrooge	The penny-pinching miser of Christmas Carol (Dickens); now any grasping, covetous person without kindness
Simon Legree	Stowe's character in 'Uncle Tom's Cabin', the character was a slaver; anyone who works others mercilessly
Uncle Tom	In Uncle Tom's Cabin, Uncle Tom was saintly old black. Today a black who is seen as quick to co-operate with whites
Uriah Heep	From Dickens's David Copperfield, a term to describe someone hypocritical.
Walter Mitty	Thurber's lovable character from 'The Secret Life of Walter Mitty'; the embodiment of fantasies.

WORDS FROM PEOPLE'S & PLACE NAMES

People's Name

America	(from the sailor Amerigo Vespucci) The two great continents
Ampere	(from Andre Ampere) Unit of electric current
Atlas	(from Greek demigod Atlas) A man holding the globe on his back; a collection of maps ;

Bowlerize	(from Dr Bowdler, an art critic) To make a book readable by deleting offensive portions
Boycott	(from Captain Boycott) Ostracize, break connections with
Braille	(from Louis Braille) A system of printing for the blind
Chauvinism	(from Nicholas Chauvin) Exaggerated loyalty or patriotism
Chimerical	(from Greek monster Chimaera) Ideal but impractical
Cologne	(from German city Cologne) Famous as "Eau de Cologne"
Derrick	(from a hangman Derrick) A hoisting apparatus
Fahrenheit	(from D.G. Fahrenheit) Unit of temperature measurement
Frankenstein	(from the person Frankenstein) A horrible monster
Galvanize	(from Dr Galvani) To electrify; to stir into action; Galvanization = electrolytic technique
Gargantua	(from Gargantua) Anything on a large scale; massive
Gordian Knot	(from king Gordian) A very tough problem
Guy	(from Guy Fawkes, a drama character) Ridiculous name for a man
Hector	(from Greek hero Hector) To quarrel, tease, bully someone
Herculean	(from Roman Hercules) Very heavy, very difficult
Hobson's Choice	(from Thomas Hobson) Really no choice at all
Jingoism	(from Jingo) Feeling of belligerence towards foreigners
Lilliputian	(from Lilliput) Very small, Miniature, Diminutive
Lothario	(from Lothario, a character) A flirt; seducer of women
Macadam	(from John Macadam) Roads made of broken stones
Malapropism	(from Mrs. Malaprop) A confusion of words; verbal blunder
Martinet	(from Gen. Martinet) A person very strict about discipline

Masochism	(from Leopold Masoch) Enjoyment through self torture
Maverick	(from Samuel Maverick) An unpredictable person
Nemesis	(from Greek God Nemesis) An agent of retribution
Nicotine	(from Jean Nicot) Poisonous drug in Tobacco
Ohm	(from GS. Ohm) Unit of electrical resistance
Procrustean	(from Greek robber Procrustes) To be able to mould
Quixotic	(from Don Quixote) Unrealistic visionary, Impractical idea
Robot	(from Robota) a machine or human totally mechanized (no soul)
Rodomontade	(from king Rodomonte) Very boasting, Blustering, Bragging
Solecism	(from Greek province Soloi) Blunder in speech or writing
Spoonerism	(from Rev. W.A. Spooner) Habit of using first letters of words in phrases
Thespian	(from Thespis, the father of Greek tragedy) An actor
Utopian	Ideal but impractical, Unfeasible, Impossible, Chimerical
Volt	(from A. Volta) Unit of electric potential
Yahoo	(from a tribe called Yahoos) A degraded human specimen

Place Name

Bohemian	Bohemia was believed to be the home of all the gypsy tribes that roamed through Europe
Canopy	Greek word konops means mosquito; canopy, a cloth covering for protection
Donny brook	The Irish fair at Donny brook, a scene of cheerful violence, a club
Dungaree	Jeans dating back to the cotton cloth woven in Dungaree
EL Dorado	"Indians in South America used to cover their king with gold dust. "He was Eldorado, the Gold-covered One; a place of incredible wealth."

Hackney	Now means a horse of English breed, a poor servant who does tiring work
Laconic	Laconic was the home of the Greek city Sparta. To be laconic is to be brief and succinct of speech, to be terse.
Limerick	The tiny town of Limerick, Ireland, gave its name to a most enjoyable verse form, the five-line limerick. Modern limericks are often suggestive
Marathon	Anything that goes on forever. The original long-distance runner fell dead at the end of his run, at Marathon, Greece.
Meander	The Asian river meanders, winds and twists its way to the sea. If a speech or movement of winds and twists it is said to meander
Sardonic	A herb from Sardinia is so bitter that if people eat it they go insane. A sardonic expression is scornful
Spartan	To live without luxuries is to follow a Spartan life. Spartans were macho men.
Utopia	Title of many books and imaginary places; unattainable
Waterloo	"To meet one's Waterloo" is to repeat Napoleon's dismal performance in his last battle and lose.

YOUNG ONES & CRISES OF ANIMALS

Young Ones of Animals

Bear	Cub
Buffalo	Calf
Cat	Kitten
Cow	Calf
Dog	Puppy
Duck	Duckling
Frog	Tadpole
Goat	Kid
Hare	Leveret
Hen	Chicken
Horse	Colt

Lion	Cub
Sheep	Lamb
Swan	Cygnets
Tiger	Cub
Eagle	Eaglet

Words Indicative of Sound

Eagle	Eaglet
Ape	Gibber
Ass	Bray
Bear	Growl
Bird	Twitter, Warble
Bull	Bellow
Camel	Grunt
Cat	Mew, Purr
Cattle	Low
Crow	Crow
Dog	Bark, Snarl, Howl, Yelp, Whine
Dove	Coo
Duck	Quack
Eagle	Scream
Elephant	Trumpet
Fly	Buzz
Frog	Croak
Fox	Yelp
Goat	Bleat
Geese	Cackle, Gaggle
Hawk	Scream
Hen	Cluck
Hog	Grunt
Horse	Neigh
Monkey	Chatter
Mouse	Squeak
Nightingale	Sing
Owl	Hoot, Screech
Parrot	Talk
Pig	Grunt
Pigeon	Coo
Puppy	Yelp
Raven	Croak
Rook	Caw
Serpent	Hiss
Tiger	Growl
Vulture	Scream
Wolf	Howl

GROUP TERMS

Deer	Herd
Fish	Shoal
Geese	Flock
Corn	Stack
Wood	Stack
People	Tribe
Birds	Flight
Wheat	Sheaf
Wolves	Pack
Hay	Bundle
Grain	Sheaf
Sticks	Bundle
Soldiers	Regiment
Plantains	Bunch
Partridges	Convoy
Ants	Nest
Bees	Hive, Swarm
Cattle	Herd
Stars	Cluster, Galaxy
Arrows	Sheaf, Quiver
Locusts	Swarm, Flight
Chickens	Brood
Pigeons	Brace
Ladies	Bevy
Sailors	Crew
Camels	String
Musicians	Band
Pistols	Brace
Cigar	Box
Arms	Pile
Swine	Herd
Sheep	Flock
Steps	Flight
Flowers	Posy
Horses	Stud
Oxen	Yoke
Hens	Brood
Birds	Flock
Savages	Hoard
Hounds	Leash
Men	Host

Puppies	Litter
Peacocks	Muster
Plants	Nursery
Oxen	Team
Fruits	Basket
People	Throng, Colony, Clique
Sticks	Faggot
Sardines	Family
Dogs	Kennel
Jurymen	Panel
Constables	Posse
Magistrates	Bench
Actors	Troupe
Guns	Battery
Actors	Company
Advisers	Council
Pictures	Gallery
Flowers	Nosegay
Boats	flotilla
Cavalry	Squadron
Shrubs	Shrubbery
Cars, ships	Fleet
Animals, Plants	Genus
Worshippers	Congregation
Merchants	Syndicate
Stars	Constellation
Fruit trees	Orchard

WORDS OF FOREIGN ORIGIN

French Words

Aplomb	Complete confidence
Au Courant	Well informed
Avant Garde	Describes artist and writers who are unorthodox and untraditional
Bete Noire	Person or subject hated
Brouhaha	Furor, Uproar
Carte Blanche	Complete authority
Cause Celebre	Situation/case arousing attention and discussion
Coup D' Etat	An unexpected action (of government) usually implying the use of force,

C'estLaVie	That's life
DejaVu	Sense of having undergone
De Mode	Out of fashion (antonym: A La Mode)
De Rigueur	Indispensable; Something absolutely required
Enfant Terrible	Unconventional person
En Passant	By the way
Espirit De Corps	Morale
Fait Accompli	An irreversible act done without warning
Faute de Mieux	For want of something better
Faux Pas	A social error
Fin De Siecle	End of century (specifically 19th century)
Haute Couture	High fashion
Haute Cuisine	Excellent cooking
Haut Monde	The elite
Laissez Faire	Policy of non-interference
Mot Juste	Perfect expression
Manque	Failing to achieve a desired goal
Noblesse Oblige	Obligation imposed on high positions
Nouveau Riche	Ostentatious person
Par Excellence	Highest degree
Parti Pris	Bias
Raison D' Etre	Reason for existence
Roman A Clef	Novel in which real characters and events figure under disguise Sang Froid Self control
Savoir Faire	Instinctive ability to act appropriately
Soi Disant	So called
Soupcon	Suggestion, Inkling
Tete-a-tete	Close conversation between two persons
Vis-a-vis	In regard to
Volte Face	Complete reversal

Latin Words

Apriori	Self evident, known independently
Ad Hominem	Directed at someone personally
Ad Libitum	Improvise (V), Off-the-cuff

AdLib	Remark(N)
Ad Nauseam	To the point of disgust
Alma Matter	One's old school, College
Alter Ego	One's other self
Alumnus	Former student ,
Amicus Curiae	Disinterested advisor
Anno Domini (Ad)	In the year of the lord
Ante Meridiem (Am)	Between midnight and noon
Annus Mirabilis	Year of wonders, Disasters
Bonafide	Genuine
Bis	Twice
Caput Mortuum	Worthless residue
Caries	Decay (of bone or teeth)
Caveat	Warning
Ceteris Paribus	Other things being equal
CuiBono	For whose good?
De Facto	Functioning in reality
De Jure	Functioning in accordance
Deus Ex Machina	Person or thing that suddenly resolves a problem
Ex Gratia	Referring to payment made as a favour
Ex Post Facto	After the event '
Ignis Fatuus	A false hope or illusion
InToto	Completely
In Extremis	In extreme circumstances
In Loco Parentis	In Place of a parent, acting as a guardian
Inter Alia	Among other things
Ipsa Facto	As an immediate consequence of a fact
Magnum Opus	A great work; Crowning achievement
Mea Culpa	An acknowledgement of guilt
Modus Vivendi	A way of getting along with another person
Non Sequitur	An illogical remark
Obiter Dictum	An incidental remark
Per Se	By itself
Persona Grata	An acceptable person
Persona Non Grata	An unacceptable person
Prima Facie	On first appearance 3'
ProRata	Proportionately

German Words

Angst	A feeling of anxiety
Blitzkrieg	Sudden attack
Ersatz	Artificial
Fraulein	Lady
Haus Frau	House wife
Kaffee	Coffee
Kaiser	Emperor
Kaput	Totally ruined
Kitsch	Worthless
Lebensraum	Living space
Putsch	Coup
Quid Pro Quo	Something given for something received
Q.E.D	That the point has been made
Sine Die	With no fixed date, Indefinitely
Sine Quanon	An indispensable condition
Status Quo	Present condition
Sui Generis	Unique; In a class by himself herself
Vade Mecum	A reference or hand book
Vox Populi	Public opinion

Spanish Words

Bonanza	A huge profit
Caballero	Gentleman
Cantina	A bar
Cortes	Parliament
Fiesta	Festival
Hidalgo	Minor nobleman
Incomunicado	Without contact with outside world
Junta	Secret Council
Manana	Soon, Tomorrow
Peccadillo	Small error
Siesta	Afternoon nap

Japanese Words

Banzai	Cheer of triumph
Bonsai	Growing miniature plants
Bushido	Samual code
Geisha	Young woman who accompanies and entertains men

Haiku	Delicate form of Japanese poetry
Hara Kiri	Suicide
Ikebana	Flower arrangement
Kamikaze	Suicide pilot (World War II)
Kimono	Long, Loose robe tied with sash
Mikado	Japanese emperor
Nisei	Children of Japanese immigrants born in the USA
Sake	Rice wine
Samurai	Warrior
Sayonara.	Good Bye
Shinto	Japanese religion
Sumo	Form of wrestling
Zaibatsu	Small group (of families) Controlling Japanese Industries
Reich	Republic
Schadenfreude	Feeling of enjoyment from misfortune of others
Strafe	Punish
Wunderkind	Prodigy
Arrivederci	Good Bye
Bravura	A show of daring
Brio	Vivacity
Carabiniere	Policeman
Chiaroscuro	Sharp contrast
Cognoscenti	Person with expert knowledge
Ciao	Used as a greeting at meeting or parting
Dilettante	Amateur or Dabbler
Dolce Far Niente	Carefree existence
Dolce Vita	The good life
Diminuendo	With gradually decreasing intensity or volume
Embroglio	Violent, Embarrassing quarrel
Lingua Franca	A common language
Palazzo	Place
Prima Donna	Leading female singer at an opera
Presto	Quickly
Verismo	Realism .
Virtuoso	An expert performer

WORD FREQUENCY

Word Frequency questions have a set of four sentences, each having a blank. Each of these sets is succeeded by four words. You have to find out the maximum number of sentences in which any one of the four given words fits. Use this number to mark your answer. If any of the four words fits into all four sentences, your answer shall be (4).

If a word fits only into three sentences (and if that is the highest frequency in comparison to all the other words) your answer shall be (3) and so on.

Let us consider an example first. This is a reproduced CAT question.

- (I) The mediator helped reach a/an ____ agreement.
 (II) The structure of most animals and insects exhibits ____ symmetry.
 (III) The two countries signed a ____ cultural exchange pact. (TV) Due to the disqualification of one of the three candidates, the election was reduced to a contest.
- (1) Amicable (2) Bilateral
 (3) Historical (4) Multilateral

Here, your answer would be (2). This is because the word 'bilateral' fits into all the four blanks. The word 'amicable' fits the blank in sentences (I) and (II) only, while the word 'multilateral*' can fit only sentence (I).

WORD ANALOGY

Questions on Word Analogy appear recurrently in aptitude tests. In these, you have to select from a number of choices, the pair of words that relates together in the same fashion as the given pair of words in the question is. Since such question demands that you should be able to analyze and identify basic relationships (apart from having a good vocabulary in English language), it is widely believed to provide a good assessment of mental ability. Every analogy gives you a pair of words represented as below:

The symbol (:) stands for 'is related to' or simply 'is to'. The base pair is followed by several choices, and the whole question is written as follows:

ORCHESTRA: INSTRUMENTAL:

- a. Trio: harmonic
 b. Choir: vocal
 c. Band: martial
 d. pianist: solo

The symbol (::) stands for 'in the same way as' or simply 'as'.

In working out an analogy, you must proceed by first determining the nature of relationship that exists between the two words in the base pair. In this example, analysis indicates that orchestra is a large group of people playing various musical instruments together.

The next step is to study the relationship between the words in the pairs of answer choices. In our example, a study of the first pair indicates that a trio is not a large group performing harmonic music; in choice 2, a 'choir' is a large group performing vocal music. In choice (3) Band is a group but is not known to perform martial music.

Similarly, in choice (4) pianist is not a group. Of all the options given, the words in the second pair match most closely with the relationship expressed in the base pair. Consequently, option (b) provides the best answer.

IMPORTANT: You must remember that the pair of words of the answer choice must match the base pair in 'form' as well as 'relationship'. If, for example, the

base pair depicts a relationship between two nouns, the correct answer will also consist of two nouns. If, moreover, the base pair is made up of two plurals, the correct answer will be made up of two plurals. Sometimes a quick scrutiny of options will help you immediately discard the choices that do not have the same form as the base pair.

Strategy For Analogy

- (i) Identify the relationship between the words in base pair by linking them using a sentence,
- (ii) Look out for contextual meaning of words,
- (iii) The relationship between the words could be vertical or horizontal.
- (iv) The sides should match; if the 'parent' is on the left side and the 'child' on the right in the question pair, then the correct answer choice should also follow the same order in sense.

CAUTION: Do read the instruction given in a test closely; sometimes the direction can be to select the option that LEAST similar to the question words.

Word analogies frequently illustrate common relationships. The following are worth noting.

COMMON TYPES OF ANALOGY

Analogy Types	Example
1 Action and its typical quality/adjective	spark plug-Car
2 Cause - effect	Clouds - rain
3 Degree / intensity	Tepid - not
4 Object / substance-natural component	Pearl – oyster
5 Object related to natural location	Camel - desert
6 Word-synonym	Suave - sophisticated
7 Word-antonym	Fertile - barren
8 Person related to tool	Carpenter - saw
9 Person related to major trait	Philosopher - wisdom
10 Person related to skill/interest	Scientists - research
11 Person related to least desirable characteristic	Stickler - approximation
12 One of a kind /class / type	Daisy-flower
13 Substance / raw material related to end product	Sheep - wool
14 Thing related to logical action.	Pestle - pound
15 ear clichéd expression	Friend - congeniality
16 Implied comparison	Clouds - sun
17 Relationship of description	Speech – articulate
18 Phenomenon/entity-measuring device/unit	Decibel - sound
19 Object is to thing with which it is made	Sweater - acrylic
20 Branch of study and subject	Ornithology-bird
21 Person is to area of specialization	Ichthyologist - marine life

22	Extract is to source	Gold-ore
23	Symbol and purpose	Lighthouse - danger
24	Legendary term - symbolic meaning	Malapropism - verbal
25	Object is to case in which it is kept	Quiver - arrows
26	Plan is to action	Agenda - meeting
27	Member is to particular group	Captain - team
28	Relationship of size	Boulder - pebble
29	Action and its typical quality/adjective	Movement - graceful
30	Relationship of structural/functional similarity	Drawbridge – castle
31	Idiomatic expression	On-off

ODD WORD OUT

“Odd word Out” is a vocabulary-based questions that test your ability to classify words based on their common attributes. As this test requires you to know the meanings of many words, your performance in such questions to a great extent depends upon your word power. These questions are better answered after strengthening your vocabulary.

IDIOMS & PHRASES

In an idiom and phrases, questions are asked to identify its literal meaning from the choices. These idioms may have their origins in other languages like Latin, French or Greek. A list of the most commonly used idioms are given in the Word Bank issued to you as part of module 1. Learn them well and look up to the dictionary for more of them. It is also interesting as well as important to collect and make a list of those idioms that you come across in your daily reading.

EXERCISE

MCQ TYPE QUESTIONS

Word Frequency

Directions for (Que. 1 - 30): Read the information carefully and answer the questions given below it.

Each of the following questions has a set of four sentences, each containing a blank.

Each set is followed by four words. Find out the maximum number of times that any of the words fits each set.

Use this number to indicate your answer.

For example, if any one of the words fits into all four sentences, your answer will be (d). However, if it fits only into three sentences (and that is also the highest frequency for any of the four words) your answer will be 3 and so on.

1. (I) This pole, erected for its aesthetic appeal, has turned into a popular ...for birds.
 (II) I did not expect that old cupboard to... such a fancy price at the auction.
 (III) The... was rusty and creaked whenever the door was closed or opened.
 (IV) The dog was trained to... the newspaper every morning.
 (a) hinge (b) perch
 (c) bring (d) fetch
2. (I) The ... adjourned all business for the day as a mark of respect for the departed leader.
 (II) For several centuries before it was proved that the earth was round, scientists believed that the earth was ...
 (III) What is referred to as an apartment in America, is called a... in England.
 (IV) Driving on a(n)... road is easier than driving on an uneven one.
 (a) even (b) flat
 (c) house (d) oral
3. (I) The troops were ordered to... the forests to locate the deserter.
 (II) Since the heavy duty equipment failed, the stones had to be moved by..... .means
 (III) The PM had to... to popular pressure in the face of documented evidence.
 (IV) This... explains how to assemble, use and maintain the system.
 (a) physical (b) comb
 (c) bow (d) manual
4. (I) It is important that your concentration should not... particularly in the reading comprehension section.
 (II) We tried t... down several passing cars in the hope of getting a lift.
 (III) Showing disrespect to the national... is an act to treason.
 (IV) The hurricane was on the... when we entered the city limits.
 (a) wane (b) flag
 (c) wave (d) emblem
5. (I) The leaning tower or Pisa is a(n)... structure.
 (II) Of the two living dodos, one died and made the other...
 (III) Tigers are protected in order to prevent them from becoming...
 (IV) The... details of his escapade were very interesting.
 (a) extinct (b) unique
 (c) rare (d) astounding
6. (I) The first day's play was washed out due to inclement...
 (II) After the accident, his thinking has become very...
 (III) His success has not come easily, he has had to... many a storm in his career.
 (IV) Because of the incessant... we could not go out.
 (a) rain (b) cloudy
 (c) weather (d) turmoil
7. (I) When the effect of the painkiller wore off, he was racked by... pain.
 (II) In any right angled triangle, at least two of the angles are...
 (III) The philanthropist was moved by the ... poverty he saw around him.
 (IV) The lawyer's commitment was... and unqualified.
 (a) total (b) prevalent
 (c) complete (d) acute
8. (I) His... in the subject stood him in good stead.
 (II) A mere... to do well will not do; it must be backed by will power and hard work.
 (III) In driving a car, you have to change gears to negotiate a/an ...
 (IV) The RBI has decreased lending ... rates by 2%,
 (a) interest (b) inclination
 (c) support (d) curve

9. (I) Why should he apologise? He has not... any mistake.
 (II) He has risen to the top only because he has been ... to his profession.
 (III) Since the supply situation is unclear, I hope you have not... any dates to the client.
 (IV) Sharma's... was there for all to see; she worked beyond the call of duty.
 (a) diligence (b) capacity
 (c) committed (d) confirmed
10. (I) Because of the transporter's strike and the consequent delay in material supply, Harsha's work schedule was way off the
 (II) ... my words; this boy is destined for greatness.
 (III) This single failure continues to be the only... on his impeccable resume.
 (IV) Manners are the... of a gentleman.
 (a) 37mark (b) stain
 (c) blot (d) impression
11. (I) When the machinery in the... broke down, the production manager was summoned immediately.
 (II) This... is a rare one because it does not produce either flowers or seeds.
 (III) This puzzle requires more than three hours to put the ... together.
 (IV) The teacher must... the idea of lateral thinking in her student
 (a) parts (b) unit
 (c) encourage (d) plant
12. (I) Because of the wide spread protests the government was forced to ... the law.
 (II) In the bilateral contest, one contestant polled 29% of the votes, giving the other an ... majority.
 (III) This countryside... with natural minerals.
 (IV) In relative terms, your salary may not be much, but in... terms, Rs.2,00,000 per month is a sizeable amount.
 (a) rescind (b) abounds
 (c) absolute (d) financial
13. (I) When Rip Van Winkle woke up, the world around him ... no resemblance to the one he had known.
 (II) This gun is a Smith and Wesson with a 0.38 ...
 (III) We avoided Ram in the office because he was such a...
 (IV) The tidal wave... down on the ship with relentless fury.
 (a) barrel (b) bore
 (c) crashed (d) tear
14. (I) His insightful... on juvenile rehabilitation has become the foundation for several laws.
 (II) No sooner did he hear the gun's ... than he collapsed.
 (III) The value of this... is inestimable.
 (IV) The company's ... into software exports has proved profitable.
 (a) essay (b) report
 (c) article (d) study
15. (I) In this era of liberation, one popular, request from local entrepreneurs is for a level playing... .
 (II) Some of the greatest scientist have had little formal training in their...
 (III) The scandal-ridden Pakistan team had to... one of its weakest teams in recent years.
 (IV) In the absence of insightful reporters, the senators had a... day at the press conference.
 (a) ground (b) field
 (c) area (d) enter
16. (I) Without the... that normally surrounded him, the musician was simple human being with his own share of weaknesses and eccentricities. :
 (II) It is amazing that the... on the vase remained even after centuries.
 (III) This text is tautological and you can grasp its meaning even if you... over it.
 (IV) The assassin was hard to trace because of the... that surrounded him.
 (a) gloss (b) glaze
 (c) web (d) skim
17. (I) The miner used his... to open up the tunnel.
 (II) Prince William had to ... at the vines to get to the castle.
 (III) This horse is a ... and cannot be used for racing purposes.
 (IV) The brick maker took the dried bricks out of the... and put them on the truck.
 (a) pick (b) hack
 (c) thoroughbred (d) chop
18. (I) The... that we normally talk about in a pencil is actually graphite.
 (II) Sherlock Homes is known for his ability to follow even a simple... to unravel a complex problem.

- (III) General Mac Arthur is a man who... by example.
- (IV) It is always a difficult task for the... speaker in a debate.
- (a) clue (b) first
- (c) lead (s) (d) principle
19. (I) As the best lawyer specializing in air disaster cases, he was without ...
- (II) During adolescence... pressure plays a major role in molding the value system.
- (III) A... resembles an apple, though it is different in taste.
- (IV) Sam asked Anitha to... out of the window and see if the guests had arrived.
- (a) social (b) peer
- (c) look (d) equal
20. (I) This is not a matrix because it has several columns but only one...
- (II) The sergeant ordered the cadets to stand in single... at shoulder lengths.
- (III) The... containing the papers on the merger was lost in transit.
- (IV) The mechanic is looking for a... to even out the rough surface.
- (a) row (b) order
- (c) file (d) hammer
21. (I) Because of the strong winds, the cargo in the ship moved unevenly and caused the ship to ...
- (II) This Ayurvedic drug is popular for its ability to... asthmatic disorders. ;
- (III) Despite the trainer's best efforts, the dog would not come to... ;
- (IV) The new pair of socks was small and would not fit ,at the... .
- (a) tilt (b) heel
- (c) foot (d) alleviate
22. (I) The annual boat race is in the month after April, that is in.....
- (II) The rate of interests works out to a/an ... 8% of given the highest rate of 10% and the lowest pf 6%.
- (III) His academic record is no ... of his intellectual caliber.
- (IV) Not helping his friend who was indisposed, was a ... act.
- (a) average (b) may
- (c) measure (d) mean
23. (I) The ... on the newly asphalted road had not dried and stuck to the wheels of our car.
- (II) The diva's voice surprisingly cracked when she moved into the song at a high ...
- (III) The cricket match had to be called off because the vandals had damaged the ...
- (IV) The advertising agency spent many days in developing its ... for the new business.
- (a) pitch (b) proposal
- (c) level (d) tar
24. (I) The ... has come to be synonymous with the psychiatrist's profession.
- (II) Despite his age, Raman had a(n)... posture.
- (III) During the last year, the government has sanctioned Rs. 1,000 to... new structures.
- (IV) The committee has taken steps to... charges of misconduct and bias.
- (a) avoid (b) erect
- (c) renovate (d) couch
25. (I) Frazier was out for the... and lost the match by K.O. to Muhammed Ali.
- (II) While coarse yams are not widely exported, yams of finer... are very much in demand.
- (III) Dracula was, in terms of social hierarchy, a ...
- (IV) To pass the bill in the assembly, the ruling party had to ,.. on the support of at least 10 opposition members.
- (a) depend (b) vampire
- (c) count (d) grade
26. (I) Trade in waste metal or... is carried out through a well organized network,
- (II) Following wide spread copying and paper leakage, the board decided to ... the test.
- (III) Tilak is a belligerent person and picks up a ... with almost everyone he meets.
- (IV) The child cut out the picture of the dinosaur for the use in his ... book.
- (a) scrap (b) perch
- (c) bring (d) fetch
27. (I) Dr. Shekar is our... representative in Bangalore.
- (II) Most dental treatment calls for only... anesthesia.
- (III) Once the foreigners left, only the... people remained.
- (IV) The invaders wrested...
- (a) area (b) local
- (c) primary (d) control

28. (I) She was a key... for the prosecution.
 (II) You should not... a book by its cover.
 (III) Thousand of people gathered to... fireworks display.
 (IV) Because he was personally involved, the lawyer could not take up the...
 (a) case (b) judge
 (c) witness (d) court
29. (I) You must ...more time if you want to make a considered decision.
 (II) Batsmen wearing rubber soles get little ...when running on the pitch.
 (III) She want to the market to... some grocery.
 (IV) The..., department in an organization is normally responsible for vendor development.
 (a) buy (b) purchase
 (c) finance (d) grip
30. (I) A...is an accepted of male office wear.
 (II) Of the two living dodos, one died and made the other...
 (III) A sailor's knot is one of the most difficult knots to...
 (IV) "... your gun from its holster", Adams said.
 (a) draw (b) bow
 (c) shirt (d) tie

Word Analogy

Directions for (Que. 31-40): Read the information carefully and answer the questions given below it.

Each question below consists of a pair of words that are related. Complete the blank with suitable word from the answer choices given to form an analogous pair.

31. Pediatrician : _____ :: Dermatologist: skin
 (a) Children (b) Feet
 (c) Plant (d) Hair
32. Blandish : Coax : : Asseverate : _____
 (a) Affirm (b) Repeat
 (c) Complain (d) Twist
33. Exigency: Emergency:: Expansive : _____
 (a) Hostile (b) Friendly
 (c) Proactive (d) Clumsy
34. Motif : decoration : : Eulogy : _____
 (a) Diatribe (b) Lament
 (c) Praise (d) Satire
35. Illustration : Clarity :: Constriction : _____
 (a) Wide (b) Upright
 (c) Swiftiness (d) Narrow
36. Cosmology: Universe :: Morphology: _____
 (a) Structure (b) Function
 (c) Radio (d) Size
37. Answer : Test :: Denouement : _____
 (a) Symphony
 (b) Mystery story
 (c) Horse race
 (d) Complete understanding
38. Astronauts: Space :: Argonauts : _____
 (a) Fire (b) Ship
 (c) Birds (d) Sea
39. Snake : Reptilian : : Fox : _____
 (a) Leopard (b) Vulpine
 (c) Tiger (d) Feline
40. Genuine: Simulated:: Unaffected: _____
 (a) Elevated (b) Bombastic
 (c) Dynamic (d) Emulated

Directions for (Que) 41 - 70): Read the information carefully and answer the questions given below it
 In each of the following questions identify the pair of words, which has a relationship similar to the pair in question

41. Wool : Acrylic

- (a) Rayon : Silk (b) Winter : Spring
 (c) Cotton : Terylene (d) Rubber : Plastic

42. Bouquet : Flower

- (a) Skin : Body (b) Chain : Link
 (c) Page : Book (d) Product : Factory

43. Germ : Disease

- (a) Man : Woman (b) War : Destruction
 (c) Doctor : Medicine (d) Owner : Shop

44. Ignominy : Disloyalty

- (a) Fame: Heroism (b) Derelict : Fool
 (c) Death : Victory (d) Martyr : Man

45. Gazelle : Swift

- (a) Horse : Slow (b) Swan : Graceful
 (c) Lion : Roar (d) Lamb : Bleat

46. Ascetic : Luxury

- (a) Capitalist : Communist
 (b) Teacher : Classroom
 (c) Misogynist : Women
 (d) Scholar : Knowledge

47. Augur : Future

- (a) Knight : Medieval (b) Post : Century
 (c) Historian : Past (d)Commuter : Present

48. Spear : Dart

- (a) Knife : Sword
- (b) Door : Window
- (c) Mountain : Molehill
- (d) Cannon : Gun

49. Buzz : Hum

- (a) Whisper : Noise (b) Echo : Sound
- (c) Crack : Whip (d) Tinkle : Clang

50. Argument : Debate

- (a) Violence : Peace
- (b) Fight : Contest
- (c) Opponent : Challenge
- (d) Abuse : Scold

51. Cow : Herd

- (a) Child : family (b) Peacock : Muster
- (c) Bald : Hair (d) Horse : Jockey

52. Bird : Ornithology

- (a) Zoology : Animal
- (b) Stars : Galaxy
- (c) Archaeology : Antiquity
- (d) Plant : Botany

53. Wood : Carve

- (a) Clay : Mould (b) Pipe : Blow
- (c) Paper : Burn (d) Trees : Sway

54. Open : Secretive

- (a) Mystery : Detective
- (b) Honest : Truth
- (c) forthright : Snide
- (d) Better : Best

55. Wave : Crest

- (a) Pinnacle : Nadir (b) Mountain : Peak
- (c) Sea : Ocean (d) High : Low

56. Letter : Word

- (a) Club : People (b) Page : Book
- (c) Product : Factory (d) Picture : Crayon

57. Spasm : Pain

- (a) Sound : Noise (b) Flash : Light
- (c) Touch : Delicate (d) Love : Hate

58. Myth : Story

- (a) Fiction : Story (b) Bonnet : Hat
- (c) Pencil : Paper (d) Novel : Tale

59. Saturnine : Mercurial

- (a) Redundant : Concise
- (b) Saturn : Venus
- (c) Heavenly : Starry
- (d) Wolf : Sly

60. Wheat : Chaff

- (a) Wine : Drugs (b) Crop : Corn
- (c) Laughter : Revelry (d) Cloud : Rain

61. Intimidate : Fear

- (a) Maintain : Satisfaction
- (b) Astonish : Wonder
- (c) Soothe : Concern
- (d) Tame : Lion

62. Handcuffs : Robber

- (a) Leash : Dog (b) Rope : Tie
- (c) Shoes : Feet (d) Paper : Kite

63. Celebrate : Marriage

- (a) Window : Bedroom
- (b) Lament : Bereavement
- (c) Pot : Pan
- (d) Crime : Penalty

64. Banish : Apostate

- (a) Reward : Traitor (b) Request : Assistant
- (c) Remove : Officer (d) Welcome : Ally

65. Affirm : Hint

- (a) Say : Deny (b) Assert : Convince
- (c) Confirm : Reject (d) Charge : Insinuate

66. Boiling : Lukewarm

- (a) Cold : Hot (b) Cold : Freezing
- (c) Cloudy : Evening (d) Freezing : Cold

67. Dishonesty : Distrust

- (a) Hand : Paper
- (b) Money : Pickpocket
- (c) Carelessness : Accident
- (d) Murder : Jail

68. Immigration : Entrance

- (a) Native : Foreigner
- (b) Emigration : Departure
- (c) File : Knife
- (d) Migration : Bird

69. Emulate : Mimic

- (a) Slander : Defame
- (b) Praise : Flatter
- (c) Aggravate : Promote
- (d) Express : Imply

70. Impeach : Dismiss

- (a) Arraign : Convict
- (b) Exonerate : Charge
- (c) Plant : Reap
- (d) Imprison : Police

Odd Word Out

Directions for (Que. 71 - 110): Read the information carefully and answer the questions given below it. Each Question has a set of four words. Choose the word that does not belong to the group.

- | | | | |
|-----------------------------|----------------|--------------------------|--------------------|
| 71. (a) Incapacitate | (b) Debauch | 90. (a) Audacious | (b) Skittish |
| (c) Weaken | (d) Debilitate | (c) Fidgety | (d) Nervous |
| 72. (a) Scepter | (b) Throne | 91. (a) Cygnet | (b) Ewe |
| (c) Crown | (d) Staff | (c) Kitten | (d) Fawn |
| 73. (a) Minaret | (b) Dome | 92. (a) Nylon | (b) Silk |
| (c) Porch | (d) Steeple | (c) Jute | (d) Cotton |
| 74. (l) Moat | (b) Tower | 93. (a) Coral | (b) Amethyst |
| (c) Drawbridge | (d) Piazza | (c) Diamond | (d) Ruby |
| 75. (a) Portrait | (b) Caricature | 94. (a) Pedagogue | (b) Scientist |
| (c) Cartoon | (d) Colossus | (c) Scholar | (d) Philanthropist |
| 76. (a) Decipher | (b) Destroy | 95. (a) Febrile | (b) Fecund |
| (c) Exterminate | (d) Decimate | (c) Prolific | (d) Fertile |
| 77. (a) Collie | (b) Pekinese | 96. (a) Pulpit | (b) Pew |
| (c) Mongrel | (d) Alsatian | (c) Chalice | (d) Chancel |
| 78. (a) Cantata | (b) Oratorio | 97. (a) Virus | (b) Bacteria |
| (c) Opera | (d) Sonata | (c) Algae | (d) Ant |
| 79. (a) Lobby | (b) Portico | 98. (a) Debutante | (b) Decent |
| (c) Reception | (d) Corridor | (c) Debonair | (d) Decadent |
| 80. (a) Beans | (b) Peas | 99. (a) Galley | (b) Deck |
| (c) Barley | (d) Lentils | (c) Caboose | (d) Ship |
| 81. (a) Succinct | (b) Terse | 100. (a) Editor | (b) Author |
| (c) Circumlocutory | (d) Brief | (c) Publisher | (d) Jockey |
| 82. (a) Pink | (b) Red | 101. (a) Poise | (b) Polish |
| (c) Indigo | (d) Green | (c) Sophistication | (d) Shoe |
| 83. (a) Bugle | (b) Trumpet | 102. (a) Attire | (b) Dress |
| (c) Flute | (d) Violin | (c) Kimono | (d) Monogram |
| 84. (l) Tennis | (b) Rugby | 103. (a) Poetry | (b) Haiku |
| (c) Football | (d) Hockey | (c) Sonnet | (d) Elegy |
| 85. (a) Deleterious | (b) Pernicious | 104. (l) Dine | (b) Dime |
| (c) Surreptitious | (d) Harmful | (c) Fine | (d) Line |
| 86. (l) Arrow | (b) Javelin | 105. (a) Triangle | (b) Rectangle |
| (c) Dart | (d) Spear | (c) Square | (d) Parallelogram |
| 87. (l) Soft | (b) Strident | 106. (a) Botany | (b) Sericulture |
| (c) Shrill | (d) Grating | (c) Horticulture | (d) Culture |
| 88. (a) Corporeal | (b) Somatic | 107. (a) Paradigm | (b) Carpet |
| (c) Physical | (d) Affective | (c) Pattern | (d) Framework |
| 89. (a) Snide | (b) Sneer | 108. (a) Cow | (b) Mare |
| (c) Smugness | (d) Slyness | (c) Lion | (d) Goose |
| | | 109. (a) Poise | (b) Balance |
| | | (c) Stability | (d) Motility |
| | | 110. (a) Keen | (b) Plexus |
| | | (c) Poignant | (d) Biting |

Idioms & Phrases

Directions for (Que. 111 - 140): Read the information carefully and answer the questions given below it.

Study the following idioms and choose their meanings from the answer choices given below them.

- 111.** By the seat of one's pants
 (a) Through the tough method
 (b) Using guess work or instinct
 (c) Through maintaining a secret
 (d) Using one's strength
- 112.** In good season
 (a) Perfect timing
 (b) Having a pleasant temper
 (c) Sufficiently early
 (d) Safe from danger
- 113.** In a sense
 (a) To be sane (b) To some extent
 (c) To be awake (d) In a moment
- 114.** At second hand
 (a) In a sale
 (b) Through something that is already used
 (c) By the minute
 (d) Through an intermediate source
- 115.** Set one's face against
 (a) To confront (b) To infuriate
 (c) To contradict (d) To be opposed to
- 116.** Dictum
 (a) Value (b) Theory
 (c) Order (d) Thesis
- 117.** Dojo
 (a) Extinct (b) Distinct
 (c) Idiot (d) Seminary
- 118.** Fio-ru-it
 (a) He/she flourished
 (b) It flowered
 (c) She gave birth
 (d) It bore fruit
- 119.** Fracas I
 (a) Mess
 (b) Disorderly mob
 (c) Disorderly light
 (d) Ire
- 120.** Ruckus
 (a) Brain-washing (b) Dirt
 (c) Wild (d) Uproar
- 121.** Savant
 (a) Elderly Person
 (b) Learned Person
 (c) Person with Good Taste
 (d) A ruler
- 122.** Grantee
 (a) One who gives a grant
 (b) One who receives a grant
 (c) Someone who assures one of something
 (d) Someone who assures an assurance
- 123.** Mayhem
 (a) Deliberate violence
 (b) Confusion
 (c) Self-sacrifice
 (d) Aggression
- 124.** Megalomania
 (a) Exaggerated self-importance
 (b) Fear of food
 (c) Love for food
 (d) Love for Insanity
- 125.** In Toto
 (a) In accordance to (b) In existence
 (c) In fear of (d) Completely
- 126.** Xeno
 (a) Warrior (b) Foreign
 (c) Angry (d) Feminine
- 127.** Kaput
 (a) Immobile (b) Worthless
 (c) Totally ruined (d) Royal
- 128.** Kitsch
 (a) Vague (b) Immaterial
 (c) Worthless (d) Destroyed
- 129.** Persona Grata
 (a) Grateful person
 (b) Acceptable person
 (c) Huge personality
 (d) Person with a pure character
- 130.** Diaspora
 (a) Dispersed Group (b) Collected mob
 (c) Varied interests (d) Colorfully magnificent
- 131.** Let sleeping dogs lie
 (a) Leave an existing situation as it is
 (b) To encourage someone to tell a lie
 (c) To shirk responsibility
 (d) Let somebody sleep

- 132.** To go to the dogs
(a) To be destroyed
(b) To degenerate
(c) To be eaten up
(d) To have a bad life
- 133.** To pump up
(a) To flatter
(b) To disintegrate
(c) To infuse with enthusiasm
(d) To improve something
- 134.** To put on the ritz
(a) To put on an accent
(b) To dress very well
(c) To cheat somebody
(d) To assume an attitude of wealth
- 135.** To put oneself out
(a) To commit suicide
(b) To leave the country
(c) To take pains & do something
(d) To fall sick intentionally
- 136.** To say put
(a) To remain in a single position
(b) To laze around
(c) To be in captivity
(d) To remain in order
- 137.** In the saddle
(a) In deep trouble
(b) To travel constantly
(c) In control
(d) Be in an embarrassing situation
- 138.** From scratch
(a) From absolutely nothing
(b) From waste material
(c) From strong evidence
(d) From the living
- 139.** Up to scratch
(a) Till Death
(b) Up to the standard
(c) Till one becomes useless
(d) Till one becomes irritable
- 140.** Go to sea
(a) To leave one's children
(b) To realize the 'truth' of life
(c) To lose a lot of money
(d) Set out on a voyage

ANSWERS

MCQ Type Questions

1. (b)	2. (b)	3. (b)	4. (c)	5. (b)	6. (b)	7. (c)	8. (c)	9. (c)	10. (c)
11. (c)	12. (b)	13. (d)	14. (c)	15. (d)	16. (b)	17. (d)	18. (d)	19. (b)	20. (b)
21. (b)	22. (b)	23. (d)	24. (b)	25. (d)	26. (d)	27. (c)	28. (b)	29. (c)	30. (c)
31. (a)	32. (a)	33. (b)	34. (c)	35. (d)	36. (a)	37. (b)	38. (d)	39. (d)	40. (a)
41. (c)	42. (b)	43. (b)	44. (a)	45. (b)	46. (c)	47. (c)	48. (d)	49. (d)	50. (b)
51. (a)	52. (d)	53. (a)	54. (c)	55. (b)	56. (b)	57. (b)	58. (b)	59. (a)	60. (a)
61. (c)	62. (b)	63. (b)	64. (a)	65. (b)	66. (d)	67. (c)	68. (b)	69. (b)	70. (a)
71. (b)	72. (d)	73. (c)	74. (d)	75. (d)	76. (b)	77. (a)	78. (c)	79. (d)	80. (c)
81. (c)	82. (a)	83. (d)	84. (a)	85. (c)	86. (a)	87. (a)	88. (d)	89. (c)	90. (a)
91. (b)	92. (a)	93. (a)	94. (d)	95. (a)	96. (d)	97. (d)	98. (a)	99. (d)	100. (d)
101. (d)	102. (d)	103. (a)	104. (b)	105. (a)	106. (d)	107. (b)	108. (c)	109. (d)	110. (b)
111. (c)	112. (d)	113. (a)	114. (c)	115. (d)	116. (b)	117. (d)	118. (a)	119. (a)	120. (c)
121. (b)	122. (c)	123. (b)	124. (a)	125. (d)	126. (b)	127. (c)	128. (c)	129. (b)	130. (a)
131. (a)	132. (b)	133. (c)	134. (d)	135. (c)	136. (a)	137. (c)	138. (a)	139. (b)	140. (d)

Analytical Aptitude

1

CHAPTER

Logical Reasoning Ability

REASONING

Reasoning is the process of drawing conclusions from facts. These conclusions must follow inevitably from the facts from which they are drawn. Reasoning is not concerned with some conclusion that has a good chance of being true when facts are true. Indeed, reasoning as used here refers to logical reasoning, not of commonsense reasoning or probabilistic reasoning. The only conclusions that are acceptable are those that follow logically from the supplied facts.

Types of Reasoning

Reasoning can be classified in two major parts :

- (i) Verbal Reasoning
- (ii) Non-verbal Reasoning

Non-verbal reasoning involves the ability to understand and analyze visual information and solve problems using visual reasoning. This reasoning test will test your non-verbal reasoning ability as the questions appear in diagrammatic and pictorial form. Such test is also called diagrammatic or abstract reasoning tests, this test discussed in next chapter of spatial aptitude. For example: identifying relationships, similarities and differences between shapes and patterns, recognizing visual sequences and relationships between objects, and remembering these.

VERBAL REASONING

Verbal reasoning tests words, letters and numbers, and require logical reasoning and a reasonable knowledge of the English Language. It is also necessary to be familiar with simple manipulations with figures, like addition, subtractions division and multiplication. The problems of number in *Test of Reasoning* test how logical we are i.e., how well we reason and think while carrying out simple arithmetic manipulations.

CLASSIFICATION OF VERBAL REASONING

1. SERIES COMPLETION

In verbal series, words, letters or digits are given in a specific sequence/order and we have to find out next word, letter or digit to complete the given series.

Numbers or alphabetical letters, are generally called *terms of the series*. These terms follow a

certain pattern throughout. In the questions we have to identify last one or two terms to continue the series or to find a missing term in between given terms to continue the sequence followed in the question. There is no set pattern and each question may follow a different pattern or sequential arrangement of letters or digits, which have to detect using common sense and reasoning ability.

Types of series completion Questions

Mainly following five types of question are asked.

I. Alphabet series

In this series, given alphabets follow a particular sequence or order. We have to detect the pattern from the given alphabets and find missing alphabet or the next alphabet to continue the pattern.

- There are no set rules.
- There can be omission of alphabets in an order.
- Alphabets may also be omitted in an increasing/decreasing order, which may be direct increase or decrease.
- There can also be alternate order.
- There may also be alternate sequences
- There may be several other patterns in the letter series.
- To tackle letter series questions, vary position of the alphabet and its position number in both forward and backward sequences.
- To continue the series after Z, we again begin with A. In other words, the sequence is kept in a circular order.
- In solving these questions pattern of the alphabet series should be noted.

Some Skipping Patterns.

- (i) **Regular Order** : Number of alphabets skipped remains the same.

Example. A, D, G, J, ...?

Ans. M

- (ii) **Increasing Order** : Each time the number of alphabets skipped increases in a given pattern.

Example. A C F J O ?

Ans. U

Here, each time number of letters skipped increases by one.

- (iii) **Decreasing Order** : Each time the number of letters skipped decreases in a given pattern.

Example. A G L P S ?

Ans. U

Here number of letters skipped decreases by one each time, *i.e.*, first 5, then 4, then 3, and so on.

- (iv) **Interlinked Series** : In this two or more different series are attached together. These different series follow their own different rules.

Example. A D F J M R ?

Ans. V

Here, there are two interlinked series.

II. Letter Series

This type of questions usually consist of a series of small letters which follow a certain pattern. However, some letters are missing from the series. Then these missing letters are given in a proper sequence as one of the alternatives.

In these type of questions we have letters of the alphabet given here. We have to first identify the patterns that the series of letters follow. Then we have to find the missing letter based on the pattern already identified. In number series we saw different patterns that the numbers in the series can follow- like squares, cubes. In letter series, obviously, patterns like squares, cubes will not be possible. In letter series, in increasing or decreasing difference. The position of the letters in the English alphabet is considered to be the value of the alphabet in questions on letter series. Also when we are counting after we count from A to Z we again start with A, *i.e.* we treat the letters as being cyclic in nature like in Number Series. In this type of Letter Series also, we can have a "combination" of series, *i.e.* two series are combined and given. We need to identify the patterns in the two series to find out the missing letter. Sometimes, there will be some special types of series also. Let us look at a few examples to understand question on Letter Series.

Example. aaa... b...a...

- (a) baa (b) abb (c) bab (d) aab (e) bbb

Ans. (a)

Method :

- First blank space should be filled in by 'b', so that we have two a's followed by two b's.
- Second blank space should be filled in either by 'a', so that we have four a's followed by two b's, or by 'b' so that we have three a's followed by three b's.

- Last space must be filled in by 'a'.
- Thus, we have two possible answers : 'baa' and 'bba'.

But, only 'baa' appears in the alternatives. So, the answer is (a).

Note : In case, we had both the possible answers in the alternatives, we would have chosen the one that forms a more prominent pattern, which is aabb/ aaabbbbaa. Thus, our answer would have been "bba".

Example 1 : Find the next letter in the series

D, G, J, M, P, _____

- (a) Q (b) R
(c) S (d) T

Solution (c)

Three letters are added to each letter to get the next letter in the series

i.e. D + 3, G + 3, J + 3, M + 3, P + 3, S and P = 16 + 3 = 19 and the 19th letter in the alphabet is S.

Example 2 : Find the next letter in the series

A, B, D, H, _____

- (a) L (b) N
(c) R (d) P

Solution (d)

Each letter in the given series is multiplied with 2 to get the next letter in the series. $A \times 2 \Rightarrow 1 \times 2 = 2$ And the 2nd letter is B, $B \times 2 \Rightarrow 2 \times 2 = 4$ and the 4th letter is D.

Similarly $H \times 2 \Rightarrow 8 \times 2 = 16$ and the 16th letter is P.

Example 3 : What is the next letter in the series?

B, D, G, K, P, _____

- (a) S (b) V
(c) W (d) X

Solution (b)

B + 2, D + 3, G + 4, K + 5, P + 6, _____

P + 6 = 16 + 6 = 22 and the 22nd letter is V

Example 4 : I, X, J, W, K, V, L, _____

- (a) M (b) U
(c) S (d) T

Solution (b)

The given series is an alternate series, I + 1, J + 1, K + 1, L is one series and X - 1, W - 1, V - 1, _____ is the other series.

X - 1 = W, W - 1 = V and V - 1 = 22 - 1 = 21 and the 21st letter is U

III. Number Series

For better understanding, we will classify this into the following broad categories.

- (i) Difference series
- (ii) Product series
- (iii) Squares /Cubes series
- (iv) Miscellaneous series
- (v) Combination series

(a) Difference Series

The difference series can be further classified as follows.

- (a) Number series with a constant difference.
- (b) Number series with an increasing or decreasing difference.

In the number series with a **Constant difference**, there is always a constant difference between two consecutive numbers. For example, the numbers of the series 1, 4, 7, 10, 13... are such that any number is obtained by adding a constant figure of 3 to preceding term of the series

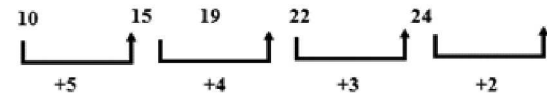
If we have to find the next number in the above series, we need to add a 3 to the last term 13. Thus, 16 is the next term of the series.

Under the series with constant difference, we can have series of odd numbers or series of even numbers also.

In the series with **increasing /decreasing difference**, the difference between consecutive terms keeps increasing (or decreasing as the case may be). For example let us try to find out the next number in the series 2, 3, 5, 8, 12, 17, 23,.....

Here, the difference between the first two terms of the series is 1; the difference between the third and the fourth terms is 3 and so on. That is the difference between any pair of consecutive terms is one more than the difference between the first number of this pair and the number immediately preceding this number. Here, since the difference between 17 and 23 is 6, the next difference should be 7. So the number that comes after 23 should be $(23 + 7) = 30$

We can also have a number series where the difference is in decreasing order (unlike in the previous example where the difference is increasing). For example, let us find out the next term of the series 10, 15, 19, 22, 24,.....

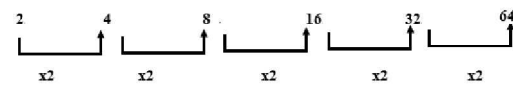


Here the difference between 1st & 2nd, 2nd & 3rd, 3rd & 4th numbers, etc are 5, 4, 3, 2 and so on since the difference between 22 and 24 is 2, the next difference should be 1. So, the number that comes after 24 should be 25.

(b) Product Series

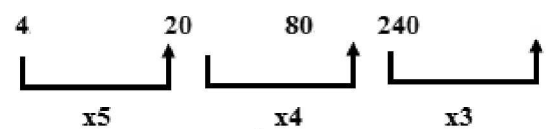
A product series is usually a number series where the terms are obtained by a process of multiplication. Here also, there can be different types of series. We will look at these through examples.

Consider the series 2, 4, 8, 16, 32, 64,.....



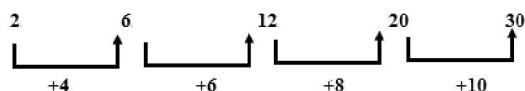
Here, each number in the series is multiplied by 2 to get the next term. So the term that comes after 64 is 128. So, each term is multiplied by a fixed number to get the next term. Similarly we can have a series where we have numbers obtained by dividing the previous term with a constant number. For example, in the series 64, 32, 16, 8.... Each number is obtained by dividing the previous number by 2 (or in other words, by multiplying the previous term by $\frac{1}{2}$). So here, the next term will be 4 (obtained 8 with 2).

Consider the series 4, 20, 80, 240,....



Here, the first term is multiplied by 5 to get the second term, the second term is multiplied by 4 to get the third term; the third term is multiplied by 3 to get the fourth term now, we have to multiply the fourth term by 2 to get the fifth term i.e. 480. So each term is multiplied by a decreasing factor (or it could also be an increasing factor) to get the next term. That is, with whatever number a particular term is multiplied to get the next term, this latest term is multiplied by a number different from the previous multiplying factor to get the next term of the series. All the multiplying factor follow a certain pattern (normally of increasing or decreasing order).

Consider the series 2, 6, 12, 20, 30....



This can be looked at a series of increasing differences. The difference of consecutive pairs of terms are 4 (between 2 and 6), 6 (between 6 and 12), 8 (between 12 and 20), 10 (between 20 and 30) and so on. Hence, the difference between 30 and the next term should be 12 and so the next terms will be 42. But this series can also be looked at as a product series

$$\begin{array}{ccccccccc}
 2 & 6 & 12 & 20 & 30 \\
 \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\
 1 \times 2 & 2 \times 3 & 3 \times 4 & 4 \times 5 & 5 \times 6
 \end{array}$$

The first term is the product of 1 and 2; the second term is the product of 2 and 3; the third term is the product of 3 and 4; the fourth term is the product of 4 and 5 and the fifth term is the product of 5 and 6. Hence the next term will be the product of 6 and that is 42.

(c) Squares/Cubes Series

There can be series where all the terms are related to the squares of numbers or cubes of numbers. With squares/ cubes of numbers as the basis, there can be many variations in the patterns of the series. Let us look at various possibilities of series based on squares/ cubes

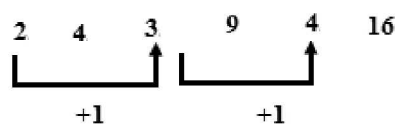
Each term of the series may be the square of a natural number such as 1, 4, 9, 16....

$$\begin{array}{ccccccc}
 1 & 4 & 9 & 16 \\
 \downarrow & \downarrow & \downarrow & \downarrow \\
 1^2 & 2^2 & 3^2 & 4^2
 \end{array}$$

The numbers are squares of 1, 2, 3, 4... respectively. The number which follows 16 (which is the square of 4) will be 25 (which is the square of 5).

The terms of the series may be the squares of odd numbers (for example, 1, 9, 25, 49....) or even numbers (for example, 4, 16, 36, 64....)

The terms of the series could be such that a number and its square are both given one after the other and such pairs are given in some specific patterns. For example, take the series 2, 4, 3, 9, 4, 16 ...



Here, 2 is followed by its square 4; then comes the number 3 (which is one more than 2)

followed by its square 9 and so on. Hence the next number in the series is 5 and the one after that is square i.e. 25.

Similarly each term could be the square root of its predecessor, For example, in the series 81, 9, 64, 8, 49, 7, 36, ... 81 is the square root of 9, 64 is the square root of 8, and so on. Therefore the next number which follow in the series should be the square root of 36 i.e. 6.

The terms of the series could be the squares of natural numbers increased or reduced by certain number. For example, in the series 3, 8, 15, 24....

We have {Squares of natural numbers-1} as the terms. The first term is $2^2 - 1$; the second term is $3^2 - 1$, the third term is $4^2 - 1$ and so on. Hence, the next term will be $6^2 - 1$ i.e. 35 [Please note that the above series can also be looked at as a series with increasing differences.]

The difference between the 1st & 2nd term, the 2nd & 3rd terms, and on are 5, 7, 9, and so on. Hence the next difference should be 11 giving us the next term as 35. There could also be series with {squares of natural numbers + some constant}.

Like we have seen with squares of numbers. We can have similar series with cubes of numbers. For example take series 1, 8, 27, 64...

$$\begin{array}{ccccccc}
 1 & 8 & 27 & 64 \\
 \downarrow & \downarrow & \downarrow & \downarrow \\
 1^3 & 2^3 & 3^3 & 4^3
 \end{array}$$

Here, all the terms are cubes of natural numbers So, the next terms will be 5^3 , i.e. 125.

Consider the series 2, 9, 28, 65....

$$\begin{array}{ccccccc}
 2 & 9 & 28 & 65 \\
 \downarrow & \downarrow & \downarrow & \downarrow \\
 1^3 + 1 & 2^3 + 1 & 3^3 + 1 & 4^3 + 1
 \end{array}$$

Here, the terms are {Cubes of natural numbers +1}. The first terms is $1^3 + 1$, the second term is $2^3 + 1$, the third term is $3^3 + 1$ and so on. Hence the next term will be $5^3 + 1$ i.e. 126.

(d) Combination Series

A number series which has more than one type of (arithmetic) operation performed or more that one series combined together is combination series. The series that are combined can be two series of the same type or could be different type of series as described above. Let us look at some example.

First let us look at those series which are formed by more than one arithmetic operation performed on the terms to get the subsequent terms.

Consider the series: 2, 6, 10, 3, 9, 13, 4, 12, ...

Here the first term 2 is multiplied by 3 to get the second term, and 4 is added to get the third term. The next term is 3 (one more than the first term 2) and it is multiplied by 3 to get 9 (which is the next terms) and then 4 is added to get the next term 13. The next term 4 (which is one more than 3) which is multiplied by 2 to get 12. Then 4 is added to this to get the next number 16.

Consider the series: 1, 2, 6, 21, 88..... Here we can observe that 88 is close to 4 times 21. It is in fact $21 \times 4 + 4$. So, if we now look at the previous term 21, it is related to the previous term 6 as $6 \times 3 + 3$. Now we get the general patterns: to get any term, multiply the previous term with k and then add k where k is a natural number with values in increasing order from 1. So, to get the second term, the first term has to be multiplied with 1 and then 1 is added. To get the third term, the second term is multiplied with 2 and then 2 is added and so, on. Hence after 88 the next term is $88 \times 5 + 5$ i.e. 445.

Now, let us look at a series that is formed by combining two (or more) series can be of the same type or of different types described above.

Consider the series 8, 12, 9, 13, 10, 14..... Here the 1st, 3rd, 5th... terms which are 8, 9, 10... form one series whereas the 2nd, 4th, 6th etc. terms which are 12, 13, 14 forms another series. Here both series that are being combined are two simple constant difference series. Therefore the missing number will be the next term of the first series 8, 9, 10 ... which is equal to 11.

Consider the series 0, 7, 2, 17, 6, 31, 12, 49, 20.... Here the series consisting of 1st, 3rd, 5th terms (i.e., the series consisting of the odd terms) which is 0, 2, 6, 12, 20,... is combined with another series consisting of 2nd, 4th, 6th terms (i.e., the series consisting of the even terms) which is 7, 17, 31, 49... The first series has the differences in increasing order 2, 4, 6, 8, 10 and so on. The second series also has the difference in increasing order 10, 14, 18... Since the last term 20 belongs to the first series a number from the second series will be obtained by adding 22 to 49 that is 71.

Consider the series 1, 1, 2, 4, 3, 9, 4, 16,... Here one series consisting of odd terms, which is 1, 2, 3, 4... is combined with the series of even terms which is 1, 4, 9, 16..... The first series is

a series of natural numbers. The second series is the squares of natural numbers. Hence, the next term is 5.

Consider the series: 1, 1, 4, 8, 9, 27 Here we have to find out the term that should be in place of the question mark. The odd terms forms one series 2, 5, 9, 14, 20 Where the difference is increasing. The difference are 3, 4, 5, 6.... This series is combined with the series of even terms 4, 9, 16... where the terms are squares of numbers 2, 3, 4.... Hence, the term that should come in place of the question mark is the next term of the second series which is 5^2 , i.e., 25.

(e) Miscellaneous Series

There are series that do not come under the other patterns and are of general nature but are important and are fairly common. Even here, sometimes there can be specific patterns in some cases.

Take the series 3, 5, 7, 11, 13 This is a series of consecutive **PRIME NUMBERS**. It is an important series and the student should look out for this as one of the patterns. The next term in this series is 17.

There can also be variations using prime numbers. Take the series 9, 25, 49, 121... In this series, the terms are squares of prime numbers. Hence the next term is 13^2 i.e. 169.

Take the series 15, 35, 77.... The first term is 3×5 ; the second term is 5×7 ; the third term is 7×11 ; here the terms are the product of two consecutive product of 11 and 13 i.e. 143.

Take the series 8, 24, 48, 120, 168.... Here the 2nd term is 3 times the first term and the 3rd term is 2 times the 2nd term, but after that it does not follow this pattern any more, if you look at the terms carefully, you will find that the terms are {one less than squares of prime numbers}. Hence the next term will be $17^2 - 1$ i.e. 288.

Consider the series 1, 4, 9, 1, 6, 2, 5, 3.... At first sight there is nothing we can say about the series. This is actually a series formed by a squares is in two or more digits, each of the digits is written as a separate term of the series. Thus the first terms are 1, 4 and 9 the squares of 1, 2 and 3 respectively. After this we should get 16 (which is the square of 4) since this different terms 1 and 6 in the series. Similarly, the next square 25 is written as two different terms 2 and 5 in the series. So, the next square 36 should be written as two terms 3 and 6. Of these 3 is already given, so the next terms of the series is 6.

Consider the series 1, 1, 2, 3, 5, 8...

$$\begin{array}{cccccc}
 1 & 1 & & 2 & & 3 & & 5 & & 8 \\
 \downarrow & & & \downarrow & & \downarrow & & \downarrow & & \\
 1+1 & & & 1+2 & & 2+3 & & 3+5 & &
 \end{array}$$

Here, each term starting with the third number, is the sum of the two preceding terms. After the first two terms as given (1 and 1), then onwards, to get any term, we need to add the two terms that come immediately before that position. Hence to get the next term of the series, we should take the two preceding terms 5 and 8 and add them up to get 13. So the next term of the series is 13. The term after this will be 21 (= 8 + 13)

A General Approach to Number Series

The best way of approaching the number series questions is to first observe the difference between terms. If the difference is constant, it is a constant difference series. If the difference is increasing or decreasing by a constant number, then it is series with a constant increasing or decreasing difference, then try out the product series approach. For this first divided the second term with the first term, third with the second, and so on. If the numbers obtained are the same then it is a product series. Alternatively try writing each term of the series as a product of two factors and see if there is any pattern that can be observed, if still there is no inference, but the difference is increasing or decreasing in a rapid manner, then check out the square series. If the increase is very high and it is not a square series, then try out the cube series. If the difference is alternatively decreasing and increasing (or increasing for some time and alternately decreasing), then it should most probably be a mixed series. Therefore test out the series with alternate numbers. If still the series is not solved, try out the general numbers.

IV. Mixed Series

In this type of number series, numbers obeying various orders of two or more different types of series are arranged alternately in a single number series.

Rules:

- Difference between consecutive numbers is same.

- Differences between consecutive numbers are in Arithmetic Progression (A.P.).
- Difference between consecutive numbers is a perfect square.
- Differences between consecutive numbers are multiples of a number.
- Differences between consecutive numbers are prime numbers.
- Difference between consecutive numbers is a perfect cube.
- Difference between consecutive numbers are in Geometric Progression (G.P.)
- Ratio between each consecutive number is the same.
- Ratio between each consecutive numbers is in Arithmetic Progression (A.P)
- Ratio between consecutive number is perfect square number.
- Ratio between consecutive number is the multiple of a number.
- Ratio between consecutive numbers is a prime number.
- Ratio between consecutive numbers is a perfect cube number.
- Ratios between consecutive number are in Geometric Progression (G.P.)

V. Correspondence Series

This type of series consists of three sequences with three different elements (usually capital letters, digits and small letters). On the basis of the similarity in positions in the three sequences, a capital letter is found to correspond with a unique digit and a unique small letter, whenever it occurs. We are required to trace out this correspondence and accordingly choose the elements to be filled in at the desired places.

2. ANALOGY

'Analogy' means 'correspondence'.

In these type of questions based on analogy, a particular relationship is given and another similar relationship has to be identified from the given alternatives.

Analogy means similarity or some kind of relationship. Analogy test, therefore tests the ability to see a relationship between two words and to recognize a similar relationship between two other words. It is a test of verbal reasoning ability rather than vocabulary. The key to analogy success is being able to express the relationship between the pair of words.

Kinds of Relationships with Examples

1. Instrument and Measurement

- Thermometer : Temperature (thermometer is an instrument used to measure temperature).
- Barometer : Pressure
- Anemometer : Wind vane
- Odometer : Speed
- Scale : Length
- Balance : Mass
- Sphygmomanometer: Blood Pressure
- Rain Gauge : Rain
- Hygrometer : Humidity
- Ammeter : Current
- Screw Gauge : Thickness
- Seismograph : Earthquakes
- Taseometer : Strains

2. Quantity and Unit

- Mass : Kilogram (kilogram is the unit of mass)
- Length : Metre
- Force : Newton
- Energy : Joule
- Resistance : Ohm
- Volume : Litre
- Angle : Radians
- Time : Seconds
- Potential : Volt
- Work : Joule
- Current : Ampere
- Luminosity : Candela
- Pressure : Pascal
- Area : Hectare
- Temperature : Degrees
- Power : Watt
- Conductivity : Mho
- Magnetic field : Oersted/Tesla

3. Individual and Group

- Soldiers : Army (group of soldiers is called an army).
- Cattle : Herd
- Flowers : Bouquet
- Grapes : Bunch
- Singer : Chorus
- Artist : Troupe
- Fish : Shoal

- Sheep : Flock
- Riders : Cavalcade
- Bees : Swarm
- Man : Crowd
- Sailors : Crew
- Nomads : Horde

4. Animal and Young one

- Cow : Calf (calf is the young one of cow)
- Horse : Pony
- Cat : Kitten
- Sheep : Lamb
- Butterfly : Caterpillar
- Insect : Larva
- Dog : Puppy
- Cock/Hen : Chicken
- Lion : Cub
- Duck : Duckling
- Man : Child

5. Male and Female

- Horse : Mare (mare is the female horse)
- Dog : Bitch
- Stag : Doe
- Son : Daughter
- Lion : Lioness
- Sorcerer : Sorceress
- Drone : Bee
- Gentleman : Lady
- Nephew : Niece
- Tiger : Tigress

6. Individual and Class

- Monkey : Mammal (monkey belongs to the class of mammal).
- Man : Mammal
- Ostrich : Bird
- Snake : Reptile
- Butterfly : Insect
- Whale : Mammal
- Rat : Rodent
- Lizard : Reptile

7. Individual and Dwelling Place

- Dog : Kennel (dog lives in a kennel)
- Bee : Apiary
- Cattle : Shed
- Lion : Den

- Poultry : Farm
- Monk : Monastery
- Fish : Aquarium
- Birds : Aviary
- Horse : Stable

8. Study and Topic

- Ornithology : Birds (ornithology is the study of birds).
- Seismology : Earthquakes
- Botany : Plants
- Onomatology : Names
- Ethnology : Human Races
- Ontology : Reality
- Herpetology : Amphibians
- Pathology : Diseases
- Astrology : Future
- Anthropology : Man
- Palaeography : Writings
- Ichthyology : Fishes
- Semantics : Language
- Nephrology : Kidney
- Concology : Shells
- Haematology : Blood
- Craniology : Skull
- Mycology : Fungi
- Entomology : Insects
- Zoology : Animals
- Occultism : Supernatural

9. Worker and Tool

- Warrior : Sword (sword is the tool used by a warrior).
- Carpenter : Saw
- Woodcutter : Axe
- Tailor : Needle
- Labourer : Spade
- Soldier : Gun
- Sculptor : Chisel
- Chef : Knife
- Doctor : Stethoscope
- Farmer : Plough
- Author : Pen

10. Tool and Action

- Needle : Sew (needle is used for sewing).
- Knife : Cut
- Sword : Slaughter
- Mattock : Dig

- Filter : Purify
- Steering : Drive
- Pen : Write
- Spanner : Grip
- Spoon : Feed
- Microscope : Magnify
- Gun : Shoot
- Shovel : Scoop
- Chisel : Carve
- Oar : Row
- Axe : Grind
- Auger : Bore
- Spade : Dig

11. Worker and Working Place

- Chef : Kitchen (chef works in a kitchen)
- Farmer : Field
- Warrior : Battlefield
- Engineer : Site
- Sailor : Ship
- Pilot : Cockpit
- Beautician : Parlour
- Artist : Theatre
- Actor : Stage
- Mechanic : Garage
- Lawyer : Court
- Scientist : Laboratory
- Teacher : School
- Doctor : Hospital
- Clerk : Office
- Servant : House
- Driver : Cabin
- Grocer : Shop
- Painter : Gallery
- Waiter : Restaurant
- Worker : Factory
- Umpire : Pitch
- Gambler : Casino

12. Worker and Product

- Mason : Wall (mason builds a wall)
- Farmer : Crop
- Teacher : Education
- Hunter : Prey
- Chef : Food
- Carpenter : Furniture

- Judge : Justice
- Author : Book
- Choreographer : Ballet
- Goldsmith : Ornaments
- Producer : Film
- Butcher : Meat
- Architect : Design
- Cobbler : Shoes
- Tailor : Clothes
- Poet : Poem
- Dramatist : Play

13. Product and Raw Material

- Prism : Glass (prism is made of glass).
- Butter : Milk
- Cloth : Fibre
- Wall : Brick
- Road : Asphalt
- Furniture : Wood
- Book : Paper
- Shoes : Leather
- Sack : Jute
- Pullover : Wool
- Omelette : Egg
- Metal : Ore
- Jewellery : Gold
- Rubber : Latex
- Linen : Flax
- Jaggery : Sugarcane
- Oil : Seed
- Wine : Grapes
- Paper : Pulp
- Fabric : Yarn

14. Part and Whole Relationship

- Pen : Nib (nib is a part of a pen).
- Pencil : Lead
- House : Kitchen
- Fan : Blade
- Class : Student
- Room : Window
- Aeroplane : Cockpit
- Book : Chapter

15. Word and Intensity

- Anger : Rage (rage is of higher intensity than Anger)

- Wish : Desire
- Kindle : Burn
- Sink : Drown
- Quarrel : War
- Error : Blunder
- Famous : Renowned
- Unhappy : Sad
- Refuse : Deny

16. Word and Synonym

- Abode : Dwelling (abode means almost the same as Dwelling)
- Blend : Mix
- Ban : Prohibition
- Assign : Allot
- Vacant : Empty
- Abduct : Kidnap
- Dearth : Scarcity
- Dissipate : Squander
- Sedate : Calm
- Brim : Edge
- House : Home
- Solicit : Request
- Presage : Predict
- Haughty : Proud
- Flaw : Defect
- Fierce : Violent
- Fallacy : Illusion
- Substitute : Replace
- Mend : Repair
- Alight : Descend
- Presume : Assume

17. Word and Antonym

- Attack : Defend (defend means opposite of Attack)
- Advance : Retreat
- Cruel : Kind
- Best : Worst
- Fresh : Stale
- Ignore : Notice
- Initial : Final
- Condense : Expand
- Chaos : Peace
- Create : Destroy
- Gradual : Abrupt
- Sink : Float

- Robust : Weak
- Gentle : Harsh
- Deep : Shallow
- Cordial : Hostile
- Affirm : Deny
- Mourn : Rejoice
- Lethargy : Alertness

18. Part

Gill : Fin (Gill and fin are two parts of a fish)

19. Type

Sword : Weapon (Sword is a type of weapon)

20. Tool & Worker

Paint Brush : Artist (Paintbrush is a tool to artist)

21. Degree

Warm : Hot (Warm is a greater degree of hot)

Fond : Doting (Fond is less extreme than doting)

22. Sign of

Grimace : Pain (Grimace is a sign of pain)

Brush : Discomfiture (A blush signifies discomfiture)

23. A place for

Miner : Quarry (A miner works in a quarry)

24. Sequence relationship

Foreward : Appendix (In a book a foreword precedes the Appendix)

25. Worker and Work

Poet : Sonnet (A sonnet is written by a poet)

Mason : Wall (A mason builds a wall)

26. Cause and Effect

Soporific : Sleepiness (Soporific causes sleepiness)

Common Types of Analogy Questions

(i) Completing Analogous Pair

In this type of questions, given two words are, which related to each other in some way. Another word is also given and we required to find out the relationship between first two words and choose the word from the given alternatives, which bears the same relationship to the third word, as the first two bear.

Example. Doctor : Nurse : : ? : Follower

- (a) Employer (b) Leader (c) Worker
(d) Manager (e) Union

Solution : (b)

Just as a nurse follows the Doctor's instructions, so also a follower works as directed by the leader.

(ii) Choosing Analogous Pair

In this type of questions, a pair of words is given, followed by four pairs of words as alternatives and we are required to choose the pair in which words bear the same relationship to each other as the words of the given pair bear.

Example. Sink : Float : : ?

- (a) Brim : Edge (b) Mend : Repair
(c) Gentle : Harsh (d) Flow : defect
(e) Solicit : Request

Solution : (b)

Just as sink is antonym of float, so also gentle is opposite of harsh.

(iii) Choosing Similar word

In this type of questions, a group of three words is given, followed by four other words as alternatives. The candidate is required to choose the alternative, which is similar to the given three words.

Example. Sitar : Guitar : Tanpura

- (a) Trumpet (b) Violin (c) Harmonium
(d) Mridanga (e) Fluet

Solution : (b)

Sitar, Guitar and Tanpura are all string instruments. Violin is also a string instrument.

(iv) Detecting Analogies

In this type of questions, we are required to trace out hidden analogy or common characteristic among the given words or to choose the word which possesses the same characteristic as the given word.

Example. Judo : Karate : Taekwando

- (a) They are names of martial arts.
(b) They can be performed by obese persons.
(c) They are performed on stage.
(d) They are important items of Asian Games.
(e) They have origin in Kerala

Solution : (a)

Judo, Karate and Taekwando are martial arts and (a) is most suitable description for all the three.

(v) Three word Analogy

In this type of questions, a group of three inter-related words is given and we are required to trace out the relationship among these three words and choose another group with similar analogy, from among the alternatives provided.

Example. Pen : Pencil : Ink

- (a) Orange : Banana : Juice
(b) Table : Chair : Wood
(c) Cow : Milk : Curd

(d) Fish : Shark : Water

(e) Car : Engine : Circle

Solution : (a)

Pen contains ink and pencil belongs to the same category as pen i.e. stationery. Similarly, orange contains juice and banana belongs to the same category as orange, i.e. fruits.

(vi) Number Analogy

It includes deals two types of questions :

- (i) Choosing a similarly related pair as the given number pair on the basis of relation between the numbers in each pair.

Example. 582 : 194 :: 258 : ?

- (a) 82 (b) 86
(c) 92 (d) 58

Solution : (b)

Just as 582 is related to 194 as 582 is 3 times 194, in the same way 258 is 3 times the 86.

- (ii) Choosing a number similar to a group of numbers on the basis of certain common properties that they possess.

Example. Which number belongs to given set of numbers?

2, 3, 5, 7, 11, 13, 17, 19, 23.....

- (a) 15 (b) 9 (c) 21 (d) 29 (e) 27

Solution : (d)

Given set of numbers belong to the prime numbers. In the given option, only prime number is 29.

(vii) Alphabet Analogy

In this type of questions, two groups of alphabets related to each other in same way, are given and we are required to find out this relationship and choose a group of alphabets which is related in the same way to a third group provided in the question.

Example. B C D : P Q R :: X Y Z : ?

- (a) FGH (b) LMN (c) RST (d) TUV (e) STU

Solution : (b)

Each letter of the first group is moved fourteen steps forward to obtain the corresponding letter of the second group. A similar relationship will exist between the third and fourth groups.

How to Answer Analogy Questions

- Figure out how the capitalized words are related.
- Create a sentence that expresses that connection.
- Test the choices with your sentence and eliminate the ones that don't work.
- If we are left with more than one answer—or no answer at all—go back and refine your sentence.

- Choose the best answer. If none of the choices fits exactly, choose the one that works best.

Plan to Solve Analogy Questions

(i) Make your Sentence More Precise

The analogies get more difficult as we work our way through each group. Use the common categories as a starting point, but be prepared to refine the relationship by making your sentence more precise.

Example. Grain : Silo ::

- (a) Pilot : plane (b) judge : courtroom
(c) water : reservoir (d) clock : time

If we apply the “place where” idea without thinking, here is what happens.

A silo is a place where we would find grain.

(a) A *plane* is a place where you would find a *pilot*.

(b) A *courtroom* is a place where you would find a *judge*.

(c) A *reservoir* is a place where you would find *water*.

(d) A *clock* is a place where you would find *time*.

We can eliminate (d), but that still leaves you with four possible answers. Now is express the relationship between silo and grain more precisely?

A silo is a place where grain is stored.

(a) A plane is a place where a pilot is stored.

(b) A courtroom is a place where a judge is stored.

(c) A reservoir is a place where water is stored.

(d) A highway is a place where automobiles are stored.

(ii) Analogies Work only if there's a clear cut connection

An analogy depends upon a necessary connection between pairs of words based on the meaning of the words. This clear connection must exist for both the original capitalized word pair and the correct answer choice, i.e. we can eliminate any answer choice for which we cannot describe a necessary relationship between the words. Use the “clear connection” test to rule out answer choices even when we don't know one of the capitalized words.

Example. Letter : Alphabet ::

- (a) note : scale
(b) ocean : merchandise

- (c) expert : automobile
(d) victory : farmland

We don't know what the capitalized words are in this analogy, but you can still eliminate choices (b), (c) and (d). There is no clear-cut connection between the words of those choices. The actual analogy above is LETTER : ALPHABET :: note : scale.

(iii) Some Analogies Work Better by Working Backward

Sometimes the capitalized words fall easily into a sentence that expresses their relationship and sometimes they don't. If we are having trouble making up a sentence that relates the two words, be prepared to shift gears. Try reversing the order of the original word pair.

Example. Ice : Glacier ::

- (a) train : trestle (b) sand : dune
(c) path : forest (d) feather : bird

If we can't come up with a sentence relating ICE to GLACIER, try relating GLACIER to ICE :

A glacier is made of ice.

If we reverse the order of the capitalized words, we must also reverse the order of the words in each answer choice.

- (a) A trestle is made up of a train.
(b) A dune is made up of sand.
(c) A forest is made up of a path.
(d) A bird is made up of a feather.

Clearly (b) exhibits the same relationship as the original pair.

(iv) Check from the Answer Choices

Many words have different meanings depending upon whether they are used as nouns or verbs or adjectives. If we are not sure how one of the capitalized words is being used, just check the answer choices. In analogies, all the answer choices will have the same grammatical structure as the capitalized words pair. That means if the answer choices are noun : noun, the capitalized pair will be NOUN : NOUN. On the other hand, if the answer choices are adjective : noun, then the capitalized pair will be ADJECTIVE : NOUN.

If we have eliminated all the answer choices with words that we know or sort of know, and we are left with a couple of choices containing words that we have never seen before, just guess and move on.

3. CLASSIFICATION

'Classification' means arrangement of given items on the basis of some common character. In this test, a group of certain items are given, out of which some are similar to in some manner and one is different from the rest. We are required to choose this one item which does not fit into the given group.

Types of Classification Questions

(i) Choosing Odd word

In this type of questions, four (or five) words are given out of which one is dissimilar to the others and students have to find that odd one from the given set.

Example. Choose the word which is least like the other words in the group.

- (a) chair (b) cupboard (c) Table
(d) paper weight (e) Bed

Ans. (d)

Here, all except paper weight are furnitures.

(ii) Choosing Odd pair of words

In this type of questions, certain pairs of words are given out of which words in all the pairs except one, bear a certain common relationship. We are required to decipher this relationship and choose the pair in which words are differently related, as the answer.

Example. Choose the odd pair of words.

- (a) Loom : Cloth (b) Table : Drawer
(c) Book : Page (d) Car : Wheel

Ans. (a)

In all other pairs, second is part of the first.

(iii) Choosing Odd numeral

In this type of questions, certain numbers are given, out of which all except one are alike in some manner while one is different and this number is to be chosen as the answer.

Example. Choose the number which is different from others in the group.

- (a) 49 (b) 64
(c) 121 (d) 144
(e) 156

Ans. (e)

Each of the numbers except 156 is complete square.

(iv) Choosing the Odd numeral pair/group

In this type of questions, certain pairs/groups of numbers are given out of which all except one are similar in some manner while one is different. The numbers in these similar

pairs may have the same property or may be related to each other according to the same rule. We are required to choose the odd pair/group.

Example. Choose the numeral pair/group which is different from others.

- (a) 83 – 75 (b) 58 – 50 (c) 49 – 42
(d) 25 – 17 (e) 131 – 123

Ans. (c)

In each of the pairs, first number is eight more than the second.

(v) Choosing the Odd letter group

In this type of questions, usually four groups of letters are given. Three of them are similar to each other in some manner while one is different and we have to the answer.

Example. Choose the group of letters which is different from others.

- (a) SU (b) PN
(c) IK (d) BD

Ans. (b)

All other are two alternate letters

4. CODING-DECODING

A CODE is a 'system of signals'. Therefore, coding is a method of transmitting a message between the sender and the receiver without a third person knowing it.

Coding and Decoding Test is set up to judge the candidate's ability to decipher the rule that codes a particular word/ message and break the code to decipher the message.

Types of Coding Decoding Questions

(i) Letter Coding

In these questions, real alphabets in a word are replaced by certain other alphabets according to a specific rule to form its code and we are required to detect the common rule and answer the questions accordingly.

Example. If TAP is coded as SZO, then how is FREEZE coded?

- (a) EQDFYG (b) ESDFYF (c) GQFDYF
(d) EQDDYD (e) EPDEZD

Solution : (d)

Each letter in the word TAP is moved one step backward to obtain the corresponding letter of the code.

	S	Z	O
-1	↑	↑	↑
	T	A	P

Thus, in FREEZE, F will be coded as E, R as Q, E as D and Z as Y. So, the code becomes EQDDYD.

(ii) Number Coding

In these questions, either numerical code values are assigned to a word or alphabetical code letters are assigned to the numbers and we are required to analyse the code as per the directions.

Example. If CHINA is written as 38126 and NEPAL is 25769, how is PLAINE is coded ?

Solution : Clearly Alphabets are coded as shown:

C	H	I	N	A	E	P	L
3	8	1	2	6	5	7	9

Hence, PLAINE is coded as 796125.

Example. In a certain code '37' means 'which class' and '583' means 'caste and class'. What is the code for 'caste' ?

- (a) 3 (b) 7 (c) 8
(d) Either 5 or 8

Solution : (d)

Given, which class' = '37' ... (i)

and 'caste and class' = '583' ... (ii)

From (i) and (ii), the common word 'class' is coded as '3'.

From (ii), 'caste' is coded as '5' or '8'.

(iii) Substitution

In this type of questions, some particular objects are assigned code names. Then a question is asked that is to be answered in the code language.

Example. If cook is called *butler*, butler is called *manager*, manager is called *teacher*, teacher is called *clerk* and clerk is called *principal*, who will teach in a class ?

- (a) Cook (b) Butler (c) Manager
(d) Teacher (e) Clerk

Solution : (e)

A 'teacher' teaches in a class and as given 'teacher' is called 'clerk'. So, a 'clerk' will teach in the class.

(iv) Mixed letter Coding

In this type of questions, three or four complete messages are given in the coded language and the code for a particular word is asked. To analyse such codes, any two messages bearing a common word are picked up. The common code word will mean that word. Proceeding similarly by picking up all possible combinations of two, the entire message can be analysed.

Example. If 'tee see pee' means 'Drink fruit juice'; 'see kee lee' means 'Juice is sweet' and 'fee ree mee' means. "He is intelligent", which word in that language means 'sweet' ?

- (a) see (b) kee (c) lee
(d) pee (e) None of these

Solution : (b)

In first and second statements, common word is 'Juice' and common code word is 'see'. So, 'see' means 'Juice'. In second and third statements, common word is 'is' and common code is 'lee'. So, 'lee' means 'is'. Thus, in second statement, remaining word 'sweet' is coded as 'kee'.

(v) Mixed number Coding

In this type of questions, a few groups of numbers each coding a certain short message, are given. Through a comparison of the given coded messages, taking two at a time, the candidate is required to find the number code for each word and then formulate the code for the message given.

Example. In a certain code, '786' means 'study very hard', '958' means 'hard work pays' and '645' means 'study and work'. Which of the following is the code for 'very' ?

- (a) 8 (b) 6 (c) 7 (d) 5

Solution : (c)

In first and second statements, common word is 'hard' and common code digit is '8'. So, '8' means 'hard'.

In the first and third statements, common word is 'study' and the common code digit is '6'. So, '6' means 'study'.

Thus, in the first statement, '7' means 'very'.

(vi) Deciphering individual letter codes by analysis

In this type of questions, certain sample words are given along with their codes. The candidate is required to decipher individual codes for different letters by comparing, taking two words at a time, and then answer the given questions accordingly.

5. BLOOD RELATIONS

In these tests, the success of a candidate depends upon the knowledge of the blood relations.

Following Blood relations help solve these Questions :

- Mother's or father's son — Brother
Mother's or father's daughter — Sister
Mother's or father's brother — Uncle
Mother's or father's sister — Aunt

- Mother's or father's father — Grand father
Mother's or father's mother — Grand mother
Son's wife — Daughter-in-Law
Daughter's husband — Son-in-Law
Husband's or wife's sister — Sister-in-Law
Husband's or wife's brother — Brother-in-Law
Brother's son — Nephew
Brother's daughter — Niece
Uncle or aunt's son or daughter — Cousin
Sister's husband — Brother-in-Law
Brother's wife — Sister-in-Law
Grandson's or Grand daughter's daughter
 — Great grand daughter

Types of Blood Relation Questions

(i) Deciphering Jumbled up descriptions

In this type of questions, a round-about description is given in the form of certain small relationships and direct relationship between the persons concerned is to be deciphered.

Example. Pointing to a photograph, a man said, "I have no brother or sister but that man's father is my father's son." Whose photograph was it ?

- (a) His own (b) His son's
(c) His father's (d) His nephew's

Solution : (a)

Since man has no brother, his father's son is he himself.

(ii) Relation Puzzle

In this type, mutual blood relations or other informations of more than two persons are mentioned and information about any two is mentioned.

Example. A and B are brothers. C and D are sisters. A's son is D's brother. How is B related to C.

- (a) Father (b) Brother (c) Grandfather
(d) Uncle (e) None of these

Solution : (d)

B is the brother of A; A's son is D's brother. This means D is the daughter of A. Since C and D are sisters, C is also the daughter of A. So, B is the uncle of C.

(iii) Coded Relations

In such questions, relationships are represented by certain codes or symbols such as +, -, ×, ÷, *, Δ etc. Then relationships between certain persons, given in the form of these codes, are to be analysed.

Example. If $A + B$ means A is the sister of B;
 $A - B$ means A is the brother of B;
 $A \times B$ means A is the daughter of B, which of
the following shows the relation that E is the
maternal uncle of D?

- (a) $D + F \times E$ (b) $D - F \times E$ (c) $D \times F + E$
(d) $D \times F \times E$ (e) None of these.

Solution : (c)

E is the maternal uncle of D means D is the
daughter of the sister (say F) of E i.e.
 $D \times F + E$.

The easiest and non-confusing way to solve these
types of problems would be to draw a family tree
diagram and increase the levels in the hierarchy
as shown below:

1 st	GRAND PARENTS	Grandfather, Grandmother, Granduncle, Grand aunt
2 nd	PARENTS & IN-LAWS	Father, Mother, Uncle, Aunt, Father-in-law, Mother in law
3 rd	SIBLINGS, SPOUSE & IN-LAWS	Brother, Sister, Cousin, Wife, Husband, Brother in law, Sister in law
4 th	CHILDREN & IN-LAWS	Son, Daughter, Niece, Nephew, Son in law, Daughter in law
5 th	GRAND CHILDREN	Grandson, Granddaughter

Mother's or Father's son	Brother
Mother's or Father's daughter	Sister
Mother's or Father's brother	Uncle
Mother's or Father's sister	Aunt
Mother's or Father's mother	Grandmother
Mother's or Father's father	Grandfather
Grandmother's brother	Grand uncle
Grandfather's sister	Grand aunt
Grandfather's brother	Grand uncle

Grandmother's sister	Grand aunt
Sister's or Brother's son	Nephew
Sister's or Brother's Daughter	Niece
Uncle or Aunt's son or daughter	Cousin
Son's wife	Daughter in law
Daughter's husband	Son-in-law

Husband's or Wife's sister	Sister-in-law
Husband's or Wife's brother	Brother-in-law
Sister's husband	Brother-in-law
Brother's wife	Sister-in-law
Children of same parents	Siblings (could be all brothers, all sisters or some brothers and some sisters)
Children	Son, Daughter
Children's Children	Grandson, Granddaughter (Grandchildren)

In addition, remember the word spouse which means
either husband or wife.

Grandfather and grandmother will come in the first
stage. Mother, father uncle and aunt will come in
the second stage: sister brother and cousin will come
at the third stage: son daughter, niece and nephew
will come in the fourth stage and finally,
granddaughters and grandsons will come. The above
stage are made from the point of view of an
individual.

In type I questions, the relationship between two
people is given through a roundabout way of
relating them through other people. We have to go
through the series of relationship can be given I in
the questions. The relationship can be given as a
simple statements or as a statement made by a
person. In the first example given below, a person is
involved in making a statement whereas in the
second question, there is no person involved in
making a statement.

Example 1 : A's father's mother-in law only
daughter's son is b. How is A related to B?

- (a) Brother (b) Sister
(c) Nephew (d) Cannot be determined

Solution (d)

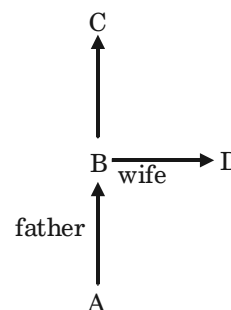
A's father's mother- in -law's only daughter is A's
mother. A's mother's son is A's brother. But A can
either brother or sister to B.

Example 2 : If A's father is B, C is the father of B
and D is A's mother, then how is C related to D?

- (a) Father (b) Grandfather
(c) Father-in law (d) Uncle

Solution (c)

A's father is B and mother is D. Therefore D is B's
wife and C is the father of B. Hence C is D's father-
in law.



Example 3 : $A + B$ means A is the son of B

$A - B$ means A is the daughter of B.

$A \times B$ means A is the father of b.

$A \div B$ means A is the mother of B.

If $M \times N + O - P \div Q$, then how is M related to?

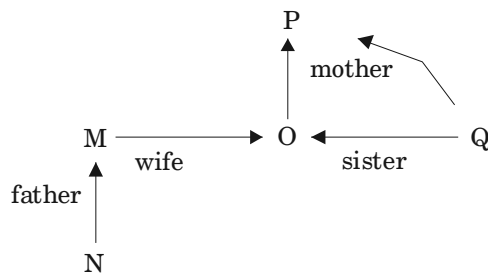
- (a) Husband (b) Cousin
(c) Brother-in law (d) Uncle

Solution (c)

$M \times N + O - P \div Q$ Means M is the father of N, N is the son of O, O is the daughter of P, P is the mother of Q.

M is the father of N and N is the son of O means m is the husband of O. O is the daughter of P and P is the mother of Q means O is the sister of Q.

M is the husband of O and O is the sister of Q means M is the brother-in law of Q.



Example 4 : $A + B$ means A is the son of B

$A - B$ means A is the daughter of B.

$A \times B$ means A is the father of B.

$A \div B$ means A is the mother of B.

Which of the following means S is the son-in law of P?

- (a) $P + Q \div R \times S - T$ (b) $P \times Q \div R - S + T$
(c) $P + Q \times R - S \div T$ (d) $P \times Q - R \div S \times T$

Solution (b)

$P + Q \div R \times S - T$ Means P is the son of Q. Q is the daughter of R, R is the father of S and S is the daughter of T. Hence S is the nephew of P.

$P \times Q \div R - S + T$ Means P is the father of Q, Q is the mother of R, R is the daughter of S and S is the son of T. Hence S is the son-in-law of P.

$R + Q \times R - S \div T$ Means P is the son of Q, Q is the father of R, R is the daughter of S and S is the mother of T. Hence S is the mother of P.

$P \times Q - R \div S \times T$ Means P is the father of Q, Q is the daughter of R, R is the mother of S and S is the father of T. Hence S is the son of P.

Example 5 : Pointing to a person, Raju said, "He is the only brother of my father's mother's daughter". How is the person related to Raju?

Solution (b)

Raju's father's mother's daughter is Raju's father's sister. Raju's father's sister's only brother is Raju's father. Hence the person is Raju's father.

6. PUZZLE TEST

This type of questions put in the form of puzzles involving certain number or items, be it persons or things and we are required to analyse the given information, condense it in a suitable form and answer the questions asked.

Type of Puzzle Tests

(i) Classification type Questions

This type consists of questions in which certain items belonging to different groups or possessing different qualities are given along with some clues with the help of which the candidate is required to group and analyse the given items and answer the questions accordingly.

Directions (1 to 5) : Read the following information carefully and answer the questions that follow:

- Five friends P, Q, R, S and T travelled to five different cities of Chennai, Calcutta, Delhi, Bangalore and Hyderabad by five different modes of transport of Bus, Train, Aeroplane, Car and Boat from Mumbai.
- The person who travelled to Delhi did not travel by boat.
- R went to Bangalore by car and Q went to Calcutta by aeroplane.
- S travelled by boat whereas T travelled by train.
- Mumbai is not connected by bus to Delhi and Chennai.

Example 1 : Which of the following combinations of person and mode is not correct ?

- (a) P — Bus (b) Q — Aeroplane
(c) R — Car (d) S — Boat
(e) T — Aeroplane

Example 2 : Which of the following combinations is true for S ?

- (a) Delhi — Bus (b) Chennai — Bus
(c) Chennai — Boat (d) Data inadequate
(e) None of these

Example 3 : Which of the following combinations of place and mode is not correct ?

- (a) Delhi — Bus
(b) Calcutta — Aeroplane
(c) Bangalore — Car
(d) Chennai — Boat
(e) Hyderabad — Bus

Example 4 : The person travelling to Delhi went by which of the following modes ?

- (a) Bus (b) Train
(c) Aeroplane (d) Car
(e) Boat

Example 5 : Who among the following travelled to Delhi ?

- (a) R (b) S
(c) T (d) Data inadequate
(e) None of these

Solutions : Given information can be analysed as follows :

Mode of Transport :

R travels by Car, Q by Aeroplane, S by Boat and T by Train.

Now, only P remains. So, P travels by Bus.

Place of Travel :

R goes to Bangalore, Q to Calcutta.

Now, bus transport is not available for Delhi or Chennai. So, P who travels by Bus goes to Hyderabad.

S travels by boat and hence, S did not go to Delhi. So, S goes to Chennai.

Now, only T remains. So, T goes to Delhi.

Person	Place	Mode
P	Hyderabad	Bus
Q	Calcutta	Aeroplane
R	Bangalore	Car
S	Chennai	Boat
T	Delhi	Train

- (e). Incorrect combination is T — Aeroplane
- (c). Correct combination for S is Chennai — Boat
- (a). Incorrect combination is Delhi — Bus
- (b). T travelled to Delhi by Train
- (c). T travelled to Delhi

(ii) Seating / Placing Arrangements

In this type of questions, some clues regarding seating or placing sequence (linear or circular) of some persons or items is given. The candidate is required to form the proper sequence using these clues and answer the questions accordingly.

Example. Four girls are sitting on a bench to be photographed. Shikha is to the left of Reena. Manju is to the right of Reena. Rita is between

Reena and Manju. Who would be second from the left in the photograph ?

- (a) Reena (b) Shikha (c) Manju
(d) Rita (e) Either Shikha or Reena

Solution : (a)

Shikha is to the left of Reena and Manju is to her right.

Rita is between Reena and Manju.

Hence order is : Shikha, Reena, Rita, Manju.

In the photograph, Reena will be second from left.

(iii) Comparison Type Questions

In this type of questions, clues are given regarding comparisons among a set of persons or things with respect to one or more qualities and we are required to analyse the whole information, form a proper ascending/descending sequence and then answer the given questions accordingly.

Directions (1 to 3) : Read the information given below and answer the questions that follow :

- There is a group of five girls.
- Kamini is second in height but younger than Rupa.
- Pooja is taller than Monika but younger in age.
- Rupa and Monika are of the same age but Rupa is tallest between them,
- Neelam is taller than Pooja and elder to Rupa.

Example 1 : If they are arranged in the ascending order of height, who will be in third position ?

- (a) Monika (b) Rupa
(c) Monika or Rupa (d) Data inadequate
(e) None of these

Example 2 : If they are arranged in the descending order of their ages, who will be in fourth position ?

- (a) Monika or Rupa (b) Kamini
(c) Monika (d) Data inadequate
(e) None of these

Example 3 : To answer the question “who is the youngest person in the group”, which of the given statements is superfluous ?

- (a) Only (i) (b) Only (ii)
(c) Only (v) (d) Either (i) or (iv)
(e) None of these

Solution : First find sequence of heights

By (iii), we have $M < P$

By (v), we have $P < N$.

Now, Rupa is tallest and Kamini is second in height, hence sequence of heights is :

$$M < P < N < K < R.$$

Now, find age sequence :

By (ii), we have $K < R$.

By (ii), we have $P < M$.

By (iv), we have $R = M$.

By (v), we have $R < N$.

Hence sequence of ages is:

$P < K < R = M < N$ or $K < P < R = M < N$

1. (e) In increasing order of height, Neelam is in third position.
2. (e) In descending order of ages, Neelam will be in fourth position (because Monika and Rupa both lie at third position).
3. (a) Only statement (i) is not necessary.

(iv) Sequential Order of Things

In this type of questions, some clues are given regarding order of occurrence of certain events and we are required to analyse the given information, frame the right sequence and then answer the questions accordingly.

Directions (1 to 4) : Read the following information carefully and answer the questions given below it :

- (i) Eight doctors P, Q, R, S, T, U, V and W visit a charitable dispensary every day except on a holiday i.e. Monday.
- (ii) Each doctor visits for one hour from Tuesday to Sunday except Saturday. The timings are 9 a.m. to 1 p.m. and 2 p.m. to 6 p.m.; 1 p.m. to 2 p.m. is lunch break.
- (iii) On Saturday, it is open only in the morning i.e. 9 a.m. to 1 p.m. and each doctor visits for only half an hour.
- (iv) No other doctor visits the dispensary before doctor Q and after doctor U.
- (v) Doctor-W comes immediately after lunch break and is followed by R.
- (vi) S comes in the same order as P in the afternoon session.

Example 1 : Doctor P visits in between which of the following pairs of doctors ?

- (a) S and V (b) U and W
- (c) R and W (d) R and U
- (e) None of these

Example 2. At what time the visit of doctor R is over on Sunday?

- (a) 1 p.m. (b) 3 p.m.
- (c) 4p.m. (d) 5p.m.
- (e) None of these

Example 3 : At what time the visit of Doctor T would be over on Saturday?

- (a) 10 a.m.
- (b) 11 a.m.
- (c) Either 10 a.m. or 11 a.m.
- (d) Data inadequate
- (e) None of these

Example 4 : If the lunch break and subsequent visiting hours are reduced by 15 minutes, at what time Doctor U is expected to attend the dispensary ?

- (a) 3.15 p.m. (b) 4 p.m.
- (c) 4.15 p.m. (d) 4.45 p.m.
- (e) None of these

Solutions : First form sequence of visit using (iv), (v) and (vi).

From (iv), Q visits first and U visits last.

From (v), W visits first after break and is followed by R.

From (vi), P visits after break.

Hence sequence of visit after break becomes W R P U

Also, S has the same position in morning session as P in afternoon session. Thus, sequence of visit before break is Q, T/V, S, V/T.

1. (d). P visits between R and U.
2. (c). Time of visit of W is 2 p.m. to 3 p.m., Time of visit of R is 3 p.m. to 4 p.m. So, the visit of doctor R is over at 4 p.m.
3. (c). T visits either second or fourth. So, the time of visit on Saturday will be either 9.30 a.m. or 10.30 a.m. Thus, T's visit will be over at either 10 a.m. or 11 a.m.
4. (b). As mentioned, lunch break will be over and doctor W will visit at 1.45 p.m., doctor R will visit at 2.30 p.m., doctor P will visit at 3.15 p.m. and U will visit at 4 p.m.

(v) Selection based on given conditions

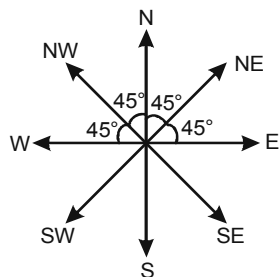
In this type of questions, a few essential criteria for selection of a group of items are given and we have to keep these conditions in mind and make the required selection as per the directions given in each question.

(vi) Jumbled problems

In this type of questions, some mixed clues regarding three or more qualities of given items or persons is given and we are required to analyse this mixed information with respect to different qualities and classify the items accordingly.

7. DIRECTION SENSE

In this type of test, the questions consist of a sort of direction puzzle. A successive follow-up of directions is formulated and we are required to ascertain final direction or the distance between two points. The test is meant to judge ability to trace and follow correctly and sense the direction correctly.



Above figure shows four main directions : North N, South S, East E, West W

Four cardinal directions : North East NE, North West NW, South East SE, South West SW

Pythagoras Theorem

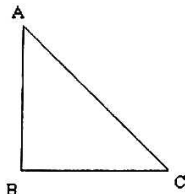
The theorem is applicable only in right angle triangles, in which from theorem one angle is of 90°

$$AC^2 = AB^2 + BC^2$$

where, AC is Hypotense

AB is Perpendicular
(Height)

BC is Base



Shortcut method

1. If 2 opposite turns are coming together, they will cancel each other

Example. left, right and right, left.

2. If similar turns are coming together they will oppose the earlier direction.

Example. if a person is going in North and taking two left turns it means now person is moving in opposite to North i.e. South.

3. Start from either left side or right side

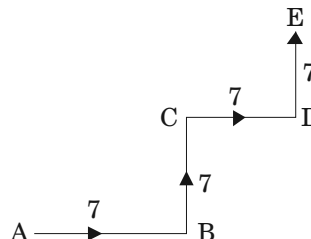
NOTE

The distance from a particular point after travelling a distance of x meters in the horizontal direction and a distance of y meters in the vertical

direction is equal to $\sqrt{x^2 + y^2}$ (please note that in common usage, North South direction is referred to as “vertical “ direction and the East West direction is referred to as the “horizontal” direction).

Example 1 : A person travels a distance of 7 km towards east from his house, then travels 7 km towards north and then a distance 7 km towards east and finally 7 km towards north. What is the vertical distance traveled by him?

Solution :



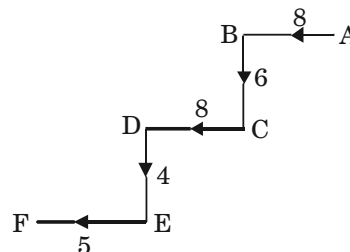
Let A and E be the initial and the final positions.

The vertical distance travelled

$$= BC + ED (7 + 7) \text{ km} = 14 \text{ km}$$

Example 2 : A person starts from his house and travels 8m towards west then he travels 6 m towards his left then 8m towards west and then 4m towards south. Finally he turns right and travels 5m. What is the horizontal distance travelled by him?

Solution :



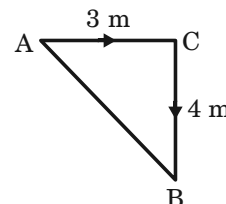
Let A and F be the initial and the final positions.

$\therefore$ Horizontal distance traveled

$$= FE + DC + BA = 5 + 8 + 8 = 21 \text{ m}$$

Example 3 : Surya travels 3 m towards east and then turns right and travels 4 m. What is the distance between the initial & the final position of Surya?

Solution :



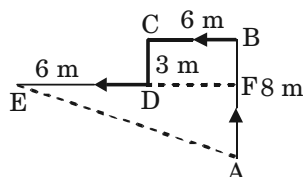
Let A and B the initial and the final positions of Surya.

$$AB = \sqrt{AC^2 + BC^2}$$

$$= \sqrt{3^2 + 4^2} = 5 \text{ m}$$

Example 4 : Starting from his house, Sachin walks a distance of 8 m towards north, then he turns left and walks 6m, then walks 3m towards south and finally travels 6m towards west to reach his office. What is the distance between his house and office and also find in which direction is his office situated with respect to his house.

Solution :

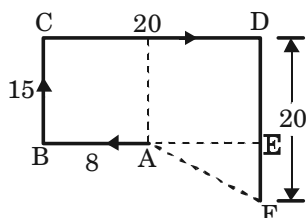


Let A be the Sachin's house and E his office. The distance between A and E

$$\begin{aligned} &= \sqrt{(EF)^2 + (AF)^2} \\ &= \sqrt{(ED + CB)^2 + (AB - FB)^2} \\ &= \sqrt{(ED + CB)^2 + (AB - CD)^2} \\ &= \sqrt{12^2 + 5^2} = 13\text{m} \end{aligned}$$

Example 5 : Starting from her house, Nisha traveled 8 m towards west then turned right and travelled 15 m. She then traveled 20 m towards east, followed by 20 m to south to reach a hostel. How far is her house from the hostel and in which direction?

Solution :



Let A and f be the initial and final positions.

$$\text{Now } AF = \sqrt{AE^2 + EF^2}$$

$$AE = CD - BA = 20 - 8 = 12\text{m}$$

$$EF = DF - DE - CB = 20 - 15 = 5\text{m}$$

$$\therefore AF = \sqrt{144 + 25} = \sqrt{169} = 13\text{m}$$

Her house is towards north- west from the hostel

8. LOGICAL VENN DIAGRAMS

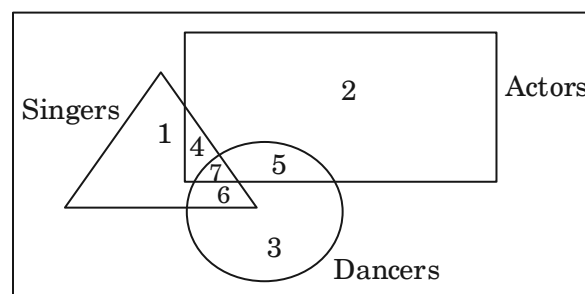
This type of questions aim at analysing ability to relate a certain given group of items and illustrate it diagrammatically.

Venn Diagrams are diagrammatic representation of sets, using geometrical figures like circles, triangles rectangles etc. Each geometrical figure

represents a group as shown in the examples. The area common to two or more figure represent those elements which are common to two or more groups. There are various models in Venn Diagrams which we will discuss with examples.

Venn Diagram Type I

In these kind of questions. There are many geometrical figures representing different groups. Let's discuss this type with the help of the following example.



This diagram consists of three groups –Singers, Actors and Dancers represented by a triangle, a rectangle and a circle respectively. There are seven regions represented by numbers from 1 to 7 where each region represents a different group

Region 1 → Represents only Singers

Region 2 → Represents only Actors

Region 3 → Represents only Dancers

Region 4 → Represents only Singers & Actors

Region 5 → Represents only Actors & Dancers

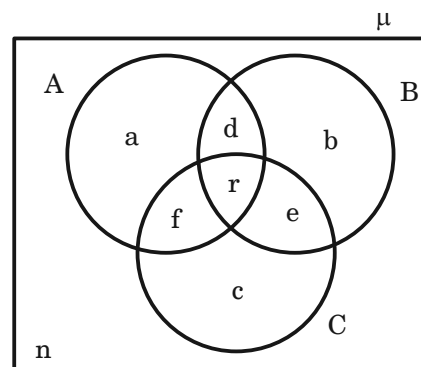
Region 6 → Represents only Dancers & Singers

Region 7 → Represents Singers, Actors & dancers.

Hence the various areas as shown represent different groups i.e. region 6 represents those Singers who are only Dancers as region 6 is the intersection of the triangle and the circle, but not the square. You may be asked questions like "Which region represents

Dancers who are only Singers?" The answer to this question, as explained above will be region 6.

Venn Diagram Type II



In this type, two three or four different groups could be given with some elements common to two or more groups. Let us observe the diagram given below.

Here A, B and C are three different groups and the various regions can be explained as given below.

Only A = a

Only B = b

Only C = c

A and B only = d

B and C only = e

C and A only = f

All the three (A, B and C) = r

Both A and B = d + r

Both B and C = e + r

Both C and D = f + r

Neither A nor B nor C = n

A, B or C and none = μ Also $\mu = (A \cup B \cup C) + n$

Here, the rectangle represents the sample space, which consists of three groups A, B and C and also n, which is the number of people belonging to neither A, nor B nor C. Some more formula are as given under.

- (i) $A' = (b + e + c) + n$ where $A' = A$ complement (not in A)
- (ii) $B' = (a + f + c) + n$; where $B' = B$ complement (not in B)
- (iii) $C' = (a + d + b) + n$ where $C' = C$ complement (not in C)
- (iv) $A - B = A - (A \cap B)$
- (v) $A \Delta B = (A - B) \cup (B - A)$
- (vi) Number of people (or things) belonging to at least one out of the three groups
 $= (A \cup B \cup C)$

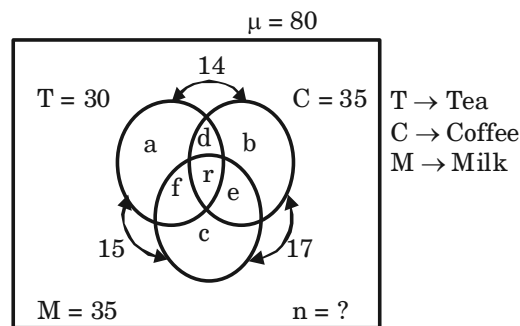
$$= (a + b + c) + (d + e + f) + r$$

$\uparrow$ $\uparrow$ $\uparrow$
 Exactly Exactly Exactly
 one two three

$$\begin{aligned} \text{(vii)} A + B + C &= (A \cup B \cup C) + (d + e + f) + 2r \\ &= (a + b + c) + 2(d + e + f) + 3r \end{aligned}$$

Example 1 : In a class of 80 students 30 students like tea, 35 students like milk and 25 students like coffee. It is also known that 15 students like both tea and milk, 17 students like both milk and coffee and 14 students like both tea and coffee 9 students like tea, coffee as well as milk.

Solution :



Based on the information given, we get the following diagram

Now $r = 9$ is given

$$\Rightarrow d = 14 - 9 = 5; e = 15 - 9 = 6; f = 17 - 9 = 8$$

$$\text{Also } a = 30 - (d + r + e) = 30 - (5 + 9 + 6) = 10$$

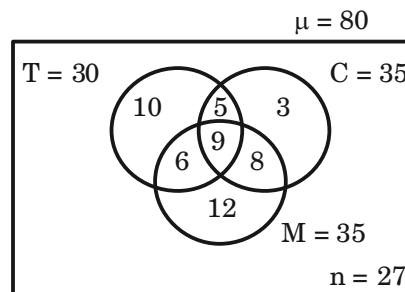
$$b = 25 - (d + r + f) = 25 - (5 + 9 + 8) = 3$$

$$c = 35 - (e + r + f) = 35 - (6 + 9 + 8) = 12$$

$$\begin{aligned} \text{Now, } T \cup C \cup M &= a + b + c + d + e + f + r \\ &= 10 + 3 + 12 + 5 + 6 + 8 + 9 \\ &= 53 \end{aligned}$$

$$\Rightarrow n = \mu - (T \cup C \cup M) = 80 - 53 = 27$$

Hence, we can now get the following diagram from the above data



Now, we can answer the questions of the following nature.

Example 2 : What is the total number of students who like neither tea nor coffee nor milk?

Solution :

$$n = \mu - (T \cup C \cup M) = 80 - 53 = 27$$

Example 3 : How many students do not like Tea?

Solution :

$$T' = b + c + n = 3 + 12 + 27 = 50$$

$$(\text{or}) T' = \mu - T = 80 - 30 = 50$$

Example 4 : What is the value of T-C?

Solution :

$$(T - C) = a + f$$

$$(\text{i.e. the regions of A excluding B})$$

$$= 10 + 6 = 16$$

Example 5 : What is the value of $C \Delta M$?

Solution :

$$\begin{aligned} C \Delta M &= (C - M) \cup (M - C) \\ &= (b + d) + (d + c) \\ &= (3 + 5) + 6 + 12 = 26 \end{aligned}$$

Example 6 : What is the total number of students who like both tea and coffee but not milk?

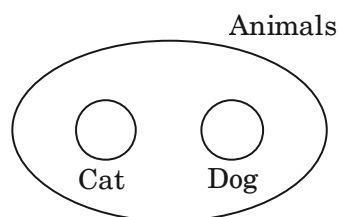
Solution :

Number of students who like both tea and coffee but not milk = $d = 5$

Venn Diagram Type III

In this type Venn diagrams are used to establish relationship between the given groups. In other words two or more groups are given and the Venn diagram, which most correctly establishes a relation between them has to be chosen out of the various Venn diagrams given in the choices. Let us look at some of the examples given below.

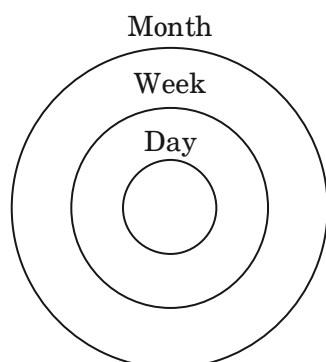
Examples 1: Animals, Cat, Dog



Solution:

Here, in animals we have many species of which cat and dog are two different kinds of species, having nothing in common. So the above diagram is the most appropriate representation of the given groups.

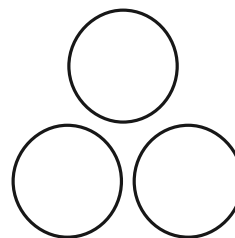
Examples 2 : Month, week, day



Solution:

We know that day is a part of the week and week is a part of the month. So the above diagram is the most appropriate representation of the given groups.

Examples 3 : Mars Earth, Jupiter



Solutions:

We know that Mars, Earth and Jupiter are three independent entities having nothing in common. So the above diagram is the most appropriate representation of the given groups.

9. ALPHABETICAL QUIBBLE

In this type of questions, we are given alphabets from A to Z. The position of a letter is given in the form of a puzzle and we are required to find this letter. However, sometimes a random letter series is given and we are required to find out how many times a letter satisfying the conditions specified in the question occurs.

Example. Following question is based on the following alphabet series.

ABCDEFGHIJKLMNOPQRSTUVWXYZ

Which letter is exactly midway between H and S in the given alphabet ?

- (a) L
- (b) M
- (c) N
- (d) O
- (e) No such letter

Ans. (e)

From given series we get that M & N are in midway between H and S. So, exactly midway is no such letter.

10. NUMBER, RANKING & TIME SEQUENCE

(i) Number Test

In this type of questions, generally you are given a long series of numbers and we are required to find out how many times a number satisfying the conditions, specified in the question, occurs.

Example : How many 5's are there in the following sequence which are immediately followed by 3 but not immediately preceded by 7 ?

8953253855687335775365335738

- (a) One
- (b) Two
- (c) Three
- (d) Four
- (e) More than four

Solution : (c)

A number which comes after a given number is said to follow it while the one which comes before the given number precedes it. Thus, numbers satisfying the given conditions, can be shown as follows. Hence there are three such numbers.

(ii) Ranking Test

In this type of test, generally ranks of a person both from top and bottom are mentioned and total number of persons is asked. However, sometimes this question is put in the form of a puzzle of interchanging seats by two persons.

Example : Anil ranked ninth from the top and thirty eighth from the bottom in a class. How many students are there in the class?

- (a) 45 (b) 46 (c) 47
(d) 48 (e) 49

Solution : (b)

Whole class consists of :

- (i) 8 students who have a rank higher than Anil
(ii) Anil; and
(iii) 37 students who have rank lower than Anil i.e., $(8 + 1 + 37) = 46$ students.

(iii) Time Sequence Test

Example : Raman remembers that his brother's birthday is after fifteenth but before eighteenth of February whereas his sister Kamla remembers that her brother's birthday is after sixteenth but before nineteenth of February. On which day in February is Raman's brother's birthday ?

- (a) 16th (b) 17th (c) 18th
(d) 19th (e) None of these

Solution : (b)

According to Raman, brother's birthday is on one of the days among 16th and 17th February

According to Kamla, the brother's birthday is on one of the days among 17th and 18th February. Hence Raman's brother's birthday is on the day common to both the above groups i.e., 17th February.

11. MATHEMATICAL OPERATIONS

This type questions are based on simple mathematical operations.

Here, four fundamental operations :

Addition, subtraction, multiplication and division and statements such as

'less than', 'greater than', 'equal to', 'not equal to', etc.

are represented by symbols, different from the usual ones.

The questions involving these operations are set using artificial symbols. The candidate has to substitute the real signs and solve the questions accordingly, to get the answer.

Type of Problems in Mathematical Operation**(i) Solving by Substitution**

In this type of questions, we are provided with substitutes for various mathematical symbols, followed by a question involving calculation of an expression or choosing the correct/ incorrect equation. We are required to put in the real signs in the given equation and then solve the questions as required.

While solving a mathematical expression, proceed according to the rule BODMAS

i.e., Brackets, Of, Division, Multiplication, Addition, Subtraction.

(ii) Deriving appropriate Conclusions

In such questions, the usual algebraic signs and mathematical symbols may be given different meanings or may even be replaced by new symbols. For instance ($\div$) may denote addition, ($\times$) may denote subtraction ($<$) may denote division and so on. The simplest way to solve such an equation is to rewrite the problem replacing the given symbols with the signs they are supposed to stand for. For instance, if :

$+$ means $-$	$\div$ means $+$
∇ means $<$	$-$ means $>$
$\times$ means $=$	$>$ means $\times$

&, $<$ means $\div$

Then equation, $65 < 5 \div 2.5 - 15$ will be rewritten as $65 \div 5 + 2.5 > 15$, and problem may be solved.

In yet another type of questions included under this system, even the digits (instead of the symbols)

0, 1, 2, 8, 9, may be replaced by alphabets and the candidate may then be asked to perform calculations and solve the given problem. For instance, codes may be used as ($a = 1$, $b = 2$, $c = 3$ etc.) or in reverse (i.e., $z = 1$, $y = 2$, $x = 3$ and so on).

Following question employs such a code.

Example 1 : If $y \times w = s$ and $u - v = z$, what is r/x ?

- (a) y (b) z (c) s (d) x

Solution : If we count backwards from z, we get y as the 2nd letter, w as the 4th letter, and s as the 8th letter. In this order u is the 6th, v is the 5th and z the 1st letter of the code.

So, $2 \times 4 = 8$ and $6 - 5 = 1$

Hence, $r = 9$ and $x = 3$.

So $\frac{r}{x} = \frac{9}{3} = 3$. So the answer is (d) x.

In certain cases, arithmetical operations may not be defined. Consider the following question.

Example 2 : If $8 + 12 = 2$, and $7 + 13 = 3$, then what is the value of $10 + 18$?

- (a) 9 (b) 6 (c) 4 (d) 7

Solution : $\frac{12-8}{2} = 2$, and $\frac{13-7}{2} = 3$. So, code employed here is special.

In $x + y$, '+' stands for $\frac{y-x}{2}$

$\therefore 10 + 18 = \frac{18-10}{2} = 4$, so the answer is (c).

12. LOGICAL SEQUENCE OF WORDS

In this type of questions, a group of words is given. The candidate is required to arrange these words in a meaningful order such as the sequence of occurrence of events, sequence from a part to the whole, sequence of increasing/decreasing size, value, intensity etc., and then choose the correct sequence accordingly.

Example 1 : Arrange the following in a meaningful sequence :

1. Consultation
 2. Illness
 3. Doctor
 4. Treatment
 5. Recovery
- (a) 2, 3, 1, 4, 5 (b) 2, 3, 4, 1, 5
(c) 4, 3, 1, 2, 5 (d) 5, 1, 4, 3, 2
(e) 2, 3, 4, 5, 1

Solution : (a)

Illness occurs first. One then goes to the doctor and after consultation with him, undergoes treatment to finally attain recovery. Thus, the correct order is 2, 3, 1, 4, 5.

Example 2 : Arrange the following in a logical order :

1. Euphoria
 2. Happiness
 3. Ambivalence
 4. Ecstasy
 5. Pleasure
- (a) 4, 1, 3, 2, 5 (b) 3, 2, 5, 1, 4
(c) 2, 1, 3, 4, 5 (d) 1, 4, 2, 5, 3
(e) 5, 4, 3, 2, 1

Solution : (b)

All the given words stand for 'Joy', but the intensity increases in the order — Ambivalence, Happiness, Pleasure, Euphoria, Ecstasy. Thus, the correct order is 3, 2, 5, 1, 4.

13. DECISION MAKING

It with questions in which we have to decide the course of action to be taken upon a candidate who has applied for a vacancy or allotment or membership to an institution, keeping in mind the essential requisites and the data given for the candidate.

Type I Decision Test

The decision about each candidate has to be made from amongst the five choices named (a), (b), (c), (d), (e) which state the courses of action to be taken as per the candidate's potentials.

Direction (1 to 5) : Read the following information to answer the given questions

Following are the criteria for selecting & marketing officer by a company. The candidate must :

- (1) be a graduate with at least 50% marks.
- (2) have secured at least 40% marks in the written test.
- (3) not be less than 24 years and more than 29 years as on 10th October, 2007.
- (4) should have work experience of at least two years as an officer. However, if a candidate:
- (5) fulfils all other criteria except at (4) above but has a diploma in Marketing Management, his/her case is to be referred to General Manager, Marketing.
- (6) fulfils all other criteria except at (3) above but has worked as Marketing Officer at least for three years, his/her case is to be referred to Director, Marketing.

Based on the above criteria and information given in each of the following questions, we have to take the decision in regard to each case. We are

not to assume anything. These cases are given as on 10th October, 2007.

Mark answer (a) if candidate is to be appointed;

Mark answer (b) if the candidate is not to be appointed;

Mark answer (c) if the data given are not sufficient to take decision;

Mark answer (d) if to be referred to General Manager - Marketing; and

Mark answer (e) if to be referred to Director – Marketing.

Example 1 : Amit, born on 5th June, 1983, has done his post-graduation in Marketing Management with first class. He has secured 50% marks in the written test. He has been working in an organisation as a Marketing Officer for the last four years.

Example 2 : Rohit has been working in an organisation as Officer for the last ten years. His date of birth is 17th February, 1964. He has secured 60% marks in the degree examination and 40% marks in the written test.

Example 3 : Manju is a first class graduate and has done a diploma in Marketing Management. She has secured 50% marks in the written test. She was 23 years old as on 5th September, 2006.

Example 4 : Nitin was born on 25th August, 1985. He has secured 60% and 50% marks in graduation and in the written test, respectively. He has been working in an organisation as Officer for the last four years.

Example 5 : Suman is a graduate with first class and has secured 60% marks in the written test. She has been working as an Officer for the last three years. She was born on 20th May, 1982.

Solutions

1. (a). All conditions of eligibility are satisfied.
2. (e). Condition (3) is violated but it can be waived by condition (6).
3. (d). The candidate satisfies all conditions except (4). But he fulfils condition (5) so that (4) is waived.
4. (e). The candidate satisfies condition (2) instead of condition (3).
5. (a). All conditions of eligibility are satisfied.

Type II Assertion and Reason

This test is meant to judge the candidate's knowledge and his ability to reason out correctly. In this test, two statements referred to as the Assertion (A) and Reason (R) respectively are

provided. Five alternative comments on these are given and the correct one is to be chosen.

Direction (1 and 2) : For the Assertions (A) and Reasons (R) below, choose the correct alternative from the following:

- (a) Both A and R are true and R is the correct explanation of A.
- (b) Both A and R are true but R is NOT the correct explanation of A.
- (c) A is true but R is false.
- (d) A is false but R is true.
- (e) Both A and R are false.

Example 1 : Assertion (A) : Moon cannot be used as a satellite for communication.

Reason (R) : Moon does not move in the equatorial plane of the earth.

Solution (a) :

Since R explains A.

Example 2 : Assertion (A) : Salt is added to cook food at higher altitudes.

Reason (R) : Temperature is lower at higher altitudes.

Solution (b) :

Since both statements A and R are separately true but R does not explain A.

Type III Situation Reaction Test

In this test, certain situation is described and the candidate is required to choose the most suitable reaction to the given situation from amongst the alternatives provided. This test judges reasoning power of the candidate and his ability to act correctly and promptly to a situation that may arise in emergency.

Direction (1 to 3) : While travelling in your car, certain persons stop you on the way asking you to take an injured child to the hospital. You would :

- (a) ask them to leave your way and then drive away.
- (b) ask them to first call the police.
- (c) immediately take the child to hospital.
- (d) get out of the car and ask some other person to help them.
- (e) None of these

Solution (c)

Situation described above demands that the person should immediately render the help asked for and take the child to the hospital.

Example 1 : While playing cricket in the school, suddenly when you hit the ball, it strikes your classmate on the forehead and blood starts oozing out. You would :

- (a) run away from the field.
- (b) start fighting with the boy why he came in the way.
- (c) blame somebody else for the accident.
- (d) take the boy to the first aid room.
- (e) None of these

Solution (d)

In the above situation, urgent need is to provide first aid to the boy so that the bleeding may stop.

Example 2 : You are visiting a place for the first time and are travelling in a bus. Suddenly you realise that the driver is taking the bus to a lonely place with no right intentions. You would :

- (a) with the help of some other passengers, try to baffle the driver and take over the bus.
- (b) sit and wait to face the repercussions.
- (c) jump out of the running bus.
- (d) console the worried passengers.

Solution. (a)

When a wrong doing is expected, immediate action to prevent it is the need.

Example 3 : You have gone to enjoy a Diwali Mela organised by a club. Suddenly you come across a lost child crying desperately.

You would :

- (a) neglect and walk away.
- (b) ask the child to find his parents.
- (c) ask him to stop crying and wait patiently for his parents.
- (d) contact with the club authorities and make an announcement for the parents.

Solution (d)

Immediate need is to find the child's parents and for this, best way is to announce the child's name and appearance so that his parents might know where the child is.

14. CUBES

A cube is a solid which has 6 faces, 8 corners and 12 edges.

Following are some of the important types based on which questions have been asked in various aptitude tests.

- (i) Number of cuts given to a cube is given, find the maximum number of identical pieces that can be produced.
- (ii) Number of pieces a cube is cut into is given, find the least number of cuts required to produce these many pieces.
- (iii) Miscellaneous types.

It must be remembered that in order to obtain the maximum number of pieces, the cuts given to a large cube must be divided as equally as possible in three different directions.

Let us discuss each of these types with the help of examples.

- (i) Number of cuts = given;

Maximum number of pieces = ?

- (a) Whenever the number of cuts made to a cube is a multiple of 3.

If 'n' is the number of pieces along each edge, and

$$n = \left(\frac{\text{Total cuts}}{3} \right) + 1, \text{ then}$$

Maximum number of identical pieces = n^3

Number of pieces with 3 faces visible = 8

Number of pieces with 2 faces visible
= $12(n - 2)$

Number of pieces with 1 face visible
= $6(n - 2)^2$

Number of pieces with no face visible
= $(n - 2)^3$

Let us take an example where 30 cuts are made to a large cube to get the maximum number of pieces.

Here, the value of 'n' will be $\frac{30}{3} + 1 = 11$

Then,

- Maximum number of pieces
= $n^3 = 11^3 = 1331$
- Number of 3 – face visible pieces = 8.
- Number of 2 – faces visible pieces
= $12(n - 2) = 12(11 - 2)$
= 108.
- Number of 1 – face visible pieces
= $6(n - 2)^2 = 6(11 - 2)^2 = 486$.
- Number of no face visible pieces
= $(n - 2)^3$
= $(11 - 2)^3 = 729$.

- (b) Whenever the number of cuts made to the large cube is NOT a multiple of 3.

In such a case, there is no direct formula for finding out the 2 - faces visible, 1 - face visible or no face visible pieces, but the maximum number of pieces produced can be found out. Let us consider that a large cube is given 4 cuts. Now 4 cuts are divided into three directions as 1 cut, 1 cut, 2 cuts. We know that the number of pieces along each edge is one more than the number of cuts along that edge (when all cuts are made in the same direction - parallel to the same pair of faces). We can represent the above as shown below:

$$\begin{array}{r} 1 \quad 1 \quad 2 \\ +1 \quad +1 \quad +1 \\ \hline 2 \times 2 \times 3 = 12 \text{ pieces.} \end{array}$$

Similarly, 11 cuts will give us maximum 100 pieces, as shown below:

$$\begin{array}{r} 3 \quad 4 \quad 4 \\ +1 \quad +1 \quad +1 \\ \hline 4 \times 5 \times 5 = 100 \text{ pieces.} \end{array}$$

- (ii) Number of pieces = given,
Least number of cuts = ?

We use the reverse process. To get the least number of cuts to produce 100 pieces, factorise 100 into 3 factors with least difference between them, and then subtract one from each of them, as shown below:

$$\begin{array}{r} 100 = 4 \times 5 \times 5 \\ \quad -1 \quad -1 \quad -1 \\ \hline 3 + 4 + 4 = 11 \text{ cuts} \end{array}$$

- (iii) Miscellaneous types:

Apart from the above two types, a cube can be coloured with one, two or more colours, and then is given some number of cuts.

Let us study and practice each of the above types with the questions given in the exercise.

15. ARRANGEMENTS

Linear Sequencing

Linear sequencing is essentially arranging the items in a sequence (in a single line.) The questions of this type are also referred to as "Seating Arrangement". The word "Seating Arrangement" should not be misconstrued- it should not be treated as questions involving only persons sitting a specified conditions. Essentially, these questions involve arranging subjects (people or things) according to the given conditions. The arrangement is done only on one "axis" and hence, the positions of the subjects assume importance in terms of order like first position second position, etc.

Directions (1 to 5) : Read the data given below carefully and answer the questions.

Seven persons Paul, Queen, Rax, Sam, Tom, Unif and Vali are sitting in a row facing us. Rax and Sam sit next to each other. There must be exactly four persons between Queen and Vali. Sam sits to the immediate right of Queen.

Example 1 : If Paul and Tom are separated exactly by two persons, then who sits to the immediate left of Vali?

- (a) Paul (b) Tom
(c) Unif (d) Rax

Example 2 : If Queen is not sitting at either extremes of the row, then who among the following has as many persons on his left as on his right ?

- (a) Sam (b) Unif
(c) Rax (d) Vali

Example 3 : If Queen sits at one extreme, then who is at the other extreme?

- (a) Paul (b) Tom
(c) Vali (d) Cannot determined

Example 4 : Tom sits to the right of Queen, and Paul is separated from Tom by exactly three persons. Then, who is sitting to the immediate left of Vali?

- (a) Sam (b) Unif
(c) Tom (d) Rex

Example 5 : In how many different ways can the seven persons sit in a row?

- (a) 3 (b) 2
(c) 10 (d) 12

Solution:

Let us write down the conditions given in short form and then represent them pictorially Also, let us treat the left of the persons sitting as "left" and their right as "right" for interpreting the conditions

Rax and Sam sit next to each other

→ RS or SR.

There are exactly 4 persons between Queen and Vali

Q ____ V or V ____ Q

Sam sits to the immediate right of Queen → SQ.

Now let us analysis the data /conditions that we are given and then put the three conditions together. Let us number the steps from our left to right as Seat 1 to Seat 7.

$$\rightarrow \mathbb{R} \qquad \mathbb{L} \rightarrow$$

1234567v

V R S Q

1234567

V R S Q

1.Solution (c)

T/P, V,U,P/T, R,S,Q

So, Unif must be sitting to the immediate left of Vali.

2.Solution (c)

If Queen is not at the extreme right, then only arrangement II mentioned above is possible . The person who has as many persons on his left as on his right can only be the person who is sitting in the middle seat i.e. seat 4. In this arrangement Rex is sitting in seat 4.

3. Solution (d)

“Queen sits at one extreme” means that we should look at arrangement I. In this arrangement I. In this arrangement, any one out of the three persons Paul, Tom and Unif can be in seat 1, i.e. extreme right

4. Solution (b)

If Tom and Paul are separated by exactly three persons, then only arrangement II is possible. So Tom and Paul have to be in seats 3 and 7. Since it is also given that Tom is to the right of Queen, Tom has to be in seat 3 and Paul, in seat 7. So the arrangement must be follows:

V, U, T, R, S, Q, P

5. Solution (d)

We have two possible arrangements – “Arrangement I and Arrangement II” that we have already looked at. In each arrangement, the remaining three people can sit in the remaining three seats in 6 ways. Thus, a total of 12 ways of seating the seven persons is possible.

Directions (6 to 10):

Read the data given below carefully and answer the questions that follow.

Seven boys, Rajan, Shyam, Vardhan , Mithra, Vimal, Raj and Kishan – are sitting in a row Shyam sits to the immediate left of Vardhan and third to the right of Rajan, whereas Mithra, who sits at the extreme left is next to Kishan.

Example 6 : Who is sitting to the immediate right of Shyam?

- (a) Mithra (b) Kishan
(c) Vamal (d) Vardhan

Example 7 : If Vardhan and Kishan exchange place with each other without changing the rest of the arrangement that is already done, then who will sit to the immediate left of Rajan ?

- (a) Kishan (b) Raj
(c) Vimal (d) Vardhan

Example 8 : If only Shyam sits between Raj and Vardhan , who is exactly in the middle of the row?

- (a) Raj (b) Vardhan
(c) Vimal (d) Rajan

Example 9 : Which of the following cannot confirm the seating arrangement of all the boys?

- Raj is to the immediate right of Rajan , wheres Vimal is to the left of Shyam.
- Mithra and Raj have two persons between them.
- Raj and Kishan have two persons between them.
- Rajan and Shyam have two person in between them.

Example 10 : After arranging all the boys as per the conditions given in the data, if Rajan now exchange his place with Mithra, and Vardhan exchange his place with Vimal then how many persons will be there between Vimal and Rajan ?

- (a) Three (b) Two
(c) Five (d) One

Solution:

Let us denote Left and Right as shown below:

$$\rightarrow L \qquad R \rightarrow$$

Now let us represent the data given in pictorial form (We use R for Raj and Rn for Rajan; Va for Vardhan; Vi for vimal; S for Shyam; M for Mithra and K for Kishan.)

Mithra sits at the extreme left –next to Kishan
→MK _ _ _ _

Shyam sits to the immediate left of Vardhan and
third to the right of Rajan → Rn _ _ S Va

Putting both the above condition together Va can
go only to the extreme right position. Thus, We have
the arrangement as M K Rn _ _ S Va.

Raj and Vimal occupy the two vacant seats between
Rajan and Shyam.

6. Solution (d)

From the seating arrangement figure above
Vardhan is to the immediate right of Shyam

7. Solution (d)

If Kishan and Vardhan exchange places as can be
seen from the arrangement, the person to the
immediate left of Rajan will be Vardhan

8. Solution (c)

If Shyam sits between Raj and Vardhan then the
seating arrangement is as follows. Mithra, Kishan,
Rajan, Vimal, Raj, Shyam, Vardhan. Then, Vimal
will be exactly in the middle of the row.

9. Solution (d)

Statement (a) gives the arrangement as Mithra,
Kishan, Rajan, Raj, Vimal, Shyam, Vardhan

Statement (b) gives the seating arrangement as
Mithra, Kishan, Rajan, Raj, Vimal Shyam, Vardhan

Statement (c) gives the seating arrangement as:
Mithra, Kishan, Rajan, Vimal, Raj, Shyam, Vardhan
So only statement (d) cannot make the seating
arrangement unique while others can.

10. Solution (c)

The arrangement is M K Rn R/Vi Vi/R S Va

Rajan exchange his place with Mithra and Vimal
with Vardhan then we have the following
arrangement Rn K M R/Va Va/R S Vi

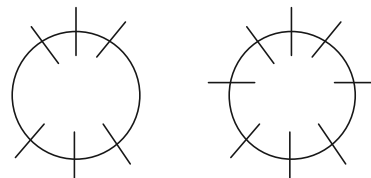
While we still do not know the exact position of
Vardhan (or which place Vimal sits), we can see that
there are five persons between Rajan and Vimal

Circular Arrangement

Questions on circular arrangement involve seating
people around a table or arrangement of things in
a circular manner (for example, different colors
beads string to form a necklace). In case of people
sitting around a table, the table could be of any
shape – rectangular, square, circular or any other.
The data given in such sets of questions specify the
positions of some or all of the individuals (or things)
in the arrangement. The positions are specified
through conditions involving specified persons
sitting (or not sitting) opposite each other or a
particular person sitting to the right or left of
another person etc.

Once you read the data, first draw the shape
specified in the data and then draw the slots in
the seating arrangement.

Six people around a circular table Eight people
around a circular table



Statement like “A and B are sitting farthest from
each other” or “A and B sit across the table” imply
that A and B sit opposite each other.

On the other hand, you should remember that
unlike in straight –line arrangement, the words
“immediately” and “directly” do not play any role
in circular arrangement. In general, there is no
left side or right side (unless we are talking of
‘immediate right’ or ‘immediate left’).

So, if it is given that C sits to right of B, then it is
clear that C must be to the immediate right of B. Go
“anti- clockwise” if anybody’s right has to be located
and go “clockwise” if somebody left has to be located.

16. CLOCKS

The hour hand and the minute hand of a clock
move in relation to each other continuously and
at any given point of time, they make an angle
between 0° and 180° with each other.

If the time shown by the clock is known, the angle
between the hands can be calculated. Similarly,
if the angle between two hands is known, the
time shown by the clock can be found out.

When we say angle between the hands, we
normally refer to the acute/obtuse angles (upto
 180°) between the two hands and not the reflex
angles ($> 180^\circ$).

For solving the problems on clocks, the following
points will be helpful.

- (1) Minute hand covers 360° in 1 hour, ie. in 60°
minutes. Hence **Minute Hand Covers 6°
Per Minute.**
- (2) Hour hand covers 360° in 12 hours. Hence
hour hand covers 30° per hour. Hence, **Hour
Hand Covers $1/2^\circ$ Per Minute.** The
following additional points also should be
remembered, in a period of 12 hours, the
hands make an angle of
 - (a) 0° with each other (ie. they coincide with
each other) 11 times.
 - (b) 180° with each other (i.e. they point
exactly in opposite directions) 11 times.
 - (c) 90° or any other angle with each other
22 times.

NOTE

- (1) We can also solve the problems on clocks using the method of "Relative Velocity".
- (2) In 1 minute Hand covers 6° and Hour hand covers $1/2^\circ$
- (3) Therefore, Relative Velocity = $6 - 1/2 = 5\frac{1}{2}^\circ$ per minute. Alternately, in 1 hour hand covers 5 minute divisions.
 $\therefore$ Relative Speed = $60 - 5 = 55$ minutes per hor. However, adopting the approach of actual angles covered is far simplest and does not create any confusion.
- (4) Any angle other than (0° and 180°) is made 22 times in a period of 12 hours.
- (5) In a period of 12 hours there are 11 coincidences of the two hands, when the two hands are in a straight line facing opposite directions.
- (6) The time gap between any two coincidences is $12/11$ hours or $65\frac{5}{11}$ minutes
- (7) If the hands of a clocks (which do not show the correct time) coincide every 'p' minutes. Then, If $65\frac{5}{11}$ then the watch is going slow or losing time.
 If then the watch is going fast or gaining time.
- (8) To calculate the angle ' θ ' between the hands of a clocks, we use the following formula (where m = minutes and h = hours)

$$(i) \theta = \frac{11}{2}m - 30h \left(\text{when } \frac{11}{2}m > 30h \right)$$

$$(ii) \theta = 30h - \frac{11}{2}m \left(\text{when } 30h > \frac{11}{2}m \right)$$

Example 1 : What is the angle between the minute hand and the hour hand of a clock at 3 hour 40 minutes?

- (a) 20° (b) 70°
 (c) 90° (d) 130°

Solution (d)

The angle between the hands can be calculated by

$$\theta = \left| \frac{11}{2}m - 30h \right| \text{ where m is minutes and h is hours.}$$

Here, m = 40 and h = 3

$$\therefore \theta = \left| \frac{11}{2} \times 40 - 30 \times 3 \right| = |220 - 90| = 130^\circ$$

The angle between the two hands is 130°

Example 2 : Find the time between 2 and 3 O'clock at which the minute hand and the hour hand

- (a) Make an angle of 60° with each other
- (b) Overlap.
- (c) are perpendicular to each other.

- (d) are on the same straight line but are facing opposite directions.

Solution

- (1) In the formula

$$\theta = \left| \frac{11}{2}m - 30h \right|$$

$$\theta = 60^\circ \text{ and } h = 2$$

$$\therefore 60 = \frac{11}{2}m - 30 \times 2$$

$$\frac{11}{2}m = 120$$

$$m = \frac{240}{11} = 21\frac{9}{11} \text{ min past 2}$$

$$(or) 60 = 30 \times 2 - \frac{11}{2}m$$

$$\therefore \frac{11}{2}m = 0$$

$$m = 0$$

Therefore, the angle between the hour hand and the hour hand and the minute hand is at

2 O'clock and at $21\frac{9}{11}$ minutes past 2 O'clock.

- (2) When the two hands overlap, the angle between them is

$$\theta = \left| \frac{11}{2}m - 30h \right|$$

$$\therefore \theta = 0^\circ \text{ and } h = 2$$

$$0 = \frac{11}{2}m - 30 \times 2$$

$$m = \frac{120}{11} = 10\frac{10}{11} \text{ min past 2.}$$

- (3) When two hands are perpendicular $\theta = 90^\circ$ and h = 2

$$\therefore \theta = \left(\frac{11}{2}m - 30h \right) \text{ or } \left(30h - \frac{11}{2}m \right)$$

$$90 = \frac{11}{2}m - 30 \times 2$$

$$\frac{11}{2}m = 150$$

$$m = \frac{300}{11} = 27\frac{3}{11} \text{ minutes past 2}$$

$$(Or) 90 = 30 \times 2 - \frac{11}{2}m$$

$$\frac{11}{2}m = 30$$

As m cannot be negative, this case is not possible. So, the hands are perpendicular to each other only once i.e. at $27\frac{3}{11}$ minutes past 2 O'clock.

- (4) When two hands are pointing opposite directions and are on a straight line the angle between them would be 180° ie. $\theta = 180^\circ$ and $h = 2$

$$180^\circ = \frac{11}{2}m - 3h$$

$$\frac{11}{2}m = 180 + 60 = 240$$

$$m = \frac{480}{11} = 43\frac{7}{11}$$

So, at $43\frac{7}{11}$ minutes past 2 O'clock the hands will be at 180° .

17. CALENDARS

Suppose you are asked to find the day of the week on 30th June 1974 it would be a tough job to find because you do not know the method. The method of finding the day of the week lies in the number of "odd day"

NOTE

Every 7th day will be the same day count wise, i.e. if today is Monday then 7th day counting from Tuesday onwards will once again be Monday. Odd days are the days remaining after completion of an exact number of weeks. An Odd day is the remainder obtained on dividing the total number of days with seven.

Example: $52 \text{ days} \div 7 = 3 \text{ odd days}$

Leap and Non-Leap Year

A non-leap year has 365 days whereas a leap year has one extra day because of 29 days in the month of February. Every year which is divisible by 4 is called leap year. Leap year consists of 366 days, (52 complete weeks + 2 days). The extra two days are the odd days. So, a leap year has two odd days.

A non-leap year consists of 365 days (52 complete weeks + 1 day). The extra one day is the odd day.

NOTE

Every century, year which is a multiple of 400, is a leap year. A century year which is not divisible by 400 is a non-leap year

Example:

400, 800, 1200, 1600... are leap years

500, 700, 900, 1900.. are non-leap years

Counting the Number of Odd Days

100 years consist of 24 leap years + 76 ordinary years. (100 years when divided by 4 we get 25. But at the 100th year is not a leap year, hence only 24 leap year).

$$\begin{aligned} &= 2 \times 24 \text{ odd days} + 1 \times 76 \text{ odd days} \\ &= 124 \text{ days} \\ &= 17 \text{ weeks} + 5 \text{ days} \end{aligned}$$

The extra 5 days are the odd days.

So, 100 years contains 5 odd days.

Similarly, for 200 years we have 10 extra days (1 week + 3 day)

$\therefore$ 200 years contains 3 odd days.

Similarly 300 years contains 1 odd day and 400 years contains 0 odd days.

Counting of Number of Odd Days, When only one Date is Given

Here we take January 1st AD as the earlier date and we assume that this day is a Monday. We take its previous day i.e. Sunday as the reference day. After this the above mentioned method is applied to count the number of odd days and find the day of the week for the week for the given date.

Counting Number of Odd Days, When Two Dates Are Given

Any month which has 31 days has 3 odd days ($\because 31 \div 7$ leaves 3 as remainder) and any month which has 30 days has 2 odd days ($30 \div 7$ leaves 2 as remainder).

Then, the total numbers of odd days are calculated by adding the odd days for each month. The value so obtained is again divided by 7 to get the final number of odd days. The day of the week of the second date is obtained by adding the odd days the day of the week of the earlier date.

Example 1 : If you were born on 14th April 1992 which was a Sunday then on which day of the week does your birthday fall in 1993?

- (a) Monday (b) Tuesday
(c) Wednesday (d) cannot be determined

Solution (a)

14th April 1992 to 14th April 1993 is a complete year, which has 365 days.

Hence the number of odd days from 14th April 1992 to 14th April 1993 is 1.

Hence 14th April 1993 is one day after Sunday i.e. Monday

Example 2 : If 1st Jan 1992 is a Tuesday then on which day of the week will 1st Jan 1993 fall?

- (a) Wednesday (b) Thursday
(c) Friday (d) Saturday

Solution (b)

Since 1992 is a leap year there are 2 odd days. Hence 1st January 1992 is two days after Tuesday i.e. Thursday.

Example 3 : If 1st April 2003 was Monday, then which day of the week will 25th December of the same year be?

- (a) Tuesday (b) Wednesday
(c) Thursday (d) Friday

Solution (b)

The number of days from 1st April to 25th December
(29 + 31 + 30 + 31 + 31 + 30 + 31 + 30 + 25) days = 268 days.

$$= \frac{268}{7} = 38 + 2 \text{ odd days.}$$

Hence 25th December is two days after Monday ie. Wednesday

Example 4 : On which day of the week does 4th June 2001 fall?

- (a) Monday (b) Tuesday
(c) Wednesday (d) Thursday

Solution (a)

4th June 2001 $\Rightarrow$ (2000) year + 1st January to 4th June 2001

We know that 2000 years have zero odd days. The number of odd days from 1st January to 4th June 2001

Month: Jan + Feb + Mar + Apr + May + June

Odd day: 3 + 0 + 3 + 2 + 3 + 4

$$\frac{15}{7} = 1 \text{ odd day.}$$

Hence 4th June 2001 was a Monday.

Example 5 : Which year will have the same calendar as that of 2005?

- (a) 2006 (b) 2007
(c) 2008 (d) 2011

Solution (d)

Year: 2005 + 2006 + 2007 + 2008 + 2009 + 2010

Odd days: 1 + 1 + 1 + 2 + 1 + 2 + 1

Total numbers of odd days from 2005 to 2010 are 7 = 0 odd days.

Hence 2011 will have the same calendar as that of 2005

Example 6 : What day of the week was 18th April 1901?

- (a) Monday (b) Tuesday
(c) Wednesday (d) Thursday

Solution (d)

18th April 1901 $\Rightarrow$ (1600 + 300) years + 1st January to 18th April 1901

1600 years have = 0 odd days

300 years have – 1 odd day. The number of days from 1st January 1901 to 108 days $\cong$ 3 odd days

$\therefore$ Total number of odd days = 3 + 1 = 4

Hence 18th April 1901 is Thursday

18. DEDUCTIONS

In number of competitive exams there are a few questions on “deductions”. Typically here, each question consists of two statements—on the basis of which a deduction has to be made. The answer has to be chosen from the given four (or five) choices and that will be the deduction made.

(If no conclusion can be drawn or the answer is not obtained then the choice which is normally worded as “none of the above” has to be marked).

These questions can be answered by representing the given statements by Venn Diagrams. However, here we will look at arriving at the deduction by using some simple rules.

First let us look at some basic terms used in the rules and understand what they mean. The two statements given in the question are called ‘premises’ & the answers are conclusion.

Eg. All dogs are cats—(i)

All cats are pigs —(ii)

The premises normally start with the words All, No Some and Some –not

The word “All” has its synonyms as –Every, Any, Each, whereas the word “Some “ can also be replaced by Many, Few, A little, Most of, Much of, More etc.

These words are referred to as qualifiers (also termed as quantifiers).

A premise consists of a subject and a predicate wherein the first term [e.g. “dogs” in statement (i)] is the subject and the second term [e.g. “cat” in statement (i)] the predicate. Similarly, in statement (ii), “cats” is called the subject and “pigs” is the predicate.

The word that occurs in both the premises is known as the ‘middle term’ (“cat” in the example, give above). The answer or “conclusion” should consist of the other two words (“dog and “pig” in the above example) and the middle term should not appear in the answer.

The premises can be divided into

- (a) Universal statements and
(b) Particular statements.

This classification of the premises into the above categories is dependent on the qualifier used in the premise. For example, the statements where “All” is used are called Universal statements and the statements where “Some” is used are called Particular statements. Premises can also be divided into

- (i) Positive (affirmative) statements and
(ii) Negative statements)

If there is a negative term like “not” or “no” in the statement it is called a negative premise. Otherwise it is called a positive premise or an affirmative statement. The combination of the two different categories of classification leads to four different premises as given in Table I below.

Table I

	Affirmative	Negative
Universal	All A	No E
Particular	Some many I	Some not many O

The subject and the predicate are either distributed ($\sqrt{}$) Or not distributed ($\times$) depending on what kind of a statement it is (particular affirmative etc.)

Table II shows the distribution Pattern of the subject and the predicate.

Table II

	Subject	Predicate
Universal affirmative	$\sqrt{}$	$\times$
Universal negative	$\sqrt{}$	$\sqrt{}$
Particular affirmative	$\times$	$\times$
Particular negative	$\times$	$\sqrt{}$

Note: ($\sqrt{}$) indicate distributed ($\times$) indicates undistributed.

Rules for Deductions

- (1) Every deduction should contain three and only three distinct terms.
- (2) The middle term must be distributed at least once in the premises.
- (3) If one premise is negative, then the conclusion must be negative.
- (4) If one premise is particular then the conclusion must be particular
- (5) If both the premises are negative, no conclusion can be drawn
- (6) If both the premises are particular, no conclusion can be drawn
- (7) No term can be distributed in the conclusion, if it is not distributed in the premises.

We take example of each type and look at them in details.

Example 1 :

All dogs are cats — (i)

All cats are pig — (ii)

As the first statement is a Universal affirmative statement. The subject (dog) has to be distributed ($\sqrt{}$) and the predicate (cats) is not distributed ($\times$). As the

second statement is also Universal affirmative, the subject cat is distributed ($\sqrt{}$) and predicate pigs is not distributed ($\times$). The above answer/ logic is arrived at on the basis of Table II.

The middle term (“cat is the middle term as it occurs in both the premises) is distributed once in the premises. Hence it satisfies Rule[2] As “dogs” is distributed in the premise and “pigs” is undistributed in the deduction also, they should appear accordingly. The type of statement that satisfies both of them is Universal affirmative statement i.e. a statement with “All” Hence the answer will be

“All dogs are pigs”

The answer cannot be “All pigs are dogs”, because Rule [7] states that no term can be distributed in the conclusion if it is not distributed in the premises. As “pigs” is not distributed in the premise. It cannot be distributed in the conclusion (because if we take “All pigs are dogs”, then the subject “pigs” will be distributed). Hence the conclusion “All pigs will dog” is wrong.

Example 2 :

All cats are dogs — (i)

All cats are pigs - - (ii)

Statement (i) is Universal affirmative and hence the subject “cats” is distributed and the predicate “dogs” is not distributed as per Table II.

Statement (ii) is also Universal affirmative and hence the subject “cats” is distributed and the predicate “pigs” is not distributed as per Table II.

Here, the middle term “cats” (“cats” is the middle term as it is occurring in both the premises) is distributed; hence we can draw a conclusion.

The answer should contain the terms “dogs” and “pigs” and both the terms are not distributed Referring to Table II ,we find that this is possible only in Particular affirmative [the conclusion cannot start with qualifier ‘All’ as the subject in “All” should be distributed]. According to Rule 7 a term cannot be distributed in the conclusion if it is not distributed in the premises. So the answer will be

“Some dogs are pigs” Or “Some pigs are dogs”

Example 3 :

All dogs are cats — (i)

All pig are cats — (ii)

Statements (i) is a Universal affirmative and hence the subject “dogs” is distributed and the predicate “cats” is not distributed. In statement (ii) which is also a Universal affirmative the subject “Pigs” is distributed

and the predicate “cats” is not distributed this is arrived at one the basis of Table II.

The middle term “cats” [“cats” is the middle term as it occurs in both the statements] is not distributed in either of the two statements From Rule [2] which states that the middle term should be distributed at least once in the premises for drawing a conclusion we cannot draw any conclusion in this case.

Example 4 :

All cats are dogs. — (i)

Some cats are pigs — (ii)

The first statement is a Universal affirmative premise and hence the subject “cats” is distributed and the predicate “dogs” is not distributed(x). The second statement is Particular affirmative and hence both the subject “cats” and the predicate “pigs” are not distributed (x) as per Table II. As we have a particular premise, the conclusion. Should also be a particular one as per Rule [4]. The middle term is distributed hence we can draw a conclusion .So the answer will be

“Some dogs are pigs” Or “Some pigs are dogs”

Example 5 :

All dogs are cats — (i)

No cats are pigs — (ii)

As the first premise is a Universal affirmative, the subject (dogs) is distributed and the predicate (cats) is not distributed. In the second premise which is a Universal negative, the first term (cats) and the second term (pigs) are both distributed (as per Table II) As the middle term is distributed at least once in the premises. Rule [2] is satisfied and hence we can draw a conclusion.

From Rule 3] which states that if one of the premises is negative the conclusion should be negative, the conclusion should be negative and as both the terms “dogs” and “pigs” are distributed, the conclusion should be a Universal negative statement. Hence the answer will be

“No dogs are pigs” Or “No pigs are dogs”

Example 6 :

All dogs are cats — (i)

Some cats are not pigs — (ii)

Since the first statement is a Universal affirmative “dogs” is distributed and “cats” is not distributed. Since the second statement is a Particular negative, “cats” is not distributed and “pigs” is distributed (as per Table II)

In the above given example no conclusion can be drawn , as rule [2] states that the middle term(“cats” in the above example as it occurs in both the premises) should be distributed at least once in the premises. This is not satisfied.

Example 7 :

All cats are dogs — (i)

Some cats are not pigs — (ii)

The first statement is a universal affirmative and hence “cats” is distributed and “dogs” is not distributed The second statement is a Particular negative and hence “cats” is not distributed and “pigs” is distributed (as per Table II)

Here, the middle term (cats) is distributed and hence we can draw a conclusion.

The conclusion should be particular negative as Rule [3] states that if a premise is negative. Also Rule[4] state that if a premise is Particular, the conclusion should also be Particular. Hence the conclusion should be a particular negative.

In particular negative, we know that the subject is not distributed and the predicate is distributed.

The terms “dogs” and “pigs” should come in the conclusion. Also since “dogs” is not distributed in the premise, it cannot be distributed in the conclusion, as per Rule [7]

As per the above reasoning only “pigs” can be the predicate in the conclusion and hence “dogs” will be the subject

Thus the answer will be “Some dogs are not pigs”

Example 8 :

No dogs are cats — (i)

No cats are pigs — (ii)

We cannot draw any conclusion, as Rule [5] states that if both premises are negative, we cannot draw any conclusion.

Example 9 :

No dogs are cats

Some cats are not pigs.

As both the premises are negative hence as per Rule [5] we cannot draw any conclusion (Please note that the first premise is a Universal negative and hence the subject (dogs) is distributed and the predicate (cats) is also distributed as per Table II.

The second statement is a Particular negative and hence the subject (cats) is not distributed and the predicate (pigs) is distributed a per Table II)

Example 10 :

Some cats are not pigs — (i)

Some cats are dogs — (ii)

As the first premise is a Particular negative the subject (cats) is not distributed and the predicate (pigs) is distributed. In the second premise both the subject and the predicate (cats and dogs respectively) are not distributed since the premise is a particular affirmative (as per Table II).

No conclusion can be drawn as both the premises are particular as per Rule [6]

Example 11 :

Some cats are not dogs — (i)

Some cats are not pigs — (ii)

We cannot get an answer from the two premises as Rule [5] states that from two negative premises, no conclusion can be drawn. Also Rule [6] states that from two particular premises no conclusion can be drawn.

19. DATA SUFFICIENCY

Question on Data Sufficiency may consist of two statements or three statements. The questions asked on Data Sufficiency are either based on Mathematical or Reasoning. A firm grasp over the basis of all topics being covered under Quantitative Ability and Reasoning is the prerequisite for doing well in Data Sufficiency.

Data Sufficiency, as the name suggests, tests the ability of the student to identify whether the data given are sufficient to answer a question or not.

Hence, in the question given on Data Sufficiency, the student must stop at the stage of determining the sufficiency of the data and he/she is not expected to solve the problem completely. For that reason most of the question given on data sufficiency will be such that the student is not required to do any calculation but can answer the question by observation. However, in some cases, it is difficult to say whether the data are sufficient or not unless complete calculations are done.

In Data Sufficiency questions, typically, there will be a question followed by some data. The data are given in the form of two statements (statement 1 and 2 or A and B) or three statements. (Statements 1, 2 and 3 or statements A, B and C).

The student has to decide the answer choice for the question depending on whether the data given in the statements are sufficient to answer the

questions or not. Direction is provided to the students as to how to arrive at the answer choice. While, as mentioned above, though there can be three statements, giving data, it is question with two statements that are very common.

Before we understand the solving part of questions on data sufficiency, let us look at the direction given on the Data Sufficiency.

Each question is followed by two statements giving certain data. You have to decide whether the information provided in the statements is sufficient for answering the question.

Choose A:

If the question can be answered by using one of the statements alone, but cannot not be answered by using the other statement alone.

Choose B:

if the question can be answered by using either statement alone.

Choose C:

if the question can be answered by using both statements together, but cannot be answered by using either statement alone.

Choose D:

If the question cannot be answered even by using both the statements together.

For solving any Data Sufficiency with two statements, one has to follow four steps in sequential order.

The steps involves are as follows.

Step1 :

Look at the sufficiency of the statement 1 alone.

Step2 :

Look at the sufficiency of statement 2 alone.

Step3 :

If anyone or either of the statements sufficient, then stop.

Step4 :

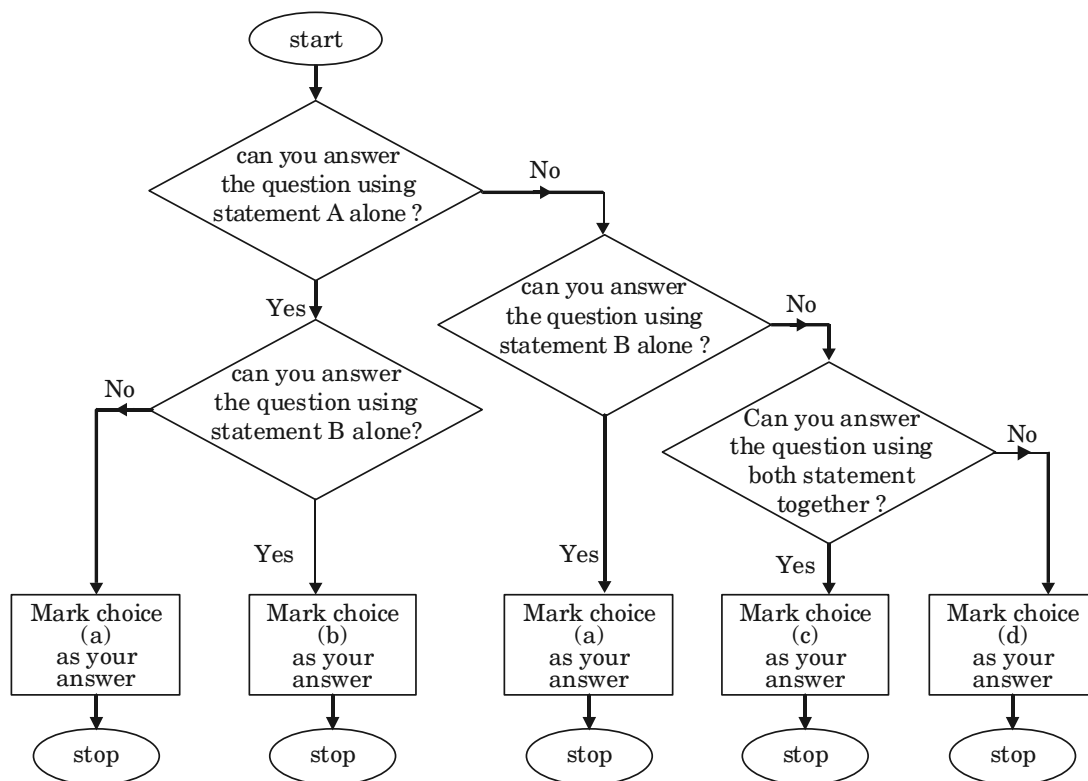
If each of the statement is independently insufficient, then combine them.

Step5 :

See whether the combination of the statements is sufficient or not.

The following flow chart explains pictorially this set of directions and shows how one can go about deciding the answer choice for any question.

Flow Chart



Example 1 : what is the age of mother?

- Two years ago, the father's age was six times the son's age.
- After 18 years, the ratio of the ages of son, mother and father will be 5:9:10.

Solution:

From statement 1, let the present age of the father and the son be f and s respectively. Then we have

$$(f - 2) = 6(s - 2) \rightarrow (1)$$

Using statement 2, after 18 years father's age is twice son's age. $(f + 18) = 2(s + 18)$ (2)

Each statement is independently not sufficient.

Combining both the statements, father's age and son's age can be obtained by simplifying (1) and (2). Using the ratio of their ages, the mother's age also can be obtained. Choice (A)

Example 2 : what is the two digit number xy ?

- The sum of the units digit and the tens digit of xy is 15.
- Difference between the digits of xy is 1.

Solution:

Given two digit number is xy .

From statement 1, we have $x + y = 15$ from which we cannot find x, y uniquely.

$\therefore$ Statement 1 is not sufficient to answer.

From statement 2, we have $x - y = 1$ or $y - x = 1$.

This alone is not sufficient to answer.

Combining both the statements, we have

$$x + y = 15 \quad \text{---(1)}$$

$$x - y = 1 \quad \text{---(2)}$$

$$\text{or } y - x = 1 \quad \text{---(3)}$$

From (1) and (2), we have $x = 8$ &

$$y = 7.$$

From (1) and (3), we have $x = 7$ and $y = 8$

Both the statements together also are not sufficient Choice (D)

Example 3 : what is the measure of angle C in triangle ABC?

- Angle A, B and C is in the ratio 2 : 1 : 3.
- $A + B = 130^\circ$

Solution:

Statement 1 alone is sufficient to answer as we know the ratio of the angle.

From statement 2, as $A + B = 130^\circ$ and as sum of the angles of any triangle is 180, we have

$$= 180^\circ - 130^\circ = 50^\circ$$

Statement 2 alone is sufficient to answer the question. Choice (B)

EXERCISE

MCQ TYPE QUESTIONS

1. Given two numbers x and y , define

Addition	$A(x, y) = x + y$
Subtraction	$S(x, y) = x - y$
Multiplication	$M(x, y) = x \cdot y$
Division	$D(x, y) = x/y$

$S[M(D(A(a, b), 2), D(A(a, b), 2)),$
 $M(D(S(a, b), 2), D(S(a, b), 2))]$ is equal to

- (a) $(a - b)^2$ (b) ab
 (c) $(a + b)^2$ (d) None of these

Directions Q. 2 – 5 :

Kamal Babu came home just after judging a beauty contest where there were four semi-finalists, Ms. Andhra Pradesh, Ms. Uttar Pradesh, Ms. Maharashtra and Ms. West Bengal. His wife was very keen on knowing who the winner was and Kamal Babu replied immediately that it was the one wearing the yellow saree. When his wife asked for more details, he gave the following information:

- * The four girls were wearing sarees of different colours (yellow, red, green, white) and the runner-up was wearing green.
- * The four girls were sitting in a row, and Ms. West Bengal was not sitting at either end.
- * There was only one runner-up and she was sitting next to Ms. Maharashtra.
- * The girls wearing yellow and white sarees occupied the seats at either end.
- * Ms. West Bengal was neither the winner nor the runner-up.
- * Ms. Maharashtra was wearing white.
- * The winner and the runner-up were not sitting next to each other.
- * The girl wearing the green saree was not Ms. Andhra Pradesh.

Answer the following questions based on the above information:

2. What was the colour of the saree that Ms. Andhra Pradesh was wearing?
 (a) White (b) Yellow
 (c) Red (d) Cannot be determined
3. Between which two was Ms. West Bengal sitting?
 (a) Ms. Andhra Pradesh and Ms. Maharashtra
 (b) Ms. Andhra Pradesh and Ms. Uttar Pradesh
 (c) Ms. Uttar Pradesh and Maharashtra
 (d) Cannot be determined
4. Who was wearing the red saree?
 (a) Ms. Andhra Pradesh
 (b) Ms. Uttar Pradesh
 (c) Ms. West Bengal
 (d) Cannot be determined

5. What was the colour of the saree that Ms. Uttar Pradesh was wearing?

- (a) White (b) Green
 (c) Red (d) Yellow

Directions Q. 6 – 7 :

A, B, C, D, E and F are a group of friends from a club. There are two housewives, one lecturer, one architect, one accountant and one lawyer in the group. There are two married couples in the group. The lawyer is married to D who is a housewife. No lady in the group is either an architect or an accountant C. The accountant, is married to F who is a lecturer. A is married to D and E is not a housewife.

6. What is E ?

- (a) Lawyer (b) Architect
 (c) Lecturer (d) Accountant

7. How many members of the group are males?

- (a) 2 (b) 3
 (c) 4 (d) none of these

Directions Q. 8 – 9 :

Seven university cricket players are to be honoured at a special luncheon. the players will be seated on the dais along one side of a single rectangular table. A and G have to leave for the lunch early and must be seated at the extreme right end of the table which is closest to the exit.

B will receive the Man of the Match award and must be in the center chair.

C and D, who are bitter rivals for the position of wicket keeper, dislike each other and should be seated as far apart as possible.

E and F are best friends and want to sit together.

8. Which of the following may not be seated at either end of the table ?

- (a) C (b) D
 (c) G (d) F

9. Which of the following pairs may not be seated together ?

- (a) E & A (b) B & D
 (c) C & F (d) G & D

Directions Q. 10 – 13 :

A study was conducted to ascertain the relative importance that employees in five different countries assigned to five different traits in their Chief Executive Officers. The traits were compassion (C), decisiveness (D), negotiation skills (N), public visibility (P), and vision (V). The level of dissimilarity between two countries is the maximum difference in the ranks allotted by the two countries to any of the five traits. The following table indicates the rank order of the five traits for each country.

Country					
RANK	INDIA	CHINA	JAPAN	MALAYSIA	THAILAND
1	C	N	D	V	V
2	P	C	N	D	C
3	N	P	C	P	N
4	V	D	V	C	P
5	D	V	P	N	D

10. Which of the following pairs of countries are most dissimilar?
- (a) China and Japan
(b) India and China
(c) Malaysia and Japan
(d) Thailand and Japan
11. Which of the following countries is least dissimilar to India?
- (a) China (b) Japan
(c) Malaysia (d) Thailand
12. Which amongst the following countries is most dissimilar to India?
- (a) China (b) Japan
(c) Malaysia (d) Thailand
13. Three of the following four pairs of countries have identical levels of dissimilarity. Which pair is the odd one out?
- (a) Malaysia and China
(b) China and Thailand
(c) Thailand and Japan
(d) Japan and Malaysia

Directions Q. 14 – 17 :

Four families decided to attend the marriage ceremony of one of their colleagues. One family has no kids, while others have at least one kid each. Each family with kids has at least one kid attending the marriage.

Given below is some information about the families, and who reached when to attend the marriage.

The family with 2 kids came just before the family with no kids.

Shanthi who does not have any kids reached just before Sridevi's family.

Sunil and his wife reached last with their only kid.

Anil is not the husband of Joya.

Anil and Raj are fathers.

Sridevi's and Anita's daughters go to the same school.

Joya came before Shanthi and met Anita when she reached the venue.

Raman stays the farthest from the venue.

Raj said his son could not come because of his exams.

14. Which woman arrived third ?

- (a) Shanthi (b) Sridevi
(c) Anita (d) Joya

15. Name the correct pair of husband and wife ?

- (a) Raj and Shanthi (b) Sunil and Sridevi
(c) Anil and Sridevi (d) Raj and Anita

16. Of the following pairs, whose daughters go to the same school ?

- (a) Anil and Raman (b) Sunil and Raman
(c) Sunil and Anil (d) Raj and Anil

17. Whose family is known to have more than one kid for certain ?

- (a) Raman's (b) Raj's
(c) Anil's (d) Sunil's

Directions Q. 18 – 20 :

Answer the following questions based on the statements given below:

- (i) There are three houses on each side of the road.
(ii) These six houses are labeled as P, Q, R, S, T and U.
(iii) The houses are of different colours, namely, Red, Blue, Green, Orange, Yellow and White.
(iv) The houses are of different heights.
(v) T, the tallest house, is exactly opposite to the Red coloured house.
(vi) The shortest house is exactly opposite to the Green coloured house.
(vii) U, the Orange coloured house, is located between P and S.
(viii) R, the Yellow coloured house, is exactly opposite to P.
(ix) Q, the Green coloured house, is exactly opposite to U.
(x) P, the White coloured house, is taller than R, but shorter than S and Q.

18. What is the colour of the house diagonally opposite to the Yellow coloured house?

- (a) White (b) Blue
(c) Green (d) Red

19. Which is the second tallest house?

- (a) P (b) S
(c) Q (d) cannot be determined

20. What is the colour of the tallest house?

- (a) Red (b) Blue
(c) Green (d) Yellow

Directions for questions 21 to 25:

Select the correct alternative from the given choices

- 21.** If five cuts are made on a cube, what is the minimum number of pieces obtained?
 (a) 18 (b) 6
 (c) 16 (d) 25
- 22.** If six cuts are made on a cube, what is the maximum number of identical pieces obtained?
 (a) 16 (b) 18
 (c) 36 (d) 27
- 23.** If two, three and four cuts are made parallel to different faces of a cube, then what is the number of identical pieces obtained?
 (a) 60 (b) 30
 (c) 48 (d) 24
- 24.** What is the minimum number of cuts required to cut a cube into 216 identical pieces obtained?
 (a) 36 (b) 18
 (c) 15 (d) 12
- 25.** What is the minimum number of cuts required to cut a cube into 24 identical pieces?
 (a) 6 (b) 8
 (c) 9 (d) 12

Directions for questions 26 to 28:

These questions are based on the following information

A cube, painted on all its faces, is cut into 125 identical smaller cubes.

- 26.** How many smaller cubes have no face painted?
 (a) 100 (b) 81
 (c) 64 (d) 27
- 27.** How many smaller cubes have only one face painted?
 (a) 36 (b) 54
 (c) 64 (d) 108
- 28.** How many smaller cubes have exactly two faces painted?
 (a) 36 (b) 54
 (c) 48 (d) 60

Directions for questions 29 to 31:

These questions are based on the following information

Each of 216 small identical cubes are painted blue on all faces and all these cubes are arranged to form a large cube. Now all the faces of the large cube are painted pink.

- 29.** How many small cubes have only one color on them?
 (a) 96 (b) 125
 (c) 64 (d) 48

- 30.** How many small cubes have exactly two faces painted pink?
 (a) 36 (b) 48
 (c) 64 (d) 80

- 31.** How many small cubes have exactly three faces painted blue?
 (a) 8 (b) 4
 (c) 2 (d) 6

Directions for questions 32 to 34:

These questions are based on the following information

A cube is painted in black and green, each on three faces such that any two faces with same color are adjacent to each other. Now this cube is cut into 60 identical pieces using 2, 3, and 4 cuts parallel to different faces.

- 32.** How many smaller pieces have exactly two faces painted in black color?
 (a) 5 (b) 9
 (c) 18 (d) 27
- 33.** How many smaller pieces have both the colors on them?
 (a) 9 (b) 18
 (c) 6 (d) 24
- 34.** How many smaller pieces have no face painted?
 (a) 6 (b) 9
 (c) 11 (d) 1

Directions: Q. No 35 to 39: These questions are based on the information given below:

Four ladies A, B, C and D and four gentlemen E, F, G and H are sitting in a circle round a table facing each other.

- (a) No two ladies or two gentlemen are sitting side by side.
 (b) C who is sitting between G & E is facing D.
 (c) F is between D and A is facing G.
 (d) H is to the right of B
- 35.** Who is sitting to the left of A?
 (a) E (b) F
 (c) G (d) H
- 36.** E is facing whom?
 (a) F (b) B
 (c) G (d) None of these
- 37.** Who are immediate neighbor of B?
 (a) G and H (b) E and F
 (c) E and H (d) F and H
- 38.** Who is sitting between E and G?
 (a) A (b) B
 (c) C (d) D

39. Who is sitting to the right of D?

- (a) F (b) A
(c) E (d) C

Directions Q. No 40 to 44

Study the following information carefully and answer the question given below: A, B, C, D, E, F, G and H are sitting around a circle facing the centre. B is second to the right of D who is third to the right of F, C is second to the left of A who is second to the left of F, G is third to the right of E.

40. In which of the following combinations is the first person sitting between the second and the third persons?

- (a) GCD (b) FGH
(c) EFH (d) ABE

41. Who is third to the right of H?

- (a) Data inadequate
(b) D
(c) C
(d) G

42. Who is the immediate right of A?

- (a) Data inadequate
(b) E
(c) F
(d) B

43. What is H's position with respect to B?

- (a) Fifth to the right
(b) Third to the left
(c) Third to the right
(d) None of these

44. Who is the immediate left of G?

- (a) H
(b) F
(c) Data inadequate
(d) D

Directions for questions 45 to 49 :

Read the following information and answer the questions that follow.

P, Q, R, S and T sit around a table. P sits two seats to the left of R and Q sits two seats to the right of R.

45. If S sits in between Q and R, who sits to the immediate right of P?

- (a) T (b) S
(c) Q (d) R

46. Which of the following cannot be the correct seating arrangement of the five persons in either the clockwise direction or the anti-clockwise direction ?

- (a) P, Q, R, S, T (b) P, S, R, T, Q
(c) P, Q, S, R, T (d) P, T, R, S, Q

47. If S is not sitting next to Q, who sitting between Q and S?

- (a) R (b) P
(c) T (d) Both R and P

48. If a new person U joins the group such that the initial conditions for the seating arrangement should be observed and also a new condition that U does not sit next to R be satisfied then which of the following statements is true?

- (a) U sits to the immediate right of S
(b) U sits to the immediate left of T
(c) U sits to the immediate left of P.
(d) P is to the right of Q

49. If a new person U joins the group such that the initial conditions for the seating arrangement should be observed and also a new condition that U does not sit next to P, S or T be satisfied then who will be the neighbours of P (one either side)?

- (a) S and T (b) S and Q
(c) T and R (d) R and Q

Directions for questions 50 to 59:

Select the correct alternative from the given choices

50. What is the angle covered by the minute hand in 22 minutes?

- (a) 66° (b) 110°
(c) 132° (d) 220°

51. By how many degrees does an hour hand move in one quarter of an hour?

- (a) 5° (b) 7.5°
(c) 10° (d) 12.5°

52. By how many degrees will the minute hand move, in the same time, in which the hour hand moves ?

- (a) 54° (b) 84°
(c) 72° (d) 60°

53. What is the angle between the hands of the clocks, when it shows 40 Minutes past 6?

- (a) 40° (b) 70°
(c) 80° (d) 90°

54. When the clocks shows 3 hours 14 minutes what is the angle between the hands of the clocks?

- (a) 10° (b) 12°
(c) 13° (d) 14°

- 55.** What is the angle between the two hands of a clock when the time is 25 minutes past 7 O'clock?
 (a) $66\frac{1}{2}^\circ$ (b) $66\frac{1}{2}^\circ$
 (c) $72\frac{1}{2}^\circ$ (d) $69\frac{1}{2}^\circ$
- 56.** When the clock shows 20 minutes past 11 O'clock, what is the angle between the two hands of the clocks?
 (a) 220° (b) 120°
 (c) 230° (d) 140°
- 57.** At the time between 9 and 10 O'clock, will both the hands of the clocks coincide?
 (a) $43\frac{3}{11}$ Minutes past 9 O'clock
 (b) $45\frac{6}{11}$ Minutes past 9 O'clock
 (c) $45\frac{6}{11}$ Minutes past 9 O'clock
 (d) $49\frac{6}{11}$ Minutes past 9 O'clock
- 58.** At what time between 4 and 5 O'clock are the hands of a clock in the opposite directions?
 (a) $52\frac{3}{11}$ minutes past 4 O'clock
 (b) $54\frac{6}{11}$ minutes past 4 O'clock
 (c) $51\frac{6}{11}$ minutes past 4 O'clock
 (d) $53\frac{9}{11}$ minutes past 4 O'clock
- 59.** The angle between the hands of a clock is 20° and the hour hand is in between 2 and 3. What is the time shown by the clock?
 (a) 2 hours $7\frac{3}{11}$ minutes
 (b) 2 hours $14\frac{6}{11}$ minutes
 (c) 2 hours $15\frac{5}{11}$ minutes
 (d) Both (a) and (b)
- Directions for questions 60 to 72:**
 Select the correct alternative from the given choices.
- 60.** The number of odd days in non-leap year is
 (a) 0 (b) 1
 (c) 2 (d) 3
- 61.** How many odd days are there in 352 days?
 (a) 1 (b) 2
 (c) 3 (d) zero
- 62.** The number of odd days for the year 1996 is
 (a) 0 (b) 1
 (c) 2 (d) 3
- 63.** Which among the following years is a leap year?
 (a) 3000 (b) 3100
 (c) 3200 (d) 3300
- 64.** What will be next leap year after 2096?
 (a) 2100 (b) 2101
 (c) 2104 (d) 2108
- 65.** If 5th January 2001 was a Friday then 25th December 2001 was a
 (a) Monday (b) Tuesday
 (c) Wednesday (d) Thursday
- 66.** If 21st march 2000 was on Monday, what day of the week will be 21st March 2003?
 (a) Tuesday (b) Friday
 (c) Thursday (d) Wednesday
- 67.** If 14th February 2001 was a Wednesday then what day of the week will be 14th February 2101 (i.e. after a century)?
 (a) Friday (b) Saturday
 (c) Sunday (d) Monday
- 68.** If in a calendar year, there are 541 days and 10 days a week then how many odd days will be there in that year?
 (a) 1 (b) 2
 (c) 3 (d) 4
- 69.** If 8th February 1995 was a Wednesday then 8th February 1994 was on which day?
 (a) Wednesday (b) Thursday
 (c) Tuesday (d) Monday
- 70.** If 11th August 1985 was a Sunday then which day of the week was 13th August 1986?
 (a) Tuesday (b) Wednesday
 (c) Thursday (d) Friday
- 71.** If 1st January 2012 is a Sunday then which days of the week will the New Year is celebrated in 2016?
 (a) Friday
 (b) Sunday
 (c) Wednesday
 (d) Saturday

72. If holidays are declared only on Sundays and in a particular year 12th March is a Sunday, is 23rd September in that year a holiday?

- (a) Yes
- (b) No
- (c) Yes, if it is a leap year
- (d) No, if it is a leap year

Direction for Q. 73 to 75:

Read the following information and answer the questions that follow.

Abhishek, Bharat, Chetan, Dinesh Eklavya and Franklin are friends married to Richa, Stuti, Urmila, Varsha, Trupti and Wanda not necessarily in same order.

Following facts are known about them-

- (1) Richa and Stuti are Abhishek's sisters
- (2) Neither Richa nor Trupti are wives of Chetan
- (3) Wanda is wife of Eklavya and Varsha is wife of Bharat
- (4) Dinesh is not married to Richa, Stuti or Trupti

73. Who is Abhishek's wife?

- (a) Richa
- (b) Urmila
- (c) Trupti
- (d) cannot be determined

74. Richa is wife of

- (a) Chetan
- (b) Franklin
- (c) Bharat
- (d) Dinesh

75. Dinesh is married to

- (a) Urmila
- (b) Barsha
- (c) Trupti
- (d) Richa

Directions for Qu. 76 to 79

Read the following information and answer the questions that follow.

XYZ, an institute involved in training the students for MBA, MCA, MMS and other entrance exams runs six batches A, B, C, D, E and F, each of which is scheduled of at least once a week.

- (1) Sunday is a holiday.
- (2) The B batch meets all days except Friday and Saturday.
- (3) The C batch meets 4 days in succession.
- (4) The F batch meets only from Monday to Thursday.
- (5) The E batch is scheduled every day, but not on Thursday and Saturday.
- (6) On alternate days, the A batch is scheduled.
- (7) The A and D batches never meet on the same days.
- (8) The C batch does not meet on Mondays and Tuesday.
- (9) The D batch is scheduled only once a week on either Wednesday or Friday.

76. Which batches meet for the same number of classes during the week?

- (a) B, A, F only
- (b) E, B, C only
- (c) E, F, A only
- (d) None of these

77. If a certain class of D is scheduled on the same today as that of B batch, then how many batches meet on Friday?

- (a) 1
- (b) 2
- (c) 3
- (d) 4

78. If there is one such day that all batches are to meet, then which batch would gain an additional day?

- (a) A
- (b) B
- (c) D
- (d) E

79. Which two batches can never be conducted on the same day?

- (a) C and D
- (b) C and E
- (c) A and D
- (d) None of these

Directions for Qu. 80 to 82

Read the following information and answer the questions that follow.

Abhita is bright girl who has a CD box on her study table. The box has three horizontal row each can be filed with 4 CD cases. She has also put her "Jewellery box" in the CD box which consume the space of two CD ceases.

The following information is known

- (1) The jewellery box is in the bottom row.
- (2) The CD cases with "Window XP" and "Windows 2000" are horizontally adjacent to each other.
- (3) The CD case with Linux is in the bottom row.
- (4) "Utility tools" is placed horizontally adjacent to "Game1" CD.
- (5) A vertical columns containing "Linux" and "music" also contains "Movies"
- (6) "Net tools" is just below the "Windows XP" and "CS II" is just above "Game 2"
- (7) "Net tools" is not at the corner.

80. Which CD is just below "Net tools"?

- (a) Utility tools
- (b) Games 1
- (c) CS II
- (d) None of these

81. If "Movie" is in second row, then how many CDs are between "Movies" and "Game2"?

- (a) 0
- (b) 1
- (c) 2
- (d) Cannot be determined

82. If "Music is in top row, how many CDs are between "Music" and "Utility tools".

- (a) 0
- (b) 1
- (c) 2
- (d) Cannot be determined

Directions:

In each of the question below are given two statements followed by two conclusions numbered I and II. You have to take the two given statements to be true even if they seem to be at variance from commonly known facts and decide which of the given conclusion (s) logically follow (s) from the two given statements, disregarding commonly known facts. Given answer as

- (a) If only conclusion I follows
- (b) If only conclusion II follows
- (c) If either I or II follows
- (d) If neither I nor II follows
- (e) If both I and II follow

83. Statements:

- (1) Some players are singers.
- (2) All singers are tall.

Conclusions:

- I. Some players are tall.
- II. All players are tall.

84. Statements:

- (1) Some vegetables are fruits.
- (2) No fruit is black.

Conclusions:

- I. Some fruits are vegetables.
- II. No vegetable is black.

85. Statements:

- (1) Some stone are bricks.
- (2) Some bricks are trees.

Conclusions:

- I. Some stones are trees.
- II. Some trees are bricks.

86. Statements:

- (1) All tigers are ships.
- (2) Some ships are cupboards.

Conclusions:

- I. Some tigers are cupboards.
- II. Some cupboards are tigers.

87. Statements:

- (1) Some stones are cups.
- (2) Some cups are black.

Conclusions:

- I. Some blacks are not cups.
- II. Some cups are stones.

Directions: Each question is followed by two statements giving certain data. You have to Decide whether the information provided in the statements is sufficient for answering the question.

Choose A : if the question can be answered by using one of the statements alone, but cannot not be answered by using the other statement alone.

Choose B : if the question can be answered by using either statement alone.

Choose C : if the question can be answered by using both statements together, but cannot be answered by using either statement alone.

Choose D : if the question cannot be answered even by using both the statements together.

88. What is the age of Ramesh?

- 1. The average age of Ramesh, Ramu and Ravi is 76 years.
- 2. Ramesh is three years younger to both, Ramu and Ravi.

89. Is $(n - 1)$ even, given that n is a prime number?

- 1. $(n - 2)$ is a whole number.
- 2. $(n + 1)$ is an odd number.

90. How many coins are there in the purse that has coins in the denominations of 25 paisa, 50 paisa and one rupee?

- 1. The value of the coins in the respective denominations is in the ratio 1:2:5.
- 2. The total amount in the three denominations is 64 Rs.

91. What is the monthly salary of Q?

- 1. Q earns 6,000 Rs more than R who earns 3,000Rs less than P.
- 2. The total monthly salary of P and Q is 27,000.

92. Does Raju makes Profit on selling two articles?

- 1. On one article Raju makes a profit of 10% and on the other Raju incurs a loss of 10%.
- 2. Both the articles are sold at the same price.

NUMERICAL TYPE QUESTIONS**1. Given two numbers x and y , define**

Addition	$A(x, y) = x + y$
Subtraction	$S(x, y) = x - y$
Multiplication	$M(x, y) = x \cdot y$
Division	$D(x, y) = x/y$

If $x = 2$, $y = 3$ and $z = 5$, then value of $M[A(S(x, y), M(y, z)), S(A(x, z), S(z, y))]$ is _____

Directions Q. 2 - 3 :

These questions are based on the following data

A, B, C, D are standing in a queue. It is known that C and D are not standing adjacent to each other and that B is not in the third place.

- 2. If A is not in the second place and B is in the first place, then which is the place for D is _____
- 3. If A is not standing adjacent to B and D is the last in the queue, then the place for C is _____

Directions Q. 4 – 5 :

A boy is asked to put in a basket one mango when ordered 'One', one orange when ordered 'Two', one apple when ordered 'Three' and is asked to take out from the basket one mango and an orange when ordered 'Four'. A sequence of orders is given as 1 2 3 3 2 1 4 2 3 1 4 2 2 3 3 1 4 1 1 3 2 3 4

4. The _____ total oranges were in the basket at the end of the above sequence.
5. The _____ total fruits will be in the basket at the end of the above order sequence.

Directions Q. 6 – 7

Each of the 11 letters A, H, I, M, O, T, U, V, W, X and Z appears same when looked at in a mirror. They are called symmetric letters. Other letters in the alphabet are asymmetric letters.

6. The _____ number of four-letter computer passwords can be formed using the symmetric letters (no repetition allowed)

7. The _____ number of three-letter computer passwords can be formed (no repetition allowed) with at least one symmetric letter.

Directions Q. 8 – 9 :

A string of three English letters is formed as per the following rules:

- (a) The first letter is any vowel
 - (b) The second letter is m, n or p.
 - (c) If the second letter is m then the third letter is any vowel which is different from the first letter
 - (d) If the second letter is n then the third letter is *e* or *u*
 - (e) If the second letter is p then the third letter is the same as the first letter.
8. The _____ strings of letters can possible be formed using the above rules.
 9. The _____ strings of letters can possibly be formed using the above rules such that the third letter of the string is *e*.

ANSWERS

MCQ Type Questions

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (b) | 2. (b) | 3. (b) | 4. (c) | 5. (b) | 6. (b) | 7. (b) | 8. (d) | 9. (a) | 10. (a) |
| 11. (b) | 12. (d) | 13. (d) | 14. (a) | 15. (b) | 16. (c) | 17. (b) | 18. (d) | 19. (d) | 20. (b) |
| 21. (b) | 22. (d) | 23. (a) | 24. (c) | 25. (a) | 26. (d) | 27. (b) | 28. (a) | 29. (c) | 30. (b) |
| 31. (a) | 32. (b) | 33. (b) | 34. (a) | 35. (b) | 36. (d) | 37. (a) | 38. (c) | 39. (a) | 40. (d) |
| 41. (a) | 42. (a) | 43. (d) | 44. (c) | 45. (a) | 46. (a) | 47. (b) | 48. (c) | 49. (a) | 50. (c) |
| 51. (b) | 52. (c) | 53. (a) | 54. (c) | 55. (c) | 56. (d) | 57. (c) | 58. (b) | 59. (d) | 60. (b) |
| 61. (b) | 62. (c) | 63. (c) | 64. (c) | 65. (b) | 66. (c) | 67. (d) | 68. (a) | 69. (c) | 70. (b) |
| 71. (a) | 72. (b) | 73. (c) | 74. (b) | 75. (a) | 76. (d) | 77. (b) | 78. (a) | 79. (c) | 80. (d) |
| 81. (c) | 82. (b) | 83. (a) | 84. (a) | 85. (b) | 86. (d) | 87. (b) | 88. (c) | 89. (a) | 90. (c) |
| 91. (c) | 92. (b) | | | | | | | | |

Numerical Type Questions

- | | | | | | | | | |
|-------|--------|------|------|-------|---------|----------|-------|-------|
| 1. 70 | 2. 105 | 3. 2 | 4. 2 | 5. 11 | 6. 7920 | 7. 12870 | 8. 35 | 9. 10 |
|-------|--------|------|------|-------|---------|----------|-------|-------|

EXPLANATIONS

MCQ TYPE QUESTIONS

1. $S[M(D(A(a, b)2, D(A(a, b)2)), M(D(S(a, b)2)) D(s(a, b)2)]$

$$= S \left[\left| M \frac{a+b}{2}, \frac{a+b}{2} \right| \left| M \frac{a-b}{2}, \frac{a-b}{2} \right| \right]$$

$$= \left(\frac{a+b}{2} \right)^2 - \left(\frac{a-b}{2} \right)^2 = ab$$

6 – 7.

A → Lawyer → Male
 B → Housewife → Female
 C → Accountant → Male
 D → Housewife → Female
 E → Architect → Male
 F → Lecturer → Female
 Married Couple : AD & CF

- 8—9. The seating arrangement is shown below:

Left			Right		
7	6	5	4	3	2
1					1
D/C	E/F	F/E	B	C/D	G/A

10 – 13.

Level of Dissimilarity is equal to Maximum difference in ranks allotted to any two countries on any of the five traits.

For example, dissimilarity between India and China.

Trait	India's Rank	China's Rank	Difference
C	1	2	1
P	2	3	1
N	3	1	2
V	4	5	1
D	5	4	1

From the above table, maximum difference is 2.

So dissimilarity level between India and China is 2.

10. Calculating the level of dissimilarity for India Vs remaining countries.

China : For the parameter N $(3 - 1) = 2$

Japan : For the parameter D $(5 - 1) = 4$

Malaysia : For the parameter C $(4 - 1) = 3$

Thailand : For the parameter V $(4 - 1) = 3$

Hence least level of dissimilarity is 2 which is for China.

Following table gives the desired details

Order	Women	Men	No. of Kids
1	Anita	Anil	at least 1
2	Joya	Raj	2
3	Shanthi	Raman	0
4	Sridevi	Sunil	1

11. From above calculations, highest level of dissimilarity is 4 which is for Japan.

12. The level of dissimilarity between

China and Japan is for the parameter D
 $(4 - 1) = 3$

India and China is for the parameter N
 $(3 - 1) = 2$

Malaysia and Japan is for the parameter V
 $(4 - 1) = 3$

Thailand and Japan is for the parameter D
 $(5 - 1) = 4$

13. The level of dissimilarity between

Malaysia and China is for the parameter V
 $(5 - 1) = 4$

China and Thailand is for the parameter V
 $(5 - 1) = 4$

Thailand and Japan is for the parameter D
 $(5 - 1) = 4$

Japan and Malaysia is for the parameter V
 $(4 - 1) = 3$

14 to 17 :

The given set of questions is based on distribution and sequencing.

Firstly, let us put the names of women and men,

Women : Shanthi, Sridevi, Joya, Anita

Men : Sunil, Anil, Raj, Raman

It is given that (2 kids family) $\leftarrow$ came before (No kid family)

Also Shanti (0 kids) $\xrightarrow{\text{came before}}$ Sridevi

As Sunil and his wife came with their only kid, hence Sunil cannot be the husband of Shanthi.

As Anil and Raj are fathers, hence neither of them is the husband of Shanthi. Hence Raman is the husband of Shanthi,

Also, Joya came before Shanthi and met Anita on the venue, this gives the following order of their reaching the venue. [Anita, Joya, Shanthi, Sridevi]

As Sunil reached last, Sunil is the husband of Sridevi. This means that Raj is Joya's husband and Anil is Anita's husband.

14. Shanthi arrived third.
15. Out of the given choices. Sunil and Sridevi is the correct pair.
16. The daughters of Anita and Sridevi go to the same school. Anita's husband is Anil and Sridevi's husband is Sunil. Hence, Sunil and Anil is the correct answer.
17. Raj's family consists of two kids for certain (Anil has at least one kid, hence could be having exactly one kid also).

18 to 20 :

From statements (vii), (viii) and (ix), we get

P	U	S
	Orange	
R	Q	
Yellow	Green	

Then, T is opposite S and the colour of S is red.

From (x), the colour of house P is white, hence the colour of house T is blue.

P	U	S
White	Orange	Red
R	Q	T
Yellow	Green	Blue

Now, from (v) and (x), we get

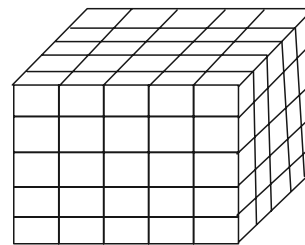
$$T > S, Q > P > R$$

From (vi), U is the shortest.

i.e., the order of houses in the descending order of their heights is T, S/Q, Q/S, P, R, U.

18. R is yellow coloured house and the house diagonally opposite R is S. And the colour of house S is Red.
19. The second tallest house is either S or Q.
20. The tallest house is T and its colour is blue.
21. If all the five cuts are made parallel to one face, the number of pieces obtained is $5 + 1 = 6$
22. If two cuts are made parallel to each perpendicular plane, then the number of pieces obtained is $3^3 = 27$
23. $(2 + 1) \times (3 + 1) \times (4 + 1) = 3 \times 4 \times 5 = 60$
Identical pieces are formed
24. $(n + 1)^3 = 216$ (n = number of cuts parallel to each face) $\Rightarrow n + 1 = 6$
 $n = 5$
 $\therefore 3n = 15$ cuts are required
25. $24 = 2 \times 3 \times 4 = (1 + 1)(2 + 1)(3 + 1)$
 $\therefore 1 + 2 + 3 = 6$ cuts are required

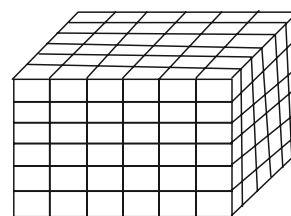
Solutions for questions 26 to 28:



Here $n^3 = 125$
 $\Rightarrow n = 5$

26. Cubes with no face painted:
 $(n - 2)^3 = (5 - 2)^3 = 27$
27. Cubes with one face painted:
 $6 \times (n - 2)^2 = 6(3)^2 = 54$
28. Cubes with two face painted:
 $12(n - 2) = 12(3) = 36$

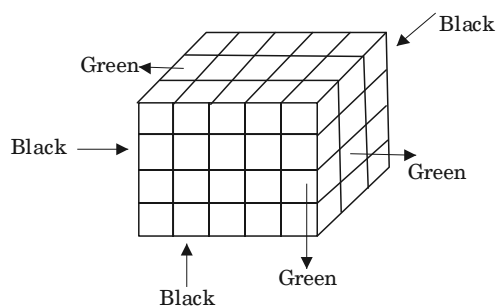
Solutions for questions 29 to 31:



Here $n^3 = 216$
 $\Rightarrow n = 6$

29. Cubes with exactly one color
 $= (n - 2)^3 = 4^3 = 64$
30. Cubes with exactly two faces painted pink
 $= 12(n - 2) = 12 \times 4 = 48$
31. The cubes at the corners have exactly three faces painted blue i.e., 8 cubes

Solutions for questions 32 and 34:



Comers (8)	Edges (12)	Faces (6)
BBB-1	BB-3	B-3
GGG-1	GG-3	G-3
GGB-3	BG-6	
BBG-3		

32. At the corners, three such pieces are there.

On the edges, the number of such pieces = 1 (along the plane with 2 cuts) + (along the plane with 3 cuts) + 3 (along the plane with 4 cuts).

$$\therefore \text{Required number of pieces} = 3 + 6 = 9$$

33. Except two corners, all the corners have both the colors $\Rightarrow$ 6 pieces.

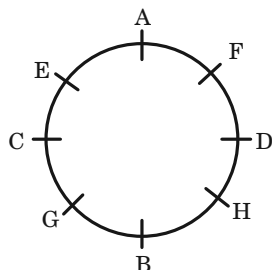
On the edges the number of pieces having both the colors according to the different cuts in each plane.

$\therefore$ Required number of pieces = $6 + 12 = 18$

34. The number of pieces with no face painted = 1 (along the plane with 2 cuts) + 2 (along the plane with 3 cuts) + 3 (along the plane with 4 cuts)

Solution for questions 35 to 39:

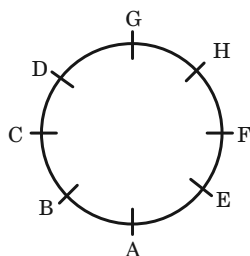
All information can be drawn in a circle



35. F is sitting the left of A.
 36. E is facing H
 37. G and H are immediate neighbors' of B
 38. C sitting between E and G
 39. F is sitting to the right of D.

Solution for questions 40 to 44:

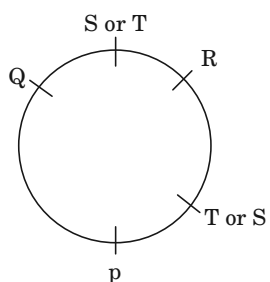
All information can be drawn in a circle



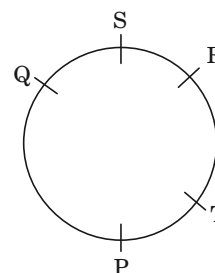
40. A is sitting between B and E.
 41. C is thief to the right of H.
 42. E is to the immediate right of A.
 43. H is to the fourth to the left of B.
 44. H is the immediate left of G.

Solution for questions 45 to 49:

P sits two seats the left of R, and Q sits two seats to the right of R. We can represent this information in the diagram below.

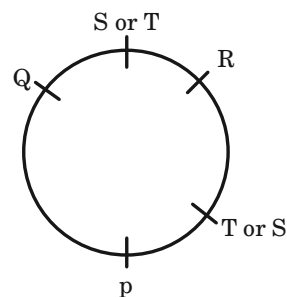


45. If S sits between Q and R, then the arrangements is as follows



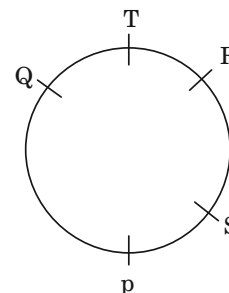
As can be seen from the diagram, T is to the immediate right of P.

46. We will take each choice and see whether it fits in the arrangement that we represented through a diagram in the analysis of the data (the same diagram is reproduced below).



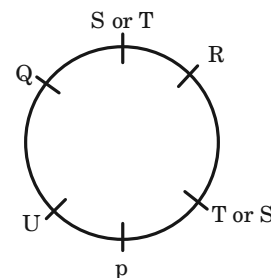
We can see that the arrangement given in choice (a) is not possible and hence the answer choice is (a)

47. If S is not next to Q, then the seating arrangement is fixed as follows



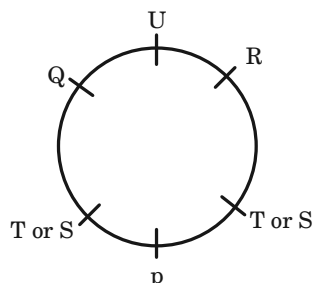
Now, P between Q and S.

48. On the basis of the diagram that we drew, we find that to accommodate U we have to create a new space between P and Q



Hence, Choice (c) is the correct answer.

49. We create a new slot for the sixth person B. But since U will not sit next to P, S or T, he will have to sit between R and Q. The arrangement will then look as follows:



As we can see from the diagram, the neighbors of P will be T and S.

50. The angle covered by the minute hand in 22 minutes is $22 \times 6^\circ = 132^\circ$
 51. The hour hand covers 360° in 12 hours.

$\therefore$ It covers $\frac{1^\circ}{2}$ in one minute. In quarter of an hour i.e., in 15 minutes the hour hand will move $15 \times \frac{1^\circ}{2} = 7.5^\circ$

52. The hour hand will move by 6° in 12 minutes, so minute hand will move $12 \times 6^\circ = 72^\circ$ in 12 minutes, as the minute hand moves by 6° in one minute

53. Angle will be $\theta = \left(\frac{11}{2}m - 30h \right)$

$$\left(\frac{11}{2} \times 20 - 30 \times 6 \right) = 40^\circ$$

54. The angle between the hands will be

$$\theta = \left| \frac{11}{2}m - 30h \right| \text{ Here } h = 3 \text{ and } m = 14$$

$$\theta = \frac{11}{2} \times 14 - 30 \times 3$$

$$\theta = |77 - 90| = 13^\circ$$

55. Angle between two hands is given by

$$\theta = \left| \frac{11}{2}m - 30h \right| \text{ Here } h = 7 \text{ and } m = 25$$

$$\theta = \left| \frac{11}{2} \times 25 - 30 \times 7 \right| = \frac{275 - 210}{2} = \frac{145^\circ}{2} = 72\frac{1}{2}$$

56. Angle between the two hands is given by

$$\theta = \left| \frac{11}{2}m - 30h \right| \text{ here } m = 20 \text{ and } h = 11$$

$$\Rightarrow \theta = \left| \frac{11}{2} \times 20 - 30 \times 11 \right| = 220$$

As angle is more than 180° the angle must be
 $360^\circ - 220^\circ = 140^\circ$

57. When hands coincide with each other the angle between them is 0. Therefore angle between hands is given by

$$\theta = 30 - \frac{11}{2}m \left(30h > \frac{11}{2}m \right) \text{ here } h = 9$$

$$270 \times \frac{2}{11} = m$$

$$\therefore m = 49\frac{1}{11} = 12 \text{ minutes past 9 hours.}$$

58. When hands of a clock are in opposite direction the angle between them is 180°

$$\text{Therefore } \theta = \left| \frac{11}{2}m - 30h \right| \text{ where } \theta = 180^\circ \text{ and } h = 4$$

$$180 = \frac{11}{2}m - 120$$

$$\frac{11}{2}m = 300$$

$$m = \frac{600}{11} = 54\frac{6}{11} \text{ minutes}$$

So, at 54 minutes past 4 the hands are in opposite direction.

59. Given $\theta = 20^\circ$ and $h = 2$

$$\theta = \frac{11}{2}m - 30h \text{ or } 30h - \frac{11}{2}m$$

$$20 = \frac{11}{2}m - 30 \times 2; \frac{11}{2}m = 80$$

$$m = \frac{160}{11} = 14\frac{6}{11} \text{ minutes}$$

$$\text{or } 20 = 30 \times 2 - \frac{11}{2}m; \frac{11}{2}m = 40$$

$$m = \frac{80}{11} = 7\frac{3}{11} \text{ minutes}$$

Therefore the angle between the hands will be

20° at hours $14\frac{6}{11}$ minutes past 2 and $7\frac{3}{11}$ minutes past 2.

60. Total numbers of days in an ordinary year are 365. Number of odd days

$$\frac{365}{7} = 52 \text{ weeks} + 1 \text{ odd day}$$

Therefore number of odd days is 1.

61. The number of days in 352 days

$$= \frac{352}{7}$$

= 50 weeks + 2 odd days.

Hence, the total number of odd days is 2

62. The year 1996 is a leap year. Total number of days in a leap year are 366 days. Number of odd

$$\text{days} = \frac{366}{7} = 52 \text{ weeks} + 2 \text{ odd days.}$$

63. Century years which are divisible by 400 are leap years. As 300 and 3300 are not divisible by 400 they are not leap years, but 3200 is a leap year.

64. For a century year to be a leap year it should be divisible by 400. As 2100 is not divided by 400 it is not a leap year. The next leap year is 2104

65. Total numbers of odd days from 5th January 2001 to 25th Dec 2001 are

Months: Jan + Feb + Mar + Apr + May + June + July + Aug + Sep + Oct + Nov + Dec

Odd days: 26 + 0 + 3 + 2 + 3 + 2 + 3 + 3 + 2 + 3 + 2 + 25 = 74

$$\frac{74}{7} = 10 \text{ Weeks} + 4 \text{ odd days}$$

4 days from Friday is Tuesday.

66. 21st March, 2000 is Monday and the year is a leap year. So, none of the next 3 year is the leap year. So the day of the week will be 3 days beyond Monday i.e. Thursday

67. A century has 5 odd days

∴ 5 day beyond Wednesday, Thursday, Friday, Saturday, Sunday, Monday.

Hence 14th February 2101 will be on Monday

68. To find the number of odd days, we have to find

remainder of $\frac{541}{10}$ the remainder is one. Hence

there is one odd day.

69. 1994 is not a leap year.

∴ It has only 1 odd day.

∴ 8th Feb 1995 is one day before Wednesday.

Hence 8th Feb 1994 is a Tuesday

70. It is given that 11th August 1985 was Sunday then 13th August 1985 is Tuesday. 1985 is not a leap year hence it has only one odd day. So 13th August 1986 is one day after Tuesday i.e. Wednesday

71. The total number of years from 2012 to 2016 is four out of which 2013, 2014 and 2015 are non leap years. Hence there is 1 odd day in each of these years 2012 is a leap year, hence it has 2 odd days.

∴ The total number of odd days in these four years is 5. 1st Jan 2016 is five days to Sunday i.e. Friday

72. The total number of odd days from 12th March to 23rd September

Months : M + A + M + J + J + A + S

Odd day : 5 + 2 + 3 + 2 + 3 + 3 + 2 = 20days.

$$\frac{20}{7} = 6 \text{ odd day}$$

Hence 23rd September is 6 days to Sunday i.e. Saturday, so 23rd September is not a holidays.

Solution for 73 to 75:

From the given information we can conclude Richa and Stuti are not Abhishek's wives Further Wanda and Varsha are wives of Eklavya and Bharat respectively. In this manner we analyses the information.

Wife \ Husband	Richa	Stuti	Trupti	Urmila	Varsha	Wanda
Abhishek	×	×				
Bharat					✓	
Chetan	×		×			
Dinesh	×	×	×			
Eklavya						
Franklin						

Now as Varsha and Wanda are married to Bharat and Eklavya they cannot be wives of anyone else Further, as every husband has one wife.

Dinesh is married to Urmila, Chetan, is married to stuti, Abhishek is married to Trupti and Franklin is married to Richa.

Wife \ Husband	Richa	Stuti	Trupti	Urmila	Varsha	Wanda
Abhishek	×	×	✓	×	×	×
Bharat	×	×	×	×	✓	×
Chetan	×	✓	×	×	×	×
Dinesh	×	×	×	✓	×	×
Eklavya	×	×	×	×	×	✓
Franklin	✓	×	×	×	×	×

Now all the question can be answered

Solutions for 76 to 79:

On the basis of the given information we get the following schedule:

Batch	Mon	Tue	Wed	Thu	Fri	Sat
A	×	✓	×	✓	×	✓
B	✓	✓	✓	✓	×	×
C	×	×	✓	✓	✓	✓
D	×	×		×		×
E	✓	✓	✓	×	✓	×
F	✓	✓	✓	✓	×	×

Here Batch D meets either on Wednesday or on Friday

Solution for questions 80 to 82:

CS II	UT or Game 1	UT or Game 1	Music or Movies
Game 2	Win 2000	Win XP	Music or Movies
Jewellery Box		Net tools	Linux

Case 2

Music or movies	UT or Game 1	UT or Game 1	CS II
Music or Movies	Win XP	Win 2000	Game 2
Linux	Net Tools	Jewellery Box	

Case 3

83. Statements:

- Some ^xplayers are ^xsingers
- All [✓]singers are ^xtall

Conclusion: Some players are tall.

84. Statements:

- Some ^xvegetables are ^xfruit
- No [✓]fruit is [✓]black

Conclusion: Some fruits are vegetable.

85. Statements:

- Some ^xstones are ^xbricks
- Some ^xbricks are ^xtrees

Conclusion: Some trees are bricks

86. Statements:

- All [✓]tigers are ^xship
- Some [✓]ships are ^xcupboards

Conclusion: No conclusion

87. Statements:

- Some ^xstones are ^xcups
- Some ^xcups are ^xblack

Conclusion: Some cups are stones

88. From statement I, we have

Ramesh + Ramu + Ravi = 228 years

But from this we cannot find any one of their ages.

From statement II, we have

Ramesh = Ramu-3 and Ramesh = Ravi-3

Again to find Ramesh's age, we should know either Ramu's or Ravi's age.

Using both the statements, we have

Ramesh + Ramesh + 3 + Ramesh + 3 = 228

∴ Ramesh age is 74 years

89. If n is a prime number, then n is either even prime (2) or odd prime (in all other cases). So n-1 is either even prime or odd. Using statement 2 alone, as n-1 is odd so n-1 is not an even number.

90. Using statement 1 alone let the total value of the each variety of coin be Rs. x, Rs. 2x and Rs. 5x. Using statement 2 alone is not sufficient.

Combining both the statements,

$$x + 2x + 5x = 64 \Rightarrow x = 8$$

The value of each variety of coins is Rs. 8, Rs. 16 and Rs. 40

Since the denominations and the total value of the coins I each denomination is known, the number of each dominations can be found out. Then, we can find the total numbers of coins.

91. Statement 1 alone is not sufficient as the relation between the CP of the two articles or the SP of the two articles is not known. Statement 2 alone is not sufficient as the relation between the CP of the two articles or the SP of the two articles is not known. Using both the statements, as both the article are sold at the same price and at equal percent of profit and loss on the two respectively, loss is made on selling an article. Hence question can be answered using both the statements.

92. From statement 1, ABCD may be square or rhombus, but since all the squares are rhombus, therefore statement 1 alone is sufficient to answer the question.

From statement 2, as $AC \perp BD$ and $AC^2 \neq BD^2$ and opposite sides are equal.

∴ The quadrilateral is a rhombus.

∴ Statement 2 alone is sufficient to answer the question.

NUMERICAL TYPE QUESTIONS

- $M[A(S(x, y), M(y, z)), S(A(x, z), S(z, y))]$
 $= M[A((2 - 3), (3, \times 5)), S((2 + 5), (5 - 3))]$
 $= M[A(-1, 15), S(7, 2)]$
 $= M[14, 5] = 70.$
- Since B is in 1, and C and D are not adjacent, the only places possible for them are 2 and 4.
- Since B is not in the third place and A is not adjacent to B, and D is last in the queue. The position of A will be beside D and B will be at the other corner. C will be between B and A. Hence the positions will be BCAD. Place of C is 2, hence choice (b)

The sitting position is as follows.

White Green Red Yellow

Maharashtra U.P. West Bengal A.P.

From the above table,

Ms. Andhra Pradesh $\longrightarrow$ Yellow

- (1) (2) 3 3 2 1 4 (2) 3 (1) 4 (2) (2) 3 3 (1) 4 1 (1) (3) (2) 3 4
Total no. oranges were in basket = 2
- Total fruits in the basket = 11
- Total number of four letter password formed
 $= {}^{11}C_4 \times 4 = 11 \times 10 \times 9 \times 8 = 7920$
- Required number of passwords
 $= {}^{26}C_3 \times 3 - {}^{15}C_3 \times 3$
 $= 26 \times 25 \times 24 - 15 \times 14 \times 13 = 12870$

- 1st Letter
2nd letter
3rd letter

There are 5 vowels, a, e, i, o, u .

hence 1st letter can be any of the 5 vowels ;

$\therefore$ 5 different methods.(i)

The second letter can be any of the m, n , or p .
But each has its own rule for the third letter.
Hence, each case to be considered separately.



In the above case; third letter is one of the remaining 4 vowels

Hence there are $5 \times 4 = 20$ ways of forming such a string.(ii)

If the second letter is n , the third is either e or u , i.e. 2 different ways.

$\therefore$ total of 5×2 ways = 10 ways(iii)

When 2nd position is filled by P, the third letter is same as the first letter. As the 1st letter is already selected, third can be filled only one way.

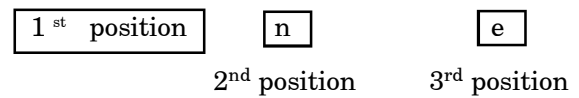
Hence, there are $5 \times 1 = 5$ ways(iv)

From equations (ii), (iii) and (iv),

Total number of strings that can be formed
 $= 20 + 10 + 5 = 35$

- Third letter find is 'e'.

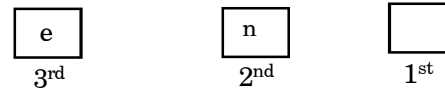
Once again considering each case of the second position being filled by m, n or p , we have



The 3rd position shall be different form vowel in first position, i.e. vowel for 1st position can be selected out of the vowels a, i, o, u only.

Hence 4 different ways(i)

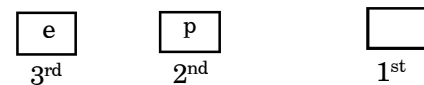
When 2nd possible is filled by n , occupying 2nd position, does not create any restriction on the vowel is position 1.



Hence, 1st position can be occupied by any of the 5 vowels.

Hence, 5 different ways(ii)

When 2nd position is filled by p ,



In this case, letters in 3rd position out 1st position shall be the same.

Hence, 1st position shall be filled by 'e'.

i.e. only one way of filling(iii)

Total number of possible strings = $4 + 5 + 1 = 10$

Spatial Aptitude

1

CHAPTER

Spatial Aptitude

TRANSFORMATIONS

In a plane, you can slide, flip, turn, enlarge, or reduce figures to create new figures. These corresponding figures are frequently designed into wallpaper borders, mosaics, and artwork. Each figure that you see will correspond to another figure. These corresponding figures are formed using transformations.

A **transformation** maps an initial image, called a preimage, onto a final image, called an image.

The geometric **transformation** is a bijection of a set that has a geometric structure by itself or another set. If a shape is transformed, its appearance is changed. After that, the shape could be congruent or similar to its preimage. The actual meaning of transformations is a change of appearance of something.

Types of Transformations

When the size of a shape is increased or reduced then the image of the shape will be similar to the pre-image. The **similar figures** have dimensions equal in proportion.

But in the case of congruent, the transformation of objects is done by using rotation, reflection or translation. The shape is turned or flipped to transform into another shape.

There are four major types of transformations namely:

- Rotation
- Translation
- Dilation
- Reflection

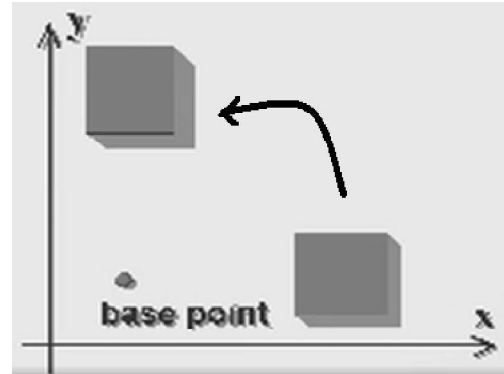
Rotation

The *rotation* transformation leaves the transformed object the same size, but turned relative to the original position.

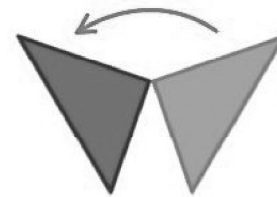
The rotation is controlled by a *base point* (which does not move) and an angle. As illustrated in the figure, if the base point is away from the object being transformed, the effect of the rotation may well include “movement” of the original shape.

In three dimensional rotations, the angle is often (but not always) measured in one of the primary coordinate system planes.

Remember, when performing multiple rotations, that Rotation “A” followed by Rotation “B” does not produce the same result as Rotation “B” followed by Rotation “A”.



This type of transformation has an object about a fixed point without changing its size or shape.



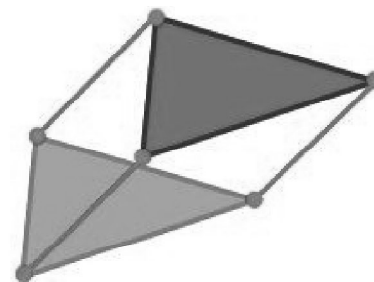
In the above figure, you can see, that the shape is rotated to form its image.

To describe you need:

- **direction** (clockwise or counterclockwise)
- **degree**
- **center point of rotation** (this is where compass point goes)

Translation

This type of translation is defined as moving the object in space by keeping its size, shape or orientation constant. In a translation, each point of the shape must be moved in the same direction and for the same distance. When you are doing a translation, the primary object is called the pre-image, and the object after the translation is called the image.



To describe you need:

- **direction** (left/right/up/down)
- **magnitude** (number of units)

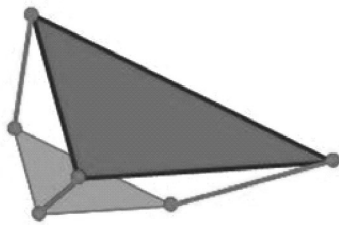
Coordinate Notation:

$$(x, y) \rightarrow (x \pm a, y \pm b)$$

Dilation

A **dilation** is a transformation that produces an image that is the same shape as the original, but is a different size. (The image is similar to the original object). Dilation is a transformation in which each point of an object is moved along a straight line. The straight line is drawn from a fixed point called the **center of dilation**. The distance the points move depends on the scale factor. The center of dilation is the only invariant point.

This type of translation expands or contracts the object by keeping its orientation or shape the same. This is also known as resizing.



$$\text{Scale factor} = \frac{\text{image length}}{\text{original length}}$$

$$= \frac{\text{distance of image from center of dilation}}{\text{distance of object from center of dilation}}$$

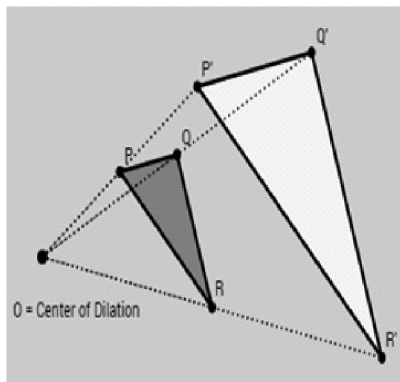
If the scale factor is greater than 1, the image is an **enlargement**.

If the scale factor is between 0 and 1, the image is a **reduction**.

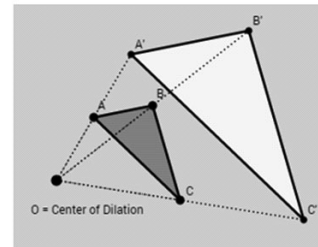
Example:

The figure shows two similar triangles PQR and P'Q'R'. Triangle P'Q'R' is a dilation of triangle PQR. We say that triangle PQR is transformed onto triangle P'Q'R' by a dilation with center at O and scale factor $\frac{OP'}{OP}$.

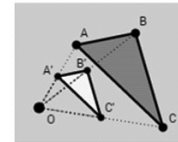
by a dilation with center at O and scale factor $\frac{OP'}{OP}$



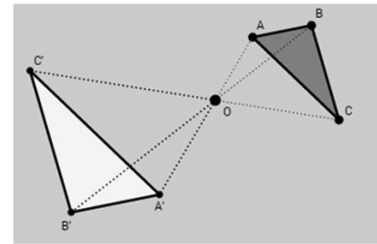
The following diagrams show the triangle ABC dilated with different scale factors. Scroll down the page for more examples and explanations of dilations.



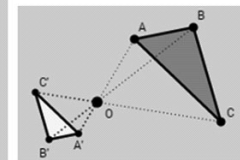
Scale Factor of 2



Scale Factor of 0.5



Scale Factor of -1.5



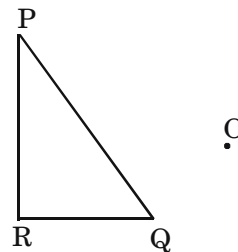
Scale Factor of -0.5

Dilation with scale factor > 1

We will first look at enlargements which are dilations with scale factors greater than 1

Example:

Enlarge triangle PQR with O as the center of dilation and a scale factor of 2.

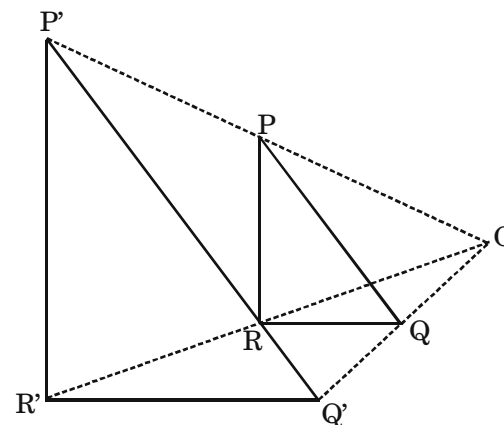
**Solution:**

Step 1: Measure OP.

Step 2: Extend the line OP to the point P' such that $OP' = 2OP$.

Step 3: Repeat the steps for all the vertices: point Q to get Q' and point R to get R'.

Step 4: Join the points P'Q'R' to form the image.



To describe you need:

- **Center point of the dilation**
- **Scale factor**

Coordinate Notation (if centered at the origin):

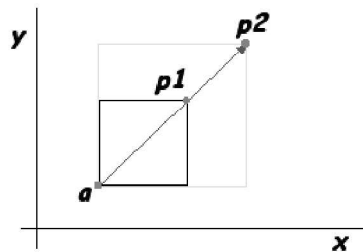
$$(x, y) \rightarrow (ax, by)$$

We find the scale factor by the help of scaling.

Scaling

An object which is **scaled** remains the same shape, but is reduced or enlarged in size in a proportionate fashion. The scaling operation shown in the figure involves enlarging the black square. Notice how the point **p1** is transformed into **p2**. Notice also that there is an anchor point (**a**) in the figure below, which remains in the same position.

This point is often one point of the object being scaled, but need not be.



The simplest scaling operation simply multiplies all coordinates of the scaled object by a uniform amount. Since this changes all non-zero coordinates, it also effectively **moves** the object, an unintended side-effect.

The “anchor point” for such a scaling operation is the origin itself, which remains unmoved.

Scaling process:

In the scaling process, we either compress or expand the dimension of the object.

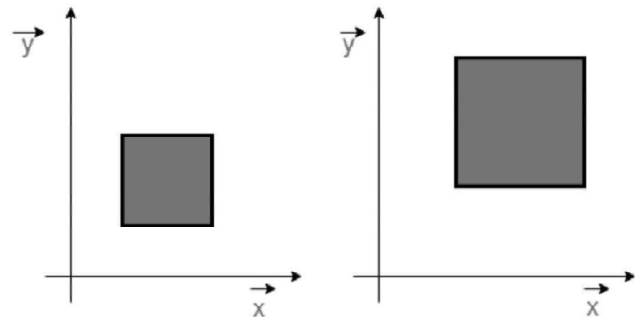
Scaling operation can be achieved by multiplying each vertex coordinate (x, y) of the polygon by scaling factor s_x and s_y to produce the transformed coordinates as (x', y') .

$$\text{So, } x' = x * s_x \text{ and } y' = y * s_y.$$

The scaling factor s_x, s_y scales the object in X and Y direction respectively. So, the above equation can be represented in matrix form:

$$\begin{bmatrix} X' \\ Y' \end{bmatrix} = \begin{bmatrix} S_x & 0 \\ 0 & S_y \end{bmatrix} \begin{bmatrix} X \\ Y \end{bmatrix}$$

$$\text{Or } P' = S.P$$



If the scaling factor S is less than 1, then we reduce the size of the object. If the scaling factor S is greater than 1, then we increase size of the object.

Algorithm:

1. Make a 2x2 scaling matrix S as:

$$S = \begin{bmatrix} S_x & 0 \\ 0 & S_y \end{bmatrix}$$
2. For each point of the polygon.
 - (i) Make a 2x1 matrix P , where $P[0][0]$ equals to x coordinate of the point and $P[1][0]$ equals to y coordinate of the point.
 - (ii) Multiply scaling matrix S with point matrix P to get the new coordinate.
3. Draw the polygon using new coordinates.

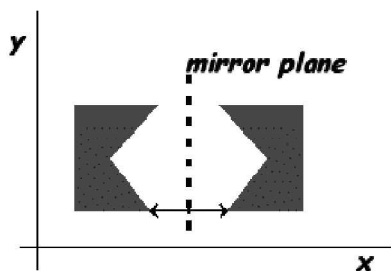
Reflection

The image of an **object** as seen in a mirror is its **mirror reflection** or mirror image. In such an image, the right side of the object appears on the left side and vice versa. A mirror-image is therefore said to be laterally inverted and the **phenomenon** is called the lateral inversion.

Some letters don't change upon reflection. In other words, they are the same as that of their Mirror Images. The letters having identical mirror images are A, H, I, M, O, T, U, V, W, X, Y. Similarly, in small letters we have i, l, o, v, w, and x that have the same Mirror Image as that of their original images. The number 1 and 8 are the only two numbers with their identical mirror images.

Any object that has a bilateral **symmetry** will have its mirror image the same as that of the object. By that, we mean that if you were to draw a line from top to bottom that splits the object into two parts, such that both the parts can be superimposed onto each other, the object's Mirror Image will be same as that of the object's image.

This type of translation is called reflection or mirroring because it flips the object across a line by keeping its shape or size constant.



In a *mirroring* or *reflection* operation, a shape is “flipped over” the mirror line (2D) or plane (3D). Each point in the original is repositioned relative to the mirror line/plane, ending up an equal distance away, but on the opposite side.

The easiest mirror operation uses one of the primary coordinate system axes as the plane and simply multiplies all x coordinates (or all y-coordinates, or z-coordinates) by -1 .

More general mirroring operations usually rely on the user selecting or indicating a line (or plane) in space. Because a line is somewhat easier to establish than a plane, a line is sometimes used, assuming that the mirror plane is perpendicular to the line, passing through its midpoint.

Topologically, mirroring turns objects inside-out, so programs maintaining consistently ordered surface orientations must compensate for the inversion by reversing the ordering of the faces which have been mirrored.

To describe you need:

- the equation of a line

Example 1:

Directions: In each of the following questions, there are four alternatives, which follow the **word** or number marked as a question. You have to select one alternative, which exactly matches with the mirror of the word/number in the question.

Which among the following illustrations specifies the correct mirror image of P R E C A R I O U S?

- (a) 2UOIRACERP (b) SUOIRACERP
(c) SUOPRECARI (d) SPRECARIOU

Ans.: (a)

Example 2:

Directions: In each of the following questions, there are four alternatives, which follow the **word** or number marked as a question. You have to select one alternative, which exactly matches with the mirror of the word/number in the question.

Which among the following illustrations specifies the correct mirror image of P E R F E C T I O N?

- (a) NOITCEFERP (b) RPEFECTION
(c) NOITCEFFERP (d) ERPFECTION

Ans.: (c)

NOTES:

Rigid transformations preserve lens measures, perimeter, and area. The image and preimage are CO. These transformations include:

- Rotations
- Reflections
- Translations

Non-Rigid transformations preserve angle measure only. The side lengths and perimeter are not equal, but are in proportion. The image and preimage are similar. This transformation is a:

- Dilation

GEOMETRICAL TRANSFORMATIONS

- Conversion of One Shape to Another.
- Assembling
- Grouping

Conversion from one Shape to Another

Each and every solid that exists occupies some volume. When you convert one solid shape to another, its volume remains the same, no matter how different the new shape is. In fact, if you melt one big cylindrical candle to 5 small cylindrical candles, the sum of the volumes of the smaller candles is equal to the volume of the bigger candle.

Hence, when you convert one solid shape to another, all you need to remember is that the volume of the original, as well as the new solid, remains the same. Let us discuss some examples to understand this better.

Conversion of Shapes Examples

Example 1:

An iron ball of radius 21 cm is melted and recast into 27 spherical balls of the same radius. Find the radius of each spherical ball.

Solution:

$$\text{Volume of the iron ball} = \left(\frac{4}{3}\right) \pi r^3$$

$$= \left(\frac{4}{3}\right) \times \left(\frac{22}{7}\right) \times 21 \times 21 \times 21$$

$$= 38,808 \text{ cm}^3$$

Let us assume that the radius of the smaller balls is r .

$$\text{So volume of 27 smaller balls} = 27 \times \left(\frac{4}{3}\right) \pi r^3$$

$$= 36\pi r^3$$

Thus we have, the volume of the big iron ball = volume of 27 smaller balls

$$\Rightarrow 38,808 \text{ cm}^3 = 36\pi r^3$$

$$\Rightarrow r^3 = 343 \text{ cm}^3$$

$$\text{So } r = 7 \text{ cm}$$

Therefore, the radius of each spherical ball is 7 cm.

Example 2:

How many cylindrical candles, 1.75 cm in diameter and of thickness 2 mm, must be melted to form a cuboid candle of dimensions 5.5 cm × 10 cm × 3.5 cm?

Solution:

$$\text{Radius of cylindrical candle} = \frac{1.75}{2} = 0.875 \text{ cm}$$

$$\text{Volume of one cylindrical candle} = \pi r^2 h$$

$$= \pi \times (0.875)^2 \times (0.02) \text{ cm}^3$$

$$= \pi \times 0.0153125$$

$$= 0.048125 \text{ cm}^3$$

$$\text{Volume of cuboid candle} = 5.5 \times 10 \times 3.5$$

$$= 192.5 \text{ cm}^3$$

Thus, number of cylindrical candles

$$= \frac{\text{Volume of cuboid candle}}{\text{Volume of one cylindrical candle}}$$

$$= \frac{192.5}{0.048125} = 4000$$

Example 3:

A copper rod with a diameter 1 cm and length 8 cm is drawn into a wire of length 18 m of uniform thickness. Determine the thickness of the wire.

Solution:

Given that, Copper rod diameter = 1 cm.

Length of the copper rod = 8 cm

Length of new wire = 18 m = 18 × 100

= 1800 cm.

We know that the rod should be in the cylindrical shape.

$$\text{Hence, the volume of the rod} = \pi \times \left(\frac{1}{2}\right)^2 \times 8$$

$$= 2\pi \text{ cm}^3$$

Therefore, the volume of the copper rod = $2\pi \text{ cm}^3$

If “r” is the radius of cross-section of the wire, then the volume of the wire is given as:

$$\text{The volume of the wire} = \pi \times r^2 \times 1800$$

Since the volume of the copper rod and the volume of the new wire should be equal, then we can write

$$\Rightarrow \pi \times r^2 \times 1800 = 2\pi$$

$$\Rightarrow r^2 = \frac{2\pi}{1800\pi}$$

$$\Rightarrow r^2 = \frac{1}{900}$$

$$\Rightarrow r = \frac{1}{30}$$

Hence, the thickness of the wire should be the diameter of the cross-section of the new wire.

$$\text{Thickness} = \left(\frac{1}{30}\right) \times 2 = \frac{1}{15} \text{ cm.}$$

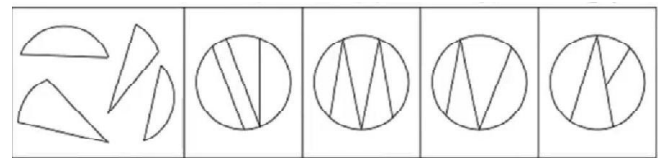
Thus, the thickness of wire is approximately equal to 0.067 cm.

Assembling:

Assembling is a process of finding the combined image with the help of visualization of it's different-different shapes.

Example 1:

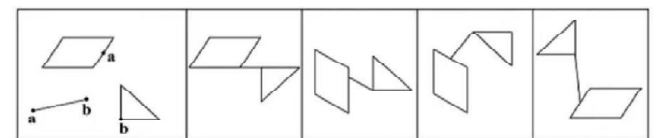
(a) (b) (c) (d)



Answer. (c)

Example 2:

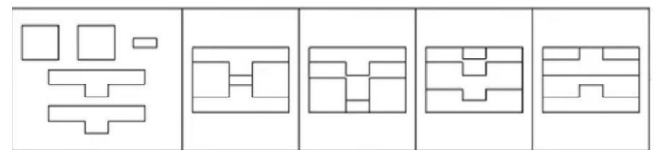
(a) (b) (c) (d)



Answer. (c & d)

Example 3:

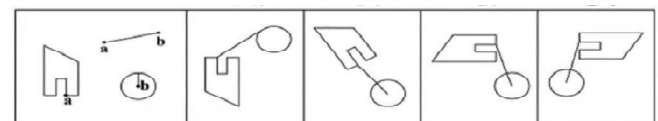
(a) (b) (c) (d)



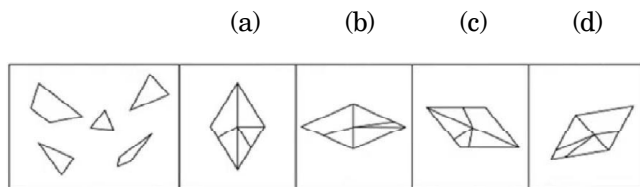
Answer. (a)

Example 4:

(a) (b) (c) (d)



Answer. (b, c & d)

Example 5:**Answer.** (d)**Grouping of Figures**

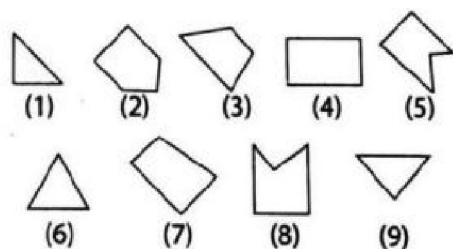
In Grouping of Figures, different figures are put together on the basis of a certain rule, pattern or property. Here in this section on the grouping of figures, we will solve many questions that are similar to the previous year questions and concepts. In these questions, a given set of figures will be featured. Some of these figures can be grouped together while as others can't be. The task is to find out the rules and/or similarities between each individual set.

In this type of questions, you are given a set of usually 6, 7 or 9 etc. figures, which are numbered. The candidate is required to analyse these figures and classify them into groups consisting of figures having more or less the same properties.

The candidate is required to analyse these figures and classify them into groups consisting of figures having more or less than the same properties. The best answer is to be selected from a given set of fairly close alternatives.

Example 1:

Directions: In the following question, group the given figures into three classes using each figure only once.

Problem Figure:

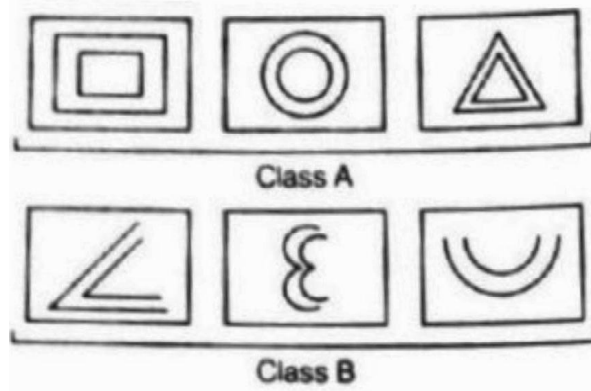
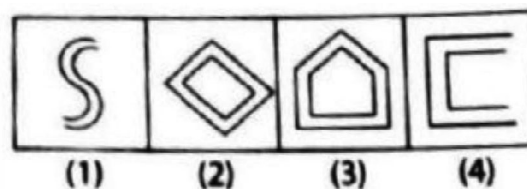
- (a) (7, 8, 9) (2, 4, 3) (1, 5, 6)
 (b) (1, 3, 3) (4, 5, 7) (6, 8, 9)
 (c) (1, 6, 8) (3, 4, 7) (2, 5, 9)
 (d) (1, 6, 9) (3, 4, 7) (2, 5, 8)
 (e) none of the above

Answer:

The figures (1), (6) and (9) are all triangles while as the figures (3), (4) and (7) are all **quadrilaterals**. Also, the figures (2), (5) and (8) are all pentagons. So these groups are identical and the correct answer is thus D.

Example 2:

Directions: There are two classes of three figures each. Class 'A' figures differ in a certain way from the figures in class 'B'. Which of the four answer **figures** belong to class 'A'?

Problem Figure:**Answer Figures:**

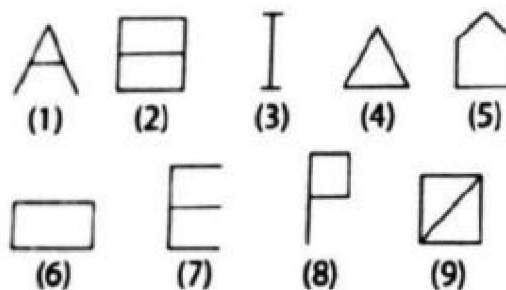
- (a) Both (1) and (3)
 (b) (1) and (2) both
 (c) Both (2) and (4)
 (d) (2) and (3) only
 (e) None of the above

Answer:

The answer here is very simple. Each figure in class A consists of two similar closed figures, which are placed one inside the other. Therefore the answer is that (2) and (3) both are correct and the correct option is (d). (2) and (3) only.

Example 3:

In the following question, group the given figures into three classes using each figure only once.

Problem Figure:

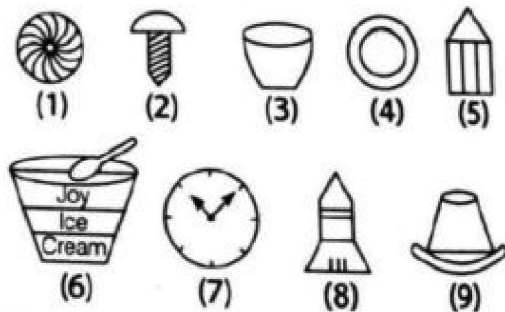
- (a) (1, 3, 4) (2, 5, 9) (6, 7, 8)
 (b) (1, 2, 3) (4, 5, 6) (7, 8, 9)
 (c) (1, 5, 9) (2, 4, 7) (3, 6, 8)
 (d) (3, 7, 8) (1, 6, 5) (4, 2, 9)
 (e) none of the above

Answer:

If we start grouping the figures on the basis of sides, we have:
 The figures that are made by three lines are (1, 3, 4)
 Also, the figures which are made by four **lines** are (6, 7, 8)
 Similarly, the figures made by the five lines are (2, 5, 9).
 These three groups are present in the option A. Thus the answer is (a).

Example 4:

In the following question, group the given figures into three classes using each figure only once.

Problem Figure:

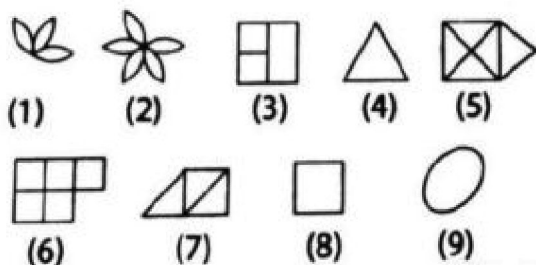
- (a) (1, 4, 7) (2, 5, 8) (3, 6, 9)
 (b) (1, 3, 6) (2, 5, 8) (4, 7, 9)
 (c) (1, 2, 4) (3, 5, 8) (6, 7, 9)
 (d) (1, 4, 9) (2, 5, 8) (3, 6, 7)
 (e) none of the above

Answer:

There are figures that have one small and one big circle are figure 1, figure 4, and figure 7. Figures which have one sharp point are the figure 2, figure 5, and figure 8. Also, the figures that have **shapes** like buckets are the figure 3, figure 6, and figure 9.

Example 5:

In the following question, group the given figures into three classes using each figure only once.

Problem Figure:

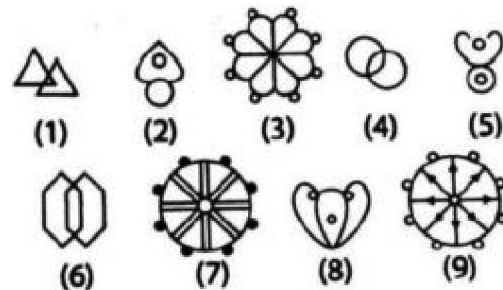
- (a) (1, 2, 3) (4, 8, 9) (5, 7, 6)
 (b) (4, 5, 7) (3, 1, 2) (7, 8, 9)
 (c) (1, 2, 7) (8, 9, 4) (2, 3, 6)
 (d) (3, 5, 6) (8, 7, 4) (9, 1, 2)
 (e) none of the above

Answer:

The figures 1, 3 and 7 have three petals, three rectangles and three triangles respectively. So, they should be grouped in one group figures 4, 8 and 9 have one triangle, one **rectangle** and one circle respectively. So, they should be grouped into one group. On the other hand, the figures between 2, 5 and 6 have five petals, five triangles and five **squares**, respectively. So, they should be grouped into one group. Therefore the correct answer is (c). (1, 2, 7) (8, 9, 4) (2, 3, 6).

Example 6:

In the following question, group the given figures into three classes using each figure only once.

Problem Figure:

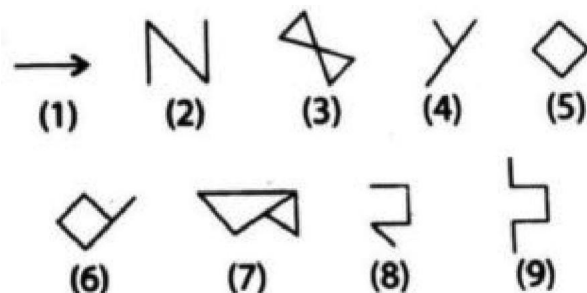
- (a) (1, 4, 8) (2, 5, 7) (3, 9, 6)
 (b) (1, 4, 6) (2, 5, 8) (3, 7, 9)
 (c) (1, 4, 6) (2, 5, 7) (3, 8, 9)
 (d) (1, 2, 3) (4, 5, 6) (7, 8, 9)

Answer:

The figures 1, 4 and 6 have two **triangles**, two circles and two **hexagons**, respectively. Figures 2, 5 and 8 are similar in shape. In figures 3, 7 and 9 there are eight small **circles** around a big circle. Now let us move on to more solved examples on the Grouping of Figures.

Example 7:

In the following question, group the given figures into three classes using each figure only once.

Problem Figure:

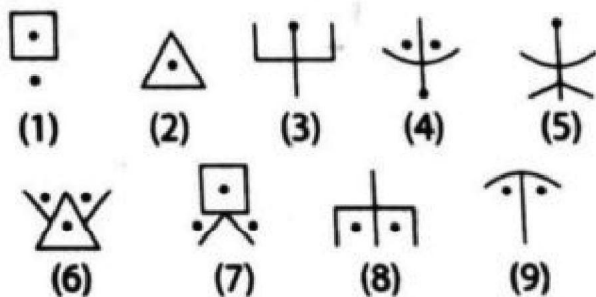
- (a) (1, 5, 6) (2, 3, 4) (7, 8, 9)
 (b) (1, 2, 4) (3, 5, 8) (6, 7, 9)
 (c) (5, 6, 7) (1, 2, 4) (3, 8, 9)
 (d) (1, 2, 4) (3, 5, 7) (6, 8, 9)
 (e) none of the above

Answer:

In the question figure we see that the figures that we label 1, 2 and 4 all consist of three lines. Also, the figures 3, 5 and 8 all consist of four lines each. Similarly, we can see that the figures 6, 7 and 9 all consist of five lines. Therefore this gives us the grouping of the figures. Thus the correct option is (b).

Example 8:

In the following question, group the given figures into three classes using each figure only once.

Problem Figure:

- (a) (1, 7, 8) (2, 6, 5) (3, 4, 9)
 (b) (1, 8, 9) (2, 3, 5) (4, 6, 7)
 (c) (2, 3, 5) (1, 7, 8) (4, 6, 9)
 (d) (2, 6, 7) (1, 3, 4) (5, 8, 9)
 (e) none of the above

Answer:

As is clear from the question figure, the figures 2, 3 and 5 have one point or in other words, we can say that they are a one-pointed system. Also, we can see that the figures 1, 8 and 9 are made up of four lines. Below we have some practice problems on the Grouping of Figures.

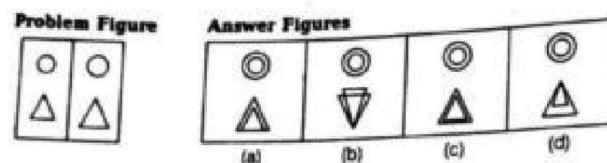
PAPERFOLDING AND CUTTING

The Paper Folding and Cutting involves a **process** in which a transparent sheet is folded and then some cuts and/or folds are made. In the questions based on Paper Folding and Cutting a few **figures** are given showing the way in which a piece is to be folded and then cut from a particular section. The dotted line is the reference line along which the paper is to be folded and the arrow indicates the **direction** of the fold.

Thus, these figures indicate the **sequence** in which the paper is to be folded. The designs from the cut will appear on each one of the folds made on the paper. In questions based on paper folding and cutting it can be asked to find either the folded or unfolded pattern of the sheet. Let us see some examples below. We also have some really tricky problems that have been put under the higher **order** thinking section.

Example 1:

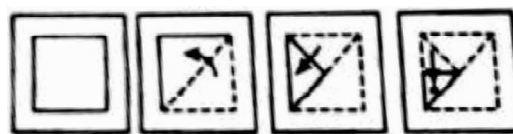
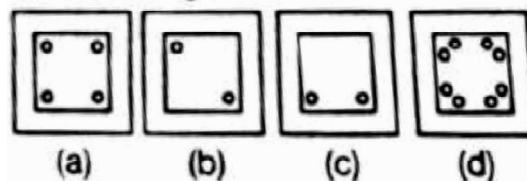
A square transparent sheet with a pattern is given. Figure out from among the four alternatives as to how the pattern would appear when the transparent sheet is folded at the middle line shown?

**Answer:**

If you look at the options, you would see that the option (a) can't be the answer because if the sheet was folded at the middle the **triangles** will be joined at the vertex. Similarly, you can discard all the other options except (d). The correct option is (d).

Example 2:

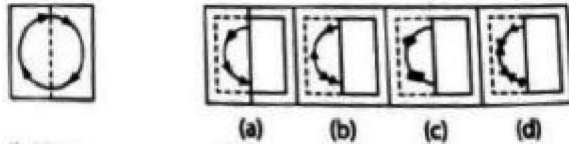
A piece of paper is folded and punched as shown below in the question figures. from the given answer figures, indicate how it will appear when opened?

Problem Figures**Answer Figures****Answer:**

In the last problem figure, we see that the paper is folded in the form of a right triangle and the cut is made near the midpoint of the hypotenuse inside the triangle. Unfolding the triangle would give us another right triangle which has marks near the two sides apart from the hypotenuse. At this point, we can say that the answer could be either (a) or (d). Similarly, if we keep unfolding, we will see that the correct answer is (d).

Example 3:

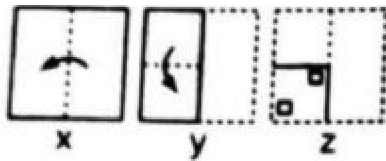
Directions: In each of the following problems, a **square** transparent sheet with a pattern is given. Figure out from amongst the four alternatives as to how the pattern would appear when the transparent sheet is folded along the dotted line

Transparent Sheet Answer Figures**Answer:**

The right half of the transparent sheet is being folded along the dotted line and placed on the left half of the sheet. The figure thus obtained resembles the answer figure (c).

Example 4:

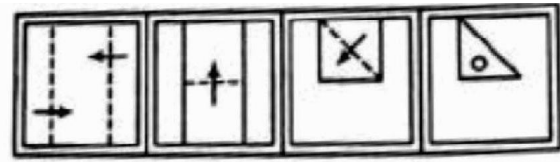
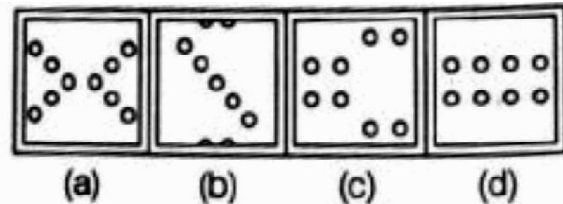
Directions: In the following question, a piece of paper is folded and cut as shown below in the question figures. From the given answer figures, indicate how it will appear when opened?

Problem Figures**Answer Figures**

Answer: In these types of questions, the easiest thing to do is to start with the last picture and undo the steps taken in the question. Let us see, if we start from the figure z, and move to y, we will get a figure that has square cuts as half diagonals in the left half only. From the figure y we have to now move to the figure x, that would give a figure similar to the one in (a). Hence the correct answer here is (a).

Example 5:

A piece of paper is folded and punched as shown below in the question figures. From the given answer figures, indicate how it will appear when opened?

Problem Figures**Answer Figures****Answer:**

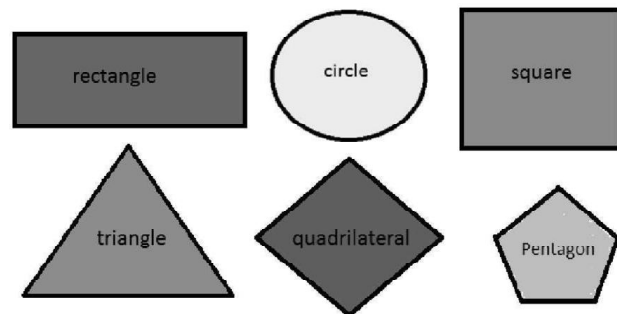
As we have done with the other questions, we will begin with the last problem figure and move on to the first one, while tracing out all the spots on each step. Whence we do that we will see that out of the four options, the only one that represents the answer figure is the one in option (a).

PATTERNS IN 2 AND 3 DIMENSIONS**2D Shape's Pattern:**

In geometry, a shape or a figure that has a length and a breadth is a 2D shape pattern. In other words, a plane object that has only length and breadth is 2 dimensional. Straight or **curved lines** make up the sides of this shape. Also, these figures can have any number of sides. In general, plane figures made of lines are known as polygons. For example, triangle and square are **polygons**.

Examples of 2D shapes

Rectangle, circle, square, triangle, quadrilateral and pentagon are some examples of 2D shapes.

**2D Rotation:**

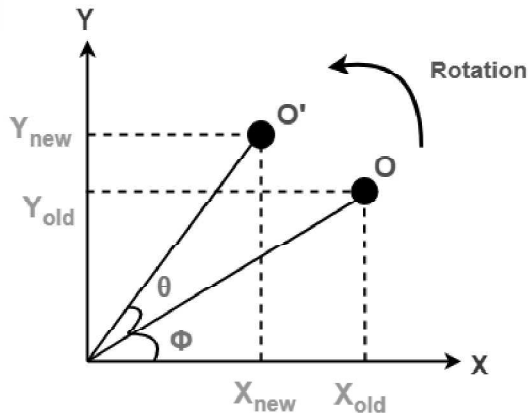
2D Rotation is a process of rotating an object with respect to an angle in a two dimensional plane.

Consider a point object O has to be rotated from one angle to another in a 2D plane.

Let-

- Initial coordinates of the object O = (X_{old}, Y_{old})

- Initial angle of the object O with respect to origin = ϕ
- Rotation angle = θ
- New coordinates of the object O after rotation = $(X_{\text{new}}, Y_{\text{new}})$



2D Rotation

This rotation is achieved by using the following rotation equations-

- $X_{\text{new}} = X_{\text{old}} \times \cos\theta - Y_{\text{old}} \times \sin\theta$
- $Y_{\text{new}} = X_{\text{old}} \times \sin\theta + Y_{\text{old}} \times \cos\theta$

Example:

Given a line segment with starting point as (0, 0) and ending point as (4, 4). Apply 30-degree rotation anticlockwise direction on the line segment and find out the new coordinates of the line.

Solution:

We rotate a straight line by its end points with the same angle. Then, we re-draw a line between the new end points.

Given-

- Old ending coordinates of the line = $(X_{\text{old}}, Y_{\text{old}}) = (4, 4)$
- Rotation angle = $\theta = 30^\circ$

Let new ending coordinates of the line after rotation = $(X_{\text{new}}, Y_{\text{new}})$.

Applying the rotation equations, we have-

X_{new}

$$= X_{\text{old}} \times \cos\theta - Y_{\text{old}} \times \sin\theta$$

$$= 4 \times \cos 30^\circ - 4 \times \sin 30^\circ$$

$$= 4 \times \left(\frac{\sqrt{3}}{2}\right) - 4 \times \left(\frac{1}{2}\right) = 2\sqrt{3} - 2$$

$$= 2(\sqrt{3} - 1) = 2(1.73 - 1)$$

$$= 1.46$$

Y_{new}

$$= X_{\text{old}} \times \sin\theta + Y_{\text{old}} \times \cos\theta$$

$$= 4 \times \sin 30^\circ + 4 \times \cos 30^\circ$$

$$= 4 \times \left(\frac{1}{2}\right) + 4 \times \left(\frac{\sqrt{3}}{2}\right) = 2 + 2\sqrt{3}$$

$$= 2(1 + \sqrt{3}) = 2(1 + 1.73) = 5.46$$

Hence, New ending coordinates of the line after rotation = (1.46, 5.46).

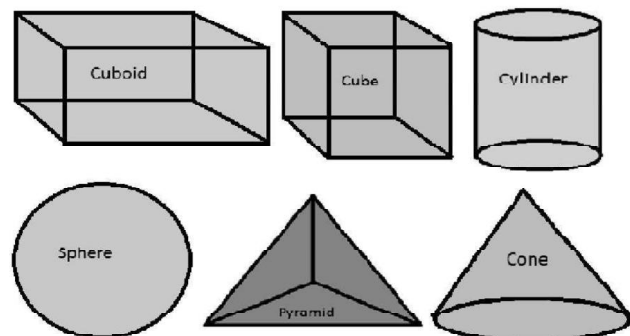
3d Shapes Pattern:

We come across a number of objects of different shapes and sizes in our day-to-day life. There are golf balls, doormats, ice-cream cones, coke cans, and so on. These objects have different characteristic properties such as length, breadth, diameter, etc. which set them apart from one another. But no matter how different their dimensions are, all of them occupy space and have three dimensions. So, they are referred to as three-dimensional or solid shapes. In this article, we will discuss the various 3D shapes, surface area and volumes, and the process of making 3D shapes using nets with the help of 2D Shapes.

In Geometry, 3D shapes are known as three-dimensional shapes or solids. 3D shapes have three different measures such as length, width, and height as its dimensions. The only difference between 2D shape and 3D shapes is that 2D shapes do not have a thickness or depth. Usually, 3D shapes are obtained from the rotation of the 2D shapes. The faces of the solid shapes are the 2D shapes. Some examples of the 3D shapes are a cube, cuboid, cone, cylinder, sphere, prism and so on. The 3D shapes consist of both curved shaped solid and the straight-sided polygon called the polyhedron. The polyhedrons are also called the polyhedrons, which are based on the 2D shapes with straight sides. Now, let us discuss the details about the polyhedrons and curved solids.

Examples of 3D Shapes

Cuboid, cube, cylinder, sphere, pyramid and cone are a few examples of 3D shapes



3D Rotation:

3D Rotation is a process of rotating an object with respect to an angle in a three-dimensional plane.

Consider a point object O has to be rotated from one angle to another in a 3D plane.

Let-

- Initial coordinates of the object O = $(X_{old}, Y_{old}, Z_{old})$
- Initial angle of the object O with respect to origin = ϕ
- Rotation angle = θ
- New coordinates of the object O after rotation = $(X_{new}, Y_{new}, Z_{new})$

In 3 dimensions, there are 3 possible types of rotation-

- X-axis Rotation
- Y-axis Rotation
- Z-axis Rotation

For X-Axis Rotation-

This rotation is achieved by using the following rotation equations-

- $X_{new} = X_{old}$
- $Y_{new} = Y_{old} \times \cos \theta - Z_{old} \times \sin \theta$
- $Z_{new} = Y_{old} \times \sin \theta + Z_{old} \times \cos \theta$

For Y-Axis Rotation-

This rotation is achieved by using the following rotation equations-

- $X_{new} = Z_{old} \times \sin \theta + X_{old} \times \cos \theta$
- $Y_{new} = Y_{old}$
- $Z_{new} = Y_{old} \times \cos \theta - X_{old} \times \sin \theta$

For Z-Axis Rotation-

This rotation is achieved by using the following rotation equations-

- $X_{new} = X_{old} \times \cos \theta - Y_{old} \times \sin \theta$
- $Y_{new} = X_{old} \times \sin \theta + Y_{old} \times \cos \theta$
- $Z_{new} = Z_{old}$

Example:

Given a homogeneous point (1, 2, 3). Apply rotation 90 degree towards X, Y and Z axis and find out the new coordinate points.

Solution:

Given-

- Old coordinates = $(X_{old}, Y_{old}, Z_{old}) = (1, 2, 3)$
- Rotation angle = $\theta = 90^\circ$

For X-Axis Rotation-

Let the new coordinates after rotation = $(X_{new}, Y_{new}, Z_{new})$.

Applying the rotation equations, we have-

- $X_{new} = X_{old} = 1$
- $Y_{new} = Y_{old} \times \cos \theta - Z_{old} \times \sin \theta$
 $= 2 \times \cos 90^\circ - 3 \times \sin 90^\circ$
 $= 2 \times 0 - 3 \times 1 = -3$
- $Z_{new} = Y_{old} \times \sin \theta + Z_{old} \times \cos \theta$
 $= 2 \times \sin 90^\circ + 3 \times \cos 90^\circ$
 $= 2 \times 1 + 3 \times 0 = 2$

Thus, New coordinates after rotation = (1, -3, 2).

For Y-Axis Rotation-

Let the new coordinates after rotation = $(X_{new}, Y_{new}, Z_{new})$.

Applying the rotation equations, we have-

- $X_{new} = Z_{old} \times \sin \theta + X_{old} \times \cos \theta$
 $= 3 \times \sin 90^\circ + 1 \times \cos 90^\circ$
 $= 3 \times 1 + 1 \times 0 = 3$
- $Y_{new} = Y_{old} = 2$
- $Z_{new} = Y_{old} \times \cos \theta - X_{old} \times \sin \theta$
 $= 2 \times \cos 90^\circ - 1 \times \sin 90^\circ$
 $= 2 \times 0 - 1 \times 1 = -1$

Thus, New coordinates after rotation = (3, 2, -1).

For Z-Axis Rotation-

Let the new coordinates after rotation = $(X_{new}, Y_{new}, Z_{new})$.

Applying the rotation equations, we have-

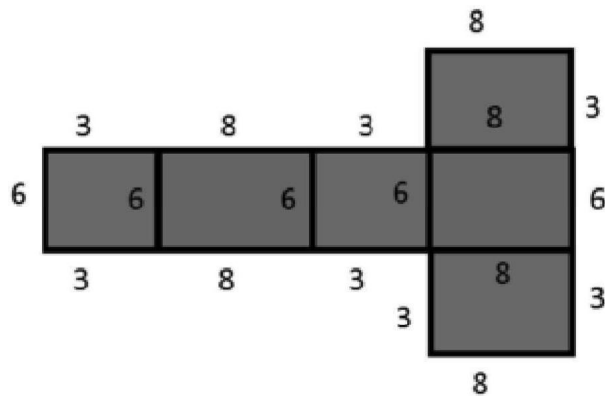
- $X_{new} = X_{old} \times \cos \theta - Y_{old} \times \sin \theta$
 $= 1 \times \cos 90^\circ - 2 \times \sin 90^\circ$
 $= 1 \times 0 - 2 \times 1 = -2$
- $Y_{new} = X_{old} \times \sin \theta + Y_{old} \times \cos \theta$
 $= 1 \times \sin 90^\circ + 2 \times \cos 90^\circ$
 $= 1 \times 1 + 2 \times 0 = 1$
- $Z_{new} = Z_{old} = 3$

Hence, New coordinates after rotation = (-2, 1, 3).

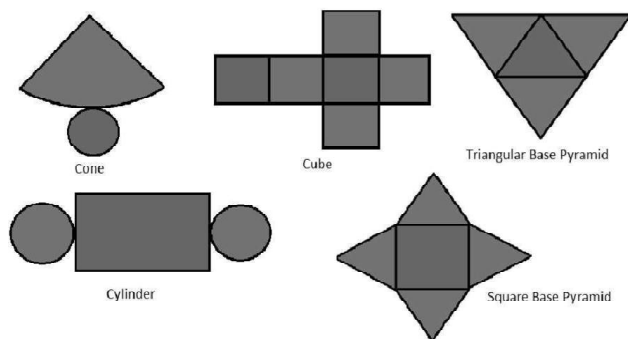
Nets for Building 3-D Shape Pattern:

A geometric net is a two-dimensional 'pattern' for a three-dimensional object that is unfolded along its edges. It represents each face of the figure in two dimensions. In other words, a net is a pattern made when the surface of a three-dimensional figure is laid out. Thus, showing each face of the figure. A solid may have different nets.

For example, a box is solid. It's a 3D object with the shape of a cuboid. Below is a net pattern for a box. Copy an enlarged version of the net and try to make the box by folding and gluing the faces together. You may use suitable units.



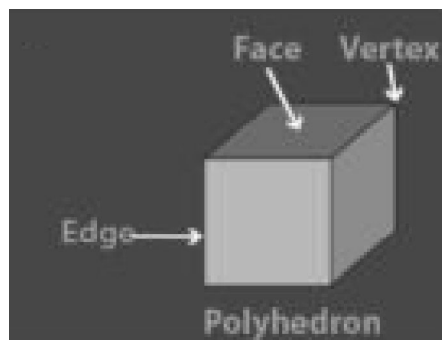
Furthermore, different shapes have different nets.



Example of 3D Shape Pattern:

Polyhedron

A solid shape bound by polygons forms a **polyhedron**. The word polyhedral is the plural of word polyhedron.



- **Faces:** Polygons forming a polyhedron are its faces.
- **Edges:** Line segments common to intersecting faces of a polyhedron are its edges.
- **Vertices:** Points of intersection of edges of a polyhedron are its vertices.

By Euler's Formula:

For Polyhedron

$$F + V = E + 2$$

Where,

F = Number of faces of polyhedron

V = Number of vertices of polyhedron

E = Number of Edges of polyhedron

A net is a 2- dimensional representation of a 3- dimensional figure that is unfolded along its edges so that each face is in 2- dimension. So, a given polyhedron can have different nets.

Polyhedrons are straight-sided solids, which has the following properties:

- It should have flat sides are called the faces
- It must have the corners, called vertices
- Polyhedrons should have straight edges.

Like polygons in two-dimensional shapes, polyhedrons are also classified into regular and irregular polyhedrons and convex and concave polyhedrons. The most common examples of polyhedral are cube, cuboid, pyramid, prism. Some other examples of regular polyhedron are tetrahedron, octahedron, dodecahedron, icosahedron, and so on. These regular polyhedrons are also known as the platonic solids, whose faces are identical to each face.

For example, the most commonly used example of a polyhedron is a cube, which has 6 faces, 8 vertices, and 12 edges.

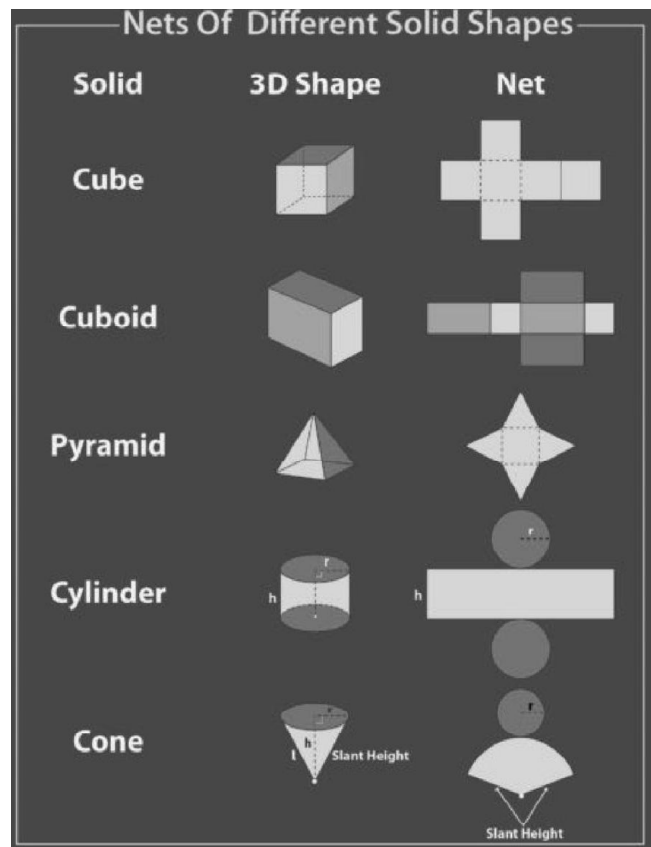
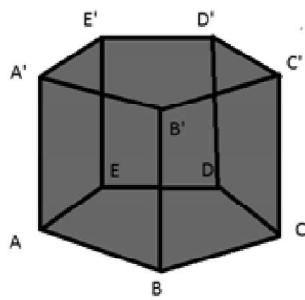
In a polyhedron, three or more edges meet at a point to form a vertex. Some examples of polyhedrons are cuboid, cube, pyramid, and triangular pyramid.

Curved Solids

Apart from polyhedrons, there are 3D shapes with curved shapes such as the sphere, cone, cylinder, etc. For example, cones have a circular base that narrows smoothly from the circular base to the point called the apex. All these shapes have curved faces, and hence they are called curved solids.



A prism is a solid, whose side faces are **parallelograms** and whose bases are congruent parallel rectilinear figures.



Shape Matching in 2D & 3D Pattern:

1. Shape Matching – Two-Dimensional pattern:

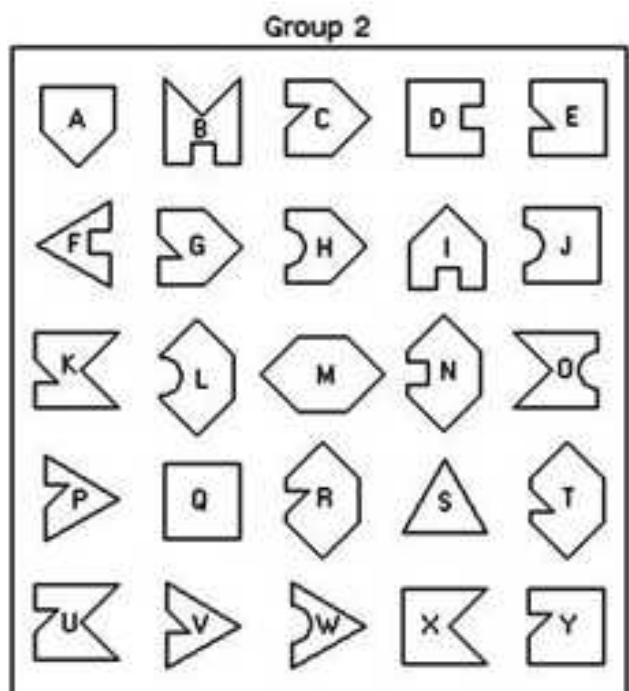
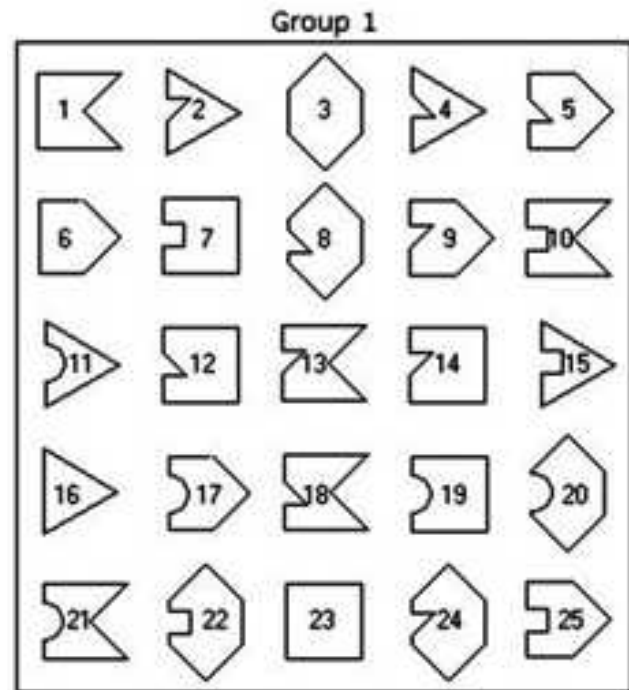
Let take an example, you are asked to look at two groups of simple, flat objects and find pairs that are exactly the same size and shape.

Each group has 25 small drawings of these two-dimensional objects. The objects in the first group are labelled with numbers and are in **numerical** order. The objects in the second group are labelled with letters in alphabetical order.

Each drawing in the first group is exactly the same as a drawing in the second group. The objects in the second group have been **moved** and some have been **rotated**.

Example:

Which shape in Group 2 corresponds to a shape in Group 1?



Answers:

- | | | | | |
|--------|--------|--------|--------|--------|
| 1. X; | 2. P; | 3. M; | 4. V; | 5. G; |
| 6. A; | 7. D; | 8. T; | 9. C; | 10. B; |
| 11. W; | 12. E; | 13. U; | 14. Y; | 15. F; |
| 16. S; | 17. H; | 18. K; | 19. J; | 20. L; |
| 21. O; | 22. N; | 23. Q; | 24. R; | 25. I. |

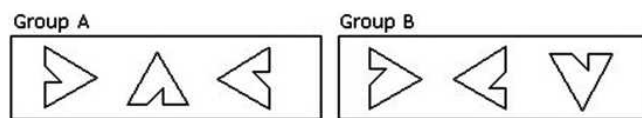
This is very much a speed test, as you will be given far more of these questions than you can comfortably answer in the given time.

These questions use a large number of shapes that are presented close together. Some people find this very distracting and find it easier to work through the shapes in the second group systematically, rather than trying to look at the whole group at once.

In some questions of this type, there may not be a one-to-one match and some of the shapes in the first group may not appear in the second. The way that the question is worded will make this clear.

You should be especially careful to look out for reflections in the second group as these will be incorrect. These are often put in by the test designers to trap the unsuspecting.

For example:



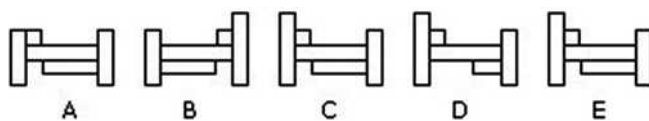
The shapes in Group A are rotations of each other. The shapes in Group B are also rotations of each other. However, none of the shapes in Group A are rotations of those in Group B.

2. Visual Comparison – Two Dimensional pattern:

In these types of question, you will be presented with several objects, only two of which are identical. Once again, these are speed questions and you will need to work quickly and attempt to answer as many as possible in the given time.

Example:

Which two pictures are identical?



Answer:

C and E are the only two pictures that are identical.

The best strategy for these types of questions is to begin with the shape on the left and work through the shapes to the right of it, systematically looking for an exact match. If there isn't one then move on to the second shape and repeat the process.

It can be quite difficult to discipline yourself to adopt this systematic approach, as you may think that it is quicker just to look at all of the shapes until the

answer 'jumps out' at you. The problem with this is that if the answer doesn't 'jump out' fairly quickly then panic sets in and you usually resort to the systematic approach anyway.

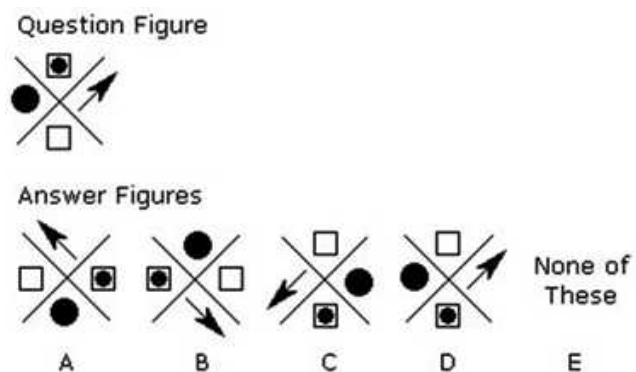
When you are confronted with a question like this that asks you to compare several different shapes, the other answer options can be distracting. It is much easier and less mentally tiring to cover the other shapes and concentrate only on the two shapes that you are comparing.

3. Group Rotation – Two Dimensional pattern:

This is slightly more complex than the rotations in the two-dimensional shape matching question.

Example:

Which of the answer figures is a rotation of the question figure?



Answer: C

Once again, you need to be careful that you don't identify reflections. The best strategy is to choose the most asymmetrical shape in the group – in this case, the arrow. Then determine how the shape would look going clockwise, anticlockwise and opposite.

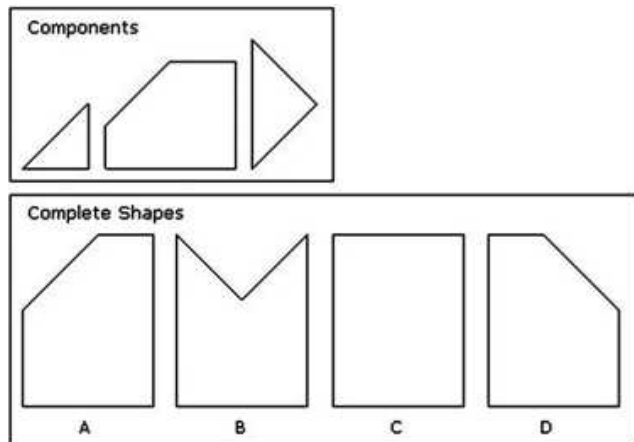
In the example above, the white square is clockwise from the arrow. This means that A, B and D cannot be rotations of the question figure. This leaves only C as a possibility, which can quickly be checked element by element.

4. Combining Two-Dimensional Pattern:

These questions show you a series of two-dimensional pattern. One of the shapes has been cut up into pieces. The questions present you with the pieces and you are asked to work out which of the shapes has been cut-up.

Example:

Which of the complete shapes can be made from the components shown?

**Spatial Ability****Answer: B**

The best strategy for answering these questions is to look at the complete shapes and see if there are any distinct features that would make it impossible to construct such a shape from the components. Sometimes there are one or two shapes that can be immediately discounted based on size alone.

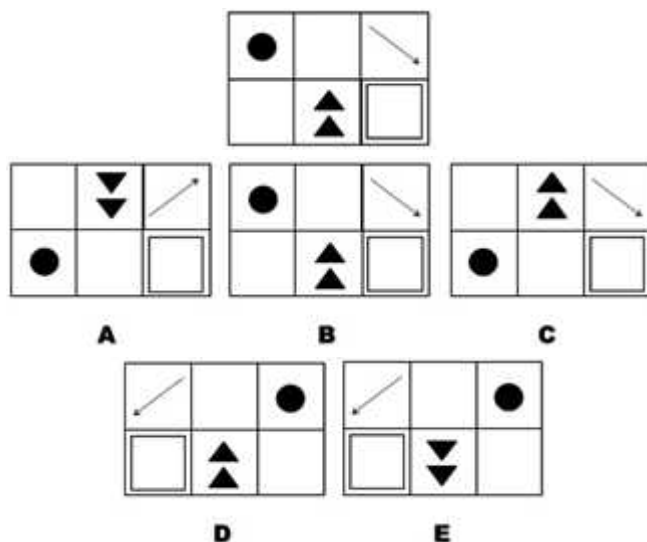
One thing to remember is that if the complete shapes don't have any bits sticking out (they usually don't) then the components must fit together so that sides of the same length are together. This reduces the number of combinations considerably.

5. Two-Dimensional Mirror Reflections:

These questions will assess how you view two-dimensional objects from a different perspective. You will be presented with an image and will be asked to identify its mirror image.

Example:

Which answer shows a reflection of the image below?

**Answer: D**

First, check the answer options to see if any elements have changed position. If so, you can eliminate these answers. This will eliminate answer A straight away as the square and arrow have swapped places.

Then look for any elements which have been manipulated differently. Answer E appears to be a mirror image, but the triangles have been incorrectly inverted.

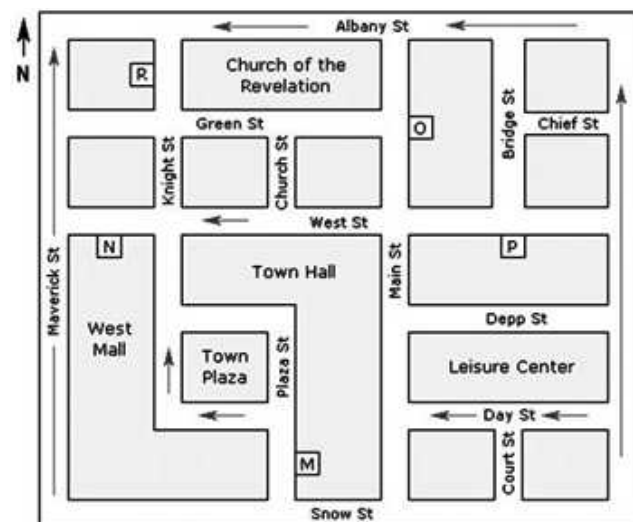
Answer C is an example where the square and arrow have changed position, but the triangles are also incorrectly inverted. Therefore, this can not be a reflection and can be eliminated.

Answer B is exactly the same as the question image, it hasn't been reflected at all. This answer can also be eliminated.

6. Two-Dimensional Maps

Map questions will assess your ability to follow basic instructions and visualise a route. Normally, you will be given a two-dimensional plan or map and a series of directions via two or three questions that all relate to the same map.

You just need to use the instructions to navigate the map. As long as you have a basic sense of direction, this will seem relatively straightforward, but you will need to do so quickly and accurately.

Example:

Officer Wilkinson is in Depp St and can see the Town Hall to her right. What direction is she facing?

- A B C D
North South East West

She turns and walks to the junction with Main St. She turns left and proceeds two blocks before turning right, then taking the next right, and walking half a block. Which location is nearest to her current position?

- A B C D
M N R P

Answers: B and A

The secret to getting the answers correct is imagining yourself walking down the street, making sure you are facing the correct way when deciding whether to turn left or right. You should also revise compass directions and make sure you are confident with each one, as well as the intermediate points such as north-west and south-east.

The first part of this question is pretty straightforward if you know your compass directions. You should immediately spot that to have the Town Hall to her right she must be facing south (answer B).

The next part of the question is a simple case of following the directions given. Remember that you have to see the map from the walker's perspective when choosing left or right.

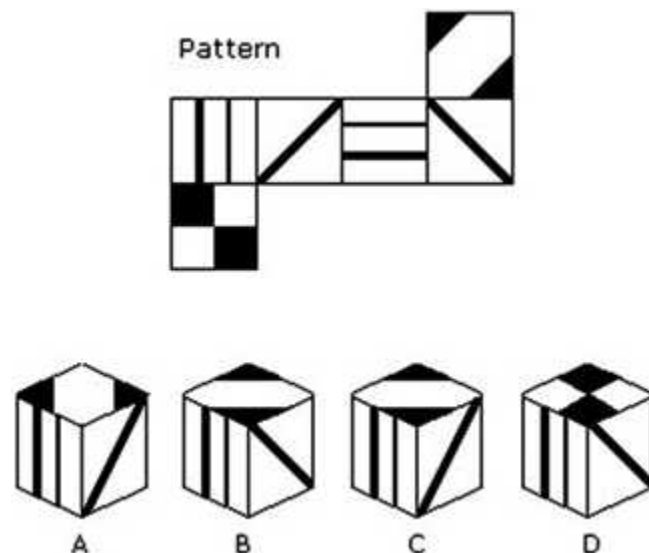
7. Cubes in Two and Three-Dimension's Pattern:

These questions show a flat (two-dimensional) pattern that can be folded to make a cube. The answer options will usually be four three-dimensional cubes.

The pattern and the cubes have symbols or markings on each face. You need to look at the pattern and decide which of the cubes, if any, could be made from it.

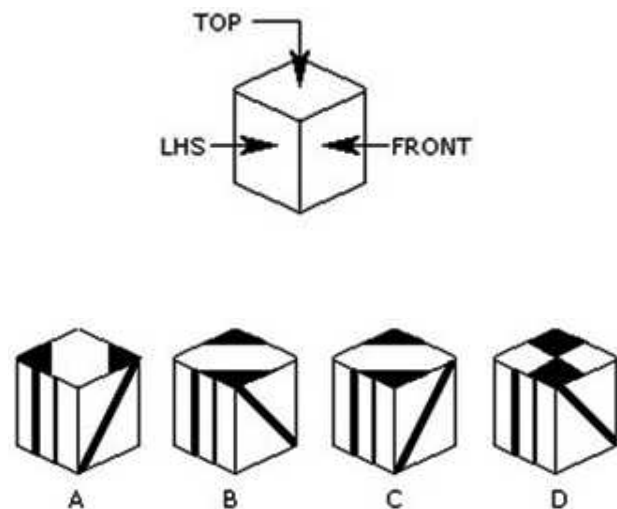
Example:

Which of the cubes shown could be made from the pattern?

**Answer: A**

The key to these questions is to remember that only three faces of the cube can be shown in the illustration, this means that you need only consider the relationship between the **three visible elements** on each cube and see if the same relationship exists in the pattern.

The best strategy for this type of question is to call one face of the cube the 'front' and then name the other faces of the cube in relation to it.

For example:

Obviously, this is an arbitrary decision as you could look at a cube from any side. However, thinking of the problem this way makes it much easier to see the relationship between the faces of the cube.

You can then:

- Look at the front of the cube
- Find that face on the pattern
- Use the pattern to identify the 'top' and eliminate any options that do not match
- Use the pattern to identify the other face that touches the 'front' (in this example LHS) and eliminate any options that do not match

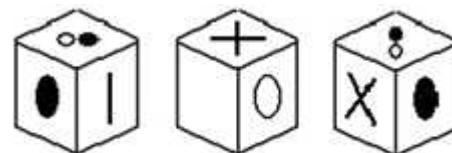
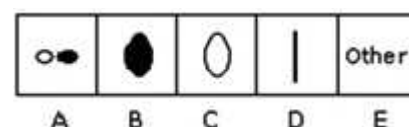
The problem may also be posed the other way around. Here you would have a single three-dimensional cube and several two-dimensional patterns, only one of which, when folded, will make the cube.

8. Cube Views in Three Dimensions pattern:

These questions show you several (usually three) views of a three-dimensional cube with unique symbols or markings on each face and then asks you a question about it. For example, which symbol is on the opposite face?

Example:

Three views of the same cube are shown above. Which symbol is opposite the X?

**Answers:**

Answer: D

Some people seem to have a natural talent for imagining objects in three dimensions and find these questions straightforward. However, if you're not one of them and you find thinking in three dimensions difficult, there are other ways to get to the answer.

In the question above, for example, you can simply use a process of elimination. If you can see a symbol next to the 'X' on the cube, then it cannot be opposite.

The second and third cubes eliminate A, B and C. This leaves only D and 'other' as possibilities. D has edges shared with A and B, which would be consistent with the third cube illustrated. Therefore, D is correct.

Although it is not usually specified in the instructions, it is almost always true that in these questions each symbol is used only once. This means that even in cases where elimination is not possible, it is sometimes quite easy to see the solution without mentally manipulating the cube too much.

In the example above, you can simply compare the first and third illustrations. The third illustration shows a 90-degree anticlockwise rotation (looking at the cube from above) of the first illustration. Therefore, D must be opposite the 'X'.

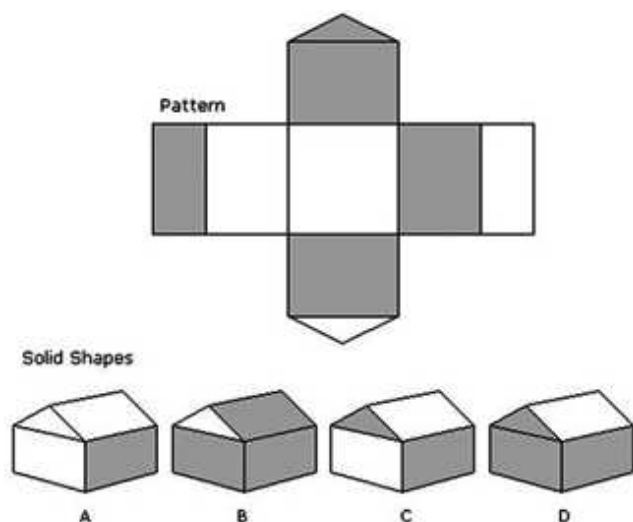
9. Other Solids in Two- and Three-Dimension's Pattern:

These questions are similar to the cube questions above but rather than cubes, they use other solid shapes which may be irregular.

In some respects, these questions are easier than the cube questions as there are more relationships to work with. In other words, each face of the solid shape has a shape of its own rather than just being square.

Example:

Which of the solid shapes shown could be made from the pattern?

**Answer: D**

Shape A can be eliminated because it shows an unshaded face below a triangular face, both of these faces (below the triangular face) are shaded on the pattern. Shape C can be eliminated for the same reason.

Shape B can be eliminated as it has a shaded roof above the shaded side, which does not appear on the pattern.

Once again, you can reduce these problems to the relationship between the visible elements on the three-dimensional picture. This makes things easier because even though the solid shape may have more faces than a cube, it is unusual for more than four faces to be shown. This means that you need to consider the relationship between the four visible faces, paying particular attention to shading or other patterns on them.

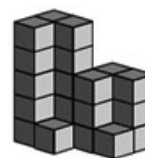
10. Block Counting in Three Dimensions Pattern:

In these questions, you will be given a three-dimensional drawing of a shape formed from a group of blocks, some of which may be hidden.

You will be asked to identify how many blocks have been used to make up the shape. You will need to visualize the blocks you cannot see.

Example:

How many blocks make up the shape below?



- | | | | | |
|----|----|----|----|----|
| 28 | 18 | 25 | 27 | 26 |
| A | B | C | D | E |

Answer: E

Try to break up the blocks in your mind. For example, in the exercise above, there is a block of three 5-cube pillars (which together makes 15 blocks), a group of three 3-cube pillars (which together makes 9 blocks), and two single blocks.

Adding together $15 + 9 + 2$ gives 26

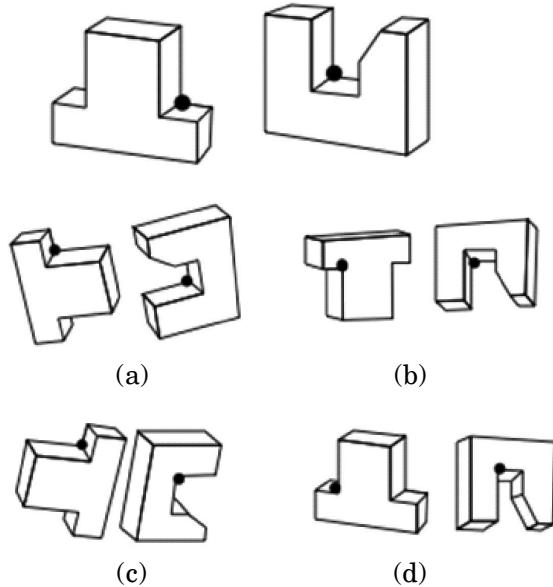
This is quicker and produces easier sums, leading to more robust answers.

EXERCISE

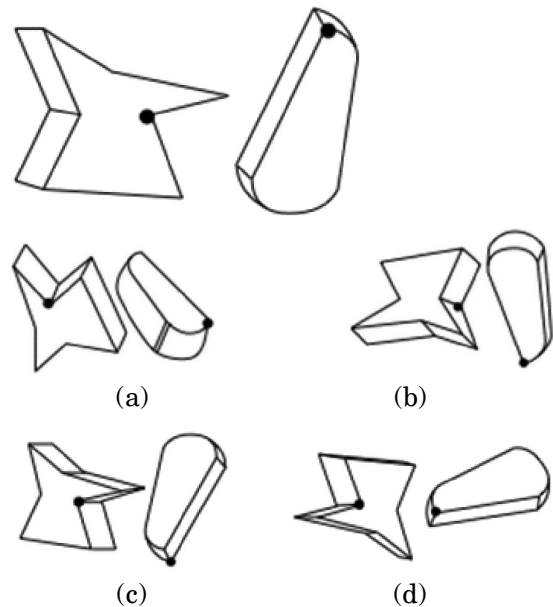
MCQ TYPE QUESTIONS

ROTATION

1. Below are two separate objects, each has a dot placed in one corner, and each has five possible rotations. Choose the answer option in which the dots are placed in the same corners of the objects as in the model.

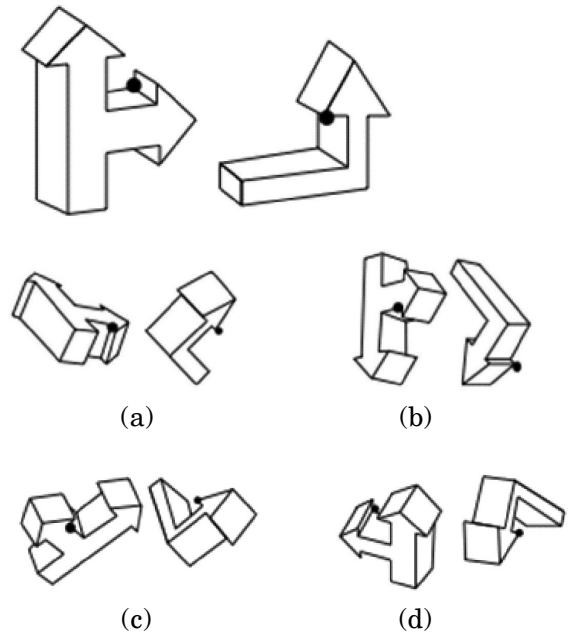


2. Below are two separate objects, each has a dot placed in one corner, and each has five possible rotations. Choose the answer option in which the dots are placed in the same corners of the objects as in the model



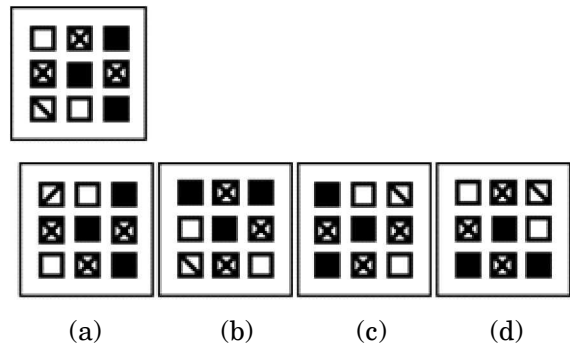
3. Below are two separate objects, each has a dot placed in one corner, and each has five possible rotations. Choose the answer option in which the

dots are placed in the same corners of the objects as in the model.



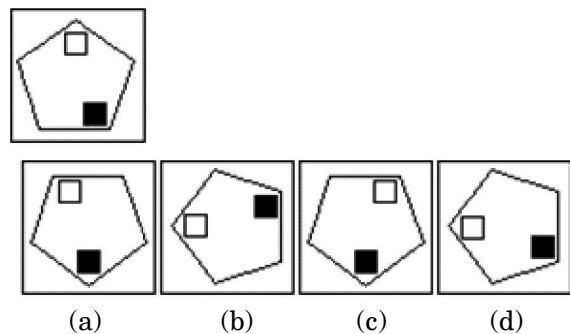
4. In the figures shown below, one of the shapes (a-d) is identical to the given figure but has been rotated.

Which figure is identical to the given figure?



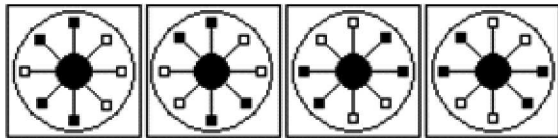
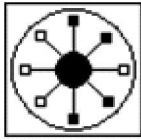
5. In the figures shown below, one of the shapes (a-d) is identical to the given figure but has been rotated.

Which figure is identical to the given figure?



6. In the figures shown below, one of the shapes (a-d) is identical to the given figure but has been rotated.

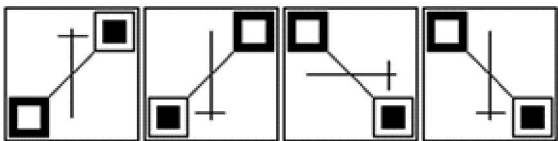
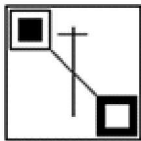
Which figure is identical to the given figure?



(a) (b) (c) (d)

7. In the figures shown below, one of the shapes (a-d) is identical to the given figure but has been rotated.

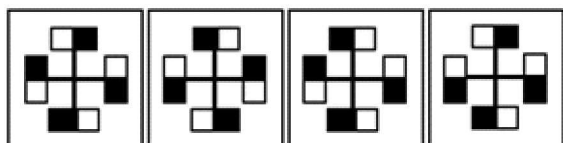
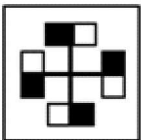
Which figure is identical to the given figure?



(a) (b) (c) (d)

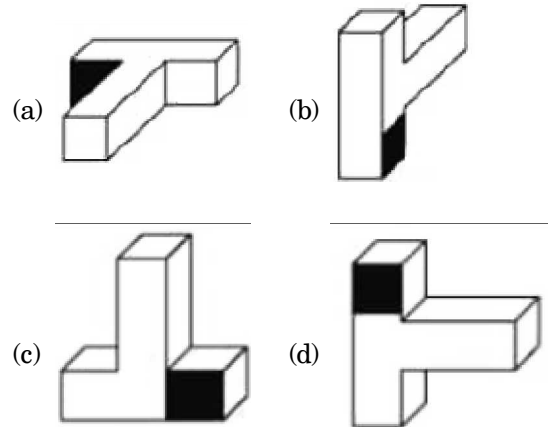
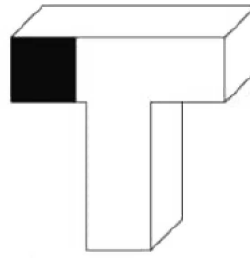
8. In the figures shown below, one of the shapes (a-d) is identical to the given figure but has been rotated.

Which figure is identical to the given figure?

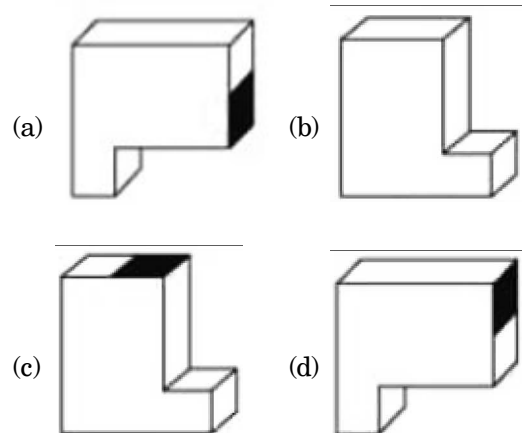
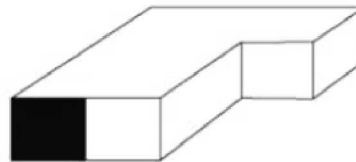


(a) (b) (c) (d)

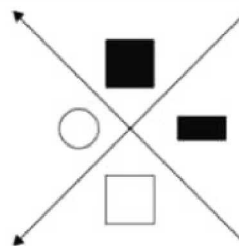
9. Which figure is a rotation of the object?

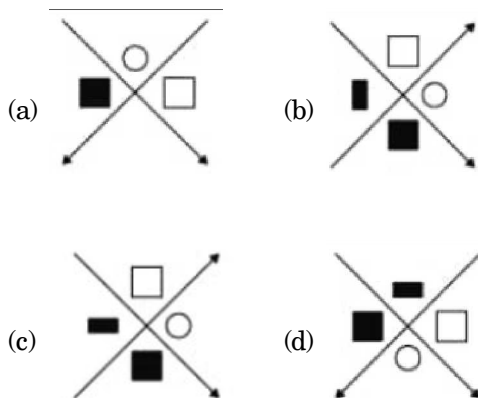


10. Which figure is a rotation of the object?

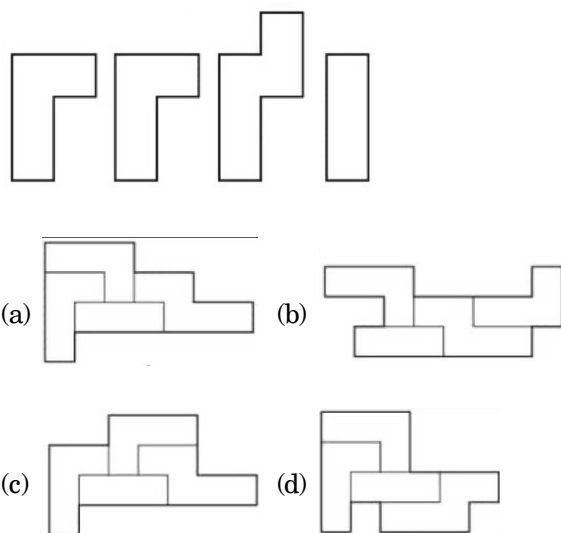


11. Which of the 4 figures presented (A, B, C, D) is a rotation of the first?



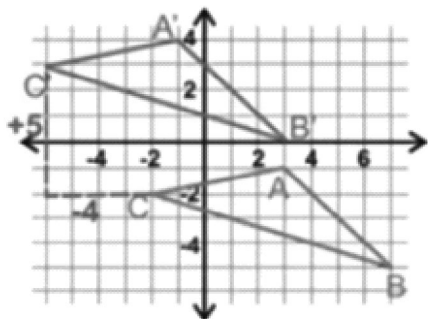


12. Which figure can be formed using only 2D rotation?



TRANSLATION

13. Triangle $\triangle ABC$ has coordinates $A(3, -1)$, $B(7, -5)$ and $C(-2, -2)$. Translate $\triangle ABC$ to the left 4 units and up 5 units. Determine the coordinates of $\triangle A'B'C'$.

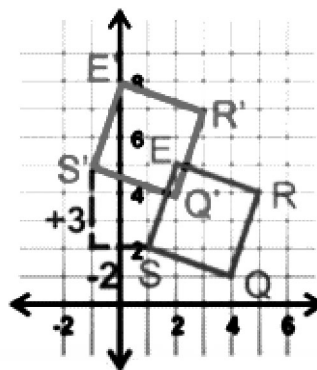


- (a) $(-1, 4)$, $(3, 0)$, $(-6, 3)$ (b) $(1, 4)$, $(0, 3)$, $(-4, 3)$
 (c) $(-1, 5)$, $(3, 0)$, $(-6, 3)$ (d) None of these.

14. Using the translation $(x, y) \rightarrow (x + 2, y - 5)$, what is the image of $A(-6, 3)$, $B(2, 5)$, $C(3, 1)$?

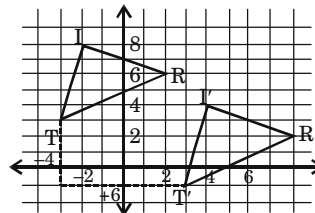
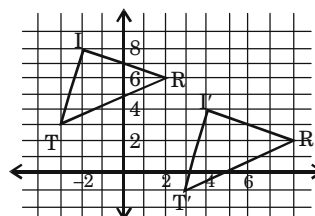
- (a) $(-4, -2)$, $(4, 0)$, $(5, -4)$
 (b) $(4, 2)$, $(0, 4)$, $(-5, 4)$
 (c) $(-4, -2)$, $(4, 0)$, $(-5, -4)$
 (d) None of these.

15. Graph square $S(1, 2)$, $Q(4, 1)$, $R(5, 4)$ and $E(2, 5)$. Find the image after the translation $(x, y) \rightarrow (x - 2, y + 3)$. Then graph and label the image. We are going to move the square to the left 2 and up 3.



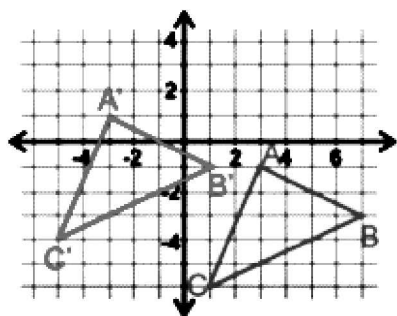
- (a) $S'(-1, 5)$, $Q'(2, 4)$, $R'(3, 7)$, $E'(0, 8)$
 (b) $S'(-2, 5)$, $Q'(2, 4)$, $R'(3, 7)$, $E'(1, 8)$
 (c) $S'(-1, 5)$, $Q'(2, 4)$, $R'(1, 7)$, $E'(2, 8)$
 (d) None of these.

16. Find the translation rule for $\triangle TRI$ to $\triangle T'R'I'$. Look at the movement from T to T' . The translation rule is



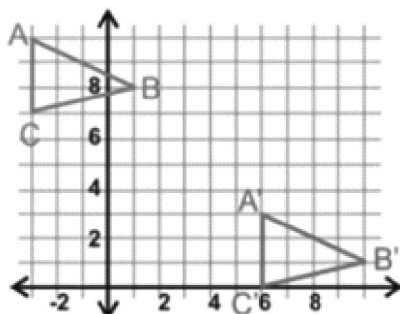
- (a) $(x, y) \rightarrow (x + 6, y - 5)$.
 (b) $(x, y) \rightarrow (x + 5, y - 3)$.
 (c) $(x, y) \rightarrow (x + 6, y - 4)$
 (d) None of these.

17. Find the translation rule for:



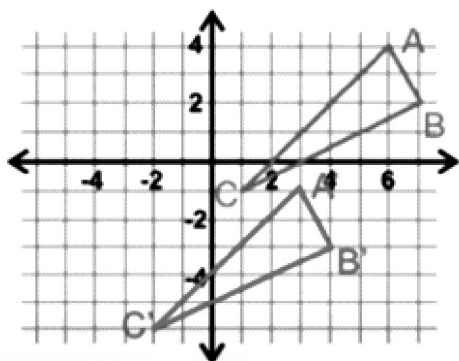
- (a) $(x, y) \rightarrow (x - 7, y + 4)$.
- (b) $(x, y) \rightarrow (x - 6, y + 2)$.
- (c) $(x, y) \rightarrow (x - 5, y + 1)$.
- (d) None of these.

18. Find the translation rule for:



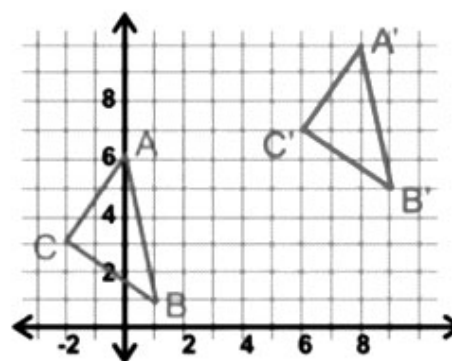
- (a) $(x, y) \rightarrow (x - 7, y + 7)$.
- (b) $(x, y) \rightarrow (x + 7, y + 9)$.
- (c) $(x, y) \rightarrow (x + 9, y - 7)$.
- (d) None of these.

19. Find the translation rule for:



- (a) $(x, y) \rightarrow (x - 3, y - 5)$.
- (b) $(x, y) \rightarrow (x + 7, y + 9)$.
- (c) $(x, y) \rightarrow (x + 9, y - 7)$.
- (d) None of these.

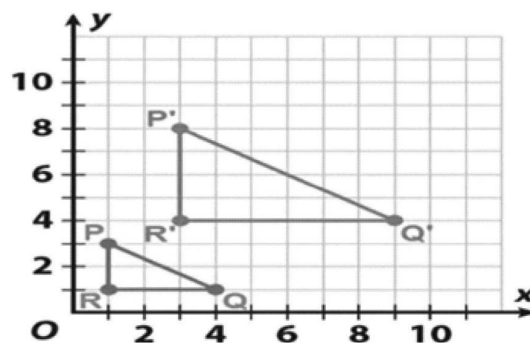
20. Find the translation rule for:



- (a) $(x, y) \rightarrow (x + 6, y - 5)$.
- (b) $(x, y) \rightarrow (x + 7, y + 9)$.
- (c) $(x, y) \rightarrow (x + 8, y + 4)$.
- (d) None of these.

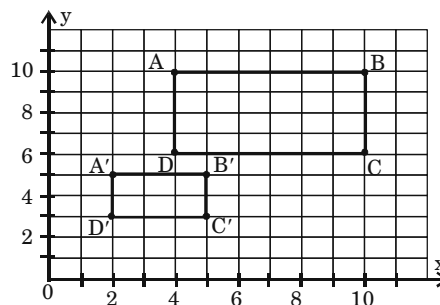
DILATION OR SCALING

21. An art supply store sells several sizes of drawing triangles. All are dilations of a single basic triangle. The basic triangle and one of its dilations are shown on the grid. Find the scale factor of the dilation.



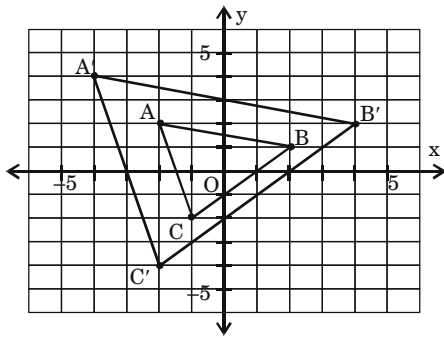
- (a) $x \rightarrow 2, y \rightarrow 2$
- (b) $x \rightarrow 1.5, y \rightarrow 2$
- (c) $x \rightarrow 2, y \rightarrow 1.5$
- (d) $x \rightarrow 2, y \rightarrow 1.5$

22. Find the scale factor of the dilation.



- (a) $x \rightarrow 2, y \rightarrow 2$
- (b) $x \rightarrow 2, y \rightarrow 3$
- (c) $x \rightarrow 3, y \rightarrow 2$
- (d) $x \rightarrow 3, y \rightarrow 3$

23. Find the scale factor of the dilation.



- (a) $x \rightarrow 2, y \rightarrow 3$ (b) $x \rightarrow 2, y \rightarrow 2$
 (c) $x \rightarrow 3, y \rightarrow 2$ (d) None of these
24. As given, the area of shaded portion of image (a) is 24m^2 and scale factor is 3, then find the area of image (b).

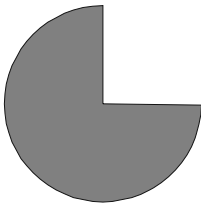


Fig. (a)

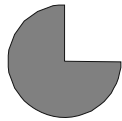
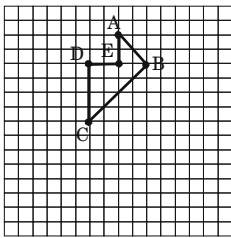


Fig. (b)

- (a) 2.66 m^2 (b) 3.66 m^2
 (c) 4.66 m^2 (d) None of these.
25. Pentagon $A'B'C'D'E'A$, prime, B, prime, C, prime, D, prime, E, prime is the image of pentagon A, B, C, D, E under a dilation with a scale factor of $\frac{1}{2}$.



What is the relation between $C'D'$ and CD ?

- (a) $C'D' = \frac{1}{2} CD$ (b) $C'D' = 2 CD$
 (c) $C'D' = 4 CD$ (d) None of these.
26. As given, for image (a); length $l = 10$, breadth $b = 6$, height $h = 8$, and scale factor is 0.5, then find the volume of image (b)

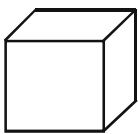


Fig. (a)

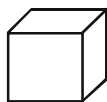


Fig. (b)

- (a) 40 (b) 50
 (c) 60 (d) 70

27. As given, for image (a); small side length $l = 8$, large side length $l = 10$, and scale factor is 0.75, then find the area of image (a) & (b).

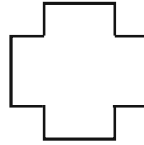


Fig. (a)

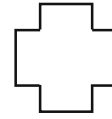


Fig. (b)

- (a) 320, 136.25 (b) 420, 236.25
 (c) 520, 336.25 (d) None of these
28. As given, for image (a); small side length $l = 8$, large side length $l = 10$, and scale factor is 0.75, then find the area of image (b).

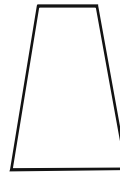


Fig. (a)

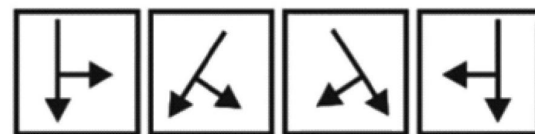
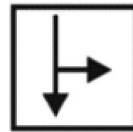


Fig. (b)

- (a) 140 (b) 150
 (c) 160 (d) 170

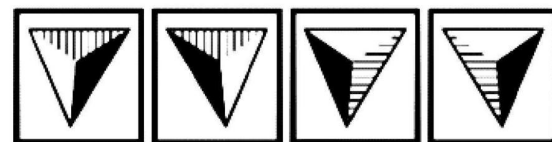
MIRRORING

29. Choose the alternative which is closely resembles the mirror image of the given figure.



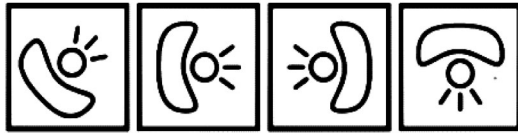
- (a) (b) (c) (d)

30. Choose the alternative which is closely resembles the mirror image of the given fig



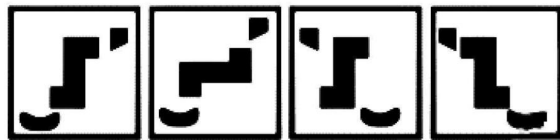
- (a) (b) (c) (d)

31. Choose the alternative which is closely resembles the Mirror image of the given figure.



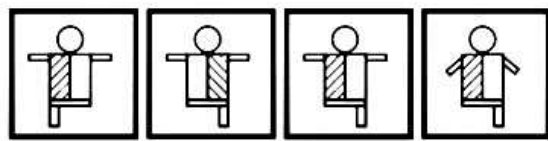
(a) (b) (c) (d)

32. Choose the alternative which is closely resembles the Mirror image of the given figure.



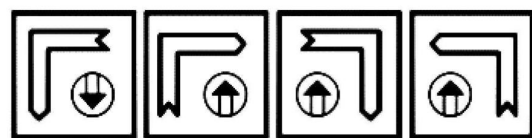
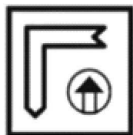
(a) (b) (c) (d)

33. Choose the alternative which is closely resembles the mirror image of the given figure.



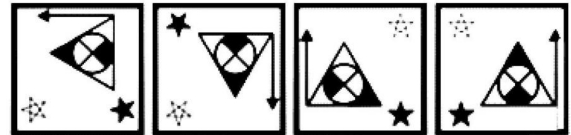
(a) (b) (c) (d)

34. Choose the alternative which is closely resembles the Mirror image of the given figure.



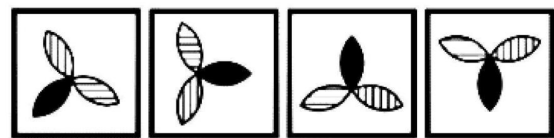
(a) (b) (c) (d)

35. Choose the alternative which is closely resembles the Mirror image of the given figure.



(a) (b) (c) (d)

36. Choose the alternative which is closely resembles the Mirror image of the given figure.



(a) (b) (c) (d)

37. Choose the alternative which is closely resembles the mirror-image of the given combination.

LETTERS

- (a) LETTERS (b) LETTEYS
(c) 2ЯETTEL (d) 2ЯETTEL

38. Choose the alternative which is closely resembles the mirror-image of the given combination.

NiCaRaGuT

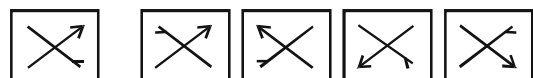
- (a) TuGaRaCiN (b) TuGsRgCiN
(c) TnGsRgCiN (d) TnGsRgCiN

39. Choose the alternative which is closely resembles the mirror-image of the given combination.

INFORMATION

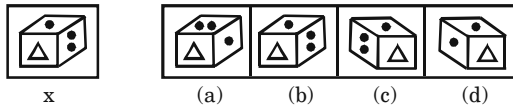
- (a) 2NOITAMЯOTNI (b) 2NOITAMЯOTNI
(c) INEOBVALIONS (d) 2NOITAMЯOTNI

40. **Directons:** Find the mirror image of following words/numbers from amongst the given alternatives.

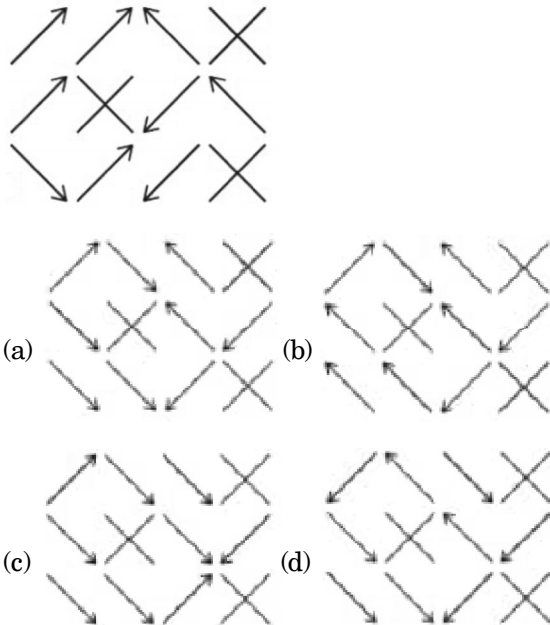


x (a) (b) (c) (d)

41. Choose the alternative which is closely resembles the mirror image of the given figure.



42. Which figure is a vertical reflection across the x axis?



Directions for questions 43 to 57: If a mirror is placed, then which of the answer figures is the mirror image of the given figure?

43. REST

(a) TSEЯ (b) TSEЯ
(c) 12EЯ (d) TSEЯ

44. 15824

(a) 12854 (b) 42851
(c) 42851 (d) 42851

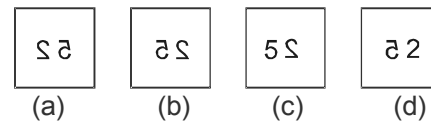
45. M35KG1

(a) M35KG1 (b) 1GK53M
(c) 1GK53M (d) 1GK53M

46. P b

(a) P p (b) P p (c) P b (d) P b

47. 25



- 48.

(a) (b) (c) (d)

- 49.

(a) (b) (c) (d)

- 50.

(a) (b) (c) (d)

- 51.

(a) (b) (c) (d)

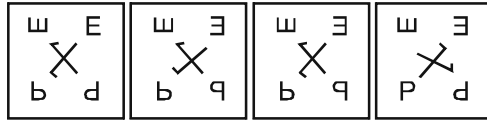
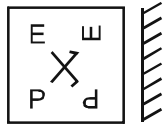
- 52.

(a) (b) (c) (d)

- 53.

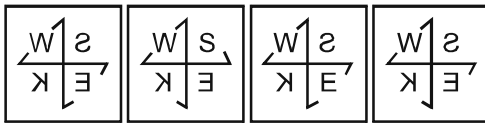
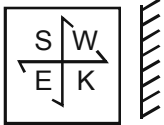
(a) (b) (c) (d)

54.



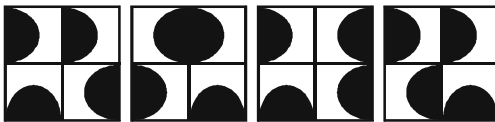
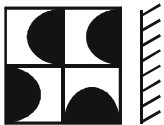
(a) (b) (c) (d)

55.



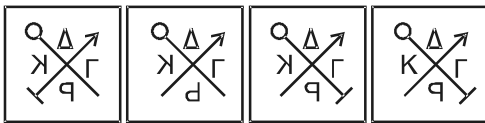
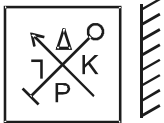
(a) (b) (c) (d)

56.



(a) (b) (c) (d)

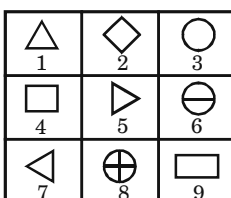
57.



(a) (b) (c) (d)

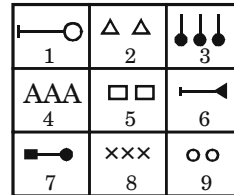
GROUPING

58. Group the following figures into three classes with identical properties and select the correct alternative



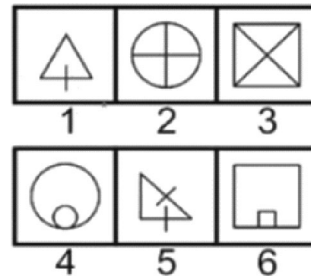
- (a) 1, 5, 8; 2, 4, 7; 3, 6, 9
 (b) 1, 4, 7; 2, 5, 8; 3, 6, 9
 (c) 1, 5, 7; 2, 4, 9; 3, 6, 8
 (d) 1, 5, 7; 3, 4, 9; 2, 6, 8

59. Group the following figures into three classes with identical properties and select the correct alternative.



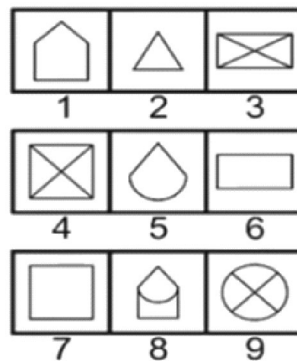
- (a) 1, 2, 3; 6, 5, 4; 7, 9, 8
 (b) 1, 6, 7; 2, 5, 9; 3, 4, 8
 (c) 1, 6, 7; 2, 5, 8; 3, 4, 9
 (d) 1, 3, 7; 2, 5, 9; 4, 6, 8

60. Choose the correct sequence which are the group of identical image.

**Options**

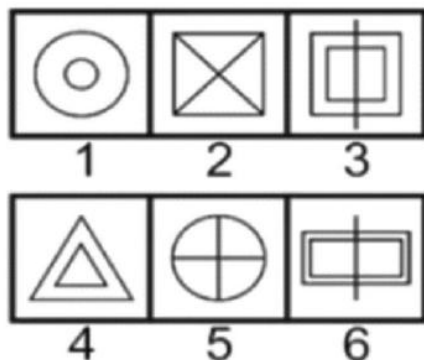
- (a) 2, 3; 4, 6; 5, 1 (b) 1, 5; 2, 3; 4, 6
 (c) 1, 4; 3, 2; 5, 6 (d) 1, 3; 4, 2; 5, 6

61. Choose the correct sequence which are the group of identical images.

**Options**

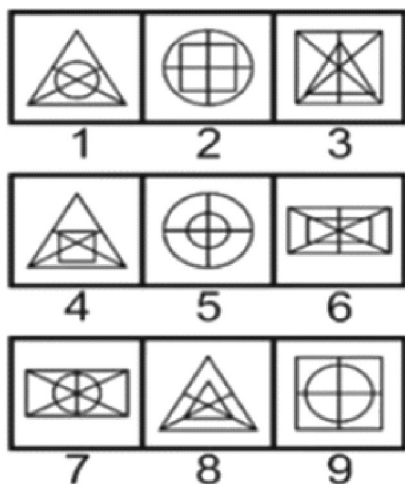
- (a) 1, 5, 8; 2, 6, 7; 3, 4, 9
 (b) 1, 2, 3; 4, 5, 6; 7, 8, 9
 (c) 4, 5, 6; 7, 8, 9; 1, 2, 3
 (d) 1, 4, 7; 2, 5, 8; 3, 6, 9

62. Choose the correct sequence which are the group of identical images.



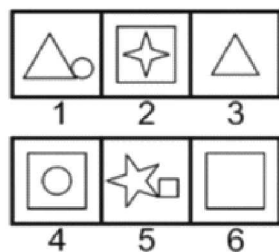
Options

- (a) 1, 2; 3, 4; 5, 6 (b) 1, 4, 6, 3; 2, 5
(c) 1, 3; 4, 5; 2, 6 (d) 1, 5; 2, 3; 4, 6
63. Choose the correct sequence which are the group of identical images.



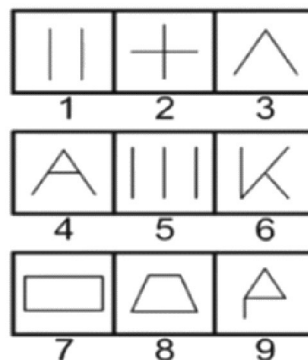
Options

- (a) 1, 3, 5; 2, 4, 6; 7, 8, 9
(b) 1, 4, 8; 2, 5, 9; 3, 6, 7
(c) 1, 2, 3; 4, 6, 7; 8, 9, 5
(d) 1, 6, 7; 2, 3, 4; 5, 8, 9
64. Choose the correct sequence which are the group of identical images.



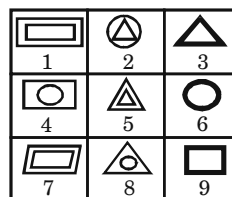
- (a) 1, 3; 2, 4; 5, 6 (b) 1, 4; 3, 5; 2, 6
(c) 1, 6; 3, 4, 2, 5 (d) 1, 5; 2, 4; 3, 6

65. Choose the correct sequence which are the group of identical images.

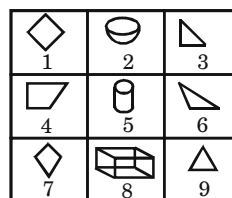


Options

- (a) 3, 2, 4; 1, 5, 6; 7, 8, 9
(b) 1, 2, 3; 4, 5, 6; 7, 8, 9
(c) 7, 1, 2; 3, 4, 5; 6, 8, 9
(d) 3, 8, 9; 1, 2, 7; 4, 5, 6
66. Group the given figures into three classes using each figure only once.



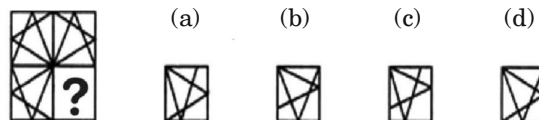
- (a) 1, 5, 7; 2, 4, 6; 3, 9, 8
(b) 1, 5, 7; 2, 4, 8; 3, 6, 9
(c) 1, 4, 7; 2, 5, 8; 3, 6, 9
(d) 1, 7, 9; 3, 5, 8; 2, 4, 6
67. Group the given figures into three classes using each figure only once.



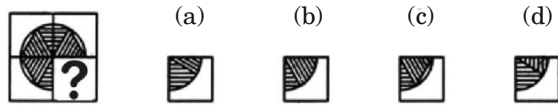
- (a) 1, 4, 7; 2, 5, 8; 3, 6, 9
(b) 1, 4, 7; 2, 5, 9; 3, 6, 7
(c) 1, 3, 4; 2, 5, 8; 6, 7, 9
(d) 1, 2, 3; 4, 5, 6; 7, 8, 9

2D & 3D Shapes

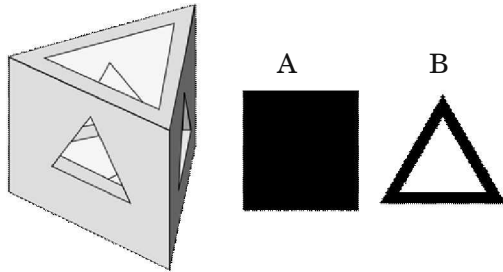
68. Identify the figure that completes the pattern.



69. Identify the figure that completes the pattern.



70. Which of the shadows on the right can result from casting light on one of the 3D-shape's sides?

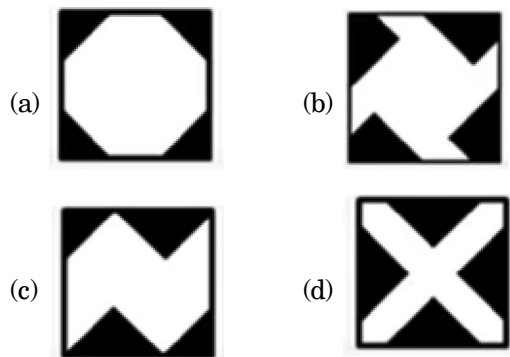


- (a) A (b) B
(c) Both A & B (d) None of these.

71. Choose the image that completes the pattern:



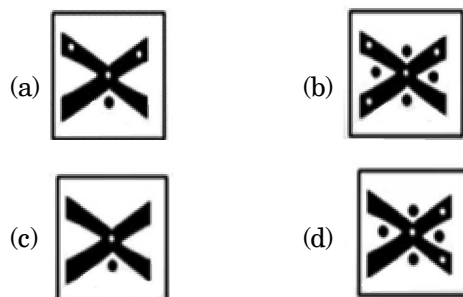
Which of the shape below continues the sequence:



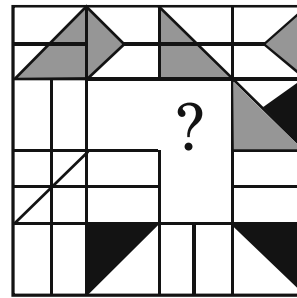
72. Choose the image that completes the pattern:



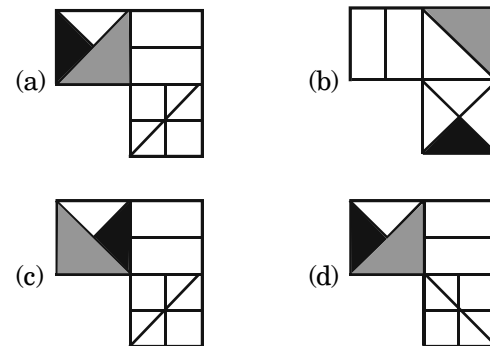
Which of the shapes below continues the sequence:



73. Choose the image that completes the pattern:



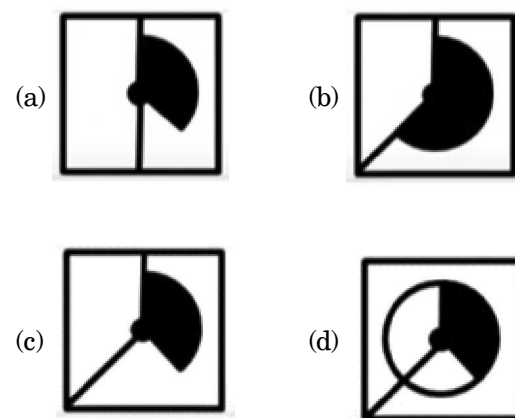
Which of the shapes below replaces the missing box above?



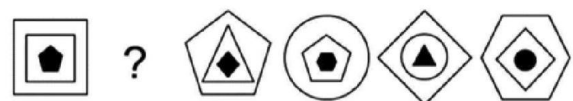
74. Choose the image that completes the pattern:



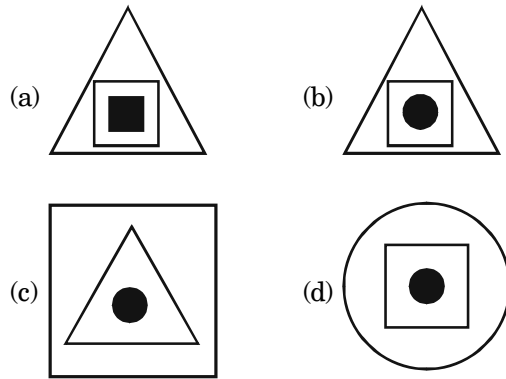
Which of the shapes below continues the sequence:



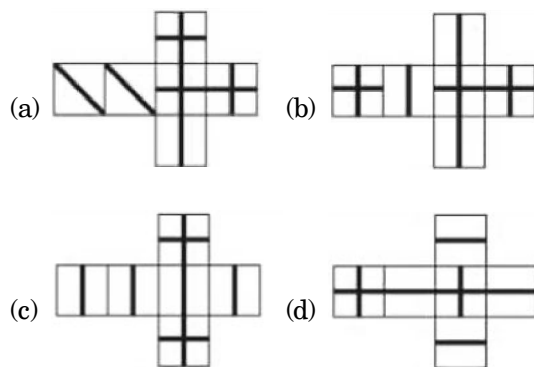
75. Choose the image that completes the pattern:



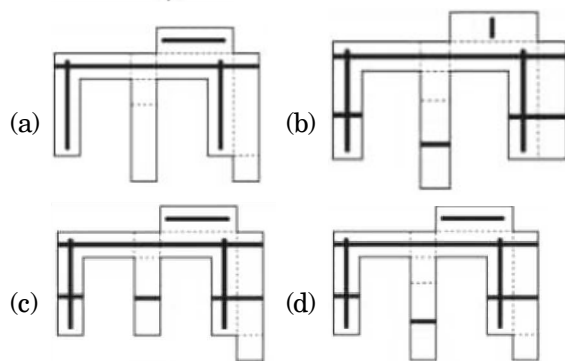
Which of the shapes below replaces the missing box above?



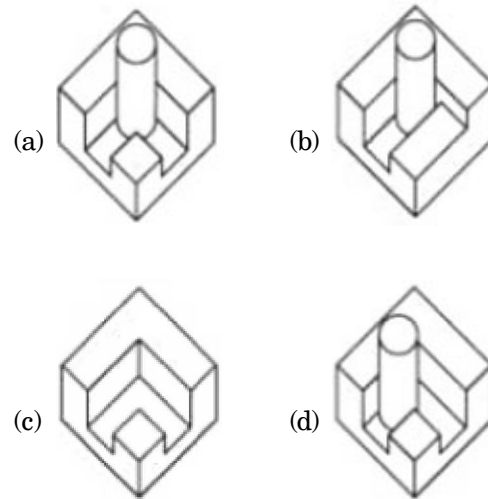
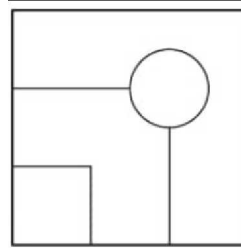
76. Which piece corresponds to the given solid?



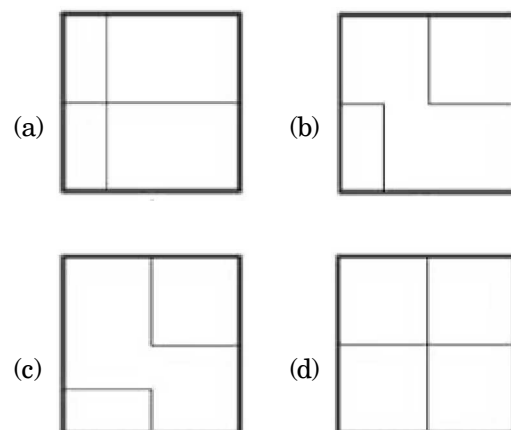
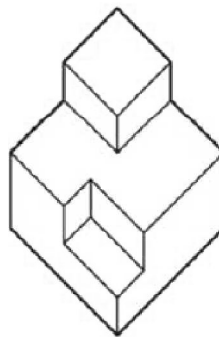
77. Which piece corresponds to the given solid?



78. To which object does the given top view correspond?



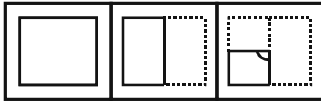
79. Which of these images is the top view of the given object?



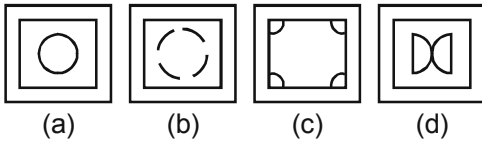
PAPER FOLDING AND CUTTING

Directions for questions 80 to 104: In each of the following questions a paper is folded and cut as shown below in the question figure. From the given answer figures, indicate how it will appear when opened?

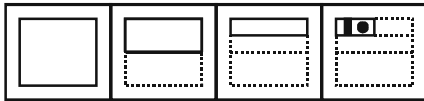
80. Question Figures



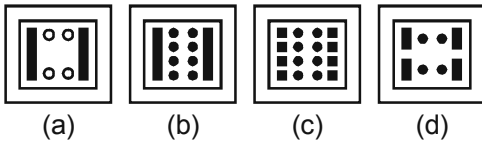
Answer Figures



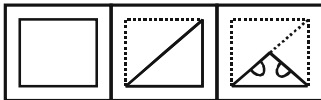
81. Question Figures



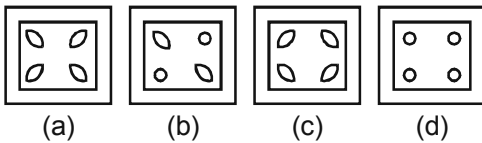
Answer Figures



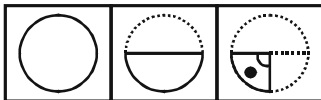
82. Question Figures



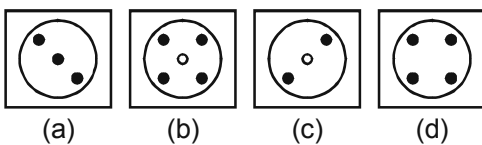
Answer Figures



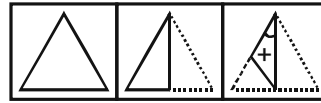
83. Question Figures



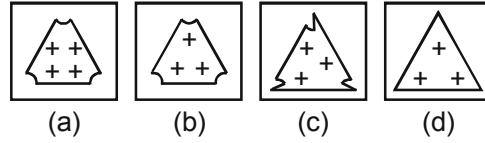
Answer Figures



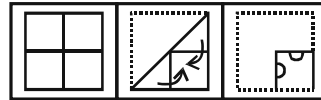
84. Question Figures



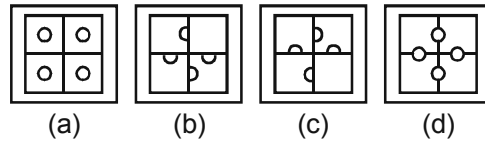
Answer Figures



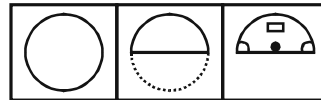
85. Question Figures



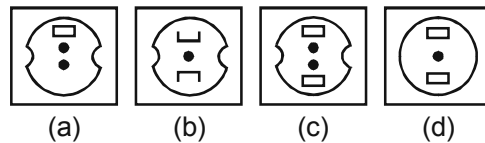
Answer Figures



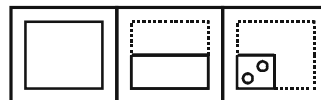
86. Question Figures



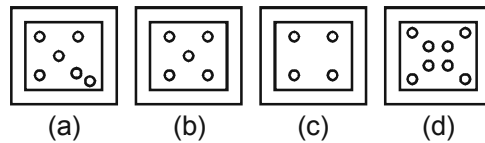
Answer Figures



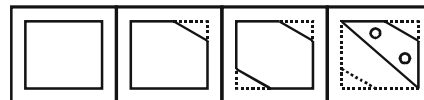
87. Question Figures



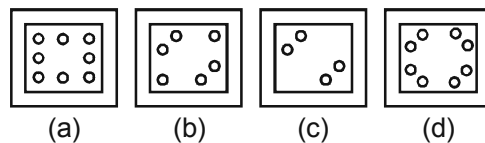
Answer Figures



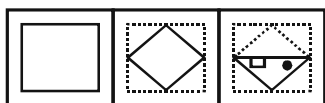
88. Question Figures



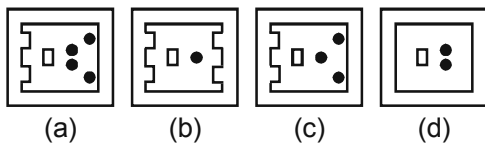
Answer Figures



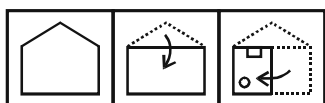
89. Question Figures



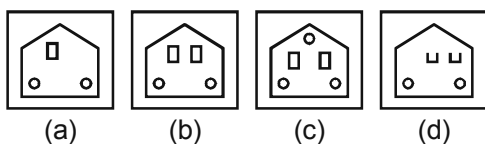
Answer Figures



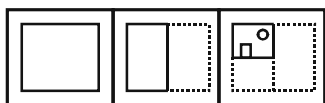
90. Question Figures



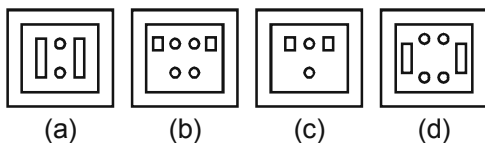
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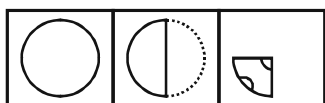
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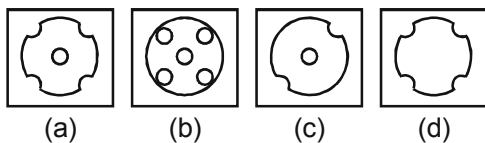
Answer Figures



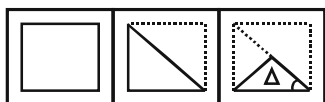
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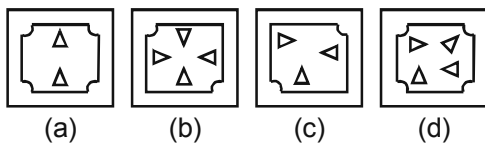
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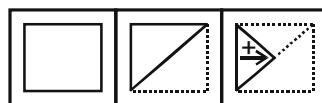
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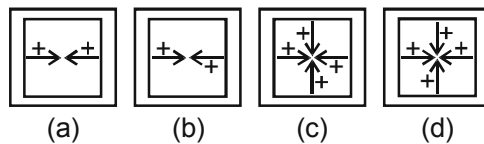
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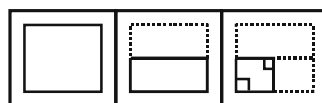
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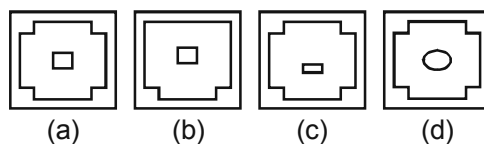
Answer Figures



95. Question Figures



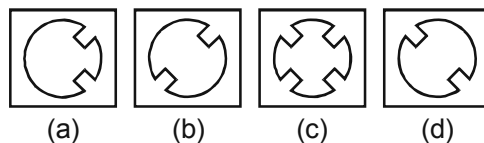
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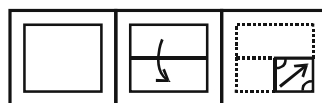
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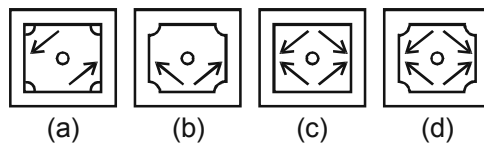
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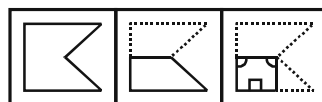
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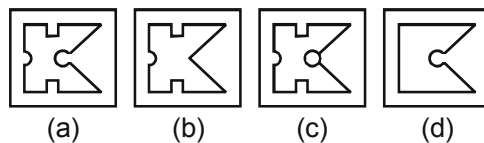
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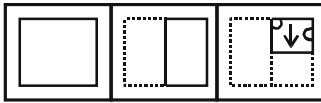
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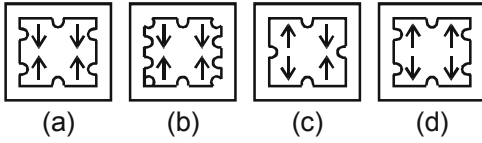
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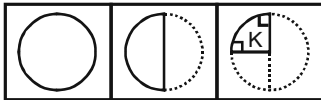
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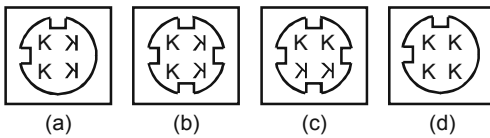
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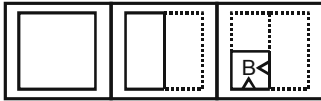
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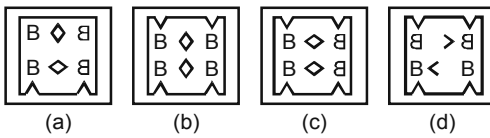
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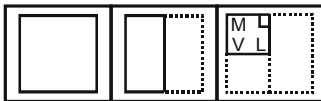
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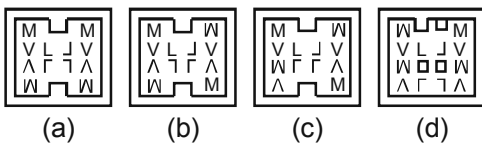
Answer Figures



102. Question Figures



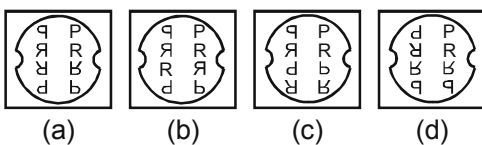
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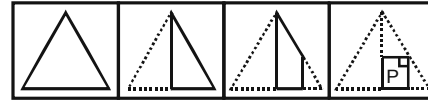
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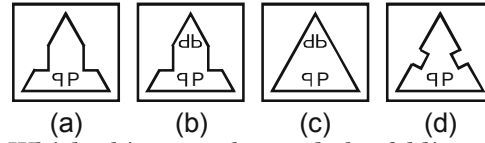
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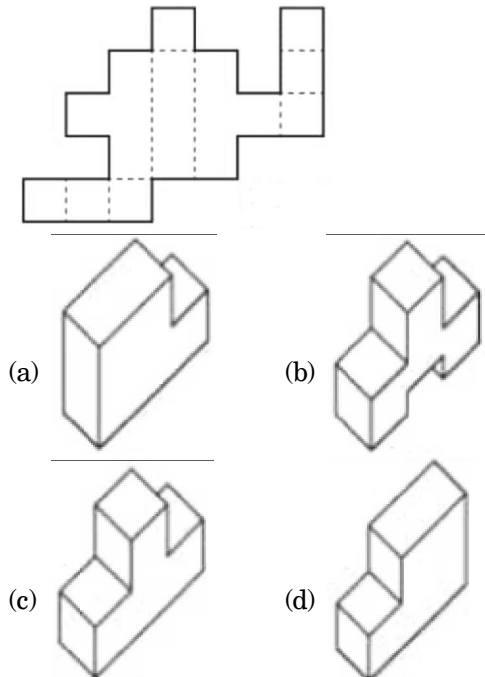
104. Question Figures



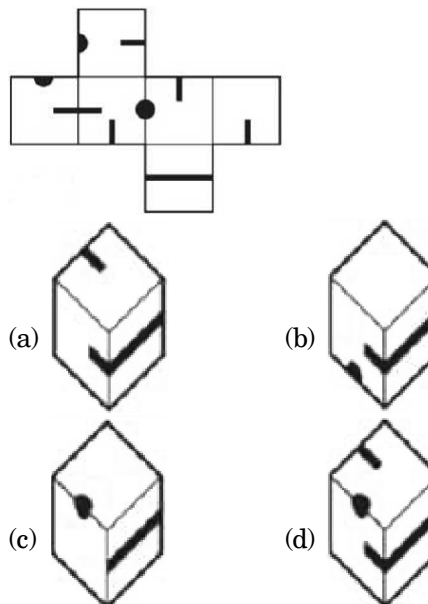
Answer Figures



105. Which object can be made by folding the given shape?



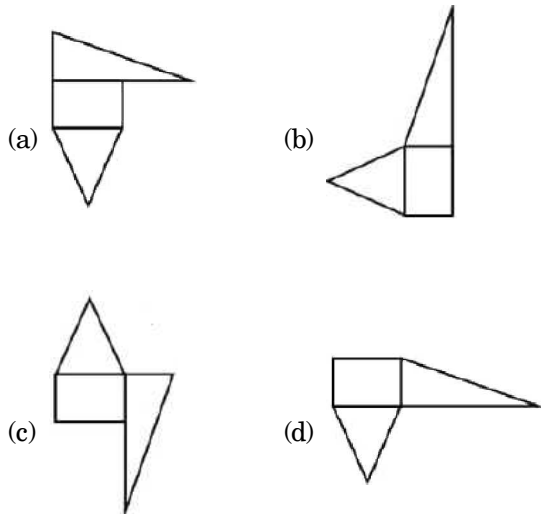
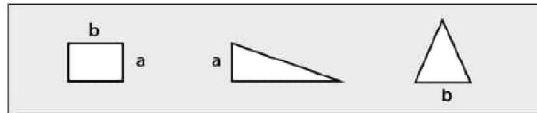
106. Which cube can be formed by folding the given shape?



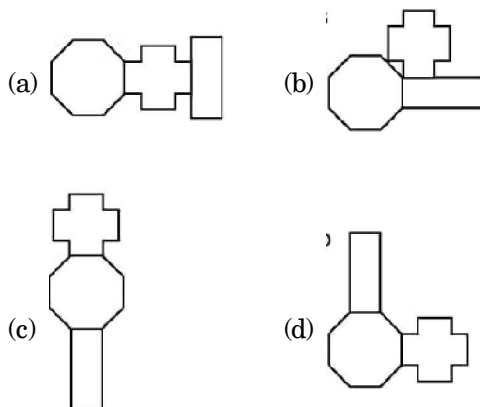
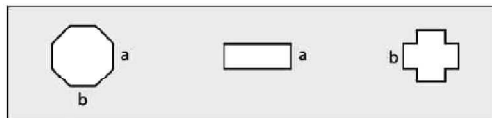
ASSEMBLING

Directions 107-138: During the following spatial reasoning question your task is to look at the given shapes and decide which of the examples match the shape when joined together by the corresponding letters.

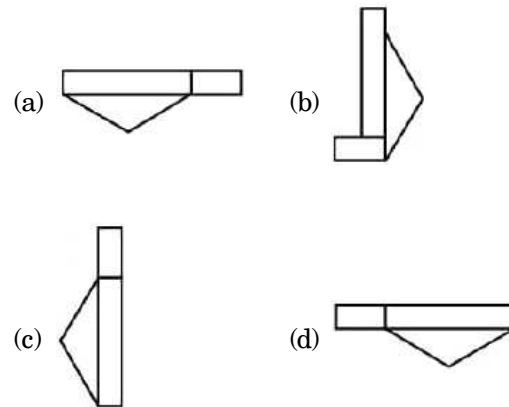
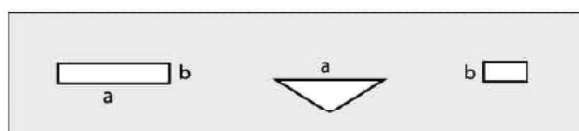
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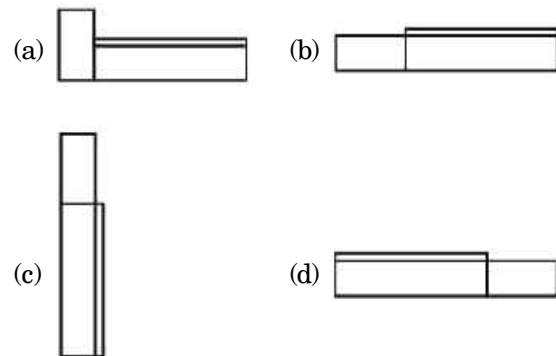
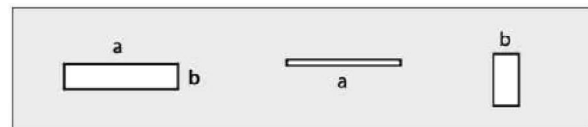
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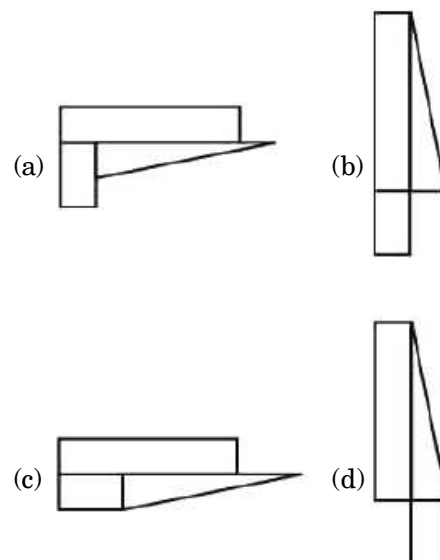
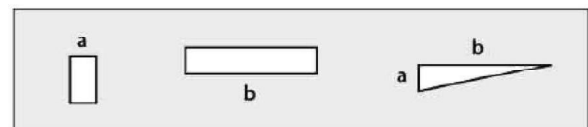
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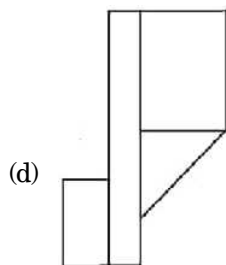
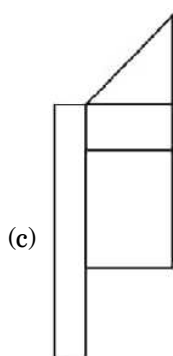
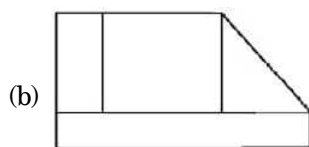
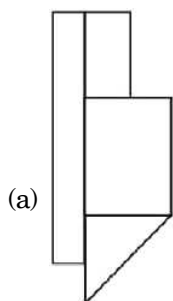
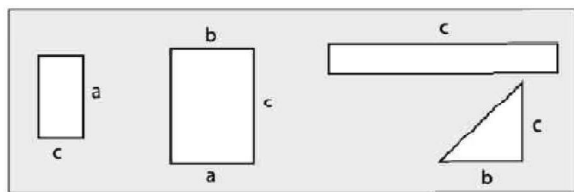
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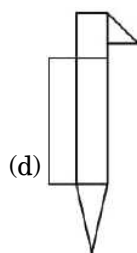
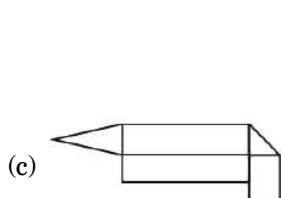
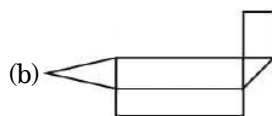
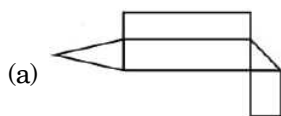
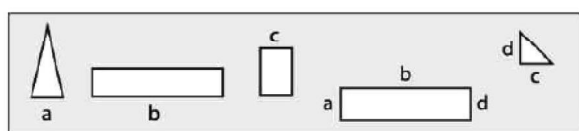
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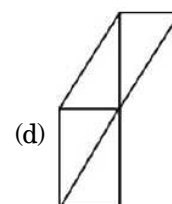
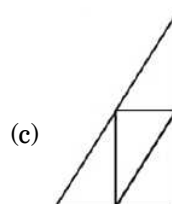
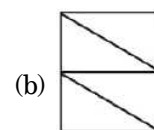
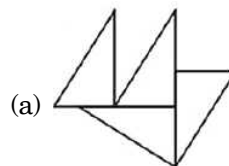
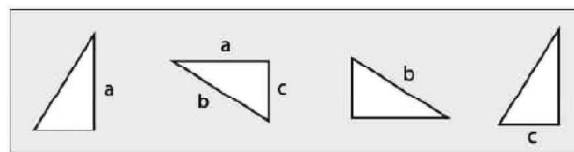
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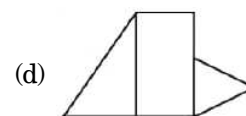
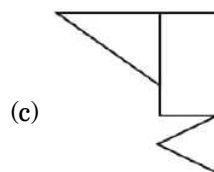
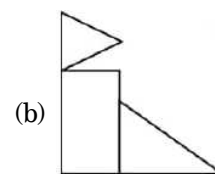
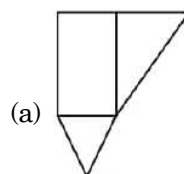
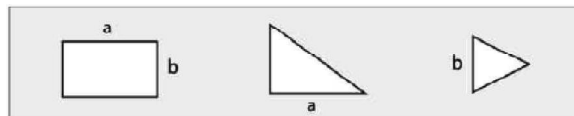
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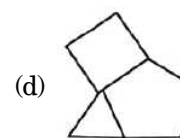
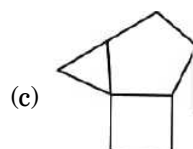
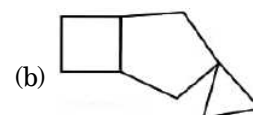
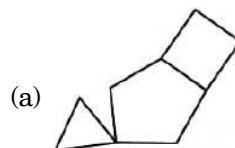
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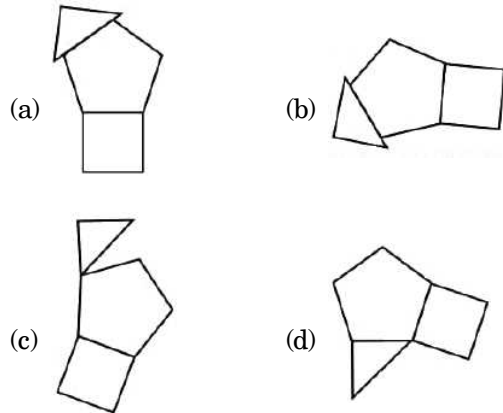
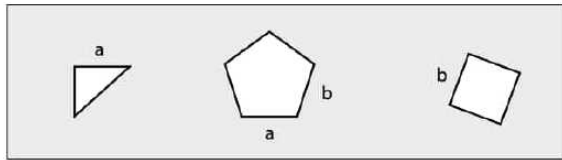
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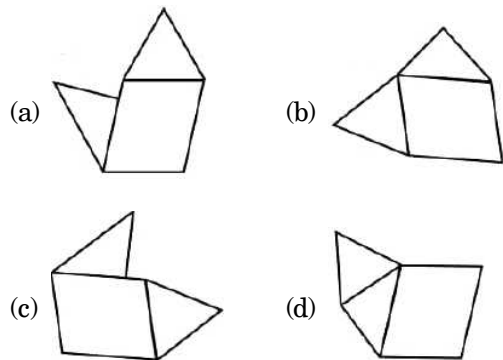
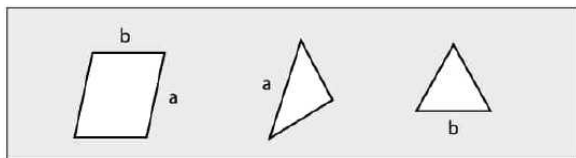
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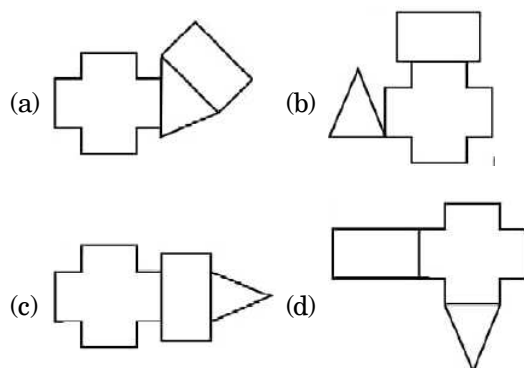
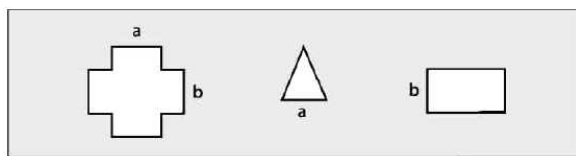
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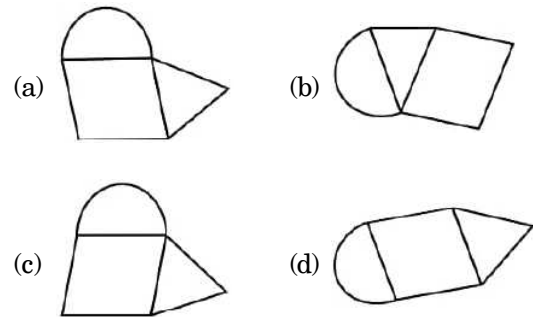
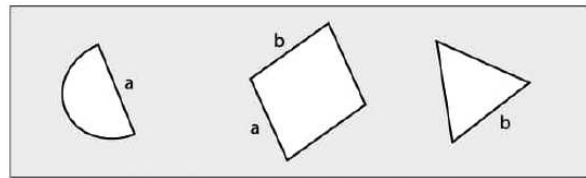
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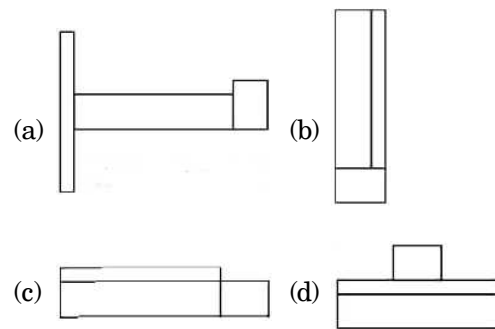
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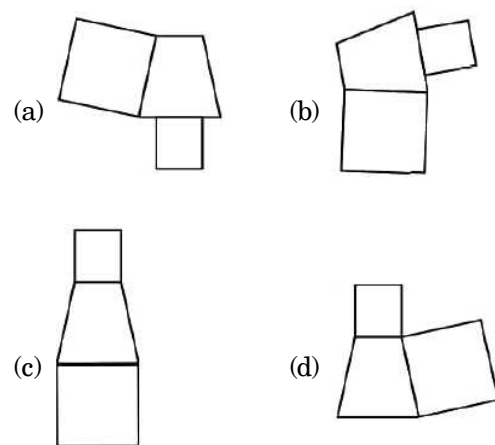
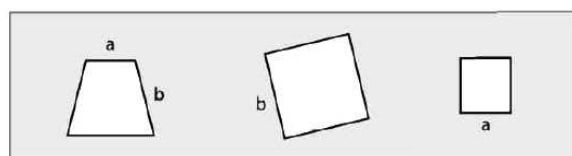
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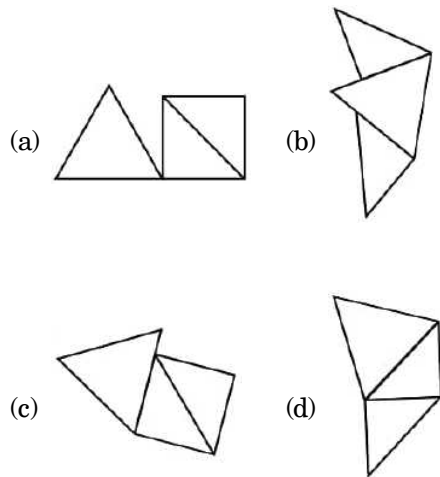
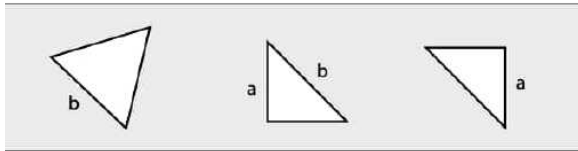
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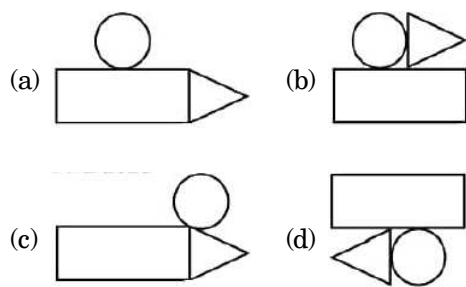
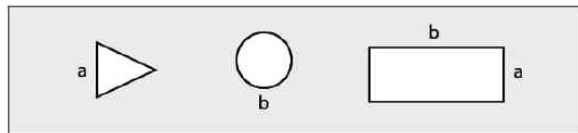
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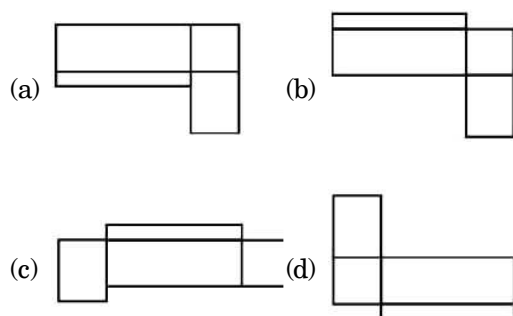
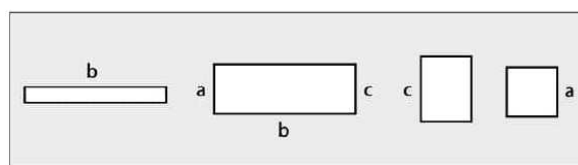
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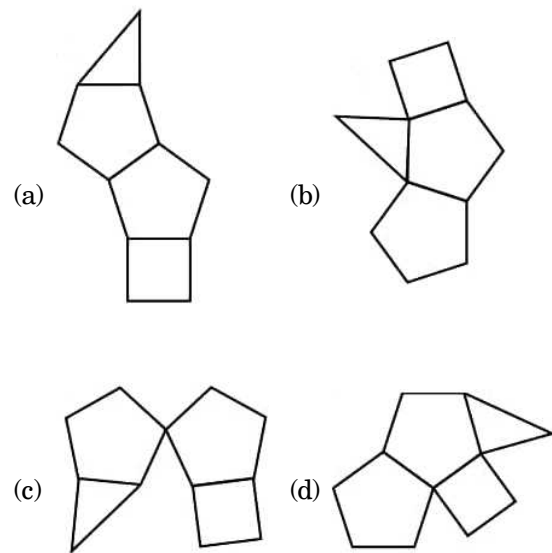
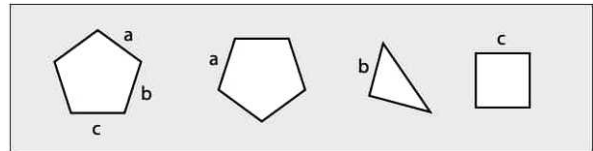
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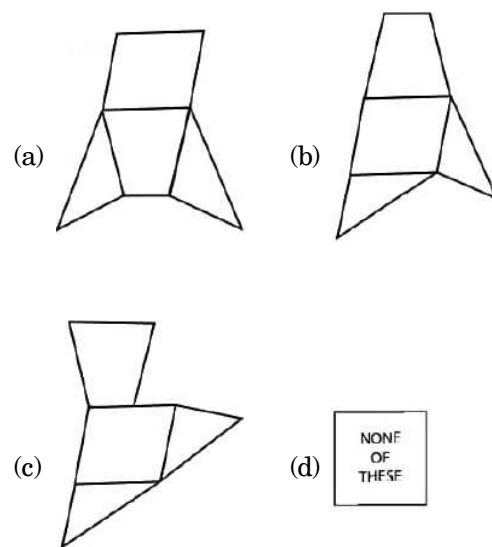
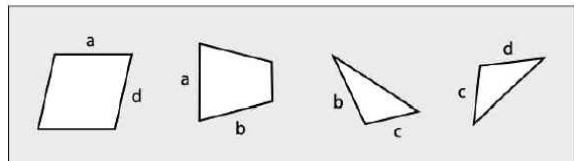
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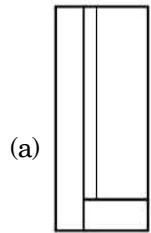
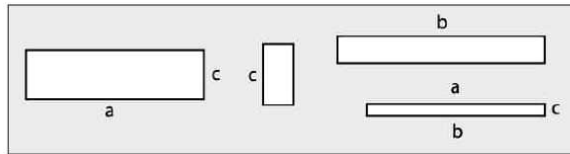
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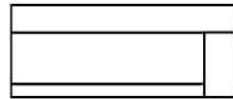
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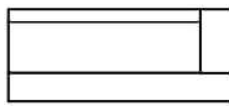
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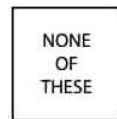
(b)



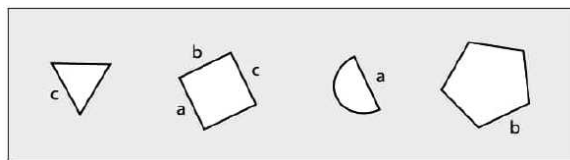
(c)



(d)



129.



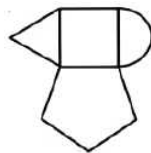
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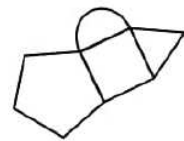
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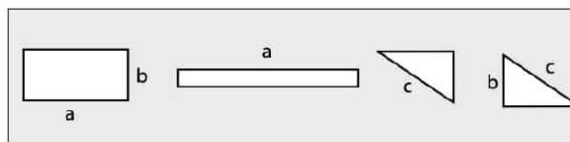
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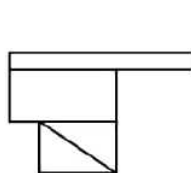
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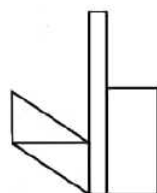
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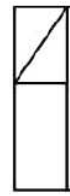
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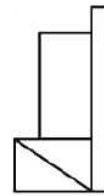
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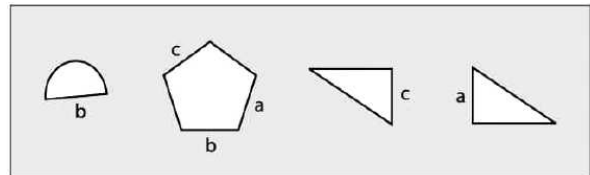
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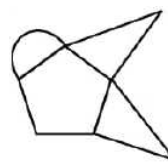
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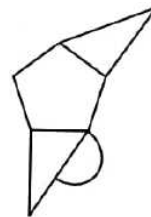
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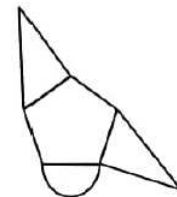
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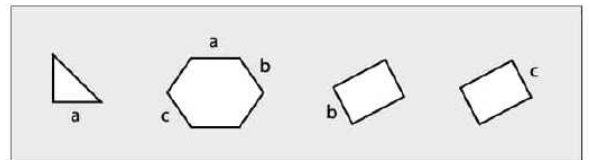
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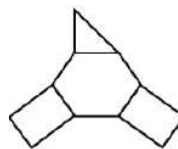
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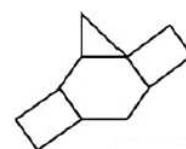
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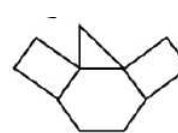
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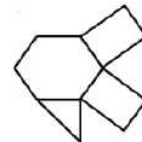
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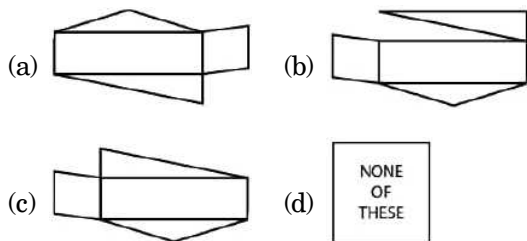
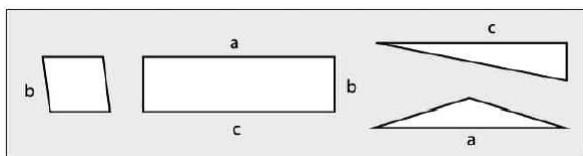
(c)



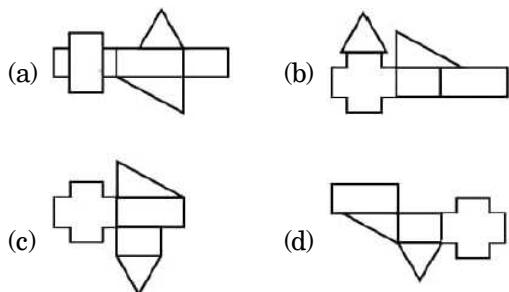
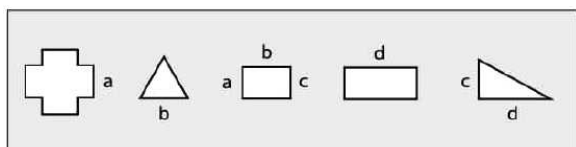
(d)



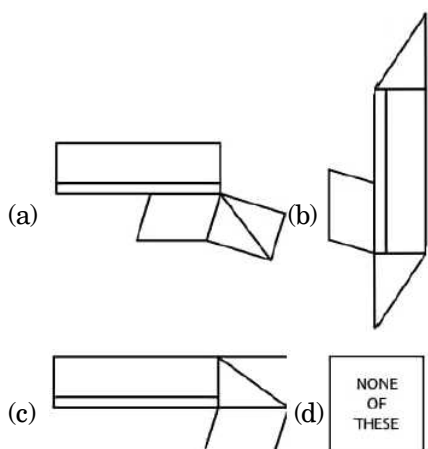
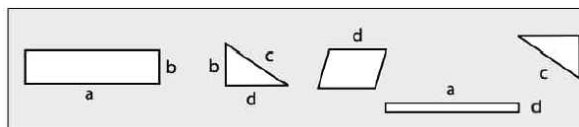
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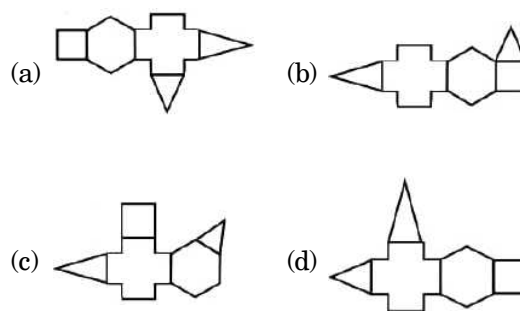
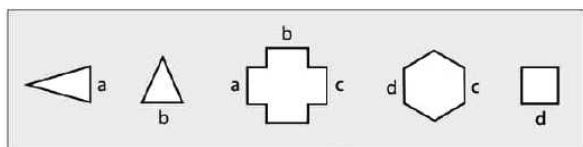
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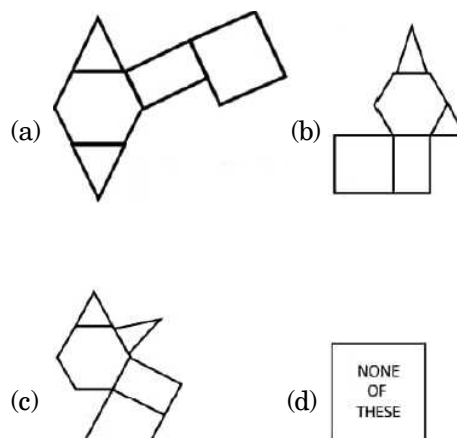
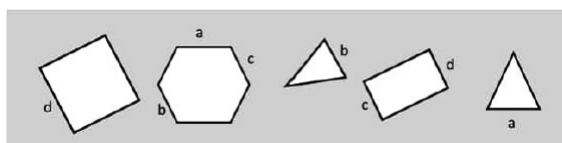
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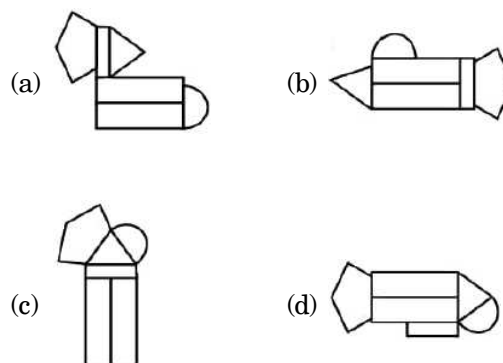
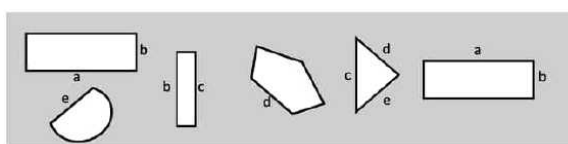
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137.



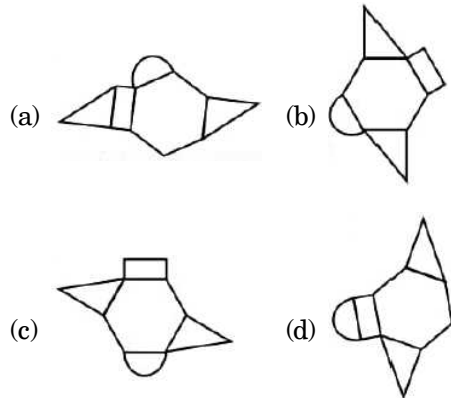
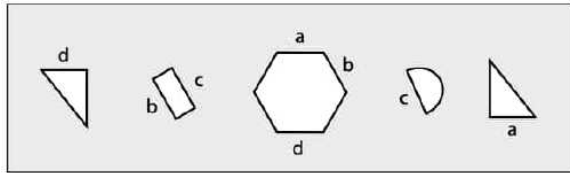
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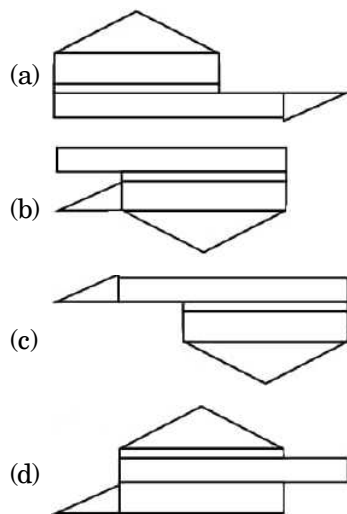
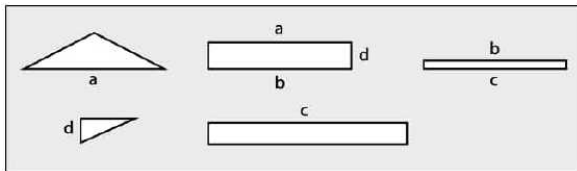
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Spatial Aptitude

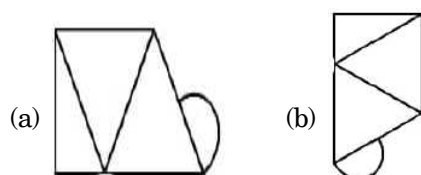
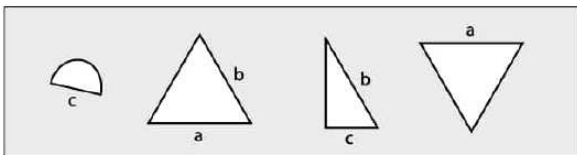
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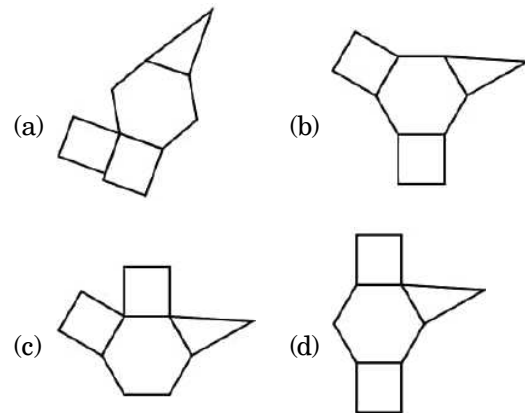
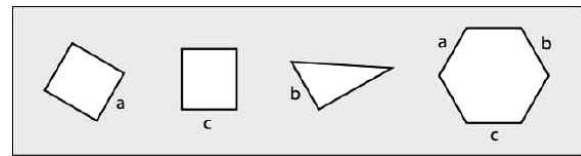
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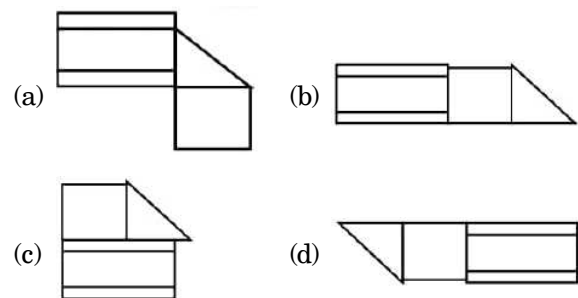
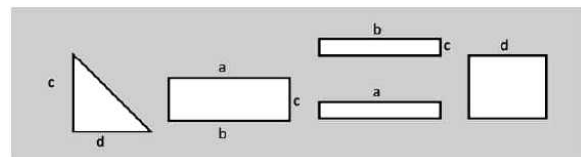
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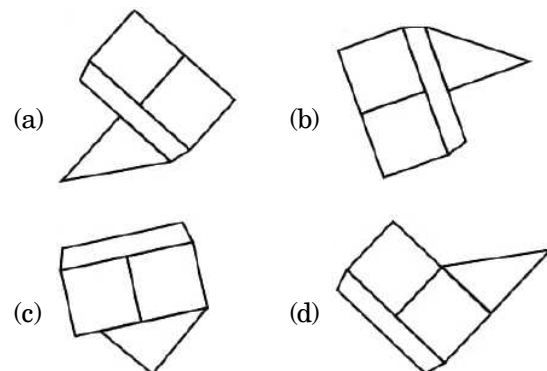
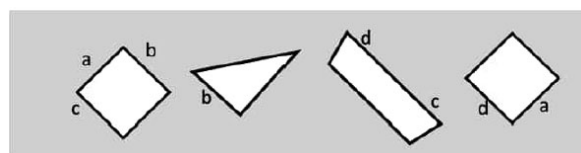
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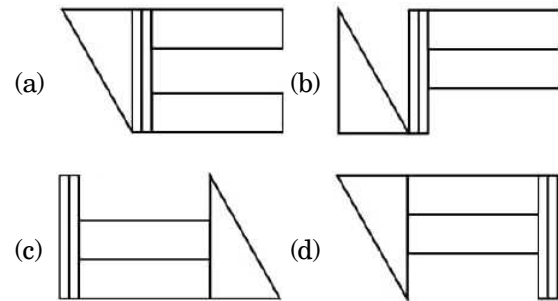
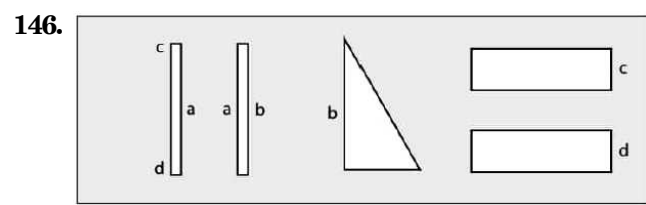
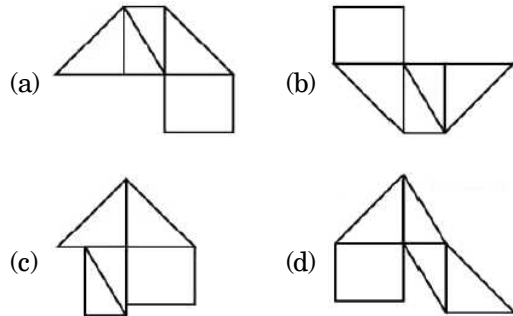
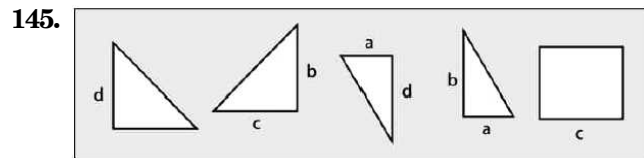


143.



144.





ANSWERS

MCQ Type Questions

- | | | | | | | | | | |
|-----------|-----------|----------|----------|-----------|-----------|----------|----------|-----------|-----------|
| 1. (d) | 2. (d) | 3. (c) | 4. (c) | 5. (b) | 6. (a) | 7. (d) | 8. (b) | 9. (b) | 10. (d) |
| 11. (c) | 12. (a) | 13. (a) | 14. (a) | 15. (a) | 16. (c) | 17. (b) | 18. (c) | 19. (a) | 20. (c) |
| 21. (a) | 22. (a) | 23. (b) | 24. (a) | 25. (a) | 26. (c) | 27. (b) | 28. (c) | 29. (d) | 30. (b) |
| 31. (b&c) | 32. (c&d) | 33. (b) | 34. (c) | 35. (b&d) | 36. (c&d) | 37. (c) | 38. (b) | 39. (c&d) | 40. (c&d) |
| 41. (c) | 42. (a) | 43. (b) | 44. (c) | 45. (d) | 46. (a) | 47. (b) | 48. (c) | 49. (a) | 50. (d) |
| 51. (c) | 52. (c) | 53. (a) | 54. (c) | 55. (a) | 56. (a) | 57. (c) | 58. (c) | 59. (b) | 60. (a&b) |
| 61. (a) | 62. (b) | 63. (b) | 64. (d) | 65. (b) | 66. (b) | 67. (a) | 68. (d) | 69. (c) | 70. (b) |
| 71. (d) | 72. (b) | 73. (d) | 74. (c) | 75. (b) | 76. (c) | 77. (d) | 78. (a) | 79. (b) | 80. (a) |
| 81. (b) | 82. (c) | 83. (b) | 84. (a) | 85. (d) | 86. (c) | 87. (d) | 88. (c) | 89. (a) | 90. (b) |
| 91. (d) | 92. (a) | 93. (b) | 94. (c) | 95. (a) | 96. (c) | 97. (d) | 98. (a) | 99. (a) | 100. (b) |
| 101. (c) | 102. (a) | 103. (a) | 104. (b) | 105. (c) | 106. (d) | 107. (b) | 108. (d) | 109. (a) | 110. (d) |
| 111. (d) | 112. (b) | 113. (a) | 114. (c) | 115. (a) | 116. (c) | 117. (d) | 118. (b) | 119. (d) | 120. (a) |
| 121. (c) | 122. (d) | 123. (d) | 124. (a) | 125. (c) | 126. (b) | 127. (d) | 128. (d) | 129. (c) | 130. (c) |
| 131. (d) | 132. (b) | 133. (d) | 134. (d) | 135. (d) | 136. (a) | 137. (d) | 138. (c) | 139. (d) | 140. (b) |
| 141. (c) | 142. (b) | 143. (a) | 144. (d) | 145. (d) | 146. (a) | | | | |

EXPLANATIONS

MCQ TYPE QUESTIONS

- 43.** Letter – Mirror image

R	–	Я
E	–	Ǝ
S	–	Ƨ
T	–	T

Hence, option (b) is correct.

- 44.** Number – Mirror image

1	–	1
5	–	2
8	–	8
2	–	5
4	–	4

Hence, option (c) is correct.

- 45.** Letter/

Number – Mirror image

M	–	M
3	–	ε
5	–	2
K	–	⅂
G	–	9
1	–	1

Hence, option (d) is correct.

- 46.** This case is like water-image. Hence, option (a) is correct.

- 47.** Mirror image of 2 is 5 and of 5 is 2

Hence, answer is 25.

- 58.** Here, figures 1, 5, 7 contain similar properties. Fig. 2, 4 and 9 are similar as they are made of four straight lines. Similarly, Figures 3, 6, and 9 contain a circle as a base.

Hence, (c) in the answer.

- 59.** Here we observe that the figure 1, 6, 7 have only elements, the figures 2, 5 and 9 have two elements while figures 3, 4 and 8 consist of three elements.

Hence, (b) is the correct alternative.

- 70.** When you look at the shape from above or below, you will see a shadow identical to image B.

When you look at the shape from the side, you will see a shadow in the form of a dark square with lit triangles in it (B.N. the lit triangle is not identical to the one shown in shape itself!).

Quantitative Aptitude

1

CHAPTER

Numbers, Algebra and Data Interpretation

NUMBERS

Natural Numbers

All positive integers are natural numbers.

e.g. 1, 2, 3, 4, 5,.....

There are infinite natural numbers and number 1 is the *least natural number*.

Types of Natural Numbers : Based on divisibility there are following two types of natural numbers :

1. Prime Numbers:

A natural number larger than unity is a prime number if it does not have other divisors except itself and unity.

Note: Unity i.e 1 is not a prime number.

To Check A Number is Prime or not:

- Take square root of the number.
- Round of the square root to the next highest integer and call this number as Z.
- Check for divisibility of the number N by all prime numbers below Z. If there is no number below the value of Z which divides N then the number will be prime.

e.g. 241 is prime or not?

$\sqrt{241}$ lies between 15 or 16. Hence take the value of $Z=16$.

Prime numbers less than 16 are 2, 3, 5, 7, 11 and 13.

241 is not divisible by any of these. Hence we can conclude that 241 is a *prime number*.

2. Composite Numbers:

The numbers which are not prime are known as composite numbers.

Note: 1 is neither prime not composite

Co-Primes: Two numbers a and b are said to be co-primes, if their H.C.F is 1.

e.g. (2, 3), (4, 5), (7, 9), (8, 11).....

Tests for Divisibility

A number is divisible by

- 2, when its unit's digit is even or 0.
- 3, when the sum of its digits is divisible by 3.
- 4 when number formed by last two right hand digits is divisible by 4, or if last two digits are 0's.
- 5, when its unit's digit is 5 or 0.
- 6, when it is divisible by 2 and 3 both.
- 8 when number formed by last three right hand digits is divisible by 8, or when last three digit are 0's.
- 9, when sum of its digits is divisible by 9.

(8) 10, when its unit's digit is 0.

(9) 11, when difference between sum of the digits in the odd and sum of the digits in the even places is 0 or a multiple of 11.

Note: When any number with an even number of digits is added to its reverse, the sum is always a multiple of 11.

(10) 12, when it is divisible by 3 and 4 both.

NUMERICAL RELATION

Let **A** and **B** be two non-empty sets. Then the relation which is usually denoted by **R** from **A** to **B** is defined as the set of ordered pairs (in each of which the first element is from **A** and the second element is from **B**) satisfying a certain condition.

The set of all first elements of ordered pairs of **R** is called "Domain" of **R**.

The set of all second elements of ordered pairs is called "range" of **R**.

Example:

If **A** = {1,4,6}, **B** = {1, 2, 3} then construct a relation **R** from **A** to **B** describing "is a multiple of".

Since 1 "is a multiple of" 1, (1, 1) is in **R**.

Since 4 "is a multiple of" 2 (4, 2) is in **R**.

Similarly, we can find all the ordered pairs of **R**.

Hence **R** = {(1, 1) (4, 1) (4, 2) (6, 1) (6, 2) (6, 3)}

Here, Domain of **R** = {1, 4, 6} and

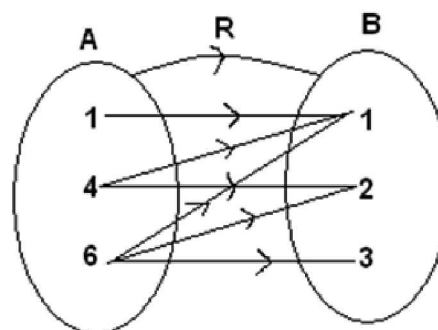
Range of **R** = {1, 2, 3}

Note : Though (4, 3) is an ordered pair, it won't be there in **R** because 4 is not a multiple of 3.

Graphical Representation of a Relation

A relation is graphically represented as follows.

If **A** = {1, 4, 6}, **B** = {1, 2, 3} then a relation **R** = {(1, 1) (4, 2) (6, 2) (6, 3)} from **A** to **B** is represented through a graph as follows.



Types of Relations

The types of relations are nothing but their properties. There are different types of relations namely **reflexive, symmetric, transitive and anti symmetric** which are defined and explained as follows through real life examples.

Reflexive Relation :

A relation R is said to be reflexive over a set A if $(a, a) \in R$ for every $a \in R$.

Example:

If A is the set of all males in a family, then the relation "is brother of" is not reflexive over A . Because any person from the set A cannot be brother of himself.

Symmetric Relation:

A relation R is said to be symmetric if $(a, b) \in R \Rightarrow (b, a) \in R$

Example:

If A is the set of all males in a family, then the relation "is brother of" is symmetric over A .

Because if a is brother of b then b is brother of a .

Transitive Relation:

A relation R is said to be symmetric if $(a, b) \in R, (b, c) \in R \Rightarrow (a, c) \in R$.

Example:

If A is the set of all males in a family, then the relation "is brother of" is transitive over A .

Because if a is brother of b and b is brother of c then a is brother of c .

Anti Symmetric Relation:

A relation R is said to be anti-symmetric if $(a, b) \in R$ and $(b, a) \in R \Rightarrow a = b$.

Example:

The relation " $\leq$ " on the set of real numbers is anti-symmetric because if $a \leq b$ and $b \leq a$ then $a = b$.

HCF AND LCM

Greatest Common Divisor (GCD)

It is the greatest divisor common to two or more given numbers. It is also called HCF (Highest Common Factor).

Method to find HCF of given Numbers.

1. By method of factorization:

Express each given number as product of primes. Now take product of common factors, which is HCF.
e.g., Find HCF of 136, 144, 168.

$$136 = 2 \times 2 \times 2 \times 17 = 2^3 \times 17$$

$$144 = 2 \times 2 \times 2 \times 2 \times 3 \times 3 = 2^4 \times 3^2$$

$$168 = 2 \times 2 \times 2 \times 3 \times 7 = 2^3 \times 3 \times 7$$

$$\text{HCF} = 2^3 = 8$$

2. By division method:

Suppose two numbers are given.

Divide greater number by the lesser one; divide

lesser by the remainder; divide first remainder by the new remainder, and so on till there is no remainder. Last divisor is the required H.C.F.

e.g., Find H.C.F. of 12 and 15.

$$\begin{array}{r} 12 \overline{) 15} (1 \\ \underline{-12} \\ 3 \end{array} \quad \begin{array}{r} 15 \overline{) 12} (4 \\ \underline{-12} \\ 0 \end{array}$$

$$\therefore \text{HCF} = 3$$

In case more than two numbers are given, then choose any two of them and find their HCF. HCF of these two and third gives HCF of three numbers and so on.

Note: If 'Z' is HCF of 'X' and 'Y', then Z is also a factor of $aX + bY$ where a, b are integers.

Least Common Multiple (LCM)

LCM of two or more given number is the least number which is exactly divisible by each of the given numbers.

Methods to find LCM of given Numbers.

1. By method of factorization:

Resolve each one of the given numbers into prime factors, then their LCM is product of highest powers of all factors, that occur in these numbers.

e.g., Find LCM of 136, 144, 168

$$136 = 2^3 \times 17$$

$$144 = 2^4 \times 3^2$$

$$168 = 2^3 \times 3 \times 7$$

$$\text{LCM} = 2^4 \times 3^2 \times 17 \times 7 = 17136$$

2. Using the formula:

Product of numbers = HCF $\times$ LCM

If two numbers are given, find their HCF.

$$\text{LCM} = \frac{\text{Product of two numbers}}{\text{HCF}}$$

e.g., Find LCM of 136, 144.

$$\text{LCM} = \frac{136 \times 144}{8} = 2448$$

If more than two numbers are given, find LCM of any two of them. Then LCM of this LCM and third number gives LCM of these three numbers and so on.

3. Short cut method using factorization:

e.g. Find LCM of 18, 28, 108, 105.

2	18,	28,	108,	105
2	9,	14,	54,	105
3	9,	7,	27,	105
3	3,	7,	9,	35
3	1,	7,	3,	35
5	1,	7,	1,	35
7	1,	7,	1,	7
	1,	1,	1,	1

$$\text{LCM} = 2 \times 2 \times 3 \times 3 \times 3 \times 5 \times 7 = 3780$$

Alternatively : We can strike out 18 at once, since any multiple of 108 must contain 18 as a factor.

2	28, 108, 105
2	14, 54, 105
3	7, 27, 105
7	7, 9, 35
	1, 9, 5

$$\text{LCM} = 2 \times 2 \times 3 \times 7 \times 9 \times 5 = 3780$$

Note: For any given numbers, HCF is necessarily a factor of their L.C.M.

HCF and LCM of Fractions.

$$\text{HCF of Fractions} = \frac{\text{HCF of Numerators}}{\text{LCM of Denominators}}$$

$$\text{LCM of Fractions} = \frac{\text{LCM of Numerators}}{\text{HCF of Denominators}}$$

e.g. Find HCF and LCM of $\frac{4}{5}, \frac{8}{35}, \frac{2}{15}, \frac{6}{25}$

$$\begin{aligned} \therefore \text{HCF} &= \frac{\text{HCF of } 4, 8, 2, 6}{\text{LCM of } 5, 35, 15, 25} \\ &= \frac{2}{5 \times 7 \times 3 \times 5} = \frac{2}{525} \\ \text{LCM} &= \frac{\text{LCM of } 2, 8, 4, 6}{\text{HCF of } 5, 35, 15, 25} = \frac{24}{5} \end{aligned}$$

PROGRESSIONS

Arithmetic Progression (A.P.)

Let 'a' be the first term of an arithmetic progression; 'd' the common difference and 'n' the number of terms in the progression.

$$T_n = n^{\text{th}} \text{ term} = a + (n - 1)d$$

$$S_n = \text{Sum of } n \text{ terms} = \frac{n}{2} \times [2a + (n - 1)d]$$

Progression can be represented as a, a + d, a + 2d,, [a + (n - 1) d]. Here, quantity d is to be added to any chosen term to get next term of the progression.

When there are n terms in an arithmetic progression, a is first term and {a + (n - 1) d} is last term.

$$\text{Then } S_n = \frac{n}{2} \times [\text{First Term} + \text{Last Term}]$$

Geometric Progression (G.P.)

Geometric progression can be represented as a, ar, ar², where a is first term and r is common ratio of the geometric progression.

nth term of the geometric progression = arⁿ⁻¹.

$$\begin{aligned} \text{Sum to } n \text{ terms} &= \frac{a(1 - r^n)}{1 - r} \text{ or } \frac{a(r^n - 1)}{r - 1} \\ &= \frac{r \times \text{last term} - \text{First term}}{r - 1} \end{aligned}$$

Infinite Geometric Progression.

If $-1 < r < 1$ or $|r| < 1$, then sum of a geometric progression does not increase infinitely; it "converges" to a particular value. Such a G.P. is called infinite geometric progression.

$$\text{Sum of an infinite geometric progression, } S_{\infty} = \frac{a}{1 - r}$$

Remember

$$\text{Sum of first } n \text{ natural numbers, } = S_n = \sum n = \frac{n(n+1)}{2}$$

Sum of squares of first n natural numbers,

$$\sum n^2 = \frac{n(n+1)(2n+1)}{6}$$

Sum of cubes of first n natural numbers,

$$\sum n^3 = \left[\frac{n(n+1)}{2} \right]^2 = \frac{n^2(n+1)^2}{4} = \left[\sum n \right]^2$$

ALGEBRAIC FORMULAE AND THEIR APPLICATION

An equation is a statement that *two algebraic expressions are equal*. If an equation is satisfied by any value of the variable, then equation is said to be an *identity*.

- $(a + b)^2 = a^2 + 2ab + b^2 = (a - b)^2 + 4ab$
- $(a - b)^2 = a^2 - 2ab + b^2 = (a + b)^2 - 4ab$
- $(a + b)^2 + (a - b)^2 = 2(a^2 + b^2)$
- $(a + b)^2 - (a - b)^2 = 4ab$
- $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$
- $(a + b + c + d)^2 = a^2 + b^2 + c^2 + d^2 + 2a(b + c + d) + 2b(c + d) + 2cd$
- $(a + b)(a - b) = a^2 - b^2$
- $(x + a)(x + b) = x^2 + (a + b)x + ab$
- $(x + a)(x + b)(x + c) = x^3 + (a + b + c)x^2 + (ab + bc + ca)x + abc$
- $(a + b)^3 = a^3 + 3ab(a + b) + b^3$
- $(a - b)^3 = a^3 - 3ab(a - b) - b^3$
- $a^3 + b^3 = (a + b)^3 - 3ab(a + b) = (a + b)(a^2 - ab + b^2)$
- $a^3 - b^3 = (a - b)^3 + 3ab(a - b) = (a - b)(a^2 + ab + b^2)$
- $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - ac - bc)$
If $a + b + c = 0$ then $a^3 + b^3 + c^3 = 3abc$
- $a^n - b^n = (a - b)(a^{n-1} + a^{n-2}b + a^{n-3}b^2 + \dots + b^{n-1})$ for all n.
- $a^n + b^n = (a + b)(a^{n-1} - a^{n-2}b + a^{n-3}b^2 - \dots - b^{n-1})$
if n is even
- $a^n + b^n = (a + b)(a^{n-1} - a^{n-2}b + a^{n-3}b^2 + \dots + b^{n-1})$
if n is odd.
- $a^4 + a^2b^2 + b^4 = (a^2 + ab + b^2)(a^2 - ab + b^2)$

POLYNOMIALS

Factors of polynomials: An expression is said to be resolved into factors when expressions of its product are found.

HCF of polynomials: When two or more polynomials are factorized, then product of all the common factors is HCF of the polynomials.

LCM of polynomials: When two or more polynomials are factorized, then product of all the factors with highest power is LCM of the polynomials.

INEQUATIONS

Properties

1. An inequality will still hold after each side has been increased, diminished, multiplied or divided by the same positive quantity.

i.e., if $a > b$ and $c > 0$

$$a + c > b + c$$

$$ac > bc$$

$$\frac{a}{c} > \frac{b}{c}$$

2. In an inequality, any term may be transposed from one side to the other if its sign is changed.

i.e., if $a - c > b$, then $a > b + c$ or $c < a - b$

3. If sides of an inequality be multiplied by same negative quantity, then sign of inequality must be reversed.

i.e., if $a > b$ and $c < 0$, then $ac < bc$

4. If $a > b$, $a, b \geq 0$ then $a^n > b^n$ and $\frac{1}{a^n} < \frac{1}{b^n}$, or $a^{-n} < b^{-n}$; if n is a positive quantity.

5. Square of every real quantity is positive and therefore must be greater than zero

i.e., for $a \neq b$, $(a - b)^2 > 0$; $a^2 + b^2 > 2ab$;

Similarly, if $x > 0$, $y > 0$, then

$$\frac{x+y}{2} \geq \sqrt{xy}$$

Hence, Arithmetic mean of two positive quantities is greater than or equal to their Geometric mean.

6. If sum of two positive quantities is given, their product is greatest when they are equal; and if product of two positive quantities is given, their sum is least when they are equal.

7. If $a, b, c, \dots, k$ are n unequal quantities, then

$$\left(\frac{a+b+c+\dots+k}{n} \right)^n > a \times b \times c \times d \dots \times k$$

$$\text{i.e., } \frac{a+b+c+\dots+k}{n} > (a \times b \times c \times \dots \times k)^{1/n}$$

Note: Arithmetic mean of any number of positive quantities is greater than their Geometric mean.

8. If a and b are positive and unequal, then

$$\frac{a^m b^m}{2} > \left(\frac{a+b}{2} \right)^m \text{ except when } m \text{ is a positive proper fraction.}$$

If m is a positive integer or any negative quantity,

$$\text{then } \frac{a^m b^m}{2} > \left(\frac{a+b}{2} \right)^m$$

If m is positive and less than 1, then $\frac{a^m b^m}{2} < \left(\frac{a+b}{2} \right)^m$

If there are n positive quantities $a, b, c, \dots, k$, then

$$\left(\frac{a^m + b^m + c^m + \dots + k^m}{n} \right) > \left(\frac{a+b+c+\dots+k}{n} \right)^m$$

unless m is a positive proper fraction.

9. If a, b, c are positive and not all equal, then

$$(a+b+c)(ab+bc+ca) > 9abc$$

$$\text{and, } (b+c)(c+a)(a+b) > 8abc.$$

10. If x is positive and $a < b$, then $\frac{a+x}{b+x} > \frac{a}{b}$

$$\text{If } x \text{ is positive and } a > b, \text{ then } \frac{a+x}{b+x} < \frac{a}{b}$$

11. If x is positive and $a > b > x$, then $\frac{a-x}{b-x} > \frac{a}{b}$

$$\text{If } x \text{ is positive and } x < a < b, \text{ then } \frac{a-x}{b-x} < \frac{a}{b}$$

12. $\frac{a+c+e+\dots}{b+d+f+\dots}$ is less than greatest and greater than

least of the fractions $\frac{a}{b}, \frac{c}{d}, \frac{e}{f}, \dots$

13. For positive numbers, if $a > x, b > y, c > z$, then $a + b + c + \dots > x + y + z + \dots$ and $abc \dots > xyz \dots$

14. $a^2 + b^2 + c^2 \geq bc + ca + ab$

15. $(n!)^2 > n^n$, for $n > 2$.

16. For any positive integer n , $2 \leq \left(1 + \frac{1}{n} \right)^n \leq 3$

17. $a^2b + b^2c + c^2a \geq 3abc$

18. $\frac{a}{b} + \frac{b}{c} + \frac{c}{d} + \frac{d}{a} \geq 4$

19. $a^4 + b^4 + c^4 + d^4 \geq 4abcd$

QUADRATIC EQUATION WITH APPLICATION TO INEQUALITIES

In a quadratic equation $ax^2 + bx + c = 0$, if $b^2 - 4ac > 0$, then

$ax^2 + bx + c = a(x - \alpha)(x - \beta)$, where α and β are two roots of the equation with $\alpha < \beta$.

$(x - \alpha)(x - \beta)$ is always positive when $x < \alpha$ or $x > \beta$ i.e., x does not lie between α and β .

$(x - \alpha)(x - \beta)$ is always negative when $\alpha < x$ or $x > \beta$ i.e., x lies between α and β .

DATA INTERPRETATION

Data can be organized in a number of ways so that larger volume of data can be presented in a more compact and precise form. Data thus presented has to be deciphered correctly by the user of the data. The deciphering of the data from its compactly presented form is called Data Interpretation.

Representation of Data

Numerical data can be presented in one or more of the following ways

- (1) Data Table
- (2) Pie Chart
- (3) Line Graph
- (4) Bar Chart
- (5) Others

The “Others” category covers miscellaneous form like descriptive case format etc. customized for the situation. Data can also be presented by using a combination of two or more of the above forms.

While some data can be presented in many different forms, some other may be amenable to be presented only in few ways. In real life situation, the style of data presentation is based on the end-objective. In certain situation data has to be presented as a combination of two or more forms of data presentation. Let us understand each of the above form of data presentation with an example.

Data Table

Here data is presented in the form of table. Where any type of data can be presented in tabular form, that too is a very accurate manner, interpreting the data in table form becomes more difficult and time consuming than the other modes, all of which are basically pictorial or graphical in presentation.

Year	A		Year	B	
	Appeared	Qualified		Appeared	Qualified
2002	250	720	2002	1750	460
2003	2750	810	2003	1860	490
2004	3000	890	2004	2000	520
2005	3250	910	2005	2100	640
2006	3720	1050	2006	2400	830

Year	C		Year	D	
	Appeared	Qualified		Appeared	Qualified
2002	100	120	2002	800	120
2003	1120	200	2003	1000	220
2004	1250	300	2004	1200	300
2005	1500	600	2005	1210	340
2006	1650	780	2006	1440	480

Year	E	
	Appeared	Qualified
2002	2000	370
2003	2200	420
2004	2500	510
2005	2750	680
2006	3440	960

Data tables can be of a number of types. They can be of a single –table variety or combination of tables. One such example of table is given above.

The above table shows the number of students appeared and qualified in an entrance test from five districts A, B, C, D and E of a state.

From the above table, we can obtain the following data;

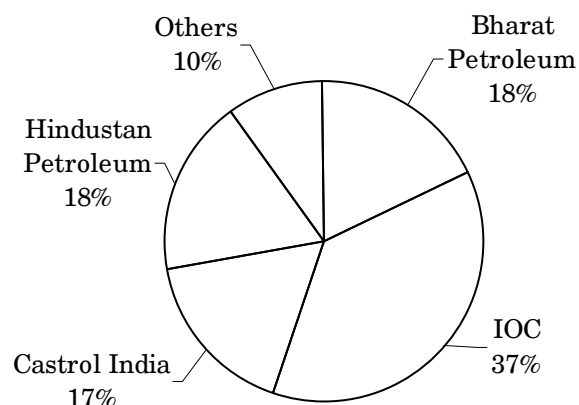
- (1) Total number of students appeared and qualified from the given districts in each of the years.
- (2) Percentage increase in the number of students appeared or qualified in the district over the years.
- (3) Average number of students appeared or qualified

Pie-Charts

This is probably the simplest of all pictorial forms of data presentation. Here, total quantity to be shown is distributed over one complete circle or 360 degree. In pie-charts, data is essentially presented shares of various elements as proportion and percentage of the total quantity. Each element or group in a pie-chart is represented in the terms of quantity (or value, as the case may be) or as the angle made by the sector representing the elements or as a proportion of the total or as a percentage of the total.

Chart 1 gives distribution of sales of different companies.

CHART – 1
Total sales Rs. 6,000 crores



From the above pie chart, we can calculate the following;

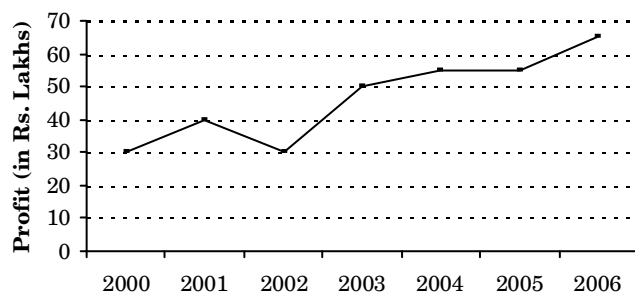
- (1) Total sales of each of the companies.
- (2) Sales of a company as a percentage of the other.
- (3) Conversion of these percentage values into angles for each zone.

Pie-chart are also very frequently used in combination with other forms of data or along with other pie-charts.

Two-Dimensional Graphs

This is essentially used for continuous data but can also be used for depicting discrete data provided we understand the limitation. Also known as Cartesian Graphs, they represent variation of the one parameter with respect to another parameter each shown on a different axis. These types of graphs are useful in studying the rate of change or understanding the trends through extrapolations.

These graphs can be of various types and a few of them are shown below:

CHART - 2

The graphs in Chart 2 show the changes in the profit of the company during a period time. One can find out trends and the growth in the profit over the years.

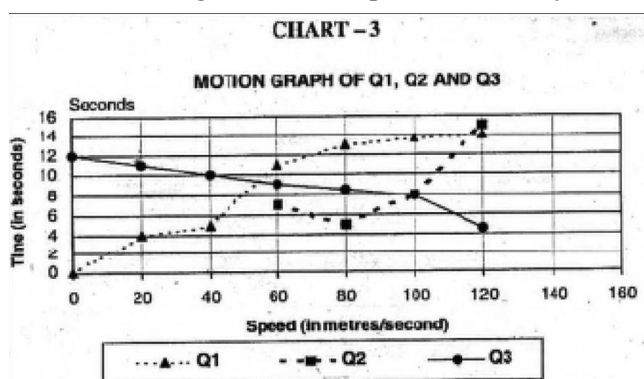


Chart 3 presents another type of two-dimensional graph which is mostly used to depict scientific data like speed, velocity, vectors etc. In the graph speed trends of three bodies Q1, Q2, Q3 is given along with their actual path of motion.

Bar Chart

This is a type of graph used mostly to depict data in a discrete way. They are accurate and comparison of variable is very convenient.

CHART-4

Import and Export of XYZ Company

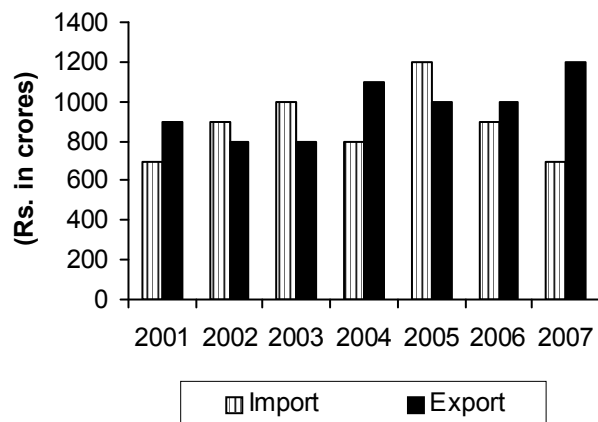


Chart 4 shows import and export of a company over different years. From this graph we can obtain the following:

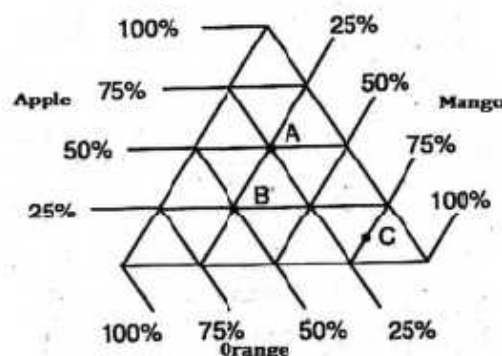
- (1) Percentage contribution of imports or exports to the company's total trade for different years.
- (2) Relative increase or decrease in the share of imports or exports
- (3) Percentage growth/average annual growth in imports or exports during given period.

Three-Dimensional Graph

The data in a triangular graph are given on each side of the triangle. Each point represents three different values, one each in each direction.

CHART 5

Students	Percentage					
A	82	66	59	76	62	65
B	76	72	65	84	74	75
C	56	78	71	66	86	70
D	64	80	68	72	66	80
E	48	68	83	88	56	60
F	60	74	79	64	80	85



The graph in chart 5 represents the percentage of people who like the three Fruit-Apple, Mango and Orange in three cities A, B and C. For example, in city B 25% like Mango, 50% like Orange and 25% like Apple.

EXERCISE

MCQ TYPE QUESTIONS

- If $x = b + c$, $y = c - a$, $z = a - b$, then $x^2 + y^2 + z^2 - 2xy - 2xz + 2yz$ is equal to
 (a) $a + b + c$ (b) $4b^2$
 (c) abc (d) $a^2 + b^2$
- HCF of $y^3 - 4y$ and $4y(y^3 + 8)$ is
 (a) $y(y + 2)$ (b) $y - 2$
 (c) $2(y + 2)$ (d) $(y + 2)(y^2 - 2y + 4)$
- A tiled floor of a room has dimensions $m \times m$ sq.m. Dimensions of the tile used are $n \times n$ sq.m. All tiles used are green tiles except diagonal tiles which are red. After some years some green tiles are replaced by red tiles to form an alternate red and green tile pattern. How many green tiles are removed?
 ($m \neq n$ and total number of tiles are odd).
 (a) $\frac{m^2 - 4mn + 2n^2}{2n^2}$ (b) $\frac{(m - 2n)^2 - n^2}{2n^2}$
 (c) $\frac{m^2 - 4mn - n^2}{2n^2}$ (d) $\frac{m^2 - 4mn - 2n^2}{2n^2}$
- If $\frac{x - 4}{-x^2 - 6} \leq \frac{2}{-x^2 - 6}$ then
 (a) $x \leq 2$ (b) $x \geq 2$
 (c) $x \leq 6$ (d) $x \geq 6$
- Which of the following is true?
 (a) $\sqrt{2} + \sqrt{5} = \sqrt{7}$ (b) $\sqrt{2} + \sqrt{5} \leq \sqrt{7}$
 (c) $\sqrt{2} + \sqrt{5} < \sqrt{7}$ (d) $\sqrt{2} + \sqrt{5} > \sqrt{7}$
- If $6 \geq x \geq -2$ and $4 \geq y \geq -4$, then limits for $\frac{y}{x}$, where x and y are non zero integers, is
 (a) $\frac{y}{x} \geq 2, \frac{y}{x} \leq \frac{2}{3}$ (b) $\frac{y}{x} \geq \frac{-2}{3}, \frac{y}{x} \leq 2$
 (c) $\frac{y}{x} \geq \frac{-2}{3}, \frac{y}{x} \leq \frac{1}{4}$ (d) $\frac{y}{x} \geq -4, \frac{y}{x} \leq 4$
- What are the limits of $\frac{2^n(n+1)^n}{n^n}$; where n is a positive integer?
 (a) $2 \leq \frac{2^n(n+1)^n}{n^n} \leq 3$
 (b) $2^{n+1} \leq \frac{2^n(n+1)^n}{n^n} \leq 3.2^n$
 (c) $2^n \leq \frac{2^n(n+1)^n}{n^n} \leq 3.2^n$
 (d) None of these
- If $g = 10^{100}$ and $H = 10^g$, then in which interval does $g! = 1.2.3 \dots 10^{100}$ lie?
 (a) $10H < g! < H$ (b) $H < g! < 10^H$
 (c) $10H < g! < 10^H$ (d) $10^H < g!$
- Which of the following is true?
 (a) Sum of four consecutive even numbers is always divisible by 8.
 (b) Sum of four consecutive odd numbers is always divisible by 8.
 (c) Product of any n consecutive natural numbers may not be divisible by $n!$.
 (d) Product of 4 consecutive odd numbers is always divisible by 15.
- If a and b are prime numbers, which of the following is true?
 I. a^2 has three positive integer factors.
 II. ab has four positive integer factors.
 III. a^3 has four positive integer factors.
Codes :
 (a) I and II only (b) II and III only
 (c) All of these (d) None of these
- If $7x + 6y = 420$, x and y are natural numbers, then what can be said about x ?
 (a) x is always odd
 (b) x is always even.
 (c) x is even only if y is odd.
 (d) x is odd if y is even.
- If a, b, c, d, p and q are non-zero, unequal integers and $\frac{a + bi}{c + di} = \frac{p}{q}$, then $\frac{a^2 + b^2}{c^2 + d^2}$ equals
 (a) $\frac{p}{q}$ (b) $\frac{p^2}{q^2}$
 (c) 1 (d) None of these
- An Egyptian fraction has a numerator equal to 1, and its denominator is a positive integer. What is the maximum number of different Egyptian fraction such that their sum is equal to 1, and their denominators are equal to 10 or less?
 (a) 3 (b) 5 (c) 7 (d) 9
- What is the least number which must be subtracted from 1936 so that the remainder when divided by 9, 10, 15 will leave in each case the same remainder 7?
 (a) 32 (b) 53
 (c) 46 (d) 39
- What is the greatest number consisting of six digits which on being divided by 6, 7, 8, 9, 10 leaves 4, 5, 6, 7, 8 as remainders respectively?
 (a) 997920 (b) 997918
 (c) 999999 (d) 997922
- What is the least number which on being divided by 5, 6, 8, 9, 12 leaves in each case a remainder 1 but when divided by 13 leaves no remainder?
 (a) 2987 (b) 3601
 (c) 3600 (d) 2986

17. A group of four numbers has only one prime number amongst them. Which of the following must be true about the group?
- HCF of the four numbers of the group is either 1 or equal to that prime number.
 - LCM of the four numbers of the group is same as product of the prime number and LCM of the remaining three numbers.
 - Product of four numbers is equal to product of the prime number $\times$ HCF of the group $\times$ LCM of the group.
- I only
 - II only
 - I and II only
 - All of these
18. If $(a, n)!$ is defined as product of n consecutive numbers starting from a , where a and n are both natural numbers, and if H is the HCF of $(a, n)!$ and $n!$, then what can be said about H ?
- $H = a!$
 - $H = n!$
 - $H \geq n!$
 - $H \geq a \times n$
19. HCF and LCM of two numbers is given. It is possible to find out the two numbers uniquely if
- either sum or difference between the two numbers is known.
 - HCF of two numbers = LCM of two numbers.
 - $\frac{\text{LCM}}{\text{HCF}} = \text{Prime number}$.
- I and II only
 - II only
 - II and III only
 - I, II and III
20. A ball is dropped from a height of 12 m and it rebounds $\frac{1}{2}$ of the distance it falls. If it continues to fall and rebound in this way, how far will it travel before coming to rest?
- 36 m
 - 30 m
 - 48 m
 - 60 m
21. There are N questions in an exam. For $i = 1, 2, \dots, N$, there are 2^{N-i} students who answered 1 or more questions wrongly. If total number of wrong answers is 8191, then N will be
- 12
 - 11
 - 10
 - 13
22. Two new charity organizations C_1 and C_2 were formed, with x members each, on January 1, 2003. On first day of each, subsequent month, in C_1 , number of members increases by a certain number a , while in C_2 , number of members increases in such a way that ratio of the number of members in a month to the preceding month bear a ratio equal to b . On May 1, 2003, both organizations had the same number of members. If $a = 20x$, then b will be
- 2
 - 3
 - 2.5
 - 3.5
23. What is the ratio of common differences d_1 and d_2 of two arithmetic progressions if respective n^{th} terms are in the ratio of $2n + 3 : n - 11$?
- 1 : 2
 - 2 : 3
 - 2 : 1
 - 1 : 3
24. What is the sum upto 20 terms of the series $\frac{2}{3} + \frac{8}{9} + \frac{26}{27} + \frac{80}{81} + \dots$?
- $\frac{19.3^{20} + 1}{3^{20}}$
 - $\frac{19.3^{20} + 1}{2.3^{20}}$
 - $\frac{19.3^{20} - 1}{2.3^{20}}$
 - $\frac{39.3^{20} + 1}{2.3^{20}}$
25. What is the sum to infinity of the series, $3 + 6x^2 + 9x^4 + 12x^6 + \dots$ given $|x| < 1$?
- $\frac{3}{(1+x^2)}$
 - $\frac{3}{(1-x^2)^2}$
 - $\frac{3}{(1+x^2)^2}$
 - $\frac{3}{(1-x^2)}$
26. A school has 5 divisions in a class IX having 60, 50, 55, 62 and 58 students. Mean marks obtained in a History test were 56, 64, 72, 63 and 50 by each division respectively. What is overall average of the marks per students?
- 56.8
 - 58.2
 - 62.4
 - 60.8
27. Average age of a committee of seven trustees is the same as it was five years ago, a younger man having been substituted for one of them. How much younger was he than trustee whose place he took?
- 32 years
 - 35 years
 - 33 years
 - 34 years
28. Six men A, B, C, D, E, F agree with a seventh man G to provide a sum of money among them. A, B, C, D, E, F are to subscribe ₹10 each, and G is to pay ₹3 more than the average of the seven. What is the whole sum to be provided?
- ₹73.50
 - ₹74
 - ₹73
 - ₹72.50

29. Which of the following is/are true?

- I. When Arithmetic Mean of a set of values and number of values in the set are known, then Geometric Mean of the set can be found out.
- II. When Arithmetic Mean of a set of values and number of values in the set are known, sum of all the elements in the set can be found out.
- III. When Geometric Mean of a set of values and number of values in the set are known, then Arithmetic Mean of the set can be found out.

Codes :

- (a) Only I (b) Only II
(c) Only I and II (d) All except II

30. Which of the following statement/s is/are true?

- I. Average of a set of values will always lie between lowest and the largest of these values.
- II. If each of the values in a set is increased by a constant k, then new average of the set is increased by k.
- III. If each of the values in a set is multiplied by a constant m, then new average will also be 'm' times the old average.

Codes :

- (a) Only I and II (b) Only II and III
(c) Only I and III (d) I, II and III

31. Three people A, B and C weigh themselves in a particular order. First A, B, C weigh themselves individually and then AB, BC CA and ABC together respectively. Recorded weight for the last measured is 180 kgs. Then average of the 7 measures is

- (a) 320 kgs (b) $\frac{360}{7}$ kgs
(c) $\frac{720}{7}$ kgs (d) Cannot be determined

32. Average age of a team of 15 employees is 36. Youngest of them is 20 years old and eldest is 36 years old. Two of them with average age 28 leave the team. If one of the two comes back on the condition that he will be made the team leader, then which of the following can possibly be average age of the new team so formed?

- (a) 35 (b) 36 (c) 38 (d) 39

Directions for Q.33 to Q.37:

Study the following table carefully and answer the questions given below:

Annual Income of Various Individuals over the Years
(Income in Lakhs)

Year	Individual					
	A	B	C	D	E	F
2002	5.5	3.2	4.8	6.35	6.15	3.5
2003	5.65	3.25	4.85	6.55	6.25	3.65
2004	5.7	3.725	4.95	7.155	6.425	3.75
2005	6.2	4.25	5.1	7.35	7.15	4
2006	6.5	4.5	5.2	7.4	7.25	4.25
2007	6.75	5	5.25	7.48	7.285	4.8
2008	7	5.35	5.3	8	7.3	5.1

33. What is the approximate average monthly income of A over the years?

- (a) Rs. 44762 (b) Rs.48348
(c) Rs. 53216 (d) none of these

34. What is the ratio of the annual income of B to that of C in the year 2005?

- (a) 6 : 5 (b) 6 : 7
(c) 5 : 6 (d) 7 : 6

35. What is the difference between the total annual income of all the individuals in the year 2003 and the total annual income of all the individuals in the year 2007?

- (a) Rs. 736500 (b) Rs.645400
(c) Rs. 636500 (d) Rs.743300

36. The annual income of E in the year 2004 is approximately what per cent of the average annual income earned by E over the years?

- (a) 98 (b) 86
(c) 88 (d) 94

37. What is the average annual income of F over the years?

- (a) Rs. 416000 (b) Rs.425000
(c) Rs.401500 (d) none of these

Directions for Q.38 to Q.42:

Study the pie-charts carefully to answer the questions that follow:

Chart I gives : Percentage distribution of candidates studying in various disciplines of a college.

Total Number of students = 4800

Chart II gives : Percentage of breakup of girl students in each discipline. Total number of girls = 2000

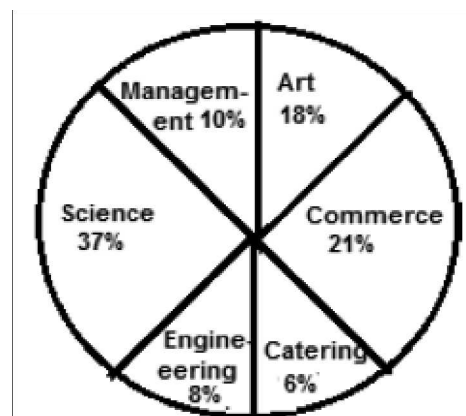


Chart I

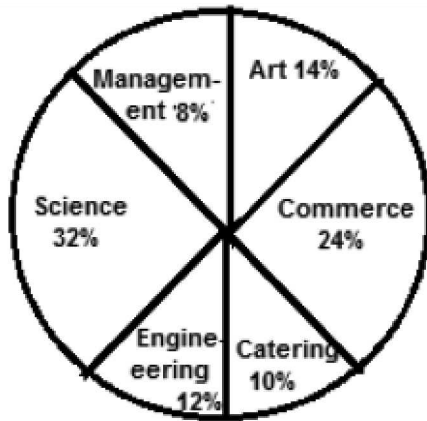


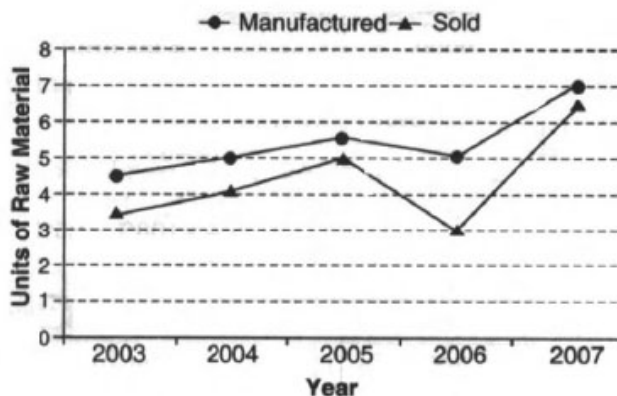
Chart II

38. What is the number of girls studying in Engineering discipline?
 (a) 180 (b) 220
 (c) 320 (d) none of these
39. Which discipline has the highest number of boys studying in it?
 (a) Science (b) Commerce
 (c) Arts (d) None of these
40. The number of boys studying in Commerce discipline forms approximately what per cent of the total number of students studying in that discipline?
 (a) 40 (b) 52
 (c) 36 (d) 28
41. What is the respective ratio of number of boys studying in Management discipline to the boys studying in Science discipline?
 (a) 71:20 (b) 18:67
 (c) 20:71 (d) None of these
42. The number of girls studying in Catering discipline forms what per cent of the total number of students in the college? (rounded off two digits after decimal)
 (a) 3.28 (b) 5.61
 (c) 3.95 (d) 4.17

Directions for Q.43 to Q.44:

Study the following graphs carefully to answer the questions given below.

Unit of raw material manufactured and sold by a company over the years (Units in crores)



43. What is the average number of unit sold over the years?

(a) 440000000 (b) 4400000
 (c) 440000 (d) 44000000

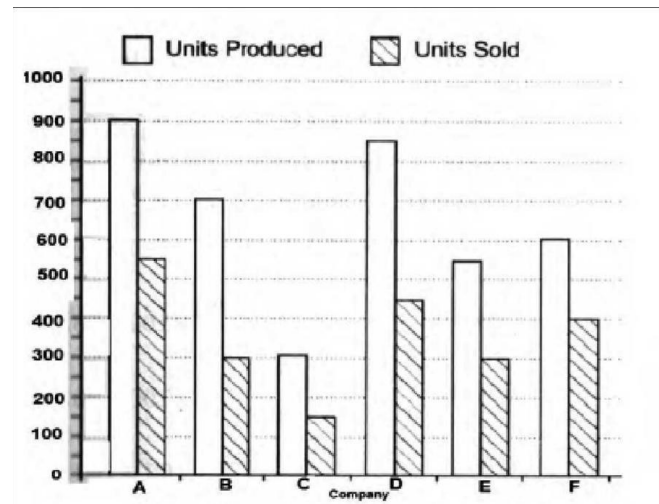
44. What is the ratio of the difference between the number of units manufactured and sold in the year 2005 to the difference between the number of units manufactured and sold in the year 2006?

(a) 2 : 3 (b) 1 : 2
 (c) 1 : 4 (d) 3 : 5

Directions for Q.45 to Q.49:

Study the following graph carefully to answer the questions that follow:

Production and sale of printers of various companies in a month



45. What is the average number of units sold by all the companies together?

(a) 360 (b) 390
 (c) 375 (d) 410

46. Which company had the highest percentage of sale with respect to its production?

(a) D (b) B
 (c) E (d) None of these

47. What is the average number of units produced by all the companies together?

(a) 675 (b) 650
 (c) 625 (d) 600

48. The total unit sold by the companies A, B and C together is approximately what per cent of the total units produced by these companies?

(a) 62 (b) 50
 (c) 76 (d) 58

49. What is the ratio of the total production of companies D and E to the total sale of the same companies?

(a) 28 : 15 (b) 9 : 5
(c) 15 : 11 (d) None of these

NUMERICAL TYPE QUESTIONS

- If $x - y = 1$, then $x^3 - y^3 - 3xy$ equals to _____
- When $x + y + z = 9$ and $xy + yz + zx = 11$, then $x^3 - y^3 - z^3 - 3xyz$ equals to _____
- If $x + y + z = 0$, then $\frac{x^2}{yz} + \frac{y^2}{zx} + \frac{z^2}{xy}$ equals to _____
- In an election for the President, if 261 valid votes are cast, for the 5 contestants then least number of votes a candidate requires to receive to win the election is _____
- A number when divided by sum of 555 and 445 gives two times their difference as quotient and 30 as remainder. The number is _____
- Sum of three prime numbers is 100. If one of them exceeds another by 36, then one of the numbers is _____
- Sum of all odd numbers up to 100 is _____
- $A = 0.a_1a_1a_1\dots$ and $B = 0.a_2a_2a_2\dots$, where a_1 and a_2 are multiples of 3 and also, a_1 and a_2 are distinct integers from 0 to 8. Then value of $A + B$ is _____
- Anil wants to divide ₹100 into a number of bags so that one can ask for any amount between ₹1 and ₹100, he can give the proper amount by giving certain number of these bags without taking out the amount from them. The minimum number of bags he will require if each bag has whole number of rupees is _____
- The remainder when $2050 \times 2071 \times 2095$ is divided by 23 is _____
- The smallest number which when increased by 3 is divisible by 27, 35, 25 and 21 is _____
- _____ is the least number which on being divided by 12, 21 and 35 will leave in each case the same remainder 6
- A gardener had a number of shrubs to plant in horizontal rows. At first he tried to plant 5 shrubs in each row, then 6, then 8 and then 12, but had always 1 left. On trying 13, in one row he had none left. The smallest number of shrubs that he could have had is _____
- Sum of an infinite geometric series is 3 and sum of an infinite geometric series formed from squares of the terms of the original series is 6. First term of the first series will be _____
- The maximum sum of the series 60, 58, 56, 54, 52, ... is _____
- On an average my income for 15 days ₹ 7, average for the first 5 days was ₹ 6 and average for the last 9 days was ₹ 8. The income on the sixth day was _____
- The harmonic mean of two numbers whose geometric mean and arithmetic mean is 8 and 5 respectively is _____
- Average marks of 15 students in a class is 145, maximum marks being 150. If two lowest scores are removed, the average increases by 5. Also, two lowest scores are consecutive multiples of 9. The lowest score in the class is _____

ANSWERS

MCQ Type Questions

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (b) | 2. (a) | 3. (b) | 4. (d) | 5. (d) | 6. (d) | 7. (b) | 8. (c) | 9. (b) | 10. (c) |
| 11. (b) | 12. (b) | 13. (a) | 14. (d) | 15. (b) | 16. (b) | 17. (a) | 18. (b) | 19. (d) | 20. (a) |
| 21. (d) | 22. (b) | 23. (c) | 24. (d) | 25. (b) | 26. (d) | 27. (b) | 28. (a) | 29. (b) | 30. (d) |
| 31. (c) | 32. (b) | 33. (d) | 34. (c) | 35. (c) | 36. (d) | 37. (d) | 38. (d) | 39. (a) | 40. (b) |
| 41. (c) | 42. (d) | 43. (c) | 44. (d) | 45. (b) | 46. (c) | 47. (a) | 48. (c) | 49. (d) | |

Numerical Type Questions

- | | | | | | | | | | |
|----------|---------|---------|----------|-----------|-------|----------|---------|------|-------|
| 1. 1 | 2. 432 | 3. 3 | 4. 53 | 5. 220030 | 6. 67 | 7. 2500 | 8. 1 | 9. 7 | 10. 6 |
| 11. 4722 | 12. 426 | 13. 481 | 14. 12/5 | 15. 930 | 16. 3 | 17. 12.8 | 18. 108 | | |

EXPLANATIONS

MCQ TYPE QUESTIONS

1. $x^2 + y^2 + z^2 - 2xy - 2xz + 2yz = (x - y - z)^2$
 $= (b + c - c + a - a + b)^2 = 4b^2$

2. $y^3 - 4y = y(y^2 - 4)$
 $= y(y - 2)(y + 2)$
 $4y(y^3 + 8) = 4y(y + 2)(y^2 - 2y + 4)$
 $\therefore \text{HCF} = y(y + 2).$

3. Total red tiles initially $= 2\left(\frac{m}{n}\right) - 1$

Total number of tiles $= \frac{m^2}{n^2}$

As number of red tiles > number of green tiles
 (since room has odd number of tiles and corner tiles are red)

Hence total red tiles required to form an alternate pattern

$$= \frac{\left(\frac{m}{n}\right)^2 + 1}{2} = \frac{m^2 + n^2}{2n^2}$$

$\therefore$ Number of green tiles removed

$$= \frac{m^2 + n^2}{2n^2} - 2\left(\frac{m}{n}\right) + 1$$

$$= \frac{m^2 + n^2 - 4mn + 2n^2}{2n^2}$$

$$= \frac{(m - 2n)^2 - n^2}{2n^2}$$

4. $\frac{x - 4}{-x^2 - 6} \leq \frac{2}{-x^2 - 6}$

i.e. $\frac{x - 4}{-(x^2 + 6)} \leq \frac{2}{-(x^2 + 6)}$

i.e. $x - 4 \geq 2$...(cancelling a negative term reverses sign of the inequality)

i.e. $x \geq 6.$

5. $(\sqrt{2} + \sqrt{5})^2 = 2 + 2\sqrt{10} + 5$
 $= 7 + 2\sqrt{10}$

$\therefore \sqrt{2} + \sqrt{5} > \sqrt{7}$

7. $\frac{2^n(n+1)^n}{n^n} = 2^n\left(\frac{n+1}{n}\right)^n = 2^n\left(1 + \frac{1}{n}\right)^n$

Now, $2 \leq \left(1 + \frac{1}{n}\right)^n \leq 3$

$\therefore 2 \cdot 2^n \leq 2^n \left(1 + \frac{1}{n}\right)^n \leq 3 \cdot 2^n$

i.e., $2^{n+1} \leq 2^n \left(1 + \frac{1}{n}\right)^n \leq 3 \cdot 2^n$

8. Choose k, so that

$$10^H = g^k = 10^{100k},$$

$$k = \frac{H}{100}$$

$\therefore 10^H = g^{\frac{H}{100}} > g^g > g!$

On the other hand,

$$10^H = 10 \times 10^g = 10^{g+1} < g!$$

(most factors are much bigger than 10)

$$10^H < g! < 10^H$$

9. (a) Four consecutive even numbers can be written as $2n, 2n + 2, 2n + 4$ and $2n + 6$, where n is any natural number.

$$\begin{aligned}\therefore \text{Sum} &= 2n + 2n + 2 + 2n + 4 + 2n + 6 \\ &= 8n + 12 \\ &= 4(2n + 3) \text{ not always divisible by 8.}\end{aligned}$$

Thus, (a) is not true.

- (b) Four consecutive odd numbers can be written as $2n + 1, 2n + 3, 2n + 5, 2n + 7$ where n is a natural number.

$$\text{Sum} = 2n + 1 + 2n + 3 + 2n + 5 + 2n + 7 = 8n + 16$$

divisible by 8

Thus, (b) is true

- (c) In product of n consecutive natural numbers atleast one is divisible by n , atleast one by $n - 1 \dots$ till 1.

$$\text{Thus product is atleast divisible by } n \times (n - 1) \times (n - 2) \dots 1 = n!$$

Thus, (c) is not true.

- (d) Take four consecutive odd numbers as $7 \times 9 \times 11 \times 13$ which is not divisible by 15.

Thus, (d) is not true.

10. Factors of a^2 are 1, a and a^2 .

Factors of ab are 1, a , b and ab .

Factors of a^3 are 1, a , a^2 and a^3 .

11. $7x + 6y = 420$

Equation is of the form:

$$7x + \text{even number} = \text{even number.}$$

$\therefore 7x$ has to be even

Hence x has to be even.

12. $\frac{a + bi}{c + di} = \frac{p}{q}$

$$\therefore qa + qbi = pc + pdi$$

Equating real and imaginary parts.

$$qa = pc \text{ and } qb = pd.$$

$$\Rightarrow a = \frac{p}{q}c \text{ and } b = \frac{p}{q}d$$

$$\therefore \frac{a^2 + b^2}{c^2 + d^2} = \frac{\frac{p^2}{q^2}c^2 + \frac{p^2}{q^2}d^2}{c^2 + d^2} = \frac{p^2}{q^2}$$

13. We ignore $\frac{1}{7}$ and $\frac{1}{9}$ because no sum of other numbers is going to give 7th or 9th in the denominator.

Also, $\frac{1}{5}$ and $\frac{1}{10}$ are not enough to add up to

anything $\left(\frac{1}{10}, \frac{2}{10}, \text{ and } \frac{3}{10}\right)$ are going to leave tenths left over no matter what else you add).

What's left is $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{6}, \frac{1}{8}$.

Sum total of these is $\frac{11}{8}$. So we need all of them

except $\frac{1}{8}$, which means $\frac{1}{2} + \frac{1}{3} + \frac{1}{6}$. Which is the only way to do this with egyptian fractions whose denominators are 10 or less. Hence maximum number of Egyptian fractions needed is 3.

14. L.C.M. of 9, 10 and 15 = 90

$$\therefore 1936 = 90 \times 21 + 46 \text{ (Remainder).}$$

But a part of this remainder is 7.

so $46 - 7 = 39$ should be subtracted from 1936.

15. $6 - 4 = 2, 7 - 5 = 2, 8 - 6 = 2, 9 - 7 = 2, 10 - 8 = 2$

L.C.M. of 6, 7, 8, 9, 10 = 2520;

Greatest number of 6 digits = 999999

$$2520 \times 396 + 2079 = 999999$$

$$\therefore \text{Reminder} = 2079.$$

Subtract 2079 from 999999, then we get

$$999999 - 2079 = 997920.$$

Subtract 2 from this number to get required number, which is 997918 and which will give the remainders 4, 5, 6, 7, 8 when divided by 5, 6, 7, 8, 9 respectively.

16. L.C.M. of 5, 6, 8, 9 and 12 is 360

Required number = $360K + 1$

$$= (13 \times 27 + 9)K + 1$$

$$= (13 \times 27)K + (9K + 1)$$

Now this number must be divisible by 13

$$\therefore K = 10 \text{ and required number} = 3601.$$

17. I. In case all the remaining three numbers are multiples of the prime number, HCF of the group is equal to that prime number. otherwise HCF of the group is 1. Thus, I is true.

- II. **Case 1:** Let numbers be 2, 4, 6 and 8, where 2 is only prime number, then LCM = 24.

But LCM of 4, 6 and 8 = 24.

$$\text{LCM} = 24 \text{ \& } 2 = 24$$

Case 2: Let numbers be 2, 9, 81 and 27 Then LCM = 162 and LCM of 9, 81 and 27 = 81.

Thus, LCM = $81 \times 2 = 162$

Hence II is not always true.

III. Let numbers be 2, 4, 6 and 8

$$\therefore \text{LCM} = 24 \text{ and } \text{HCF} = 2$$

Product of four numbers

$$= 2 \times 4 \times 6 \times 8 = 384$$

Also, $384 \neq 24 \times 2 \times 2$.

Thus, III is not true.

18. (a. n)! = product of n consecutive natural number starting from 'a' which is atleast divisible by n!.
(n)! = product of n consecutive natural numbers.

For n = 2 :

$$(a. n)! = a(a + 1) \text{ and } n! = 2$$

a(a + 1) is divisible by 2!.

For n = 3 :

$$(a. n)! = a(a + 1)(a + 2) \text{ and } n! = 6.$$

One of the factors of a(a + 1)(a + 2) is divisible by 3 and other by 2.

Thus, proceeding in this manner, (a. n)! and n! have HCF = n!

$$\therefore \text{H} = n!.$$

19. Let HCF be h and LCM be l.

I. Let numbers be ah and bh.

$$\text{Then } abh = l \text{ and } (a + b)h = m$$

$$\Rightarrow (a - b)h = n$$

Using these ah and bh can be uniquely determined. Thus, I is true

II. If HCF = LCM, then two numbers are equal and same as HCF or LCM.

Thus, II is true.

$$\text{III. } \frac{\text{LCM}}{\text{HCF}} = a \text{ prime i.e., } \frac{l}{h} = P.$$

Then one of the numbers is equal to h and other is equal to l. Thus, III is true.

20. Total distance travelled by the ball before coming to rest

$$= 12 + 2(6 + 3 + 1.5 + \dots \text{ to } ?)$$

$$= 12 + 2 \cdot \left(\frac{6}{1 - 1/2} \right)$$

$$= 12 + 2(12) = 36\text{m}$$

21. Number of students who answered 1 or more questions wrongly = 2^{N-1} .

Number of students who answered 2 or more questions wrongly = 2^{N-2}

Hence number of students who answered 1 question wrongly

$$= 2^{N-1} - 2^{N-2} = 2^{N-2}$$

Similarly it can be shown that number of students

who answered 2 questions wrongly

$$= 2^{N-2} - 2^{N-3} = 2^{N-3}$$

Similarly we can find number of students who answered K questions wrongly where $K \geq 3$

Hence total number of questions attempted wrongly

$$s = 2^{N-2} + 2(2^{N-3}) + 3(2^{N-4}) + \dots$$

$$+ (N-1)(2^0) + N(1) \dots (i)$$

$$\therefore \frac{s}{2} = 2^{N-3} + 2(2^{N-4}) + \dots + (N-2)(2^0)$$

$$+ \frac{N-1}{2} + \frac{N}{2} \dots (ii)$$

Subtracting equations (i) from (ii)

$$\frac{s}{2} = 2^{N-2} + 2^{N-3} + \dots + 2^0 + \frac{1}{2}$$

$$\Rightarrow s = 2^{N-1} + 2^{N-2} + 2^{N-3} + \dots + 1$$

$$= 2^N - 1 = 8191$$

$$\Rightarrow N = 13$$

22. Number of members in C_1 on May 1, 2003 = $x + 4a$

Number of members in C_2 on May 1, 2003 = xb^4

$$x + 4a = xb^4 \text{ and } a = 20x$$

$$\therefore x(b^4 - 81) = 0$$

$$\text{As } x \neq 0, \quad b^4 - 81 = 0$$

$$\therefore b = 3$$

23. n^{th} terms are in the ratio of $\frac{2n+3}{n-11}$.

If $n = 1$ in $2n + 3$ series, then first term is

$$T_1 = 2(1) + 3 = 5$$

$$n = 2, \text{ then second term is } 2(2) + 3 = 7$$

$$T_2 = 7$$

$$d = T_2 - T_1 = 7 - 5 = 2$$

$$\therefore d_1 = 2$$

If $n = 1$ in $n - 11$ series, then first term is $1 - 11 = -10$

If $n = 2$, then second term is $2 - 11 = -9$

$$\therefore d_2 = T_2 - T_1 = -9 - (-10) = 1$$

$$\therefore \frac{d_1}{d_2} = \frac{2}{1} \text{ or } 2 : 1$$

24. $\frac{2}{3} + \frac{8}{9} + \frac{26}{27} + \frac{80}{81} + \dots$ up to 20 terms

$$= \left(1 - \frac{1}{3}\right) + \left(1 - \frac{1}{9}\right) + \left(1 - \frac{1}{27}\right) + \left(1 - \frac{1}{81}\right) +$$

... up to 20 terms

$$= 20 - \left(\frac{1}{3} + \frac{1}{3^2} + \frac{1}{3^3} + \frac{1}{3^4} + \dots + \frac{1}{3^{20}} \right)$$

$$= 20 - \frac{1}{3} \left(\frac{1 - \left(\frac{1}{3}\right)^{20}}{1 - \frac{1}{3}} \right)$$

$$= 20 - \frac{3^{20} - 1}{2 \cdot 3^{20}}$$

$$= \frac{40 \cdot 3^{20} - 3^{20} + 1}{2 \cdot 3^{20}} = \frac{39 \cdot 3^{20} + 1}{2 \cdot 3^{20}}$$

25. Let $s = 3 + 6x^2 + 9x^4 + 12x^6 + \dots$
 $\therefore x^2s = 3x^2 + 6x^4 + 9x^6 + 12x^8 + \dots$
 $\Rightarrow s - x^2s = 3 + 3x^2 + 3x^4 + 3x^6 + \dots$
 $\Rightarrow s(1 - x^2) = 3[1 + x^2 + x^4 + x^6 + \dots]$
 $\Rightarrow s(1 - x^2) = 3 \left[\frac{1}{1 - x^2} \right] [\because |x| < 1]$
 $\therefore s = \frac{3}{(1 - x^2)^2}$

26. Average

$$= \frac{60 \times 56 + 50 \times 64 + 55 \times 72 + 62 \times 63 + 58 \times 50}{60 + 50 + 55 + 62 + 58}$$

$$= \frac{3360 + 3200 + 3960 + 3906 + 2900}{285} = 60.8$$

27. Increase after 5 years should have been

$$7 \times 5 = 35 \text{ years.}$$

But since average remains same, difference in their age is 35 years.

28. Let average of the seven men be x .

$$\therefore x = \frac{10 \times 6 + (x + 3)}{7}$$

$$\Rightarrow 60 + x + 3 = 7x$$

$$\Rightarrow x = \frac{63}{6} = 10.5$$

$$\therefore \text{Total amount} = 10.5 \times 7 = 73.5$$

29. Consider a set of three numbers a, b and c .

I. $AM = \frac{a + b + c}{3}$

Thus, $a + b + c = 3AM$.

But $GM = \sqrt[3]{abc}$ which cannot be found out.

Thus, I is false,

II. $AM = \frac{a + b + c}{3}$

$$\therefore \text{Sum of the numbers} = a + b + c = 3AM.$$

Thus, II is true.

III. $GM = \sqrt[3]{abc}$.

But $AM = \frac{a + b + c}{3}$, which cannot be found out.

Thus, III is false.

30. I. Average of a set of numbers is greater than smallest and smaller than the greatest number of the set. Thus, I is true.

II. Consider five numbers a, b, c, d and e whose

average is $\frac{a + b + c + d + e}{5}$

Now if each of them is increased by k , then we have average

$$= \frac{a + k + b + k + c + k + d + k + e + k}{5}$$

$$= \frac{a + b + c + d + e}{5} + \frac{5k}{5} = \text{old average} + k.$$

Thus, II is true.

III. Consider three numbers x, y and z whose

average is $\frac{x + y + z}{3}$

Now if each of them is multiplied by m then

new average = $\frac{xm + ym + zm}{3}$

$$= m \left(\frac{x + y + z}{3} \right) = m \times \text{old average}.$$

Thus, III is true.

31. Order of measures is

$$A, B, C, A + B, B + C, C + A, A + B + C.$$

Given: $A + B + C = 180$

Hence, average of the 7 measures

$$= \left[\frac{(A) + (B) + (C) + (A + B) + (B + C) + (C + A) + (A + B + C)}{7} \right]$$

$$= \frac{4}{7}(A + B + C) = \frac{4}{7} \times 180 = \frac{720}{7} \text{ kgs.}$$

32. Total age = 15×36

After two left, total age = $15 \times 36 - 2 \times 28$

$$= 540 - 56 = 484 \text{ years.}$$

Let the age of the person who returns be x years.

$$\text{Then new average} = \frac{484 + x}{14}$$

Now x lies between 20 and 36 (both inclusive)

$$\therefore \text{New average min} = \frac{484 + 20}{14} = 36 \text{ years}$$

$$\begin{aligned} \text{And New average max} &= \frac{484 + 36}{14} \\ &= 37.14 \text{ years.} \end{aligned}$$

33. Total income of A = 43.3 lakh

$$\begin{aligned} \text{Monthly income of A} &= \frac{43.3}{7 \times 12} \text{ lakh} \\ &= \text{Rs. } 51547 \approx \text{Rs. } 51245 \end{aligned}$$

34. Required ratio = $\frac{425}{510} = \frac{5}{6}$ i.e. 5 : 6

35. Required difference
= (36.565 – 30.2) lakh – Rs. 636500

36. Required percentage

$$\begin{aligned} &= \frac{6425}{47810} \times 100 \times 7 \\ &= 94\% \end{aligned}$$

37. Average annual income of F

$$\begin{aligned} &= \frac{1}{7} \left(\frac{3.50 + 3.65 + 3.75 + 4 + 4.25 + 4.80 + 5.10}{7} \right) \text{ lakh} \\ &= \frac{1}{7} (29.05) = 4.15 \text{ lakh} \end{aligned}$$

38. No. of girls studying in engineering discipline

$$= \frac{12}{100} \times 2000 = 240$$

39. It is obvious from pie-charts.

40. Number of students in Commerce discipline

$$= \frac{21}{100} \times 4800 = 1008$$

Number of girls in Commerce discipline

$$= \frac{24}{100} \times 2000 = 480$$

Number of boys = 1008 – 480 = 528

Hence, required %

$$= \frac{528}{1008} \times 100 = 52\%$$

41. Number of students in Management discipline

$$= \frac{10}{100} \times 4800 = 1008$$

Number of girls in Management discipline

$$= \frac{8}{100} \times 2000 = 160$$

Number of boys = 480 – 160 = 320

Number of students in Science discipline

$$= \frac{37}{100} \times 4800 = 1776$$

Number of girls in Science discipline

$$= \frac{32}{100} \times 2000 = 640$$

Number of boys 1776 – 640 = 1136

Hence, required ratio = 320 : 1136
= 20 : 71

42. Number of girls in catering discipline

$$= \frac{10}{100} \times 2000 = 200$$

Hence, required percentage

$$= \frac{200}{4800} \times 100 = 4.1\%$$

43. Required average

$$\begin{aligned} &= \frac{1}{5} [3.5 + 4 + 5 + 3 + 6.5] \\ &= \frac{1}{5} [22] = 4.4 \text{ crore} = 44000000 \end{aligned}$$

44. Required ratio

$$= \frac{(5.5 - 5)}{(5.3)} = \frac{0.5}{2}$$

$$= \frac{1}{4} \text{ ie, } 1 : 4$$

45. Average number of units sold by all the companies together

$$\begin{aligned} &= \frac{1}{6} (650 + 300 + 150 + 450 + 300 + 400) \\ &= \frac{1}{6} \times 2250 = 375 \end{aligned}$$

46. Hence, it is maximum for company A

47. Average number of units produced by all companies together

$$= \frac{1}{6} (900 + 700 + 300 + 850 + 550 + 600)$$

$$= \frac{1}{6} \times 3900 = 650$$

48. Required percentage = $\frac{1100}{1900} \times 100 = 58\%$

49. Required ratio = $\frac{(850 + 550)}{(450 + 300)} = \frac{1400}{750}$

$$= \frac{28}{15} \text{ ie, } 28 : 15$$

NUMERICAL TYPE QUESTIONS

- $x^3 - y^3 - 3xy$
 $= x^3 - y^3 - 3xy(x - y) \dots \dots \text{ as } (x - y) = 1$
 $= (x - y)^3 = 1.$
- $x^3 - y^3 - z^3 - 3xyz$
 $= (x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx)$
 $= 9[(x + y + z)^2 - 3(xy + yz + zx)]$
 $= 9(81 - 33) = 432.$
- $\frac{x^2}{yz} + \frac{y^2}{zx} + \frac{z^2}{xy}$
 $\dots \text{ (since, } x + y + z = 0; x^3 + y^3 + z^3 = 3xyz)$
 $\frac{x^3 + y^3 + z^3}{xyz} = \frac{3xyz}{xyz} = 3.$
- Worst scenario is when other four get equal number of votes.
 Let the winning candidate get x votes.
 $\therefore x > \frac{261}{5}$
 $\Rightarrow x > 52.2$
 $\therefore x = 53$
- Number = $(555 + 445) \times (555 - 445) \times 2 + 30$
 $= (555 + 445) \times 2 \times 110 + 30$
 $= 220000 + 30 = 220030.$
- $x + (x + 36) + y = 100$
 $\Rightarrow 2x + y = 64$
 Therefore y must be even prime which is 2
 $2x + 2 = 64 \Rightarrow x = 31.$
 $\therefore$ Third prime number
 $= x + 36 = 31 + 36 = 67.$
- Given numbers are 1, 3, 5.....99.
 This is an A.P with a = 1, d = 2.

Let it contain n terms

$$\therefore 1 + (n - 1)2 = 99 \Rightarrow n = 50$$

Hence required sum = $\frac{n}{2}$ (first term + last term)

$$= \frac{50}{2} (1 + 99) = 2500.$$

8. $A = 0.a_1a_1a_1\dots$
 $\therefore 10A = a_1.a_1a_1\dots$
 $\Rightarrow 9A = a_1$
 $\Rightarrow A = \frac{a_1}{9}$

Similarly, $B = \frac{a_2}{9}.$

Now, a_1 and a_2 are multiples of 3 and are distinct.
 Also, these values are less than 8.

either $a_1 = 3$ and $a_2 = 6$ or $a_1 = 6$ and $a_2 = 3$

$$\therefore a_1 + a_2 = 9.$$

$$\therefore A + B = \frac{a_1}{9} + \frac{a_2}{9} = \frac{a_1 + a_2}{9} = \frac{9}{9} = 1.$$

9. If Anil has to give 1 rupee he needs a bag with ₹1. For 2 rupees he had two bags with ₹1 each or ₹2. bag. To have minimum bags, he has a bag with ₹2. Now with the two bags he can give ₹3. So next he will require a bag with ₹4. With these three he can give ₹5. ₹6 and ₹7 and next bag will be one containing ₹8 and so on. Thus he would have bags with ₹1. 2. 4. 8. 16. 32. Sum of which is 63 and remaining 37 can be put in the last bag. So total number of bags is 7.

10. $2050 \times 2071 \times 2095$
 $(23 \times 89 + 3) \times (23 \times 90 + 1) \times (23 \times 91 + 2)$
 $= (23[89 \times 90 + 89 + 90 \times 3] + 3) \times (23 \times 91 + 2)$
 $= (23 \times K + 3) \times (23 \times 91 + 2)$
 $= (23 \times [K \times 91 + K \times 2 + 91 \times 3] + 6)$
 $= 23 \times L + 6 \text{ (where K and L are constants).}$

Hence remainder is 6.

11. L.C.M. of 27, 35, 25 and 21 = 4725
 $\therefore$ Number = $4725 - 3 = 4722.$
12. L.C.M. of 12, 21 and 35 = 420
 $\therefore$ Number required = $420 + 6 = 426.$
13. Number is $120K + 1$

$$= ((13 \times 9 + 3)K + 1)$$

$$= (13 \times 9)K + 3K + 1,$$

which is divisible by 13.

$3K + 1$ is divisible by 13.

$$\therefore K = 4. \text{ Number} = 481.$$

$$\begin{aligned}
 14. \quad & \frac{a}{1-r} = 3 \\
 \Rightarrow & 3 - 3r = a \\
 \text{Also, } & \frac{a^2}{1-r^2} = 6 \\
 \Rightarrow & \frac{a}{1-r} \cdot \frac{a}{1+r} = 6 \\
 \Rightarrow & \frac{a}{1+r} = 2 \\
 \Rightarrow & 2 + 2r = a \\
 \therefore & 3 - 3r = 2 + 2r \\
 \Rightarrow & r = \frac{1}{5} \\
 \therefore & a = 3 - \frac{3}{5} = \frac{12}{5}
 \end{aligned}$$

15. Since $d = -2$, maximum sum of the series will be sum of all the positive terms.

Least positive term of the series is 2 or 0 (both giving equal sum)

Let, 2 be the n^{th} term

$$2 = 60 + (n - 1)(-2)$$

$$\Rightarrow n = 30$$

$$\therefore \text{Required sum} = 30/2 (60 + 2) = 930$$

16. Total income for 15 days = $15 \times 7 = ₹105$

$$\text{Total income for first 5 days} = 5 \times 6 = ₹30$$

$$\text{Total income for last 9 days} = 9 \times 8 = ₹72$$

$$\begin{aligned}
 \therefore \text{Income on sixth day} \\
 &= 105 - (30 + 72) \\
 &= 105 - 102 = ₹3
 \end{aligned}$$

17. $(GM)^2 = HM \times AM$

$$HM = \frac{8 \times 8}{5} = \frac{64}{5} = 12.8$$

18. Total marks of 15 students = $15 \times 145 = 2175$.

Average marks of 15 students (excluding two lowest scores) = $13 \times (145 + 5) = 1950$.

$$\therefore \text{Total of two lowest scores} = 2175 - 1950 = 225$$

Given that two scores are consecutive multiples of 9

$$\therefore 9x + 9x + 9 = 225$$

$$\Rightarrow 18x = 216$$

$$\Rightarrow x = \frac{216}{18}$$

$$\therefore \text{Lowest score} = 9x = \frac{216}{18} \times 9 = 108.$$

2

CHAPTER

Percentage and Its Applications

The term percentage is quite frequently used in our day to day life especially to avoid the fractions less than 1. Instead of treating the complete entity as 1, we treat as 100, and take the ratios accordingly.

PERCENTAGE

Per Cent: This term means for every hundred. A fraction whose denominator is 100 is called *percentage* and numerator of the fraction is called *rate per cent*. It is denoted by the symbol %.

To find % equivalent of a fraction.

Express fraction with the denominator 100 and numerator is the required answer.

e.g., $\frac{11}{24} = \frac{11}{24} \times 100 = 45\frac{5}{6}\%$

To find fraction equivalent of 'a %': Divide 'a' by 100.

e.g., $21\frac{7}{8}\% = \frac{21\frac{7}{8}}{100} = \frac{175}{800} = \frac{7}{32}$

To increase a number by a given %:

Multiply the number by the factor $\frac{100 + \text{rate}}{100}$

e.g., increase 20 by 15%,

i.e. $20 \times \left(\frac{100 + 15}{100}\right) = 20 \times \frac{115}{100} = 23$

or $20 \times 1.15 = 23$

To decrease a number by a given %:

Multiply the number by the factor $\frac{100 - \text{rate}}{100}$

e.g., decrease 30 by 20%, i.e. $30 \times \left(\frac{100 - 20}{100}\right)$

$$= 30 \times \frac{80}{100} = 24$$

or $30 \times 0.8 = 24$

To find the % increase of a number:

$$\begin{aligned} \% \text{ increase} &= \frac{\text{total increase}}{\text{initial value}} \times 100 \\ &= \frac{\text{Final value} - \text{Initial value}}{\text{Initial value}} \times 100 \end{aligned}$$

e.g., Population of a village in 1980 was 3000 and in 1990 was 3200. Find the % increase.

$$\% \text{ increase} = \frac{3200 - 3000}{3000} \times 100$$

To find % decrease of a number:

$$\% \text{ decrease} = \frac{\text{total decrease}}{\text{Initial value}} \times 100$$

$$= \frac{\text{Initial value} - \text{Final value}}{\text{Initial value}} \times 100$$

e.g., Cost of a bike last year was ₹ 19000. Its cost this year is ₹ 17000. Find the % decrease in its cost.

$$\begin{aligned} \% \text{ decrease} &= \frac{19000 - 17000}{19000} \times 100 \\ &= \frac{2000}{19000} \times 100 = 10.5\% \end{aligned}$$

Note:

(1) If price of a commodity increases by r%, then reduction in consumption, so as not to increase

$$\text{expenditure is } \left(\frac{r}{100 + r} \times 100\right)\%$$

(2) If price of a commodity decrease by r%, then the increase in consumption so as not to decrease

$$\text{expenditure is } \left(\frac{r}{100 - r} \times 100\right)\%$$

(3) If A's income is r% more than that of B, then B's income is less than that of A by $\left(\frac{r}{100 + r} \times 100\right)\%$

(4) If A's income is r% less than that of B, then B's income is more than that of A by $\left(\frac{r}{100 - r} \times 100\right)\%$.

Example: If the production of rice went up from 250 MT in 2005 to 300 MT in 2006, then the percentage increase in rice production from 2005 to 2006 is calculated as follows:

$$\text{Actual increase} = 300 - 250 = 50\text{MT}$$

Percentage increase:

$$\begin{aligned} &= \frac{\text{Actual increase from 2005 to 2006}}{\text{Actual production of rice in 2005}} \times 100 \\ &= \frac{50}{250} \times 100 = 20\% \end{aligned}$$

I. If the increase on a value of 350 is 15% the new quantity is $1.15 \times 350 = 402.5$ (where $1.15 = 1 + 0.15$, 0.15 being the decimal equivalent of 15%)

2.2

- II. If the production in 2005 is given as 400 MT and the increase from 2004 to 2005 is given to be 25% then the production in 2004 will be equal to

$$\frac{400}{1.25} = 320 \text{ MT (where } 1.25 = 1 + 0.25, 0.25 \text{ being}$$

the decimal equivalent of 25%)

- III. Similarly, if there is a decrease of 12% on a quantity of 225, then the new quantity will be equal to 225×0.88 (where $0.88 = 1 - 0.12$, 0.12 being the decimal equivalent of 12%). If the production in 2005 is given as 400MT and it is a decrease of 13% from 2004, then the production in 2004 will be

$$\text{equal to } \frac{400}{0.87} \text{ (where } 0.87 = 1 - 0.13, 0.13 \text{ being}$$

the decimal equivalent of 13%).

On the basis of percentage increase, we can write down how many times the old value gives the new value. For example, if the percentage increase is 100%, we can conclude that the new value is 2 times the old value and if the percentage increase is 300% then the new value is 4 times the old value. If the percentage increase is 450% then the new value is 5.5 times the old value. In general if the percentage increase is $p\%$ then the new value

$$\text{is } \left(\frac{p}{100} + 1 \right) \text{ times the old value.}$$

Conversely, if we know how many times the old value gives the new value, we can find out the percentage increase in the old value to get the new value. For example, if the new value is 3 times the old value, the percentage increase in the old value to get the new value is 200%. If the new value is 4.25 times the old value, then the percentage increase is 325%. In general if the new value is k times the old value, then the percentage increase is $(k - 1) \times 100\%$

Example: If 45% of a certain number is 990, then find the value of 54% of that number.

Solution:

Let the number be x

$$\text{Given, } \frac{45}{100}x = 990$$

$$\Rightarrow x = 2200$$

$$\therefore \text{required value} = \frac{54}{100}x = 1188$$

Example: The ratio of the monthly salaries of A in

December 2004 and in January 2005 was $3\frac{3}{5} : 4\frac{1}{2}$.

Find the percentage increase in his salary.

Percentage and Its Applications

Solution:

$$\text{The ratio of the monthly salaries } 3\frac{3}{5} : 4\frac{1}{2} = \frac{18}{5} : \frac{9}{2}$$

$$(18)(2) : (9)(5) = 4 : 5$$

Let the monthly salary of A in December 2004 = $4x$

His monthly salary in January 2005 would be = $5x$.
Required percentage increase in his salary

$$= \frac{5x - 4x}{4x} (100)\% = 25\%$$

Example: There are three numbers. The first and the second numbers are 50% less and 60% less respectively than

the third. What percentage of the first number is the second?

Solution:

Let the third number be 100

$$\text{First number} = 100 \left(1 - \frac{50}{100} \right) = 50$$

$$\text{Second number} = 100 \left(1 - \frac{60}{100} \right) = 40$$

$$\therefore \text{it is } \frac{40}{50} \text{ th i.e. } \frac{4}{5} \text{ th or } 80\% \text{ of the first number.}$$

Example: Ram got 30% in a test and failed by 10 marks. If the pass marks in the test was 70, find the maximum marks in it.

Solution:

Let the maximum mark in the test be M .

$$\text{Ram's mark} = \frac{30}{100} M$$

$$\text{Pass mark} = \frac{30}{100} M + 10$$

$$\frac{30}{100} M + 10 = 70 \text{ i.e., } M = 200$$

Example: The price of an article is decreased by 20%. By what percentage must the consumption of it be increased in order to retain the expenditure on it?

Solution:

Let the initial price be Rs. 100/gm

Let the initial consumption be 1 gm

Initial expenditure = Rs. 100

$$\text{New price} = 100 \left(1 - \frac{20}{100} \right)$$

$$= \text{Rs. } 80/\text{gm}$$

$$\text{New expenditure} = \text{Rs } 100$$

$$\therefore \text{New consumption} = \frac{100}{80} \text{ gm}$$

$$= \frac{5}{4} \text{ gm}$$

$\therefore$ Consumption must increase by

$$\frac{\left(\frac{5}{4} - 1\right) 100\%}{1} = 25\%$$

Example: The length of a rectangle increases by 14% and the breadth by 8%. What is the consequent percentage increase in area?

Solution:

Let length and breadth of the rectangle be l and b

Area of rectangle = lb

Length is increased by 14%

$$\therefore \text{New length} = 1 \times 1 \times \frac{14}{100} = (1 + 0.141) = 1.141$$

Breadth is increased by 8%

$$\therefore \text{New breadth} = b + 8\% b$$

$$= b + 0.08b = (1.08)b$$

Area of rectangle

$$= 1.141 \times 1.08b = (1.2312)lb$$

$$\text{Increase in area} = 1.2312lb - lb$$

$$\text{i.e., } 0.2312lb$$

$$\therefore \text{Percentage increase} = \frac{23.12}{100} lb$$

$$\text{i.e., } 23.12\%$$

SIMPLE INTEREST AND COMPOUND INTEREST

When a sum of money is lent by A to B, A is called *lender (creditor)*, B the *borrower (debtor)*.

The sum lent is called *principal (P)*.

Interest (I) is the extra money paid by the borrowed to the lender for the use of the money for a specified time.

The time for which the money is borrowed is called *period (N)*.

The extra amount paid per 100 rupees in a year is called *rate per cent per annum (R)*.

The sum of interest and principal is called *Amount (A)*.

$$A = P + I$$

Simple Interest (S.I.)

When interest is paid as it falls due, it is called simple interest i.e., throughout the loan period, interest is charged on the original sum (principal) borrowed.

$$\text{S.I.} = \frac{\text{PNR}}{100}$$

e.g., Find interest to be paid on ₹ 1500 at 9% per annum for a period of 9 years.

Here $P = 1500$, $R = 9\%$, $N = 3$ years

$$\therefore \text{S.I.} = \frac{\text{PNR}}{100} = \frac{1500 \times 3 \times 9}{100} = ₹ 405$$

Compound Interest (C.I.)

Money is said to be lent at compound interest when at the end of a year or other fixed period, the interest that has become due is not paid to the lender, but is added to the sum lent, and the amount thus obtained becomes the principal for the next year or period. The process is repeated until the amount for the last period has been found. The difference between final amount and the original principal is the *compound interest (C.I.)*.

In compound interest, interest is calculated on the accrued interest also.

$$\text{Amount} = P \left[1 + \frac{R}{100} \right]^N$$

$$\text{Compound Interest} = \text{Amount} - \text{Principal}$$

Note:

(1) The time period after which the interest is added each time to form a new principal is called *conversion period*. It may be annually, semi annually or quarterly.

(2) In case, interest is paid semi annually (half yearly), N is number of half years and R is rate percent per half year i.e.,

$$\text{Number of years} \times 2 = \text{Number of half years.}$$

$$\text{Rate percent per annum} \div 2 = \text{Rate percent per half year.}$$

(3) In case interest is paid quarterly, N is number of quarters and R is rate percent per quarter.

$$\text{Number of years} \div 4 = \text{Number of quarters}$$

$$\text{Rate percent per quarter} \times 4 = \text{Rate percent per year.}$$

(4) When rates are different for different years, say R_1, R_2, R_3 percent of 1st, 2nd and 3rd years respectively then

$$\text{Amount} = P \left(1 + \frac{R_1}{100} \right) \left(1 + \frac{R_2}{100} \right) \left(1 + \frac{R_3}{100} \right)$$

(5) In case the time is a fraction of a year, say $y \frac{x}{z}$ years, then

$$\text{Amount} = P \left(1 + \frac{R}{100} \right)^y \times \left(1 + \frac{\frac{x}{z} \times R}{100} \right)$$

Example: A certain sum of money at C.I. amounts to Rs. 811.25 in 2 years and to Rs. 843.65 in 3 years. Find the sum of money.

Solution:

$$\text{Since, } A = P \left[1 + \frac{R}{100} \right]^n$$

$$\Rightarrow 811.25 = P \left[1 + \frac{R}{100} \right]^2 \quad \dots (1) \text{ and}$$

$$\Rightarrow 843.65 = P \left[1 + \frac{R}{100} \right]^3 \quad \dots (2)$$

On dividing (2) by (1), we get:

$$\frac{843.65}{811.25} = 1 + \frac{R}{100}$$

$$\Rightarrow 1.04 = 1 + \frac{R}{100} \Rightarrow R = 4$$

Now, putting $R = 4$ into (1), we get

$$\Rightarrow 811.25 = P \left[1 + \frac{4}{100} \right]^2 \Rightarrow P = 750$$

$\Rightarrow$ The sum of money is Rs. 750

EQUAL ANNUAL INSTALLMENT TO PAY THE DEBT (BORROWED) AMOUNT

Let, the value of each equal annual installment = Rs. a .

Rate of interest = $R\%$ p.a.

Number of installments per year = n .

Number of years = T .

$\therefore$ Total number of installments = $n \times T$.

Borrowed amount = B . Then,

$$a \left[\frac{100}{100+R} + \left(\frac{100}{100+R} \right)^2 + \dots + \left(\frac{100}{100+R} \right)^{n \times T} \right] = B$$

Example: What annual installment will be required to repay a borrowed amount of Rs. 1,32,400 in 3 years at 10% per annum compounded annually?

Solution:

Let each annual installment be Rs. a .

By using the formula,

$$132400 = a \left[\frac{100}{100+10} + \left(\frac{100}{100+10} \right)^2 + \left(\frac{100}{100+10} \right)^3 \right]$$

$$\Rightarrow 132400 = a \left[\frac{10}{11} + \left(\frac{10}{11} \right)^2 + \left(\frac{10}{11} \right)^3 \right]$$

$$\Rightarrow 132400 = a \times \frac{10}{11} \left[1 + \frac{10}{11} + \frac{100}{121} \right]$$

$$\Rightarrow 132400 = a \times \frac{10}{11} \times \frac{331}{121}$$

$$\Rightarrow a = 132400 \times \frac{11}{10} \times \frac{121}{331}$$

$$\Rightarrow a = \text{Rs. } 53,240$$

Example: A loan of Rs. 2000 is to be paid back in 3 equal annual installments. How much is each Installment to the nearest whole rupee, if the interest

is compounded annually at $12\frac{1}{2}\%$ p.a.?

Solution:

$$2000 = a \left\{ \frac{100}{112.5} + \left(\frac{100}{112.5} \right)^2 + \left(\frac{100}{112.5} \right)^3 \right\}$$

= Rs. 840 (approx.)

Population Formula: The original population of a town is P and annual increase is $r\%$, then

population in n years is $\left(1 + \frac{R}{100} \right)^n$ and if annual decrease is $r\%$, then population in n years is given by a change of sign in the formula

$$\text{i.e.,} \quad \left(1 - \frac{R}{100} \right)^n$$

e.g., If annual increase in the population of a town is 4% and present population is 15625, what will be the population in 3 years?

Required population = $15625 (1.04)^3 = 17576$

Hire Purchase: In a hire purchase plan, a customer can make use of the goods while paying for them. The amount paid at the time of purchase is called *down payment*. The remainder is paid in equal installments and each is the monthly installment. The difference between total amount to be paid and the cash price is called *installment charge*.

Monthly Installment

$$= \frac{\text{Amount to be paid} - \text{down payment}}{\text{Number of instalments}}$$

e.g., If a transistor is available at ₹ 400 each or ₹ 100 down payment and ₹ 70 per month for 5 months, find total amount paid for it, and the installment charge

Amount paid = $100 + 70 \times 5 = ₹ 450$

Installment charge = $450 - 400 = ₹ 50$

PROFIT AND LOSS

Cost Price (CP): The price for which an article is bought is called its *cost price*.

Selling Price (SP): The price at which an article is sold is called its *selling price*.

Profit (Gain): The difference between selling price and cost price is called *profit*. For profit, selling price should be greater than cost price.

Loss: The difference between cost price and the selling price is called *loss*. When cost price is greater than the selling price, there is a loss.

Profit and loss is generally represented as a percent of the cost price, unless otherwise stated.

Overhead charges: If an individual has to spend some money on transportation etc., then this extra expenditure is called *overhead charges*.

Marked price (MP): The price on the label is called *marked price* or *list price*.

Discount: The reduction made on the 'marked price' of an article is called *discount*. When no discount is given, 'selling price' is the same as 'marked price'.

List of Formulae

- (1) Profit = SP – CP ... (SP > CP)
- (2) Loss = CP – SP ... (CP > SP)
- (3) % Profit = $\frac{\text{Profit}}{\text{CP}} \times 100 = \frac{\text{SP} - \text{CP}}{\text{CP}} \times 100$
- (4) % Loss = $\frac{\text{Loss}}{\text{CP}} \times 100 = \frac{\text{CP} - \text{SP}}{\text{CP}} \times 100$
- (5) Profit = $\frac{\text{Profit \%} \times \text{CP}}{100}$
- (6) Loss = $\frac{\text{Loss \%} \times \text{CP}}{100}$
- (7) SP = $\frac{100 + \text{Profit \%}}{100} \times \text{CP}$
- (8) SP = $\frac{100 - \text{Loss \%}}{100} \times \text{CP}$
- (9) SP = Marked Price – Discount
- (10) CP = SP $\times \left(\frac{100}{100 + \text{Profit \%}} \right)$
- (11) CP = SP $\times \left(\frac{100}{100 - \text{Loss \%}} \right)$
- (12) Discount % = $\frac{\text{Discount}}{\text{Marked Price}} \times 100$

Some important point:

- (1) If two items are sold, each at ₹ X, one at a gain of p% and the other at a loss of p%, there is an overall loss given by $\frac{p^2}{100}\%$. The absolute value of the loss is given by $\frac{2p^2x}{100^2 - p^2}$
- (2) If CP of two items is the same and % Loss and % Gain on the two items are equal, then net loss or net profit is zero.
- (3) Buy x get y free i.e., if x + y articles are sold at cost price of x articles, then
percentage discount = $\frac{x}{x+y} \times 100$.
- (4) By using false weight, if a substance is sold at cost price, then
overall gain % = $\frac{100 + \text{Gain \%}}{100} = \frac{\text{True Scale or Weight}}{\text{False Scale or Weight}}$
- (5) In case of successive discount a% and b%,
effective discount = $\left(a + b - \frac{ab}{100} \right)\%$.

Example: Two shirts were having a cost price of Rs. 200 each. One was sold at a profit of 15% and the other was sold at a loss of 15%. Find the net profit or loss.

Solution:

$$\text{SP}_1 = 200 \times \frac{(100 + 15)}{100} = \text{Rs. } 230.$$

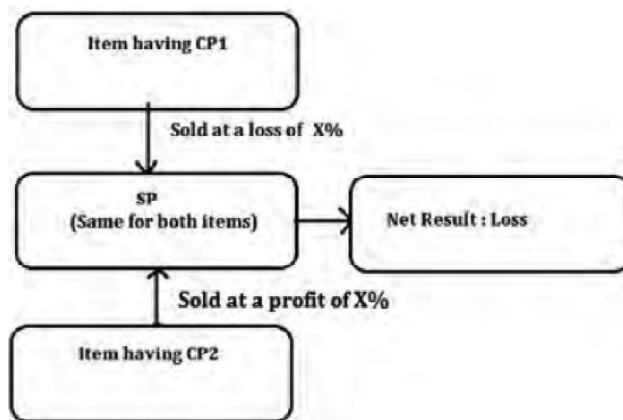
$$\text{SP}_2 = 200 \times \frac{(100 - 15)}{100} = \text{Rs. } 170.$$

$$\text{Total S.P. received} = \text{S.P.}_1 + \text{S.P.}_2 = 230 + 170 = \text{Rs. } 400.$$

$$\text{Total C.P.} = 200 + 200 = 400.$$

Net result: No profit, No Loss.

Note: If two items are **SOLD**, each at rupees S, one at a gain of X% and other at a loss of X%, then the net result is always a loss.



Example: Two articles were sold at Rs. 100 each. After selling it was realized that on one, a profit of 10% was made, and on the other, a loss of 10% was made. What is the net result?

Solution:

	Item 1	Item 2
Selling Price	Rs. 100	Rs. 100
Profit % 10	-	
Loss %	-	10
Cost Price	$\frac{100}{1.1}$	$\frac{100}{0.9}$
	= 90.90	= 111.11

$$\text{Total S. P. received} = 100 + 100 = \text{Rs. } 200$$

$$\text{Total C. P.} = 90.90 + 111.11 = \text{Rs. } 202.01$$

$$\text{Loss \%} = \frac{200 - 202.01}{202.01} \times 100$$

$$= \frac{-2.01}{202.01} \times 100 \approx -1\%$$

The same calculation can be done by a very simple formula

$$\text{Loss \%} = \frac{X^2}{100} \text{ and Value of loss} = \frac{2X^2S}{100^2 - X^2}.$$

Where X is the percentage profit and loss made on each of the items and S is the common selling price received on both. In case of discounts being offered, the price on which the discount was offered is known as the marked price and the price that is finally received is known as the selling price.

2.6

Example: A merchant purchases an item for Rs. 500. He marks the item at a price of Rs. 700 but allows a discount of 10% on cash payment. What is the total profit in terms of amount and percentage made by the merchant?

Solution:

C.P. = Rs. 500, M. P. = Rs. 700.

$$\text{Hence S.P.} = 700 \left(1 - \frac{10}{100} \right) = \text{Rs. } 630$$

Thus, profit = Rs. 630 – Rs. 500 = Rs. 130.

$$\text{Profit \%} = \left(\frac{130}{500} \right) \times 100 = 26\%.$$

If a trader uses a false scale for selling his goods, then the overall gain made by him in this process will be

$$\text{Gain\%} = \left[\frac{\text{Error}}{\text{True Value} - \text{Error}} \times 100 \right]$$

Example: A milkman claims to sell milk at the cost price but uses a measure of 800 ml instead of a litre. Find the net profit made by him.

Solution:

Using the formula given above

$$\text{Gain \%} = \left[\frac{200}{1000 - 200} \times 100 \right] = \frac{200}{800} \times 100 = 25$$

If a tradesman defrauds (by means of a false balance or otherwise) to the tune of X% in buying and also defrauds to the tune of X% in selling, his overall percentage gain will be

$$\left[\frac{(100 + \text{common gain \%})^2}{100} - 100 \right] \%$$

Example: A trader defrauds the seller by 10% when he purchases goods from him, and while selling the same to a customer, he defrauds once again by 10%. Find the net gain made by the trader.

Solution:

The required answer is

$$\left[\frac{(100 + 10)^2}{100} - 100 \right] \% = \frac{12100}{100} - 100$$

$$= 121 - 100 = 21\%$$

If a tradesman defrauds (by means of a false balance or otherwise) to the tune of X% in buying and also defrauds to the tune of Y% in selling, his overall percentage gain will be

$$\left[\frac{(100 + X\%)(100 + Y\%)}{100} - 100 \right] \%$$

Example: A milkman defrauds by means of a false measure to the tune of 20% in buying and also

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defrauds to the tune of 25% in selling. Find his overall % gain.

Solution:

The milkman defrauds 20% in buying and also defrauds 25% in selling, so his overall % gain will be

$$\left[\frac{(100 + 20\%)(100 + 25\%)}{100} - 100 \right] \% = 50\%$$

If selling price of X articles is equal to the cost price of Y articles, then the net profit percentage is given

$$\text{by } \frac{Y - X}{X} \times 100$$

Example: The cost price of 20 pens is equal to the selling price of 25 pens. What is the net loss percentage?

Solution:

Here, cost price of 20 pens = selling price of 25 pens. So

$$\text{the net loss percentage} = \frac{20 - 25}{25} \times 100 = -20\%$$

Here minus sign indicates the loss.

PARTNERSHIP

It is an association of two or more persons who invest their money in order to carry on a certain business. A partner who manages the business is called *working partner* and the one who simply invests the money is called *sleeping partner*.

If capitals of the partners are invested for the same time, then partnership is called *simple*, and if invested for different periods, the partner is called *compound*.

If period of investment is the same for each partner, then profit or loss is divided in the ratio of their investments.

e.g., (i) If A and B are partners in a business, then

$$\frac{\text{Investment of A}}{\text{Investment of B}} = \frac{\text{Profit of A}}{\text{Profit of B}}$$

$$\text{or } \frac{\text{Investment of A}}{\text{Investment of B}} = \frac{\text{Loss of A}}{\text{Loss of B}}$$

(ii) If A, B and C are partners in a business, then

$$\frac{\text{Investment of A}}{\text{Investment of B}} = \frac{\text{Investment of B}}{\text{Investment of C}}$$

$$= \text{Profit of A} : \text{Profit of B} : \text{Profit of C}$$

$$\text{OR Loss of A} : \text{Loss of B} : \text{Loss of C}$$

Monthly Equivalent Investment: It is the product of the capital invested and the period for which it is invested.

If period of investment is different, then profit or loss is divided in the ratio of their Monthly Equivalent Investment.

e.g., If A and B are partners in a business, then,

$$\frac{\text{Monthly Equivalent Investment of A}}{\text{Monthly Equivalent Investment of B}}$$

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$$\frac{\text{Profit of A}}{\text{Profit of B}} = \frac{\text{Loss of A}}{\text{Loss of B}}$$

$$\text{i.e., } \frac{\text{Investment of A} \times \text{Period of Investment of A}}{\text{Investment of B} \times \text{Period of Investment of B}} = \frac{\text{Profit of A}}{\text{Profit of B}} = \frac{\text{Loss of A}}{\text{Loss of B}}$$

If A, B and C are partners in a business, then

Monthly Equivalent Investment of A

: Monthly Equivalent Investment of B

: Monthly Equivalent Investment of C = Profit of A

: Profit of B : Profit of C

or

Monthly Equivalent Investment of A : Monthly Equivalent Investment of B

: Monthly Equivalent Investment of C = Loss of A

: Loss of B : Loss of C

Example: A and B together invested Rs. 12000 in a business. At the end of the year, out of a total profit of Rs. 1800, A's share was Rs. 750. What was the investment of A?

Solution:

Since profits are shared in the ratio of their investments

$$\therefore \frac{\text{A's investment}}{\text{B's investment}} = \frac{\text{Profit share of A}}{\text{Profit share of B}}$$

(Money invested by A & B for the same period)

$$= \frac{750}{1800 - 750} = \frac{750}{1050} = \frac{5}{7}$$

$$\therefore \text{Investment of A} = \frac{5}{5+7} \times 12000 = \text{Rs. } 5000$$

Example: In a business A, B and C invested Rs.380, Rs. 400 & Rs. 420 respectively. Divide a new profit of Rs. 180 among the partners.

Solution:

A's Profit. B's Profit. C's Profit

= A's investment : B's investment : C's investment

= A's investment : B's investment : C's investment

= 380 : 400 : 420 = 19 : 20 : 21

$$\text{Profit share of A} = \frac{19}{60} \times 180 = \text{Rs. } 57.$$

$$\text{Profit share of B} = \frac{20}{60} \times 180 = \text{Rs. } 60.$$

$$\text{Profit share of C} = \frac{21}{60} \times 180 = \text{Rs. } 63.$$

Example: If the partners invest different amounts and for different period of time, then the profits at the end of the year are shared in the ratio of products

2.7

of (Investment x time period) calculated for each partner.

A started a business with a capital of Rs. 10000. Four months later, B joined him with a capital of Rs. 5000. What is the share of A in a total profit of Rs. 2000 at the end of the year?

Solution:

$$\frac{\text{Profit of A}}{\text{Profit of B}} = \frac{\text{Amount} \times \text{No. of months}}{\text{Amount} \times \text{No. of months}}$$

$$= \frac{10000 \times 12}{5000 \times 8} = \frac{3}{1}.$$

$$\therefore \text{profit share of A} = \frac{3}{3+1} \times 2000 = \text{Rs. } 1500.$$

Example: A, B and C enter into a partnership. A contributes Rs. 320 for 4 months, B contributes Rs. 510 for 3 months and C contributes Rs. 270 for 5 months. If the total profit is Rs. 208, then find the profit share of each of the partners.

Solution:

A's Profit : B's Profit : C's Profit

$$= 320 \times 4 : 510 \times 3 : 270 \times 5$$

$$= 1280 : 1530 : 1350 = 128 : 153 : 135$$

$$\therefore \text{profit of A} = \frac{128}{128+153+135} \times 208$$

$$= \frac{128}{416} \times 208 = \text{Rs. } 64$$

$$\therefore \text{profit of B} = \frac{153}{128+153+135} \times 208$$

$$= \frac{153}{416} \times 208 = \text{Rs. } 76.50$$

$$\therefore \text{profit of C} = \frac{135}{128+153+135} \times 208$$

$$= \frac{135}{416} \times 208 = \text{Rs. } 67.50$$

STOCKS AND SHARES

Face value: The price of a share is printed on the share certificate is called face value of the share. (Nominal value).

Dividend: Every shareholder of a company is entitled to a proportionate share of the profits of the company. The amount of profits on each share is called *dividend*. Dividend is expressed as a percentage of the face value.

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Market Value: In the market, selling price of the share may be different from its face value. Thus, market value of a share is *fluctuable price*.

Shares: *In the market, three types of shares are available :*

- (i) **Premium share:** If market price of a share is more than nominal value, then it is called *premium share*.
- (ii) **Par share:** If market price of a share is equal to the nominal value, then it is called *par share*.
- (iii) **Discount share:** If market price of a share is less than the nominal value, then it is called a *discount share*.

Types of shares: *There are two types of shares:*

- (i) **Preference shares:** On these shares a fixed rate of dividend is paid to share holders irrespective of whether the company makes any profits or whether it runs into loss. In case the company is not able to pay the agreed dividend to preference shareholders, the dividend on cumulative preference shares goes on accumulating and is paid as and when the company can do so.
- (ii) **Equity shares:** Profit of the company that is left over after its distribution to the preference shareholders is distributed among the equity share holders. This dividend on equity share may sometimes be more or sometimes less than the dividend on the preference share.

Stock

In order to meet expenses of a certain plan, the Government of India sometimes raises a loan from the public at a certain fixed rate of interest. Bonds or promissory notes called *stocks of a fixed value* are used for sale to the public.

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Note:

- (1) Income per share = Rate of dividend
× Nominal value of 1 share.
- (2) Annual income = Income per share
× Number of shares.

- (3) Number of shares

$$= \frac{\text{Nominal value of all the shares}}{\text{Nominal value of one shares}}$$

$$= \frac{\text{Market value of the shares (Investment)}}{\text{Market value of one share}}$$

$$= \frac{\text{Total Dividend}}{\text{Dividend per share}}$$

- (4) Rate of interest on the Invest (yield)

$$= \frac{\text{Total income} \times 100}{\text{Total investment}}$$

- (5) When stock/share is purchased, brokerage is added to cost price.
- (6) When stock/share is sold, brokerage is subtracted from selling price.
- (7) In most problems, face value of a share is taken to be ₹, 100 unless specified.

EXERCISE

MCQ TYPE QUESTIONS

- The Manager (M) of the company along with his Assistant Manager (AM) and a Business Analyst (BA) invest ₹ 55000, ₹ 45000 and ₹ 60000 respectively in a partnership firm for six months. The Manager being the highest authority among the three gets one-tenth of the total profit and Assistant Manager being second highest authority gets $\frac{1}{20}$ th of the remaining and the remaining profit is divided among the three in the ratio of their investments. If the total profit was ₹ 40000, who gets the minimum share and how much?
 - BA, ₹ 11400
 - AM, ₹ 11419
 - AM, ₹ 12500
 - None of these
- In a business partnership of three people A, B and C, B's share in total profit is ₹ 250 per ₹ 2000. At this profit his return on investment is 15%. If B earns total of ₹ 900, what is the total investment made by A and C in business? Assume that each of A, B and C have invested their money for the entire year and profits are shared in the ratio of their investments.
 - ₹ 48000
 - ₹ 42000
 - ₹ 6000
 - Data insufficient
- Abhay and Bimal started a business in partnership. Abhay invested a capital of ₹ 50000. Bimal agrees to pay Abhay an interest on half the capital at 8% per annum. Also Bimal receives a salary of ₹ 500 per month for managing the business. After six months Abhay withdraws ₹ 20000 and Bimal pitches in an equal amount. Bimal continues to pay Abhay interest at 8% per annum for the balance amount. At the end of the year they divide the profit equally and found that total earning of Bimal in the year was exactly thrice that of Abhay. The total profit?
 - ₹ 7200
 - ₹ 6800
 - ₹ 5400
 - ₹ 4800
- How much stock must be bought in 3 percent $89\frac{1}{8}$ in order that by selling it at $91\frac{3}{8}$ a profit of ₹ 45 may be made?
 - 20
 - 30
 - 40
 - 50
- Which is the better investment (the security being equal), 3 percents perpetually at par, or 5 percents at 124, the latter stock being redeemable in 20 years at par? (Simple interest to be reckoned).
 - 3% at par
 - 5% at 124
 - Both are same
 - Cannot be determined
- What rate percent per annum does a man get by buying ₹100 share (₹ 70 paid-up) at ₹ 60 when dividend is 5% per annum?
 - 4.28%
 - 7.14%
 - $8\frac{1}{3}\%$
 - $5\frac{5}{6}\%$
- A rejects 0.08% of the meters as defective .How many will he examine to reject 2?
 - 2500
 - 1250
 - 625
 - 5000
- (a% of b) + (b% of a) is equal to
 - a% of b
 - 2% of ab
 - 20% of ab
 - 2% of 100 ab
- Ryan's stock of goods is worth \$9462, which is 15% more than Harland's, and 15% less than Lambert's. What is the value of stock carried by Harland and Lambert respectively?
 - \$ 8218, \$ 10032
 - \$ 1419, \$ 10881
 - \$ 8042, \$ 10881
 - \$ 8228, \$ 11132
- Numerator of a fraction increases by 25%. What can be said about the resultant fraction?
 - It is 67% less than original.
 - It is equal to the original fraction.
 - It is 67% more than the original fraction.
 - The change cannot be determine.
- Which of the following statement/s is/are true?
 - If two numbers are such that one is 25% more than the second, then sum of the two numbers is nine times that of the difference of the two numbers.
 - If 56% of a number A is 24% of another number B then 49% of A is 21% of B.

Codes:

 - Only I
 - Only II
 - Only I and II
 - None of these
- In the Annual General Meeting of the Rotary club a chairman is to be selected for which A and B is the only two candidates. After the first round, each of them gave a speech on their policy and work plan for the coming year after which another round of election was held. The number of votes not cast for A increased by 25% in the second round over those not cast for him in the first round and A lost second round to B by twice as many votes as that by which he had won in the first round. If total 260 people voted each time what was the percent decrease in votes received by A in the second round assuming that no vote was void in any of the rounds?
 - 18%
 - 25%
 - 21.43%
 - 30%

13. A watermelon weighs 5000 gm. 99% of its weight is water. It is kept in a drying room and after some time it turns out that it is only 98% water by weight. What is its weight now?
- (a) 2500 gm
(b) 4500 gm
(c) 4950 gm
(d) None of these
14. A sum of ₹ 5000 is divided into two parts A and B such that simple interests at the rate of 20% p.a. on A and B after 2 and 3 years respectively are equal. Which of the following is false about A and B?
- (a) Ratio of A to B is 2 : 3
(b) Ratio of A to B is 3 : 2
(c) A exceeds B by ₹ 1000
(d) Total interest earned on A and B is ₹ 2400
15. Which amongst the following two offers is the better one? Consider with respect to interest accrued.
- I. Investing an amount compounded annually at 1% per annum for 100 years.
II. Investing the amount compounded annually at 100% per annum for 1 year.
- Codes:**
- (a) First offer
(b) Second offer
(c) Both are same
(d) Cannot be determined
16. The population of rats in a locality x increases by 20% in one year. Observing this, the pest control committee decided to use a special kind of pesticide 'xyz' which effectively kills 160 rats in 3 months. Just after 2 years, what is the net increase or decrease in the population of rats if, initially the population of rats is 3200 and pesticide is used effectively?
- (a) Increase of 128 rats.
(b) Decrease of 128 rats.
(c) Neither an increase nor a decrease in the population.
(d) None of these.
17. A man travels from A to B to buy goods which he can get 10% cheaper in B than in A. If expenses of the journey are ₹ 15 and he makes a clear saving of ₹ 10, how much does he pay for the goods?
- (a) ₹ 225
(b) ₹ 200
(c) ₹ 150
(d) Cannot be determined
18. By selling 5 dozen mangoes for ₹ 156, it was found that $\frac{3}{10}$ th of the outlay was gained. What should the retail price per mango be in order to gain 60%?
- (a) ₹ 4 (b) ₹ 2 (c) ₹ 3.2 (d) ₹ 4.2
19. A man sells sugar to a tradesman at a profit of 20% but the tradesman becoming bankrupt pays only 80 paise in the rupee. How much percentage does the man gain or lose by his sale?
- (a) 2.5% (b) 3% (c) 4% (d) 5.2%
20. A trader allows a discount of 5 percent to his customers. What price should he mark on a article the cost price of which is ₹ 800 so as to make clear profit of 25 percent on his outlay?
- (a) ₹ 1000 (b) ₹ 1053
(c) ₹ 1200 (d) ₹ 1123
21. A camera costing ₹550 is marked to be sold at a price, which gives a profit of 30%. What will be its selling price in a sale when 20% is taken off the marked price?
- (a) ₹ 600 (b) ₹ 572
(c) ₹ 635 (d) ₹ 605
22. Buy two and get one free. Which of the following is true?
- (a) The person is selling at a loss.
(b) The person is selling at a profit.
(c) The loss or gain depends on the profit from sale of two articles.
(d) The person is selling at the cost price.
- Directions (Q.23–25):** Refer to the data below and answer the questions that follow.
- A profit-maximizing producer has the option of discriminating between markets A and B (i.e., charging different prices in markets A and B for the same product) for a product X. Demand for the product in each of the markets A and B is given by
- $$Q_A = 42 - 0.2 P_A$$
- $$Q_B = 100 - 0.8 P_B$$
- Total cost in any market is given by
- $$C = 1000 + 5Q, \text{ where } Q \text{ is quantity sold in that market.}$$
- Consider two cases:
- Case 1** in which she discriminates between the markets and
- Case 2** in which she does not discriminate.

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23. What is the approximate ratio of the price in market A to price in market B in Case 1?
(a) 1.65 (b) 2.00
(c) 2.25 (d) 2.50
24. What is the ratio of the total quantity of product X sold in Case 1 to the total quantity sold in Case 2?
(a) 0.50 (b) 1
(c) 1.10 (d) 1.05
25. What is the difference in total profit in Case 1 and the total profit in Case 2?
(a) ₹ 289 (b) ₹ 829 (c) ₹ - 289 (d) ₹ - 829
7. A school has only three classes which contain 40,50,60 students respectively. The pass percent of these classes are 10, 20 and 10 respectively. Then pass percent in the school will be _____
8. For a sphere of radius 10cm, the numerical value of surface area is _____ percent of the numerical value of its volume.
9. At an election, there are two candidates only, a candidate who gets 43 per cent of the votes is rejected by a majority of 420 votes. Then total number of votes recorded assuming that there was no void vote are _____

NUMERICAL TYPE QUESTIONS

1. A and B enter into a partnership. A puts in the whole capital of ₹ 45000 on the condition that the profits will be equally divided after which B will pay A interest on half the capital at 10% p.a. and receive ₹ 60 per month from A for carrying on the concern. The yearly profit in rupees, if B's income is half of A's income is _____
2. A, B and C started a business by investing ₹ 72000, ₹ 27000 and ₹ 81000. It was decided that servant's salary ₹ 2000 per month should be given from the profit. If profit is ₹ 25000 in the first month and ₹ 20000 in the second month and average profit of remaining 10 months is ₹ 21000, then share of C is _____
3. Three friends started a business of renting out air conditioners by investing ₹ 20000, ₹ 24000 and ₹ 16000, respectively. C gets 20% of total profit for repair and maintenance of the air conditioner. If in a particular year, C gets ₹ 487.50 less than total earnings of the other two, then total profit for the year in ₹ is _____
4. A person holds forty ₹ 500 shares in a concern, which pays dividend at the rate of 6 percent per annum. When the shares are at ₹ 675, he sells out and invests half the proceeds in 4 percent stock at 90. With the other half he buys a house. For which he receives an annual rent of ₹ 1440 subject to a deduction of 25P per rupees for repairs and taxes. Alteration in rupees in his annual income will be _____.
5. A person invests ₹ 34539 in three percent at 87. After receiving one year's dividend, he sells out at 89. He then invests the whole in railway stock paying 5 percent at 115. The difference in his income will be _____
6. Difference of two numbers is 1660. If 7.5 % of one number is 12.5% of the other number, then two numbers are _____
10. If price of wheat be raised 30%, then by _____ percent a householder must reduce his consumption of the article so as not to increase his expenditure?
11. A fraction in reduced form is such that when it is squared and then its numerator is reduced by $33\frac{1}{3}\%$ and denominator is reduced to 20%, its result is twice the original fraction. Sum of the numerator and the denominator is _____
12. A person borrows two equal sums at the same time at 5 and 4 percent respectively and finds that if he repays the former sum with simple interest on certain date 6 months before the latter, he will have to pay in each case the same amount, viz., ₹ 1100. The amount borrowed in rupees is _____
13. Effective annual rate corresponding to a nominal rate of 6 percent per annum, payable half-yearly is _____
14. A tradesman puts two prices on his goods, one for ready money and the other for 6 month's credit interest being calculated at $12\frac{1}{2}\%$ per annum. If credit price on an article be ₹ 26.56, then its cash price in rupees is _____
15. An article costing ₹ 9000 is sold at a discount which is equal to the simple interest on ₹ 3000 for N months. If rate of discount is same as the rate of interest, the N will be _____ months
16. A lent B ₹ 900 for a certain time at a certain rate percent per annum, which is equal to the square root of the number of the months of the time. After the time B wanted to return the money, but A, instead of taking the interest which amounted to ₹ 8 asked for a certain sum of money at the same rate for a certain number of years which is equal to the square root of the rate percent. Sum of money in rupees so that none of them should be the loser will be _____ (Take simple interest)

17. A shopkeeper sells a microwave at a discount on its marked price of ₹ 2500. But in a clearance sale he gives an additional discount of 20% thus selling it at ₹ 1500. Then first discount was _____%
18. A sports goods shop owner has 6 packets of balls each containing the same number of balls. If he sells each ball at ₹ 8 he gains ₹ 96. If there were 1 ball less in each packet and he sells each ball at ₹ 9 then he gains ₹ 90 in the transaction. The cost price of each packet of balls in rupees is _____
19. One merchant correctly calculates his percentage profit on the cost price; another wrongly calculates it on the selling price. The difference in actual profits in rupees if both claim to make $17\frac{1}{2}\%$ profit on goods sold at ₹ 3760 is _____
20. A manufacturer marks his goods in advance at 80 percent more than the cost price, but he allows 15 articles to the dozen and also 10 percent discount for cash. The profit on his outlay does he obtain from a customer who pays cash is _____%
21. A trader buys a certain amount of goods worth ₹ 22520. He decides to make a profit of 5.36% on the sale of goods worth ₹ 5000 and increase the profit percent by 3.14% for sales upto ₹ 15000 and then increase the profit percent for the sale of remaining lot such that he is able to make a profit of 25% on the sale of the full lot. Then profit in rupees that he makes on the third lot of goods is _____
22. Three shepherds A, B and C rented a pasture for a year. A grazed 22 sheeps for 4 months, B grazed 16 sheeps for 8 months and C grazed 32 sheeps for 6 months. If C's share of rent is ₹ 600, then total rent in rupees for the year was _____
23. Three bachelors, Amar, Akbar and Anthony rented a house for a year. But, Amar left after 4 months, Akbar stayed for 8 months and only Anthony stayed for the entire year. If the annual rent was ₹ 6000, then share of Akbar is _____
24. Ram, Shyam and Ghanshyam invest ₹ 2000, ₹ 10000 and ₹ 5000 to set up a stall. Ram gets 15% of the total profit for running the stall. Ram gets ₹ 500 less than Shyam and Ghanshyam together, then total profit of that year is _____
25. x, y and z start a joint venture wherein they make an annual profit of ₹ 3600. x invested one-third of the capital for one-fourth of the time, y invested one-fourth of the capital for one-half of the time, while z invested the remainder of the capital for the entire year. Then share of x in the profit is _____.

ANSWERS

MCQ Type Questions

1. (b) 2. (b) 3. (a) 4. (a) 5. (b) 6. (d) 7. (a) 8. (b) 9. (d) 10. (c)
11. (c) 12. (c) 13. (a) 14. (a) 15. (a) 16. (c) 17. (a) 18. (c) 19. (c) 20. (b)
21. (b) 22. (c) 23. (a) 24. (b) 25. (a)

Numerical Type Questions

1. 9180 2. 103950 3. 2812.50 4. 480 5. 345.2 6. (4150, 2490)
7. $13\frac{1}{3}$ 8. 30 9. 3000 10. $23\frac{1}{13}$ 11. 8 12. 1000
13. 6.09 14. 25 15. 4 16. 600 17. 25 18. 48
19. 98 20. 29.6 21. 4512 22. 1275 23. 2000 24. 1000
25. 480

EXPLANATIONS

MCQ TYPE QUESTIONS

1. The ratio of their investments is 11 : 9 : 12.

So, Manager gets $\frac{1}{10}(40000) = 4000$ + his share

Remaining amount = 36000.

Asstt. Manager gets $\frac{1}{20}(36000) = 1800$ + his share

Remaining amount = 36000 - 1800 = 34200

$$\begin{aligned}\therefore \text{M's share} &= \frac{11}{32} \times 34200 + 4000 \\ &= 11756.25 + 4000 \\ &= ₹ 15756.25 \approx ₹ 15756\end{aligned}$$

$$\begin{aligned}\text{AM's share} &= \frac{9}{32} \times 34200 + 1800 \\ &= 9618.75 + 1800 \\ &= ₹ 11418.75 \approx ₹ 11419\end{aligned}$$

$$\text{BA's share} = \frac{12}{32} \times 34200 = ₹ 12825.$$

2. B's share in the profit = 250 per 2000

If B gets ₹ 900, then

$$\text{total profit} = 2000 \times \frac{18}{5} = 7200$$

B's return on investment = 15%

$$\therefore \frac{900}{\text{B's investment}} \times 100 = 15$$

$$\therefore \text{B's investment} = 6000$$

Hence profits are shared in the ratio of their investments.

$$\text{Total investments} = 6000 \times \frac{7200}{900} = 48000$$

$$\therefore \text{Investment made by A and C} = 48000 - 6000 = 42000.$$

3. Interest paid by Bimal to Abhay for first six months

$$= \frac{50000}{2} \times \frac{8}{100} \times \frac{1}{2} = ₹ 1000$$

For last six months

$$\begin{aligned}&= \frac{30000 - 20000}{2} \times \frac{8}{100} \times \frac{1}{2} \\ &= ₹ 200\end{aligned}$$

$$\therefore \text{Total salary paid to Bimal} = 12 \times 500 = ₹ 6000$$

Let total profit be ₹ x

$$\therefore \text{Abhay's earning} = 1200 + \frac{x - 6000}{2}$$

$$\text{Bimal's earning} = 6000 + \frac{x - 6000}{2} - 1200$$

$$\therefore 3 \left(1200 + \frac{x - 6000}{2} \right) = 4800 + \frac{x - 6000}{2}$$

$$\Rightarrow 3600 + \frac{3x}{2} - 9000 = 4800 + \frac{x}{2} - 3000$$

$$\Rightarrow \frac{3x}{2} - \frac{x}{2} = 7200;$$

$$\therefore x = ₹ 7200$$

$$4. \text{Profit per stocks} = 91 \frac{3}{8} - 89 \frac{1}{8} = ₹ 2 \frac{2}{8}$$

$$\therefore \frac{45}{2 \frac{1}{4}} = 20 \text{ stocks}$$

5. Income from a stock at 3% in 20 years at par

$$= \frac{100}{100} \times 3 \times 20 = 60$$

Income from a stock at 5% at 124 in 20 years

$$= \frac{100}{124} \times 5 \times 20 = 80.64.$$

6. He invests ₹ 60 and gets ₹ 100 share

Income = ₹ 5

$$\text{Required \%} = \frac{5}{60} \times 100(0.70) = 5.83\% = 5 \frac{5}{6} \%$$

7. Let number of meters to be examined be x.

Then 0.08% of x = 2.

$$\Rightarrow \frac{0.08}{100} \times x = 2$$

$$x = \frac{2 \times 100}{0.08} = \frac{2 \times 100 \times 100}{8} = 2500$$

$$8. \left(\frac{a}{100} \times b \right) + \left(\frac{b}{100} \times a \right) = \frac{ab}{100} + \frac{ab}{100} = \frac{2ab}{100}$$

$$= \frac{2}{100} \times ab = 2\% \text{ of } ab.$$

9. Ryan's stock = 1.15 × Harland's stock

$$\therefore \text{Harland's stock} = \frac{9462}{1.15} = \$8228$$

Ryan's stock = 0.85 × Lambert's stock

$$\therefore \text{Lambert's stock} = \frac{9462}{0.85} = \$11132.$$

10. Let the fraction be $\frac{x}{y}$.

$$\text{Let } \begin{aligned} x_1 &= x + 0.25x = 1.25x \\ y_1 &= y - 0.25x = 0.75x \end{aligned}$$

$$\therefore \frac{x_1}{y_1} = \frac{1.25x}{0.75x} = \frac{5}{3} \frac{x}{y}$$

$$\therefore \text{Increase} = \frac{\frac{5}{3} \frac{x}{y} - \frac{x}{y}}{\frac{x}{y}} \times 100 = \frac{2}{3} \times 100 = 66.67\%$$

Thus, resultant fraction is more than the original fraction by 67%. Thus III is true.

11. I. Let first number be x .

$$\therefore \text{Second number} = 1.25x$$

$$\text{Sum} = x + 1.25x = 2.25x$$

$$\text{Difference} = 1.25x - x = 0.25x$$

$$\text{Also, } 9 \times 0.25x = 2.25x$$

Thus I is true.

$$\text{II. } \frac{56}{100} \times A = \frac{24}{100} \times B$$

Multiplying by $\frac{49}{56}$ on both the sides.

$$\therefore \frac{49}{56} \times \frac{56}{100} \times A = \frac{49}{56} \times \frac{24}{100} \times B$$

$$\Rightarrow \frac{49}{100} A = \frac{21}{100} B$$

$$\Rightarrow 49\% \text{ of } A = 21\% \text{ of } B.$$

Thus, II is true.

12. In the first round, let A get 'a' votes and B get 'b' votes.

$$\text{Then, } a + b = 260$$

$$\text{i.e., } a = 260 - b \quad \dots (i)$$

In second round, B gets

$$b \times \frac{125}{100} = \frac{5b}{4}$$

$$\text{and A gets } 260 - \frac{5b}{4}$$

$$\text{Also, } \frac{5b}{4} - \left(260 - \frac{5b}{4}\right) = 2(a - b)$$

$$\frac{10b}{4} - 260 = 2(260 - b - b)$$

$$\frac{10b}{4} - 260 = 520 - 4b$$

$$4b + \frac{10b}{4} = 520 + 260$$

$$b = 120$$

Hence first round, A gets 140 and B gets 120.

In second round A gets 110 and B gets 150.

$\therefore$ % decrease in votes received by A

$$= \frac{140 - 110}{140} \times 100 = 21.43\%$$

13. Non-water matter is 1% of 5000 = 50 gm.

After drying, this matter becomes 2% of the weight of the water-melon. Hence weight of the watermelon now is 2500 gm.

14. Let $A = ₹ x$ and $B = ₹ (5000 - x)$

$$\text{Then, } \frac{x \times 2 \times 20}{100} = \frac{(5000 - x) \times 3 \times 20}{100}$$

$$\Rightarrow 2x = 15000 - 3x$$

$$\Rightarrow 5x = 15000$$

$$\Rightarrow x = 3000$$

$$A = x = ₹ 3000$$

$$\begin{aligned} \text{and } B &= 5000 - x \\ &= 5000 - 3000 \\ &= ₹ 2000 \end{aligned}$$

$$\therefore \frac{A}{B} = \frac{3000}{2000} = \frac{3}{2}$$

Thus, (a) is not true and (b) is true.

$$A - B = 3000 - 2000 = ₹ 1000.$$

Thus, (c) is true.

(d) The total interest

$$\begin{aligned} &= \frac{3000 \times 2 \times 20}{100} + \frac{2000 \times 3 \times 20}{100} \\ &= ₹ 2400. \end{aligned}$$

Thus, (d) is true.

15. First investment gives a return

$$P \left(1 + \frac{1}{100}\right)^{100} - P$$

$$\text{First term is } P \left(1 + \frac{1}{100}\right)^{100} = X,$$

$$\text{such that } 2P < X \leq 3P$$

Hence return $X - P$ lies between P and $2P$

$$\text{i.e., } 2P - P < X - P \leq 3P - P.$$

$$\text{i.e., } P < X - P \leq 2P.$$

Percentage and Its Applications

Whereas second offer.

$$\begin{aligned} &= \text{Return} = P \left(1 + \frac{100}{100} \right)^1 - P \\ &= 2P - P = P. \end{aligned}$$

Alternatively

Option of investing at 1% for 100 years is obviously better than investing at 100% for 1 year, because total compound interest for 100 years will be at least 100% of the total amount (1% for each year), plus whatever the compounding effect will have.

16. Growth rate of rat population in 3 months

$$= 20 \times \frac{3}{12} = 5\%.$$

Increase in first 3 months = $3200 \times 1.05 = 3360$

Also, net decrease in 3 months = 160

$\therefore$ Rat population = $3360 - 160 = 3200$

In the same way, after every 3 months, rat population remains the same.

Hence, even after 3×8 months i.e., 2 years, the population is maintained

17. Let CP in A be ₹ x

$\therefore$ CP in B = ₹ 0.9x

$\therefore$ $0.9x + 15 + 10 = x$

$\Rightarrow$ $0.1x = 25$

$\Rightarrow$ $x = 250$

$\therefore$ Price in B = $250 \times 0.9 = ₹ 225$

18. Let CP of 5 dozen mangoes be ₹ x.

SP = ₹ 156 and Gain = 0.3x

$\therefore$ $156 - x = 0.3x$

$\Rightarrow$ $x = 120$

$\therefore$ SP of 60 mangoes = $120 \times 1.6 = 192$

$\therefore$ SP per mango = ₹ 3.2

19. Let CP be ₹ x ; SP = 1.2x

But he gets only $1.2x \times 0.8 = 0.96x$

$\therefore$ Loss = $0.04x$

$\therefore$ Loss = 4 %

20. Selling price = $800 \times 1.25 = ₹ 1000$

$\therefore$ Marked price = $₹ \frac{1000}{0.95} = 1053$

21. CP = ₹ 550

$\therefore$ Marked price = $1.3 \times 550 = ₹ 715$

$\therefore$ Sales price = $0.8 \times 715 = ₹ 572$.

22. Profit or loss of the sale depends upon how much profit the person is earning on the sale of two articles.

If SP > CP of 1 article, there is a profit.

If SP < CP of 1 article, there is a loss.

2.15

If SP = CP of 1 article, there is not profit no loss.

Thus, only (c) is true.

23. Following two key points need to be noted :

(A) Producer is profit maximizing which implies she will set prices that maximize the profit function.

(B) Quantity demanded in market A is independent of quantity demanded in market B.

CASE 1:

In this case, producer charges different prices in markets A and B. This implies that she will maximize profits individually.

Equations can be re-written as

$$P_A = 210 - 5Q_A$$

$$P_B = 125 - 1.25 Q_B$$

Profit in Market A = π_A

= Revenue - Cost

$$= Q_A \times P_A - C_A$$

$$= Q_A \times (210 - 5Q_A) - (1000 + 5Q_A)$$

At the maximum profit level, differential π_A with respect to Q_A is 0

or

$$\therefore 210 - 10Q_A - 5 = 0$$

$$\Rightarrow Q_A = 20.5, P_A = 107.5$$

and total revenue in market A

$$= 20.5 \times 107.5 = ₹ 2203.75$$

Similarly

Profit in Market B = π_B

= Revenue - Cost

$$= Q_B \times P_B - C_B$$

$$= Q_B \times (125 - 1.25 Q_B) - (1000 + 5Q_B)$$

At maximum profit level, differential of π_B with respect to Q_B is 0

$$\text{or } 125 - 2.5Q_B - 5 = 0$$

$$\Rightarrow Q_B = 48, P_B = 65$$

and total revenue in market B

$$= 48 \times 65 = ₹ 3120$$

Using the above, we get

$$\frac{P_A}{P_B} = \frac{107.5}{65} \sim 1.65.$$

24. Data for 1 is available above. Consider case 2. In this case, prices across the two markets are the same. Hence, quantity equations can be added to give

$$Q = 142 - P$$

where Q is total quantity and P is price in both markets

$$\Rightarrow P = 142 - Q$$

2.16

$\therefore \pi = \text{Profit} = Q \times (142 - Q) - (1000 + 5Q)$
At the maximum profit level, differential of π with respect to Q is 0

$\therefore Q = 68.5, P = 73.5$
and Total Revenue = 5034.75

Given this, we note that quantities sold in both Case 1 and Case 2 are the same. Hence, the ratio is exactly 1.

25. Profit Difference between case 1 and case 2
= Revenue Difference (since quantity sold and hence, cost is the same)
 $\therefore$ Profit difference = 2203.75 + 3120 - 5034.75
= 289

NUMERICAL TYPE QUESTIONS

- Interest on ₹ 22500 = $0.1 \times 22500 = ₹ 2250$
Charges for managing the concern
= $60 \times 12 = ₹ 720$
If yearly profit is ₹ x , then
$$\text{B's share} = \frac{\text{A's Share}}{2}$$

$$\therefore \frac{x}{2} - 2250 + 720 = \frac{1}{2} \left(\frac{x}{2} - 720 + 2250 \right)$$

 $\Rightarrow x = ₹ 9180.$
- Total annual profit
= $25000 + 20000 + 21000 \times 10$
= ₹ 255000
Actual profit
= $255000 - (2000 \times 12) = ₹ 231000$
Total investment
= $72000 + 27000 + 81000 = ₹ 180000$
A's share = $\frac{72000}{180000} \times 231000 = ₹ 92400$
B's share = $\frac{27000}{180000} \times 231000 = ₹ 34650$
C's share = $\frac{81000}{180000} \times 231000 = ₹ 103950.$
- 80% of the total profit is divided in the ratio
 $20000 : 24000 : 16000 = 5 : 6 : 4$
80% of total profit = $5x + 6x + 4x = 15x$
Total profit = $\frac{15x}{80\%} = 18.75x$
 $\therefore$ Share of C in profit = $4x + 20\%$ of $18.75x$
= $4x + 3.75x = 7.75x$
Share of A in profit = $5x$
Share of B in profit = $6x$

Percentage and Its Applications

- $\therefore (6x + 5x) - 7.75x = 487.50$
 $\Rightarrow 3.25x = 487.50$
 $\Rightarrow x = 150$
 $\therefore$ Total profit = $18.75 \times 150 = ₹ 2812.50.$
- Selling price of 40 shares
= $675 \times 40 = ₹ 27000$
Income from half the proceeds
$$= \frac{13500}{90} \times 4 = ₹ 600$$

Income from house = $1440 \times 0.75 = 1080$
Present total income = $1080 + 600 = ₹ 1680$
Earlier Income = $40 \times 500 \times \frac{6}{100} = ₹ 1200$
Increase in income = ₹ 480
 - One years dividend = $\frac{34539}{87} \times 3 = ₹ 1191$
Selling Price = $\frac{34539}{87} \times 89 = ₹ 35333$
Number of railway stocks bought
$$= \frac{35333}{115} = 307.24$$

Income from railway stocks
= $307.24 \times 5 = ₹ 1536.2$
Difference in income = 345.2
 - Let two numbers be x and y .
7.5% of $x = 12.5\%$ of y
 $\therefore 75x = 125y$
 $\Rightarrow 3x = 5y$
 $\Rightarrow x = \left(\frac{5}{3} \right) y.$
Now $x - y = 1660$
 $\Rightarrow \left(\frac{5}{3} \right) y - y = 1660$
 $\Rightarrow \left(\frac{2}{3} \right) y = 1660$
 $\Rightarrow y = 2490$
 $\therefore x = 2490 + 1660 = 4150.$
Hence numbers are 4150, 2490.
 - Number of passed candidates
$$= \frac{10}{100} \times 40 + \frac{20}{100} \times 500 + \frac{10}{100} \times 60$$

= $4 + 10 + 6 = 20$
Total students in school
= $40 + 50 + 60 = 150$

∴ Required percentage

$$= \frac{20}{150} \times 100 = \frac{40}{3} = 13\frac{1}{3}\%$$

8. Surface area = $4 \times \frac{22}{7} \times r^2$

$$= \frac{3}{r} \left(\frac{4}{3} \times \frac{22}{7} \times r^3 \right) = \frac{3}{r} \times \text{Volume}$$

where $r = 10$ cm

So we have

$$S = \frac{3}{10} V = \frac{3}{10} \times 100 \% \text{ of } V = 30 \% \text{ of } V$$

So surface area is 30 % of volume.

9. Difference in % of votes = $57 - 43 = 14\%$

∴ 14% is represented by 420

$$\therefore \text{Total number of votes} = 420 \times \frac{100}{14} = 3000.$$

10. % reduction in consumption = $\left(\frac{30}{100+30} \times 100 \right) \%$

$$= \left(\frac{30}{130} \times 100 \right) \% = 23\frac{1}{13} \%$$

11. Let the fraction be $\frac{a}{b}$

$$\therefore \frac{a^2}{b^2} \times \frac{2}{\frac{1}{5}} = 2 \frac{a}{b} \frac{a}{b} \neq 0 \text{ if } a \neq 0,$$

then $\frac{a}{b} \times \frac{2}{3/5} = 2$

$$\Rightarrow \frac{a}{b} = \frac{3}{5}$$

$$\therefore a + b = 8.$$

12. $1100 = P + \frac{P \times 5 \times \left(T - \frac{1}{2} \right)}{100}$

$$1100 = P + \frac{P \times 4 \times T}{100}$$

$$\therefore \frac{P \times 5 \times \left(T - \frac{1}{2} \right)}{100} = \frac{P \times 4 \times T}{100}$$

$$\Rightarrow 5T - \frac{5}{2} = 4T$$

$$\therefore T = \frac{5}{2} \text{ years}$$

$$\therefore 1100 = P \left(1 + \frac{4 \times \frac{5}{2}}{100} \right) = P \left(\frac{110}{100} \right)$$

$$\therefore P = ₹ 1000$$

13. $\left(1 + \frac{3}{100} \right)^2 = 1.0609$

∴ Effective annual rate = 6.09%

14. Let cash price be ₹P

$$\therefore 26.56 = P + \frac{P \times 12.5 \times 0.5}{100}$$

$$\Rightarrow P \approx ₹ 25.$$

15. Let R be the rate of interest.

$$\text{Discount} = \frac{9000 \times R}{100}$$

$$\text{Interest} = \frac{3000 \times R \times T}{100}$$

By the given information:

$$\frac{9000 \times R}{100} = \frac{3000 \times R \times T}{100}$$

$$\Rightarrow T = \frac{1}{3} \text{ years} = 4 \text{ months.}$$

16. Let N be the number of years

$$\therefore R\% = \sqrt{N \times 12}$$

$$\Rightarrow 48 = \frac{900 \times \sqrt{12N} \times N}{100}$$

Squaring both sides

$$N^2 = \frac{48 \times 48}{9 \times 9 \times 12} = \frac{16 \times 4}{9 \times 3} = \frac{64}{27}$$

$$\therefore R = \sqrt{12 \times \frac{4}{3}} = 4\%$$

$$\therefore 48 = \frac{P \times 4 \times 2}{100}$$

$$\Rightarrow P = ₹ 600$$

17. Let first discount be x%

$$\therefore \frac{(100-20)}{100} \times \frac{100-x}{100} \times 2500 = 1500$$

$$\Rightarrow \frac{80}{100} \times \frac{100-x}{100} \times 2500 = 1500$$

$$\Rightarrow 100 - x = \frac{1500 \times 100 \times 100}{80 \times 2500}$$

$$\Rightarrow 100 - x = 75;$$

$$\Rightarrow x = 25$$

Alternatively,

$$\text{Successive discount} = \left(a + b - \frac{ab}{100} \right) \%$$

$$\therefore \left(\frac{2500 - 1500}{2500} \right) \times 100 = a + 20 - \frac{20a}{100}$$

$$\Rightarrow \frac{1000}{2500} \times 100 = a + 20 - \frac{a}{5}$$

$$\Rightarrow 40 - 20 = a - 0.2a$$

$$\Rightarrow a = \frac{20}{0.8} = 25\%$$

18. Let there be n number of balls in each packet.

$$\therefore 6n \times 8 = \text{CP} + 96$$

$$\Rightarrow 48n = \text{CP} + 96 \quad \dots(i)$$

$$\text{Also, } 6(n-1) \times 9 = \text{CP} + 90$$

$$\Rightarrow 54n - 54 = \text{CP} + 90$$

$$\Rightarrow 54n = \text{CP} + 144 \quad \dots(ii)$$

Solving equations (i) and (ii), we get,

$$n = 8$$

$$\therefore \text{CP} = 48 \times 8 - 96 = 384 - 96 = ₹ 288$$

Each packet cost = ₹48.

19. CP of the merchant who calculates his % profit on

$$\text{CP} = \frac{3760}{1.175} = ₹ 3200$$

$$\therefore \text{His profit} = 0.175 \times 3760 = ₹ 658$$

Profit of the merchant who calculates his % profit on SP

$$= 0.175 \times 3760 = ₹ 658$$

$$\therefore \text{Difference in profit} = 658 - 560 = ₹ 98.$$

20. Let CP be ₹ x

$$\therefore \text{MP} = 1.8x$$

$$\text{SP of 15 articles} = 12 \times 1.8x = 21.6x$$

$$\therefore \text{Discount} = 0.9 \times 21.6x = 19.44x$$

$$\therefore \text{Profit on 15 articles} = (19.44 - 15)x = 4.44x$$

$$\therefore \text{Profit \%} = \frac{4.44x}{15x} \times 100 = 29.6\%$$

21. C.P. of goods for the trader = ₹ 22520

$$\text{S.P.} = \frac{125 \times 22520}{100} = ₹ 28150$$

$$\therefore \text{Profit} = ₹ 5630$$

$$\text{Now, } 5630 = 0.0536 \times 5000 + 0.085 \times 10000$$

$$+ \frac{x}{100} \times 7520$$

$$\Rightarrow 5630 = 268 + 850 + 75.2x$$

$$\Rightarrow 75.2x = ₹ 4512,$$

Which is nothing but profit from sale of third lot.

Percentage and Its Applications

22. A's Monthly Equivalent Rent = 22×4

$$\text{B's Monthly Equivalent Rent} = 16 \times 8$$

$$\text{C's Monthly Equivalent Rent} = 32 \times 6$$

$$\frac{\text{B's Monthly Equivalent Rent}}{\text{C's Monthly Equivalent Rent}} = \frac{\text{Rent paid by B}}{\text{Rent paid by C}}$$

$$\text{Rent paid by B} = \frac{16 \times 8}{32 \times 6} \times 600 = ₹ 400$$

Similarly, rent paid by A

$$= \frac{22 \times 4 \times 600}{32 \times 6} = ₹ 275$$

$$\therefore \text{Total rent} = 400 + 600 + 275 = ₹ 1275$$

23. Rate in which the rent is to be divided = 4 : 8 : 12

$\therefore$ Akbar's share of rent

$$= \frac{8}{24} \times 6000 = ₹ 2000.$$

24. 85% of the total profit is divided in the ratio 2 : 10 : 5.

$$85\% \text{ of total profit} = 2x + 10x + 5x = 17x$$

$$\text{Total profit} = \frac{17x}{85\%} = 20x$$

$\therefore$ Share of Ram in the profit

$$= 2x + 15\% \text{ of } 20x = 2x + 3x = 5x$$

Share of Shyam in the profit = $10x$

Share of Ghanshyam in the profit = $5x$

$$\text{Now, } (10x + 5x) - 5x = 500$$

$$\Rightarrow 10x = 500$$

$$\Rightarrow x = 50$$

$$\therefore \text{Total profit} = 20 \times 50 = ₹ 1000.$$

25. Let capital invested be C .

Monthly Equivalent Investment of

$$x = \frac{C}{3} \times \frac{T}{4}$$

Monthly Equivalent Investment of

$$y = \frac{C}{4} \times \frac{T}{2}$$

Monthly Equivalent Investment of

$$z = \frac{5C}{12} \times T$$

$\therefore$ Profit is to be shared in the ratio

$$\frac{CT}{12} : \frac{CT}{8} : \frac{5CT}{12}$$

i.e., 2 : 3 : 10

$\therefore$ Share of x in the profit

$$= \frac{2}{15} \times 3600 = ₹ 480$$

3

CHAPTER

Speed, Time and Work

SPEED

If the location of an object changes with time then it is said to be in motion. A bus running on a road, an ant crawling on a floor, etc. all are examples of objects in motion because the locations of these objects keep on changing w.r.t. their surroundings.

Let an object move from a point A to the point B through any path, then the actual length of the path followed by the object is called the distance travelled by the object.

The rate at which any moving body covers a particular distance is called its speed.

$$\text{Speed} = \frac{\text{Distance travelled}}{\text{Time taken}},$$

if the distance is constant, $\text{Speed} \propto \frac{1}{\text{Time}}$

if the time is constant, $\text{Distance} \propto \text{Speed}$

if the speed is constant, $\text{Distance} \propto \text{Time}$.

We can say that for constant distance travelled, speed is inversely proportional to the time taken. This can be explained by a simple example. To cover a distance of 100 km, if a person goes at the speed of 25 kmph, he will require 4 hours to complete the journey and travelling at a speed of 50 kmph, 2 hours will be required.

Unit of Measurement

Time is measured in seconds (s), minutes (min) or hours (hr). Distance is usually measured in metres (m), kilometres (km), miles, yards or feet.

Speed is usually measured in metres per second (mps), kilometres per hour(kmph) or miles per hour (mph).

Conversion of Units

1 hour = 60 minutes = 60 × 60 seconds.

1 kilometre = 1000 metres.

1 kilometre = 0.6214 mile.

1 mile = 1.609 kilometre.

i.e. 8 metres ≈ miles.

1 yard = 3 feet.

$$1\text{km} / \text{hr} = \frac{5}{18} \text{m} / \text{sec}$$

To convert kmph to m/s we multiply by 5/18

$$1 \frac{\text{m}}{\text{sec}} = \frac{18}{5} \text{km} / \text{hr}.$$

To convert m/s to kmph we multiply by 18/5

$$1 \frac{\text{km}}{\text{hr}} \approx \frac{5 \text{ miles}}{8 \text{ hr}}$$

$$1 \frac{\text{miles}}{\text{hr}} = \frac{22 \text{ ft}}{15 \text{ sec}}$$

Note: If a man changes his speed in the ratio m : n, then the ratio of time taken becomes n : m

Example: Walking at 5/6 of his usual rate, a boy reaches his school 12 min late. Then find his usual time taken to reach the school.

Solution: If the boy is walking at 5/6th of his usual speed, then the time taken should become 6/5 times the original time required. Thus, we can say that the boy is going to require 1/5th more/extra time as compared to the usual time required.

$$\left(\text{as } \frac{6}{5}T - 1T = \frac{1}{5}T \right)$$

$$\text{Thus, } \frac{1}{5}T \rightarrow 12 \text{ min}$$

$$1T \rightarrow X$$

Solve to get X = 60 min.

Example: Walking at $\frac{13}{11}$ th of his usual rate, a boy takes 3 min less to reach his school. Then find his usual time taken to reach the school.

Solution: If the boy is walking $\frac{13}{11}$ th of his usual

speed, then the time taken should become $\frac{11}{13}$ times

the original time required. Thus, we can say that the

boy is going to required $\frac{2}{13}$ th less time as compared

to the standard time required

$$\text{Thus, } \frac{2}{13}T \rightarrow 3 \text{ min}$$

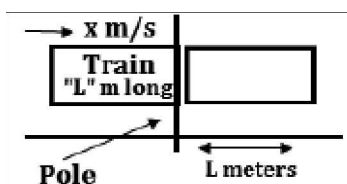
$$1T \rightarrow X$$

Solve to get X = 39/2 = 19.5 minutes.

If three men cover the same distance with speeds in the ratio $a : b : c$, the times taken by these three will be respectively in the ratio $1/a : 1/b : 1/c$.

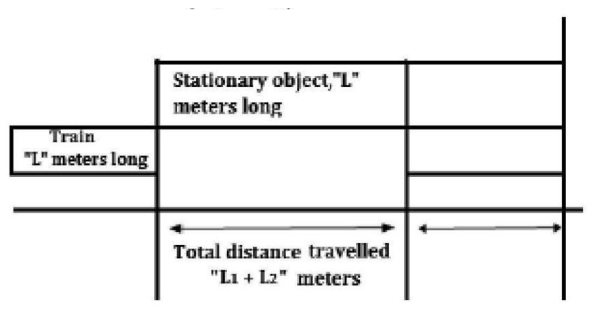
Note:

- (1) The time taken by a train L meters long, travelling at a speed of X m/sec in passing a signal post or a telegraph pole (or a standing man) is the same as the time taken by the train to cover a distance of L meters i.e. its own length because the length of the pole is nearly zero with respect to the length of the train though its height may be significant.



$$\text{Thus, } T = \frac{L}{X} \text{ seconds}$$

- (2) The time taken by a train " L_1 " meters long, travelling at a speed of X m/sec in passing a stationary object (a bridge or a tunnel or a train at rest or a platform) of length " L_2 " meters is the same as the time taken by the train to cover a distance of $(L_1 + L_2)$ meters.



$$T = \frac{L_1 + L_2}{X} \text{ seconds}$$

- (3) If two trains of lengths L_1 and L_2 meters respectively, are moving in the same direction with a speed of X and Y m/sec (where $X > Y$), then $(X - Y)$ m/sec is called **Relative Speed** and time taken by faster train to pass the slower train

$$= \frac{L_1 + L_2}{X - Y} \text{ seconds.}$$

- (4) If two trains of lengths L_1 and L_2 meters respectively, are moving in opposite directions (towards each other or away from each other) with a speed of X and Y m/sec, then $(X + Y)$ m/sec is called **Relative Speed** and time taken by the trains to pass each other.

$$= \frac{L_1 + L_2}{X + Y} \text{ seconds.}$$

- (5) If a man running at a speed X m/sec in the same direction in which a train of length L meters is running at a speed Y m/sec, then $(Y - X)$ m/sec is called the speed of the train relative to man. Then the time taken by the train to cross the man.

$$= \frac{L}{Y - X} \text{ seconds.}$$

- (6) If a man running at a speed X m/sec in a direction opposite to that in which a train of length L meters is running with a speed Y m/sec, then $X + Y$ is called the speed of the train relative to man. Then the time taken by the train to cross the man.

$$= \frac{L}{Y + X} \text{ seconds.}$$

- (7) If two trains or moving bodies are crossing each other such that

Length of the first train or object = L_1 meters,

Length of the second train or object = L_2 meters,

Time taken by the two when crossing each other in opposite direction = X sec,

Time taken by the two when crossing each other in same direction = Y sec,

Then speed of the faster train

$$= \frac{L_1 + L_2}{2} \left(\frac{1}{X} + \frac{1}{Y} \right);$$

$$\text{Speed of the slower train} = \frac{L_1 + L_2}{2} \left(\frac{1}{X} - \frac{1}{Y} \right).$$

Example: Two trains 110 m and 88 m long respectively are running in same direction. The first runs at the rate of 35.2 kmph and the second at the rate of 44 kmph. How long will they take to cross each other?

Solution: It is clear that the trains will cross each other when they have travelled a distance equal to the sum of their lengths = $110 + 88 = 198$ m. Since they are moving in the same direction, we can find the relative speed = $44 - 35.2 = 8.8$ km/hr.

$\Rightarrow$ Time required = distance/speed

$$= \frac{198}{8.8} \times \frac{18}{5} \text{ second} = 81 \text{ sec}$$

Example: A train in motion, 66 m long overtakes a train 88 m long travelling @ 30 kmph in the opposite direction in 0.168 min. The speed of the first train is

- (1) 60 kmph (2) 25 kmph
(3) 45 kmph (4) 36 kmph

Solution:

$$30 \text{ kmph} = \frac{30 \times 5}{18} \times 8.33 \text{ m/sec}$$

Let the speed of the first train be x m/sec.

Now total distance = $66 + 88 = 154$ m.

$$\text{Time} = \frac{\text{Distance}}{\text{Speed}}$$

$$\Rightarrow 0.168 \times 60 = \frac{154}{8.33 + x}$$

$$\Rightarrow 10.08(8.33 + x) = 154$$

$$\Rightarrow x = 6.95 \text{ m/sec} = 25 \text{ kmph}$$

Ans. (2)

Example: A train travelling with constant speed crossed a 96 m long platform in 12 sec and another 141 m long platform in 15 sec. Find the length and speed of the train.

Solution: If X be the length of train in meters and Y be the speed of the train in m/s, then

$$(X + 96)/Y = 12, (X + 141)/Y = 15$$

Solve to get $X = 84$, $Y = 15$.

$$15 \text{ m/sec} = 15 \times 18/5 = 54 \text{ kmph.}$$

Hence the required answer is 84m & 54kmph.

Average Speed: If a body travels $d_1, d_2, d_3 \dots d_n$ distances, with speeds $s_1, s_2, s_3 \dots s_n$ in time $t_1, t_2, t_3 \dots t_n$ respectively, then average speed of the body through the total distance is given by

$$\begin{aligned} \text{Average Speed} &= \frac{\text{Total distance travelled}}{\text{Total time taken}} \\ &= \frac{d_1 + d_2 + d_3 + \dots + d_n}{t_1 + t_2 + t_3 + \dots + t_n} \\ &= \frac{s_1 t_1 + s_2 t_2 + s_3 t_3 + \dots + s_n t_n}{t_1 + t_2 + t_3 + \dots + t_n} \\ &= \frac{d_1 + d_2 + d_3 + \dots + d_n}{\frac{d_1}{s_1} + \frac{d_2}{s_2} + \frac{d_3}{s_3} + \dots + \frac{d_n}{s_n}} \end{aligned}$$

Some Important Points:

- (1) If a body covers the distance d_1 and d_2 at a speed of s_1 and s_2 km/hr, respectively, in time t_1 and t_2 , then

$$\text{Total time taken, } T = t_1 + t_2 = \frac{d_1}{s_1} + \frac{d_2}{s_2}$$

$$\text{Total distance covered, } d = d_1 + d_2 = s_1 t_1 + s_2 t_2$$

- (2) While travelling a certain distance d , if a man changes his speed in the ratio $m : n$, then ratio of time taken becomes $n : m$.

- (3) If a certain distance (d), say from A to B, is covered at ' a ' km/hr and same distance is covered again

say from B to A in ' b ' km/hr, then average speed during the whole journey is given by

$$\text{Average speed} = \left(\frac{2ab}{a+b} \right) \frac{\text{km}}{\text{hr}} \dots$$

(which is harmonic mean of a and b)

Also, if t_1 and t_2 is taken to travel from A to B and B to A respectively, then distance ' d ' from A to B is given by

$$d = (t_1 + t_2) \left(\frac{ab}{a+b} \right); \quad d = (t_1 - t_2) \left(\frac{ab}{a-b} \right);$$

$$d = (a-b) \left(\frac{t_1 t_2}{t_1 - t_2} \right)$$

- (4) If a body travels a distance ' d ' from A to B with speed ' a ' in time t_1 and travels back from B to A i.e., same distance with $\frac{m}{n}$ of the usual speed ' a ', then change in time taken to cover the same distance is given by

$$\text{Change in time} = \left(\frac{n}{m} - 1 \right) \times t_1; \text{ for } n > m$$

$$= \left(1 - \frac{n}{m} \right) \times t_1; \text{ for } m > n$$

- (5) If two persons A and B start at the same time in opposite directions from two points and meet at any point in ' a ' and ' b ' hr, respectively after having met, then,

$$\frac{\text{A's Speed}}{\text{B's Speed}} = \frac{\sqrt{b}}{\sqrt{a}}$$

Relative speed.

- (1) Time taken by a moving object ' x ' meters long in passing a stationary object of negligible length from the time they meet is same as the time taken by the moving object to cover ' x ' meters with its own speed.
- (2) Time taken by a moving object ' x ' meters long in passing a stationary object ' y ' meters long from the time they meet, is same as the time taken by the moving object to cover ' $x + y$ ' meters with its own speed.
- (3) If two objects of length ' x ' and ' y ' meters move in the same direction at ' a ' and ' b ' m/s, then time taken to cross each other from the time they meet

$$= \frac{\text{Sum of their length}}{\text{Relative speed}}$$

$$\text{i.e., } \frac{x+y}{a-b} \text{ if } a > b \text{ or else, } \frac{x+y}{b-a}$$

- (4) If two objects of length 'x' and 'y' meters, move in the opposite direction at 'a' and 'b' m/s, then time taken to cross each other from the time they meet

$$= \frac{\text{Sum of their length}}{\text{Relative speed}} = \frac{x + y}{a + b}$$

- (5) If speed of a boat in still water is x km/hr and speed of the stream is y km/hr, then

Relative speed while travelling with the stream i.e., speed downstream = (x + y) km/hr.

Relative speed while travelling against the stream i.e., speed upstream = (x - y) km/hr.

- (6) Also speed of the boat in still water = $\frac{1}{2}$ (Speed with stream + Speed against stream)

Speed of the river = $\frac{1}{2}$ (Speed with stream - Speed against stream)

BOATS AND STREAMS

A Few Important Terminologies

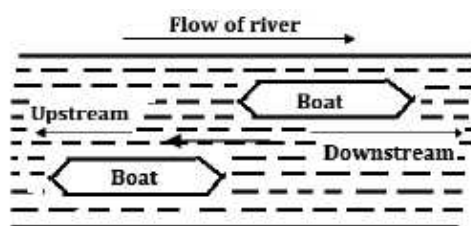
The following terms will be used often while discussing boats and streams.

Stream: It implies that the water in the river is moving or flowing.

Upstream: Going against the flow of the river.

Downstream: Going with the flow of the river.

Still Water: It implies that the speed of water is zero (generally in a lake).



Downstream (With The Stream) Rowing

It indicates that the stream favors the man's rowing (or boating). i.e. direction of rowing and direction of flow (stream) is same.

Upstream (Against The Stream) Rowing

It indicates that the stream flows against the man's rowing (or boating) i.e. direction of rowing and direction of stream (current) are opposite.

Note: Let the speed of a boat (or man) in still water be X m/sec and the speed of the stream (or current) be Y m/sec.

Speed of boat with the stream (or Downstream of D/S) (X + Y) m/sec.

Speed of boat against the stream (or Upstream or U/S) = (X - Y) m/sec.

Speed of man/boat in still water

$$\text{Speed of the stream } Y = \frac{(X + Y) - (X - Y)}{2}$$

$$= \frac{\text{Downstream} - \text{Upstream}}{2}$$

Example: A boat is rowed down a river 28 km in 4 hours and up a river 12 km in 6 hours. Find the speed of the boat and the river.

Solution:

Downstream speed is $\frac{28}{4} = 7$ kmph,

Upstream speed is $\frac{12}{6} = 2$ kmph,

Speed of Boat = $\frac{1}{2}$ [downstream + upstream speed]

$$= \frac{1}{2} (7 + 2) = 4.5 \text{ kmph.}$$

Speed of current = $\frac{1}{2}$ [downstream - upstream]

$$= \frac{1}{2} (7 - 2) = \frac{5}{2} = 2.5 \text{ kmph}$$

Example: A man rows 18 km down a river in 4 hours with the stream and returns in 12 hours. Find his speed and also the speed of the stream.

Solution:

Speed with the stream = $18/4 = 4.5$ kmph.

⇒ Speed against the stream = $18/12 = 1.5$ kmph.

⇒ Speed of the stream = $1/2 \{(4.5 - 1.5)\} = 1.5$ kmph.
and speed of the man = $4.5 - 1.5 = 3$ kmph.

If a man capable of rowing at the speed of X m/sec in still water, rows the same distance up and down a stream flowing at a rate of Y m/sec, then his average speed throughout the journey is

$$= \frac{\text{Speed Upstream} \times \text{Speed Downstream}}{\text{Man's rate in still water}}$$

$$= \frac{(X - Y)(X + Y)}{X}$$

Note: When downstream distance = upstream distance, then

$$\frac{\text{Man's rate in still water}}{\text{Speed of stream}} = \frac{t_{\text{up}} + t_{\text{down}}}{t_{\text{up}} - t_{\text{down}}}$$

Average speed for total journey (UP + DOWN)

$$= \frac{\text{Upstream Rate} \times \text{Downstream rate}}{\text{Man's rate in still water}}$$

Total Journey time ($t_{\text{up}} + t_{\text{down}}$)

$$\frac{\text{Man's rate in still water} \times \text{Total distance}}{\text{Upstream Rate} \times \text{Downstream rate}}$$

Example: A man rows 10 km upstream and back again to the starting point in 55 min. If the speed of stream is 2 kmph, then find the speed of rowing in still water.

Solution: Let x be the speed of rowing in still water.
 y = speed of stream = 2 kmph.

$$\text{Total time } T = \frac{55}{60} \text{ h.}$$

Hence,

$$\text{Total Time} = \frac{\text{Speed in still water} \times \text{Total distance}}{\text{Upstream Rate} \times \text{Downstream rate}}$$

$$\Rightarrow \frac{55}{60} = \frac{x \times 2 \times 10}{(x+2)(x-2)} \Rightarrow \frac{55}{60}(x^2 - 2^2) = 2 \times x \times 10$$

$$\Rightarrow 11x^2 - 240x - 44 = 0$$

$$\Rightarrow (x - 22)(11x + 2) = 0$$

$\therefore x = 22$, since $(-)$ ve value of x is not admissible.

Total Distance = Downstream distance + Upstream distance = 2 x any one side distance.

Example: A man who can swim 48 m/min in still water swims 200m against the current and 200 m with the current. If the difference between these two times is 10 min, then find the speed of the current in m per min.

Solution: Let the speed of stream be x kmph. The equation becomes

$$\frac{200}{48-x} - \frac{200}{48+x} = 10$$

$$\Rightarrow 200(48+x) - 200(48-x) = 10[48x^2 - 2^2]$$

$$\Rightarrow x^2 + 40x - 2304 = 0$$

On solving it we get $x = 32$ and $x = -72$ (not acceptable)
 i.e. speed of stream is 32 m/min.

LINEAR AND CIRCULAR RACES

The terms given below are commonly used in this topic, and a clear understanding will help the student get a good grip on the subject.

(1) **Race:** A contest of speed between participants is called a race.

(2) **Starting Point:** The point from where a race begins is called the starting point.

(3) **Race Course:** It means the path or ground on which races are run.

(4) **Finishing Point:** The point where the race finishes is called the winning post or finishing point or a goal.

(5) **Dead Heat Race:** A race in which no one is the winner because all the runners reach the winning post at the same time, is called a dead heat race.

(6) **Winner:** The person who first reaches the finishing point is called the winner.

(7) **Winner's Time:** The time taken by the winner to complete the race is called winner's time. If A and B both start from the same place, then winner's distance = Length of the race.

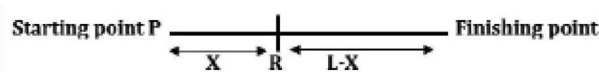
(8) **Loser's time:** The time taken by the loser to complete the race is called loser's time.

Linear Races

Suppose A & B are two contestants in race. If before the start of the race, A is at the starting point and B is ahead of A by 10 meters, then A is said to have given B a start of 10 meters. To cover a race of 300 meters in this case, A will have to cover a distance of 300 meters and B will have to cover $(300 - 10) = 290$ meters only.

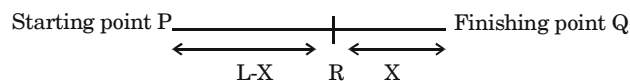
Suppose A & B are two contestants in race. At the end of the race, if A is at the finishing point and B is x meters away from the finishing point, and then A is said to have beaten B by x meters in a race.

(1) **A gives B a start of x meters** implies that, if the distance between the starting point and finishing point is L meters, A covers L meters while B covers $L - x$ meters.



From the figure given above, it is clear that A starts at point P, but B starts at R at the same moment. For example in a 100 meters race, A gives B a start of 10 meters means, while A runs 100 meters, B runs 90 meters.

(2) **A beats B by x meters** implies that, if the distance between the starting point and finishing point is L meters, A wins the race by covering L meters while B covers $L - x$ meters only.



(3) **A gives B a start of t seconds**, implies that A starts the race t seconds after B starts from the starting point.

(4) **A beats B by t seconds**, implies that, A and B start together from the starting point, but A reaches the finishing point t seconds before B reaches.

- (5) Start distance is the distance between the two contestants at start if they are not starting from the same position.
- (6) If A & B are starting from the same point, A beats B by 'x' meters or 't' seconds means, B runs 'x' meters in 't' seconds.

Example: In a km race, A beats B by 40 meters or 7 seconds. Find A's time over the course.

Solution: Here B runs 40 meters in 7 seconds.

$$\therefore \text{B runs 1000 m in } \frac{1000 \times 7}{40} = 175 \text{ seconds.}$$

Hence, A's time over the course $-(175 - 7) = 168$ sec

Short-cut:

By formula,

$$\frac{\text{Winner's time}}{\text{Loser distance}} = \frac{\text{Beat time} + \text{Start time}}{\text{Beat distance} + \text{Start distance}}$$

$$\therefore \text{A's time} = \frac{7}{40} \times 960 = 168 \text{ seconds.}$$

If a race ends in a dead heat, then beat time = 0 and beat distance = 0.

Circular Races

When two or more persons start from the same place at the same time and travel round a circle in the same direction or in opposite directions, then they will be first together at the starting point again after an interval of time which is the LCM of the times in which each of them makes one complete round.

Example: A and B walk around a circular path of circumference 1200 meters. A walks @ 150 m/min and B @ 70 m/min. If they start from the same point and walk in the same direction, then when will they be first together again at the starting point?

Solution: Time for 1 round by A = $1200/150$
= 8 minutes and for B
= $1200/70 = 120/7$ minutes.

$\Rightarrow$ Time taken to meet again at the starting point
= LCM of 8 and $120/7 = 120$ minutes = 2 hrs.

They will be first together again after an interval of time which is the LCM of the times in which one of the persons gains one complete round over each of the others.

Example: Assuming the data given in previous example, when will they be together again anywhere else on the course.

Solution: Now, in 1 minute, A gains over B $(150 - 70)$
= 80 meters.

To gain 1200 meters over B, A will take $1200/80 \times 1$
= 15 min.

They will be first together again after an interval of time which is required to complete a round with the relative speed.

Example: Assuming the data given in previous example, when will they be together supposing they walk in opposite directions?

Solution:

Relative speed = $150 + 70 = 220$ m/min.

So, they will be together after $1200/220$
= $60/11$ minutes.

We hope that the concept of circular race is clear to you. If not, please go through the following example. It will help you to understand the concept thoroughly.

Example: Along with A and B, C is also running across the circular track at the rate of 130 m/min in the same direction. When will all the three people meet (use data of previous example)

- 1) at the starting point?
- 2) at any other point on the track?
- 3) At what distance (nearest) from the starting point do the three meet?

Solution:

- (1) Time for 1 round by A = 8 min.

Time for 1 round by B = $120/7$ min.

Time for 1 round by C = $120/13$ min.

So, all the three people will meet at the starting point after the time.

= The LCM of 8, $120/7$ and $120/13$ i.e. 120 minutes.

- (2) A gains 80 meters per minute over B.

So, they will meet every $1200/80 = 15$ min

A gains 20 meters per minute over C.

So they will meet every $1200/20 = 60$ min

So, all the three people will meet after the time which is equal to the LCM of 15 and 60 i.e. after 60 minutes or 1 hour.

- (3) They are meeting after 60 minutes from the start.

Distance covered by A in 60 minutes

= $150 \times 60 = 9000$ m = 600m from the starting point.

Distance covered by B in 60 minutes

= $70 \times 60 = 4200$ m = 600m from the starting point.

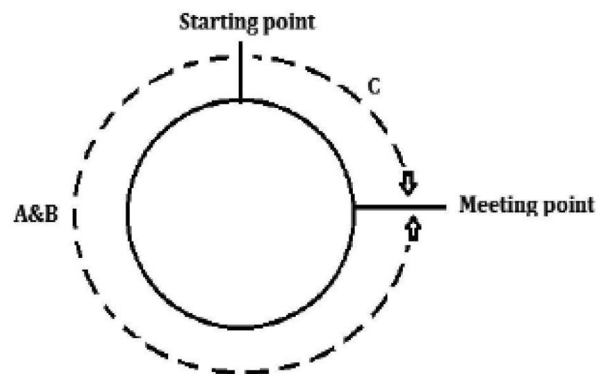
Distance covered by C in 60 minutes

= $130 \times 60 = 7800$ m = 600m from the starting point.

So, the three people will meet at 600 m from the starting point.

Example: Now, if we assume that C starts running in the opposite direction, then the three will once again meet at the starting point after the same time interval. But can they meet anywhere else? If Yes, after what time interval and what minimum distance from the starting point?

Solution: C starts running in the opposite direction, as shown in the figure below.



The three will once again meet at the starting point after the same time interval i.e. 2 hours.

Now A & C will meet after $1200/280 = 30/7$ minutes from the starting point. B & C will meet after $1200/200 = 6$ minutes from the starting point.

So, in this case the three people will meet after the time which is equal to the LCM of $30/7$ and 6 i.e. after 30 minutes from the starting point.

Distance covered by A in 30 minutes
 $= 150 \times 30 = 4500\text{m} = 900\text{m}$ from the starting point.

Distance covered by B in 30 minutes
 $= 70 \times 30 = 2100\text{m} = 900\text{m}$ from the starting point.

Distance covered by C in 30 minutes
 $= 130 \times 30 = 3900\text{m} = 300\text{m}$ from the starting point.

The minimum distance at which they will meet = 300 m from the starting point.

Note: Throughout this chapter, units of quantities have a crucial importance. e.g. while solving a problem if we take distance in meters, we should take speed in m/sec and time in seconds. If proper units are not used and conversions not effected, then you'll find yourself in trouble.

WORK & TIME

In most of the problems based on Work and Time, either the amount of time taken to finish a given job or the amount of work done is to be calculated.

Unless otherwise specified, the amount of work done is generally taken as unity (1). Also, if it is given that a person X can finish a job in D days, then it implies that X alone can do the job in D days.

If a man can do a piece of work in N days (or hours or any other unit of time), then the work done by him in one day will be $1/N$ of the total work.

Example: Anish can finish a piece of work by himself in 14 days. Then calculate the amount of work done by him in 1 day and 11 days.

Solution: The amount of work done by Anish, working alone in 14 days = 1 unit of work. So, the amount of work done by Anish, working alone in 1 day = $1/14$ unit of work and in 11 days = $11/14$ units of work.

Note: If A is twice as good a workman as B, then A will take half the time B takes to finish a piece of work.

Example: Suresh can finish a piece of work by himself in 42 days. Mahesh, who is $1/5$ times more efficient as Suresh, requires X days to finish the work by working all by himself. Then what is the value of X?

Solution: Suresh, working alone 42 days = 1 unit of work.

Mahesh is $1/5$ time more efficient than Suresh. So Mahesh is $6/5$ times as efficient as Suresh. Hence Mahesh should require $5/6$ th of the time, the time taken by Suresh.

Therefore time taken by Mahesh = $5/6 \times 42 = 35$ days.

Measurement Of Work

Work can be measured by many units. Some of them are

(1) Man-Days

A certain number of men working for a certain number of days.

Example: 5 men can finish a piece of work in 13 days. This implies that the work is of 5×65 Man-days, i.e. if one man alone works to finish the given work then he is going to take 65 days to finish it and if 65 men work together, then the work would be finished in 1 day.

(2) Man-Hours

A certain number of men working for a certain number of hours.

Example: 3 men can finish a piece of work in 13 hours. This implies that the work is of $3 \times 39 = 39$ Man-hours, i.e. if one man alone works to finish the given work then he is going to take 39 hours to finish it and if 39 men work together, then the work would be finished in 1 hour.

Some others are, Man-minutes, Machine-hours etc.

Note: If A, B and C can finish a piece of work in X, Y and Z days respectively, while working alone and together they require M days to finish the work, then the amount of work done by A is M/X , B is M/Y & C is M/Z .

Concept Of Work Equivalence

In order to solve certain types of questions, there is a very logical method that can be used.

This is the WORK EQUIVALENCE method which works on a simple premise-make the LHS equal to the RHS on the basis of SAME UNITS in terms of which work can be measured.

Some Important Points:

- (1) 'A gives B a start of x meters', implies that, if distance between starting point and finishing point is L meters, then A covers L meters while B covers $L - x$ metres.
e.g., In a 100 metre race. A gives B a start of 10 metres means, while A runs 100 metres, B runs $100 - 10 = 90$ metres.
- (2) 'A beats B by x metres', implies that, if distance between starting point and finishing point is L meters, then A wins the race by covering L metres, while B covers $L - x$ metres only.
- (3) 'A gives B a start of t seconds', implies that A starts the race t seconds after B starts from the starting point.
- (4) 'A beats B by t seconds', implies that, A and B start together from the starting point, but A reaches the finishing point t seconds before B finishes.

Note: (iii) and (iv) both imply that B takes t seconds more than A to finish the distance.

- (5) A beats B by ' x ' metres or ' t ' seconds means, B runs ' x ' metres in ' t ' seconds.
- (6) Winner's distance = Length of the race.
- (7) Distance covered by loser = Winner's distance - (Beat distance + Start distance)
- (8) Time taken by winner = Time taken by loser - (Beat time + Start time)
- (9)
$$\frac{\text{Winner's Time}}{\text{Loser's Distance}} = \frac{\text{Loser's time}}{\text{Winner's distance}}$$
$$= \frac{\text{Beat time} + \text{Start time}}{\text{Beat distance} + \text{Start distance}}$$
- (10) If a race ends in a dead heat, then beat time = 0 and beat distance = 0.
- (11) Two persons starting at the same time and from the same point along a circular path will be together again for the first time. When faster gains one complete round over the other, then time taken by faster person to complete one round over the other
$$= \frac{\text{Length of race course}}{\text{Relative Speed}}$$
- (12) Two persons, starting at the same time from the same point along a circular path, will be together again for the first time at the same starting point, at a time which is LCM of the time taken by each to complete a round.

- (13) Three persons, starting at the same time and from the same point along a circular path, will be together for the first time after the start at a time which is equal to LCM of the time taken by the fastest to gain a complete round over each of the other two.

- (14) A overtakes B $\frac{1}{n}$ th of x^{th} round means, when A has completed $\left(x - \frac{1}{n}\right)$ rounds, B has completed $\left[(x - 1) - \frac{1}{n}\right]$ round.

$$\text{Also, } \frac{\text{A's Speed}}{\text{B's Speed}}$$

$$= \frac{\text{Rounds completed by A in a given time}}{\text{Rounds completed by B in the same time}}$$

e.g., A overtakes B in the middle of the 4th round implies, when A has completed $3\frac{1}{2}$ rounds, B has completed $2\frac{1}{2}$ rounds. $2 \frac{\text{A's Speed}}{\text{B's Speed}} = \frac{3\frac{1}{2}}{2\frac{1}{2}}$

- (15) If a can do a work in ' a ' number of days, then in one day $\frac{1}{a}$ work is done.

Conversely, if a man does $\frac{1}{a}$ of a work in 1 day, then he can complete the work in $1 \div \frac{1}{a} = a$ days.

- (16) If A is ' x ' time as good a workman as B, then he will take $\frac{1}{x}$ of the time taken by B to do the same work.
- (17) If A and B can do a piece of work in ' x ' and ' y ' days respectively. then working together, they will take $\frac{xy}{x+y}$ days to finish the work and in one day.

They finish $\left(\frac{x+y}{xy}\right)^{\text{th}}$ part of the work.

- (18) To compare the work done by different people, first find the amount of work each can do in the same time.
- (19) If number of men to do a job is changed in the ratio $a : b$, then time required to do the work will be in the ratio $b : a$, assuming amount of work done by each of them in the given time is the same, or they are identical.
- (20) If two men A and B together can finish a job in ' x ' days and if A working alone takes ' a ' days more than A and B working together and B working alone takes ' b ' days more than A and B working together, then $x = \sqrt{ab}$.

- (21) To do a piece of work, the number of men employed and the number of days required to do the work are in inverse proportion. Also, the number of men employed and the hours worked per day are in inverse proportion.

Example: A group of soldiers can completely destroy an enemy bunker in 7 days. However 12 soldiers fell ill. The remaining now can do the job in 10 days. Find the original group strength.

Solution: Here, first of all, let us see how WORK can be defined. It is obvious that work can be measured a “destruction of the enemy bunkers.”

In the first case, let us say that there were S numbers of soldiers in the group. So they had to work for 7 days for the work which we call W .

$$\Rightarrow S \times 7 = W \quad \dots(1)$$

Now 12 fell ill and the remaining did the work in 10 days. Hence the new equations

$$(S - 12) \times 10 = W \quad \dots(2)$$

Just compare the two equations to get the answer.

$$S \times 7 = (S - 12) \times 10$$

$$\Rightarrow 7S = 10S - 120$$

$$\Rightarrow 120 = 3S$$

$$\Rightarrow S = 40 \text{ soldiers}$$

Hence, there were 40 soldiers in the group initially.

Example: A group of 20 cows can graze a field of 3 acres in size in 10 days. How many cows can graze a field twice as large in 8 days?

Solution: Here, first of all, let us see how WORK can be defined. It is obvious that work can be measured as “acres grazed”. In the first case, there were 20 cows in the group.

They had to work for 10 days to do the work which we call W (which = 3).

$$\Rightarrow 20 \times 10 = 3 \quad \dots(1)$$

Do not be worried about the numerical values on either side. The point is that logically this equation is consistent as the LHS indicates “Cowdays” and the RHS indicates “Acres”, both of which are correct ways of measuring work done.

Now the field is twice as large. Hence the new equation is

$$\Rightarrow C \times 8 = 6 \quad \dots(2)$$

Just divide (2) by (1) to get the answer.

$$8C/200 = 6/3$$

$$\Rightarrow 8C = 2 \times 200$$

$$\Rightarrow C = 400/8 = 50 \text{ cows}$$

Hence, there were 50 cows in the second group.

PIPES AND CISTERNS

A pipe connected with a cistern is called an *inlet*, if it fills the cistern.

A pipe connected with a cistern is called an *outlet*, if it empties the cistern.

filling or emptying a cistern can be considered as work done.

Some Important Points:

- (1) If an inlet pipe fills a cistern in ‘ a ’ hours, then

$$\frac{1^{\text{th}}}{a} \text{ part is emptied in 1 hour.}$$

Similarly, if an outlet pipe empties a cistern in

$$‘a’ \text{ hours, then } \frac{1^{\text{th}}}{a} \text{ part is emptied in 1 hour.}$$

- (2) If pipe A is ‘ x ’ times bigger than pipe B, then

$$\text{pipe A will take } \frac{1^{\text{th}}}{x} \text{ of the time taken by pipe B to fill the cistern.}$$

- (3) If A and B fill a cistern in ‘ m ’ and ‘ n ’ hours,

$$\text{respectively, then together they will take } \frac{mn}{m+n} \text{ hours to fill the cistern and in one hour}$$

$$\frac{m+n}{mn} \text{ part of the cistern will be filled.}$$

Similarly, ‘A’ and ‘B’ empty a cistern in ‘ m ’ and ‘ n ’ hours respectively, then, together they will

$$\text{take } \frac{mn}{m+n} \text{ hours to empty the cistern and in}$$

$$\text{one hour } \frac{m+n}{mn} \text{ part of the cistern will be}$$

empty.

- (4) If an inlet pipe fills a cistern in 'm' hours and an outlet pipe empties the cistern in 'n' hours, then the net part filled in 1 hour when both the pipes

are opened is $\left(\frac{1}{m} - \frac{1}{n}\right)$ i.e., $\frac{n-m}{mn}$ and the

cistern will get filled in $\left(\frac{mn}{n-m}\right)$ hours.

For the cistern to get filled, $m < n$. If $m > n$, the cistern will never get filled.

In general,

Net part filled of a cistern = (Sum of work done by inlets) – (Sum of work done by outlets)

- (5) If an inlet pipe fills a cistern in 'a' minutes, takes 'x' minutes longer to fill the cistern due to a leak in the cistern, then the time in which the leak will empty the cistern is given by $a \times$

$$\left(1 + \frac{a}{x}\right).$$

- (6) If two pipes A and B can fill a cistern in 'x' minutes and if A alone can fill it in 'a' minutes more than 'x' minutes and B alone can fill it in 'b' minutes more than 'x' minutes, then $x = \sqrt{ab}$.

Example: If an inlet pipe can fill a tank in 4 hours and an outlet pipe empties the full tank in 5 hours, then what is the net part filled in 1 hour when both the pipes are opened?

Solution: The net part filled in 1 hour when both the pipes are opened = $(1/4) - (1/5) = (1/20)$.

So, time required to fill the tank = $1/(1/20) = 20$ hrs.

Example: An inlet pipe can fill the tank in 6 hours and an outlet pipe can empty the tank in 4 hours. When the tank is full and both the pipes are open, find the new part emptied in 1 hour.

Solution: The net part emptied in 1 hour

$$= \frac{1}{4} - \frac{1}{6} = \frac{1}{12}.$$

{ $\therefore$ Time required to empty the full tank is 12 hours.}

Two pipes A and B can fill a cistern in X and Y hours respectively, while working alone. If both the pipes are opened together, then the time taken to fill the

cistern will be $\frac{X \times Y}{X + Y}$.

Example: Two pipes A and B can fill a cistern in 4 and 5 hours respectively, while working alone. If both the pipes are opened together, then find the time taken to fill the cistern.

Solution: Pipe A can fill the cistern in 4 hours and Pipe B can fill the cistern in 5 hours.

When the pipes are opened together, the time taken

to fill the cistern will be $\frac{4 \times 5}{4 + 5} = \frac{20}{9}$ hrs.

Three pipes A, B and C can fill a cistern in X, Y and Z hours respectively, while working alone. If all the three pipes are opened together, the time taken to

fill the cistern will be $\frac{X \times Y \times Z}{XY + YZ + ZX}$.

Example: Three pipes A, B and C can fill a cistern in 10, 12 and 15 hours respectively, while working alone. If all the three pipes are opened together, then find the time taken to fill the cistern.

Solution: The pipes can fill the cistern in 10, 12 and 15 hours respectively.

When the three pipes are opened together, the time taken to fill the cistern

$$= \frac{10 \times 12 \times 15}{10 \times 12 + 12 \times 15 + 10 \times 15} = 4 \text{ hrs.}$$

EXERCISE

MCQ TYPE QUESTIONS

1. A barrel full of beer has 2 taps, one midway which draws a litre in 6 minutes and the other at the bottom which draws a litre in 4 minutes. The lower tap is normally used after the level of beer in the barrel is lower than midway. The capacity of the barrel is 36 litres. A new assistant opens the lower tap when the barrel is full and draws out some beer. As a result the lower tap has been used 24 minutes before the usual time. For how long was the beer drawn out by the new assistant?
(a) 10 mins. (b) 8 mins.
(c) 16 mins. (d) 32 mins.
2. Pipe A can fill a cistern in 36 minutes and B in 48 minutes. If both the pipes are opened together, when should pipe B be closed so that the cistern may be just full in 24 minutes?
(a) 8 mins. (b) 9 mins.
(c) 12 mins. (d) 16 mins.
3. Pipes P, Q and R are attached to a tank and each can act as either an inlet or outlet pipe. Pipes P, Q and R respectively take 8, 10 and 12 hours to fill the empty tank or empty the full tank. In the first hour, pipes P and R work as inlet and Q work as outlet. In the second hour, pipes P and Q work as inlet and pipe R as outlet. In the third hour pipes Q and R work as inlet and pipe P as outlet and the process goes on like this. When will the cistern be filled?
(a) In the 8th hour. (b) In the 9th hour.
(c) In the 10th hour. (d) In the 11th hour.
4. There are 2 inlet pipes and an outlet pipe. The efficiency of one of the inlet pipes is double than that of the other. Also, the efficiency of the outlet pipe is half that of the lesser efficient inlet pipe. The empty tank gets filled in 16 hours when all the pipes are opened. How many hours will be taken to fill the empty tank when the 'lesser' efficiency inlet pipe is plugged and the rest kept opened?
(a) $33\frac{1}{3}$ hours (b) $26\frac{2}{3}$ hours
(c) 100 hours (d) 80 hours
5. A train overtakes 2 persons walking at 2 km/hr and 4 km/hr respectively in the same direction and completely passes them in 9 seconds and 10 seconds. Length of the train and its speed in km/hr respectively are
(a) 75 m, 18 km/hr (b) 80 m, 21 km/hr
(c) 60 m, 20 km/hr (d) 50 m, 22 km/hr
6. A policeman goes after a thief who has a 100 m start. The policeman runs 1 km. in 6 mins. and the thief 1 km in 10 mins. How far did the thief go before he was overtaken?

- (a) 100 m (b) 125 m
(c) 150 m (d) 250 m

7. Everyday Sachin exercises for $2\frac{1}{2}$ hours. He exercises by walking, jogging and running. His speeds while walking, jogging and running in the ratio 1 : 2 : 4. Which of the following statements are true?
I. If he spends equal time on the three activities, then he runs twice the distance that he jogs.
II. If he runs at 8 km per hour and distance covered by running and jogging together is 12 km, then he walks a distance of 1 km in 1 hour.
III. If the time for which he walks, jogs and runs is in the ratio 2 : 2 : 1 and he covers a total distance of 10 km, then his speeds while walking, jogging and running are 2 km/hr, 4 km/hr and 8 km/hr respectively.

Codes:

- (a) I and II (b) II and III
(c) I and III (d) I, II and III

8. A monkey climbing up a greased pole ascends 10 metres and slips down 2 metres in alternate minutes. If the pole is 64 metres high, how long will it take him to reach the top?
(a) 16 min
(b) 14 min and 48 sec
(c) 12 min
(d) 14 min
9. A train travelling at 10 m/sec from A to B at 7 a.m. meets a train leaving B at 7:20 a.m. and coming to A at a speed $\frac{1}{3}$ times faster than the first train. If the distance from A to B is 68 km. then, at what distance from A will the two trains meet?
(a) 72 km (b) 36 km
(c) 60 km (d) 50 km
10. Two trains A and B start from stations X and Y towards each other. B leaves station Y half an hour after train A leaves station X. Two hours after train A has started, the distance between trains A and B is $\frac{19}{30}$ th of the distance between stations X and Y. How much time would it take each train (A and B) to cover the distance X to Y, if train A reaches half an hour later to its destination as compared to B?
(a) 8 hrs, 6 hrs (b) 5 hrs, 4 hrs
(c) 10 hrs, 9 hrs (d) 9 hrs, 8 hrs

11. Two men, A and B, run a 4 km race on a circular course of $\frac{1}{4}$ km. If their speeds are in the ratio of 5 : 4, how often does the winner pass the other?
- (a) Once (b) Twice
(c) Thrice (d) Four times
12. A and B start from the same point to run in opposite directions round a circular path 550 yards in length, A giving B a start of 100 yards. They pass each other when A has run 250 yards. Who will come first to the starting point and at what distance will they be apart?
- (a) A, 20 yards (b) B, 10 yards
(c) B, 20 yards (d) A, 10 yards
13. Two runners run in the same direction along a circular track 2 km long. The faster runner overtakes the slower one every 30 minutes. What are their speeds, if faster one completes a round one minute sooner than the other?
- (a) 24 km/hr and 20 km/hr
(b) 44 km/hr and 28 km/hr
(c) 20 km/hr and 18 km/hr
(d) 20 km/hr and 16 km/hr
14. To do certain work alone 'A' takes 4 hrs, 'B' takes 5 hrs and 'C' takes 6 hrs. How long would A and B together take to do a work which C can do in $4\frac{1}{2}$ hrs.?
- (a) 2 hrs 10 mins. (b) 2 hrs 30 mins.
(c) 1 hr 20 mins. (d) 1 hr 40 mins.
15. A can do piece of work in 8 days which B and C can do in 12 and 16 days respectively. If they work on alternate days, then which of the following is true?
- (a) If A started the work on the first day followed by B and c, then the work will be finished on the 10th day.
(b) If B started the work on the first day followed by C and A, then the work will be finished on the 11th day.
(c) If C started the work on the first day followed by A and B, then the work will be finished on the 11th day.
(d) None of these

NUMERICAL TYPE QUESTIONS

1. Meera paints walls for a living. She sometimes has Ashok to work with her. Meera takes 2 days to paint a 10 ft by 10 ft wall. Ashok takes 3 days to do the same job. Meera has to pay Ashok for his work on a per day basis, so she'd rather do

without him. Meera has recently got a contract to paint the walls and ceiling of a room of dimensions 10 ft by 15 ft and a height of 10 ft. She had promised to do the job within 9 days. The _____ number of days does she need to employ Ashok?

2. 3 pipes can fill a reservoir in 10, 15 and 20 hours respectively. If the three taps are opened one after another in the given order, with a certain fixed time gap between them, the reservoir fills in 5 hours. The time gap will be _____ hr.
3. Two pipes A and B can fill a tank in 8 hours. If only pipe A is open, then it would take 4 hours longer to fill the tank. The _____ hours it would take if only pipe B is open.

Directions (Q.4 – 5): Refer to the data below and answer the questions that follow.

The boiler tank in a chemical factory holds 10^5 litres. 5 tanks each having one-fifth the capacity of the boiler tank fill in 'hard water' at same rates in the boiler tank in 2 hours. The outlet of the two of smaller tanks work as inlet pipes and other two work as outlet and the fifth tank fill in the main 'boiler' at half its efficiency.

4. The number of hours taken to fill in the main boiler when all the pipes are opened at once is _____
5. Only three pipes are working, two at their full efficiency and the third one with half of its efficiency and all three are acting as an inlet. If all the three are opened alternately with the latter pipe being first to be opened to fill the boiler such that each smaller tank fills 'hard water' in the main boiler for equal time, then number of hours taken to fill the boiler to half its capacity is _____
6. A contract is to be completed in 46 days and 117 men were set to work, each working 8 hours a day. After 33 days, $\frac{4}{7}$ of the work is completed. The _____ number of additional men may be employed so that the work may be completed in time, each man now working 9 hours a day?
7. A train travelling at 42 km/hr. passes a cyclist going in the same direction in 9 secs.; if cyclist had been going in the opposite direction, the train would have passed him in 5 secs. Length of the train is _____ metres
8. A man rows upstream 13 km. and downstream 28 km. taking 5 hrs. each time. Velocity of the current in km/hr _____ is

Directions (Q. 9 – 11): Refer to the data below and answer the questions that follow.

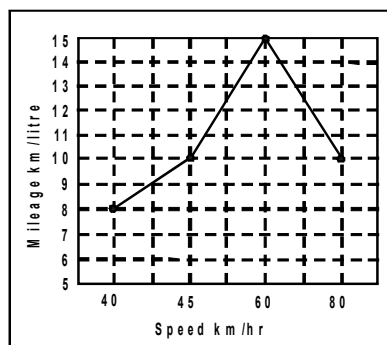
The variation in the speed of a car on a particular day at the respective times is shown in the table below:

s(km/hr)	t(hr)
0	11.00 am
40	11.30 am
50	1.00 pm
85	1.30 pm
10	3.30 pm
10	4.30 pm

9. The distance travelled by the car from 11 a.m. to 1 p.m. is _____ km
10. If car maintains the speed it has at 4.30 p.m. then at _____ time will the car cover the same distance as it had covered from 11.00 a.m. to 1.00 p.m.?
11. Average speed of the car from 11.00 a.m. to 4.30 p.m. is _____ kmph
12. A cheetah chase a deer which is 100 m ahead. The time in which the deer takes 10 leaps the cheetah takes only 6 leaps. In one leap, the deer covers 1 m while the cheetah covers 2 m. In _____ leaps would the cheetah catch up the deer.
13. A hare pursued by a hound who is 50 of her own leaps before him. When the hare takes 4 leaps, the hound takes 3. In one leap, the hare goes $1\frac{3}{4}$ metres and the hound $2\frac{3}{4}$ metres. In how _____ leaps will the hound overtake the hare.

Directions (Q. 14 – 15): Refer to the graph below and answer the questions that follow.

The engine of the new car Palio brought by Fiat in the market follows certain mileage characteristics as shown in the diagram below:



Vijay bought the new car Palio and filled exactly 10 litres in his car. He goes at a speed of 40 km/hr for first 20 km and then at a speed of 60 km/hr for next 60 km and driving at a constant speed reaches his destination which was 115 km away from his starting point.

14. The time in hours taken by Vijay if he reaches his destination in minimum possible time and finishes all the fuel.
15. In the above question, if Vijay had travelled at a constant average speed rather than changing his speed, taking the same minimum time, then the approximate saving in the fuel consumed is _____ litres
16. In a 2 km race on a circular course of $\frac{1}{4}$ of a km. A overlaps B in the middle of his 6th round. Then _____ distance (in km) will A win at the same rate of running
17. A runs $1\frac{3}{8}$ times faster than B. A gives B a start of 120 metres. _____ metre must the winning post be so that it may be a dead heat.
18. The work done by a man, a woman and a child is in the ratio of 3 : 2 : 1. There are 20 men, 30 women and 36 children in a factory. Their weekly wages amount of ₹ 780, which is divided in the ratio of work done by the men, women and children. The wages of 15 men, 21 women and 30 children for 2 weeks is _____
19. A can do a work in 6 days and B in 8 days. With the help of a boy, the three complete the work in 3 days and get ₹ 200. The boy's share in rupees, if the money is distributed in the ratio of the work done is _____
20. A can finish a job in 12 days and B when working at twice his efficiency finishes a job in 9 days. The _____ number of days will they take if they work for two days alternately working at their standard rate given that A starts first.

ANSWERS

MCQ Type Questions

1. (c) 2. (d) 3. (c) 4. (b) 5. (d) 6. (c) 7. (c) 8. (b) 9. (b) 10. (c)
11. (c) 12. (d) 13. (a) 14. (d) 15. (c)

Numerical Type Questions

1. 6 2. $\frac{1}{2}$ 3. 16 4. 20 5. 6 6. 81 7. 75 8. 1.5 9. 77.5
10. 12.15 a.m. 11. 39.3 12. 300 13. 210 14. 2 15. $2\frac{1}{3}$ 16. $\frac{4}{11}$
17. 440 18. 1170 19. 25 20. 14

EXPLANATIONS

MCQ TYPE QUESTIONS

1. Top tap is operational till 18 litres is drawn out.
 $\therefore$ Time after which the lower tap is usually open
 $= 18 \times 6 = 108$ minutes.
 $\therefore$ Time after which it is open now
 $= 108 - 24 = 84$ minutes.
 $\therefore$ Litres drawn $= \frac{84}{6} = 14$ litres
 $\therefore$ $18 - 14 = 4$ litres
 were drawn by the new assistant.
 $\therefore$ Time $= 4 \times 4 = 16$ minutes.

Alternatively

Upper tap was not used for 24 minutes.

- $\therefore$ Litres drawn $= \frac{24}{6} = 4$ litres.
 $\therefore$ Beer was drawn for $4 \times 4 = 16$ minutes.
2. Let pipe B be turned-off after x minutes.

$$\text{Part filled in } x \text{ minutes} = x \left(\frac{1}{36} + \frac{1}{48} \right) = \frac{7x}{144}.$$

$$\text{Part filled in } 24 - x \text{ minutes} = \frac{24 - x}{36}$$

$$\therefore \frac{7x}{144} + \frac{24 - x}{36} = 1$$

$$\Rightarrow \frac{7x + 96 - 4x}{144} = 1$$

$$\Rightarrow x = 16 \text{ minutes}$$

3. In a cycle of 3 hours : pipes P, Q and R are working as inlet pipes for 2 hours each and they are working as outlet pipes for an hour each. So part of tank filled in 3 hours.

$$= 2x \left(\frac{1}{8} + \frac{1}{10} + \frac{1}{12} \right) - \left(\frac{1}{8} + \frac{1}{10} + \frac{1}{12} \right)$$

$$= \frac{1}{8} + \frac{1}{10} + \frac{1}{12} = \frac{15 + 12 + 10}{120} = \frac{37}{120}$$

$$\text{Hence they will take } \frac{120 \times 3}{37} = 9\frac{27}{37} \text{ hours.}$$

Thus, tank will be filled in 10th hour.

4. Let number of hours taken by the outlet and the 2 inlet pipes be $4x$, $2x$ and x respectively. So in 1 hour the part of empty tank filled is

$$\frac{1}{x} + \frac{1}{2x} - \frac{1}{4x} = \frac{1}{16}$$

$$\frac{5}{4x} = \frac{1}{16}$$

$$\Rightarrow x = 20$$

Hence, inlet pipe with higher efficiency fills the tank in 20 hours.

So in one hour, the desired pipes fill

$$\frac{1}{x} - \frac{1}{4x} = \frac{3}{4x} \text{ th of the tank.}$$

Tank gets filled in

$$\frac{4x}{3} = \frac{80}{3} \text{ hours} = 26\frac{2}{3} \text{ hours.}$$

5. In each case, train has to travel its own length, to pass each man.

If x km/hr is speed of the train, then

$$\text{Length} = \text{Relative speed} \times \text{Time}$$

$$= (x - 2) \times \frac{9}{60 \times 60} = (x - 4) \times \frac{10}{60 \times 60}$$

$$\therefore 9x - 18 = 10x - 40$$

$$\Rightarrow x = 22 \text{ km/hr.}$$

$$\therefore \text{Length} = (22 - 2) \times \frac{9}{3600}$$

$$= \frac{20 \times 9}{3600} \text{ km} = 50 \text{ m.}$$

6. Policeman gains $= \frac{1}{6} - \frac{1}{10} = \frac{4}{60} \text{ km/min}$

$$= 66 \frac{2}{3} \text{ m/min.}$$

Hence to gain 100 m, time required

$$= \frac{100}{66 \frac{2}{3}} = 1 \frac{1}{2} \text{ min.}$$

Hence thief has gone ahead by

$$1 \frac{1}{2} \times \frac{1}{10} \times 1000 \text{ metres} = 150 \text{ m.}$$

7. I. Ratio of speed = 1 : 2 : 4

If equal time is spent on each of these, then ratio of distance = 1 : 2 : 4

So, Sachin runs twice the distance that he jogs.

Thus, I is true.

- II. He runs at 8 km/hr and therefore walks at 2 km/hr.

Hence II is not true.

- III. Ratio of speed = 1 : 2 : 4

Ratio of time = 2 : 2 : 1

Ratio of distance

$$= 2 \times 1 : 2 \times 2 : 4 \times 1 = 2 : 4 : 4$$

If Sachin covers a total distance of 10 km, then he

$$\text{Walks for } \frac{2}{2+4+4} \times 10 = 2 \text{ km,}$$

$$\text{Joggs for } \frac{4}{2+4+4} \times 10 = 4 \text{ km.}$$

$$\text{Runs for } \frac{4}{2+4+4} \times 10 = 4 \text{ km.}$$

$$\text{Total time taken is } 2 \frac{1}{2} \text{ hours} = \frac{5}{2} \text{ hours.}$$

$$\text{Thus, he walks for } \frac{2}{2+2+1} \times \frac{5}{2} = 1 \text{ hour.}$$

$$\text{jogs for } \frac{2}{2+2+1} \times \frac{5}{2} = 1 \text{ hour.}$$

$$\text{runs for } \frac{1}{2+2+1} \times \frac{5}{2} = \frac{1}{2} \text{ hour.}$$

So speed while,

$$\text{Walking} = \frac{2}{1} = 2 \text{ km/hr}$$

$$\text{Jogging} = \frac{4}{1} = 4 \text{ km/hr}$$

$$\text{Running} = \frac{4}{\frac{1}{2}} = 8 \text{ km/hr.}$$

Thus, III is true.

8. In 1 minute, monkey ascends 10 metres but he takes 1 minute to slip down 2 metres. Thus, at the end of 2 minutes, net ascending of the monkey is $= 10 - 2 = 8$ metres.

Thus, to have a net ascending of 8 metres, process of ascending and then slipping happens once. So, to cover 64 metres, above process is repeated $\frac{64}{8}$ or 8 times.

It is clear that in 8 such happenings, the monkey will slip 7 times, because 8th time, he will ascend to the top. Thus, in climbing 7 times and slipping 7 times, he covers (7×8) or 56 metres.

Time taken to cover 56 metres

$$= \frac{56 \times 2}{8} = 14 \text{ minutes}$$

Remaining distance = $64 - 56 = 8$ metres

Time taken to ascend 8 metres

$$= \frac{8}{10} \text{ min} = \frac{4}{5} \text{ min.}$$

$$\text{Total time taken} = 14 \text{ minutes} + \frac{4}{5} \text{ min}$$

$$= 14 \text{ mins. 48 sec.}$$

9. If they meet t hrs after 7 a.m.,

$$10 \times \frac{18}{5} \times t + 10 \times \frac{18}{5} \times \frac{4}{3} \left(t - \frac{1}{3} \right) = 68$$

$$\Rightarrow 36t + 48t - 16 = 68$$

$$\Rightarrow t = 1 \text{ hr.}$$

They meet at a distance of

$$10 \times 60 \times 60 = 36000 \text{ m}$$

i.e., 36 km from A.

10. Two equations are

$$T_1 - T_2 = \frac{1}{2} + \frac{1}{2} = 1 \quad \dots(i)$$

where T_1 and T_2 are time taken by trains A and B to cover the whole distance

$$\text{and } \frac{2}{T_1} + \frac{1.5}{T_2} = \frac{11}{30} \quad \dots(ii)$$

Solving equations (i) and (ii), we get

$$T_1 = 10 \text{ hrs. and } T_2 = 9 \text{ hrs.}$$

11. When A runs 5 rounds, B runs 4 rounds (ratio of speeds)

$\therefore$ A passes B each time A has run 5 rounds or

$$5 \times \frac{1}{4} = \frac{5}{4} \text{ km.} = 1 \frac{1}{4} \text{ km.}$$

$$1 \frac{1}{4} \text{ km is contained in 4 km.}$$

3 times.

Hence A passes B thrice.

12. When A has run 250 yards, B should have run

$$550 - 250 = 300 \text{ yards.}$$

$\therefore$ B runs $300 - 100 = 200$ yards in the same time taken by A to run 250 yards.

$\therefore$ Ratio of the speeds of A and B

$$= 250 : 200 = 5 : 4$$

If speed of A is $5x$ yards per minute, then speed at B = $4x$ yards/min.

Time taken by B to reach the starting point

$$= \frac{450}{4x} = \frac{112.5}{x} \text{ min.}$$

Time taken by A to reach the starting point

$$= \frac{550}{5x} = \frac{110}{x} \text{ min.}$$

$\therefore$ A wins by $\frac{2.5}{x}$ min

$\therefore$ Distance travelled by B in $\frac{2.5}{x}$ min.

$$= \left(\frac{2.5}{x} \right) \times (4x) = 10 \text{ yards}$$

Hence B is 10 yards behind A.

13. Let n_1 and n_2 be the speeds in km/hr and let n_1 be the greater speed i.e., $n_1 > n_2$.

$$\therefore \frac{2}{n_2} = \frac{2}{n_1} + \frac{1}{60}.$$

$$\Rightarrow n_1 n_2 = 120(n_1 - n_2)$$

Faster runner covers $(n_1 - n_2)$ km. more than the slower runner in 1 hour.

Hence to cover one round of 2 km, he will take

$$\frac{2}{n_1 - n_2} \text{ or } \frac{1}{2} \text{ hour.}$$

$$\therefore \frac{2}{n_1 - n_2} = \frac{1}{2}$$

$$\Rightarrow n_1 - n_2 = 4$$

$$\Rightarrow n_1 = 4 + n_2$$

$$\Rightarrow n_2(4 + n_2) = 120 \times 4$$

$$\Rightarrow n_2^2 + 4n_2 - 480 = 0$$

$$\Rightarrow n_2 = 20 \text{ or } -24 \text{ (rejected)}$$

$$\therefore n_1 = 4 + n_2 = 24 \text{ km/hr}$$

Hence speeds are 24 km/hr and 20 km/hr

Alternatively

Options may also be substituted to find the result.

14. Work done by A and B in 1 hr

$$= \frac{1}{4} + \frac{1}{5} = \frac{9}{20} \text{ part.}$$

$\therefore$ Ratio of work done by (A + B)

$$\text{and } C = \frac{9}{20} : \frac{1}{6}$$

$\therefore$ Ratio of time taken

$$= \frac{1}{6} : \frac{9}{20} = 10 : 27$$

If C takes 27 hrs. (A + B) will take 10 hrs.

If C takes $4\frac{1}{2}$ hrs. (A + B) will take

$$\frac{10 \times 9}{2 \times 27} \cong 1 \text{ hr. 40 mins.}$$

15. I. Work done by A, B and C together in 3 days

$$\frac{1}{8} + \frac{1}{12} + \frac{1}{16} = \frac{13}{48} \text{ th of the work.}$$

Thus, in 9 days, total work done will be

$$3 \times \frac{13}{48} = \frac{39}{48} \text{ th of the work}$$

$\therefore$ Work remaining = $1 - \frac{39}{48} = \frac{9}{48}$ th of the

work is left of which A will finish $\frac{6}{48}$ th of work on 10th day and B will finish remaining work on 11th day.

Thus, I is false

- II. Work done by A, B, C together in 9 days = $\frac{39}{48}$

$$\text{Work left} = 1 - \frac{39}{48}$$

$$= \frac{9}{48} \text{ th of the whole.}$$

On the 10th day B will finish $\frac{1}{12}$ th of the whole.

$$\therefore \text{Work left} = \frac{9}{48} - \frac{1}{12}$$

$$= \frac{5}{48} \text{ th of the whole.}$$

On 11th day C will finish $\frac{1}{16}$ th of the work

$$\therefore \text{Work left} = \frac{5}{48} - \frac{1}{16}$$

$$= \frac{2}{48} \text{ th of the whole}$$

which will be done by A on the 12th day. Thus, work will be finished on the 12th day. Thus, II is false.

- III. Work done by A, B and C in 9 days = $\frac{39}{48}$

Work left after 9 days

$$= 1 - \frac{39}{48} = \frac{9}{48} \text{ th of the whole.}$$

On the tenth day $\frac{1}{16}$ th will be finished by C.

$$\begin{aligned}\therefore \text{Work left} &= \frac{9}{48} - \frac{1}{16} \\ &= \frac{6}{48} \text{ th of the whole,}\end{aligned}$$

which will be finished by A on the eleventh day. Thus, III is true.

NUMERICAL TYPE QUESTIONS

1. Total area to be painted

$$\begin{aligned}&= 10 \times 15 \text{ (ceiling)} + 2 \times 10 \times 10 \text{ (wall)} \\ &\quad + 2 \times 10 \times 15 \text{ (wall)} \\ &= 650 \text{ sq.ft}\end{aligned}$$

In one day Meera can paint

$$10 \times \frac{10}{2} = 50 \text{ sq.ft}$$

Ashok can paint

$$10 \times \frac{10}{3} = \frac{100}{3} \text{ sq.ft,}$$

So, together they can paint $\frac{250}{3}$ sq.ft per day.

Let number of days Ashok works be x.

$$\text{Work done in 9 days} = 50(9 - x) + \frac{250}{3}x = 650$$

$$\therefore 450 - 50x + \frac{250x}{3} = 650$$

$$\Rightarrow \frac{100}{3}x = 200$$

$$\Rightarrow x = 6 \text{ days.}$$

2. Let fixed interval be x

$$\therefore \frac{5}{10} + \frac{5-x}{15} + \frac{5-2x}{20} = 1$$

$$\Rightarrow (5)6 + (5-x)4 + (5-2x)3 = 60$$

$$\Rightarrow 30 + 20 - 4x + 15 - 6x = 60$$

$$\Rightarrow x = \frac{1}{2} \text{ hour.}$$

3. Let pipe B alone take b hours longer than if both A and B were open.

$$8 = \sqrt{4 \times x}$$

$$\Rightarrow 64 = 4 \times x$$

$$\Rightarrow x = 16 \text{ hours.}$$

Alternatively

$$\text{Part filled if both A and B are open} = \frac{1}{8}$$

$$\text{Part filled if only A is open} = \frac{1}{12}$$

$$\therefore \text{Part filled if only B is open} = \frac{1}{8} - \frac{1}{12} = \frac{1}{24}$$

$\therefore$ B alone would take 24 hours, i.e., B alone would take $24 - 8 = 16$ hours longer to fill the tank.

4. Since 5 tanks of same efficiency fill the boiler in 2 hours each tank individually takes

$$2 \times 5 = 10 \text{ hours to fill it.}$$

Tank with half the efficiency will take

$$10 \times 2 = 20 \text{ hours.}$$

In one hour, fraction of boiler getting filled when 4 fully efficient pipes with two as inlet and two as outlet and 5th acts at 50% efficiency as an inlet is

$$\frac{1}{10} + \frac{1}{10} + \frac{1}{20} - \frac{1}{10} - \frac{1}{10} = \frac{1}{20}$$

Hence it will take 20 hours to fill the boiler tank.

5. Here fraction of tank filled in 1 hour

$$\frac{1}{10} + \frac{1}{10} + \frac{1}{20} = \frac{5}{20} = \frac{1}{4} \text{ th}$$

of the tank. If they are opened alternately, then tank gets filled in $4 \times 3 = 12$ hours

Hence half of the tank gets filled in $\frac{12}{2} = 6$ hours.

6. $\frac{4}{7}$ of work is completed.

$$\text{Remaining work} = 1 - \frac{4}{7} = \frac{3}{7}$$

Remaining period = $46 - 33 = 13$ days

Less work, less men (direct proportion)

Less days, more men (Indirect proportion)

More hours/day, less men (Indirect proportion)

work $\frac{4}{7} : \frac{3}{7}$

Days $13 : 33 :: 117 : x$

hrs/day $9 : 8$

$$\frac{4}{7} \times 13 \times 9 \times x = \left(\frac{3}{7}\right) 33 \times 8 \times 117$$

$$\begin{aligned}x &= (3 \times 33 \times 8 \times 117) / (4 \times 13 \times 9) \\ &= 198 \text{ men}\end{aligned}$$

So, additional men to be employed

$$= 198 - 117 = 81$$

7. Let L is length of train in km and x is speed of the cyclist, Then

$$(i) \frac{3600}{9} L = 42 - x = 400L$$

$$(ii) \frac{3600}{5} L = 42 + x = 720L$$

On adding, we get

$$1120L = 84$$

$$\Rightarrow L = \frac{84 \times 1000}{1120} = 75 \text{ m}$$

$$8. \quad \frac{13}{\text{Speed upstream}} = 5$$

$$\text{and} \quad \frac{28}{\text{Speed downstream}} = 5$$

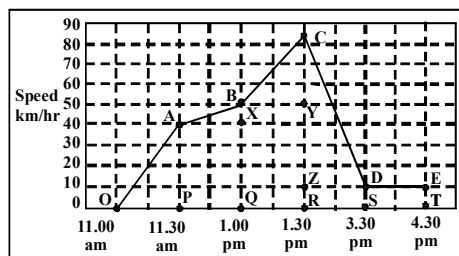
$$\therefore \text{Speed upstream} = \frac{13}{5} \text{ km/hr.}$$

$$\text{and Speed downstream} = \frac{28}{5} \text{ km/hr.}$$

$$\therefore \text{Speed of river} = \frac{1}{2} \left(\frac{28}{5} - \frac{13}{5} \right)$$

$$= \frac{1}{2} \times \frac{15}{5} = 1.5 \text{ km/hr}$$

9. Graph of speed vs time is plotted as shown. Area under the graph and time axis gives the distance.



Required distance

$$= A(\triangle OAP) + A(\triangle BXA) + A(\square APQX)$$

$$= \frac{1}{2} \times \left(\frac{1}{2} \right) \times 40 + \frac{1}{2} \times 10 \times \left(\frac{3}{2} \right) + \left(\frac{3}{2} \right) \times 40$$

$$= 10 + \frac{30}{4} + 60 = 77.5 \text{ km.}$$

10. Distance covered are the same.

Speed in the latter is maintained at 10 kmph.

$$\therefore 77.5 = 10 \times x$$

$$\Rightarrow x = 7.75 \text{ hours.}$$

Hence, clock time will be 4.30 + 7 hours 45 mins
= 12.15 a.m.

11. Average speed = $\frac{\text{total distance travelled}}{\text{total time taken}}$

Distance travelled from 1.00 pm to 4.30 p.m.

$$= A(\square BQRY) + A(\triangle BYC) + A(\triangle CZD) + A(\square ZRTE)$$

$$= 50 \times \left(\frac{1}{2} \right) + \left(\frac{1}{2} \right) \times \left(\frac{1}{2} \right) \times 35 + \frac{1}{2} \times (2) \times 75 + 3 \times 10$$

$$= \frac{50}{2} + \frac{35}{4} + 75 + 30$$

$$= 130 + 8.75$$

$$= 138.75 \text{ km.}$$

$$\text{Total distance travelled} = 77.5 + 138.75$$

$$= 216.25 \text{ km.}$$

$$\therefore \text{Average speed} = \frac{216.25}{5.5} = 39.3 \text{ kmph.}$$

12. While deer makes 10 leaps, cheetah makes 6 leaps.

$$\therefore \text{Distance covered in 10 leaps by deer}$$

$$= 10 \times 1 = 10 \text{ m}$$

Distance covered in 6 leaps by cheetah

$$= 6 \times 2 = 12 \text{ m}$$

Hence in six leaps, cheetah gains

$$12 - 10 = 2 \text{ m over the deer.}$$

Cheetah has to gain 100 m over the deer.

So, total leaps required by cheetah

$$= \frac{6}{2} \times 100 = 300.$$

13. 50 leaps of the hare $50 \times 1 \frac{3}{4} = \frac{175}{2} = 87.5 \text{ m.}$

So, hound should gain $\frac{175}{2} \text{ m}$ over the hare.

When hound travels $3 \times 2 \frac{3}{4} \text{ m}$, then

hare travels $4 \times 1 \frac{3}{4} \text{ m.}$

Hence in 3 leaps of the hound the hound gains

$$\frac{33}{4} - \frac{28}{4} = \frac{5}{4} \text{ metres}$$

$$\therefore \text{Number of leaps required} = \frac{175}{\frac{5}{4}} \times 3$$

$$= \frac{175}{2} \times \frac{4}{5} \times 3 = 210 \text{ leaps.}$$

14. Fuel required for first 20 km = $\frac{20}{8} = 2.5 \text{ litres.}$

Fuel required for next 60 km = $\frac{60}{15} = 4 \text{ litres.}$

$$\therefore \text{Petrol left} = 10 - 6.5 = 3.5 \text{ litres.}$$

Now, he has to travel = $115 - (60 + 20)$

$$= 35 \text{ km in 3.5 litres.}$$

So he can go at a speed of either 45 km/hr or 80 km/hr as mileage is 10 km/litre in both cases. But since he reaches in minimum time, he travels at 80 km/hr

$$\therefore \text{Time taken} = \frac{20}{40} + \frac{60}{60} + \frac{35}{80} = \frac{155}{80}$$

$$= 1 + \frac{75}{80} \text{ hours}$$

$$= 1 \text{ hours } 56 \text{ minutes and } 25 \text{ seconds.}$$

15. Total distance travelled = 115 km

$$\text{Total time taken} = \frac{155}{80} \text{ hours}$$

$$\therefore \text{Average speed} = \frac{115}{\frac{155}{80}} = 60 \text{ km/hr}$$

$$\therefore \text{Fuel consumption} = \frac{115}{15} = 7\frac{2}{3} \text{ litres.}$$

$$\text{So, fuel saved} = 10 - 7\frac{2}{3} = 2\frac{1}{3} \text{ litres.}$$

16. A overlaps B in the middle of the 6th round

So, when A runs $5\frac{1}{2}$ rounds, B runs only $4\frac{1}{2}$ rounds.

$$\therefore \frac{\text{A's Speed}}{\text{B's Speed}} = \frac{5\frac{1}{2}}{4\frac{1}{2}} = \frac{11}{9}$$

So when A runs 11 km B runs 9 km.

$$\text{Thus, when A runs 2 km, B runs } 2 \times \frac{9}{11} = \frac{18}{11} \text{ km.}$$

$$\therefore \text{A wins by } \left(2 - \frac{18}{11}\right) = \frac{4}{11} \text{ km.}$$

17. Speeds of A and B are as

$$1\frac{3}{8} : 1 = \frac{11}{8} : 1 = 11 : 8$$

In a race of 11 metres, A gain

$$11 - 8 = 3 \text{ metres over B.}$$

To gain 120 metres, race must be of

$$120 \times \frac{11}{3} = 440 \text{ metres.}$$

18. Work Number

Men 3 20

Women 2 30

Children 1 36

$$\text{Ratio of wages} = (3 \times 20) : (2 \times 30) : (1 \times 36) \\ = 5 : 5 : 3$$

$$\text{Total wages of 20 men} = \frac{5}{13} \times 780 = ₹ 300$$

$$\therefore \text{Wages of a man} = ₹ 15.$$

Similarly, wages of 30 women

$$\frac{5}{13} \times 780 = ₹ 300$$

$$\text{Wages of a woman} = ₹ 10$$

$$\text{Wages of 36 children } \frac{3}{13} \times 780 = ₹ 180$$

$$\text{Wages of a child} = ₹ 5$$

$$\therefore \text{Total wages of 15 men, 21 women and 30 children}$$

$$= 15 \times 15 + 21 \times 10 + 30 \times 5 \\ = ₹ 585$$

$$\therefore \text{Total wages for 2 weeks} = ₹ 1170.$$

19. Work done by A in 3 days

$$= 3 \times \frac{1}{6} = \frac{1}{2} \text{ of the work.}$$

Work done by B in 3 days

$$= 3 \times \frac{1}{8} = \frac{3}{8} \text{ of the work.}$$

$\therefore$ Work done by the boy in 3 days

$$= 1 - \left(\frac{1}{2} + \frac{3}{8}\right) \\ = \frac{1}{8} \text{ of the work.}$$

$\therefore$ Ratio of work done

$$= \frac{1}{2} : \frac{3}{8} : \frac{1}{8} = 4 : 3 : 1$$

$$\therefore \text{Boy's share} = \frac{1}{8} \times ₹ 200 = ₹ 25.$$

20. A finishes the job at standard rate in 12 days,

$$\text{So in 1 day, the part of job done by A} = \frac{1}{12}$$

Similarly the part of job done by B in 1 day

$$= \frac{1}{2 \times 9} = \frac{1}{18}$$

If A works for 2 days followed by 2 days of B at standard rate.

$$\text{Part of work done} = \frac{2}{12} + \frac{2}{18} = \frac{5}{18}$$

$$\text{So, in (2 alternate days work of 'A' and 'B') } \\ 3 \times 4 = 12 \text{ days}$$

$$\text{They will finish } 3 \times \frac{5}{18} = \frac{15}{18} \text{ th of the work.}$$

$$\text{Work remaining} = 1 - \frac{15}{18} = \frac{3}{18} = \frac{1}{6}$$

$$\text{This is finished by A alone in } \frac{1}{\frac{1}{12}} = 2 \text{ days.}$$

$$\text{Hence, total number days} = 12 + 2 = 14 \text{ days.}$$

4

CHAPTER

Ratio, Proportion and Mixtures

RATIO

Ratio of two terms 'a' and 'b' is denoted by $a : b$

(read as 'a is to b') and is measured by $\frac{a}{b}$.

Numerator 'a' is called '*antecedent*' and denominator 'b' is called '*consequent*'.

While comparing two quantities in terms of ratio :

- Two quantities must be of the same kind.
- Units of measurement of the two quantities must be the same.
e.g., if $a = ₹ 2$ and $b = 50$ paise,
then $a : b = 200 : 50 = 4 : 1$.
- Ratio is a pure number, i.e., without any unit of measurement as it denotes how many times a given quantity is in comparison to another.
- Ratio would stay unaltered even if both antecedent and the consequent are multiplied or divided by the same number.

$$\text{e.g., } a : b = \frac{a}{b} = \frac{a \times m}{b \times m} = \frac{\frac{a}{m}}{\frac{b}{m}}.$$

Properties of Ratios

- Invertendo :** If $\frac{a}{b} = \frac{c}{d}$, then $\frac{b}{a} = \frac{d}{c}$ i.e., inverse ratios of two equal ratios are equal.
- Alternendo :** If $\frac{a}{b} = \frac{c}{d}$, then $\frac{a}{c} = \frac{b}{d}$ i.e., ratios of antecedents and consequents of two equal ratios are equal.
- Componendo :** If $\frac{a}{b} = \frac{c}{d}$, then $\frac{a+b}{b} = \frac{c+d}{d}$ i.e., adding 1 to both sides.
- Dividendo :** If $\frac{a}{b} = \frac{c}{d}$, then $\frac{a-b}{b} = \frac{c-d}{d}$ i.e., subtracting 1 from both sides.
- Componendo-Dividendo :** If $\frac{a}{b} = \frac{c}{d}$, then $\frac{a+b}{a-b} = \frac{c+d}{c-d}$ i.e., dividing results of componendo by dividendo.

6. **Equal Ratio :** If $\frac{a}{b} = \frac{c}{d} = \frac{e}{f}$; then each ratio

$$= \frac{\text{Sum of Numerators}}{\text{Sum of Denominators}} = \frac{a + c + e \dots}{b + d + f \dots}$$

The principle can also be applied after multiplying Numerator and Denominator of any fraction by the same number.

If $\frac{a}{b} = \frac{c}{d} = \frac{e}{f} = \dots$, then each ratio is equal to

$$\left(\frac{pa^n + qc^n + re^n + \dots}{pb^n + qd^n + rf^n + \dots} \right)^{1/n}, \text{ where } p, q, r, n \dots \text{ may have}$$

any values except that they must not all be zeros.

In the ratio $a : b$, a is called the antecedent (the first term) and b is called the consequent (the second term)

Note:

The ratio of two numbers a and b , written as $a : b$, is

the fraction $\frac{a}{b}$ provided $b \neq 0$. Ratio is always reduced

to its simplest form. Thus $a : b = \frac{a}{b}$, $b \neq 0$. if $a = b \neq 0$, the ratio is $1 : 1$ or $1/1 = 1$

$$1. \text{ The ratio of 4 to 6} = 4 : 6 = \frac{4}{6} = \frac{2}{3}$$

$$2. \frac{2}{3} : \frac{4}{5} = \frac{2/3}{4/5} = \frac{4}{6}$$

$$3. 5x : \frac{3y}{4} = \frac{5x}{3y/4} = \frac{20x}{3y}$$

Type of Ratio

Compounded Ratio

When two or more ratios are multiplied term wise, the ratio thus obtained is called their compounded ratio. For the ratios $a : b$ and $c : d$, the compounded ratio is $ac : bd$.

Example: What is the compounded ratio, for $2 : 3$ and $4 : 5$?

Solution: The compounded ratio is $2 \times 4 : 3 \times 5$ or $8 : 15$.

Duplicate Ratio

It is the compounded ratio of two equal ratios. Thus the duplicate ratio of

$$a : b \text{ is } a^2 : b^2 \text{ or } a^2 : b^2$$

Example: Find the duplicate ratio of 4 : 5.

Solution: The duplicate ratio of 4 : 5 is 16 : 25.

Triplicate Ratio

It is the compounded ratio of three equal ratios. Thus the triplicate ratio of $a : b$ is $a^3 : b^3$ or $a^3 : b^3$

Example: Find the triplicate ratio of 4 : 5.

Solution: The triplicate ratio of 4 : 5 is 64 : 125.

Sub-Duplicate Ratio

For any ratio $a : b$ its sub-duplicate ratio is defined as $\sqrt{a} : \sqrt{b}$.

Example: Find the sub-duplicate ratio of 16 : 25.

Solution: Sub-duplicate ratio of 16 : 25 is $\sqrt{16} : \sqrt{25}$, i.e. 4 : 5.

Sub-Triplicate Ratio

For any ratio $a : b$, its sub-triplicate ratio is defined as $\sqrt[3]{a} : \sqrt[3]{b}$.

Example:

Find the sub-triplicate ratio of 27 : 64.

Solution:

Sub-triplicate ratio of 27 : 64 is $\sqrt[3]{27} : \sqrt[3]{64}$ i.e. 3 : 4.

Comparing Ratios

To compare two ratios, we express them as fractions and then compare.

Example:

Which is greater 3 : 4 or 4 : 5?

Solution:

$$3 : 4 = \frac{3}{4} \text{ and } 4 : 5 = \frac{4}{5}.$$

$$\frac{3}{4} = \frac{3 \times 5}{4 \times 5} = \frac{15}{20}$$

$$\frac{4}{5} = \frac{4 \times 4}{5 \times 4} = \frac{16}{20}$$

$$\text{Since } \frac{16}{20} > \frac{15}{20} \text{ or, } \frac{4}{5} > \frac{3}{4},$$

So, 4 : 5 > 3 : 4.

Note: A ratio is said to be in its simplest form if the HCF of the antecedent and the consequent is 1.

Example:

Divide 2400 in the ratio 3 : 5.

Solution:

The first part is 3 units and the second part is 5 units. The total of both the parts = 3 units + 5 units = 8 units.

Here, 8 units = 2400,

$$\text{So, 1 unit} = \frac{2400}{8} = 300.$$

The first part = 3 units = 3 × 300 = 900.

The second part = 5 units = 5 × 300 = 1500.

Example:

A sum of money is divided between Vinod and Lokesh in the ratio of 3 : 7. Vinod gets Rs. 240. What does Lokesh get?

Solution:

Vinod gets 3 units = Rs. 240.

$$\text{So, 1 unit} = \frac{240}{3} = 80.$$

Therefore, 7 units = 7 × 80 = 560.

Thus, Lokesh gets Rs. 560.

Problems Leading to the Application of Ratios

Example:

The ratio of the number of boys to the number of girls in a school of 1638 is 5 : 2. If the number of girls increased by 60, then what must be the decrease in the number of boys to make the new ratio of boys to girls as 4 : 3?

Solution:

$$\text{Number of boys} = \frac{5}{7} \times 1638 = 1170$$

$$\text{Number of girls} = \frac{2}{7} \times 1638 = 468$$

Number of girls after increase

$$= 468 + 60 = 528$$

Total number of boys as per new ratio

$$= 528 \times \frac{4}{3} = 704$$

∴ The number of boys to be decreased

$$= 1170 - 704 = 466.$$

Example:

If $a : b = 3 : 4$ and $b : c = 6 : 13$, then find $a : b : c$.

Solution:

The best way to solve such questions is to make b common in the two ratios.

Thus, we can write $a : b = 9 : 12$ and $b : c = 12 : 26$.

Now that b is equal in both the ratios, we can write the same as

$$a : b : c$$

$$9 : 12$$

$$12 : 26$$

Thus, we can write $a : b : c = 9 : 12 : 26$.

Using formula directly, we can get

$$a : b : c = (3 \times 6) : (4 \times 6) : (4 \times 13) = 9 : 12 : 26$$

PROPORTION

When ratio of two terms is equal to the ratio of two other terms, then these four terms are said to be in **proportion**, i.e., if $a : b = c : d$, then a, b, c and d are in proportion.

a, b, c and d are called first, second, third and fourth proportionals respectively. The terms a and d are called **extremes**, while b and c are called **means**.

If $a : b = c : d$, then $ad = bc$ i.e., product of extremes is equal to product of means.

Continued Proportion :

When $\frac{a}{b} = \frac{b}{c}$, then a, b and c are said to be in continued proportion and b is called *geometric mean* or *mean proportional* between a and c .

$$\text{Also, } b^2 = a \times c, \quad \Rightarrow \quad b = \sqrt{ac}$$

If $\frac{a}{b} = \frac{b}{c} = \frac{c}{d} = \frac{d}{e} = \dots$, then $a, b, c, d, e \dots$ are said to be in continued proportion.

Concept of continued proportion is useful in the following situation.

- (i) If four quantities a, b, c and d and $a : b, b : c$ and $c : d$, are known, then we can find $a : d$ as

$$\frac{a}{d} = \frac{a}{b} \times \frac{b}{c} \times \frac{c}{d}$$

- (ii) If a, b, c and d are four quantities and $a : b, b : c, c : d$ are known, then we can find $a : b : c : d$.

e.g., If $a : b = 2 : 3; b : c = 4 : 5, c : d = 7 : 9$,

find $a : b : c : d$.

$a : b = 8 : 12, b : c = 12 : 15$ (LCM of 3 and 4 is 12)

$\therefore a : b : c = 8 : 12 : 15$

$a : b : c = 56 : 84 : 105, c : d = 105 : 135$ (LCM of 7 and 15 is 105)

$\therefore a : b : c : d = 56 : 84 : 105 : 135$

Example:

Find the fourth proportional to $12a^2, 9a^2b$ and $6ab^2$.

Solution:

Let x be the fourth proportional, then

$$\frac{12a^2}{9a^2b} = \frac{6ab^2}{x} \Rightarrow x = \frac{6ab^2 \cdot 9a^2b}{12a^2} = \frac{9}{2}ab^3$$

Example:

Find the mean proportional between $9a^2b$ and $25b^3$.

Solution:

Let x be the mean proportional.

$$\therefore \frac{9a^2b}{x} = \frac{x}{25b^3} \Rightarrow x^2 = 9a^2b \cdot 25b^3$$

$$\Rightarrow x = \pm 15ab^2$$

VARIATION

Most of us would still remember statements like “The distance travelled varies directly as the speed of the vehicle.” These and similar statements have precise mathematical meanings and they represent a specific type of function called **variation functions**.

The three general types of variation functions are direct variation, inverse variation and joint variation.

Direct Variation

If two quantities X & Y are related such that any increase or decrease in ‘ Y ’ produces a proportionate increase or decrease in ‘ X ’ or vice versa, then the two quantities are said to be in direct proportion.

X is directly proportional to Y is written as $X \propto Y$ or $X = KY$.

In other words $X : Y = \frac{X}{Y} = K$. Here K is a constant whose value for a particular variation is same.

Consider $X_1 = KY_1$ and $X_2 = KY_2$, dividing the two we

$$\text{get } \frac{X_1}{X_2} = \frac{Y_1}{Y_2}.$$

Thus, the chances of your success in the test are directly proportional to the number of hours of sincere work devoted every day.

Example:

If $X \propto Y$ and $x = 9$ when $y = 30$, then find the relation

between x and y . Find x when $y = 7\frac{1}{2}$ and y when $x = 6$.

Solution:

Let $x = ky$, then $9 = k(30)$, $k = \frac{3}{10}$, i.e. $x = \frac{3}{10}y$.

When $y = 7\frac{1}{2}$, $x = \frac{3}{10} \left(7\frac{1}{2} \right) = 2\frac{1}{4}$.

When $x = 6$, $x = \frac{3}{10}y \therefore y = 20$.

Example:

Different sizes of the car have different models. The weight of a car model varies directly as the cube of its length. The weight of a car model of length 3 cm is 10 gm. What is the weight of a car model of length 12 cm?

Solution:

Let W gm be the weight of a car model and L cm be its length.

$\therefore W \propto L^3$ or $W = kL^3$ (where k is a constant)

$$\Rightarrow 10 = k(3)^3 \Rightarrow k = \frac{10}{27} \text{ i.e. } W = \frac{10}{27} L^3$$

$$\text{When } L = 12, W = \frac{10}{27} (12)^3 = 640 \text{ gm.}$$

$\therefore$ The required weight is 640 gm.

Inverse Variation

Here two quantities X & Y are related such that, any increase in X would lead to a decrease in Y or any decrease in X would lead to an increase in Y. Thus the quantities X & Y are said to be inversely related

and X is inversely proportional to Y is written as $X \propto \frac{1}{Y}$

$$\text{or } X = \frac{k}{Y} \text{ or } XY = k \text{ (Constant)}$$

Thus, $X_1 Y_1 = X_2 Y_2$

Or the product of two quantities remains constant.

Thus, the chances that you will be able to cheat in a test are inversely proportional to the smartness of the invigilator.

Example:

If y varies inversely as x, and y = 3 when x = 2, then find x when y = 21.

Solution:

$$y \propto \frac{1}{x} \text{ or } y = \frac{k}{x} \text{ (where k is a constant)}$$

$$\text{then } 3 = \frac{k}{2}, k = 6 \text{ i.e. } y = \frac{6}{x}$$

$$\text{When } y = 21, 21 = \frac{6}{x}, x = \frac{2}{7}$$

Joint Variation

(a) A varies jointly as B and C and is denoted by $A \propto BC$ Or $A = kBC$ (where k is a constant).

(b) A varies directly as B and inversely as C and is denoted by $A \propto \frac{B}{C}$ Or $A = \frac{kB}{C}$ (where k is a constant).

(c) If A varies as B when C is constant, and if A varies as C when B is constant then A varies as BC when B and C both vary.

$$\therefore A \propto BC \text{ Or } A = kBC \text{ (where k is a constant).}$$

Example: Given, a varies as b when c is constant, and as c^2 when b is constant. If a = 770, then b = 15 & c = 7, and when c = 3 & a = 132 find b.

Solution:

As $a \propto bc^2$ or $a = kbc^2$ (where k is a constant)

$$\therefore 770 = k(15)(7)^2$$

$$\Rightarrow k = \frac{22}{21} \text{ i.e. } a = \frac{22}{21} bc^2. \text{ When } c = 3, a = 132.$$

$$\therefore 132 = \frac{22}{21} (b)(3)^2 \quad \therefore b = 14$$

Note:

1. If $A \propto B$ and $B \propto C$, then $A \propto C$
2. If $A \propto C$ and $B \propto C$, then $(A \pm B) \propto C$ $\sqrt{AB} \propto C$
3. If $A \propto BC$, then $\frac{A}{C} \propto B$ and $\frac{A}{B} \propto C$.
4. If $A \propto B$ and $C \propto D$, then $AC \propto BD$.
5. If $A \propto B$, then $A^n \propto B^n$.
6. If $A \propto B$ and $A \propto C$, then $A \propto (B - C)$ and $A \propto (B + C)$
7. If $A \propto B$, then $AP \propto BP$ where P is any quantity, constant or variable.

MIXTURES AND ALLIGATIONS

Simple Mixture : When two different ingredients are mixed together, it is called *simple mixture*.

Compound Mixture : When two or more simple mixtures are mixed together to form another mixture, it is called compound mixture.

Alligation Rule

Alligation rule states that "When different quantities of the same or different ingredients, of different costs (one cheap and other dear) are mixed together to produce a mixture of a mean cost, then ratio of their quantities is inversely proportional to the difference in their cost from the mean cost."

Quantity of Cheap

Quantity of Dear

$$= \frac{\text{Price of Dear} - \text{Mean Price}}{\text{Mean Price} - \text{Price of Cheap}}$$

Application of Alligation Rule :

- (1) To find mean value of a mixture when prices of two or more ingredients, which are mixed together and the proportion in which they are mixed are given.
- (2) To find proportion in which ingredients at given prices must be mixed to produce a mixture at a given price.

Some Important Results :

- (1) When x_1 quantity of ingredient A of cost C_1 and x_2 quantity of ingredient B of cost C_2 are mixed, the cost of the mixture,

$$C_m = \frac{C_1 x_1 + C_2 x_2}{x_1 + x_2}$$

- (2) Similarly, when more than two ingredients are

$$\text{mixed, then } C_m = \frac{C_1x_1 + C_2x_2 + C_3x_3 + \dots C_nx_n}{x_1 + x_2 + x_3 + \dots x_n}$$

- (3) When two mixtures M_1 and M_2 each containing ingredient A and B in the ratio $a : b$ and $x : y$, respectively, are mixed, then proportion of the ingredients A and B i.e., $q_A : q_B$, in the compound mixture is given by.

$$\frac{q_A}{q_B} = \frac{M_1 \times \left(\frac{a}{a+b}\right) + M_2 \times \left(\frac{x}{x+y}\right)}{M_1 \times \left(\frac{b}{a+b}\right) + M_2 \times \left(\frac{y}{x+y}\right)}$$

and, quantity in which M_1 and M_2 is to be mixed when quantity of A and B i.e., q_A and q_B in the compound mixture is known is given by

$$\frac{\text{Quantity of } M_1}{\text{Quantity of } M_2} = \frac{\left(\frac{x}{x+y}\right) - \left(\frac{q_A}{q_A + q_B}\right)}{\left(\frac{q_A}{q_A + q_B}\right) - \left(\frac{a}{a+b}\right)}$$

- (4) When a mixture of three ingredients A, B and C is given, take any two ingredients such that cost of the mixture is between costs of the two chosen ones and find the ratio. Once again, take two more ingredients and find their ratio. Then find combined ratio. This will give infinite number of solutions.
- (5) If a vessel contains 'a' litres of liquid A, and 'b' litres be withdrawn and replaced by liquid B, then if 'b' litres of mixture be withdrawn and replaced by liquid B, and operation repeated 'n' times in all, then

$$\frac{\text{Liquid A left after } n^{\text{th}} \text{ operation}}{\text{Initial quantity of liquid A in vessel}}$$

$$= \frac{\left(\frac{a-b}{a}\right)^n}{1 - \left(\frac{a-b}{a}\right)^n}$$

$$\frac{\text{Liquid A left after } n^{\text{th}} \text{ operation}}{\text{Liquid B left after } n^{\text{th}} \text{ operation}}$$

$$= \frac{\left(\frac{a-b}{a}\right)^n}{1 - \left(\frac{a-b}{a}\right)^n}$$

Example: In what ratio should tea @ 35 per kg be mixed with tea @ 27 per kg so that mixture may cost Rs .30 per kg?

Solution:

$$\frac{\text{Quantity of cheaper}}{\text{Quantity of dearer}} = \frac{35 - 30}{30 - 27} = \frac{5}{3}$$

Hence the two should be mixed in the ratio 3 : 5, the above ratio calculated is 3 : 5 and not 5 : 3, i.e. tea at 35 per kg to have 3 parts and tea at 27 per kg to have 5 parts. This may be checked mentally by the following simple rule: "If mean price is closer to cheaper C.P., then quantity of cheaper will be more and vice versa."

Example: In what ratio should two different types of mixtures containing milk and water in the ratio of 5 : 1 and 2 : 1 respectively be mixed to obtain a final mixture containing milk and water in the ratio 3 : 1?

Solution: Strength of milk in mixture 1 = 5/6 (because there are 5 parts of milk for every 1 part of water. Hence there are 5 parts of milk for a total of 6 parts of the mixture). Strength of milk in mixture 2 = 2/3 (because there are 2 parts of milk for every 1 part of water. Hence there are 2 parts of milk for total of 3 parts of the mixture).

Strength of milk in the final mixture required = 3/4 (because there are 3 parts of milk for every 1 part of water. Hence there are 3 parts of milk for a total of 4 parts of the mixture). As we can see, the mixture 1 is strong in milk as the dearer mixture, mixture 2 as the cheaper mixture and the final mixture as the mean mixture.

$$\frac{\text{Quantity of cheaper}}{\text{Quantity of dearer}} = \frac{\frac{5}{6} - \frac{3}{4}}{\frac{3}{4} - \frac{2}{3}} = \frac{1}{1}$$

Hence, both the types of mixtures will have to be mixed in the ratio of 1:1.

Two mixtures of same ingredients mixed (Compound mixture)

Mixture 1 has ingredients (A and B) in $a : b$.

Mixture 2 has same ingredients (A and B) in $x : y$.

$$\Rightarrow \frac{\text{quantity of ingredient A}}{\text{quantity of ingredient B}} = \frac{X}{Y}$$

Now 'M' unit of Mixture 1 and 'N' units of Mixture 2 are mixed to form a resultant mixture with ingredients (A and B) in the ratio $q_A : q_B$

Case 1:

When q_A and q_B are to be found out. By alligation rule in the resultant mixture,

$$\frac{\text{Quantity of ingredient A}}{\text{Quantity of ingredient B}} = \frac{q_A}{q_B}$$

$$= \frac{M \times \left(\frac{a}{a+b}\right) + N \times \left(\frac{x}{x+y}\right)}{M \times \left(\frac{b}{a+b}\right) + N \times \left(\frac{y}{x+y}\right)}$$

Then, amount of ingredient A in the resultant mixture

$$= \frac{q_A}{(q_A + q_B)} \times (M + N)$$

Amount of ingredient B in the resultant mixture

$$= \frac{q_B}{(q_A + q_B)} \times (M + N)$$

Case 2:

When M and N are to be found out (i.e. amount of mixtures)

Consider the quantity of any ingredients (Say, A) in all the three mixtures, i.e. quantity of ingredients a in

$$\text{Mix 1} = \frac{a}{a + b}, \text{ i.e. } a \text{ out of } (a + b)$$

$$\text{Mix 2} = \frac{x}{x + y}, \text{ in the resultant mix} = \frac{q_A}{q_A + q_B}$$

By alligation rule we get, in the resultant mixture,

$$\frac{\text{Quantity of 1st mixture}}{\text{Quantity of 2nd mixture}} = \frac{Q_1}{Q_2} = \frac{\left(\frac{x}{x + y}\right) - \left(\frac{q_A}{q_A + q_B}\right)}{\left(\frac{q_A}{q_A + q_B}\right) - \left(\frac{a}{a + b}\right)}$$

Then, amount of 1st mixture in the resultant mixture

$$= \frac{Q_1}{(Q_1 + Q_2)} \times (M + N)$$

Amount of 2nd mixture in the resultant mixture

$$= \frac{Q_2}{(Q_1 + Q_2)} \times (M + N)$$

Example: A tea merchant buys two kinds of tea, the price of the first kind being twice that of the second. He sells the mixture at Rs.14/kg there by making a profit of 40%. If the ratio of the first to second kind of tea in the mixture is 2 : 3, then find the cost price of each kind of tea.

Solution:

$$\text{The cost of mixture} = \frac{14 \times 100}{140} = \text{Rs. } 10/\text{kg.}$$

Ratio in which the cheaper and dearer is mixed = 3:2.

Let the price of cheaper tea be Rs. x/kg and dearer tea be Rs. 2x/kg.

∴ Applying the alligation rule, we get

$$\frac{3}{2} = \frac{2x - 10}{10 - x}$$

$$\therefore 30 - 3x = 4x - 20$$

$$\therefore 7x = 50$$

$$\therefore x = \frac{50}{7} \text{ and cost of dearer tea} = \text{Rs. } 14 \frac{2}{7}$$

EXERCISE

MCQ TYPE QUESTIONS

- A cask of wine when fully filled holds 10 litres. 2 litres of wine is removed and filled with water. Then 4 litres in the solution is replaced with water. Then, 6 and 8 litres respectively. At the end of the 4th operation, the ratio of wine to water is

(a) $\frac{4!}{(5)^4}$

(b) $\frac{8!}{(10)^4}$

(c) $\frac{4!}{5^4 - 4!}$

(d) $\frac{8!}{(10)^4 - 8!}$

- A 20 litre vessel is filled with alcohol. Some of the alcohol is poured out into another vessel of an equal capacity, which is then completely filled by adding water. Then mixture obtained is poured into the first vessel to capacity. Then $6\frac{2}{3}$ litres is poured from the first vessel into second. Both vessels now contain an equal amount of alcohol.

How much alcohol was originally poured from the first vessel into the second?

(a) 9 litres

(b) 10 litres

(c) 12 litres

(d) 12.5 litres

- What is the ratio compounded of $x - y : x + y$ and $y^2 + xy : x^2 - xy$?

(a) $y^2 : x^2$

(b) $xy : 1$

(c) $y : x$

(d) $\sqrt{x} : \sqrt{y}$

- If ratio of two natural numbers x and y is 'a' and that of y and x is 'b', then value of 'a + b' is

(a) greater than 2

(b) lesser than 1

(c) greater than 1

(d) data insufficient.

- The charges of a goldsmith is partly fixed and partly variable with the amount in grams of gold. The charge is ₹ 300 for 20 grams of gold and ₹ 550 for 45 grams of gold. Find the charge of work for 100 grams of gold.

- (a) ₹ 900 (b) ₹ 1000
(c) ₹ 1100 (d) ₹ 9000

Directions (Q.6–7) : Refer to the data below and answer the questions that follow.

The ratio of white collar to blue collar employees in a firm is 8 : 3. Also ratio of male employees to female employees is 7 : 4. It is observed that 60% of the white collar employees are males.

6. What is the ratio of female white collar employees to male blue collar employees?
(a) 16 : 11 (b) 24 : 11
(c) 3 : 1 (d) Data insufficient
7. If there are 48 female blue collar workers, what is the difference between male white collar employees and female blue collar employees?
(a) 288 (b) 240
(c) 220 (d) Data insufficient
8. In a chemical experiment, two NaOH solution bottles are used. Bottle A contains salt and water in the ratio 7 : 3 and bottle B contains salt and water in the ratio 4 : 3. In what proportion should the quantities be taken from A and B to give the 2 : 1 NaOH solution?
(a) 2 : 1 (b) 10 : 7
(c) 20 : 7 (d) 1 : 2
9. A triswitch mixer grinder operates in three stages of speed in revolutions per min i.e., rpm. The speed of the motor with some load in the second stage varies as a square of the speed in stage 1 and that in stage 3 varies as a cube of that in stage 2. If mixer motor makes 50 revolutions in 3 minutes in stage 1, its speed in stage 2 and 3 with the same load is 2500 and 12500 rpm respectively. What will be the speed of motor (in rpm) in stage 3, if at the same load motors speed in rpm is 25 revolutions in 6 minutes in stage 1?
(a) $\frac{3125}{16}$ rpm (b) $\frac{5^5}{2^{10}}$ rpm
(c) $\frac{5^{10}}{2^5}$ rpm (d) None of these
10. Two bags have certain number of mangoes. If half the mangoes from bag 1 are transferred to bag 2, the ratio of mangoes in bag 1 to bag 2 gets reversed. Now one-third and two-third of the mangoes from the first and the second bag are

transferred to the second and first bag respectively. The ratio of mangoes in bag 1 to bag 2 now is

- (a) 2 : 1 (b) 2 : 2
(c) 4 : 5 (d) 5 : 4

NUMERICAL TYPE QUESTIONS

- Five litres are drawn from a cask full of wine and it is then filled with water. Five litres of the mixture are drawn and the cask is again filled with water. The quantity of wine now left in the cask to that of the water in it is in the ratio 361 : 39. The cask hold _____ litres.
- In a mixture of wheat and barley, wheat is 60%. To 400 kg of the mixture a quantity of barley is added and then the wheat is $53\frac{1}{3}\%$. The _____ kg of barley are added?
- Two jars of capacity 3 and 5 litres are filled with mixtures of alcohol and water. In the smaller jar 25% of the mixture is alcohol and in the larger 25% of the mixture is water. The jars are emptied into a 9 litre cask and remaining volume is filled up with water. Then percentage of alcohol in the cask will be _____
- A man buys milk at ₹ 5 a litre and after adding water, sells it at ₹ 6 a litre, thereby making a profit of $33\frac{1}{3}\%$. The proportion of water to milk in the mixture is _____
- The percentage of alcohol in two wine tanks is 20 and 40 respectively and rest is grape juice. If a wine glass is filled to its half by the tank 1 and 2 in the ratio 2 : 3 and the rest is filled again by pure alcohol, then the ratio of the grape juice to alcohol in the wine glass is _____
- In _____ ratio must a grocer mix two varieties of tea worth ₹ 60 a kg and ₹ 65 a kg so that by selling the mixture at ₹ 68.20 a kg he may gain 10%?
- A container contains 40 lit of milk. From this container 4. lit of milk was taken out and replaced by water. This process was repeated further two times. The _____ litres of milk is now contained by the container?
- A merchant has 1000kg of sugar, part of which he sells at 8% profit and the rest at 18% profit. He gains 14% on the whole. The quantity sold at 18% profit is _____ kg

9. Volumes in the proportion 3 : 4 : 7 of three different substances are mixed together. The densities of equal volume of the substances are in the ratio 5 : 2 : 6 respectively. The weight of the third substance contained in 130 lbs. of the mixture is _____ lbs
10. In a bag, there are 50 paise coins, 25 paise coins and rupee coins, proportional to the numbers 3, 4 and $2\frac{1}{2}$. If they amount to ₹ 210, then number of rupee coins is _____

ANSWERS

MCQ Type Questions

1. (c) 2. (b) 3. (c) 4. (c) 5. (c) 6. (a) 7. (b) 8. (c) 9. (b) 10. (a)

Numerical Type Questions

1. 100 2. 50 3. 50 4. 1 : 9 5. 17 : 33 6. 3 : 2 7. 29.16 8. 600 9. 84 10. 105

EXPLANATIONS

MCQ TYPE QUESTIONS

1. Stage	Wine left Original quantity
0	$\frac{10}{10}$
1	$\frac{8}{10}$
2	$\frac{8}{10} \times \frac{6}{10}$
3	$\frac{8}{10} \times \frac{6}{10} \times \frac{4}{10}$
4	$\frac{8}{10} \times \frac{6}{10} \times \frac{4}{10} \times \frac{2}{10} \Rightarrow \frac{4 \times 3 \times 2 \times 1}{5^4} = \frac{4!}{5^4}$
$\therefore$	$\frac{\text{Wine}}{\text{Water}} = \frac{\frac{4!}{5^4}}{1 - \frac{4!}{5^4}} = \frac{4!}{5^4 - 4!}$

2.		
	(1)	(2)

	C_1	C_2
Operation 1	20 litres alcohol	Empty
Operation 2	$(20 - x)$ alcohol	x litres alcohol $(20 - x)$ litres water
Operation 3	$(20 - x)$ litres alcohol + x litres mixture i.e., $\left(\frac{x^2}{20}\right)$ litres alcohol	$(20 - x)$ litres mixture

Now $\frac{1}{3}$ rd of C_1 poured into C_2

Quantity of alcohol in $C_1 = C_2 = 10$ litres

$$\text{In } C_1 : \quad \frac{2}{3} \left[20 - x + (x) \left(\frac{x}{20} \right) \right] = 10$$

Solving we get, $x = 10$ litres

Hence 10 litres of alcohol was initially poured from C_1 to C_2 .

$$3. \text{ Required ratio} = \frac{x-y}{x+y} \times \frac{y^2+xy}{x^2-xy} = \frac{y}{x} = y : x$$

$$4. \text{ Given : } \frac{x}{y} = a \text{ and } \frac{y}{x} = b$$

$$\text{If } \frac{x}{y} > 1, \text{ then } \frac{y}{x} < 1,$$

$$\Rightarrow \frac{x}{y} + \frac{y}{x} > 1.$$

Hence option (c) is correct.

5. Total charge = fixed + variable

$$T_c = m + kx$$

where m and k are constants.

$$\therefore 300 = m + k \times 20$$

$$\text{Also, } 550 = m + k \times 45$$

Solving for m and k , we get

$$m = 100; k = 10.$$

Hence, for 100 grams,

$$\text{total charge} = 100 + 10 \times 100 = ₹ 1100.$$

Q. 6 and 7.

Using given data, we can draw following table

	Male	Female	
White Collar	4.8x	3.2x	8x
Blue Collar	2.2x	0.8x	3x
	7x	4x	11x

Now, all the questions can be answered.

$$6. \frac{\text{Female white collar employess}}{\text{Male blue collar employees}} = \frac{3.2x}{2.2x} = \frac{16}{11}$$

$$7. \text{ Since } 0.8x = 48 \\ \Rightarrow x = 60$$

$$\therefore \text{ Required difference} = 4.8x - 0.8x \\ = 4.8 \times 60 - 0.8x = 240$$

8. Let $10x$ be taken from bottle A and $7y$ be taken from bottle B.

$$\text{Total salt taken} = \left(\frac{7}{10} \times 10x\right) + \left(\frac{4}{7} \times 7y\right) \\ = 7x + 4y$$

$$\text{Total water taken} = 3x + 3y.$$

$$\therefore \frac{7x + 4y}{3x + 3y} = \frac{2}{1}$$

$$\Rightarrow 7x + 4y = 6x + 6y$$

$$\Rightarrow x = 2y$$

$$\therefore \frac{x}{y} = \frac{2}{1}$$

$$\therefore \text{ Ratio of quantities} = \frac{10x}{7y}$$

$$= \frac{10}{7} \times \frac{2}{1} = \frac{20}{7}$$

9. Let speed of the motor in stage 1 be x rpm. Then its speed in stage 2 be $(x^2) = ax^2$ (say) and in stage 3 be $(ax^2)^3 = b.a^3x^6$ (say)

$$\text{Given that } x = \frac{50}{3} \quad \dots(i)$$

$$ax^2 = 2500 \quad \dots(ii)$$

$$ba^3x^6 = 12500 \quad \dots(iii)$$

Substituting value of x in (ii), we get

$$a = \frac{2500}{(50)^2} \times 3^2 = 9$$

Again, substituting values of a and x , we get

$$b = \frac{12500}{(2500)^3} = \frac{1}{125} \times 10^{-4}$$

$$\text{Given : } x = \frac{25}{6} \text{ rpm}$$

$$\therefore \text{ Speed in stage 3} = \frac{10^{-4}}{125} \times (9)^3 \times \left(\frac{25}{6}\right)^6 \\ = \frac{10^{-4}}{5 \times 25} \times (3)^6 \times \frac{(25 \times 5) \times (5 \times 25^4)}{(2)^6 \cdot (3)^6} \\ = \frac{5 \times 25^4}{2^6 \times 10^4} = \frac{5^5}{2^{10}} \text{ rpm}$$

10. Let bag 1 and 2 have

x mangoes y mangoes

$$\text{Step 1} \quad \frac{x}{2} \quad y + \frac{x}{2}$$

$$\text{But } \frac{x}{2} : y + \frac{x}{2} = y : x$$

$$\text{Step 2} \quad \frac{2}{3}y + \frac{2}{3}x \quad \frac{1}{3}y + \frac{1}{3}x$$

$$\therefore \text{ Required ratio} = \frac{\frac{2}{3}y + \frac{2}{3}x}{\frac{1}{3}y + \frac{1}{3}x} = \frac{2\left(\frac{x+y}{3}\right)}{1\left(\frac{x+y}{3}\right)} \\ = 2 : 1$$

NUMERICAL TYPE QUESTIONS

1. Let quantity of wine be a , i.e., capacity of the cask be ' a '

$$\text{i.e., } \frac{\text{Quantity of wine}}{\text{Quantity of water}} = \frac{\left(1 - \frac{5}{a}\right)^2}{1 - \left(1 - \frac{5}{a}\right)^2}$$

$$\therefore \frac{361}{39} = \frac{\left(1 - \frac{5}{a}\right)^2}{1 - \left(1 - \frac{5}{a}\right)^2}$$

$$\Rightarrow \frac{361}{39} - \frac{361}{39} \left(1 - \frac{5}{a}\right)^2 = \left(1 - \frac{5}{a}\right)^2$$

$$\Rightarrow \left(1 - \frac{5}{a}\right) = \sqrt{\frac{361}{400}} = \frac{19}{20}$$

$$\Rightarrow \frac{5}{a} = \frac{1}{20}$$

$$\therefore a = 100 \text{ litres}$$

2. Quantity of wheat in 400 kg of mixture

$$= 0.6 \times 400 = 240 \text{ kg}$$

$$\text{Quantity of barley} = 400 - 240 = 160 \text{ kg}$$

Let x kg of barley be added to 400 kg of the mixture

$$\therefore \frac{240}{160 + x} = \frac{53\frac{1}{3}}{46\frac{2}{3}} = \frac{160}{140} = \frac{8}{7}$$

$$\Rightarrow 240 \times 7 = 8 \times 160 + 8x$$

$$\Rightarrow x = 50 \text{ kg}$$

3. Alcohol content in 1st jar = $0.25 \times 3 = 0.75$ litre

$$\text{Alcohol content in 2nd jar} = 0.75 \times 5 = 3.75 \text{ litres}$$

$$\therefore \text{ Total alcohol in 9 litre cask} = 0.75 + 3.75 \\ = 4.5 \text{ litres}$$

∴ Percentage of alcohol in the cask

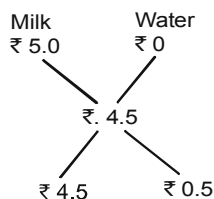
$$= \frac{4.5}{9} \times 100 = 50\%$$

4. Cost price of milk = ₹ 5/litre

$$\text{Cost price of mixture} = \frac{100}{133\frac{1}{3}} \times 6 = ₹ 4.5/\text{litre}$$

$$\therefore \frac{\text{Quantity of water}}{\text{Quantity of milk}} = \frac{5 - 4.5}{4.5 - 0} = \frac{1}{9}$$

i.e.,



Milk : Water = 9 : 1

5.

	Alcohol	Grape juice	Alcohol : Grape
Tank 1	20%	80%	1 : 4
Tank 2	40%	60%	2 : 3

Ratio of grape juice to alcohol in half filled wine glass

$$= \frac{2 \times \frac{4}{5} + 3 \times \frac{3}{5}}{2 \times \frac{1}{5} + 3 \times \frac{2}{5}} = \frac{17}{8}$$

So, half capacity of glass = 17 + 8 = 25

So, full capacity = 25 × 2 = 50

Now rest 25 parts has to be filled with alcohol.

So, ratio of the grape juice to alcohol = 17 : (8 + 25) = 17 : 33.

Alternatively,

As the wine glass is filled to half by taking two parts from tank 1 and three parts from tank 2 and second half is equivalent to five parts which is pure alcohol,

percentage concentration of grape juice

$$= \frac{2 \times 80 + 3 \times 60 + 5 \times 10}{2 + 3 + 5} = 34\%$$

Hence ratio of grape juice to alcohol

$$= 34 : 66 = 17 : 33.$$

6. S.P of 1 kg of mixture = ₹ 68.20, Gain = 10%

$$\text{S.P of 1 kg of mixture} = ₹ \left(\frac{100}{110} \times 68.2 \right) = ₹ 62.$$

Cost of 1 kg tea of

1st kind

60

Cost of 1 kg tea of

2nd kind

65

Mean Price ₹ 62

3

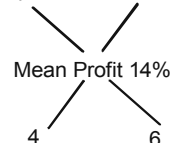
2

∴ Required ratio = 3 : 2.

7. Amount of milk left after 3 operations

$$\begin{aligned} &= 40 \left(1 - \frac{4}{40} \right)^3 \\ &= 40 \times \frac{9}{10} \times \frac{9}{10} \times \frac{9}{10} \\ &= 29.16 \text{ lit} \end{aligned}$$

8. Profit on 1st part 8% Profit on 2nd part 18%



Ratio of 1st and 2nd parts = 4 : 6 = 2 : 3.

Quantity of 2nd part = $\frac{3}{5} \times 1000 \text{ kg} = 600 \text{ kg}.$

9. Ratio of volumes = $\frac{3}{14} : \frac{4}{14} : \frac{7}{14}$

Ratio of densities = $\frac{5}{13} : \frac{2}{13} : \frac{6}{13}$

$$\begin{aligned} \therefore \text{Ratio of weights} &= \frac{3}{14} \times \frac{5}{13} : \frac{4}{14} \times \frac{2}{13} : \frac{7}{14} \times \frac{6}{13} \\ &= 15 : 8 : 42 \end{aligned}$$

... (Mass = Volume × Density)

Weight of the third substance = $\frac{42}{65} \times 130 = 84 \text{ lbs.}$

10. Let number of 50p coins be 3x, number of 25p coins be 4x and number of rupee coins be $2\frac{1}{2}x.$

$$\therefore 2\frac{1}{2}x + \frac{1}{2} \times 3x + \frac{1}{4} \times 4x = 210$$

$$\Rightarrow \frac{5}{2}x + \frac{3}{2}x + x = 210$$

$$\Rightarrow 5x = 210$$

$$\Rightarrow x = 42$$

$$\therefore \text{Number of rupee coins} = \frac{5}{2} \times 42 = 105.$$

$$\text{Number of goats} = 416 \times \frac{13}{32} = 169$$

$$\therefore \text{Total number of hens} = 65 + 65 = 130$$

$$\text{Total number of ducks} = 91 + 182 = 273$$

$$\text{Total number of goats} = 169 + 169 = 338$$

$$\therefore \text{The new ratio of hens: ducks: goats} = 130: 273: 338 = 10: 21: 26.$$

5 CHAPTER

Permutations-Combinations, Elementary Statistics and Probability

PERMUTATIONS-COMBINATION

Permutations

Each of the arrangements which can be made by taking some or all of a number of items is called a *Permutation*. Permutation implies “*arrangement*” or that “*order of the items*” is important.

Permutations of three items a, b and c taken two at a time are ab, ba, ac, ca, cb and bc. Since the order in which items are taken is important, ab and ba are counted as two different permutations. The words “permutation” and “arrangement” are synonymous and can be used interchangeably.”

Number of permutations of n things taking r at a time is denoted by ${}^n P_r$ (and read as “nPr”)

Combinations

Each of the groups or selections which can be made by taking some or all of a number of items is called *Combination*. In combinations, order in which the items are taken is not considered as long as the specific things are included.

Combinations of three items a, b and c taken two at a time are ab, bc and ca. Here, ab and ba are not considered separately because the order in which a and b are taken is not important but it is only required that a combination including a and b is what is to be counted. Words “*combination*” and “*selection*” are synonymous.

Number of combinations of n things taking r at a time is denoted by ${}^n C_r$ (and read as “nCr”)

$${}^n P_r = \frac{n!}{(n-r)!}, \quad {}^n C_r = \frac{n!}{r!(n-r)!}$$

Number of ways in which n things may be arranged taking them all at a time, when p of the things are exactly alike of one kind, q of them exactly alike of another kind, r of them exactly alike of a third kind,

and the rest all distinct is $\frac{n!}{p!q!r!}$

Number of ways of selecting one or more items from n given items = $2^n - 1$

Number of ways of dividing (p + q) items into two

groups of p and q items respectively is $\frac{(p+q)!}{p!q!}$.

Number of ways of dividing 2p items into two equal

groups of p each is $\frac{(2p)!}{(p!)^2}$ where two groups have

distinct identity.

Number of ways of dividing 2p items into two equal

groups of p each is $\frac{(2p)!}{2!(p!)^2}$ where two groups do not

have distinct identity.

Number of ways in which (p + q + r) things can be divided into three groups containing p, q and r things

respectively is $\frac{(p+q+r)!}{p!q!r!}$

Number of circular arrangements of n distinct items is (n – 1)!. If there is a DIFFERENCE between clockwise and anticlockwise arrangements and (n – 1)!/2 if there is NO DIFFERENCE between clockwise and anticlockwise arrangements.

For $x_1 + x_2 + x_3 + \dots + x_n = s$ where $s \geq 0$, number of positive integral solutions (when $s \geq n$) is ${}^{s-1}C_{n-1}$ and number of non-negative integral solution is ${}^{n+s-1}C_{n-1}$.

NOTES:

The Number of Diagonals in an N-Sided Regular Polygon

An n-sided regular polygon has n vertices. Joining any two vertices we get a line of the polygon which are ${}^n C_2$ lines, n of them are sides. Hence diagonals

$$\text{are } {}^n C_2 = \frac{n(n-3)}{2}$$

The following points are useful in solving the problems.

- 1) ${}^n C_0 = {}^n C_n = 1$
- 2) ${}^n C_1 = {}^n C_{n-1} = n$
- 3) ${}^n C_r = {}^n C_{n-r}$
- 4) if ${}^n C_r = {}^n C_s$ then $r = s$ or $n = r + s$
- 5) ${}^n C_r + {}^n C_{r-1} = {}^{n+1} C_r$

Example:

Find the number of 4 –digit numbers that can be formed using the digits 0, 2, 5, 7, 9, 6 where a digit can be repeated more than once.

Solution:

Among the given digits 0, 2, 5, 7, 9, 6 zero cannot be in thousands place 2, 5, 7, 9, 6 are to be used for the thousands place of a four digit number, which can be done in 5 ways.

Since the repetition of digits is allowed, for each of the hundreds, tens and the units place, all the given digits can be used which can be done in $6(6)(6)$ or 216 ways. $\therefore$ The total number of 4 digit numbers that can be formed is $5(216)$ or 1080

Example:

If ${}^nP_4 = 7920$, then find nC_4

Solution:

$$\begin{aligned} {}^nP_4 &= n(n-4)(n-2)(n-3) \\ &= 10(4)(198) \\ &= 8(9)(10)(11) \end{aligned}$$

$$\therefore n = 11 \therefore {}^nP_4 = {}^{11}C_4 = {}^nP_4 \left(\frac{1}{4!} \right)$$

$$= \frac{1}{4!} 7920 = 330$$

Example:

In how many ways can the letters of the word RECRUDESCENCE be arranged?

Solution:

The given word contains 4E's, 2R's, 3C's, 1U, 1D, 1S, 1N. Here the total numbers of letters = 13

$$\therefore \text{Number of arrangements} = \frac{13!}{1!2!3!}$$

Example:

Find the number of ways in which six boys and five girls can be seated in a row such that each girls has a boy sitting on either side.

Solution:

There are 6 boys and 5 girls

Possible seating arrangement : BGBGBGBGBGB

$$\therefore \text{Total number of sitting arrangements} = 5!6!$$

Example:

Find the number of ways of drawing four cards, all of different suits from pack of 52 playing cards. (Diamonds, spades, hearts and clubs are the suits in a pack)

Solution :

A pack of 52 cards contains 4 different suits. Number of ways of drawing four cards each $= {}^{13}C_1 \cdot {}^{13}C_1 \cdot {}^{13}C_1 \cdot {}^{13}C_1 = 13^4$

ELEMENTARY STATISTICS

If you find 'elementary' statistical formulae meaningless, or merely wish to refresh our understanding,

The term 'statistic' can refer to several rather different things.

A statistic summarizes or represents a set of information - most commonly as a single number. The term statistic is used both for the value and for the mathematical 'function' (usually an equation) used to obtain that value.

Many functions are available to summarize information. For example, a salesman could equally truthfully provide the most typical cost as 'on average' or give the maximum ('up to...') or the minimum ('from...') just \$ 300. The 'average', 'maximum' and 'minimum' are all statistics.

Note, summary statistics of a sample are often used as estimates for the population at large - for instance when you are told 'the average man has 1.8 children' that result was found in a sample of men - it is usually impossible to check every man.

Statistics are also used to describe how sets of results varied, or to infer how they are liable to vary, or to infer how their summary statistic might be expected to vary. Inferential statistics are variously used to indicate how reliable an outcome is, or the probability it occurred by simple chance - given a simple (hopefully plausible) set of assumptions.

When used in the plural, 'statistics' also describes 'the study of the collection, organization, analysis, interpretation and presentation of data'.

Humans, of course, use non-numerical summaries all of the time. For example when you say 'cats are smaller than dogs' you are probably describing the average situation - however some people assume you mean every cat is smaller than every dog.

Humans also use non-numerical estimates of probability, using a simple scale, ranging from impossible to certain. Research shows most people divide that scale into surprisingly few levels - seldom more than 7 - and have problems in dealing with very small probabilities.

Statistics is a mathematical science including methods of collecting, organizing and analyzing data in such a way that meaningful conclusions can be drawn from them. In general, its investigations and analyses fall into two broad categories called descriptive and inferential statistics.

Descriptive statistics deals with the processing of data without attempting to draw any inferences from it. The data are presented in the form of tables and graphs. The characteristics of the data are described in simple terms. Events that are dealt with include everyday happenings such as accidents, prices of goods, business, incomes, epidemics, sports data, population data.

Inferential statistics is a scientific discipline that uses mathematical tools to make forecasts and projections by analyzing the given data.

Measures of Central Tendency in Elementary Statistics

Average of a distribution is the value of the variable which is representative of the entire distribution.

The following are the measures of central tendency.

- (i) Arithmetic mean
- (ii) Geometric mean
- (iii) Harmonic Mean
- (iv) Median
- (v) Mode

(i) **Arithmetic Mean:** If $x_1, x_2, \dots, x_n$ are n values of variable X , then the arithmetic mean of these values is given by

$$\bar{X} = \frac{1}{n}(x_1 + x_2 + \dots + x_n) = \frac{1}{n} \left[\sum_{i=1}^n x_i \right]$$

In case of a frequency distribution $n_i/f_i, i = 1, 2, \dots, n$ where f_i is the frequency of the variable x_i :

$$\bar{X} = \frac{f_1 x_1 + f_2 x_2 + \dots + f_n x_n}{f_1 + f_2 + \dots + f_n} = \frac{1}{N} \left(\sum_{i=1}^n f_i x_i \right) \dots (i)$$

$$\text{where } N = f_1 + f_2 + \dots + f_n = \sum_{i=1}^n f_i$$

$$\bar{X} = A + \frac{1}{N} \sum_{i=1}^n f_i d_i$$

e.g., Find arithmetic mean of 10, 15, 17, 20.

$$x = \frac{10 + 15 + 17 + 20}{4} = 15.5$$

When large numbers are involved, an Arithmetic Mean is generally assumed to facilitate calculation. If a is assumed Arithmetic mean, then true Arithmetic mean $\bar{x}$ of group of n numbers $x_1, x_2, x_3, \dots, x_n$ is

$$\bar{x} = a + \frac{\sum_{i=1}^n d_i}{n},$$

where $d_i = x_i - a$ is deviation from the assumed mean. e.g., Find the Arithmetic mean of 42, 46, 48, 50, 55, 60, 70

Let assumed mean be 50

x_i	d_i
42	$42 - 50 = -8$
46	$46 - 50 = -4$
48	$48 - 50 = -2$
50	$50 - 50 = 0$
55	$55 - 50 = 5$
60	$60 - 50 = 10$
70	$70 - 50 = 20$

$$\therefore \bar{x} = 50 + \frac{\sum_{i=1}^7 d_i}{7} = 50 + \frac{21}{7} = 53.$$

Weighted Arithmetic Mean: If values $x_1, x_2, \dots, x_n$ are assigned weights $w_1, w_2, \dots, w_n$ respectively, then

$$\text{weighted arithmetic mean, } x_w = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i}$$

The term 'weight' stands for relative importance that is attached to the different values.

'w' can also be replaced by 'f' in a frequency distribution where 'f' denotes frequency or simply the number of times a particular observation occurs.

e.g.,

x_i	w_i	$w_i x_i$
20	3	60
30	1	30
40	5	200
50	1	50

$$\Sigma w_i = 10; \Sigma x_i w_i = 340$$

$$\therefore \bar{x}_w = \frac{\sum_{i=1}^4 w_i x_i}{\sum_{i=1}^4 w_i} = \frac{340}{10} = 34$$

When two or more groups are combined, then combined average of the groups is

$$x_c = \frac{x_1 n_1 + x_2 n_2 + x_3 n_3 + \dots + x_n n_n}{n_1 + n_2 + n_3 + \dots + n_n}$$

where $x_1, x_2, x_3, \dots, x_n$ individual averages of the groups and $n_1, n_2, \dots, n_n$ are number of observations or values in each of the groups.

e.g., Mean marks scored by 50 boys are 70 and those scored by 100 girls are 55. Find average marks scored by this group.

$$x_c = \frac{n_1 x_1 + n_2 x_2}{n_1 + n_2} = \frac{70 \times 50 + 100 \times 55}{150} = \frac{9000}{150} = 60$$

Properties of Arithmetic Mean

- (i) Algebraic sum of the deviations of a set of values from their arithmetic mean is zero.

$$\begin{aligned}\sum_{i=1}^n f_i (x_i - \bar{X}) &= \sum_{i=1}^n f_i x_i - \bar{X} \sum_{i=1}^n f_i \\ &= N\bar{X} - N\bar{X} \left[\because \bar{X} = \frac{1}{N} \sum_{i=1}^n f_i x_i \right] = 0\end{aligned}$$

- (ii) The sum of the squares of the deviations of a set of values is minimum when taken about mean.

- (iii) Mean of the composite series: If $\bar{X}_1, \bar{X}_2, \dots, \bar{X}_K$ are the mean of K series of sizes $n_1, n_2, \dots, n_K$ respectively then the mean $\bar{X}$ of the composite series is given by

$$\bar{X} = \frac{n_1 \bar{X}_1 + n_2 \bar{X}_2 + \dots + n_K \bar{X}_K}{n_1 + n_2 + \dots + n_K}.$$

- (ii) **Geometric Mean:** Geometric mean of a set of n observations is the n^{th} root of their product.

$$G = \text{antilog} \left(\frac{1}{N} \sum_{i=1}^n f_i \log x_i \right)$$

e.g., Geometric mean of 2, 3 and 4 is $\sqrt[3]{2 \times 3 \times 4}$
 $= \sqrt[3]{24} = 2. \sqrt[3]{3}$

Geometric mean of a and b is $\sqrt{ab}$

e.g., Find Geometric mean of 9 and 4.

$$\text{G.M.} = \sqrt{9 \times 4} = 6$$

Geometric mean is generally used to calculate rate of growth.

If geometric mean of one group of 'a' numbers is 'x' and that of another group of 'b' numbers is 'y', then

Geometric mean of the combined groups

$$= (x_a + y_b) \frac{1}{a + b}$$

Propertie of Geometric Mean

If G_1 and G_2 are the geometric means of two series of sizes n_1 and n_2 respectively, then the geometric mean G of the combined series is given by

$$\log G = \frac{n_1 \log G_1 + n_2 \log G_2}{n_1 + n_2}$$

- (iii) **Harmonic Mean:** Harmonic mean of a observations is the reciprocal of the AM of the reciprocals of the given values. Thus, harmonic mean H of n non-zero observations $x_1, x_2, \dots, x_n$ is

$$H = \frac{1}{\frac{1}{n} \sum_{i=1}^n \left(\frac{1}{x_i} \right)} \quad \dots(i)$$

In case of a frequency distribution

$x_i | f_i \quad i = 1, 2, \dots, n$

$$H = \frac{1}{\frac{1}{N} \sum_{i=1}^n \left(\frac{f_i}{x_i} \right)} \quad \dots(ii)$$

e.g., Find Harmonic mean of 1, 0.5, 10.

x_i	$\frac{1}{x_i}$
1	1
0.5	2
10	0.1

$$\sum \frac{1}{x_i} = 3.1$$

$$\text{H.M.} = \frac{3}{3.1} = \frac{30}{31}$$

This is an important result and is generally used to find average speed when equal distances are covered at different speeds. (when different distances are covered in the same time, arithmetic mean is used to determine the average speed).

e.g., A person travels from A to B at 6 kmph and returns at 4 kmph, find his average speed.

Average Speed = Harmonic Mean of 6 and 4

$$= \frac{2 \times 6 \times 4}{6 + 4} = \frac{48}{10} = 4.8 \text{ kmph}$$

Note: This concept will again be covered in time, speed and distance.

Weighted Harmonic Mean: Weighted Harmonic Mean of 3 numbers x_1, x_2 and x_3 with

weights w_1, w_2 and w_3 respectively is $\frac{\sum_{i=1}^3 \frac{w_i}{x_i}}{\sum_{i=1}^3 \frac{w_i}{x_i}}$

This is used to calculate average speed when different distances are covered at different speeds.

e.g., A cyclist covers his first 5 km at an average speed of 10 kmph, another 3 km at 8 kmph and last 2 km at 5 kmph. Find his average speed during the journey.

Average Speed = Weighted HM

$$= \frac{5 + 3 + 2}{\frac{5}{10} + \frac{3}{8} + \frac{2}{5}} = 7.84 \text{ kmph}$$

- (iv) **Median and Other Partition Values**

Median of a distribution is the value of the variable which divides it into equal parts.

$$\text{Median} = l + \left(\frac{\frac{N}{2} - F}{f} \right) h$$

where l = lower limit of the median class

f = frequency of the median class

h = size (width) of the median class

F = c.f. of the class preceding the median class.

1. $N = \sum_{i=1}^n f_i$
2. As median divides a distribution into two equal parts, similarly the quantities, deciles and percentiles divide the distribution respectively into 4, 5, 10 and 100 equal parts the i^{th} quartile is given by.

$$Q_1 = l + \left(\frac{\frac{iN}{4} - F}{f} \right) h, i = 1, 2, 3$$

3. Q_1 is the lower quartile, Q_2 is the median and Q_3 is called the upper quartile.

The i^{th} decile is given by

$$D_i = l + \left(\frac{\frac{iN}{10} - F}{f} \right) h, i = 1, 2, \dots, 9$$

D_5 is the median.

e.g., Find median of the numbers 40, 41, 33, 48, 41, 39 and 38.

- (v) **Mode :** The mode or modal value of a distribution is that value of the variable for which the frequency is maximum.

In case of a discrete frequency distribution the value of the mode is determined by the method of grouping.

In case of a grouped or continuous frequency distribution, mode is given by the formula.

$$\text{Mode} = l + \frac{f - f_1}{2f - f_1 - f_2} \times h$$

where l = lower limit of the modal class.

h = width of the modal class.

f_1 = frequency of the class proceeding the modal class

f_2 = frequency of the class following the modal class.

f = frequency of modal class.

e.g., Find mode of the set of numbers 2, 4, 6, 6, 5, 4, 3, 6, 2, 6, 4, 1, 4, 5, 4.

Relation between A.M., G.M. and H.M.

1. Consider two number a and b .

$$\text{Arithmetic Mean} = \frac{a+b}{2}; \text{Geometric mean}$$

$$= \sqrt{ab} \quad \text{Harmonic mean} = \frac{2ab}{a+b}$$

$$\therefore (GM)^2 = AM \times HM$$

e.g., If Arithmetic mean of two numbers is 6.5 and their Geometric mean is 6, what is their Harmonic Mean?

$$HM = \frac{(G.M.)^2}{A.M.} = \frac{6 \times 6}{6.5} = 5.54$$

2. Arithmetic Mean > Geometric Mean > Harmonic Mean

e.g., Arithmetic Mean of 2 and 4 is 3.

Geometric Mean of 2 and 4 is 2.82.

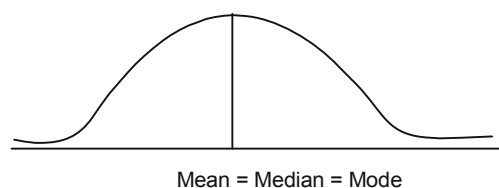
Harmonic Mean of 2 and 4 is 2.66

$$\therefore 3 > 2.82 > 2.66$$

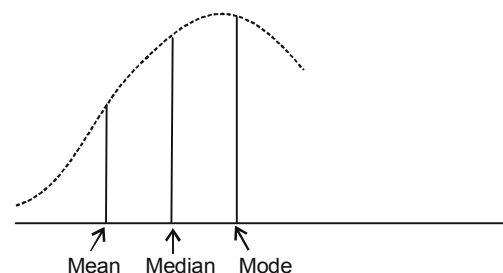
Symmetric Distribution

A distribution is a symmetric distribution if the value of mean, mode and median coincide. In a symmetric distribution frequencies are symmetrically distributed on both side of the centre be point of the frequency curve.

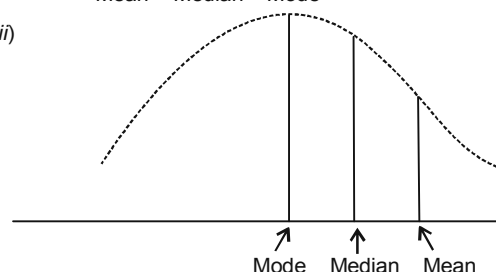
(i)



(ii)



(iii)



We have the following empirical relation between them:

$$\begin{aligned} \text{Mean} - \text{Mode} &= 3(\text{Mean} - \text{Median}) \\ \Rightarrow \text{Mode} &= 3 \text{ Median} - 2 \text{ Mean} \end{aligned}$$

Measure of Dispersion

The measure of dispersion commonly used are:

- (i) Range
- (ii) Quartile deviation or the semi-interquartile range
- (iii) Mean-deviation and
- (iv) standard deviation.

(i) **Range:** If A and B are the greatest and the smallest observations respectively in a distribution, then its range = A – B.

The coefficient of range (or scatter) = $\frac{A - B}{A + B}$.

(ii) **Quartile Deviation or semi-interquartile range (Q.D.):** If Q_1 and Q_3 be the lower and upper quartiles.

Then quartile deviation $Q = \frac{1}{2}(Q_3 - Q_1)$

coefficient of quartile deviation

$$= \frac{(Q_3 - Q_1)/2}{(Q_3 + Q_1)/2} = \frac{Q_3 - Q_1}{Q_3 + Q_1}.$$

(iii) **Mean Deviation (M.D.):** If x_i/f_i , $i = 1, 2, \dots, n$ is the frequency distribution, then mean deviation from an average A (median, modern, mode, or Am)

is given by $M.D = \frac{1}{N} \sum_{i=1}^n f_i |x_i - A|$, where $\sum_{i=1}^n f_i = N$.

Mean coefficient of dispersion

$$= \frac{\text{Mean deviation from the mean}}{\text{Mean}}$$

Median coefficient of dispersion

$$= \frac{\text{Mean deviation from the median}}{\text{Median}}$$

Mode coefficient of dispersion

$$= \frac{\text{Mean deviation from the mode}}{\text{Mode}}.$$

(iv) **Standard Deviation (S.D) and Variance**

The standard deviation of variation of variate X is the square root of the AM of the square of all deviation of X from the AM of the observation and it is denoted by the Greek letter small sigma (σ).

Thus, if x_i/f_i , $i = 1, 2, \dots, n$ is the frequency distribution, then,

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^n f_i (X_i - \bar{X})^2} \quad \dots(i)$$

Where $\bar{X}$ is the A.M of the distribution and

$$N = \sum_{i=1}^n f_i.$$

Variance of the Combined Series

If n_1, n_2 are the sizes, $\bar{X}_1, \bar{X}_2$ the means and σ_1, σ_2 the standard deviation of two series, is given by

$$\sigma^2 = \frac{1}{n_1 + n_2} [n_1 (\sigma_1^2 + d_1^2) + n_2 (\sigma_2^2 + d_2^2)]$$

where $d_1 = \bar{X}_1 - \bar{X}$, $d_2 = \bar{X}_2 - \bar{X}$ and $\bar{X} = \frac{n_1 \bar{X}_1 + n_2 \bar{X}_2}{n_1 + n_2}$

Skewness: As we have seen that a distribution is skewness if (1) mean $\neq$ median $\neq$ mode.

- (i) Quartiles are not equidistant from the median and
- (ii) the frequency curve drawn is stretched more to one side than to the other.

Measures of Skewness

Measures of skewness tell us the direction and extent of asymmetry in a series. There are two types of measures of skewness (i) absolute (ii) relative.

Absolute Measures of Skewness

Various measures of skewness are

- (i) $S_K = M - M_d$
- (ii) $S_K = M - M_0$
- (iii) $S_K = Q_3 + Q_1 - 2 M_d$.

where M_d = median, M_0 = Mode, M = mean.

Relative Measures of Skewness

There are four relative measures of skewness.

- (i) The Karl Pearson's coefficient of skewness.
- (ii) The Bowley's coefficient of skewness.
- (iii) The Kelly's coefficient of skewness and
- (iv) Measures of skewness based upon moments.

PROBABILITY

If in an experiment there are 'n' occurrences out of which 'm' occurrences are favourable to a particular event E, then probability of the event E [written as P(E)] is defined as $P(E) = m/n$.

For the event E, non-occurrence of the event is denoted by $\bar{E}$. Out of a total n occurrences, if m occurrences are favourable to the event E, it means that (n – m) occurrences are not favourable to the event $\bar{E}$, i.e. (n – m) occurrences are favourable to the event $\bar{E}$.

Hence, probability of the event $\bar{E}$ will be $P(\bar{E}) = \frac{n - m}{n}$ (complementary event).

We can also see that $P(\bar{E}) + P(E) = 1$.

In some problems when we are asked to find out the probability of an event $\bar{E}$, it becomes easier to calculate probability of E and subtract that probability from 1 to get $P(\bar{E})$.

Note:

(i) Probability $[P(E)]$ of the happening of an event E is called *probability of success* and probability

$[P(\bar{E})]$ of non-happening of the event is called *probability of failure*.

(ii) If $P(E) = 1$, then event is called a *certain event* and if $P(E) = 0$, then event is called *impossible event*.

(iii) Instead of saying that the chance of happening of an event is m/n , we can also say that odds in favour of the event are m (to $n - m$). Similarly, if m/n is probability of the occurrences of an event, since $(n - m)/n$ is probability of non-occurrence of the event, we can also say that the odds against the event are $(n - m)$ to m .

Addition Theorem on Probability

If there are two sets A and B , we know that the number of elements in $A \cup B$ is given by

$$n(A \cup B) = n(A) + n(B) - n(A \cap B).$$

A similar relationship exists in probability theory (called Addition Theorem of Probability) as $P(A \cup B) = P(A) + P(B) - P(A \cap B)$, where A and B are two events.

If A and B are mutually exclusive events (i.e, A and B are disjoint sets), then $P(A \cap B) = 0$

If there are two mutually exclusive events A and B , then $P(A \cup B) = P(A) + P(B)$.

Independent Events

Two events, E_1 and E_2 are said to be independent, if occurrence of the event E_2 is not affected by the occurrence or non-occurrence of the event E_1 .

If A and B are two independent events, then

$$P(A \cap B) = P(A) \times P(B)$$

Example:

If dice is rolled, then find the probability of getting a number divisible by 2.

Solution:

When a dice is rolled $n(S) = 6$

Favorable outcomes are $\{2, 4, 6\}$ i.e, 3

$\therefore$ Required probability $= 3/6 = 1/2$

Example:

When a dice is rolled, what is the probability of getting a composite number?

Solution:

Number of favorable outcomes $= 2(\{4, 6\})$

Total outcomes $= 6$

Required probability $= \frac{2}{6} = \frac{1}{3}$

Example:

When two coins are tossed together, find the probability of getting exactly two tails?

Solution:

Number of favorable outcomes $= 1(TT)$

Total outcomes $= 4(HH, HT, TH, TT)$

$\therefore$ Required probability $= \frac{1}{4}$

Example:

If two dice are tossed, the probability that the sum is 8 is

Solution:

Two dice are rolled; the number of possibilities is 36.

The favorable cases are $(2, 6), (6, 2), (3, 5), (5, 3), (4, 4)$

$\therefore$ Required probability $= \frac{5}{36}$

Example:

Suppose, seven coins are tossed then the probability of getting atleast one head is

Solution:

Probability of getting at least one head

$= 1 - (\text{probability of getting no heads}).$

$$= 1 - \frac{1}{2^7} = 1 - \frac{1}{128} = \frac{127}{128}$$

Example: A and B play a game where each is asked to select a number from 1 to 8. If both of them select the same number, then they win a prize. Find the probability that they will not win the prize.

Solution:

The total number of ways of selecting numbers for both A and B is $= 8(8) = 64$

They win the prize if they select same numbers.

The total possibilities are 8, i.e. $(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6), (7, 7), (8, 8)$

$\therefore$ Probability of winning prizes $= \frac{8}{64} = \frac{1}{8}$

The probability of not getting a prize $= 1 - \frac{1}{8} = \frac{7}{8}$.

EXERCISE

MCQ TYPE QUESTIONS

Directions (Q. 1 – 2) :

If A and B are two possible events of an experiment such that $P(A \cap B) = 0.6$ and $P(A) = 0.3$ then find $P(B)$ given that

1. A and B are mutually exclusive events

(a) $\frac{7}{10}$ (b) $\frac{4}{10}$

(c) $\frac{3}{10}$ (d) $\frac{1}{2}$

2. A and B are independent events

(a) $\frac{4}{7}$ (b) $\frac{9}{10}$ (c) $\frac{3}{10}$ (d) $\frac{3}{7}$

3. Raju throws a fair dice. He is promised an amount (in rupees), which is twice the number showing up if that number is odd and an amount thrice the number showing up, if it is even. What is the maximum amount Raju would be willing to pay each time to throw the dice, if in the long run he wants to make an average profit of ₹ 7 per throw?

(a) ₹ 6 (b) ₹ 7 (c) ₹ 3 (d) ₹ 2

Directions (Q. 4 – 5) :

7 letters are to be placed in seven addressed envelopes. If the letters are placed at random into the envelopes, the probability that

4. All of them are placed in the corresponding envelopes is

(a) 1

(b) $\frac{1}{6!}$

(c) $\frac{1}{7!}$

(d) None of these

5. Exactly six letters are placed in their corresponding envelopes is

(a) $1 - \frac{1}{7!}$ (b) $\frac{1}{7!}$ (c) 1 (d) 0

Directions (Q. 6 – 8) :

If three cards are drawn at random, from a well shuffled pack of cards, then what is the probability that

6. All of them are from the same suit?

(a) $\frac{4({}^{13}C_3)}{{}^{52}C_3}$ (b) $\frac{({}^{13}C_3)^3}{{}^{52}C_3}$

(c) $\frac{4({}^{13}C_3)^2}{{}^{52}C_3}$ (d) $\frac{{}^{13}C_3}{{}^{52}C_3}$

7. All of them are from different suits?

(a) $\frac{13}{{}^{52}C_3}$ (b) $\frac{4(13)^3}{{}^{52}C_3}$

(c) $\frac{{}^{39}C_3}{{}^{52}C_3}$ (d) $\frac{(13)^4}{{}^{52}C_3}$

8. Two of them are number cards from the same suit and the remaining number card is from a different suit?

(a) $\frac{{}^9C_2 {}^9C_1}{{}^{52}C_3}$ (b) $\frac{{}^9C_2 {}^{27}C_1}{{}^{52}C_3}$

(c) $\frac{4({}^9C_2 {}^{27}C_1)}{{}^{52}C_3}$ (d) $\frac{4({}^9C_2 {}^9C_1)}{{}^{52}C_3}$

9. A bag contains 5 white balls and 7 red balls. If 5 balls are drawn at random, then probability that 3 balls are red and the rest are white is

(a) $\frac{175}{198}$ (b) $\frac{175}{396}$

(c) $\frac{38}{132}$ (d) $\frac{221}{396}$

10. A bag contains 6 five rupee coins, 5 two rupee coins and 4 one rupee coins. If 5 coins are selected at random from the bag, then odds in favour of the draw yielding minimum possible amount are

(a) 1:3002 (b) 5:2998

(c) 2:3003 (d) 5:3003

11. In how many ways can 6 girls and 6 boys sit around a circular table so that no two boys sit together?

(a) $(5!)^2$ (b) $(6!)^2$

(c) $5! 6!$ (d) $11!$

12. How many ten-digit numbers can be formed using all the digits of 1324642119 such that the even digits appear only in odd places?

(a) $(5!)^2$ (b) $\frac{(5!)^2}{3!}$

(c) $5! 3! 2!$ (d) $\frac{(5!)^2}{(2!)^2 3!}$

Directions (Q. 13 – 14) :

A man has 12 friends whom he wants to invite for lunch.
In how many ways can he invite

13. At least one of them?

- (a) 4096 (b) 4095
(c) 2047 (d) 2048

14. At least 10 of them?

- (a) 66 (b) 140
(c) 153 (d) None of these

15. In how many ways can 6 prizes be distributed among 3 boys, if each boy is eligible to receive one or more prizes?

- (a) 3^6 (b) 6^3
(c) 6C_3 (d) 6P_3

Directions (Q. 16 – 17) :

Find the number of ways of dividing 16 different books equally

16. Among 4 boys

- (a) $\frac{16!}{(4!)^4}$ (b) $\frac{16!}{(4!)^3}$
(c) $\frac{16!}{(4!)^5}$ (d) $(4!)^4$

17. Into 4 parcels (Parcels are not distinct)

- (a) $\frac{16!}{(4!)^4}$ (b) $\frac{16!}{(4!)^3}$
(c) $\frac{16!}{(4!)^5}$ (d) $(4!)^4$

18. How many linear arrangements can be made using all the letters of the word ILLUSTRATE which begin with S and end with a vowel?

- (a) $\frac{8!}{(2!)^2}$ (b) $8!$
(c) $\frac{8!}{2!}$ (d) $\frac{8!}{(2!)^3}$

19. From a group of 10 professors and 6 assistant professors, a management institute desires to send to delegation of 8 persons consisting of 5 professors and 3 assistant professors to the IIMs annual meet. If Prof. Balamurali, a science professor refuses to be in the delegation if assistant Prof. Sheshdari, an arts professor is included in the delegation, then in how many ways can the delegation be formed?

- (a) ${}^9C_4 {}^4C_3$
(b) ${}^9C_5 {}^4C_2$
(c) ${}^{10}C_5 {}^6C_3 - {}^9C_4 {}^5C_2$
(d) ${}^9C_4 {}^4C_3 + {}^9C_5 {}^4C_2$

20. Two red pencils, three black pencils and two white pencils are to be arranged in a row such that

- No two adjacent pencils are of the same colour and
- Pencils at the two ends of the row are of same colour.

- (a) 12 (b) 8 (c) 9 (d) 10

NUMERICAL TYPE QUESTIONS
Directions (Q. 1 – 2) :

The odds against an event are 4 to 5 and the odds in favour of another independent event are 3 to 7. The probability that

- Exactly one of them will occur is _____
- Neither of them will occur is _____

Directions (Q. 3 – 5) :

Three persons Shiva, Jagan and Rohit aim at a target. Their respective probabilities of hitting the target are $\frac{2}{3}$, $\frac{5}{7}$ and $\frac{3}{8}$. The probability that

- None of them hits the target is _____
- At least two of them hit the target is _____
- Exactly one of them hits the target is _____
- If three consecutive letters are selected at random from the English alphabet, then the probability that at least one letter is a vowel is _____
- Eight unbiased coins are tossed together. The probability that the number of heads is equal to the number of tails is _____
- In a biased coin, Head occurs three times as frequently as tail occurs. If the coin is tossed 3 times, then the probability of getting two heads is _____

Directions (Q. 9 – 10) :

Two cards are drawn at random from a well shuffled pack of cards. Given that both are black, the probability that the cards have

- Same honour on them is _____
- Different honours and belong to different suits is _____

Directions (Q. 11 – 13) :

Three mountaineers Akil, Nikil, and Sunil are climbing up a mountain with their respective probability of reaching the summit being $\frac{2}{3}$, $\frac{5}{8}$ and $\frac{4}{7}$, respectively. The probability that

- None of them reach the summit is _____
- Exactly two of them reaches the summit is _____
- Atleast two of them reach the summit is _____

Directions (Q. 14 – 15) :

There are 12 points in a plane of which 4 are on a straight line and no three of the other points lie on a straight line.

14. The _____ number straight lines can be formed by joining these points
15. The _____ number of triangles can be formed by joining these points
16. The _____ number of non-negative integral solutions does the equation $x_1 + x_2 + x_3 + x_4 = 10$ have?
17. A certain number of students of a school participated in the chess tournament of their annual sports meet. Each player played 1 game against each of the other player. It was found that in 66 games both the players were girls, and in 240 games one was a girl and the other was a boy. The number of games in which both the players were boys is _____
18. Nine points are marked on a straight line and 10 points are marked on another line which is parallel to the first line. The _____ number of triangles can be formed with these points as vertices?
19. The _____ number of positive integers less than 100,000 and divisible by 125 can be formed using the digits 0, 1, 2, 5 and 8, if repetition is allowed?
20. There is an unlimited supply of identical red, blue, and green coloured balls. In _____ number of ways can 12 balls be selected from the supply.
21. Number of positive integral solutions of the equation $a + b + c = 15$ is _____
22. If six unbiased coins are tossed together, then probability that the number of heads exceeds the number of tails is _____

ANSWERS**MCQ Type Questions**

1. (c) 2. (d) 3. (d) 4. (c) 5. (d) 6. (a) 7. (b) 8. (c) 9. (b) 10. (b)
11. (c) 12. (d) 13. (b) 14. (d) 15. (a) 16. (a) 17. (c) 18. (b) 19. (c) 20. (d)

Numerical Type Questions

1. $\frac{47}{90}$ 2. $\frac{14}{45}$ 3. $\frac{5}{84}$ 4. $\frac{107}{168}$ 5. $\frac{17}{56}$ 6. $\frac{13}{24}$ 7. $\frac{35}{128}$ 8. $\frac{27}{64}$ 9. $\frac{4}{325}$ 10. $\frac{12}{325}$
11. $\frac{3}{56}$ 12. $\frac{37}{84}$ 13. $\frac{19}{28}$ 14. 61 15. 216 16. 286 17. 190 18. 765 19. 99 20. 91
21. 91 22. $\frac{57}{64}$

EXPLANATIONS**MCQ TYPE QUESTIONS****1 and 2**

Given : $P(A \cup B) = 0.6$

and $P(A) = 0.3$

1. If A and B are mutually exclusive events, then

$$P(A \cap B) = 0$$

$$\therefore P(A \cup B) = P(A) + P(B)$$

$$0.6 = 0.3 + P(B)$$

$$\Rightarrow 0.3 = P(B)$$

Hence $P(B) = 0.3$

2. If A and B are independent events, then

$$P(A \cap B) = P(A) \cdot P(B)$$

$$\begin{aligned} \therefore P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ &= P(A) + P(B) - P(A) \cdot P(B) \end{aligned}$$

$$P(A \cap B) = P(A) + P(B) [1 - P(A)]$$

$$\Rightarrow 0.6 = 0.3 + P(B) (1 - 0.3)$$

$$0.3 = P(B)(0.7)$$

$$\Rightarrow \frac{0.3}{0.7} = P(B)$$

$$\therefore P(B) = \frac{3}{7}$$

3. When a die is rolled, for each number the probability of getting it is $\frac{1}{6}$.

When number on the dice is 1, 3 or 5 Raju receives ₹ 2, ₹ 6 and ₹ 10 respectively.

When the number on the die is 2, 4 or 6, then Raju receives ₹ 6, ₹ 12 and ₹ 18 respectively.

$$\therefore \text{Expected value} = \frac{1}{6} [2 + 6 + 10 + 6 + 12 + 18]$$

$$= \frac{54}{6} = ₹ 9$$

Hence to make an average profit of ₹ 7 per throw, Raju must pay ₹ 2 (9 – 7) for each time to throw the die.

4. 7 letters can be arranged in 7 addressed envelopes in 7! ways.

$$\therefore n(S) = 7!$$

We can arrange 7 letters into corresponding 7 addressed envelopes in only one way.

$$\therefore \text{Required probability} = \frac{1}{7!}.$$

5. If six letters can be placed in their corresponding addressed envelopes, then seventh letter is also in the correct envelope.

Hence required probability is 0.

6, 7 and 8.

Three cards can be drawn from 52 cards in ${}^{52}C_3$ ways

$$n(S) = 52c_3$$

6. Since we have 4 suits i.e., Diamonds, Spades, Clubs and Hearts and each suit contains 13 cards. Three cards can be selected from 13 cards in ${}^{13}C_3$ ways.

$$\therefore \text{Number of favorable outcomes} = 4 \cdot {}^{13}C_3$$

$$\text{Required probability} = 4 \cdot \frac{{}^{13}C_3}{{}^{52}C_3}$$

7. From four suits we can select 3 suits in 4C_3 ways. One card can be selected from each suit in ${}^{13}C_1$.

$$\therefore \text{Number of favorable outcomes} = {}^4C_3 \cdot {}^{13}C_1 \cdot {}^{13}C_1 \cdot {}^{13}C_1$$

$$\therefore \text{Required probability} = \frac{4(13)^3}{{}^{52}C_3}$$

8. Each suit contains 9 number cards. 2 number cards can be drawn from 9 cards in 9C_2 ways..

Third card can be selected from the remaining 27 number cards in ${}^{27}C_1$ ways.

Hence total number of favorable outcomes is $4({}^9C_2)({}^{27}C_1)$

$$\text{Required probability} = \frac{4({}^9C_2)({}^{27}C_1)}{{}^{52}C_3}$$

9. Total number of balls in the bag = 5 + 7 = 12.

5 balls can be drawn from 12 balls in ${}^{12}C_5$ ways

Number of ways of drawing 3 red balls and 2 white balls from 5 white balls and 7 red balls is ${}^5C_2 \cdot {}^7C_3$

$$\begin{aligned} \therefore \text{Required probability} &= \frac{{}^5C_2 \cdot {}^7C_3}{{}^{12}C_5} \\ &= \frac{10 \times 35}{792} = \frac{175}{396} \end{aligned}$$

10. Total number of coins in bag = 6 + 5 + 4 = 15.

5 coins can be drawn from 15 coins in ${}^{15}C_5$ ways.

When 4 coins are one rupee coins and fifth coin is a two rupee coin the amount will be minimum.

$$\therefore \text{Number of favourable cases} = {}^4C_4 \cdot {}^{45}C_1 = 5$$

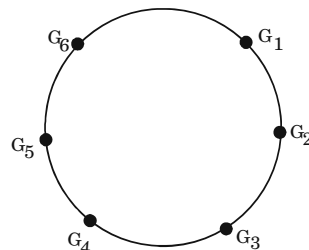
Probability that the amount will be minimum is

$$P(E) = \frac{5}{{}^{15}C_5} = \frac{5}{3003}$$

$$\therefore P(\bar{E}) = 1 - P(E) = 1 - \frac{5}{3003} = \frac{2998}{3003}$$

$$\therefore \text{Odds in favour of the required event} = 5 : 2998$$

11.



Six girls can be arranged around a circle in 5! ways.

Now, the six boys can be arranged in the six places in between the girls in 6! ways.

Hence, required number of ways = 5 ! 6!

12. Of the 10 places available, there are 5 even places.

Digits 2, 4, 6, 4 and 2 can be placed in these 5 even places in $\frac{5!}{2!2!}$ ways.

Remaining digits 1, 3, 1, 1 and 9 can be placed in

the remaining 5 places in $\frac{5!}{3!}$ ways.

Hence, required number of ways

$$= \frac{5!}{2!2!} \frac{5!}{3!} = \frac{(5!)^2}{(2!)^2 3!}$$

13. Number of ways of inviting at least one friend is

$$2^{12} - 1 = 4095$$

14. Number of ways of inviting at least 10 friends

$$= {}^{12}C_{10} + {}^{12}C_{11} + {}^{12}C_{12} = 66 + 12 + 1 = 79$$

15. Each prize can be distributed in 3 ways, so, 6 prizes can be distributed in 3^6 ways.

16. Number of ways of dividing 16 books equally among 4 boys

$$= \frac{16!}{4!4!4!4!} = \frac{16!}{(4!)^4}$$

17. Number of ways of dividing 16 books equally into 4 parcels

$$= \frac{16!}{4!(4!)^4} = \frac{16!}{(4!)^5}$$

18. There are four vowels viz., U, A, I, E in the word "ILLUSTRATE".

First letter must be S and the last letter can be any one of the vowels U, A, I and E.

Remaining 8 places can be filled with the remaining 8 letters in $\frac{8!}{2!2!}$ ways.

Hence, required number of arrangements is

$$= 4 \times \frac{8!}{2!2!} = 8!$$

19. Prof. Balamurali and Asst. B of Sheshadri cannot be included together in the delegation.

Hence, number of ways of forming delegation

$$= {}^{10}C_5 - {}^9C_4 = {}^5C_2$$

20. Possible ways of arranging the pencils are:

RBWBWR, WBRBRBW, BWBRWRB, BRBWRWB, BWRBWRB, BWRBRWB, BRWBWRB, BRWBWRB, BWRBWRB and BRWBWRB

i.e. a total of 10 possible arrangements.

NUMERICAL TYPE QUESTIONS

1 and 2 :

Let A and B be the two events.

$$\text{Given: } P(\bar{A}) : P(A) = 4 : 5$$

$$\Rightarrow P(\bar{A}) = \frac{4}{9}; \quad P(A) = \frac{5}{9}$$

$$\text{Also, } P(B) : P(\bar{B}) = 3 : 7$$

$$\Rightarrow P(B) = \frac{3}{10} \quad \text{and} \quad P(\bar{B}) = \frac{7}{10}$$

1. $A \cap \bar{B}$ and $\bar{A} \cap B$ denote events that exactly one of them will occur.

$$P(A \cap \bar{B}) = P(A) P(\bar{B}) = \frac{5}{9} \cdot \frac{7}{10} = \frac{7}{18}$$

$$= P(\bar{A} \cap B) = P(\bar{A}) P(B)$$

$$= \frac{4}{9} \cdot \frac{3}{10} = \frac{2}{15}$$

Required probability is

$$= \frac{7}{18} + \frac{2}{15} = \frac{35+12}{90} = \frac{47}{90}$$

2. P (neither of them occurs)

$$= P(\bar{A} \cap \bar{B}) = P(\bar{A}) P(\bar{B}) = \frac{7}{10} \left(\frac{4}{9} \right) = \frac{14}{45}$$

- 3, 4 and 5.

Let A, B and C be the events that Shiva, Jagan and Rohit hit the target respectively.

$$\therefore P(A) = \frac{2}{3}; \quad P(\bar{A}) = \frac{1}{3}$$

$$P(B) = \frac{5}{7}; \quad P(\bar{B}) = \frac{2}{7}$$

$$P(C) = \frac{3}{8}; \quad P(\bar{C}) = \frac{5}{8}$$

3. P (None of them hit the target)

$$= P(\bar{A} \cap \bar{B} \cap \bar{C}) = P(\bar{A}) P(\bar{B}) P(\bar{C})$$

$$= \left(\frac{1}{3} \right) \cdot \left(\frac{2}{7} \right) \cdot \left(\frac{5}{8} \right) = \frac{5}{84}$$

4. Event $(A \cap B \cap \bar{C}) \cup (A \cap \bar{B} \cap C) \cup (\bar{A} \cap B \cap C)$ denotes that exactly two persons hit the target, and $A \cap B \cap C$ is the event that all the three persons hit the target.

$$P(A \cap B \cap \bar{C}) = P(A) P(B) P(\bar{C})$$

$$= \frac{2}{3} \cdot \left(\frac{5}{7} \right) \cdot \left(\frac{5}{8} \right) = \frac{25}{84}$$

$$P(A \cap \bar{B} \cap C) = P(A) P(\bar{B}) P(C)$$

$$= \frac{2}{3} \cdot \left(\frac{2}{7} \right) \cdot \left(\frac{3}{8} \right) = \frac{1}{14}$$

$$P(\bar{A} \cap B \cap C) = P(\bar{A}) P(B) P(C)$$

$$= \frac{1}{3} \cdot \left(\frac{5}{7} \right) \cdot \left(\frac{3}{8} \right) = \frac{5}{56}$$

$$P(A \cap B \cap C) = P(A) P(B) P(C)$$

$$= \frac{2}{3} \cdot \left(\frac{5}{7} \right) \cdot \left(\frac{3}{8} \right) = \frac{5}{28}$$

Hence required probability

$$\begin{aligned}
 &= P(A \cap B \cap \bar{C}) + P(A \cap \bar{B} \cap C) \\
 &\quad + P(\bar{A} \cap B \cap C) + P(A \cap B \cap C) \\
 &= \frac{25}{84} + \frac{1}{14} + \frac{5}{56} + \frac{5}{28} = \frac{50+12+15+30}{168} = \frac{107}{168}
 \end{aligned}$$

5. Exactly one of them hits the target

$$(A \cap \bar{B} \cap \bar{C}) \cup (\bar{A} \cap B \cap \bar{C}) + (\bar{A} \cap \bar{B} \cap C)$$

denotes that exactly one of them hits the target.

$$P(A \cap \bar{B} \cap \bar{C}) = P(A) \cdot P(\bar{B}) \cdot P(\bar{C})$$

$$= \frac{2}{3} \times \frac{2}{7} \times \left(\frac{5}{8}\right) = \frac{5}{42}$$

$$P(\bar{A} \cap B \cap \bar{C}) = P(\bar{A}) \cdot P(B) \cdot P(\bar{C}) = \frac{1}{3} \cdot \frac{2}{7} \cdot \frac{3}{8} = \frac{1}{28}$$

$$P(\bar{A} \cap \bar{B} \cap C) = P(\bar{A}) \cdot P(\bar{B}) \cdot P(C) = \frac{1}{3} \cdot \frac{5}{7} \cdot \frac{5}{8} = \frac{25}{168}$$

Hence required probability is

$$\begin{aligned}
 &= P(A \cap \bar{B} \cap \bar{C}) + P(\bar{A} \cap B \cap \bar{C}) + P(\bar{A} \cap \bar{B} \cap C) \\
 &= \frac{5}{42} + \frac{25}{168} + \frac{1}{28} = \frac{20+25+26}{168} = \frac{51}{168} = \frac{17}{56}
 \end{aligned}$$

6. Three consecutive letters can be selected from English alphabet in 24 ways.

$$\therefore n(S) = 24$$

Since, no two consecutive letters are vowels, the number of ways that among three letters one letter is vowel is 13.

$$\therefore \text{Required probability} = \frac{13}{24}$$

7. Eight coins are tossed, so $n(S) = 2^8$

Let E be the favorable event. It is required to get 4 heads and 4 tails which is possible in the favorable out come HHHHTTTT.

This can be arranged in $\frac{8!}{4!4!}$ ways.

$$n(E) = \frac{8 \times 7 \times 6 \times 5 \times 4!}{4 \times 3 \times 2 \times 1 \times 4!} = 70$$

$$\therefore \text{Required probability} = \frac{n(E)}{n(S)} = \frac{70}{2^8} = \frac{35}{128}$$

8. Given : $P(H) = 3P(T)$

We know

$$P(H) + P(T) = 1$$

$$3P(T) + P(T) = 1$$

$$P(T) = \frac{1}{4}$$

$$P(H) = \frac{3}{4}$$

Since the coin is tossed 3 times, two heads may occur in 1st and 2nd trial, 2nd and 3rd trial or 1st and 3rd trial.

$\therefore$ Required probability

$$\begin{aligned}
 &= \frac{3}{4} \times \frac{3}{4} \times \frac{1}{4} + \frac{1}{4} \times \frac{3}{4} + \frac{3}{4} \times \frac{1}{4} \times \frac{3}{4} \\
 &= \frac{9}{64} + \frac{9}{64} + \frac{9}{64} = \frac{27}{64}
 \end{aligned}$$

9. Two black cards can be selected in ${}^{26}C_2$ ways.

A pack contain 8 black honours.

Hence number ways of selecting two same honours is 4.

$$\therefore \text{Required probability} = \frac{4}{{}^{26}C_2} = \frac{4}{325}$$

10. Let one card is king, then second card can be any of the other 3 cards from the other suit and it can be selected in $4 \times 3 = 12$ ways. Since there are four honour cards, hence

$$\therefore \text{Required probability} = \frac{12}{325}$$

11, 12 and 13

Probability that Akil reach the summit

$$= P(A) = 2/3$$

$$\therefore P(\bar{A}) = 1/3$$

Probability that Nikil reach the summit is

$$P(N) = 5/8$$

$$\therefore P(\bar{N}) = 3/8$$

Probability that Sunil reach the summit

$$P(S) = \frac{4}{7}$$

$$\therefore P(\bar{S}) = \frac{3}{7}$$

11. Probability that none of them reaches the summit

$$= P(\bar{A} \cap \bar{N} \cap \bar{S}) = P(\bar{A})P(\bar{N})P(\bar{S})$$

$$= \frac{1}{3} \times \frac{3}{8} \times \frac{3}{7} = \frac{3}{56}$$

12. Probability that exactly two of them reaches the summit

$$= P(A \cap S \cap \bar{N}) + P(A \cap N \cap \bar{S}) + P(\bar{A} \cap N \cap S)$$

$$= P(A)P(S)P(\bar{N}) + P(A)P(N)P(\bar{S}) + P(\bar{A})P(N)P(S)$$

$$= \frac{2}{3} \cdot \frac{4}{7} \cdot \frac{3}{8} + \frac{2}{3} \cdot \frac{5}{8} \cdot \frac{3}{7} + \frac{1}{3} \cdot \frac{5}{8} \cdot \frac{4}{7} = \frac{37}{84}$$

13. Probability that atleast two of them reaches the summit

$$= \text{Exactly two of them reach summit} \\ + \text{three of them reach the summit}$$

$$= \frac{37}{84} + \frac{5}{21} = \frac{57}{84} = \frac{19}{28}$$

14. 2 points determine a line.

Hence 12 points can form ${}^{12}C_2$ straight lines.

But 4 points are collinear, i.e they give only one line.

$$\text{Number of straight lines that can be formed} \\ = {}^{12}C_2 - {}^4C_2 + 1 = 66 - 6 + 1 = 61$$

15. Three points determine a triangle.

$\therefore$ 12 non-collinear points can give ${}^{12}C_3$ triangles.

But 4 points are collinear, so, these 4 points do not give any triangles.

$$\text{Hence, number of triangles} = {}^{12}C_3 - {}^4C_3 = 216$$

16. Number of non-negative integral solutions of the equation

$$x_1 + x_2 + \dots + x_k = n \text{ is } {}^{(n+k-1)}C_{k-1}.$$

Hence number of non-negative integral solutions of

$$x_1 + x_2 + x_3 + x_4 = 10 \text{ is } {}^{13}C_3 = 286.$$

17. Let m, n be the number of girls and the number of boys respectively. Then

$${}^mC_2 = 66$$

$$\Rightarrow \frac{m(m-1)}{2!} = 66 \Rightarrow m = 12$$

$$\text{Also, } {}^mC_1 \times {}^nC_1 = 240 \Rightarrow n = 20$$

Hence, number of games in which both the players

$$\text{were boys} = {}^{20}C_2 = \frac{20(19)}{2} = 190$$

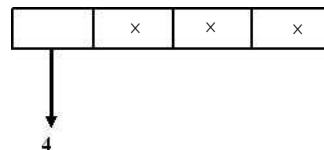
18. Points lying on a straight line do not form triangles. Hence, number of triangles that can be formed by using the given points

$$= {}^{19}C_3 - ({}^9C_3 + {}^{10}C_3) = 765$$

19. A number is divisible by 125 if and only if number formed by its last three digits is divisible by 125. So, possible combinations for last three digits are: 000, 125, 250 and 500.

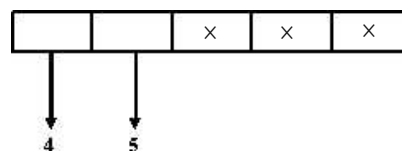
Now, number of three-digit number divisible by 125 is 3.

$$\text{Number of four-digit numbers divisible by 125} \\ = 4(4) = 16.$$



Number of five-digit numbers divisible by 125

$$= 4(4)(5) = 80.$$



$$\text{Hence, required number of numbers} \\ = 3 + 16 + 80 = 99.$$

20. Let the number of red, blue and green balls selected from the supply be x_1, x_2 and x_3 respectively.

$$\text{Then, } x_1 + x_2 + x_3 = 12.$$

Number of non-negative integral solutions of the above equation

$$= {}^{(n+k-1)}C_{k-1} = {}^{14}C_2 = 91.$$

21. Number of positive integral solutions of equation

$$x_1 + x_2 + \dots + x_k = n \text{ is } {}^{n-1}C_{k-1}.$$

Number of positive integral solutions of the equation

$$\text{Hence } a + b + c = 15$$

$${}^{15-1}C_{3-1} = {}^{14}C_2 = 91$$

22. When 'n' coins are tossed together, then

$$\text{probability of getting exactly 'r' tails} = \frac{{}^nC_r}{2^n}$$

Hence probability of getting atleast two tails

$$= 1 - (\text{probability of getting no tail}$$

$$+ \text{probability of getting exactly one tail})$$

$$= 1 - \left\{ \frac{{}^6C_0}{64} + \frac{{}^6C_1}{64} \right\} = 1 - \frac{7}{64} = \frac{57}{64}$$

6

CHAPTER

LINEAR EQUATIONS

An equation where the maximum power of any variable is unity (one) is a linear equation. A **linear equation** is of the form:

$$AX + BY + C = 0$$

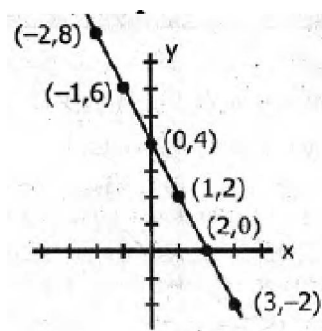
Where A and B are coefficient and C is a constant

The equation is called **LINEAR** because the graph of the equation on the X-Y Cartesian plane is a straight line. The sets of values of X & Y satisfying any equation(s) are called its solution(s).

Consider the equation $2x + y = 4$. Now, if we substitute $x = -2$ in the equation, we obtain $2(-2) + y = 4$ or $-4 + y = 4$ or $y = 8$. Hence $(-2, 8)$ is a solution. If we substitute $x = 3$ in the equation, we obtain $2(3) + y = 4$ or $6 + y = 4$ or $y = -2$. Hence $(3, -2)$ is a solution. The following table lists six possible values for x and the corresponding values for y, i.e. six solutions of the equation.

x	-2	-1	0	1	2	3
y	8	6	4	2	0	-2

If we plot the solutions of the equation $2x + y = 4$ which appear in the table above then we see that they all lie on the same line. We call this line the graph of the equation since it corresponds precisely to the solution set of the equation.



Two Linear Equations in two Unknowns Variables

We now consider a system of two linear equations, in two unknowns x and y:

$$a_1x + b_1y = c_1$$

$$a_2x + b_2y = c_2$$

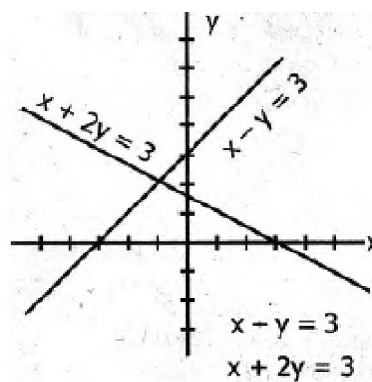
A pair of numbers which satisfies both equations is called a simultaneous solution of the given equations or a solution of the system of equations. There are

Miscellaneous

three cases which can be described geometrically. (Here we assume that the coefficients of x and y in each equation are not both zero.)

1. The system has exactly one solution:

Here the lines corresponding to the linear equations intersect in one point as shown in fig. below.

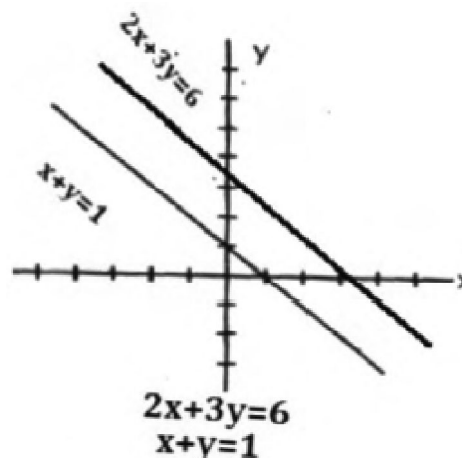


The system $a_1X + b_1Y = c_1$ and $a_2X + b_2Y = c_2$ has

a unique solution, if $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$.

2. The system has no solution:

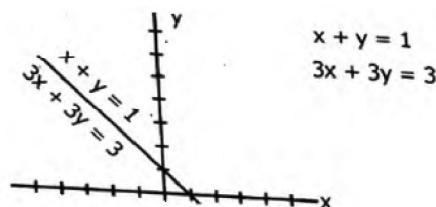
Here the lines corresponding to the linear equations are parallel as shown in fig. below.



The system $a_1X + b_1Y = c_1$ & $a_2X + b_2Y = c_2$ has no

solution if $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$.

3. **The system has an infinite number of solution:** Here the lines corresponding to the linear equations coincide as shown in fig. below.



The system $a_1X + b_1Y = c_1$ & $a_2X + b_2Y = c_2$ has

infinitely many solution if $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

System of Linear Equations

Consistent System: A system (of 2 or 3 or more equations taken together) of linear equations is said to be consistent, if it has at least one solution.

Inconsistent System: A system of simultaneous linear equations is said to be inconsistent, if it has no solutions at all.

e.g. $X + Y = 9$; $3X + 3Y = 8$

Clearly there are no values of X & Y which simultaneously satisfy the given equations. So the system is inconsistent.

Example: Find k for which the system $6x - 2y = 3$, $kx - y = 2$ has a unique solution.

Solution: The given system will have a unique solution

if $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$; i.e. $\frac{6}{k} \neq \frac{-2}{-1}$ or $k \neq 3$.

Example: What is the value of k for which the system $x + 2y = 3$, $5x + ky = -7$ is consistent?

Solution: The given system will be inconsistent if

$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$ i.e.

if $\frac{1}{5} = \frac{2}{k} \neq \frac{3}{-7}$. Hence $k = 10$.

Simultaneous Equations Involving Three Unknown Quantities

Method of solving: Suppose we are given three equations containing three unknown quantities X , Y and Z . Mark these equations (1), (2) and (3). Now, from any two equations say (1) and (2), try to eliminate (or get rid of) one unknown quantity, say Z , i.e., to say, multiply (1) and (2) by such suitable number which will make the coefficients of Z in (1) and (2) equal; then add or subtract to cancel Z . Mark this equation (4). After this, take any two equations, say (2) and (3), and from this, eliminate Z in a similar way. Mark this equation (5). Now, solving (4) and (5) by any of the

methods given previously, we will find X and Y . Substituting the value of X and Y in any of the equations (1), (2) and (3), we will get the value of Z .

Example: Solve $X + Y + Z = 6$, $2X + 2Y + 3Z = 13$, $3X + 4Y + 5Z = 22$

Solution:

We have

$$X + Y + Z = 6 \quad \dots (1)$$

$$2X + 2Y + 3Z = 13 \quad \dots (2)$$

$$3X + 4Y + 5Z = 22 \quad \dots (3)$$

Multiply (1) by 3 and subtract (2) and we have

$$3(X + Y + Z) - (2X + 2Y + 3Z) = 18 - 13$$

$$\text{or } X + Y = 5 \quad \dots (4)$$

Again, multiply (1) by 5 and subtract (3), then

$$5(X + Y + Z) - (3X + 4Y + 5Z) = 30 - 22$$

$$\text{or } 2X + Y = 8 \quad \dots (5)$$

Subtracting (4) from (5), we have $X = 3$.

Hence, from (4), $Y = 2$.

Substituting these values of X and Y in (1), we have $3 + 2 + Z = 6$ or $Z = 1$.

QUADRATIC EQUATION

A quadratic polynomial of the form $ax^2 + bx + c$, where $a \neq 0$ and a, b, c are real numbers, is called a quadratic equation

when $ax^2 + bx + c = 0$.

Here a and b are the coefficients of x^2 and x respectively and ' c ' is a constant term.

Any value is a solution of a quadratic equation if and only if it satisfies the quadratic equation.

Quadratic formula: The roots, i.e., α and β of a quadratic equation $ax^2 + bx + c = 0$ are given by

$$\frac{-b \pm \sqrt{D}}{2a} \text{ or } \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \text{ provided } b^2 - 4ac \geq 0.$$

Here, the value $b^2 - 4ac$ is known as the discriminant and is generally denoted by D . ' D ' helps us to determine the nature of roots for a given quadratic equation.

Hence, $D = b^2 - 4ac$.

Quadratic Equation Rules are:

1. If $D = 0 \Rightarrow$ The roots are Real and Equal.
2. If $D > 0 \Rightarrow$ The two roots are Real and Unequal.
3. If $D < 0 \Rightarrow$ No Real roots exist.

If α and β are the roots of the quadratic equation, then Quadratic equation is $x^2 - (\alpha + \beta)x + \alpha\beta = 0$ Or $x^2 - (\text{sum of roots})x + \text{product of roots} = 0$

where, Sum of roots $(\alpha + \beta) = -\text{coefficient of } x / \text{coefficient of } x^2 = -b/a$

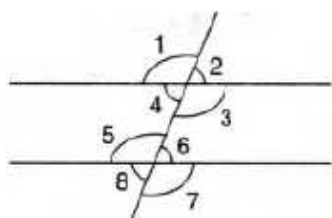
Product of roots $(\alpha \times \beta) = \text{constant term} / \text{coefficient of } x^2 = c/a$

GEOMETRY AND MENSURATION**STRAIGHT LINE**

Parallel lines: - Two straight lines are parallel if they lie on the same plane and do not intersect however far produced.

Transversal :- It is a straight line that intersects two parallel lines. When a transversal two parallel lines then

- (1) Corresponding angles are equal, ? (that is : For the figure below)
- (2) Alternate interior angles are equal, that is (Refer figure below.) $4 = 6$; $5 = 3$
- (3) Alternate exciter or angles are equal, that is $2 = 8$; $1 = 7$
- (4) Interior angles on the same side of transversal add up to 180° , that is $4 + 5 = 33 + 6 = 180^\circ$

**Polygons**

Polygons are plane figures formed by a closed series of rectilinear (straight) segments.

Example:

Triangle, Rectangles.

Polygons can broadly be divided into two types:

- (a) Regular polygons: Polygons with all the sides and angles equal.
- (b) Irregular polygons: Polygons in which all the sides or angles are not of the same measure.

Triangles (Δ)

A triangle is a polygon having three sides. Sum of all the angles of a triangle $= 180^\circ$

Types:

- (1) Acute angle triangle: Triangles with all the angles acute (less than 90°)
- (2) Obtuse angle triangle: Triangles with one of the angles obtuse (more than 90°)

Note: We cannot have more than one obtuse angle triangle.

- (3) Right angle triangle: Triangle with one of the angles equal to 90°
- (4) Equilateral triangle: Triangle with all sides equal. All the angles in such a triangle measure 60°

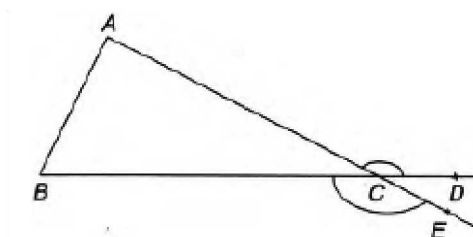
- (5) Isosceles triangle: Triangle with two of its sides equal and consequently the angles opposite the equal sides are also equal.

- (6) Scalene triangle: Triangle with none of the sides equal to any other side.

Properties (General)

- (1) Sum of the length of any two sides of a triangle has to be always greater than the third side.
- (2) Difference between the lengths of any two sides of a triangle has to be always lesser than the third side.
- (3) Side opposite to the greatest angle will be the greatest and the side opposite to the smallest angle the smallest.
- (4) The exterior angle is equal to the sum of two interior angles not adjacent to it.

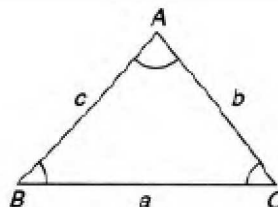
$$\angle ACD = \angle BCE = \angle A + \angle B$$

**AREA:**

- (1) Area $= \frac{1}{2}$ base $\times$ height or $\frac{1}{2}bh$.
Height = Perpendicular distance between the base and vertex opposite to it
- (2) Area $= \sqrt{s(s-a)(s-b)(s-c)}$ (Heros's formula)

Where $S = \frac{a+b+c}{2}$ (a, b and c being the length of the sides)

- (3) Area $= r \times S$ (where r is in radius)
- (4) Area $= \frac{1}{2} \times$ product of two sides $\times$ sine of the included angle.
 $= \frac{1}{2} ac \sin B$
 $= \frac{1}{2} ab \sin C$
 $= \frac{1}{2} bc \sin A$



- (5) Area $= \frac{abc}{4R}$
Where R = circum radius

CONGRUENCY OF TRIANGLES

Two triangles are congruent if all the sides of one are equal to the corresponding sides of another, It follows that all the angles of one are equal to the corresponding angles of another. The notation for congruency is $\cong$

1. **SAS congruency:** If two sides and an included angle of one triangle are equal to two sides and an included angle of another, the two triangles are congruent. (See figure below.)

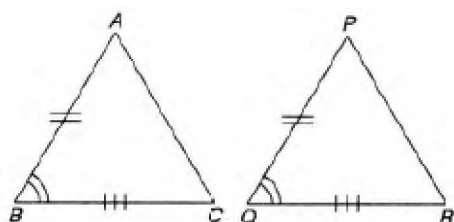
Here,

$$AB = PQ$$

$$BC = QR$$

$$\text{And } \angle B = \angle Q$$

$$\text{So } \triangle ABC \cong \triangle PQR$$



2. **ASA congruency:** If two angles and the included side of one triangle is equal to two angles and the included side of another, the triangles are congruent.

(See figure below.)

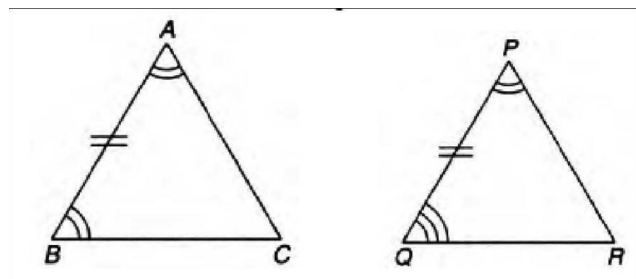
Here,

$$\angle A = \angle P$$

$$\angle B = \angle Q$$

$$\text{And } AB = PQ$$

$$\text{So } \triangle ABC \cong \triangle PQR$$



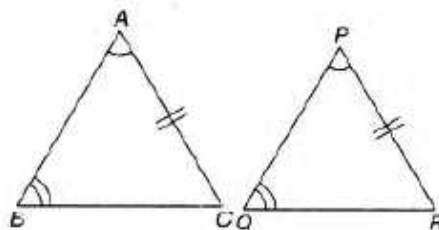
3. **AAS congruency:** If two angles and side opposite to one of the angle is equal to the corresponding angles and the side of another triangle, the triangles are congruent. In the figure below:

$$\angle A = \angle P$$

$$\angle B = \angle Q$$

$$\text{And } AC = PR$$

$$\text{So } \triangle ABC \cong \triangle PQR$$



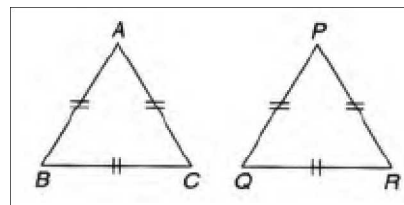
4. **SSS congruency:** If three sides of one triangle are equal to three sides of another triangle, the two triangles are congruent. In the figure below:

$$AB = PQ$$

$$BC = QR$$

$$AC = PR$$

$$\therefore \triangle ABC \cong \triangle PQR$$

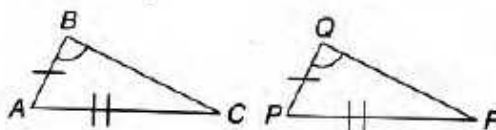


5. **SSA congruency :** If two sides and the angle opposite the greater side of one triangle are equal to the two sides and the angle opposite to the greater side of another triangle, then the triangles are congruent. The congruency doesn't hold if the equal angles lie opposite the shorter side. In the figure below, if the below, if

$$AB = PQ$$

$$AC = PR$$

$$\angle B = \angle Q$$



Then the triangles are congruent.

$$\text{i.e., } \triangle ABC \cong \triangle PQR$$

SIMILARITY OF TRIANGLES

Similarity of triangles is a special case where if either of the conditions of similarity of polygons holds if either of the conditions of similarity of polygons holds, the other will hold automatically.

Type of Similarity

- (1) **AAA similarity:** If in two triangles corresponding angles are equal, that is the two triangles are equiangular then triangles are similar. Corollary (AA similarity) If two angles of another triangle then the two triangles are similar. The reason being, the third angle becomes equal automatically.
- (2) **SSS similarity:** If the corresponding sides of two triangles are proportional then they are similar. For $\triangle ABC$ to be similar to $\triangle PQR$, $AB/PQ = BC/QR = AC/PR$, must hold true.
- (3) **SAS similarity:** If in two triangles, one pair of corresponding sides are proportional and the included angles are equal then the two triangles are similar.

$$\triangle ABC \sim \triangle PQR$$

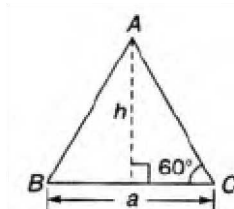
$$\text{If } AB/BC = PQ/QR \text{ and } \angle B = \angle Q$$

EQUILATERAL TRIANGLES (OF SIDE A)

$$(1) \quad h = \frac{a\sqrt{3}}{2} (\because \sin 60^\circ \sqrt{3} / 2 = h / \text{side})$$

$$(2) \quad \text{Area } 1/2 (\text{base}) \times (\text{height})$$

$$= \frac{1}{2} \times a \times \frac{a\sqrt{3}}{2} = \frac{a^2\sqrt{3}}{4}$$



$$(3) \quad R (\text{circum radius}) = \frac{2h}{3} = \frac{a}{\sqrt{3}}$$

$$(4) \quad R (\text{in radius}) = \frac{h}{3} = \frac{a}{2\sqrt{3}}$$

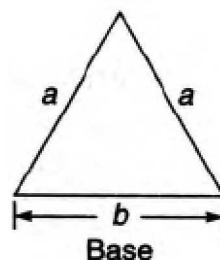
Properties

- The in centre and circum centre lies at a point that divides the height in the ratio 2 : 1
- The circum radius is always twice the in radius. $[R = 2r]$
- Among all the triangles that can be formed with a given perimeter, the equilateral triangle will have the maximum area.
- An equilateral triangles in a circle will have the maximum area compared to other triangles inside the same circle.

ISOSCELES TRIANGLE

$$\text{Area} = \frac{b}{4} \sqrt{4a^2 - b^2}$$

In an isosceles triangle, the angles opposite to the equal sides are equal.

**RIGHT- ANGLED TRIANGLE
PYTHAGORAS THEOREM**

In the case of a right angled triangle, the square of the hypotenuse is equal to the sum of the squares of the other two sides. In the figure below, for triangle ABC, $a^2 = b^2 + c^2$

Area = $1/2$ (product of perpendicular sides)

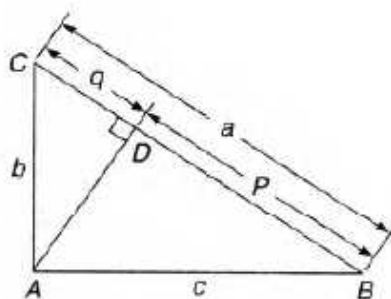
$$R (\text{circumradius}) = \frac{\text{hypotenuse}}{2}$$

$$\text{Area} = rs$$

(Where r = in radius and $s = (a + b + c)/2$ where a, b , and c are sides of the triangle)

$$\Rightarrow 1/2 bc = r (a + b + c)/2$$

$$r = (bc)/(a + b + c)/2$$



In the triangle ABC,

$$\triangle ABC \sim \triangle DBA \sim \triangle DAC$$

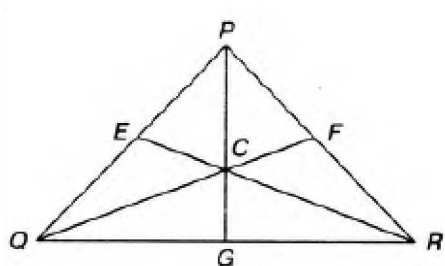
- $\triangle ABC \sim \triangle DBA$
 $\therefore AB/BC = DB/BA$
 $\Rightarrow AB^2 = DB \times BC$
 $\Rightarrow c^2 = pa$
- $\triangle ABC \sim \triangle DAC$
 $AC/BC = DC/AC$
 $\Rightarrow AC^2 = DC \times BC$
 $\Rightarrow b^2 = qa$
- $\triangle DBA \sim \triangle DAC$
 $DA/DB = DC/DA$
 $\Rightarrow DA^2 = DB \times DC$
 $\Rightarrow AD^2 = pq$

IMPORTANT TERMS OF TRIANGLE**Important terms with respect to a triangle**

1. Median: A line joining the mid-point of a side of triangle to the opposite vertex is called a median. In the figure the three medians are PG, QF and RE where G, E and F are mid-point of their respective sides

- (1) A median divides a triangle into two parts of equal area.
- (2) The point where the three medians of a triangle meet is called the centroid of the triangle.
- (3) The centroid of a triangle divides each median in the ratio 2:1

$$\text{i.e., } PC : CG = 2:1 = QC : CF = RC : CE$$

**Important formula with respect to a median**

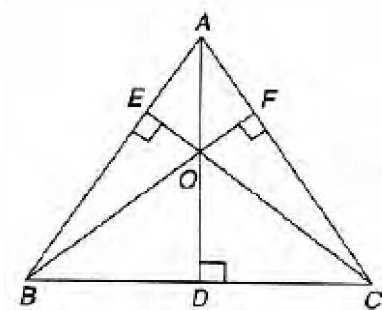
$\rightarrow 2 \times (\text{median})^2 + (1/2 \text{ the third side})^2 = \text{Sum of the squares of other two sides}$

$$\Rightarrow 2(PG)^2 + 2 \times \left(\frac{QR}{2}\right)^2$$

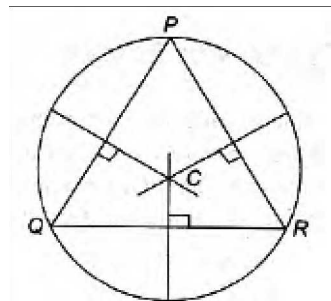
2. Altitude/ Height: A perpendicular drawn from any vertex to the opposite side is called the altitude (In the figure, AD, BF and CE are the altitudes of the triangles)

- (1) All the altitudes of a triangle meet at a point called the orthocenter of the triangle.
- (2) The angle made by any side at the orthocenter and the vertical angle make a supplementary pair (i.e., they both add up to 180°). In the figure below:

$$\angle A + \angle BOC = 180^\circ = \angle C + \angle AOB$$



3. Perpendicular Bisectors: A line that is perpendicular to a side and bisects it is the perpendicular bisector of the side.

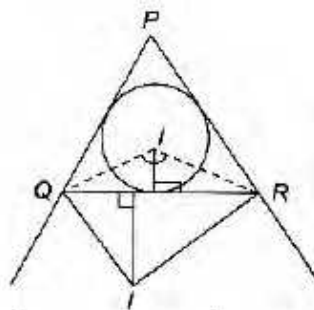


- (1) The point at which the perpendicular bisectors of the sides meet is called the circumcentre of the triangle.
- (2) The circumcentre is the centre of the circle that circumscribes the triangle. There can be only one such circle.
- (3) Angle formed by any side at the circumcentre is two times the vertical angle opposite to the side. This is the property of the circle where by angles formed the property of the circle where by angles formed by an arc at the centre are twice that of the angle formed by the same arc in the opposite arc. Here we can view this as:

$$\angle QCR = 2\angle QPR \text{ (When we consider arc QR and its opposite arc QPR)}$$

4. In center:

- (1) The lines bisecting the interior angles of a triangle are the angle bisectors of that triangle.
- (2) The angle bisectors meet at a point called the in centre of the triangle.
- (3) The in centre is equidistant from all the sides of the triangle.



- (4) From the in centre with a perpendicular drawn to any of the sides as the radius, a circle can be drawn touching all the three sides. This is called the in circle of the triangle. The radius of the circle is known as in radius.

- (5) The angle formed by any side at the incentre is always a right angle more than half the angle opposite to the side. This can be illustrated as

$$\angle QIR = 90 + 1/2 \angle P$$

- (6) If QI and RI be the angle bisectors of exterior angles at Q and R then, $\angle QIR = 90 + 1/2 \angle P$

QUADRILATERALS

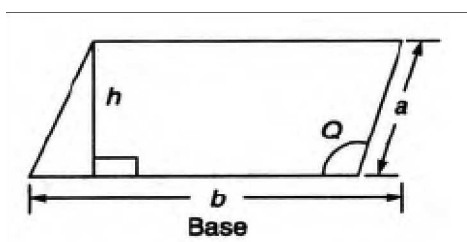
A quadrilateral is a plane figure that has four side or ages, and also have four corners or vertices.

Types of Quadrilaterals

1. Parallelogram

A parallelogram is a quadrilateral with opposite sides parallel (as shown in the figure below)

- (1) Area = Base (b) $\times$ Height (h)

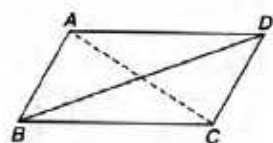


- (2) Area = product of any two adjacent sides $\times$ sine of the included angle.
 (3) perimeter = $2(a + b)$
 where a and b are any two adjacent sides.

Properties of Parallelogram

- (1) Diagonals of a parallelogram bisect each other.
- (2) Bisectors of the angles of a parallelogram form a rectangle.
- (3) A parallelogram inscribed in a circle is a rectangle.
- (4) A parallelogram circumscribed about a circle is a rhombus.
- (5) The opposite angles I a parallelogram are equal.
- (6) The sum of the squares of the diagonals is equal to the sum of the squares of the four sides in the figure:

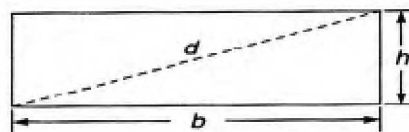
$$\begin{aligned} AC^2 + BD^2 &= AB^2 + BC^2 + CD^2 + AD^2 \\ &= 2(AB^2 + BC^2) \end{aligned}$$



2. Rectangles:

A rectangle is a parallelogram with all angles 90°

- (1) Area = Base $\times$ Height = $b \times h$



Note: Base and height are also referred to as the length and the breadth in a rectangle.

- (2) Diagonal (d) = $\sqrt{b^2 + h^2} \rightarrow$ by Pythagoras theorem

PROPERTIES OF A RECTANGLE

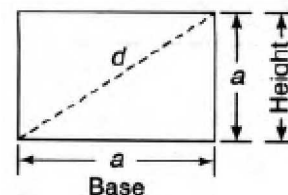
A parallelogram having all the sides equal is a rhombus.

- (1) Area = $1/2 \times$ product of diagonals $\times$ sine of the angles between them.
 $1/2 \times d_1 \times d_2 \sin 90^\circ$ (diagonals in a rhombus intersect at right angles) = $1/2 \times d_1 \times d_2$ (since $\sin 90^\circ = 1$)
- (2) Area = Product of adjacent sides $\times$ sine of the angle between them.
- (3) Diagonals bisect each other at right angles.
- (4) All rhombuses are parallelograms by the reverse is not true.
- (5) A rhombus may or may not be a square but all square are rhombus.

3. Properties of Square:

A square is a rectangle with adjacent sides equal or a rhombus with each angle 90°

- (1) Area = base $\times$ height = a^2
- (2) Area = $1/2$ (diagonal) $^2 = \frac{1}{2}d^2$ (square s a rhombus too)
- (3) Perimeter = $4a$ (a = side of the square)
- (4) Diagonal = $\sqrt{2}$
- (5) inradius = $\frac{a}{2}$



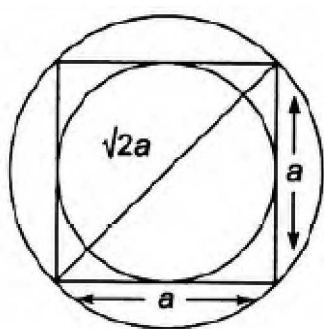
Trapezium of Square

- (1) Diagonals are equal and bisect each other at right angles.
- (2) Side is the diameter of the inscribed circle.

- (3) Diagonal is the diameter of the circumscribing circle

$$\Rightarrow \text{Diameter} = a\sqrt{2}$$

$$\text{circumradius} = a / \sqrt{2}$$



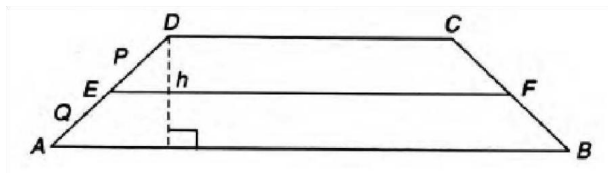
4. Trapezium:

A trapezium is a quadrilateral with only two sides parallel to each other.

- (1) Area = $1/2 \times \text{sum of parallel sides} \times \text{height} = 1/2 (AB + DC) \times h$. For the figure below.
- (2) Median = $1/2 \times \text{sum of the parallel sides}$ (median is the line equidistant from the parallel sides)

For any line EF parallel to

$$EF = \frac{\{[P \times (AB)] + [Q \times (DC)]\}}{AD}$$



Properties of Trapezium

- (1) If the non-parallel sides are equal then diagonals will be equal too.

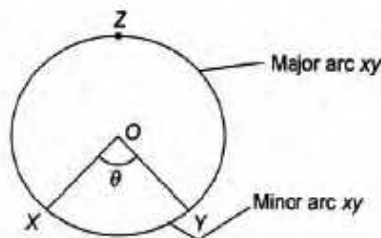
CIRCLES

- (1) Area = πr^2
- (2) Circumference = $2\pi r$ (r = radius)
- (3) Area = $1/2 \times \text{circumference} \times r$

Arc

It is a part of the circumference of the circle. The bigger one is called the major arc and the smaller one the minor arc.

- (1) Length (ArcXY) = $\frac{\theta}{360} \times 2\pi r$



- (2) Sector of a circle is a part of the area of a circle between two radii

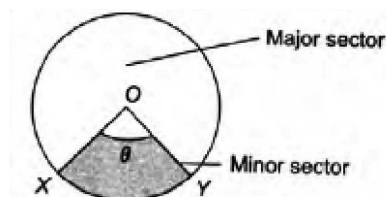
$$(3) \text{ Area of a sector} = \frac{\theta}{360} \times \pi r^2$$

(where θ is the angle between two radii)

$$= (1/2)r \times \text{length (arc xy)}$$

$$(\therefore \pi r \theta / 180 = \text{length arc xy})$$

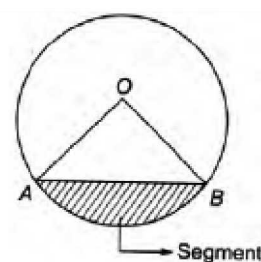
$$= \frac{1}{2} \times r \times \frac{\pi r \theta}{360}$$



- (4) **Segment:** A sector minus the triangle formed by the two radii is called the segment of the circle.

Area = Area of the sector - Area

$$\Delta OAB = \frac{\theta}{360} \times \pi r^2 - \frac{1}{2} \times r^2 \sin \theta$$



- (5) Perimeter of segment = length of the arc + of segment AB

$$= \frac{\theta}{360} \times 2\pi r + 2r \sin\left(\frac{\theta}{2}\right)$$

$$= \frac{\pi r \theta}{180} + 2r \sin\left(\frac{\theta}{2}\right)$$

- (6) **Congruency:**

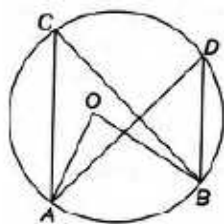
Two circles can be congruent if and only if they have equal radii.

Properties of Circle

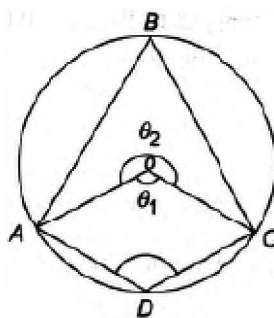
- (1) The perpendicular from the centre of circle to a chord bisects the chord. The converse is also true.
- (2) The perpendicular bisectors of two chords of a circle intersect at its centre.
- (3) There can be one and only one circle passing through three or more non-collinear points.
- (4) If two circles intersect in two points then the line through the centers is the perpendicular bisector of the common chord.
- (5) If two chords of a circle are equal, then the centre of the circle lies on the angle bisector of the two chords.
- (6) Equal chords of a circle or congruent circles are equidistant from the centre.
- (7) Equidistant chords from the centre of a circle are equal to each other in terms of their length.
- (8) The degree measure of an arc of a circle is twice the angle subtended by it at any point on the alternate segment of the circle. This can be clearly seen in the following figure:

With respect to the arc

$$AB, \angle AOB = 2\angle ACB$$



- (9) Any two angles in the same segment are equal.
Thus $\angle ACB = \angle ADB$
- (10) The angle subtended by a semi-circle is right angle. Conversely, the arc of a circle subtending a right angle at any point of the circle in its alternate segment is a semi-circle.
- (11) Any angle subtended by a minor arc in the alternate segment is acute, and any angle subtended by a major arc in the alternate segment is obtuse. In the figure below.



$\angle ABC$ is acute and

$\angle ADC$ = obtuse

$$\text{Also } \theta_1 = 2\angle B$$

$$\text{And } \theta_2 = 2\angle D$$

$$\therefore \theta_1 + \theta_2 = 2(\angle B + \angle D)$$

$$= 360^\circ = 2(\angle B + \angle D)$$

$$\text{or } \angle B + \angle D = 180^\circ$$

Or sum of opposite angles of a cyclic quadrilateral is 180°

MENSURATION

The following formulae hold true in the area of mensuration:

1. Cuboid

A cuboid is a three-dimensional box. Its volume is defined by the virtue of its length l breadth b and height h . It can be visualized as a room which has its length, breadth and height different from each other.

$$(1) \text{ Total surface area of a cuboid} = 2(lb + bh + lh)$$

$$(2) \text{ Volume of cuboid} = lbh$$

2. Cube of Side 'S'

A cube is cuboids which has all its edges equal i.e. length = breadth = height = s

$$(1) \text{ Total surface area of a cube} = 6s^2$$

$$(2) \text{ Volume of the cube} = s^3$$

3. Prism

A prism is a solid which can have any polygon at both its ends. Its dimensions are defined by the dimensions of the polygon at its ends and its height.

$$(1) \text{ Lateral surface area of a tight prism} = \text{perimeter of base} \times \text{height}$$

$$(2) \text{ Volume of a right prism} = \text{area of base} \times \text{height}$$

$$(3) \text{ Whole surface of a right prism} = \text{Lateral surface of the prism} + \text{area of the two plane ends.}$$

4. Cylinder

Is a solid which has both its ends in the form of a circle. Its dimensions are defined in the form of the radius of the base (r) and the height h . A gas cylinder is a close approximation of a cylinder.

$$(1) \text{ Curved surface of a right cylinder} = 2\pi rh \text{ where } r \text{ is the radius of the base and } h \text{ the height.}$$

$$(2) \text{ Whole surface of a right circular cylinder} = 2\pi rh + 2\pi r^2$$

$$(3) \text{ Volume of a right circular cylinder} = \pi r^2 h$$

5. Pyramid

A pyramid is a solid which can have any polygon as its base and its edges converge to a single apex. Its dimensions are defined by the dimensions of the polygon at its base and the length of its lateral edges which lead to the apex. The Egyptian pyramids are example of pyramids.

- (1) Slant surface of a pyramid = $\frac{1}{2} \times \text{Perimeter of the base} \times \text{slant height}$
- (2) Whole surface of a pyramid = Slant surface + area of the base
- (3) Volume of a pyramid = $\frac{\text{area of the base} \times \text{height}}{3}$

6. Cone

A cone is a solid which has a circle at its base and a slanting lateral surface that converges at the apex. Its dimensions are defined by the radius of the base (r) the height (h) and the slant height (l). A structure similar to a cone is used in ice cream cones

- (1) Curved surface of a cone = $\pi r l$ where l is the height
- (2) Whole surface of a cone = $\pi r l + \pi r^2$
- (3) Volume of a cone = $\frac{\pi r^2 h}{3}$

7. Sphere

Is a solid in the form of a ball with radius r.

- (1) Surface area of a sphere = $4\pi r^2$
- (2) Volume of a sphere = $\frac{4}{3}\pi r^3$

POWER AND EXPONENTS:

The exponents tell us that how many times the number should be multiplied. It is called the Exponential form. Exponential form is written like this:

$$\underbrace{10}_{\text{Base}}^9 \Rightarrow \text{Exponent}$$

Power

Here 10 is the base and 9 is the exponent and this complete number is the power. We pronounce it as 10 raised to the power 9. The exponent could be positive or negative.

This tells us that the number 10 will be multiplied 9 times, like, $10 \times 10 \times 10 \times 10 \times 10 \times 10 \times 10 \times 10 \times 10$

Powers with Negative Exponents

The exponents could be negative also and we can convert them in positive by the following method.

$$100^{-3} = \frac{1}{100^3}$$

This shows that for any non-zero negative integers a,

$$a^{-m} = \frac{1}{a^m}$$

where m is the positive integer and a^m is the multiplicative inverse of a^{-m} .

Laws of Exponents

If we have a and b as the base and m and n as the exponents, then

$$a^m \times a^n = a^{m+n}$$

$$(a^m)^n$$

$$\frac{a^m}{a^n} = a^{m-n}, m > n$$

$$a^m b^m = (ab)^m$$

$$a^0 = 1$$

$$a^1 = a$$

Use of Exponents to Express Small Numbers in Standard Form:

Sometimes we need to write the numbers in very small or large form and we can use the exponents to represent the numbers in small numbers.

1. Standard form to write the natural numbers like xyz000000.....

Step 1: First of all count the number of digits from left leaving only the first digit.

Step 2: To write it in exponent or standard form, write down the first digit.

Step 3: If there are more digits in the number then put a decimal after the first digit and then write down the other digits until the zero comes. And if there are no digits after the first digit then skip this step.

Step 4: Now place a multiplication sign and then write down the counted digits in the first step as the exponent to the base number 10.

$$\begin{array}{c} 2 \times 10^9 \\ 2.000000000 \\ \underbrace{}_{1 \ 2 \ 3 \ 4 \ 5 \ 6 \ 7 \ 8 \ 9} \\ 2,000,000,000 \end{array}$$

Example:

Express 1730000000000 in exponent form.

Solution:

In standard form, the number 1730000000000 will be written as 1.73×10^{12} .

2. Standard form to write decimal numbers like 0.00000.....xyz.

Step 1: First of all count the number of digits from the decimal point to the last digit.

Step 2: If there is only one digit after the zeros then simply write down that digit. Place a multiplication sign and write down the counted digits in step-1 with a negative sign as the exponent to base number 10.

Step 3: If there are two or more non-zero digits at the end of the number. Then, write down the digits followed by a decimal point after the first digit and the other non-zero digits.

Step 4: Now calculate the number of digits in the first step and minus the number of digits appearing after the decimal point.

Step 5: Place a multiplication sign and write down the counted digits in step-4 with a negative sign as an exponent to base number 10.

$$\begin{array}{r} 2 \times 10^{-9} \\ \text{000000002} \\ \text{9 8 7 6 5 4 3 2 1} \\ \text{0.000000002} \end{array}$$

Example:

Express 0.000000000000073 in exponent form.

Solution:

In standard form, the number will be written as 7.3×10^{-14} .

Comparing Very Large and Very Small Numbers:

To compare the very large or very small numbers we need to make their exponents same. When their exponents are the same then we can compare the numbers and check which number is large or small.

Example:

Compare the two numbers 4.56×10^8 and 392×10^7 .

Solution:

To compare these numbers, we need to make their exponents same.

$$4.56 \times 10^8$$

$$392 \times 10^7 = 39.2 \times 10^8$$

As the exponents are the same, we can easily see that which number is larger.

$$392 \times 10^7 > 4.56 \times 10^8$$

Note: To add and subtract also we need to make their exponents same and then they can be easily added or subtracted.

LOGARITHMS

Determinants

⇒ The logarithm of a given number base is then index of power to which the base must be raised in order to equal the given number.

⇒ If $a > 0$ and $a \neq 1$, then logarithm of a positive number N is defined as the Index x of that power ' a ' which equals N i.e., $\log_a N = x$ iff $a^x = N \Rightarrow a^{\log_a N} = N$, $a > 0$, $a \neq 1$ and $N > 0$

⇒ It is also known as fundamental logarithmic Identity.

⇒ Its domain is $(0, \infty)$ and range is $\mathbb{R}$, a is called the base of the logarithmic function.

⇒ When base ' e ' then the logarithmic function is called natural or Napierian logarithmic function and when is 10, then it is called common logarithmic function.

Characteristic and Mantissa

1. The integral part of a logarithm is called the characteristic and the fractional part is called mantissa.

$$\begin{array}{ccc} \log_{10} N = \text{integer} & + & \text{fraction (+ve)} \\ \downarrow & & \downarrow \\ \text{Characteristics} & & \text{Mantissa} \end{array}$$

2. If the characteristics of $\log_{10} N$ be n , then the number of digits in N is $(n + 1)$.
3. The mantissa part of $\log$ of a number is always kept positive.
4. If the characteristics of $\log_{10} N$ be $(-n)$ then there exists $(n-1)$ number of zeros after decimal part of N .

Properties of Logarithms

1. $\log_a (mn) = \log_a m + \log_a n$
2. $\log_a \left(\frac{m}{n}\right) = \log_a m - \log_a n$
3. $\log_a \left(\frac{1}{n}\right) = -\log_a n$
4. $a \log_a m = m$
5. $\log_a \left(\frac{1}{n}\right) = -\log_a n$
6. $\log_a a = 1, \log_a 1 = 0$
7. $\log_a b \cdot \log_b a = 1 \Rightarrow \log_a b = \frac{1}{\log_b a}$

Logarithmic Inequalities

1. If $a > 1, P > 1 \Rightarrow \log_a P > 0$
2. If $a > 1, 0 < P < 1 \Rightarrow \log_a P < 0$
3. If $0 < a < 1, P > 1 \Rightarrow \log_a P < 0$
4. If $0 < a < 1, 0 < P < 1 \Rightarrow \log_a P > 0$
5. If $p > a > 1 \Rightarrow \log_a P > 1$
6. If $a > p > 1 \Rightarrow 0 < \log_a P < 1$
7. $\log_p a > \log_p b = a \geq b$ if base p is positive and > 1 or $a \leq b$ if base p is positive and < 1 i.e., $0 < p < 1$
8. $\log_m a < b \Rightarrow \begin{cases} a < m^b; & \text{if } m > 1 \\ a > m^b; & \text{if } 0 < m < 1 \end{cases}$
9. If $\log_m a > b \Rightarrow \begin{cases} a > m^b; & \text{if } m > 1 \\ a < m^b; & \text{if } 0 < m < 1 \end{cases}$
10. If $0 < p < a < 1 \Rightarrow \log_a p > 1$

EXERCISE

MCQ TYPE QUESTIONS

1. If one root of a quadratic equation is $3 + \sqrt{2}$, then quadratic equation is

(a) $x^2 - 6x - 7 = 0$ (b) $x^2 + 6x + 7 = 0$
 (c) $x^2 + 6x - 7 = 0$ (d) $x^2 - 6x + 7 = 0$

2. Three consecutive positive integers such that square of their sum exceeds the sum of their squares by 214, are

(a) 5, 6, 7 (b) 10, 11, 12
 (c) 9, 10, 11 (d) 7, 8, 9

3. A and B attempt to solve a quadratic equation of the form $ax^2 + bx + c = 0$. A starts with a wrong value of b and gets the roots as -3 and -5. B starts with a wrong value of C and gets the roots as 6 and 2. The correct roots are

(a) 3, 4 (b) 3, 5
 (c) 4, 6 (d) 2, 3

4. What is the value of

$$\frac{1}{5 + \frac{1}{5 + \frac{1}{5 + \dots}}}$$

(a) $\frac{5 + \sqrt{29}}{2}$ (b) $\frac{5 - \sqrt{29}}{2}$
 (c) $\frac{5 \pm \sqrt{29}}{2}$ (d) 7

5. What is the number of triplets (a, b, c) where a, b, c are positive integers, which satisfy the simultaneous equations?

(i) $ab + bc = 44$
 (ii) $ac + bc = 23$ is/are

(a) 1 (b) 2
 (c) 4 (d) Indefinitely many

6. A man sitting beside me at a hotel ate idlis one after the other by ordering plate by plate. He said to me after drinking some water the last one I ate was my hundredth idli in last five days. Each day I ate 6 more than the previous day. How many he ate yesterday?

(a) 26 (b) 24 (c) 22 (d) 20

7. If $\log_3 \log_4 x > 0$, then

(a) $x > 1$ (b) $x > 4$
 (c) $x > 64$ (d) none of these

8. $\log \frac{1}{4} (a^2 - 1) < \log_{1/2} (a + 1)^2$ then:

(a) $a < 1$
 (b) $a < -1$
 (c) $a > 1$
 (d) none of these

9. If $\log_e \left(\frac{a+b}{2} \right) = \frac{1}{2} (\log_e a + \log_e b)$, then:

(a) $a = b$ (b) $a = \frac{b}{2}$
 (c) $2a = b$ (d) $a = \frac{b}{3}$

10. If $2 \log (x + 1) - 10 \log (x^2 - 1) = \log^2$, then x equals

(a) 1 (b) 0 (c) 2 (d) 3

11. The number $\log 2^7$ is:

(a) an integer (b) a rational number
 (c) an irrational number (d) a prime number

12. If $y = 2^{1/\log_x (8)}$, then x is equal to:

(a) y (b) y^2
 (c) y^3 (d) none of these

13. If $\log_{05} \sin x = 1 - \log_{05} \cos x$, then number of solution of $x \in [-2\pi, 2\pi]$ is:

(a) 1 (b) 2 (c) 3 (d) 4

14. If a, b, c are in G.P. then $\log ax^x, \log_{bx} x, \log_{cx} x$ are in:

(a) GP (b) HP
 (c) AP (d) none of these

15. Consider the following statements:

1. Solution of the inequality $\log_3 (x^2 - 11x + 43) < 2$ is (0, 2)

2. If $[x - 1]^{(\log_3 x^2 - 2 \log x^9)} = (x - 1)^7$ then x is 2 and 81. Which of these is/are correct?

(a) only (1) (b) only 2
 (c) both of these (d) none of these

16. $\log \cos x \sin x \geq 2$ and $x \in [0, 3\pi]$, then sin x lines in the interval:

(a) $\left[0, \frac{1}{2}\right]$ (b) $\left[\frac{\sqrt{5}-1}{2}, 1\right]$
 (c) $\left[0, \frac{\sqrt{5}-1}{2}\right]$ (d) none of these

17. If $\log_2 x + \log_2 y \geq 6$, then the least value of (x + y) is

(a) 4 (b) 9
 (c) 16 (d) 32

18. If $4^{\log 3 \sqrt{3}} + 9^{\log 2^2} = 10^{\log_x 83}$, ($x \in \mathbb{R}$), then x is:

(a) 4 (b) 9
 (c) 10 (d) none of these

19. Which of the following is not true?

(a) $\frac{1}{\log_3 \pi} + \frac{1}{\log_4 \pi} > 2$
 (b) $\log_3 5$ is an irrational number
 (c) $\log \sqrt{8x} = \frac{10}{3} \Rightarrow x = 16$
 (d) If $\log_x (a^2 + 1) < 0$, ($a \neq 0$) then $0 < x < 1$

20. If $y = \frac{1}{a^{1-\log ax}}$ and $z = \frac{1}{a^{1-\log ay}}$ then x is equal to:

- (a) $\frac{1}{a^{1+\log_a z}}$ (b) $\frac{1}{a^{z+\log_a z}}$
 (c) $\frac{1}{a^{1-\log_a z}}$ (d) none of these

21. The least value of n in order that the sum of first n terms of an infinite series $1 + \frac{3}{4} + \left(\frac{3}{4}\right)^2 + \left(\frac{3}{4}\right)^3 + \dots$ should differ from the sum of the series by less than 10^{-6} is

(Given: $\log_{10} 2 = 0.30103$, $\log_{10} 3 = 0.47712$)

- (a) 14 (b) 27
 (c) 53 (d) 57

22. Solution of the equation $x \log^{x^2} = \log 3(x+y)$ and $x^2 + y^2 = 65$ is:

- (a) $x = 8, y = 1$
 (b) $x = 1, y = 8$
 (c) $(x = 8, y = 1); (x = 1, y = 8)$
 (d) none of the above

23. The identity $\log_a n \log_b n + \log_b n \log_c n + \log_c n \log_a n$ is:

- (a) $\frac{\log_a n \log_b n \log_c n}{\log_{abc} n}$
 (b) $\frac{\log_{abc} n}{\log_a n}$
 (c) $\frac{\log_b n}{\log_{abc} n}$
 (d) none of these

24. The least value of the expression $2 \log_{10} x - \log x$ (0.01) for $x > 1$, is:

- (a) 10 (b) 2
 (c) -0.01 (d) none of the above

25. There are two numbers such that the sum of twice the first number and thrice the second number is 100 and sum of thrice the first number and twice the second number is 120. Which is the largest number?

- (a) 32 (b) 12
 (c) 14 (d) 35

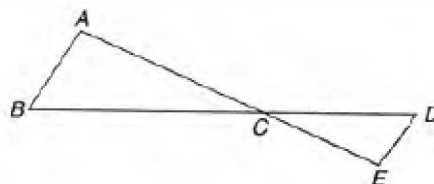
26. Three mangoes, four guavas and five watermelons cost Rs. 750. Ten watermelon, six mangoes and nine guavas cost Rs. 1580. What is the cost of six mangoes, ten watermelon and four guavas?

- (a) 1280 (b) 1180
 (c) 1080 (d) Cannot be determined

27. The sum of the digits of a two digit number is 9. The digit at the tens place is 4 less than $\frac{1}{6}$ th of the number. Find the number.

- (a) 45 (b) 54
 (c) 90 (d) None of these

28. In the figure, $\triangle ABC$ is similar to $\triangle EDC$



If we have $AB = 4\text{cm}$,
 $ED = 3\text{cm}$, $CE = 4.2$ and
 $CD = 4.8\text{cm}$,

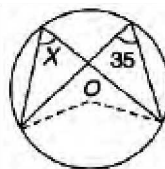
Find the value of CA and CB

- (a) 6cm, 6.4 cm (b) 4.8 cm, 6.4 cm
 (c) 5.4 cm, 6.4 cm (d) 5.6 cm, 6.4 cm

29. Two poles of height 6 m and 11 m stand vertically upright on a plane ground. If the distance between their foot is 12 m, find the distance between their tops.

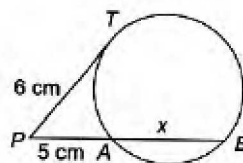
- (a) 12 cm (b) 14 cm
 (c) 13 cm (d) 11 cm

30. If O is the centre of circle, find $\angle x$



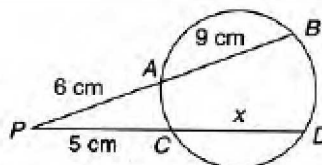
- (a) 35° (b) 30°
 (c) 39° (d) 40°

31. Find the value of x in the given figure.



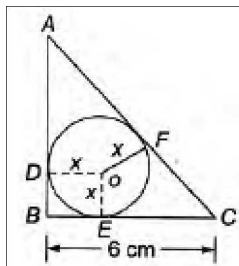
- (a) 2.2 cm (b) 1.6 cm
 (c) 3 cm (d) 2.6 cm

32. Find the value of x in the given figure.

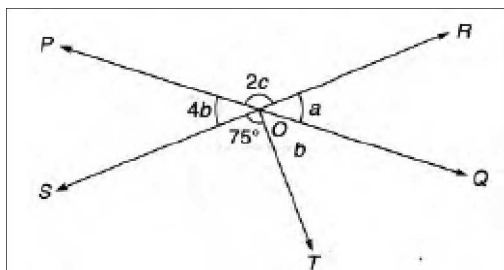


- (a) 13 cm (b) 12 cm
 (c) 16 cm (d) 15 cm

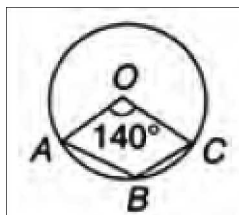
33. ABC is a right angled triangle with $BC = 6\text{ cm}$ & $AB = 8\text{ cm}$. A circle with centre O and radius x has been inscribed in $\triangle ABC$. What is the value of x .



- (a) 2.4 cm (b) 2 cm
(c) 3.6 cm (d) 4 cm
34. In the given figure find $\angle ADB$
(a) 132° (b) 144°
(c) 48° (d) 96°
35. In the given two straight line PQ and RS intersect each other at O. If $\angle SOT = 75^\circ$, find the value of a , b , and c .

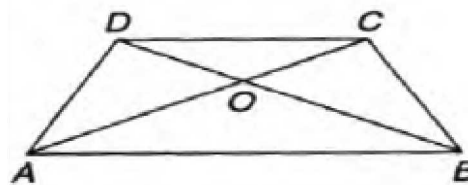


- (a) $a = 84^\circ$, $b = 21^\circ$, $c = 48^\circ$
(b) $a = 48^\circ$, $b = 20^\circ$, $c = 50^\circ$
(c) $a = 72^\circ$, $b = 24^\circ$, $c = 54^\circ$
(d) $a = 64^\circ$, $b = 28^\circ$, $c = 45^\circ$
36. In the following figure, it is given that O is the centre of the circle and $\angle AOC = 140^\circ$. Find $\angle ABC$

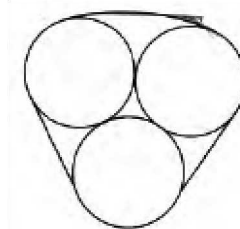


- (a) 110° (b) 120°
(c) 115° (d) 130°
37. In a triangle ABC, point D is on side AB and point E is on side AC, such that BCED is a trapezium. $DE : BC = 3:5$. Calculate the ratio of the area of $\triangle ADE$ and the trapezium BCED
- (a) 3:4 (b) 9:16
(c) 3:5 (d) 9:25

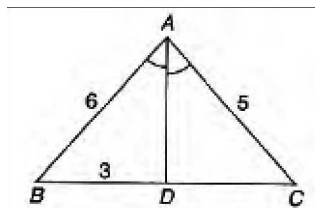
38. In the adjoining figure, ABCD is a trapezium in which $AB \parallel DC$ and $AB = 3DC$. Determine the ratio of the areas of $(\triangle AOB \text{ and } \triangle COD)$.



- (a) 9:1 (b) 1:9
(c) 3:1 (d) 1:3
39. The diagram below represents three circular garbage cans, each of diameter 2 m. The three cans are touching as shown fig, in meters, the perimeter of the rope encompassing the three cans.

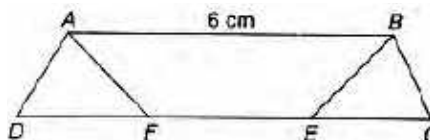


- (a) $2\pi + 6$ (b) $3\pi + 6$
(c) $4\pi + 6$ (d) $6\pi + 6$
40. In the above figure, AD is the bisector of $\angle BAC$, $AB = 6\text{ cm}$, $AC = 5\text{ cm}$ and $BD = 3\text{ cm}$. Find DC,



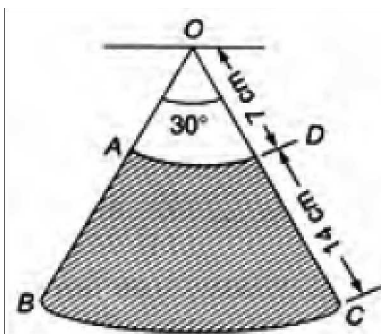
- (a) 11.3 cm (b) 2.5 cm
(c) 3.5 cm (d) 4 cm
41. ABCD is a trapezium in which AB is parallel to DC, $AD = BC$, $AB = 6\text{ cm}$, $AB = EF$ and $DF = EC$. If two lines AF and BE are drawn so that area of

ABEF is half of ABCD. Find $\frac{DF}{CD}$.



- (a) $\frac{1}{4}$ (b) $\frac{1}{3}$
(c) $\frac{2}{5}$ (d) $\frac{1}{6}$

42. The diagram represents the area swept by the wiper of a car. With the dimensions given in the figure, calculate the shaded area swept by the wiper.



- (a) 102.67 cm (b) 205.34 cm
(c) 51.33 cm (d) 208.16 cm

NUMERICAL TYPE QUESTIONS

- When $x^5 + x^4 + 5x^2 - 3$ is divided by $(x + 2)$, the remainder will be _____
- Pressure of a gas is inversely proportional to the volume. If pressure of oxygen is 84 atm when the volume is 1000 cm^3 , then volume when the pressure is 28 atm will be _____ cm^3
- The value of 'a' will make $x^2 - ax + 25$ a perfect square is _____
- The value of $\alpha^3 + \beta^3$ if α and β are roots of the equation : $2x^2 - 7x - 4 = 0$ is _____
- Sum of Anita's and Sunita's age is 43 years. 11 years hence, Anita's age will be $\frac{7}{6}$ times Sunita's age then. Sunita's present age is _____ years
- In a school trip, the students were accommodated in two rooms A and B. If 5 students are shifted from room A to room B then there will be equal number of students in both the rooms. But if 5 students from room B is shifted to room A, then the resulting number of students in room A is double the number of students left in room B. Total number of students are _____

7. A band of workers unloaded a certain number of boxes from their ship. The number of workers were between 50 and 60 (both inclusive). Each worker handled exactly 11 boxes, and each box was handled by exactly 7 workers. Exactly _____ number of boxes were there?

8. If $ax = b, by = c, cz = a$, then value of xyz is _____
9. The value of

$$\frac{1}{\left(1 + \frac{xb}{xa} + \frac{xc}{xa}\right)} + \frac{1}{\left(1 + \frac{xa}{xb} + \frac{xc}{xb}\right)} + \frac{1}{\left(1 + \frac{xb}{xc} + \frac{xa}{xc}\right)}$$

is _____

10. A palm tree was 90 cm high, when it was planted. It grows by an equal number of cm each year, and at the end of the seventh year it was one ninth taller than at the end of the sixth year. The height of tree in cm at the end of the twelfth year is _____
11. A bottle and its cork together cost ₹ 1.10, and the bottle costs ₹ 1.00 more than its cork. The price of the bottle is _____
12. The value of $(100 - 1)(100 - 2)(100 - 3) \dots (100 + 1)(100 + 2)(100 + 3)$ is _____

13. The value of
$$\frac{2 + \frac{1}{3 + \frac{4}{5}}}{2 + \frac{1}{3 + \frac{1}{1 + \frac{1}{4}}}}$$
 is _____

14. Three integers, a, b and c, are such that $\frac{b}{c} = 3$.

The fraction $\frac{b}{a}$ is in its lowest form. The _____ is the possible value of a.

ANSWERS

MCQ Type Questions

1. (d) 2. (a) 3. (b) 4. (a) 5. (b) 6. (a) 7. (b) 8. (b) 9. (a) 10. (d)
 11. (c) 12. (c) 13. (b) 14. (b) 15. (b) 16. (c) 27. (c) 18. (c) 19. (c) 20. (c)
 21. (c) 22. (a) 23. (a) 24. (d) 25. (a) 26. (b) 27. (b) 28. (d) 29. (c) 30. (a)
 31. (a) 32. (a) 33. (b) 34. (a) 35. (a) 36. (a) 37. (b) 38. (a) 39. (a) 40. (b)
 41. (b) 42. (a)

Numerical Type Questions

1. 1 2. 3000 3. 10 4. $\frac{511}{8}$ 5. 19 6. 60 7. 88 8. 1 9. 1
 10. 450 11. 1.05 12. 0 13. 1 14. 104

EXPLANATIONS

MCQ TYPE QUESTIONS

1. If one root = $3 + \sqrt{2}$, then other root = $3 - \sqrt{2}$
 $\therefore$ Sum of roots = $3 + \sqrt{2} + 3 - \sqrt{2} = 6$
 $\therefore$ Product of roots
 $= (3 + \sqrt{2})(3 - \sqrt{2}) = 9 - 2 = 7$
 Quadratic equation is
 $x^2 - (\text{Sum of roots})x + \text{Product of roots} = 0$
 i.e., $x^2 - 6x + 7 = 0$
2. Let consecutive positive integers be $x - 1$, x and $x + 1$
 $\therefore [x - 1 + x + x + 1]^2$
 $= (x - 1)^2 + x^2 + (x + 1)^2 + 214$
 $(3x)^2 = x^2 + 1 - 2x + x^2 + x^2 + 2x + 1 + 214$
 $9x^2 = 3x^2 + 216$
 $6x^2 = 216$
 $x^2 = 36$; $x = \pm 6$
 But x is a positive integer, hence $x = 6$
 So, numbers are 5, 6, and 7.
3. Equation formed by A with the roots -3 and -5 is
 $x^2 - (-8x) + 15 = 0$
 i.e., $x^2 + 8x + 15 = 0$
 But value of B is wrong.
 Hence coefficient of x is wrong.
 Equation formed by B with roots 6 and 2 is
 $x^2 - 8x + 12 = 0$
 But the value of C is wrong.
 hence constant term is wrong.
 So, correct equation is
 $x^2 - 8x + 15 = 0$.
 Hence, real roots are 3 and 5.

4. Let

$$x = 5 + \frac{1}{5 + \frac{1}{5 + \frac{1}{5 + \dots}}}$$

$$\text{i.e., } x = 5 + \frac{1}{x} = \frac{5x + 1}{x}$$

$$\Rightarrow x^2 = 5x + 1$$

$$\Rightarrow x^2 - 5x - 1 = 0$$

$$\Rightarrow x = \frac{5 \pm \sqrt{(-5)^2 - 4(-1)(1)}}{2}$$

$$= \frac{5 \pm \sqrt{25 + 4}}{2} = \frac{5 \pm \sqrt{29}}{2}$$

Since, all the terms are positive. The answer has to be $\frac{5 + \sqrt{29}}{2}$.

5. Equation (ii) is $c(a + b) = 23$, a prime number.
 So, 2 factors must be 1, 23.
 Since a, b are positive, therefore $a + b > 1$
 $\Rightarrow c = 1$; $a + b = 23$
 Put $c = 1$; $b = 23 - a$ in (i)
 $\Rightarrow a^2 - 22a + 21 = 0$
 $\Rightarrow a = 1$ or 21
 $\Rightarrow b = 22$ or 2
 $\therefore$ Solution sets are $(1, 22, 1)$; $(21, 2, 1)$.
6. First day the number of idlis he ate be x
 Second day the count is $(x+6)$
 Third day ----- $(x+12)$
 Fourth day ----- $(x+18)$

Fifth day -----(x+24)

Total is $5x + 6(1+2+3+4) = 100$

$$5x + 60 = 100$$

$$x = 8$$

Day	Idlis
1	8
2	14
3	20
4	26
5	32

So, on fourth day, number of idlis the man ate were 26.

7. We have $\log_3 \log_4 x > 0$

$$\Rightarrow \log_4 x > 1 \Rightarrow x > 4'$$

$$\Rightarrow x > 4$$

8. $\log_{1/4} (a^2 - 1) < \log_{(1/4)} (a + 1)^2$

$$\Rightarrow a^2 - 1 > (a + 1)^2$$

$$\Rightarrow a^2 - 1 > a^2 + 1 + 2a$$

$$\Rightarrow 2a + 2 < 0 \Rightarrow a < -1$$

9. $\log\left(\frac{a+b}{2}\right) = \frac{1}{2}(\log_e a + \log_e b)$

$$\Rightarrow \frac{a+b}{2} = \sqrt{ab}$$

$$\Rightarrow a + b - 2\sqrt{ab} = 0$$

$$\Rightarrow \sqrt{a} = \sqrt{b} \Rightarrow a = b$$

10. By definition $x \neq 1, = 1$

Given equation can be written as

$$\log \frac{(x+1)^2}{x^2-1} = \log 2 \Rightarrow \frac{x+1}{x-1} = 2 \Rightarrow x = 3$$

11. Let $x = \log 2^7$

$$\Rightarrow 2^x = 7$$

which is only possible for irrational number.

12. $y = 2^{1/\log x(8)}$

$$\Rightarrow y = 2^{\log 8x} \Rightarrow y = 2^{\log 2^3 x}$$

$$\Rightarrow x = y^3$$

13. $\log 0.5 \sin x + \log 0.5 \cos x = 1$

$$\Rightarrow \log 0.5 \sin x \cos x = 1$$

$$\Rightarrow \sin x \cos x = \frac{1}{2} \Rightarrow \sin 2x = 1$$

$$\Rightarrow 2x = n\pi + (-1)^n \frac{\pi}{2} \Rightarrow x = \frac{n\pi}{2} + (-1)^n \frac{\pi}{4}$$

Since $\log_{0.5} \sin x$ and $\log_{0.5} \cos x$ are real.

$\therefore \sin x$ and $\cos x$ must lie in first quadrant

$$\therefore x = \frac{\pi}{4}, -2\pi + \frac{\pi}{4}$$

Hence, the number of solutions is 2.

$$14. \log_{ax} x = \frac{1}{\log_x ax} \Rightarrow \log_{ax} x = \frac{1}{1 + \log_x a}$$

If a, b, c are in G.P

$$\Rightarrow \log_x a, \log_x b, \log_x c \text{ are in AP}$$

$$\Rightarrow 1 + \log_x a, 1 + \log_x b, 1 + \log_x c \text{ are in AP}$$

$$\Rightarrow \frac{1}{1 + \log_x a}, \frac{1}{1 + \log_x b}, \frac{1}{1 + \log_x c} \text{ are in HP.}$$

$$\Rightarrow \log_{ax} x, \log_{bx} x, \log_{cx} x \text{ are in HP.}$$

15. (1) $a \log_5 (x^2 - 11x + 43) < 2$

$$\text{and } x^2 - 11x + 43 > 0$$

$$\Rightarrow x^2 - 11x + 43 < 5^2$$

$$\text{and } \left(x - \frac{11}{2}\right)^2 + \frac{51}{4} > 0$$

$$\Rightarrow x^2 - 11x + 18 < 0$$

$$\text{and } (x-2)(x-9) < 0$$

solution is (2, 9)

(2) For domain $|x-1| \neq 0, \neq -1$

$$\text{Now, } |x-1| = 1, \Rightarrow x-1 = \pm 1$$

$$\Rightarrow x = 0, 2$$

$x = 0$ is not in the domain and $x = 2$ satisfies the given equation.

If $x-1 > 0$ i.e., $x > 1$ then the given equation becomes

$$2 \log_3 x - \frac{4}{\log_3 x} = 7 \Rightarrow x = 81, \frac{1}{\sqrt{3}}$$

But $\frac{1}{\sqrt{3}}$ being less than 1 is not valid.

Hence, $x = 2, 81$

16. $\log \cos x \sin x \geq 2 \Rightarrow \sin x \leq \cos^2 x$

$$\Rightarrow \sin x \leq 1 - \sin^2 x$$

$$\Rightarrow \sin^2 x + \sin x - 1 \leq 0$$

$$\Rightarrow \left(\sin x + \frac{1}{2}\right)^2 - \frac{5}{4} \leq 0$$

Also by definition of logarithm

$$\sin x > 0, \cos x > 0, \cos x \neq 1$$

$$\Rightarrow \sin x + \frac{1}{2} \leq \frac{\sqrt{5}}{2} \Rightarrow 0 < \sin x \leq \frac{\sqrt{5}-1}{2}$$

17. $\log_2 xy \geq 6 \Rightarrow xy \geq 2^6$

$$\text{Now } \frac{x+y}{2} \geq \sqrt{xy}$$

$$\Rightarrow \frac{x+y}{2} \geq 2^3 \Rightarrow x+y \geq 2^4 \Rightarrow x+y \geq 16$$

$$18. \quad 4^{\log_3 31/2} + 9^{\log_2 2^2} = 10^{\log x^{83}}$$

$$\Rightarrow 4^{1/2} + 9^2 = 10^{\log x^{83}}$$

$$\Rightarrow 2 + 81 = 10^{\log x^{83}}$$

$$\Rightarrow 83 = 10^{\log x^{83}}$$

$$\Rightarrow x = 10$$

$$19. (a) \frac{1}{\log_3 \pi} + \frac{1}{\log_4 \pi} = \log_\pi 3 + \log_\pi 4 = \log_\pi 12 > 2$$

$$\Rightarrow 12 > \pi^2$$

(b) $\log_3 5$ is an irrational number

$$(c) \log \sqrt{8} x = \frac{10}{3} \Rightarrow x = (\sqrt{8})^{10/3} = 2^5$$

$$\Rightarrow x = 32$$

$$(d) \log_x (a^2 + 1) < 0, a \neq 0 \Rightarrow a^2 + 1 > 1$$

So, log is negative

Hence, base is (0, 1).

20. From the given relation, we have

$$a = y^{1 - \log_a x} = z^{1 - \log_a y}$$

$$\therefore \log_a a = (1 - \log_a x) \log_a y$$

$$\text{and } \log_a a = (1 - \log_a y) \log_a z$$

$$\Rightarrow \log_a y (1 - \log_a x) = 1$$

$$\text{and } \log_a z (1 - \log_a y) = 1$$

$$\Rightarrow \log_a y = \frac{1}{1 - \log_a x}$$

$$\text{and } \log_a z = \frac{1}{1 - \log_a y}$$

$$\therefore \log_a z = \frac{1}{1 - \log_a y} = \frac{1}{1 - \frac{1}{1 - \log_a x}}$$

$$= \frac{1 - \log_a x}{-\log_a x}$$

$$\text{Now } \frac{1}{1 - \log_a z} = \frac{1}{1 + \frac{1 - \log_a x}{\log_a x}} = \log_a x$$

$$\log_a x = \frac{1}{1 - \log_a z}$$

$$x = a^{\frac{1}{1 - \log_a z}}$$

$$21. \text{ Series } 1 + \frac{3}{4} + \left(\frac{3}{4}\right)^2 + \dots + n \text{ terms} = \frac{1 - \left(\frac{3}{4}\right)^n}{1 - \frac{3}{4}}$$

$$\text{and } 1 + \frac{3}{4} + \left(\frac{3}{4}\right)^2 + \dots = \frac{1}{1 - \frac{3}{4}} = 4$$

$$\text{Now, } \frac{1 - \left(\frac{3}{4}\right)^n}{\frac{1}{4}} = 4 - 10^{-6}$$

$$\Rightarrow 1 - \left(\frac{3}{4}\right)^n = 1 - \frac{1}{4}(10^{-6})$$

$$\Rightarrow \left(\frac{3}{4}\right)^n = \frac{10^{-6}}{4}$$

$$\Rightarrow n \log_{10} \left(\frac{3}{4}\right) = \log_{10} 10^{-6} - \log_{10} 4$$

$$\Rightarrow n(0.47712 - 2 \times 0.30103) = -6 - 2 \times (0.30103)$$

$$\Rightarrow n = \frac{6.60206}{0.12494} = 53$$

22. Given that, $x^{\log x^2} = 2 = \log_3 (x + y)$

$$\Rightarrow x + y = 9 \text{ and } x^2 + y^2 = 65$$

$$\Rightarrow x = 8, y = 1 \text{ or } x = 1, y = 8$$

$$\text{But } x \neq 1$$

$$\therefore x = 8, y = 1$$

23. $\log_a n \log_b n + \log_b n \log_c n + \log_c n \log_a n$

$$= \frac{1}{\log_n a \log_n b} + \frac{1}{\log_n b \log_n c} + \frac{1}{\log_n c \log_n a} \left[\because \log_m n = \frac{1}{\log_n m} \right]$$

$$= \frac{\log_n c + \log_n a + \log_n b}{\log_n a \log_n b \log_n c}$$

$$= \frac{\log_n (abc)}{\log_n a \log_n b \log_n c} = \frac{\log_a n \log_b n \log_c n}{\log_{abc} n}$$

24. Here, $2 \log_{10} x - \log_x (10)^{-2} = 2 \log_{10} x + 2 \log x 10$

$$= 2 \log_{10} x + 2 \frac{1}{\log_{10} x}$$

$$= 2 \left\{ \log_{10} x + \frac{1}{\log_{10} x} \right\} \quad \dots (i)$$

Using AM $\geq$ GM we get

$$\frac{\log_{10} x + \frac{1}{\log_{10} x}}{2} \geq \left(\log_{10} x \frac{1}{\log_{10} x} \right)^{1/2}$$

$$\Rightarrow \log_{10} x + \frac{1}{\log_{10} x} \geq 2$$

$$\therefore 2 \log_{10} x - \log_x (0.01) \geq 4$$

$\therefore$ Least value is 4.

26. $3M + 4G + 5W = 750$... (i)

$6M + 9G + 10W = 1580$... (ii)

Now, if we look at the equation (i) and multiply it by 2, we get:

$6M + 8G + 10W = 1500$... (iii)

From equation (ii) and (iii) we get, $G = 80$

If we subtract the cost of 4 guavas from equation (iii) we would get:

$6M + 4G + 10W = 1500 - 320 = 1180$

27. Let, the two digits number be $10x + y$

$x + y = 9$... (1)

$x = \frac{1}{6}(10x + y) - 6$... (2)

Solving Eq (1) and (2) we get,

$x = 5$ and $y = 4$

Therefore, Number = 54

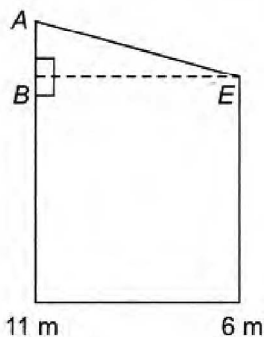
28. $\triangle ABC \sim \triangle CDE$

Then $\frac{AC}{CD} = \frac{BC}{CE} = \frac{AB}{DE}$

Then $\frac{AC}{4.8} = \frac{4}{3} = AC = 6.4$ cm and

$\frac{BC}{4.2} = \frac{4}{3} = BC = 5.6$ cm

29.



$BC = ED = 6$ m

$AB = AC - BC = 11 - 6 = 5$ m

$CD = BC = 12$ m

Then by Pythagoras theorem :

$AC^2 = AB^2 + BE^2 \Rightarrow AE = 13$ m

30. $\angle x = 35^\circ$ because angles subtended by an arc anywhere on the circumference are equal.

31. By the rule of tangents, we know :

$6^2 = (5 + n) 5$

$\Rightarrow 36 = 25 + 5x$

$\Rightarrow 11 = 5x$

$\Rightarrow x = 2.2$ cm

32. By the rule of chords, cutting externally, we get

$\Rightarrow (9 + 6)6 = (5 + x)5$

$\Rightarrow 90 = 25 + 5x$

$\Rightarrow 5x = 65$

$\Rightarrow x = 13$ cm

33. In radius = $\frac{\text{area}}{\text{semi perimeter}}$

$= \frac{24}{12} = 2$ cm

34. ADBC is a cyclic quadrilateral as well its four vertices are on the circumference of the circle. Also, the opposite angles of the cyclic quadrilateral are supplementary.

Therefore, $\angle ADB = 180 - 48^\circ = 132^\circ$

35. $\angle POS = \angle QOR$

(Vertically opposite angles)

So $a = 4b$

$\angle SOT + \angle TOQ + \angle QOR = 180^\circ$

(sum of angles on a line = 180°)

$4b + 2c = 180^\circ$

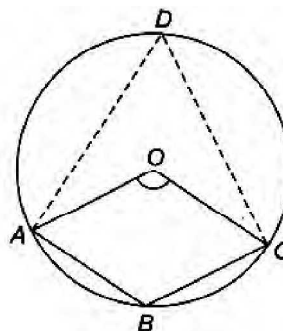
$84 + 2c = 180^\circ$

$\Rightarrow 2c = 96^\circ$

$\Rightarrow c = 48^\circ$

So $a = 84^\circ$, $b = 21^\circ$, $c = 48^\circ$

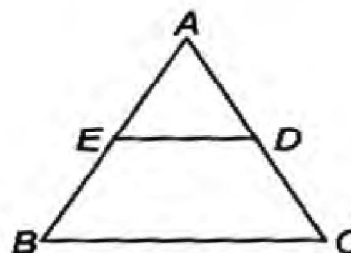
36.



$\angle ADC = \frac{140}{2} 70^\circ$ (because the angle subtended by an arc on the circumference is half of what it subtends at the centre.)

ABCD one cyclic quadrilateral So $\angle ABC = 180^\circ - 70^\circ = 110^\circ$ (because opposite angles of a cyclic quadrilateral are supplementary.)

37.



$\triangle ADE$ is similar to $\triangle ABC$ (AAA property)

$$ED : BC = 3 : 5$$

$$\text{Area of } \triangle ADE : \text{Area of } \triangle ABC = 9 : 25$$

Area of trapezium = area of $\triangle ABC$

$$\text{Area of } \triangle ADE = 25 - 9 = 16$$

$$\text{Thus, Area of } \triangle ADE : \text{Area of trapezium } EDBC = 9 : 16$$

38. $\angle DOC$ and $\triangle AOB$ are similar by AAA property)

$$AB : DC = 3 : 1$$

$$\text{So area of } \triangle AOB : \text{Area of } \triangle DOC = (3 : 1)^2$$

$$\Rightarrow 9 : 1$$

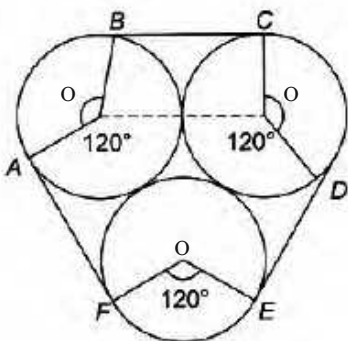
39. $\angle AOB = \angle COD = \angle FOE = 120^\circ$

Distance between 2 centers = 2 m

$$\therefore BC = DE = FA = 2\text{ m}$$

Perimeter of the figure = $BC + DE + FA +$ circumference of sector $AB, CO'B$ and $FO''E$.

But three equal sectors of $120^\circ = 1$ full circle of same radius



Therefore, perimeter of surface $2\pi r + BC + DE + FA = (2\pi + 6)m$

40. In the given figure, $\triangle ABD$ is similar to $\triangle ADC$

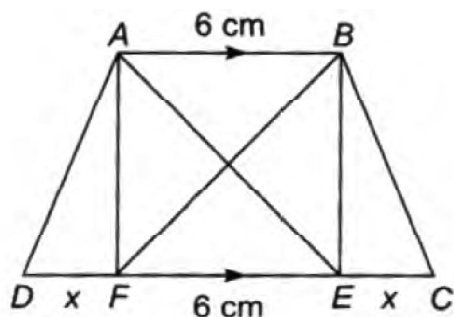
Then

$$\frac{AB}{BD} = \frac{AC}{DC}$$

$$\Rightarrow \frac{6}{3} = \frac{5}{DC}$$

$$\Rightarrow DE = 2.5 \text{ cm}$$

41.



In the above question

$$FE = AB = 6 \text{ cm}$$

$$\triangle ADF \cong \triangle BEC; \text{ so } DF = EC$$

$$\text{Let } DF = EC = x$$

Solving through options; e.g. options

$$(b) \frac{1}{3}; x = 6$$

Then by Pythagoras triplet $AF = 8$

$$\text{Area of } ABEF = 8 \times 6 = 48 \text{ cm}^2$$

$$\text{Area of } \triangle AFD + \triangle BEC$$

$$= 2 \times \frac{1}{2} \times 6 \times 8$$

$$\Rightarrow 48 \text{ cm}^2$$

$$\therefore \text{Area of } ABCD = 48 + 48 = 96 \text{ cm}^2$$

Hence the condition is proved.

42. Larger Radius (R) = $14 + 7 = 21 \text{ cm}$

$$\text{Smaller Radius } (r) = 7 \text{ cm}$$

Area of shaded portion

$$\pi R^2 = \frac{8}{360} - \frac{\pi r^2 \theta}{360}$$

$$\Rightarrow \frac{\pi \theta}{360} (21^2 - 7^2)$$

$$\Rightarrow 102.67 \text{ cm}$$

NUMERICAL TYPE QUESTIONS

$$\begin{aligned} 1. \quad p(x) &= x^5 + x^4 + 5x^2 - 3 \\ &= (-2)^5 + (-2)^4 + 5(-2)^2 - 3 \\ &= (-32) + 16 + 5 \times 4 - 3 = 1 \end{aligned}$$

$$\begin{aligned} 2. \quad p &\propto \frac{1}{v} \\ \therefore p &= \frac{k}{v} \\ \text{where, } k &= 84 \times 1000 \end{aligned}$$

$$\therefore v = \frac{k}{p} = \frac{84 \times 1000}{28} = 3000 \text{ cm}^3$$

$$\begin{aligned} 3. \quad \text{Middle term} &= 2 \sqrt{\text{first term} \times \text{last term}} \\ ax &= 2 \times x \times 5 = 10x \\ \therefore a &= 10 \end{aligned}$$

$$\begin{aligned} 4. \quad \alpha + \beta &= \frac{7}{2} \\ \alpha\beta &= -\frac{4}{2} = -2 \\ (\alpha + \beta)^3 &= \alpha^3 + \beta^3 + 3\alpha\beta(\alpha + \beta) \end{aligned}$$

$$\left(\frac{7}{2}\right)^3 = \alpha^3 + \beta^3 + 3(-2)\left(\frac{7}{2}\right)$$

$$\frac{343}{8} = \alpha^3 + \beta^3 - 21$$

$$\frac{343}{8} + 21 = \alpha^3 + \beta^3$$

$$\therefore \frac{511}{8} = \alpha^3 + \beta^3$$

5. Let Anita's present age be 'x' years and that of Sunita by 'y' years.

$$\therefore x + y = 43 \quad \dots(1)$$

$$\text{and } (x + 11) = \frac{7}{6}(y + 11)$$

$$\Rightarrow 6x + 66 = 7y + 77$$

$$\Rightarrow 6x - 7y = 11 \quad \dots(2)$$

Multiplying (1) by 6.

$$6x + 6y = 258 \quad \dots(3)$$

Subtracting (3) from (2),

$$-13y = -247$$

$$\therefore y = 19 \text{ years.}$$

6. Let the number of students in room A be x and number of students in room B be y.

$$\therefore x - 5 = y + 5$$

$$x - y = 10 \quad \dots(1)$$

$$\text{Also, } x + 5 = 2(y - 5)$$

$$\Rightarrow x - 2y = -15 \quad \dots(2)$$

Solving (1) and (2) we get,

$$x = 35 \text{ and } y = 25$$

$$\therefore x + y = 60$$

7. Let P be the number of workers, and let C be the number of boxes. Then, by counting the number of boxes handling in 2 different ways yields $11P = 7C$.

Since P and C are integers, it follows that P must be divisible by 7.

$$\text{Since } 50 \leq P \leq 60,$$

P must equal 56

$$\therefore C = 11 \times \frac{56}{7} = 88$$

8. $ax = b$

$$\Rightarrow (cz) x = b \quad \dots[\text{since } cz = a]$$

$$\Rightarrow (by) xz = b \quad \dots[\text{since } by = c]$$

$$\Rightarrow xyz = 1$$

$$\begin{aligned} 9. & \frac{1}{\left(1 + \frac{xb}{xa} + \frac{xc}{xa}\right)} + \frac{1}{\left(1 + \frac{xa}{xb} + \frac{xc}{xb}\right)} + \frac{1}{\left(1 + \frac{xb}{xc} + \frac{xa}{xc}\right)} \\ &= \frac{xa}{(xa + xb + xc)} + \frac{xb}{(xa + xb + xc)} + \frac{xc}{(xa + xb + xc)} \\ &= \frac{(xa + xb + xc)}{xa + xb + xc} = 1 \end{aligned}$$

10. Let tree grows x cm each year

Height of the tree at the end of the sixth year

$$= (90 + 6x) \text{ cm}$$

Growth in seventh year,

$$x = \frac{1}{9}(90 + 6x) \text{ cm}$$

$$x = 10 + \frac{2x}{3} \quad \therefore x = 30$$

Hence height of tree at the end of the twelfth year = $(90 + 12 \times 30) = 450 \text{ cm}$.

11. Let B = price of the bottle

C = price of the cork

$$\text{Given : } B + C = ₹ 1.10 \quad \dots(1)$$

$$\text{and } B - C = ₹ 1.00 \quad \dots(2)$$

From equations (1) and (2)

$$B = ₹ 1.05$$

$$C = ₹ 0.05.$$

12. $(100 - 1)(100 - 2)(100 - 3) \dots$

$$(100 - 100)(100 + 1)(100 + 2)(100 + 3)$$

$$\Rightarrow (100 - 1)(100 - 2)(100 - 3) \dots$$

$$(0)(100 + 1)(100 + 2)(100 + 3) = 0.$$

$$13. \frac{2 + \frac{1}{19}}{\frac{5}{2 + \frac{1}{3 + \frac{4}{5}}}} = \frac{2 + \frac{5}{19}}{2 + \frac{5}{19}} = 1$$

14. $\frac{b}{c} = 3 \Rightarrow b = 3c$, i.e., b is divisible by 3.

Also, $\frac{b}{c}$ cannot be reduced,
i.e., a is not divisible by 3.

SOLVED PAPER – 2016

1. Based on the given statements, select the appropriate option with respect to grammar and usage.

Statements :

I. The height of Mr. X is 6 feet.

II. The height of Mr. Y is 5 feet.

(a) Mr. X is longer than Mr. Y.

(b) Mr. X is more elongated than Mr. Y.

(c) Mr. X is taller than Mr. Y.

(d) Mr. X is lengthier than Mr. Y.

2. The students _____ the teacher on teachers' day for twenty years of dedicated teaching.

(a) facilitated

(b) felicitated

(c) fantasized

(d) facillitated

3. After India's cricket world cup victory in 1985, Shrotria who was playing both tennis and cricket till then, decided to concentrate only on cricket. And the rest is history.

What does the underlined phrase mean in this context?

(a) history will rest in peace

(b) rest is recorded in history books

(c) rest is well known

(d) rest is archaic

4. Given $(9 \text{ inches})^{1/2} = (0.25 \text{ yards})^{1/2}$, which one of the following statements is TRUE?

(a) 3 inches = 0.5 yards

(b) 9 inches = 1.5 yards

(c) 9 inches = 0.25 yards

(d) 81 inches = 0.0625 yards

5. S, M, E and F are working in shifts in a team to finish a project. M works with twice the efficiency of others but for half as many days as E worked. S and M have 6 hour shifts in a day whereas E and F have 12 hours shifts. What is the ratio of contribution of M to contribution of E in the project?

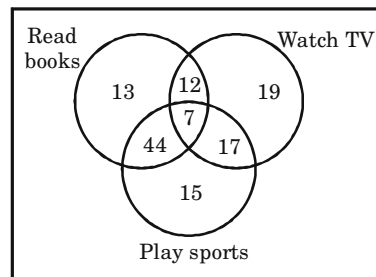
(a) 1 : 1

(b) 1 : 2

(c) 1 : 4

(d) 2 : 1

6. The Venn diagram shows the preference of the student population for leisure activities.



From the data given, the number of students who like to read books or play sports is ____.

(a) 44

(b) 51

(c) 79

(d) 108

7. Social science disciplines were in existence in an amorphous form until the colonial period when they were institutionalized. In varying degrees, they were intended to further the colonial interest. In the time of globalization and the economic rise of postcolonial countries like India, conventional ways of knowledge production have become obsolete.

Which of the following can be logically inferred from the above statements?

I. Social science disciplines have become obsolete.

II. Social science disciplines had a pre-colonial origin.

III. Social science disciplines always promote colonialism.

IV. Social science must maintain disciplinary boundaries.

(a) II only

(b) I and III only

(c) II and IV only

(d) III and IV only

8. Two and a quarter hours back, when seen in a mirror, the reflection of a wall clock without number markings seemed to show 1 : 30. What is the actual current time shown by the clock?

(a) 8:15

(b) 11:15

(c) 12:15

(d) 12:45

9. M and N start from the same location. M travels 10 km East and then 10 km North-East. N travels 5 km South and then 4 km South-East. What is the shortest distance (in km) between M and N at the end of their travel?

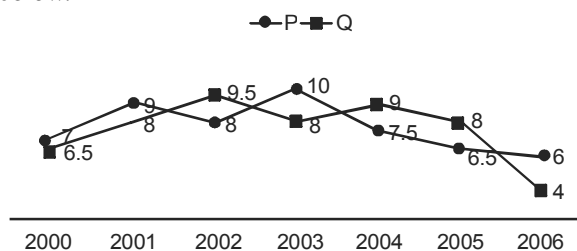
(a) 18.60

(b) 22.50

(c) 20.61

(d) 25.00

10. A wire of length 340 mm is to be cut into two parts. One of the parts is to be made into a square and the other into a rectangle where sides are in the ratio of 1:2. What is the length of the side of the square (in mm) such that the combined area of the square and the rectangle is a MINIMUM?
 (a) 30 (b) 40
 (c) 120 (d) 180
11. If I were you, I _____ that laptop. It's much too expensive.
 (a) won't buy (b) shan't buy
 (c) wouldn't buy (d) would buy
12. He turned a deaf ear to my request. What does the underlined phrasal verb mean?
 (a) ignored (b) appreciated
 (c) twisted (d) returned
13. Choose the most appropriate set of words from the options given below to complete the following sentence.
 _____ is a will, _____ is a way.
 (a) Wear, there, their
 (b) Were, their, there
 (c) Where, there, there
 (d) Where, their, their
14. $(x \% \text{ of } y) + (y \% \text{ of } x)$ is equivalent to _____.
 (a) 2 % of xy (b) 2 % of $(xy/100)$
 (c) xy % of 100 (d) 100 % of xy
15. The sum of the digits of a two digit number is 12. If the new number formed by reversing the digits is greater than the original number by 54, find the original number.
 (a) 39 (b) 57
 (c) 66 (d) 93
16. Two finance companies, P and Q, declared fixed annual rates of interest on the amounts invested with them. The rates of interest offered by these companies may differ from year to year. Year-wise annual rates of interest offered by these companies are shown by the line graph provided below.



If the amounts invested in the companies, P and Q, in 2006 are in the ratio 8 : 9, then the amounts received after one year as interests from companies P and Q would be in the ratio:

- (a) 2 : 3 (b) 3 : 4
 (c) 6 : 7 (d) 4 : 3
17. Today, we consider Ashoka as a great ruler because of the copious evidence he left behind in the form of stone carved edicts. Historians tend to correlate greatness of a king at his time with the availability of evidence today. Which of the following can be logically inferred from the above sentences?
 (a) Emperors who do not leave significant sculpted evidence are completely forgotten.
 (b) Ashoka produced stone carved edicts to ensure that later historians will respect him.
 (c) Statues of kings are a reminder of their greatness.
 (d) A king's greatness, as we know him today, is interpreted by historians.
18. **Fact 1** : Humans are mammals.
Fact 2 : Some humans are engineers.
Fact 3 : Engineers build houses.
 If the above statements are facts, which of the following can be logically inferred ?
 I. All mammals build houses.
 II. Engineers are mammals.
 III. Some humans are not engineers.
 (a) II only (b) III only
 (c) I, II and III (d) I only
19. A square pyramid has a base perimeter x , and the slant height is half of the perimeter. What is the lateral surface area of the pyramid?
 (a) x^2 (b) $0.75 x^2$
 (c) $0.50 x^2$ (d) $0.25 x^2$
20. Ananth takes 6 hours and Bharath takes 4 hours to read a book. Both started reading copies of the book at the same time. After how many hours is the number of pages **to be** read by Ananth, twice that **to be** read by Bharath? Assume Ananth and Bharath read all the pages with constant pace.
 (a) 1
 (b) 2
 (c) 3
 (d) 4

21. Which of the following is **CORRECT** with respect to grammar and usage?

Mount Everest is _____.

- (a) the highest peak in the world
(b) highest peak in the world
(c) one of highest peak in the world
(d) one of the highest peak in the world
22. The policeman asked the victim of a theft, "What did you _____?"
- (a) loose (b) lose
(c) loss (d) louse
23. Despite the new medicine's _____ in treating diabetes, it is not _____ widely.
- (a) effectiveness --- prescribed
(b) availability --- used
(c) prescription --- available
(d) acceptance --- proscribed
24. In a huge pile of apples and oranges, both ripe and unripe mixed together, 15% are unripe fruits. Of the unripe fruits, 45% are apples. Of the ripe ones, 66% are oranges. If the pile contains a total of 5692000 fruits, how many of them are apples?
- (a) 2029198 (b) 2467482
(c) 2789080 (d) 3577422
25. Michael lives 10 km away from where I live. Ahmed lives 5 km away and Susan lives 7 km away from where I live. Arun is farther away than Ahmed but closer than Susan from where I live. From the information provided here, what is one possible distance (in km) at which I live from Arun's place?
- (a) 3.00 (b) 4.99
(c) 6.02 (d) 7.01
26. A person moving through a tuberculosis prone zone has a 50% probability of becoming infected. However, only 30% of infected people develop the disease. What percentage of people moving through a tuberculosis prone zone remains infected but does not show symptoms of disease?
- (a) 15 (b) 33
(c) 35 (d) 37
27. In a world filled with uncertainty, he was glad to have many good friends. He had always assisted them in times of need and was confident that they would reciprocate. However, the events of the last week proved him wrong.

Which of the following inference(s) is/are logically valid and can be inferred from the above passage?

- (i) His friends were always asking him to help them.
(ii) He felt that when in need of help, his friends would let him down.
(iii) He was sure that his friends would help him when in need.
(iv) His friends did not help him last week.
- (a) (i) and (ii) (b) (iii) and (iv)
(c) (iii) only (d) (iv) only
28. Leela is older than her cousin Pavithra. Pavithra's brother Shiva is older than Leela. When Pavithra and Shiva are visiting Leela, all three like to play chess. Pavithra wins more often than Leela does. Which one of the following statements must be **TRUE** based on the above?
- (a) When Shiva plays chess with Leela and Pavithra, he often loses.
(b) Leela is the oldest of the three.
(c) Shiva is a better chess player than Pavithra.
(d) Pavithra is the youngest of the three.
29. If $q^{-a} = \frac{1}{r}$ and $r^{-b} = \frac{1}{s}$ and $s^{-c} = \frac{1}{q}$, the value of abc is ____.
- (a) $(rqs)^{-1}$ (b) 0
(c) 1 (d) $r + q + s$
30. P, Q, R and S are working on a project. Q can finish the task in 25 days, working alone for 12 hours a day. R can finish the task in 50 days, working alone for 12 hours per day. Q worked 12 hours a day but took sick leave in the beginning for two days. R worked 18 hours a day on all days. What is the ratio of work done by Q and R after 7 days from the start of the project?
- (a) 10:11 (b) 11:10
(c) 20:21 (d) 21:20
31. The chairman requested the aggrieved shareholders to _____ him.
- (a) bare with (b) bore with
(c) bear with (d) bare
32. Identify the correct spelling out of the given options :
- (a) Managable (b) Manageable
(c) Mangaible (d) Managible

33. Pick the odd one out in the following :

13, 23, 33, 43, 53

- (a) 23 (b) 33
(c) 43 (d) 53

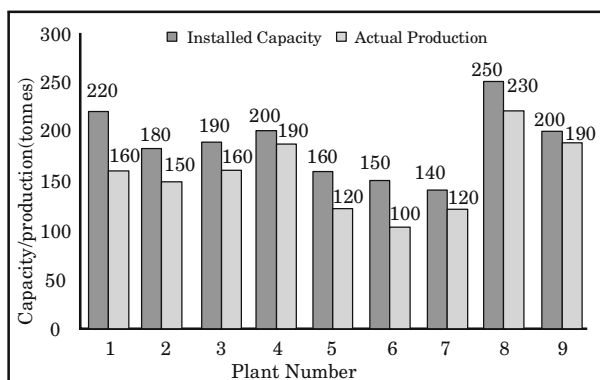
34. R2D2 is a robot. R2D2 can repair aeroplanes. No other robot can repair aeroplanes. Which of the following can be logically inferred from the above statements?

- (a) R2D2 is a robot which can only repair aeroplanes.
(b) R2D2 is the only robot which can repair aeroplanes.
(c) R2D2 is a robot which can repair only aeroplanes.
(d) Only R2D2 is a robot.

35. If $|9y - 6| = 3$, then $y^2 - \frac{4y}{3}$ is _____.

- (a) 0 (b) $+\frac{1}{3}$
(c) $-\frac{1}{3}$ (d) Undefined

36. The following graph represents the installed capacity for cement production (in tonnes) and the actual production (in tonnes) of nine cement plants of a cement company. Capacity utilization of a plant is defined as ratio of actual production of cement to installed capacity. A plant with installed capacity of at least 200 tonnes is called a large plant and a plant with lesser capacity is called a small plant. The difference between total production of large plants and small plants, in tonnes is _____.



37. A poll of students appearing for masters in engineering indicated that 60% of the students believed that mechanical engineering is a profession unsuitable for women. A research study on women with masters or higher degrees in mechanical engineering found that 99% of such women were successful in their professions.

Which of the following can be logically inferred from the above paragraph?

- (a) Many students have misconceptions regarding various engineering disciplines.
(b) Men with advanced degrees in mechanical engineering believe women are well suited to be mechanical engineers.
(c) Mechanical engineering is a profession well suited for women with masters or higher degrees in mechanical engineering.
(d) The number of women pursuing higher degrees in mechanical engineering is small.
38. Sourya committee had proposed the establishment of Sourya Institutes of Technology (SITs) in line with Indian Institutes of Technology (IITs) to cater to the technological and industrial needs of a developing country.

Which of the following can be logically inferred from the above sentence?

Based on the proposal,

- (i) In the initial years, SIT students will get degrees from IIT.
(ii) SITs will have a distinct national objective.
(iii) SIT like institutions can only be established in consultation with IIT.
(iv) SITs will serve technological needs of a developing country.
- (a) (iii) and (iv) only (b) (i) and (iv) only
(c) (ii) and (iv) only (d) (ii) and (iii) only
39. Shaquille O' Neal is a 60% career free throw shooter, meaning that he successfully makes 60 free throws out of 100 attempts on average. What is the probability that he will successfully make exactly 6 free throws in 10 attempts?
- (a) 0.2508 (b) 0.2816
(c) 0.2934 (d) 0.6000

40. The numeral in the units position of

$211^{870} + 146^{127} \times 3^{424}$ is _____.

41. Out of the following four sentences, select the most suitable sentence with respect to grammar and usage.

- (a) I will not leave the place until the minister does not meet me.
(b) I will not leave the place until the minister doesn't meet me.
(c) I will not leave the place until the minister meet me.
(d) I will not leave the place until the minister meets me.

42. A rewording of something written or spoken is a _____.

(a) paraphrase (b) paradox
(c) paradigm (d) paraffin

43. Archimedes said, "Give me a lever long enough and a fulcrum on which to place it, and I will move the world."

The sentence above is an example of a _____ statement.

(a) figurative (b) collateral
(c) literal (d) figurine

44. If 'relftaga' means carefree, 'otaga' means careful and 'fertaga' means careless, which of the following could mean 'aftercare'?

(a) zentaga (b) tagafer
(c) tagazen (d) relffer

45. A cube is built using 64 cubic blocks of side one unit. After it is built, one cubic block is removed from every corner of the cube. The resulting surface area of the body (in square units) after the removal is _____.

(a) 56 (b) 64
(c) 72 (d) 96

46. A shaving set company sells 4 different types of razors, Elegance, Smooth, Soft and Executive. Elegance sells at Rs. 48, Smooth at Rs. 63, Soft at Rs. 78 and Executive at Rs. 173 per piece. The table below shows the numbers of each razor sold in each quarter of a year.

Quarter \ product	Elegance	Smooth	Soft	Executive
Q1	27300	20009	17602	9999
Q2	25222	19392	18445	8942
Q3	28976	22429	19544	10234
Q4	21012	18229	16595	10109

Which product contributes the greatest fraction to the revenue of the company in that year?

(a) Elegance (b) Executive
(c) Smooth (d) Soft

47. Indian currency notes show the denomination indicated in at least seventeen languages. If this is not an indication of the nation's diversity, nothing else is.

Which of the following can be logically inferred from the above sentences?

(a) India is a country of exactly seventeen languages.
(b) Linguistic pluralism is the only indicator of a nation's diversity.
(c) Indian currency notes have sufficient space for all the Indian languages.
(d) Linguistic pluralism is strong evidence of India's diversity.

48. Consider the following statements relating to the level of poker play of four players P, Q, R and S.

I. P always beats Q
II. R always beats S
III. S loses to P only sometimes
IV. R always loses to Q

Which of the following can be logically inferred from the above statements?

(i) P is likely to beat all the three other players
(ii) S is the absolute worst player in the set
(a) (i) only (b) (ii) only
(c) (i) and (ii) (d) neither (i) nor (ii)

49. If $f(x) = 2x^7 + 3x - 5$, which of the following is a factor of $f(x)$?

(a) $(x^3 + 8)$ (b) $(x - 1)$
(c) $(2x - 5)$ (d) $(x + 1)$

50. In a process, the number of cycles to failure decreases exponentially with an increase in load. At a load of 80 units, it takes 100 cycles for failure. When the load is halved, it takes 10000 cycles for failure. The load for which the failure will happen in 5000 cycles is _____.

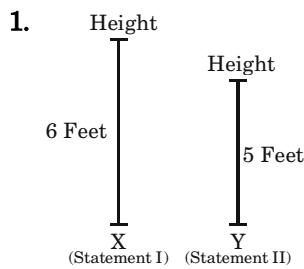
(a) 40.00 (b) 46.02
(c) 60.01 (d) 92.02

ANSWERS

1. (c) 2. (b) 3. (c) 4. (c) 5. (b) 6. (d) 7. (a) 8. (d) 9. (c) 10. (b)
 11. (c) 12. (a) 13. (c) 14. (a) 15. (a) 16. (d) 17. (d) 18. (b) 19. (d) 20. (c)
 21. (a) 22. (b) 23. (a) 24. (a) 25. (c) 26. (c) 27. (b) 28. (d) 29. (c) 30. (c)
 31. (c) 32. (b) 33. (b) 34. (b) 35. (c) 36. (120) 37. (a) 38. (c) 39. (a) 40. (7)
 41. (d) 42. (a) 43. (a) 44. (c) 45. (d) 46. (b) 47. (d) 48. (a) 49. (b) 50. (b)

EXPLANATIONS

General Aptitude



Hence from the given figure Mr. X is taller than Mr. Y by 1 foot.

4. $(9 \text{ inches})^{1/2} = (0.25 \text{ yards})^{1/2}$,

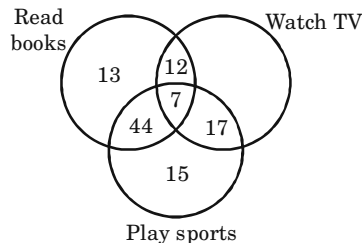
Solving we get 9 inch = 0.25 yards

(since 1 inch = 0.028 yard)

5. M is twice as efficient as E but worked for half as many days. So in this case they will do equal work if their shifts had same timings. But M's shift is for 6 hours, while E's shift for 12 hours. Hence, E will do twice the work as M.

Ratio of contribution of M : E in work is 1 : 2.

6. Given Venn diagram is

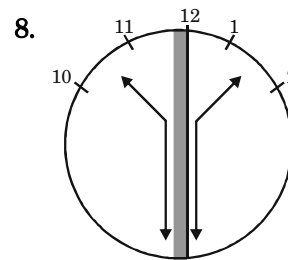


The number of students who like to read books or play sports

$$= 13 + 12 + 44 + 7 + 15 + 17$$

$$= 108$$

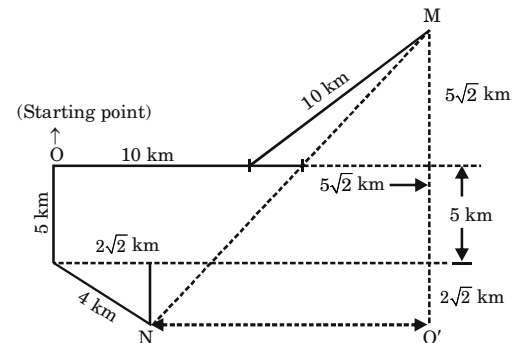
7. Social science disciplines had a pre-colonial origin.



Mirror image of 1 : 20 is 10 : 30

10 : 30 was the time two and quarter hour back so time now will be 12 : 45

- 9.



From the given figure

$$MN = \sqrt{(OM)^2 + (ON)^2}$$

$$O'M = 5\sqrt{2} + 5 + 2\sqrt{2} = 5 + 7\sqrt{2}$$

$$O'N = 10 + 5\sqrt{2} - 2\sqrt{2} = 10 + 3\sqrt{2}$$

$$MN = \sqrt{(5 + 7\sqrt{2})^2 + (10 + 3\sqrt{2})^2}$$

$$= \sqrt{25 + 98 + 70\sqrt{2} + 100 + 18 + 60\sqrt{2}}$$

$$\approx 20.61 \text{ km}$$

14. $(x\% \text{ of } y) + (y\% \text{ of } x)$

$$\frac{x}{100} \times y + \frac{y}{100} \times x = \frac{2xy}{100} = 2\% \text{ of } xy$$

∴ Option (a) is correct.

15. Let the two digit number be xy

Now sum of the digit of a two digit number is 12

$$\therefore x + y = 12 \quad \dots(i)$$

Now according to question,

$$(10y + x) - (10x + y) = 54$$

$$\therefore 9y - 9x = 54$$

$$\therefore y - x = 6 \quad \dots(ii)$$

Now solving equation (i) and (ii) we get,

$$x = 3 \text{ and } y = 9$$

Hence the original number is 39.

$\therefore$ Option (a) is correct.

16. Amount invested by Company P in 2006 = ₹ 8x

and amount invested by Company Q in 2006 = ₹ 9x

Interest from Company P

$$\begin{aligned} & \frac{\text{₹ } 8x \times 6}{100} \\ &= \frac{\text{₹ } 9x \times 4}{100} \\ &= \frac{4}{3} \end{aligned}$$

$\therefore$ Option (d) is correct.

21. Before superlative article 'the' has to be used.

"One of" the expression should take plural noun and so option 'C' and 'D' can't be the answer.

22. 'lose' is verb; to complete that sentence a verb is required so 'B' is correct.

23. 'Effectiveness' is noun and 'prescribed' is verb. These words are apt and befitting with the word 'medicine'.

24. Total no. of fruits = 5692000

Unripe type of apples = 45% of 15% of 5692000

$$\begin{aligned} &= \frac{45}{100} \times \frac{15}{100} \times 5692000 \\ &= 384210 \end{aligned}$$

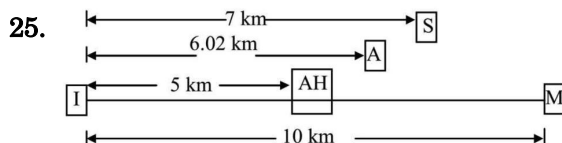
$$\text{Ripe type of apples} = \frac{34}{100} \times \frac{85}{100} \times 5692000$$

$$= 1644988$$

$\therefore$ Total number of apples

$$= 384210 + 1644988$$

$$= 2029198$$



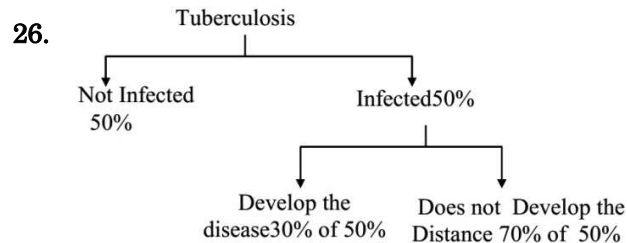
I = I live

AH = Ahmed lives

M = Michel lives

S = Susan lives

A = Arun lives



$$\frac{70}{100} \times \frac{50}{100} \Rightarrow \frac{35}{100} = 35\%$$

$\therefore$ Option (c) is correct

27. The words 'was confident that they would reciprocate' and 'last week proved him wrong' lead to statements iii and iv as logically valid inferences.

28. From given data, the following arrangement is possible

Shiva

Leela

Pavithra

Among four alternatives, option D is TRUE.

$$29. q^{-a} = \frac{1}{r} \Rightarrow \frac{1}{q^a} = \frac{1}{r} \Rightarrow q^a = r$$

$$r^{-b} = \frac{1}{s} \Rightarrow \frac{1}{r^b} = \frac{1}{s} \Rightarrow s = r^b$$

$$s^{-c} = \frac{1}{q} \Rightarrow \frac{1}{s^c} = \frac{1}{q} \Rightarrow s^c = q$$

$$q^a = r \Rightarrow (s^c)^a = r \Rightarrow s^{ac} = r$$

$$(s^{ac})^b = s \Rightarrow s^{abc} = s^1$$

$$\therefore abc = 1$$

$\therefore$ Option (c) is correct

30. Q can finish the task = 25 days, 12 hrs/day = 300 hrs

$$\Rightarrow 1 \text{ hr} = \frac{1}{300} \text{ th}$$

$$\begin{aligned} \text{R can finish the task} &= 50 \text{ days, 12 hrs/day} \\ &= 50 \times 12 = 600 \text{ hrs} \end{aligned}$$

$$\Rightarrow 1 \text{ hr} = \frac{1}{600} \text{ th}$$

$$\text{Q working hours} \Rightarrow (7-2) \times 12 = 60 \text{ hrs}$$

$$\text{R working hours} \Rightarrow 7 \times 18 = 126 \text{ hrs}$$

After 7 days, the ratio of work done by Q and R

$$\begin{array}{ccc} Q & : & R \\ \frac{60}{300} & & \frac{126}{600} \\ 20 & : & 21 \end{array}$$

33. The given number 33 is odd one out, because the remaining numbers are prime number.

35. Given, $|9y - 6| = 3$

It has two values either

$$(9y - 6) = 3 \quad \dots(i)$$

Or $-(9y - 6) = 3$

$$9y - 6 = -3 \quad \dots(ii)$$

From equation (i),

$$9y = 9$$

$$\therefore y = 1$$

Or from equation (ii),

$$9y = 3$$

$$\therefore y = \frac{1}{3}$$

Now $y^2 = \frac{4y}{3}$

put $y = 1$,

we get $= 1 - \frac{4}{3}$

$$= -\frac{1}{3}$$

and also put $y = \frac{1}{3}$

we get, $\frac{1}{9} - \frac{4}{9} = \frac{-3}{9} = \frac{-1}{3}$

So, the required value is $\frac{-1}{3}$.

36. From the given graph large plants which are having installed capacity of at least 200 tonnes are 1, 4, 8 and 9.

So, the total production of large plants

$$\begin{aligned} &= 160 + 190 + 230 + 190 \\ &= 770 \end{aligned}$$

Now the remaining plants with installed capacity is less than 200 tonnes are 2, 3, 5, 6 and 7.

So, total production of small plants

$$\begin{aligned} &= 150 + 160 + 120 + 100 + 120 \\ &= 650 \end{aligned}$$

Then difference = $770 - 650$

$$= 120$$

39. The question clearly explains that shaquille makes 60 free throws out of 100

Hence, Probability of free throw = $\frac{60}{100} = 0.6$

And probability of NOT free throw

$$= 1 - 0.6 = 0.4$$

So required probability of exactly 6 throws in 10 attempts will be given by

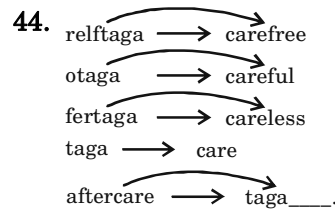
$${}^{10}C_6 \cdot 0.6^6 \times 0.4^4 = 0.2508$$

40. Unit digit of $211^{870} + 146^{127} \times 3^{424}$ is $1 + 6 \times 1 = 7$

41. I will not leave the place until the minister meets me.

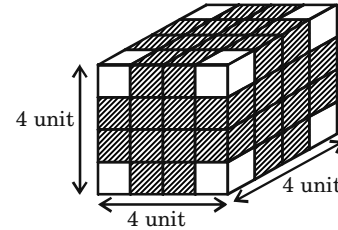
42. Rewording of something written or spoken is PARAPHRASE.

43. Figurative statement



Option (c) matches i.e. tagazen.

- 45.



$$\begin{aligned} \text{Required Surface Area} &= (6 \times 4) + (8 \times 6) + (3 \times 8) \\ &= 24 + 48 + 24 \\ &= 96 \text{ units.} \end{aligned}$$

46. Elegance $\Rightarrow (27300 + 25222 + 28976 + 21012) \times 48$
 Executive $\Rightarrow (9999 + 8942 + 10234 + 10109) \times 173$
 Smooth $\Rightarrow (20009 + 19392 + 22429 + 18229) \times 63$
 Soft $\Rightarrow (17602 + 18445 + 19544 + 16595) \times 78$
 Executive has the highest revenue.

48. (i) is followed.

(ii) is not followed.

49. $f(x) = 2x^7 + 3x - 5$

Putting $x = 1$

$$f(1) = 2 + 3 - 5 = 0$$

So, $(x - 1)$ is the factor of $f(x)$.

SOLVED PAPER – 2017

1. What is the value of x when

$$81 \times \left(\frac{16}{25}\right)^{x+2} \div \left(\frac{3}{5}\right)^{2x+4} = 144?$$

- (a) 1
(b) -1
(c) -2
(d) Cannot be determined
2. There was no doubt that their work was thorough. Which of the words is closest in meaning to the underlined word above?

- (a) pretty (b) complete
(c) sloppy (d) haphazard

3. Four cards lie on a table. Each card has a number printed on the one side and a color on the other. The faces visible on the cards are 2, 3 red and blue.

Proposition. If a card has an even value on one side then its opposite face is red.

The cards which **MUST** be turned over to verify the above proposition are

- (a) 2, red (b) 2, 3, red
(c) 2, blue (d) 2, red, blue
4. The event would have been successful if you _____ able to come.

- (a) are (b) had been
(c) have been (d) would have been

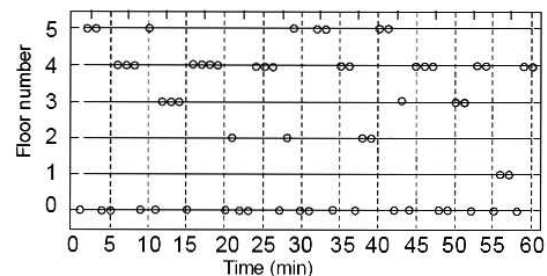
5. Two dice are thrown simultaneously. The probability that the product of the numbers appearing on the top faces of the dice is a perfect square is

- (a) $\frac{1}{9}$
(b) $\frac{2}{9}$
(c) $\frac{1}{3}$
(d) $\frac{4}{9}$

6. Bhaiclung was observing the pattern of people entering and leaving a car service centre. There was a single window where customers were being served. He saw that people inevitably came out of the centre in the order that they went in. However, the time they spent inside seemed to vary a lot some people came out in a matter of minutes while for others it took much longer.

From this what can one conclude?

- (a) The centre operates on a first-come-first-served basis, but with variable service times depending on specific customer needs.
(b) Customers were served in an arbitrary order since they took varying amounts of time for service completion in the centre.
(c) Since some people came out within a few minutes of entering the centre, the system is likely to operate on a last-come-first-served basis.
(d) Entering the centre early ensured that one would have shorter service times and most-people attempted to do this.
7. The points in the graph below represent the halts of a lift for duration of 1 minute, over a period of 1 hour.



Which of the following statements are correct?

- (i) The elevator never moves directly from any non-ground floor to another non-ground floor over the one hour period.
(ii) The elevator stays on the fourth floor for the longest duration over the one hour period.
- (a) only (i) (b) only (ii)
(c) Both (i) and (ii) (d) Neither (i) nor (ii)
8. A map shows the elevations of Darjeeling, Gangtok, Kalimpong, Pelling and Siliguri. Kalimpong is at a lower elevation than Gangtok. Pelling is at a lower elevation than Gangtok. Pelling is at a higher elevation than Siliguri. Darjeeling is at a higher elevation than Gangtok. Which of the following statements can be inferred from the paragraph above?
- (i) Pelling is at a higher elevation than Kalimpong
(ii) Kalimpong is at a lower elevation than Darjeeling

- (iii) Kalimpong is at a higher elevation than Siliguri
 (iv) Siliguri is at a lower elevation than Gangtok.
 (a) Only (ii)
 (b) Only (ii) and (iii)
 (c) Only (ii) and (iv)
 (d) Only (iii) and (iv)
9. P, Q, R, S, T and U are seated around a circular table. R is seated two places to the right of Q. P is seated three places to the left of R. S is seated opposite U. If P and U now switch seats which of the following must necessarily be true.
 (a) P is immediately to the right of R
 (b) T is immediately to the left of P
 (c) T is immediately to the left of P or P is immediately to other right of Q.
 (d) U is immediately to the right of R or P is immediately to the left of T.
10. Budhan covers a distance of 19 km in 2 hours by cycling one fourth of the time and walking the rest. The next day he cycles (at the same speed as before) for half the time and walks the rest (at the same speed as before) and covers 26 km in 2 hours. The speed in km/h at which Budhan walks is
 (a) 1 (b) 4
 (c) 5 (d) 6
11. The probability that a k-digit number does NOT contain the digits 0, 5 or 9 is
 (a) 0.3^k (b) 0.6^k
 (c) 0.7^k (d) 0.9^k
12. Find the smallest number y such that that $y \times 162$ is a perfect cube.
 (a) 24 (b) 27
 (c) 32 (d) 36
13. After Rajendra Chola returned from his voyage to Indonesia, he _____ to visit the temple in Thanjavur.
 (a) was wishing (b) is wishing
 (c) wished (d) had wished
14. Research in the workplace reveal that people work for many reasons_____.
 (a) money beside
 (b) beside money
 (c) money besides
 (d) besides money
15. Rahul, Murali, Srinivas and Arul are seated around a square table. Rahul is sitting to the left of Murali. Srinivas is sitting to the right of Arul. Which of the following pairs are seated opposite each other?
 (a) Rahul and Murali
 (b) Srinivas and Arul
 (c) Srinivas and Murali
 (d) Srinivas and Rahul
16. “The hold of the nationalist imagination on our colonial past is such that anything inadequately or improperly nationalist is just not history.”
 Which of the following statements best reflects the author’s opinion?
 (a) Nationalists are highly imaginative
 (b) History is viewed through the filter of nationalism
 (c) Our colonial past never happened
 (d) Nationalism has to be both adequately and properly imagined.
17. The expression $\frac{(x+y) - |x-y|}{2}$ is equal to
 (a) the maximum of x and y
 (b) the minimum of x and y
 (c) 1
 (d) none of the above
18. Arun, Gulab, Neel and Shweta must choose one shirt each from a pile of four shirts coloured red, pink, blue and white respectively. Arun dislikes the colour red and Shweta dislikes the colour white. Gulab and Neel like all the colours. In how many different ways can they choose the shirts so that no one has a shirt with a colour he or she dislikes?
 (a) 21 (b) 18
 (c) 16 (d) 14
19. Six people are seated around a circular table. There are at least two men and two women. There are at least three right-handed persons. Every woman has a left-handed person to her immediate right. None of the women are right-handed. The number of women at the table is
 (a) 2
 (b) 3
 (c) 4
 (d) Cannot be determined

20. A test has twenty questions worth 100 marks in total. There are two types of questions. Multiple choice questions are worth 3 marks each and essay questions are worth 11 marks each. How many multiple choice questions does the exam have ?

(a) 12 (b) 15
(c) 18 (d) 19

21. Choose the option with words that are not synonyms.

(a) aversion, dislike (b) luminous, radiant
(c) plunder, loot (d) yielding, resistant

22. There are five buildings called V, W, X, Y and Z in a row (not necessarily in that order). V is to the West of W. Z is to the East of X and the West of V. W is to the West of Y. Which is the building in the middle ?

(a) V
(b) W
(c) X
(d) Y

23. There are 3 red socks, 4 green socks and 3 blue socks. You choose 2 socks. The probability that they are of the same colour is

(a) $1/5$
(b) $7/30$
(c) $1/4$
(d) $4/15$

24. Saturn is _____ to be seen on a clear night with the naked eye.

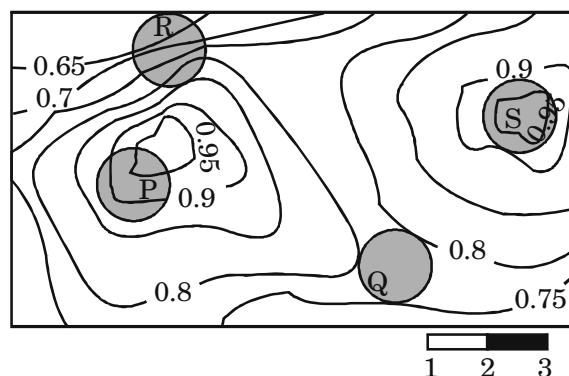
(a) enough bright
(b) bright enough
(c) as enough bright
(d) bright as enough

25. "We lived in a culture that denied any merit to literary works, considering them important only when they were handmaidens to somethings seemingly more urgent – namely ideology. This was a country where all gestures, even the most private, were interpreted in political terms."

The author's belief that ideology is not as important as literature is revealed by the word :

(a) 'culture'
(b) 'seemingly'
(c) 'urgent'
(d) 'political'

26. An air pressure contour line joins locations in a region having the same atmospheric pressure. The following is an air pressure contour plot is a geographical region. Contour lines are shown at 0.05 bar intervals in this plot.



If the possibility of a thunderstorm is given by how fast air pressure rises or drops over a region, which of the following regions is most likely to have a thunderstorm ?

(a) P (b) Q
(c) R (d) S

27. The number of roots of $e^x + 0.5x^2 - 2 = 0$ in the range $[-5, 5]$ is

(a) 0
(b) 1
(c) 2
(d) 3

28. There are three boxes. One contains apples another contains oranges and the last one contains both apples and oranges. All three are known to be incorrectly labelled. If you are permitted to open just one box and then pull out and inspect only one fruit, which box would you open to determine the contents of all three boxes?

(a) The box labelled 'Apples'
(b) The box labelled 'Apples and Oranges'
(c) The box labelled 'Oranges'
(d) Cannot be determined

29. X is a 30 digit number starting with the digit 4 followed by the digit 7. Then the number X^3 will have

(a) 90 digits
(b) 91 digits
(c) 92 digits
(d) 93 digits

30. In the summer, water consumption is known to decrease overall by 25%. A Water Board official states that in the summer household consumption decreases by 20%, while other consumption increases by 70%.

Which of the following statements is correct?

- (a) The ratio of household to other consumption is 8/17.
 (b) The ratio of household to other consumption is 1/17.
 (c) The ratio of household to other consumption is 17/8.
 (d) There are errors in the official's statement.

31. Some tables are shelves. Some shelves are chairs. All chairs are benches. Which of the following conclusions can be deduced from the preceding sentences?

- (i) At least one bench is a table
 (ii) At least one shelf is a bench
 (iii) At least one chair is a table
 (iv) All benches are chairs

- (a) only (i) (b) only (ii)
 (c) only (ii) and (iii) (d) only (iv)

32. I _____ made arrangements had I _____ informed earlier.

- (a) could have, been (b) would have, being
 (c) had, have (d) had been, been

33. 40% of deaths on city roads may be attributed to drunken driving. The number of degrees needed to represent this as a slice of a pie chart is

- (a) 120 (b) 144
 (c) 160 (d) 212

34. She has a sharp tongue and it can occasionally turn _____.

- (a) hurtful (b) left
 (c) methodical (d) vital

35. Trucks (10 m long) and cars (5 m long) go on a single lane bridge. There must be a gap of at least 20 m after each truck and a gap of at least 15 m after each car. Trucks and cars travel at a speed of 36 km/h. If cars and trucks go alternately. What is the maximum number of vehicles that can use the bridge in one hour?

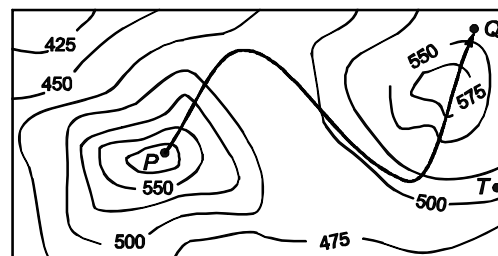
- (a) 1440 (b) 1200
 (c) 720 (d) 600

36. "If you are looking for a history of India, or for an account of the rise and fall of the British Raj, or for the reason of the cleaving of the subcontinent into two mutually antagonistic parts and the effects this mutilation will have in the respective sections, and ultimately on Asia, you will not find it in these pages; for though I have spent a lifetime in the country, I lived too near the seat of events, and was too intimately associated with the actors, to get the perspective needed for the impartial recording of these matters".

Here, the word 'antagonistic' is closest in meaning to

- (a) impartial (b) argumentative
 (c) separated (d) hostile

37. A contour line joins locations having the same height above the mean sea level. The following is a contour plot of a geographical region. Contour lines are shown at 25 m intervals in this plot.



The path from P to Q is best described by

- (a) Up-Down-Up-Down
 (b) Down-Up-Down-Up
 (c) Down-Up-Down
 (d) Up-Down-Up

38. There are 3 Indians and 3 Chinese in a group of 6 people. How many subgroups of this group can we choose so that every subgroup has at least one Indian?

- (a) 56 (b) 52
 (c) 48 (d) 44

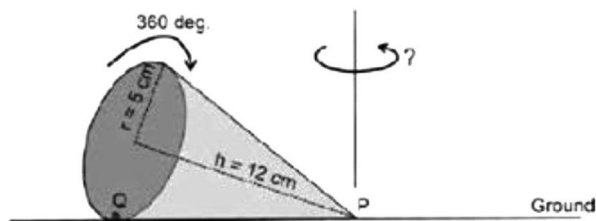
39. S, T, U, V, W, X, Y and Z are seated around a circular table. T's neighbours are Y and V. Z is seated third to the left of T and second to the right of S. U's neighbours are S and Y; and T and W are not seated opposite each other. Who is third to the left of V?

- (a) X
 (b) W
 (c) U
 (d) T

40. He was one of my best _____ and I felt his loss _____

- (a) Friend, keenly (b) Friends, keen
(c) Friend, keener (d) Friends, keenly

41. A right-angled cone (with base radius 5 cm and height 12 cm), as shown in the figure below, is rolled on the ground keeping the point P fixed until the point Q (at the base of the cone, as shown) touches the ground again.



- (a) $\frac{5\pi}{12}$ (b) $\frac{5\pi}{24}$
(c) $\frac{24\pi}{5}$ (d) $\frac{10\pi}{13}$

42. In a company with 100 employees, 45 earn ₹ 20,000 per month, 25 earn ₹ 30,000, 20 earn ₹ 40,000, 8 earn ₹ 60,000 and 2 earn ₹ 150,000. The median of the salaries is _____

- (a) ₹ 20,000 (b) ₹ 30,000
(c) ₹ 32,300 (d) ₹ 40,000

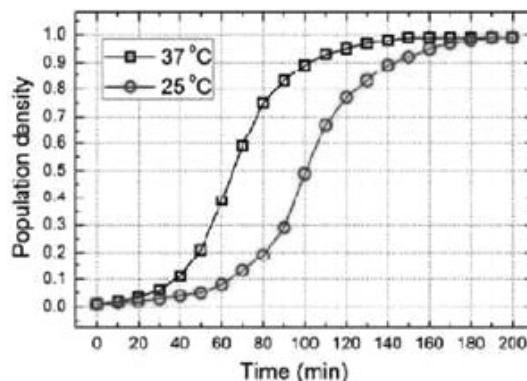
43. P, Q and R talk about S's car collection. P states that S has at least 3 cars. Q believes that S has less than 3 cars R indicates that to his knowledge, S has at least one car. Only one of P, Q and R is right. The number of cars owned by S is

- (a) 0
(b) 1
(c) 3
(d) Cannot be determined

44. As the two speakers became increasingly agitated, the debate became _____.

- (a) Lukewarm (b) Poetic
(c) Forgiving (d) Heated

45. The growth of bacteria (lactobacillus) in milk leads to curd formation. A minimum bacterial population density of 0.8 (in suitable units) is needed to form curd. In the graph below, the population density of lactobacillus in 1 litre of milk is plotted as a function of time, at two different temperature, 25°C and 37°C.



Consider the following statements based on the data shown above:

- i. The growth in bacterial population stops earlier at 37°C as compared to 25°C
ii. The time taken for curd formation at 25°C is twice the time taken at 37°C Which one of the following options is correct?

- (a) Only i (b) Only ii
(c) Both i and ii (d) Neither i nor ii

46. What is the sum of the missing digits in the subtraction problem below?

$$\begin{array}{r} 5 _ _ _ _ \\ - 48 _ 89 \\ \hline \end{array}$$

1 1 1 1

- (a) 8
(b) 10
(c) 11
(d) Cannot be determined

47. Let S_1 be the plane figure consisting of the points (x, y) given by the inequalities $|x - 1| \leq 2$ and $|y + 2| \leq 3$. Let S_2 be the plane figure given by the inequalities $x - y \geq -2$, $y \geq 1$, and $x \leq 3$. Let S be the union of S_1 and S_2 . The area of S is

- (a) 26 (b) 28
(c) 32 (d) 34

48. Two very famous sportsmen Mark and Steve happened to be brothers, and played for country K. Mark teased James, an opponent from country E, "There is no way you are good enough to play for your country." James replied, "Maybe not, but at least I am the best player in my own family." Which one of the following can be inferred from this conversation?

- (a) Mark was known to play better than James
(b) Steve was known to play better than Mark
(c) James and Steve were good friends
(d) James played better than Steve

49. “Here, throughout the early 1820s, Stuart continued to fight his losing battle to allow his sepoys to wear their caste- marks and their own choice of facial hair on parade, being again reprimanded by the commander-in-chief. His retort that “A stronger instance than this of European prejudice with relation to this country has never come under my observations “had no effect on his superiors”

According to this paragraph, which of the statements below I most accurate?

- (a) Start’s commander-in-chief was moved by this demonstration of his prejudice
- (b) The Europeans were accommodating of the sepoys’ desire to wear their caste-marks
- (c) Stuart’s losing battle’ refers to his inability to succeed in enabling sepoys to wear cast-marks
- (d) The commander-in-chief was exempt from the European prejudice that dictated how the sepoys were to dress.

ANSWERS

1. (b)	2. (b)	3. (a)	4. (b)	5. (b)	6. (a)	7. (d)	8. (c)	9. (c)	10. (d)
11. (c)	12. (d)	13. (c)	14. (d)	15. (d)	16. (b)	17. (b)	18. (d)	19. (a)	20. (b)
21. (d)	22. (a)	23. (d)	24. (b)	25. (c)	26. (c)	27. (c)	28. (a)	29. (a)	30. (d)
31. (b)	32. (a)	33. (b)	34. (a)	35. (a)	36. (d)	37. (c)	38. (a)	39. (a)	40. (d)
41. (d)	42. (b)	43. (a)	44. (d)	45. (a)	46. (8)	47. (c)	48. (b)	49. (c)	

EXPLANATIONS

$$\begin{aligned}
 1. \quad & 81 \times \left(\frac{16}{25}\right)^{x+2} \div \left(\frac{3}{5}\right)^{2x+4} = 144 \\
 \Rightarrow & \left(\frac{4}{5}\right)^{2x+4} \times \left(\frac{5}{3}\right)^{2x+4} = \frac{144}{81} \\
 \Rightarrow & \left(\frac{4}{3}\right)^{2x+4} = \frac{144}{81} \\
 \Rightarrow & \left(\frac{4}{3}\right)^{x+2} = \sqrt{\frac{144}{81}} \\
 \Rightarrow & \left(\frac{4}{3}\right)^x \times \frac{16}{9} = \frac{12}{9} \\
 \Rightarrow & \left(\frac{4}{3}\right)^x = \frac{3}{4} \\
 \Rightarrow & \left(\frac{4}{3}\right)^x = \left(\frac{4}{3}\right)^{-1} \\
 & x = -1
 \end{aligned}$$

3. Card with visible face 2 should have red on the other side.

Card with visible face 3 may have any color on other side.

Card with visible face red should have even number on other side.

Card with visible face blue may have any number on other side.

Hence, to verify the above proposition, cards with face 2 and Red must be turned.

5. Favourable outcomes = $\{(1, 1), (1, 4), (2, 2), (3, 3), (4, 1), (4, 4), (5, 5), (6, 6)\}$

No. of favourable outcomes = 8

Total outcomes = 36

$$\therefore \text{Probability} = \frac{8}{36} = \frac{2}{9}$$

7. The elevators has moved from 2nd to 5th floor between time 25 min to 30 min.

Elevation stayed at 4th floor for

$$= 3 + 4 + 3 + 2 + 3 + 2 + 2 = 19 \text{ min}$$

Elevation stayed at ground floor for

$$\begin{aligned}
 &= 1 + 2 + 1 + 1 + 1 + 1 + 2 + 1 + 2 + 1 + 1 + 1 \\
 &\quad + 1 + 2 + 1 + 1 + 1 \\
 &= 21 \text{ min.}
 \end{aligned}$$

Hence, (i) and (ii) both are false.

8. $K < G, P < G, P > S, G < D$

if $K < G$ and $G < D$

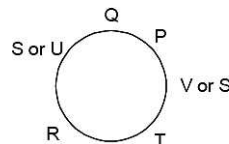
Hence, $K < D$

If $S < P$ and $P < G$

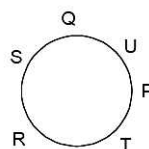
Hence, $S < G$

Hence, (ii) and (iv) are correct.

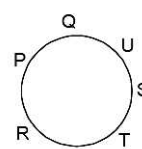
- 9.



Before switching of seats of P and U



1st possibility



2nd possibility

10. Let C = Speed by cycling (km/h)

W = Speed of walking (km/h)

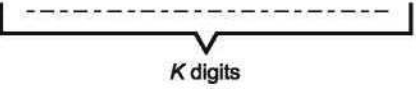
Now, by according to questions,

$$\begin{aligned}
 C \times \frac{2}{4} + W \times 2 \times \frac{3}{4} &= 19 \\
 \Rightarrow C + 3W &= 38 \quad \dots(i)
 \end{aligned}$$

$$\begin{aligned}
 \text{and, } C \times \frac{2}{2} + W \times \frac{2}{2} &= 26 \\
 \Rightarrow C + W &= 26 \quad \dots(ii)
 \end{aligned}$$

By solving equations (i) and (ii) we get,

$$W = 6 \text{ km/h}$$

11. 

Each digit can be filled in 7 ways as 0, 5 and 9 is not allowed, so each of these places can be filled by 1, 2, 3, 4, 6, 7, 8 so required probability is

$$\left(\frac{7}{10}\right)^K \text{ or } 0.7^K$$

$$\frac{fC}{TC} = \frac{7^K}{10^K} = 0.7^K$$

12. $y \times 162 = \text{Perfect cube}$

Option, $y = 24 \Rightarrow 2^3 \times 3 (2 \times 81) \neq \text{Not perfect cube}$

$y = 27 \Rightarrow 3^3 \times (2 \times 3^4) \neq \text{Not perfect cube}$

$y = 32 \Rightarrow 2^5 \times 2 \times 3^4 \neq \text{Not perfect cube}$

$y = 36 \Rightarrow 2^2 \times 3^2 \times 2 \times 3^4$

$$= 2^3 \times 3^6 = (2 \times 3^2)^3$$

is a perfect cube Hence the answer is, $y = 36$

13. Correct option is; Wished

After Rajender Chola **retured** from his voyage to Indonesia, he **wished** to visit the temple in Thanjavur.

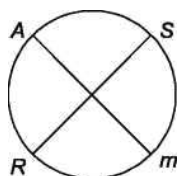
Both are events of past. Use of past perfect form is unwarranted as it reflects part of past.

14. Besides money

Research in the work place reveals the people works for many reasons **besides money**. Besides conveys the meaning of 'in addition'

Beside means 'next to'

15. Following seating arrangement can be drawn



Therefore correct option is (d).

16. History is viewed through the filter of rationalism.

17.
$$\frac{(x+y) - |x-y|}{2} \dots (i)$$

If $x > y$ $|x-y| = x-y$;

If $x < y$ $|x-y| = y-x$

Now if $x > y$, above expression (i) becomes

$$\begin{aligned} \frac{(x+y) - (x-y)}{2} &= \frac{2x}{2} \\ &= y = \text{minimum of } (x, y) \text{ as } x > y \end{aligned}$$

Now if $x < y$,

$$\begin{aligned} \frac{x+y - (y-x)}{2} &= \frac{2y}{2} \\ &= x = \text{minimum of } (x, y) \text{ as } x < y \end{aligned}$$

Therefore correct answer is option (b).

Alternate solution:

Use easy values,

$$x = 1$$

and $y = -2$

$$\text{Now, } \frac{(1-2) - |1-(-2)|}{2} = -2$$

or $x = 2$

and $y = -1$

$$\frac{(2-1) - |2-(-1)|}{2} = \frac{1-3}{2} = -1$$

which is minimum of (x, y) .

Therefore, correct answer is option (b).

18. As there are 4 people A, G, N, S and four colours so without any restriction total ways have to be $4 \times 4 = 16$.

Now, Arun $\rightarrow$ dislikes Red and Shweta dislikes white so $16 - 2 = 14$ ways.

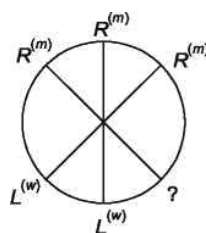
Therefore correct answer should be option (d).

Alternate solution:

Only one option is less than 16.

Therefore correct answer should be option (d).

19. Out of six people. 3 place definitely occupied by right handed people as at least 2 women are there so these two will sit adjacently. Now as only on seat is left it will be occupied by a left handed man because on right side of this seat is sitting an right handed man.



? have to be a left handed man.

Here, $R^{(m)}$ indicates right handed man and $L^{(w)}$ indicates left handed women. Therefore answer should be only 2 women.

20. Let number of multiple choice question be 'x'

$\therefore$ Total number of essay type question = $20 - x$

Now according to question

$$x \times 3 + (20 - x)11 = 100$$

$$\therefore 3x + 220 - 11x = 100$$

$$\therefore 220 - 100 = 8x$$

$$\therefore x = \frac{120}{8}$$

$$\therefore x = 15$$

$\therefore$ Option (b) is correct.

21. Aversion means a strong dislike or disinclination.

Hence aversion is synonyms of dislike.

Aversion means giving of light which is synonym of radiant plunder means loot while yielding and resistant are not synonyms of each other.

$\therefore$ Option (d) is correct.

22. From the given condition of sequence we get following order of building in

X, Z, V, W, Y

Hence 'V' building is in the middle

$\therefore$ Option (a) is correct

23. $n(R) = 3$

$$n(G) = 4$$

$$n(B) = 3$$

Probability that 2 socks chosen are of both Red or Both Green or Both blue

$$\begin{aligned}\text{Same colour} &= \frac{3}{10} \times \frac{2}{9} + \frac{4}{10} \times \frac{3}{9} + \frac{3}{10} \times \frac{2}{9} \\ &= \frac{6}{90} + \frac{12}{90} + \frac{6}{90} \\ &= \frac{24}{90} \\ &= \frac{4}{15}\end{aligned}$$

∴ Option (d) is correct.

24. Saturn is bright enough to be seen on a clear night with the naked eye.

∴ Option (b) is correct

25. The author's belief that ideology is not as important as literature is revealed by the word 'urgent' in above paragraph.

∴ Option (c) is correct.

27. $e^x + 0.5x^2 - 2 = 0$

Differentiation w.r.t to 'x' we get

$$e^x + x = 0$$

∴ Hence 'x' has total number of root which lies in the range $[-5, 5]$ is '2'.

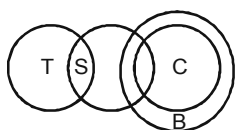
30. Let H is household consumption and P be other consumption

Now according to question,

$$H \times 0.8 + P \times 1.7 = (H + P) \times 0.75$$

From this it is concluded that ratio is negative.

31. From given condition



Only conclusion (ii) follows

32. Use of conditional sentence based on past participle form.

I could have made arrangements had I been informed earlier.

33. $40\% \text{ of } 360^\circ = \frac{40}{100} \times 360^\circ$
 $= 144^\circ$

So, the angle subtended on pie chart will be 144°

34. Hurtful means causing pain or suffering or something that is damaging or harmful. The expression 'sharp tongue' defines a bitter or critical manner of speaking.

35. According to question,

Length of Truck + gap required = $10 + 20 = 30$ m and

Length of Car + gap required = $5 + 15 = 20$ m

Alternative pairs of Truck and Car needs

$$30 + 20 = 50 \text{ m}$$

Let 'n' be the number of repetition of (Truck + Car) in 1 hour (3600 secs)

$$\begin{aligned}\text{Given speed} &= 36 \text{ km/hr} \\ &= 10 \text{ m/sec}\end{aligned}$$

$$\therefore \frac{50 \text{ m} \times n}{3600 \text{ secs}} = 36 \text{ km/hr}$$

$$\therefore \frac{50n}{3600} \text{ m/sec} = 10 \text{ m/sec}$$

$$\begin{aligned}\therefore n &= \frac{36000}{50} \\ &= 720 \text{ (Trucks + Cars)}\end{aligned}$$

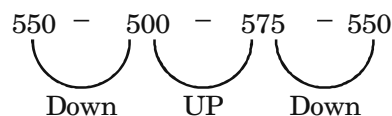
So, 720 (Trucks + Cars) passes

$$= (720 \times 2)$$

$$= 1440 \text{ vehicles}$$

36. Antagonist is a adversary or one who opposes/ contends against another. Hence the closest meaning to word "antagonistic" is hostile.

37. Contour lines can be observed to cross region with height



Down-Up-Down satisfies.

38. Subgroups containing only Indians

$$\begin{aligned}&= {}^3C_1 + {}^3C_2 + {}^3C_3 \\ &= 3 + 3 + 1 = 7\end{aligned}$$

and Subgroups containing one Indian and rest Chinese

$$\begin{aligned}&= {}^3C_1[{}^3C_1 + {}^3C_2 + {}^3C_3] \\ &= 3[3 + 3 + 1] \\ &= 21\end{aligned}$$

and Subgroups containing two Indian and remaining Chinese

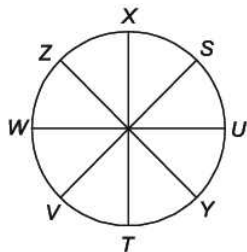
$$\begin{aligned}&= {}^3C_2[{}^3C_1 + {}^3C_2 + {}^3C_3] \\ &= 21\end{aligned}$$

and Subgroups containing three Indian and remaining Chinese

$$\begin{aligned}&= {}^3C_3[{}^3C_1 + {}^3C_2 + {}^3C_3] \\ &= 7\end{aligned}$$

Hence total number of subgroups
 $= 7 + 21 + 21 + 7$
 $= 56.$

39. Following circular seating arrangement can be drawn from the given data



From the given arrangement 'X' is the third to the left of 'V'.

42.

Salary	No. of employees	Cumulative
2000	45	45
30000	25	70
40000	20	90
60000	8	98
150000	2	100

$$\text{Median} = \frac{100 + 1}{2} = 50.5$$

From cumulative column,

Median salary = 30000

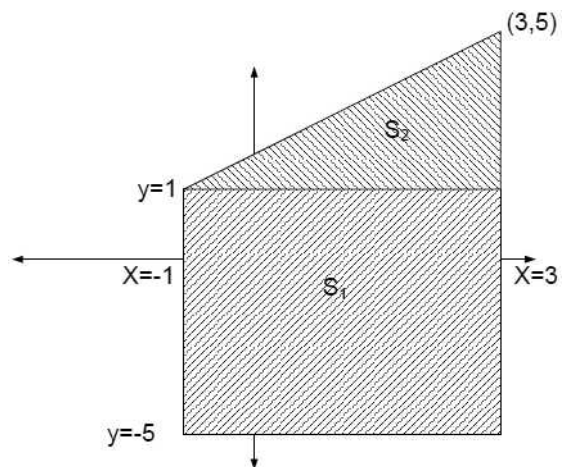
46.

$$\begin{array}{r} 5 \ 0 \ 0 \ 0 \ 0 \\ 4 \ 8 \ 8 \ 8 \ 9 \\ \hline \end{array}$$

$$1 \ 1 \ 1 \ 1$$

$$\text{sum} = 0 + 0 + 0 + 0 + 8 = 8$$

47.



$$\begin{aligned} S &= S_1 + S_2 \\ &= 6 \times 4 + \frac{1}{2} \times 4 \times 4 \\ &= 24 + 8 = 32 \end{aligned}$$

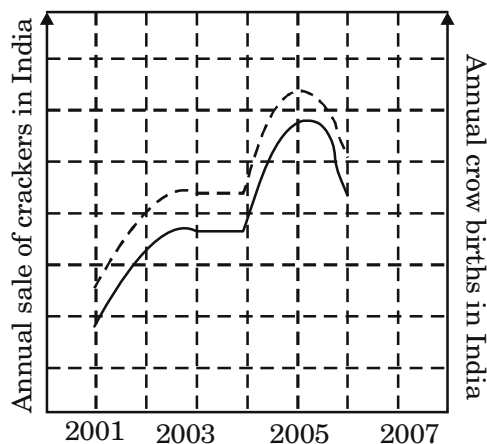
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SOLVED PAPER – 2018

GENERAL APTITUDE

SET-1

- “When she fell down the _____. She received many _____ but little help.” The words that best fill the blanks in the above sentence are
 (a) stairs, stares (b) stairs, stairs
 (c) stares, stairs (d) stares, stares
- “In spite of being warned repeatedly, he failed to correct his _____ behaviour.” The word that best fills that blank in the above sentence is
 (a) rational (b) reasonable
 (c) errant (d) good
- For $0 \leq x \leq 2\pi$, $\sin x$ and $\cos x$ are both decreasing functions in the interval _____.
 (a) $\left\{0, \frac{\pi}{2}\right\}$ (b) $\left\{\frac{\pi}{2}, \pi\right\}$
 (c) $\left\{\pi, \frac{3\pi}{2}\right\}$ (d) $\left\{\frac{3\pi}{2}, 2\pi\right\}$
- The area of an equilateral triangle is $\sqrt{3}$. What is the perimeter of the triangle?
 (a) 2 (b) 4
 (c) 6 (d) 8
- Arrange the following three-dimensional objects in the descending order of their volumes:
 (i) A cuboid with dimensions 10 cm, 8 cm and 6 cm
 (ii) A cube of side 8 cm
 (iii) A cylinder with base radius 7 cm and height 7 cm
 (iv) A sphere of radius 7 cm
 (a) (i), (ii), (iii), (iv) (b) (ii), (i), (iv), (iii)
 (c) (iii), (ii), (i), (iv) (d) (iv), (iii), (ii), (i)
- An automobile travels from city A to city B and returns to city A by the same route. The speed of the vehicle during the onward and return journeys were constant at 60 km/h and 90 km/h, respectively. What is the average speed in km/h for the entire journey?
 (a) 72 (b) 73
 (c) 74 (d) 75
- A set of 4 parallel lines intersect with another set of 5 parallel lines. How many parallelograms are formed?
 (a) 20 (b) 48
 (c) 60 (d) 72
- To pass a test, a candidate needs to answer at least 2 out of 3 questions correctly. A total of 6,30,000 candidates appeared for the test. Question A was correctly answered by 3,30,000 candidates. Question B was answered correctly by 2,50,000 candidates. Question C was answered correctly by 2,60,000 candidates. Both questions A and B were answered correctly by 1,00,000 candidates. Both questions B and C were answered correctly by 90,000 candidates. Both questions A and C were answered correctly by 80,000 candidates. If the number of students answering all questions correctly is the same as the number answering none, how many candidates failed to clear the test?
 (a) 30,000 (b) 2,70,000
 (c) 3,90,000 (d) 4,20,000
- If $x^2 + x - 1 = 0$ what is the value of $x^4 + \frac{1}{x^4}$?
 (a) 1 (b) 5
 (c) 7 (d) 9
- In a detailed study of annual crow births in India, it was found that there was relatively no growth during the period 2002 to 2004 and a sudden spike from 2004 to 2005. In another unrelated study, it was found that the revenue from cracker sales in India which remained fairly flat from 2002 to 2004, saw a sudden spike in 2005 before declining again in 2006. The solid line in the graph below refers to annual sale of crackers and the dashed line refers to the annual crow births in India. Choose the most appropriate inference from the above data.



- (a) There is a strong correlation between crow birth and cracker sales.
 (b) Cracker usage increases crow birth rate.
 (c) If cracker sale declines, crow birth will decline.
 (d) Increased birth rate of crows will cause an increase in the sale of crackers.

SET - 2

1. "The driver applied the _____ as soon as she approached the hotel where she wanted to take a _____."

The words that best fill the blanks in the above sentence are

- (a) brake, break (b) break, break
 (c) brake, brake (d) break, brake
2. "It is no surprise that every society has had codes of behavior; however, the nature of these codes is often _____."

The word that best fills the blank in the above sentence is

- (a) unpredictable (b) simple
 (c) expected (d) strict
3. Hema's age is 5 years more than twice Hari's age. Suresh's age is 13 years less than 10 times Hari's age. If Suresh is 3 times as old as Hema. How old is Hema?

- (a) 14 (b) 17
 (c) 18 (d) 19

4. Tower A is 90 m tall and tower B is 140 m tall. They are 100 m apart. A horizontal skywalk connects the floors at 70 m in both the towers. If a taut rope connects the top of tower A to the bottom of tower B, at what distance (in meters) from tower A will the rope intersect the skywalk?

5. The temperature T in a room varies as a function of the outside temperature T_0 and the number of persons in the room p , according to the relation $T = K(\theta p + T_0)$, where θ is K are constants. What would be the value of θ given the following data?

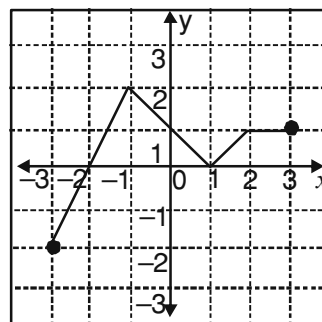
T_0	p	T
25	2	32.4
30	5	42.0

- (a) 0.8 (b) 1.0
 (c) 2.0 (d) 10.0
6. A fruit seller sold a basket of fruits at 12.5% loss. Had he sold it for Rs. 108 more, he would have made a 10% gain. What is the loss in Rupees incurred by the fruit seller?
- (a) 48 (b) 52
 (c) 60 (d) 108

7. The price of a wire made of a superalloy material is proportional to the square of its length. The price of 10 m length of the wire is Rs. 1600. What would be the total price (in Rs.) of two wires of lengths 4 m and 6 m?

- (a) 768 (b) 832
 (c) 1440 (d) 1600

8. What of the following function(s) in an accurate description of the graph for the range(s) indicated?



- (i) $y = 2x + 4$ for $-3 \leq x \leq -1$
 (ii) $y = |x - 1|$ for $-1 \leq x \leq 2$
 (iii) $y = ||x| - 1|$ for $-1 \leq x \leq 2$
 (iv) $y = 1$ for $2 \leq x \leq 3$
 (a) (i), (ii) and (iii) only
 (b) (i), (ii) and (iv) only
 (c) (i) and (iv) only
 (d) (ii) and (iv) only

9. Consider a sequence of number $a_1, a_2, a_3, \dots, a_n$

where $a_n = \frac{1}{n} - \frac{1}{n+2}$, for each integer $n > 0$. What is the sum of the first 50 terms?

- (a) $\left(1 + \frac{1}{2}\right) - \frac{1}{50}$ (b) $\left(1 + \frac{1}{2}\right) + \frac{1}{50}$
 (c) $\left(1 + \frac{1}{2}\right) - \left(\frac{1}{51} + \frac{1}{52}\right)$ (d) $1 - \left(\frac{1}{51} + \frac{1}{52}\right)$

10. Each of the letters arranged as below represents a unique from 1 to 9. The letters are positioned in the figure such that $(A \times B \times C)$, $(B \times G \times E)$ and $(D \times E \times F)$ are equal. Which integer among the following choices cannot be represented by the letters A, B, C, D, E, F or G?

A		D
B	G	E
C		F

- (a) 4 (b) 5
 (c) 6 (d) 9

SET-3

1. "From where are they bringing their books?
_____ bringing _____ books from _____."

The words that best fill the blanks in the above sentence are

- (a) Their, they're, there
(b) They're, their, there
(c) There, their, they're
(d) They're, there, there
2. "A _____ investigation can sometimes yield new facts, but typically organized ones are more successful."

The word that best fills the blank in the above sentence is

- (a) meandering (b) timely
(c) consistent (d) systematic
3. The area of a square is d . What is the area of the circle which has the diagonal of the square as its diameter?
- (a) πd (b) πd^2
(c) $\frac{1}{4}\pi d^2$ (d) $\frac{1}{2}\pi d$

4. What would be the smallest natural number which when divided either by 20 or by 42 or by 76 leaves a remainder of 7 in each case?

- (a) 3047 (b) 6047
(c) 7987 (d) 63847

5. What is the missing number in the following sequence?

2, 12, 60, 240, 720, 1440, _____, 0

- (a) 2880 (b) 1440
(c) 720 (d) 0

6. In appreciation of the social improvements completed in a town, a wealthy philanthropist decided to gift Rs. 750 to each male senior citizen in the town and Rs. 1000 to each female senior citizen. Altogether, there were 300 senior citizens eligible for this gift. However, only $\frac{8}{9}$ of the eligible men and $\frac{2}{3}$ of the eligible women claimed the gift. How much money (in Rupees) did the philanthropist give away in total?

- (a) 1,50,000 (b) 2,00,000
(c) 1,75,000 (d) 1,51,000

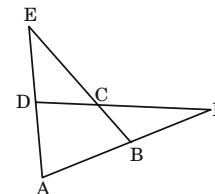
7. In $pqr \neq 0$ and $p^{-x} = \frac{1}{q}$, $q^{-y} = \frac{1}{r}$, $r^{-z} = \frac{1}{p}$, what is the value of the product xyz ?

- (a) -1 (b) $\frac{1}{pqr}$
(c) 1 (d) pqr

8. In a party, 60% of the invited guests are male and 40% are female. If 80% of the invited guests attended the party and if all the invited female guests attended, what would be the ratio of males of females among the attendees in the party?

- (a) 2 : 3 (b) 1 : 1
(c) 3 : 2 (d) 2 : 1

9. In the figure below, $\angle DEC + \angle BFC$ is equal to ____.



- (a) $\angle BCD - \angle BAD$
(b) $\angle BAD + \angle BCF$
(c) $\angle BAD + \angle BCD$
(d) $\angle CBA + \angle ADC$

10. A six sided unbiased die with four green faces and two red faces is rolled seven times. Which of the following combinations is the most likely outcome of the experiment?

- (a) Three green faces and four red faces.
(b) Four green faces and three red faces.
(c) Five green faces and two red faces.
(d) Six green faces and one red face.

SET-4

1. "By giving him the last _____ of the cake, you will ensure lasting _____ in our house today."

The words that best fill the blanks in the above sentence are

- (a) peas, piece (b) piece, peace
(c) peace, piece (d) peace, peas

2. "Even though there is a vast scope for its _____, tourism has remained a/an _____ area."

The words that best fill the blanks in the above sentence are

- (a) improvement, neglected
(b) rejection, approved
(c) fame, glum
(d) interest, disinterested

3. If the number $715 ? 423$ is divisible 3 (? denotes the missing digit in the thousandths place), then the smallest whole number in the place of ? is

- (a) 0 (b) 2
(c) 5 (d) 6

4. What is the value of $1 + \frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \frac{1}{256} + \dots$?

- (a) 2 (b) $\frac{7}{4}$
(c) $\frac{3}{2}$ (d) $\frac{4}{3}$

5. As 1.5 m in tall person is standing at a distance of 3 m from a lamp post. The light from the lamp at the top of the post casts her shadow. The length of the shadow is twice her height. What is the height of the lamp post in meters?

(a) 1.5 (b) 3
(c) 4.5 (d) 6

6. Leila aspires to buy a car worth Rs. 10,00,000 after 5 years. What is the minimum amount in Rupees that she should deposit now in a bank which offers 10% annual rate of interest, if the interest was compounded annually?

(a) 5,00,000 (b) 6,21,000
(c) 6,66,667 (d) 7,50,000

7. Two alloys A and B contain gold and copper in the ratios of 2 : 3 and 3 : 7 by mass, respectively. Equal masses of alloys A and B are melted to make an alloy C. The ratio of gold to copper in alloy C is _____.

(a) 5 : 10 (b) 7 : 13
(c) 6 : 11 (d) 9 : 13

8. The Cricket Board has long recognized John's potential as a leader of the team. However, his on-field Temper has always been a matter of concern for them since his junior days. While this aggression has filled stadia with die-hard fans, it has taken a toll on his own batting. Until recently, it appeared that he found it difficult to convert his aggression into big scores. Over the past three seasons though, that picture of John has been replaced by a cerebral, calculative and successful batsman-captain. After many years, it appears that the team has finally found a complete captain. Which of the following statements can be logically inferred from the above paragraph?

- (i) Even as a junior cricketer, John was considered a good captain.
(ii) Finding a complete captain is a challenge.
(iii) Fans and the Cricket Board have differing views on what they want in a captain.
(iv) Over the past three seasons John has accumulated big scores.

(a) (i), (ii) and (iii) only (b) (iii) and (iv) only
(c) (ii) and (iv) only (d) (i), (ii), (iii) and (v)

9. A cab was involved in a hit and run accident at right you are given the following data about the cabs in the city and the accident.

- (i) 85% of cabs in the city are green and the remaining cabs are blue.
(ii) A witness identified the cab involved in the accident as blue.

- (iii) It is known that a witness can correctly identify the cab colour only 80% of the time.

Which of the following options is closest to the probability that the accident was caused by a blue cab?

(a) 12% (b) 15%
(c) 41% (d) 80%

10. A coastal region with unparalleled beauty is home to many species of animals. It is dotted with coral reefs and unspoilt white sandy beaches. It has remained inaccessible to tourists due to poor connectivity and lack of accommodation. A company has spotted the opportunity and is planning to develop a luxury resort with helicopter service to the nearest major city airport. Environmentalists are upset that this would lead to the region becoming crowded and polluted like any other major beach resorts.

Which one of the following statements can be logically inferred from the information given in the above paragraph?

- (a) The culture and tradition of the local people will be influenced by the tourists.
(b) The region will become crowded and polluted due to tourism.
(c) The coral reefs are on the decline and could soon vanish.
(d) Helicopter connectivity would lead to an increase in tourists coming to the region.

SET-5

1. Going by the _____ that many hands make light work, the school _____ involved all the students in the task."

(a) principle, principal
(b) principal, principle
(c) principle, principle
(d) principal, principal

2. "Her _____ should not be confused with miserliness; she is ever willing to assist those in need."

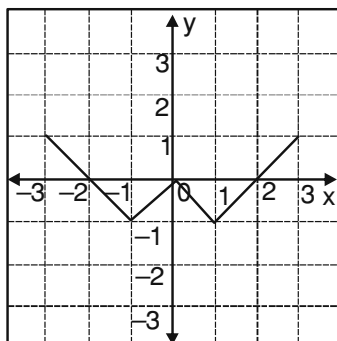
The word that best fills the blank in the above sentence is

(a) cleanliness (b) punctuality
(c) frugality (d) greatness

3. Seven machines take 7 minutes to make 7 identical toys. At the same rate, how many minutes would it take for 100 machines to make 100 toys?

(a) 1 (b) 7
(c) 100 (d) 700

4. A rectangle becomes a square when its length and breadth are reduced by 10m and 5 m, respectively, during this process, the rectangle loses 650 m^2 of area. What is the area of the original rectangle in square meters?
- (a) 1125 (b) 2250
(c) 2924 (d) 4500
5. A number consists of two digits. The sum of the digits is 9. If 45 is subtracted from the number, its digits are interchanged. What is the number?
- (a) 63 (b) 72
(c) 81 (d) 90
6. For integers a, b and c, what would be the minimum and maximum values respectively of $a + b + c$ if $\log |a| + \log |b| + \log |c| = 0$
- (a) -3 and 3 (b) -1 and 1
(c) -1 and 3 (d) 1 and 3
7. Given that a and b are integers and $a + a^2b^3$ is odd, which one of the following statements is correct?
- (a) a and b are both odd
(b) a and b are both even
(c) a is even and b is odd
(d) a is odd and b is even
8. From the time the front of train enters a platform, it takes 25 seconds for the back of the train to leave the platform, while travelling at a constant speed of 54 km/h. At the same speed, it takes 14 seconds to pass a man running at 9 km/h in the same direction as the train. What is the length of the train and that of the platform in meters, respectively?
- (a) 210 and 140 (b) 162.5 and 187.5
(c) 245 and 130 (d) 175 and 200
9. Which of the following functions describe the graph shown in the below figure.



- (a) $y = ||x| + 1| - 2$ (b) $y = ||x| - 1| - 1$
(c) $y = ||x| + 1| - 1$ (d) $y = ||x - 1| - 1|$

10. Consider the following three statements:

- (i) Some roses are red
(ii) All red flowers fade quickly
(iii) Some roses fade quickly

Which of the following statements can be logically inferred from the above statements?

- (a) If (i) is true and (ii) is false, then (iii) is false.
(b) If (i) is true and (ii) is false, then (iii) is true.
(c) If (i) and (ii) are true, then (iii) is true.
(d) If (i) and (ii) are false, then (iii) is false.

SET-6

1. "Since you have gone off the _____, the _____ sand is likely to damage the car". The words that best fill the blanks in the above sentence are
- (a) course, coarse (b) course, course
(c) coarse, course (d) coarse, coarse
2. "A common misconception among writers is that sentence structure mirrors thought; the more _____ the structure, the more complicated the ideas".
- (a) detailed (b) simple
(c) clear (d) convoluted
3. The three roots of the equation $f(x) = 0$ are $x = \{-2, 0, 3\}$. What are the three values of x for which $f(x - 3) = 0$?
- (a) -5, -3, 0 (b) -2, 0, 3
(c) 0, 6, 8 (d) 1, 3, 6
4. For what values of k given below is $\frac{(k+2)^2}{k-3}$ and integer?
- (a) 4, 8, 18 (b) 4, 10, 16
(c) 4, 8, 28 (d) 8, 26, 28
5. Functions, $F(a, b)$ and $G(a, b)$ are defined as follows:
 $F(a, b) = (a - b)^2$ and $G(a, b) = |a - b|$, where $|x|$ represents the absolute value of x. What would be the value of $G(F(1, 3), G(1, 3))$?
- (a) 2 (b) 4
(c) 6 (d) 36
6. An e-mail password must contain three characters. The password has to contain one numeral from 0 to 9, one upper case and one lower case character from the English alphabet. How many distinct passwords are possible?
- (a) 6,760 (b) 13,520
(c) 40,560 (d) 1,05,456
7. In a certain code AMCF is written as EQGJ and NKUF is written as ROYJ. How will DHLP be written in the code?
- (a) RSTN (b) TLPH
(c) HLPT (d) XSVR

8. A class of twelve children has two more boys than girls. A group of three children are randomly picked from this class to accompany the teacher on a field trip. What is the probability that the group accompanying the teacher contains more girls than boys?

- (a) 0
(b) $\frac{325}{864}$
(c) $\frac{525}{864}$
(d) $\frac{5}{12}$

9. A designer uses marbles of four different colours for his designs. The cost of each marble is the same, irrespective of the colour. The table below shows the percentage of marbles of each colour used in the current design. The cost of each marble increased by 25%. Therefore, the designer decided to reduce equal numbers of marbles of each colour to keep the total cost unchanged. What is the percentage of blue marbles in the new design?

Blue	Black	Red	Yellow
40%	25%	20%	15%

- (a) 35.75
(b) 40.25
(c) 43.75
(d) 46.25

10. P, Q, R and S crossed a lake in a boat that can hold a maximum of two persons, with only one set of oars. The following additional facts are available

- (i) The boat held two persons on each of the three forward trips across lake and one person on each of the two return trips.
(ii) P is unable to row when someone else is in the boat.
(iii) Q is unable to row with anyone else except R.
(iv) Each person rowed for at least one trip.
(v) Only one person can row during a trip.

Who rowed twice?

- (a) P
(b) Q
(c) R
(d) S

ANSWERS

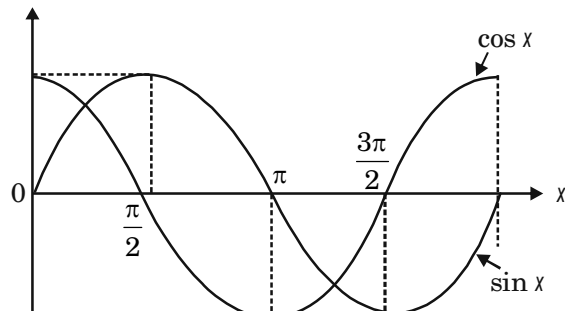
GENERAL APTITUDE

SET-1

1. (a) 2. (c) 3. (b) 4. (c) 5. (d)
6. (a) 7. (c) 8. (d) 9. (c) 10. (a)

EXPLANATIONS

1. Stares - To look at someone for a long time
Stairs - A construction designed to bridge a large vertical distance by dividing it into smaller vertical distances, called steps
2. Errant means misbehaving, exhibiting inappropriate behaviour / offending conduct.
- 3.



From the curve it is clear that $\sin x$ and $\cos x$ both are decreasing in the interval $\left(\frac{\pi}{2}, \pi\right)$

4. Area of equilateral triangle = $\frac{\sqrt{3}}{4}a^2$

$$\begin{aligned}\therefore \sqrt{3} &= \frac{\sqrt{3}}{4}a^2 \\ \therefore a^2 &= 4 \\ \therefore a &= 2\end{aligned}$$

Now perimeter of equilateral triangle

$$= 3a = 3 \times 2 = 6$$

5. (i) Volume of cuboid = $(8 \times 10 \times 6) \text{ cm}^3 = 480 \text{ cm}^3$
(ii) Volume of cube = $(8 \times 8 \times 8) \text{ cm}^3 = 512 \text{ cm}^3$
(iii) Volume of cylinder = $\pi r^2 h$

$$\left(\frac{22}{7} \times 7 \times 7 \times 7\right) \text{ cm}^3 = 1078 \text{ cm}^3$$

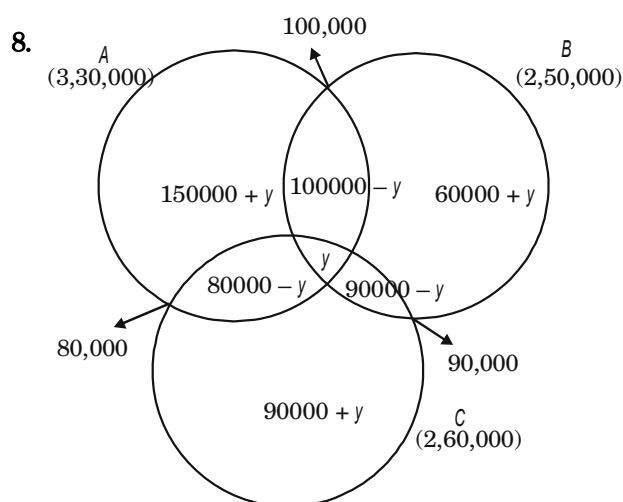
$$\begin{aligned}\text{(iv) Volume of Sphere} &= \frac{4}{3}\pi r^3 = \left(\frac{4}{3}\pi \times (7)^2\right) \text{ cm}^3 \\ &= 1436.75 \text{ cm}^3\end{aligned}$$

Hence, the descending order of their volumes is

- (iv), (iii), (ii), (i).

$$\begin{aligned}
 6. \quad \text{Average speed} &= \frac{2(S_1 \times S_2)}{S_1 + S_2} \\
 &= \frac{2(60 \times 90)}{(60 + 90)} \text{ km/h} \\
 &= \frac{2 \times 60 \times 90}{150} = 72 \text{ km/h}
 \end{aligned}$$

$$\begin{aligned}
 7. \quad \text{Number of parallelogram} &= {}^4C_2 \times {}^5C_2 \\
 &= \frac{4!}{2! \times 2!} \times \frac{5!}{3! \times 2!} \\
 &= 6 \times \frac{5 \times 4}{2} \\
 &= 60
 \end{aligned}$$



From above diagram,

$$\therefore 6,30,000 = 2y + 1,50,000 + 100,000 + 80,000 + 60,000 + 90,000 + 90,000$$

$$\therefore 63,000 = 2y + 57,000$$

$$\therefore 6,30,000 - 5,70,000 = 2y$$

$$\therefore 60,000 = 2y$$

$$\therefore y = 30,000$$

Students who failed to clear the test

$$\begin{aligned}
 &= 150,000 + 60,000 + 90,000 + 4y \\
 &= 300,000 + 4 \times 30,000 \quad (\because y = 30,000) \\
 &= 420,000
 \end{aligned}$$

Students who failed to clear the test = 420,000

$$9. \quad x^2 - 1 = -x$$

$$x - \frac{1}{x} = -1$$

$$x^2 + \frac{1}{x^2} = 0.2$$

$$\begin{aligned}
 \text{Now,} \quad x^4 + \frac{1}{x^4} &= \left(x^2 + \frac{1}{x^2}\right)^2 - 2 \\
 &= (3)^2 - 2 \\
 &= 9 - 2 \\
 &= 7
 \end{aligned}$$

ANSWERS

SET-2

1. (a) 2. (a) 3. (d) 4. 22.22 5. (b)
6. (c) 7. (b) 8. (b) 9. (c) 10. (b)

EXPLANATIONS

3. Let the age of Hari be 'x'

$$\therefore \text{Hema age} = 5 + 2x$$

$$\text{and Suresh age} = 10x - 13$$

Now Suresh is 3 times old as Hema

$$\therefore (10x - 13) = 3(5 + 2x)$$

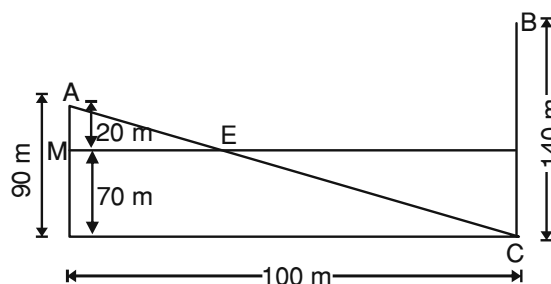
$$\therefore 10x - 13 = 15 + 6x$$

$$\therefore 4x = 28$$

$$\therefore x = 7$$

$$\begin{aligned}
 \therefore \text{Hema's age} &= 5 + 2x \\
 &= 5 + 14 \\
 &= 19
 \end{aligned}$$

4.



$$\triangle AME \approx \triangle APC$$

$$\frac{AM}{AP} = \frac{ME}{PC}$$

$$\therefore \frac{20}{90} = \frac{ME}{100}$$

$$\therefore ME = \frac{20 \times 100}{90} = 22.22$$

5. For $T = 32.4$,

$$p = 2$$

$$\text{and } T_0 = 25$$

$$T = K(\theta p + T_0)$$

$$\therefore 32.4 = K(20 + 25) \quad \dots(i)$$

$$\text{Now for } T = 42, p = 5 \text{ and } T_0 = 30$$

$$T = K(\theta p + T_0)$$

$$\therefore 42 = K(50 + 30) \quad \dots(ii)$$

Solving equation (i) and (ii) we get

$$\theta = 1$$

and $K = \frac{78}{65}$

6. $12.5\%x + 10\%x = 108$

$$x = \frac{108}{22.5}$$

So loss $108 \times \frac{12.5}{22.5} = 60$

7. Total price $(p) = p_1 + p_2$
 $= 16(4)^2 + 16(6)^2$
 $= 16[16 + 36]$
 $= 16[52] = 832$

Price $\propto L^2$

Price $= KL^2$ (where K is constant)

Now according to question

$$1600 = K(10)^2$$

$\therefore K = 16$

8. Put value and verify

(i) $y = 2x + 4$ is true in $-3 \leq x \leq -1$

On putting $x = -3, y = -2$ and $x = -2, y = 0$ and $x = 1, y = 2$

(ii) $y = |x - 1|$ is also true ($x = -1, y = 2$), ($x = 0, y = 1$) and ($x = 1, y = 0$)

(iv) $y = 1$ in ($2 \leq x \leq 3$) always true

(i), (ii) and (iv) are true.

9. Sum of series will be

$$= \left(1 - \frac{1}{3}\right) + \left(\frac{1}{2} - \frac{1}{4}\right) + \left(\frac{1}{3} - \frac{1}{5}\right) \dots \left(\frac{1}{48} - \frac{1}{50}\right) \\ + \left(\frac{1}{49} - \frac{1}{51}\right) + \left(\frac{1}{50} - \frac{1}{52}\right)$$

All like terms will cancel out and we will be left with

$$= \left(1 + \frac{1}{2}\right) - \left(\frac{1}{51} + \frac{1}{52}\right)$$

10. $A \times B \times C = B \times G \times E$
 $= D \times E \times F$
 $= 72$

$$8 \times 9 \times 1 = 9 \times 2 \times 4$$

$$= 3 \times 4 \times 6$$

$$= 72$$

Any of A, B, C, D, E, F, G cannot be 5.

ANSWERS

SET-3

1. (b) 2. (a) 3. (d) 4. (c) 5. (b)
 6. (b) 7. (c) 8. (b) 9. (a) 10. (c)

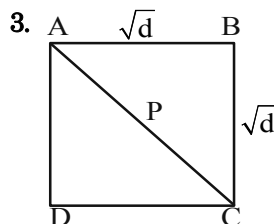
EXPLANATIONS

1. They're used for pointing group.

Their is pointing people.

There is used for place.

2. Meandering : wandering aimlessly/indirect.



Area of square $= d$

Side one side of square $= \sqrt{d}$

Now Diagonal of square $(AC) = \sqrt{d+d} = \sqrt{2d}$

Now, Area of circle $= \pi r^2$

$$= \pi \left(\frac{\sqrt{2d}}{2} \right)^2 \left(\because r = \frac{\text{diameter}}{2} \right)$$

$$= \pi \frac{d}{2} = \frac{1}{2} \pi d$$

4. Number is divided by either by 20 or 42 or by 76

$K \times \text{LCM}(20, 42, 76) + \text{constant difference}$

$= 7890K + 7$ (K is natural number)

Least number will be $7890 + 7 = 7897$.

5.
$$\begin{array}{ccccccccccc} 2 & & 12 & & 60 & & 240 & & 720 & & 1440 & & 1440 & & 0 \\ & \swarrow & & \searrow & & \swarrow & & \searrow & & \swarrow & & \searrow & & \swarrow & & \searrow \\ (2 \times 6) & & (12 \times 5) & & (60 \times 4) & & (240 \times 3) & & (720 \times 2) & & (1440 \times 1) & & (1440 \times 0) \\ & \nwarrow & & \swarrow & & \nwarrow & & \swarrow & & \nwarrow & & \swarrow & & \nwarrow \\ & -1 & & -1 & & -1 & & -1 & & -1 & & -1 & & \end{array}$$

So, 1440 must be answer.

6. Male (M) + Female (F) = 300

$$\text{Total money} = \frac{8}{9}M \times 750 + \frac{2}{3}F \times 1000 \quad \dots (i)$$

$$= \frac{6000}{9}M + \frac{6000}{9}F = \frac{6000}{9}(M + F)$$

From equation (i) we get,

$$\text{Total money} = \frac{6000 \times 300}{9} = 2,00,000$$

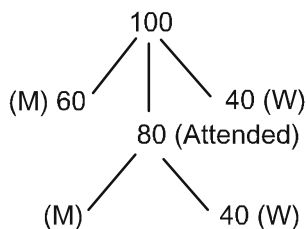
7. $p^x = q \Rightarrow x \log p = \log q$

and $q^y = r \Rightarrow y \log q = \log r$

and $z^2 = p \Rightarrow z \log r = \log p$

$\therefore xyz = \frac{\log q}{\log p} \times \frac{\log r}{\log q} \times \frac{\log p}{\log r} = 1$

8. Let total number of people are 100.

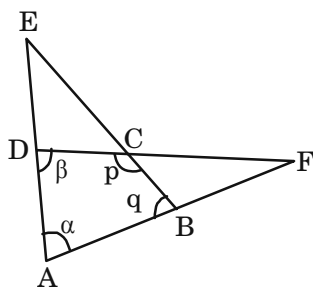


So, male attended the party $80 - 40 = 40$

Ratio of male to female who attended the party

$$\begin{aligned} M : W &= 40 : 40 \\ &= 1 : 1 \end{aligned}$$

9.



$$\angle E + \angle F = ?$$

$$\alpha + q + E = 180 \quad \dots(i)$$

$$\text{and} \quad \alpha + \beta + F = 180 \quad \dots(ii)$$

$$\text{and} \quad \alpha + \beta + p + q = 360 \quad \dots(iii)$$

Adding equations (i) and (ii) and equating it to (iii) we get

$$\alpha + q + E + \alpha + \beta + F = \alpha + \beta + p + q$$

$$\therefore E + F = p - \alpha$$

10. Four green, two red face

$$P(G) = \frac{4}{6} = \frac{2}{3}$$

$$q(R) = 1 - P(G) = \frac{1}{3}$$

$$n = 7$$

Now from options

$$\text{From option (a),} \quad P(G = 3) = {}^7C_3 \left(\frac{2}{3}\right)^3 \left(\frac{1}{3}\right)^4$$

$$\frac{35 \times 2^3}{(3)^7} = \frac{35 \times 2^3}{(3)^7}$$

$$\text{From option (b),} \quad P(G = 4) = {}^7C_4 \left(\frac{2}{3}\right)^4 \left(\frac{1}{3}\right)^3$$

$$\frac{35 \times 2^4}{(3)^7} = \frac{35 \times 2^4}{(3)^7}$$

$$\text{From option (c),} \quad P(G = 5) = {}^7C_5 \left(\frac{2}{3}\right)^5 \left(\frac{1}{3}\right)^2$$

$$\frac{21 \times 2^5}{(3)^7} = \frac{42 \times 2^4}{(3)^7}$$

$$\text{From option (d),} \quad P(G = 6) = {}^7C_6 \left(\frac{2}{3}\right)^6 \times \left(\frac{1}{3}\right)$$

$$\frac{7 \times 2^6}{(3)^7} = \frac{28 \times 2^4}{(3)^7}$$

Option c is maximum value.

ANSWERS

SET - 4

1. (b) 2. (a) 3. (b) 4. (d) 5. (b)
6. (b) 7. (b) 8. (c) 9. (a) 10. (b)

EXPLANATIONS

3. Rule of divisibility by 3 : Sum of all digits should be divisible by 3

$$715 ? 423$$

$$7 + 1 + 5 + 4 + 2 + 3 = 22$$

So, the next number after sum 22 which are divisible by 3 are 24, 27, 30 etc.

$$\text{So,} \quad 22 + 2 = 24$$

$$22 + 5 = 27$$

But according to given condition 2 is right answer.

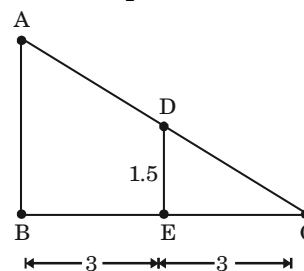
4. The given series is in G.P,

$$\text{where } a = 1 \text{ and } r = \frac{1}{4}$$

$$\text{Now, in G.P,} \quad s_n = \frac{a}{1-r}$$

$$\frac{1}{1 - \frac{1}{4}} = \frac{1}{3/4} = \frac{4}{3}$$

5.



Let 'AB' be the height of lamp post

Here $\triangle ABC$ and $\triangle DEC$ are similar triangles,

$$\therefore \frac{AB}{DE} = \frac{BC}{EC}$$

$$\therefore \frac{AB}{1.5} = \frac{6}{3}$$

$$\therefore \frac{AB}{1.5} = 2$$

$$AB = 3 \text{ meter}$$

Hence height of lamp post is 3 m.

6.

$$P : A \\ (10)^5 : (11)^5$$

$$\therefore P = \frac{(10)^6 \times (10)^5}{(11)^5} \\ = \frac{(10)^{11}}{(11)^5} \\ = 620921.323 \approx 621000$$

7.

$$\begin{array}{cc} A & B \\ G : C & G : C \\ 2 : 3 & 3 : 7 \end{array}$$

Now making the mass equal of alloy A and B.

$$\begin{array}{cc} A & B \\ G : C & G : C \\ 4 : 6 & 3 : 7 \end{array}$$

$\therefore$ Alloy of C contain

$$G : C = 4 + 3 : 6 + 7 = 7 : 13$$

8. Statement (i) is not true as nowhere it is mentioned that John was a captain in junior team. The introductory line emphasizes on the board recognizing John's potential (Latent quality/possibility) as leader of the team.

Statement (iii) also manipulates the facts mentioned in the argument.

The 3rd statement of the argument while this aggression has filled stadia with die-hard fans does not indicate fans expectations from John as a captain.

Statement (ii) The concluding statement of the para suggests that finding a completer captain is a tough task as it took John many years to become a successful and calculative batsman - captain.

Statement (iv) can be explicitly concluded from the last 4 lines of the para.

9. Probability that accident was caused by blue cab
 $= 0.15 \times 0.8 = 0.12 = 12\%$

10. (A) is beyond the scope of given information option (c) can also be discarded on the same grounds.

The argument deals with the coastal region becoming crowded and polluted because of the upcoming luxury resort. Option (b) precisely underlines the theme of the para.

ANSWERS

SET-5

1. (a) 2. (c) 3. (b) 4. (b) 5. (b)
 6. (a) 7. (d) 8. (d) 9. (b) 10. (c)

EXPLANATIONS

$$3. \frac{\text{Machine} \times \text{time}}{\text{Number of toys}} = \frac{M_1 \times t_1}{T_1}$$

$$= \frac{M_2 \times t_2}{T_2}$$

$$\therefore \frac{7 \times 7}{7} = \frac{100 \times t_2}{100}$$

$$\therefore \boxed{t_2 = 7 \text{ min}}$$

4. Let the initial length of rectangle is l and breath is b . Now if length is reduced by 10 m and breath is reduced by 5 m then it becomes square.

$$\therefore l - 10 = b - 5$$

[$\because$ In square length = breadth]

$$\therefore l - b = 5$$

Given, initial, Area (A) = $l \times b$

Now according to question,

$$(l - 10)(b - 5) = A - 650$$

$$\therefore lb - 10b - 5l + 50 = A - 650$$

$$\therefore A - 10b - 5l = A - 700$$

$$\therefore 10b + 5l = 700$$

$$\therefore 10b + 5(b + 5) = 700$$

$$\therefore 15b + 25 = 700$$

$$\therefore 15b = 700 - 25$$

$$\therefore 15b = 675$$

$$\Rightarrow b = 45 \text{ m}$$

$$\therefore l = 45 + 5 = 50 \text{ m}$$

$$\therefore \text{Area of original rectangle (A)} = (45 \times 50) \text{ m}^2 \\ = 2250 \text{ m}^2.$$

5. Let two digit number be xy .

Now sum of digit is 9

$$\therefore x + y = 9 \quad \dots(i)$$

and when '45' is subtracted from number then digit gets interchanged

$$\therefore 10x + y - 45 = 10y + x$$

$$\therefore x - 5 = y$$

$$x - y = 5 \quad \dots(ii)$$

Now solving equation (i) and (ii) we get

$$x = 7 \text{ and } y = 2$$

Hence the number is 72.

6. $\log |a| + \log |b| + \log |c| = 0$

It is possible only,

when $|a|$, $|b|$ and $|c|$ all are equal to 1.

$\therefore a, b, c$ may be $\pm 1, \pm 1, \pm 1$ respectively.

Now for minimum value of all three will be negative.

$$\therefore \text{minimum value} = -3$$

and maximum value of all three will be positive.

$$\therefore \text{maximum value} = +3$$

7. From option (a),

$$\text{Let } a = 1, b = 3$$

$$\therefore a + a^2b^3 = 1 + 27$$

$$= 28 \text{ (Even)}$$

$\therefore$ Option (a) is wrong

From option (b)

$$\text{Let } a = 2, b = 4$$

$$\therefore a + a^2b^3 = 2 + 4 \times 64$$

$$= 258 \text{ (Even)}$$

From option (c),

$$\text{Let } a = 2, b = 1$$

From option (d),

for $x = 1$

$$\begin{aligned} y &= |x-1|-1 \\ &= |-1| = 1 \end{aligned}$$

$\therefore$ Option (c) is wrong.

Now from option (b),

for $x = 1$

$$\begin{aligned} y &= |x|-1-1 \\ &= |1|-1-1 \\ &= -1 \end{aligned}$$

$\therefore$ Option (b) is correct.

8. Train speed (S_T) = 54 km/h

Time = 25 sec for travelling length of train and length of platform

Man speed (S_M) = 9 km/h

Speed of train to man = 45 km/h

Time = 14 sec

So, length of train = time $\times$ speed

$$= 14 \times 45 \times \frac{5}{18}$$

Length of train (L_T) = $(35 \times 5 \text{ m}) = 175 \text{ m}$

Length of platform (L) + length of train (L_T) = speed $\times$ time

$$= 54 \times \frac{5}{18} \times 25 = 15 \times 25 = 375 \text{ m}$$

$\therefore$ Length of platform (L) = $375 - 175 = 200 \text{ m}$

9. Here from graph for $x = 1, y = -1$

Checking the same from options.

From option (a)

for $x = 1$

$$\begin{aligned} y &= ||1|+1|-2 \\ &= 2-2=0 \end{aligned}$$

$\therefore$ Option (a) is wrong

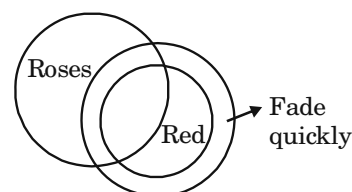
From option (c)

for $x = 1$

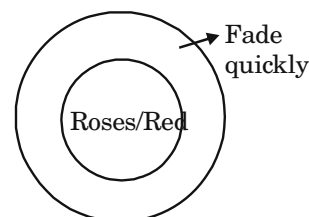
$$\begin{aligned} y &= ||x|+1|-1 \\ &= |1+1|-1=1 \end{aligned}$$

$\therefore$ Option (c) is wrong.

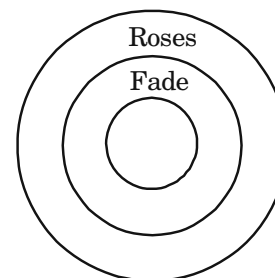
10.



Case-I



Case-II



Case-III

ANSWERS

SET-6

- | | | | | |
|--------|--------|--------|--------|---------|
| 1. (a) | 2. (d) | 3. (d) | 4. (c) | 5. (a) |
| 6. (c) | 7. (c) | 8. (b) | 9. (c) | 10. (c) |

EXPLANATIONS

3. $f(x) = 0$

for, $x = \{-2, 0, 3\}$

$\therefore f(-2) = 0, f(0) = 0, f(3) = 0$

$f(x-3) = 0$

Now from option (d),

$$f(1-3) = f(-2) = 0$$

$$f(3-3) = f(0) = 0$$

$$f(6-3) = f(3) = 0$$

$$\left. \begin{array}{l} \text{4. For } K = 4 \Rightarrow \frac{(4+2)^2}{4-3} = 36 \\ K = 8 \Rightarrow \frac{(8+2)^2}{8-3} = \frac{100}{5} = 20 \\ \text{For } K = 28 \Rightarrow \frac{(28+2)^2}{28-3} = \frac{900}{25} = 36 \end{array} \right\} \text{Integers}$$

$\therefore$ Option (c) is correct.

5. Here $F(a, b) = (a - b)^2$

and $G(a, b) = |a - b|$

Now $G[F(1, 3), G(1, 3)]$

$$= G[4, 2]$$

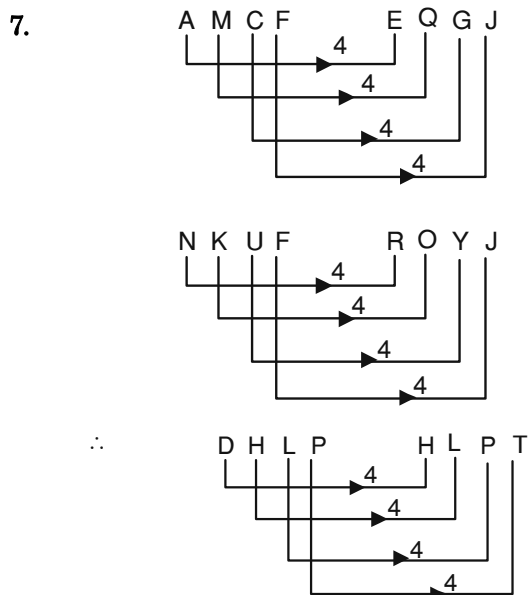
$$= |4 - 2| = 2$$

6. Numeral can be selected in 10 ways (0 - 9).

Each of upper case and lower case alphabets can be done in 26 ways each.

All three choosen (1 numeral and 2 alphabet) can be arrange in $3!$ ways.

$\therefore$ total number of ways – $(10 \times 26 \times 26) 3! = 40560$ ways



8. There are 7B and 5G in class

Now according to question 3 students are taken at random.

Hence possible ways are

G	B
2	1
3	0

$$\therefore \text{ Required probability} = \frac{{}^5C_2 + {}^7C_1 + {}^5C_3 + {}^7C_0}{{}^{12}C_3}$$

$$= \frac{10 \times 7 + 10 \times 1}{\frac{12 \times 10 \times 11}{6}} = \frac{80}{220} = \frac{4}{11} \approx 0.3636$$

9. Let the number of marbles be 100

Now Cost of marbles increased by Rs. 25

$\therefore$ New cost = Rs. 125

$$\text{Number of marbles in Rs. 100} = \frac{100}{125} \times 100 = 80 \text{ marbles}$$

Now according to question

$$(40 - x) + (25 - x) + (20 - x) + (15 - x) = 80$$

$$\therefore 100 - 4x = 80$$

$$\therefore x = 5$$

$$\begin{aligned} \% \text{ blue marbles in new design} &= \left[\frac{(40 - 5)}{80} \times 100 \right] \% \\ &= \left(35 \times \frac{5}{4} \right) \% = 43.75\% \end{aligned}$$

10. (i) Q and R moves first.

In forward trip Q rowed.

In return trip R rowed

(ii) P and R moves in second trip.

R rowed in forward trip

P rowed in return trip

(iii) P and S moves in last trip

S rowed in forward trip.

R rowed twice.

SOLVED PAPER – 2019

GENERAL APTITUDE

SET-1

(Q.1 – 5) : Carry One Mark Each

1. The lecture was attended by quite _____ students, so the hall was not very _____.
(a) few, quite (b) few, quiet
(c) a few, quiet (d) a few, quite
2. They have come a long way in _____ trust among the users.
(a) creating (b) create
(c) created (d) creations
3. If $E = 10$; $J = 20$; $O = 30$; and $T = 40$, what will be $P + E + S + T$?
(a) 51 (b) 120
(c) 82 (d) 164
4. The CEO's decision to quit was as shocking to the Board as it was to _____.
(a) myself (b) I
(c) my (d) me
5. On a horizontal ground, the base of a straight ladder is 6m away from the base of a vertical pole. The ladder makes an angle of 45° to the horizontal. If the ladder is resting at a point located at one-fifth of the height of the pole from the bottom, the height of the pole is _____ meters.
(a) 35
(b) 30
(c) 15
(d) 25

(Q.6 – 10) : Carry Two Marks Each

6. "The increasing interest in tribal characters might be a mere coincidence, but the timing is of interest. None of this, though, is to say that the tribal here has arrived in Hindi cinema, or that the new crop of characters represents the acceptance of the tribal character in the industry. The films and characters are too few to be described as a pattern."
What does the word „arrived. mean in the paragraph above?
(a) went to a place
(b) came to a conclusion
(c) attained a status
(d) reached a terminus

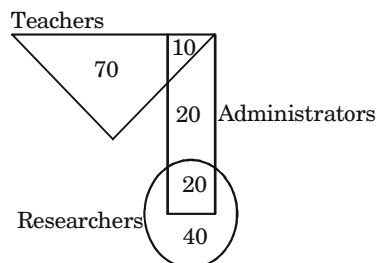
7. In a sports academy of 300 people, 105 play only cricket, 70 play only hockey, 50 play only football, 25 play both cricket and hockey, 15 play both hockey and football and 30 play both cricket and football. The rest of them play all three sports. What is the percentage of people who play at least two sports?
(a) 28.00 (b) 23.30
(c) 50.00 (d) 25.00
8. P, Q, R, S and T are related and belong to the same family. P is the brother of S. Q is the wife of P. R and T are the children of the siblings P and S respectively. Which one of the following statements is necessarily FALSE?
(a) S is the aunt of T
(b) S is the aunt of R
(c) S is the sister-in-law of Q
(d) S is the brother of P
9. The new cotton technology, Bollgard-II, with herbicide tolerant traits has developed into a thriving business in India. However, the commercial use of this technology is not legal in India. Now with standing that, reports indicate that the herbicide tolerant Bt cotton had been purchased by farmers at an average of Rs 200 more than the control price of ordinary cotton, and planted in 15% of the cotton growing area in the 2017 Kharif season.
Which of the following statements can be inferred from the given passage?
(a) Farmers want to access the new technology by paying high price
(b) Farmers want to access the new technology if India benefits from it
(c) Farmers want to access the new technology for experimental purposes
(d) Farmers want to access the new technology even if it is not legal
10. A square has sides 5 cm smaller than the sides of a second square. The area of the larger square is four times the area of the smaller square. The side of the larger square is _____ cm.
(a) 18.50 (b) 10.00
(c) 15.10 (d) 8.50

SET-2**Q. 1 to Q. 5 : Carry One Mark Each.**

- Ten friends planned to share equally the cost of buying a gift for their teacher. When two of them decided not to contribute, each of the other friends had to pay Rs. 150 more. The cost of the gift was Rs. _____.
(a) 666 (b) 6000
(c) 3000 (d) 12000
- A court is to a judge as _____ is to a teacher.
(a) a student (b) a syllabus
(c) a school (d) a punishment
- The search engine's business model _____ around the fulcrum of trust.
(a) bursts (b) revolves
(c) plays (d) sinks
- Two cars start at the same time from the same location and go in the same direction. The speed of the first car is 50 km/h and the speed of the second car is 60 km/h. The number of hours it takes for the distance between the two cars to be 20 km is _____.
(a) 2 (b) 1
(c) 6 (d) 3
- The expenditure on the project _____ as follows: equipment Rs. 20 lakhs, salaries Rs. 12 lakhs, and contingency Rs. 3 lakhs.
(a) break (b) break down
(c) breaks down (d) breaks

Q. 6 to Q. 10 : Carry Two Marks Each.

- In the given diagram, teachers are represented in the triangle, researchers in the circle and administrators in the rectangle. Out of the total number of the people, the percentage of administrators shall be in the range of _____.



- 0 to 15
- 31 to 45
- 46 to 60
- 16 to 30

- In a college, there are three student clubs. Sixty students are only in the Drama club, 80 students are only in the Dance club, 30 students are only in the Maths club, 40 students are in both Drama and Dance clubs, 12 students are in both Dance and Maths clubs, 7 students are in both Drama and Maths clubs, and 2 students are in all the clubs. If 75% of the students in the college are not in any of these clubs, then the total number of students in the college is _____.
(a) 975 (b) 1000
(c) 225 (d) 900

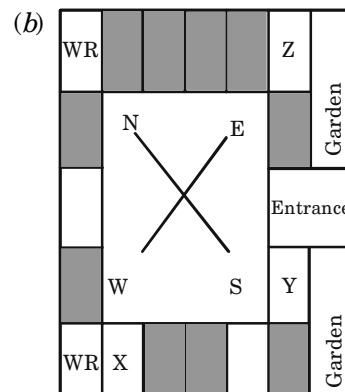
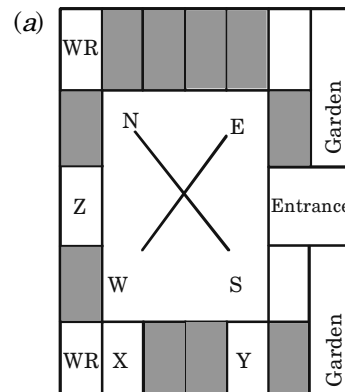
- Three of the five students allocated to a hostel put in special requests to the warden. Given the floor plan of the vacant rooms, select the allocation plan that will accommodate all their requests.

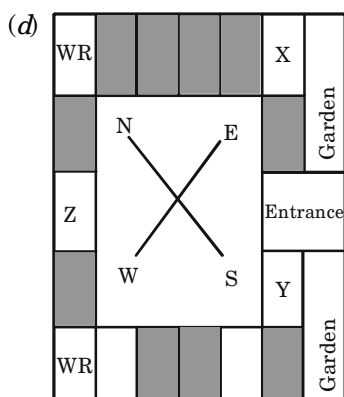
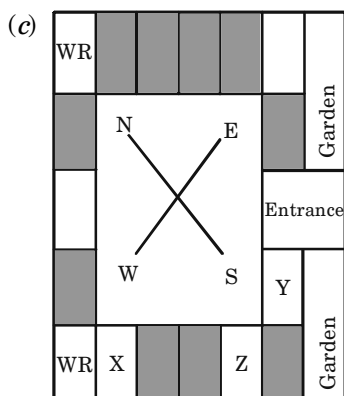
Request by X : Due to pollen allergy, I want to avoid a wing next to the garden.

Request by Y : I want to live as far from the washrooms as possible, since I am very sensitive to smell.

Request by Z : I believe in Vaastu and so want to stay in the South-west wing.

The shaded room are already occupied. WR is washroom.





9. The police arrested four criminals – P, Q, R and S. The criminals knew each other. They made the following statements :
- P says “Q committed the crime.”
- Q says “S committed the crime.”
- R says “I did not do it.”
- S says “What Q said about me is false.”
- Assume only one of the arrested four committed the crime and only one of the statements made above is true. Who committed the crime?
- (a) R (b) P
- (c) Q (d) S
10. “A recent High Court judgement has sought to dispel the idea of begging as a disease — which leads to its stigmatization and criminalization — and to regard it as a symptom. The underlying disease is the failure of the state to protect citizens who fall through the social security net.”
- Which one of the following statements can be inferred from the given passage?
- (a) Begging is an offence that has to be dealt with firmly
- (b) Beggars are lazy people who beg because they are unwilling to work
- (c) Beggars are created because of the lack of social welfare schemes
- (d) Beggings has to be banned because it adversely affects the welfare of the state

SET-3

Q. 1 to Q. 5 : Carry One Mark Each.

- The passengers were angry _____ the airline staff about the delay.
(a) towards (b) on
(c) with (d) about
- The missing number in the given sequence 343, 1331, _____, 4913 is
(a) 4096 (b) 3375
(c) 2744 (d) 2197
- Newspapers are a constant source of delight and recreation for me. The _____ trouble is that I read _____ many of them.
(a) only, too (b) only, quite
(c) even, too (d) even, quite
- It takes two hours for a person X to mow the lawn. Y can mow the same lawn in four hours. How long (in minutes) will it take X and Y, if they work together to mow the lawn?
(a) 60 (b) 80
(c) 120 (d) 90
- I am not sure if the bus that has been booked will be able to _____ all the students.
(a) deteriorate (b) sit
(c) accommodate (d) fill

Q. 6 to Q. 10 : Carry Two Marks Each.

- Given two sets $X = \{1, 2, 3\}$ and $Y = \{2, 3, 4\}$, we construct a set Z of all possible fractions where the numerators belong to set X and the denominators belong to set Y. The product of elements having minimum and maximum values in the set Z is _____.
(a) $\frac{1}{12}$ (b) $\frac{3}{8}$
(c) $\frac{1}{8}$ (d) $\frac{1}{6}$
- Consider five people-Mita, Ganga, Rekha, Lakshmi and Sana. Ganga is taller than both Rekha and Lakshmi. Lakshmi is taller than Sana. Mita is taller than Ganga.
Which of the following conclusions are TRUE?
1. Lakshmi is taller than Rekha
2. Rekha is shorter than Mita
3. Rekha is taller than Sana
4. Sana is shorter than Ganga
(a) 3 only (b) 1 only
(c) 2 and 4 (d) 1 and 3

8. How many integers are there between 100 and 1000 all of whose digits are even?
 (a) 60 (b) 100
 (c) 90 (d) 80
9. An award-winning study by a group researchers suggests that men are as prone to buying on impulse as women but women feel more guilty about shopping.
 Which one of the following statements can be inferred from the given text?
 (a) Many men and women indulge in buying on impulse
 (b) All men and women indulge in buying on impulse
 (c) Few men and women indulge in buying on impulse
 (d) Some men and women indulge in buying on impulse
10. The ratio of the number of boys and girls who participated in an examination is 4 : 3. The total percentage of candidates who passed the examination is 80 and the percentage of girls who passed is 90. The percentage of boys who passed is _____.
 (a) 90.00
 (b) 80.50
 (c) 55.50
 (d) 72.50
3. It would take one machine 4 hours to complete a production order and another machine 2 hours to complete the same order. If both machines work simultaneously at their respective constant rates, the time taken to complete the same order is _____ hours.
 (a) $\frac{2}{3}$ (b) $\frac{7}{3}$
 (c) $\frac{4}{3}$ (d) $\frac{3}{4}$
4. When he did not come home, she _____ him lying dead on the roadside somewhere
 (a) concluded (b) pictured
 (c) notice (d) looked
5. The strategies that the company _____ to sell its products _____ house-to-house marketing.
 (a) uses, include (b) use, includes
 (c) uses, including (d) used, includes

Q. 6 to Q. 10 : Carry Two Marks Each.

6. "Indian history was written by British historians—extremely well documented and researched, but not always impartial. History had to serve its purpose: Everything was made subservient to the glory of the Union Jack. Latter-day Indian scholar presented a contrary picture."

From the text above, we can infer that :

Indian history written by British historians ____.

SET - 4

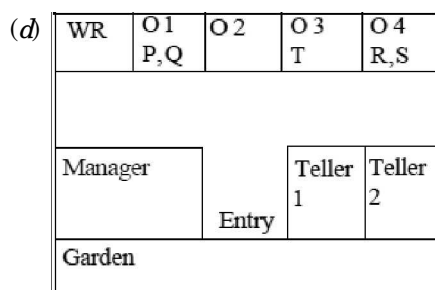
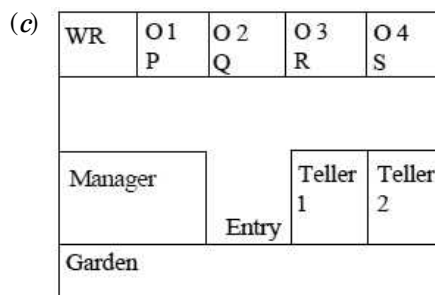
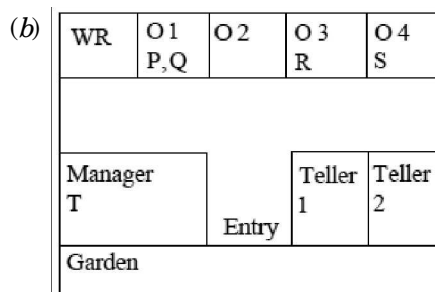
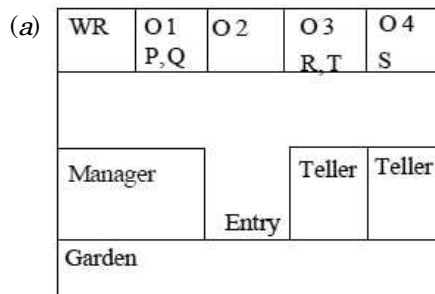
Q. 1 to Q. 5 : Carry One Mark Each.

1. Five different books (P, Q, R, S, T) are to be arranged on a shelf. The books R and S are to be arranged first and second, respectively from the right side of the shelf. The number of different order in which P, Q and T may be arranged is _____.
 (a) 2
 (b) 120
 (c) 6
 (d) 12
2. The boat arrived _____ dawn.
 (a) on
 (b) at
 (c) under
 (d) in
7. Two design consultants, P and Q, started working from 8 AM for a client. The client budgeted a total of USD 3000 for the consultants. P stopped working when the hour hand moved by 210 degrees on the clock. Q stopped working when the hour hand moved by 240 degrees. P took two tea breaks of 15 minutes each during her shift, but took no lunch break. Q took only one lunch break for 20 minutes, but no tea breaks. The market rate for consultants is USD 200 per hour and breaks are not paid. After paying the consultants, the client shall have USE _____ remaining in the budget.
 (a) 000.00 (b) 433.33
 (c) 166.67 (d) 300.00

8. Five people P, Q, R, S and T work in a bank. P and Q don't like each other but have to share an office till T gets a promotion and moves to the big office next to the garden. R, who is currently sharing an office with T wants to move to the adjacent office with S, the handsome new intern.

Given the floor plan, what is the current location of Q, R and T?

(O = Office, WR = Washroom)



9. Four people are standing in a line facing you. They are Rahul, Mathew, Seema and Lohit. One is an engineering, one is a doctor, one a teacher and another a dancer. You are told that:

- Mathew is not standing next to Seema
- There are two people standing between Lohit and the engineer

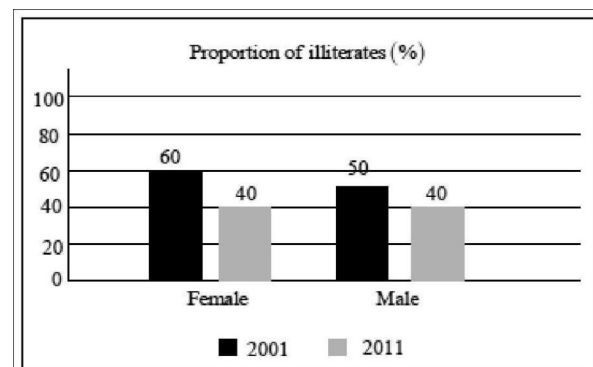
- Rahul is not a doctor
- The teacher and the dancer are standing next to each other.
- Seema is turning to her right to speak to the doctor standing next to her.

Who amongst them is an engineer?

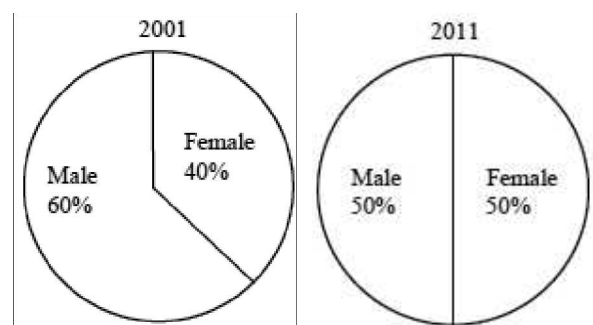
- Rahul
- Mathew
- Seema
- Lohit

10. The bar graph in Panel (a) shows the proportion of male and female illiterates in 2001 and 2011. The proportions of males and females in 2001 and 2011 are given in Panel (b) and (c), respectively. The total population did not change during this period.

The percentage increase in the total number of literate from 2001 to 2011 is _____.



Panel (a)



Panel (b)

Panel (c)

- 33.43
- 35.43
- 34.43
- 30.43

SET-5**Q. 1 to Q. 5 : Carry One Mark Each.**

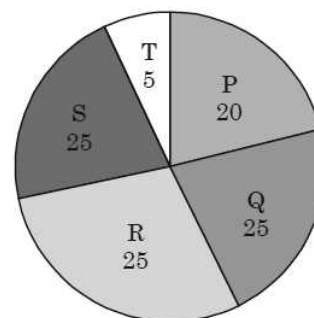
- John Thomas, an ___ writer, passed away in 2018.
(a) imminent (b) dominant
(c) eminent (d) prominent
- The minister avoided any mention of the issue of women's reservation in the private sector. He was accused of _____ the issue.
(a) collaring (b) belting
(c) tying (d) skirting
- The sum and product of two integers are 26 and 165 respectively. The difference between these two integers is _____.
(a) 4 (b) 2
(c) 6 (d) 3
- _____ I permitted him to leave, I wouldn't have had any problem with him being absent _____. I?
(a) Have, would (b) Have, wouldn't
(c) Had, would (d) Had, wouldn't
- A worker noticed that the hour hand on the factory clock had moved by 225 degrees during her stay at the factory. For how long did she stay in the factory ?
(a) 8.5 hours
(b) 3.75 hours
(c) 4 hours and 15 minutes
(d) 7.5 hours

Q. 6 to Q. 10 : Carry Two Marks Each.

- A person divided an amount of Rs. 100,000 into two parts and invested in two different schemes. In one he got 10% profit and in the other he got 12%. If the profit percentages are interchanged with these investments he would have got Rs. 120 less. Find the ratio between his investments in the two schemes.
(a) 9 : 16 (b) 47 : 53
(c) 11 : 14 (d) 37 : 63
- Under a certain legal system, prisoners are allowed to make one statement. If their statement turns out to be true then they are hanged. If the statement turns out to be false then they are shot. One prisoner made a statement and the judge had no option but to set him free. Which one of the following could be that statement ?
(a) I committed the crime
(b) You committed the crime
(c) I will be shot
(d) I did not commit the crime

- M and N had four children P, Q, R and S. Of them, only P and R were married. They had children X and Y respectively. If Y is a legitimate child of W, which one of the following statements is necessarily FALSE ?
(a) R is the father of Y
(b) W is the wife of R
(c) M is the grandmother of Y
(d) W is the wife of P
- Congo was named by Europeans. Congo's dictator Mobuto later changed the name of the country and the river to Zaire with the objective of Africanising names of persons and spaces. However, the name Zaire was a Portuguese alteration of Nzadi o Nzere, a local African term meaning 'River that swallows Rivers'. Zaire was the Portuguese name for the Congo river in the 16th and 17 centuries.
Which one of the following statements can be inferred from the paragraph above ?
(a) The term Nzadi o Nzere was of Portuguese origin
(b) As a dictator Mobuto ordered the Portuguese to alter the name of the river to Zaire
(c) Mobuto's desire to Africanise names was prevented by the Portuguese
(d) Mobuto was not entirely successful in Africanising the name of his country

- A firm hires employees at five different skill levels P, Q, R, S, T. The shares of employment at these skill levels of total employment in 2010 is given in the pie chart as shown. There were a total of 600 employees in 2010 and the total employment increased by 15% from 2010 to 2016. The total employment at skill levels P, Q and R remained unchanged during this period. If the employment at skill level S increased by 40% from 2010 to 2016, how many employees were there at skill level T in 2016 ?

Percentage share of skills in 2010

- 72
- 60
- 30
- 35

ANSWERS AND EXPLANATIONS

SET-1

ANSWERS

1. (c) 2. (a) 3. (b) 4. (d) 5. (b)
6. (c) 7. (d) 8. (a) 9. (a) 10. (b)

EXPLANATIONS

3. Given $E = 10$, $J = 20$, $O = 30$ and $T = 40$

$$P + E + S + T = ?$$

A	B	C	D	E	F	G	H	I	J	K	L	M
1	2	3	4	5	6	7	8	9	10	11	12	13
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
14	15	16	17	18	19	20	21	22	23	24	25	26

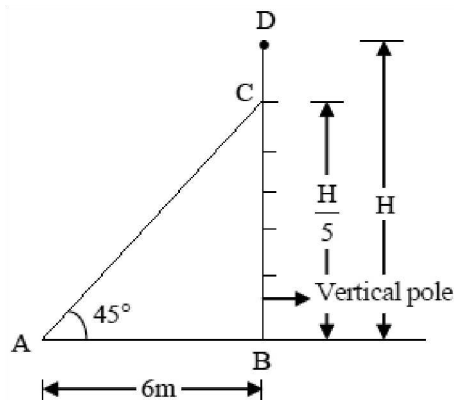
$$\text{If } E = 10 [\because 2 \times 5]; J = 20 [10 \times 2]; O = 30 [15 \times 2];$$

$$T = 40 [20 \times 2]; \text{ then}$$

$$P = 32 [16 \times 2]; S = 38 [19 \times 2]; T = 40 [20 \times 2]$$

$$\therefore P + E + S + T = 32 + 10 + 38 + 40 = 120.$$

5.



H denotes height of the pole from the bottom.

In $\triangle ABC$;

$$\tan 45^\circ = \frac{H/5}{6}$$

$$\Rightarrow 1 = \frac{H/5}{6} \quad [\because \tan 45^\circ = 1]$$

$$\Rightarrow \frac{H}{5} = 6 \Rightarrow H = 30 \text{ m.}$$

7. Given,

Play only cricket $\rightarrow 105$

Play only hockey $\rightarrow 70$

Play only football $\rightarrow 50$

Play both cricket and hockey $\rightarrow 25$

Play both hockey and football $\rightarrow 15$

Play both cricket and football $\rightarrow 30$

Total number of people in a sports academy = 300

Number of people who can play only one sport

$$= 105 + 70 + 50 = 225$$

Number of people who can play at least two sports

$$= \text{Total} - (\text{people who can play only one sport})$$

$$= 300 - 225 = 75$$

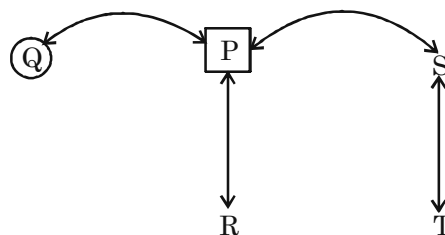
$$\therefore \text{Percentage} = \left[\frac{75}{300} \times 100 \right] \% = 25\%$$

8. Given,

P is the brother of S ,

Q is the wife of P ,

R and T are the children of the siblings P and S respectively.



○ $\rightarrow$ Female

□ $\rightarrow$ Male

$\therefore T$ is son/daughters of S .

$\therefore S$ is the aunt of T is false.

10. Let, Side of smaller square = x cm

Side of larger square = $(x + 5)$ cm

Area of small square = x^2

Area of larger square = $(x + 5)^2$

Given,

$(x + 5)^2 = 4x^2$ [$\because$ Area of the large square is four times the area of the smaller square]

$$\Rightarrow x^2 + 25 + 10x = 4x^2$$

$$\Rightarrow 3x^2 - 10x - 25 = 0$$

$$\Rightarrow x = 5; -\frac{5}{3} \rightarrow \text{which is not possible.}$$

$\therefore$ Side of larger square 10 cm.

SET-2

ANSWERS

1. (b) 2. (c) 3. (b) 4. (a) 5. (c)
6. (b) 7. (d) 8. (c) 9. (a) 10. (c)

EXPLANATIONS

1. Let share of each student be Rs. x

Now, cost of gift = $10 \times x$

According to question

$$x = 8(x + 150)$$

$$x = 600$$

$$\therefore \text{cost of gift} = 10 \times 600 = (6000)$$

2. Court is a place where a judge works.

Similarly, school is a place where a teacher works.

3. Fulcrum is any thing that plays a central or an essential role in an activity, event, or situation.

Hence among the given options, 'revolves' is the most appropriate.

4. Speed of 1st car = 50 km/hr

Speed of 2nd car = 60 km/hr

Since, both cars A and B are moving in same direction, the relative speed = $(60 - 50) = 10$ km/hr

Distance to be covered = 20 km

$$\therefore \text{Time} = \frac{\text{Distance}}{\text{Speed}} = \frac{20}{10} = 2 \text{ hrs}$$

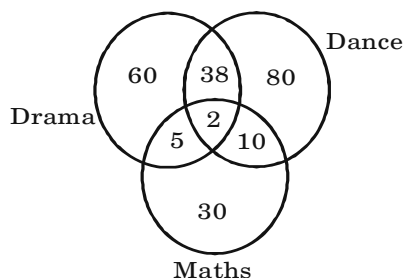
5. 'Breaks down' means to divide something such as a total amount into separate parts.

6. Percentage of Administrators

$$= \left[\frac{\text{Administrators}}{\text{Total number of people}} \times 100 \right] \%$$

$$= \left[\frac{50}{160} \times 100 \right] \% = 31.25\%$$

7. According to question, following venn diagram is formed



$$\begin{aligned} \text{Total number of students} &= 60 + 80 + 30 \\ &+ 38 + 5 + 10 + 2 = 225 \end{aligned}$$

$$25\% = 225$$

If $25\% = 225$

$$\therefore 100\% = \frac{225}{25} \times 100 = 900$$

9. Suffix C as criminal, NC as not criminal.

Case I :

Criminals	P	Q	R	S
Assumption	T	F	F	F
Result	Q_C	S_{NC}	R_C	S_C

Q and R are criminal in the result is impossible because only one person committed the crime.

Case II :

Criminals	P	Q	R	S
Assumption	F	F	T	F
Result	Q_{NC}	S_{NC}	R_{NC}	S_C

S_{NC} and S_C in the result which is contradiction. [S committed crime and same time not committed crime which is contradiction]

Case III :

Criminals	P	Q	R	S
Assumption	F	T	F	F
Result	Q_{NC}	S_C	R_C	S_C

S and R are criminal in the result is impossible because only one person committed the crime.

Case IV :

Criminals	P	Q	R	S
Assumption	F	F	F	T
Result	Q_{NC}	S_{NC}	R_C	S_{NC}

R is criminal in the result.

Hence this case satisfies only one person committed the crime.

So option (a) is correct.

10. The passage states that the underlying disease behind begging is the failure of the state to protect citizens who fall through the social security net.

Only option (b) can be inferred from this.

SET-3**ANSWERS**

1. (c) 2. (d) 3. (a) 4. (b) 5. (c)
 6. (b) 7. (c) 8. (b) 9. (a) 10. (d)

EXPLANATIONS

2. $7^3 = 343$

$11^3 = 1331$

$17^3 = 4913$

Hence the given series is cube of prime number

$\therefore (13)^3 = 2197$

4. Time taken by working together

$$= \left(\frac{4 \times 2}{4 + 2} \right) \text{hr}$$

$$= \frac{8}{6} \text{hr} = \left(\frac{8}{6} \times 60 \right) \text{min} = 80 \text{ min}$$

6. Given two sets

$X = \{1, 2, 3\}$ and $Y = \{2, 3, 4\}$

set of all possible fraction is given by Z

$$\therefore Z = \left\{ \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{2}{2}, \frac{2}{3}, \frac{2}{4}, \frac{3}{2}, \frac{3}{3}, \frac{3}{4} \right\}$$

set of all possible fractions; where the numerators belong to set X and the denominators belong to set Y.

$\therefore$ minimum value in Z is $\frac{1}{4}$ and maximum

value in Z is $\frac{3}{2}$

Now according to question

$$\therefore \text{Required product} = \frac{1}{4} \times \frac{3}{2} = \frac{3}{8}$$

7. From the given information,
 Mita > Ganga > Rekha & Lakshmi > Sana
 8. Integers between 100 and 1000 all of the whose digits are even = 100; since
 from 100 to 199 → No integers
 from 200 to 300 → 25 integers
 Similarly; 400 to 500 → 25 integers; 600 to 700 → 25 integers; 800 to 900 → 25 integers

$\therefore$ Total number of integers between 100 to 1000 all of whose digits are even

$$= 25 + 25 + 25 + 25 = 100$$

10. Given, ratio of number of boys to girls who participated in the exam are 4 : 3

Let, total no. of students who participated in the examination be $7x$;

Given, total pass percentage = 80%

$\therefore$ Total number of students passed in the

$$\text{examination} = \left(7x \times \frac{80}{100} \right) = 5.6x$$

and the percentage of girls who passed = 90%
 i.e., number of girls who passed

$$= \left(3x \times \frac{90}{100} \right) = 2.7x$$

$\therefore$ Number of boys who passed in the exam

$$\therefore 5.6x - 2.7x = 2.9x$$

$$\therefore \text{Required percent} = \left(\frac{2.9x}{4x} \times 100 \right) \% = 72.50\%$$

SET-4**ANSWERS**

1. (c) 2. (b) 3. (c) 4. (b) 5. (a)
 6. (c) 7. (c) 8. (a) 9. (b) 10. (d)

EXPLANATIONS

1. $\therefore$ The number of different orders in which P, Q and T arranged = $3! \times 2 \times 1 = 6$

3. Time taken = $\left(\frac{4 \times 2}{4 + 2} \right) \text{hr} = \frac{4}{3} \text{ hours}$

7. Given, P and Q started working from 8 A.M. for a client

Total budget = USD 3000

P worked exactly 7 hours and took 30 min break in between.

$$\Rightarrow \text{'P' worked number of hours} = 6.5 \text{ hours}$$

'Q' worked exactly 8 hours and took 20 min break in between.

$$\Rightarrow \text{'Q' worked number of hours} \approx 7.67 \text{ hours}$$

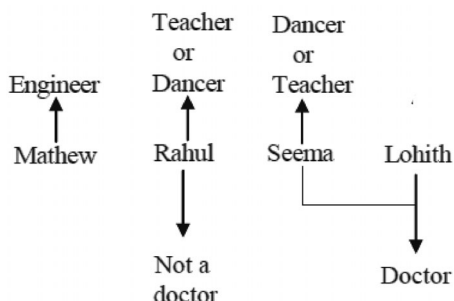
Client amount paid by for both P and Q

$$= \text{USD } 200/\text{hr}$$

$$\begin{aligned}\therefore \text{Total USD Paid} &= (6.5 \times 200 + 7.67 \times 200) \\ &= (1300 + 1534) = 2834\end{aligned}$$

$$\begin{aligned}\text{Remaining amount left} \\ &= 3000 - 2834 \approx 166 \text{ (approx)}\end{aligned}$$

9. According to the given data;



$\therefore$ Mathew must be an Engineer.

10. Let us assume, that population = 100 [2001–2011]

From panel (a);	From panel (b & c)
Percentage of male literates in 2001 = 50%	Number of males in 2001 = 60
Percentage of female literates in 2001 = 40%	Number of females in 2001 = 40
Percentage of male literates in 2011 = 60%	Number of males in 2011 = 50
Percentage of female literates in 2011 = 60%	Number of females in 2011 = 50

$\therefore$ Number of male literates in 2001

$$= 60 \times \frac{50}{100} = 30$$

Number of male literates in 2011

$$= 50 \times \frac{60}{100} = 30$$

Number of female literates in 2001

$$= 40 \times \frac{40}{100} = 16$$

Total number of literates in 2001

$$= 30 + 16 = 46$$

Number of female literates in 2011

$$= 50 \times \frac{60}{100} = 30$$

Total number of literates in 2011

$$= 30 + 30 = 60$$

$\therefore$ Percentage increase in the total number of literates from 2001–2011

$$= \left[\frac{60 - 46}{46} \times 100 \right] \% = \left[\frac{14}{46} \times 100 \right] \% = 30.43\%$$

SET-5

ANSWERS

1. (c) 2. (d) 3. (a) 4. (c) 5. (d)
6. (b) 7. (c) 8. (d) 9. (a) 10. (b)

EXPLANATIONS

3. From prime factorisation of

$$165 = 3 \times 5 \times 11$$

$$165 = 15 \times 11$$

So the two numbers whose sum is 26 and multiplication is 165 is 15 and 11

$$\text{Now, difference} = 15 - 11 = 4$$

5. Since, we know that

$$360^\circ = 12 \text{ hours}$$

$$\therefore 10 = \frac{12}{360} \text{ hours}$$

$$\therefore 225^\circ = \frac{12 \times 225}{360} \text{ hours}$$

$$= 7.5 \text{ hours}$$

6. Let the first part = x

So second part = 100,000x

According to question,

$$[0.1x + 0.12(100,000 - x)] - [0.12x + 0.1(100,000 - x)] = 120$$

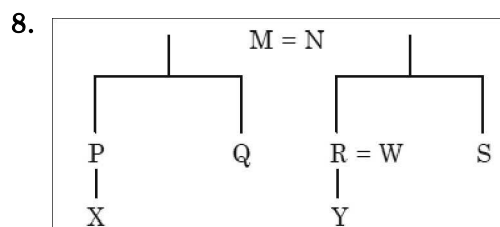
$$\Rightarrow 0.2x - 0.24x + 1200 - 1000 = 120$$

$$\Rightarrow -0.04x = -1880$$

$$x = 47000$$

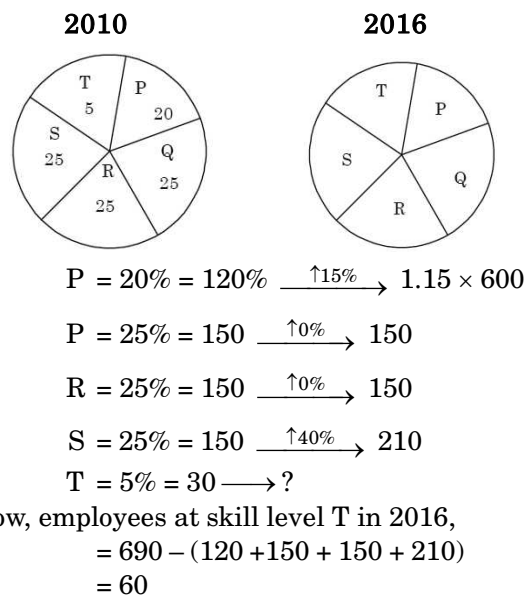
$$\frac{x}{100,000 - x} = \frac{47000}{53000} = \frac{47}{53}$$

7. If prisoner make the statement “I will be shot” then only judge have no option and prisoner will be set to free.



From the above family chart of ‘M’ and ‘N’, we can conclude ‘W’ is the wife of P.

10.



SOLVED PAPER – 2020

GENERAL APTITUDE

SET-1

Q 1 – Q 5 carry one mark each

1. It is a common criticism that most of the academicians live in their ____, so, they are not aware of the real life challenges.
(a) homes (b) ivory towers
(c) glass palaces (d) big flats
2. His hunger for reading is insatiable. He reads indiscriminately. He is most certainly a/an ____ reader.
(a) all-round (b) precocious
(c) voracious (d) wise
3. Select the word that fits the analogy:
Fuse : Fusion :: Use: _____.
(a) Usage (b) User
(c) Uses (d) Usion
4. If 0, 1, 2,, 7, 8, 9 are coded as O, P, Q, V, W, X, then 45 will be coded as _____.
(a) TS (b) ST
(c) SS (d) SU
5. The sum of two positive numbers is 100. After subtracting 5 from each number, the product of the resulting numbers is 0. One of the original numbers is _____.
(a) 80 (b) 85
(c) 90 (d) 95

Q 6 – Q 10 carry two marks each

6. The American psychologist Howard Gardener expounds that human intelligence can be subcategorised into multiple kinds, in such a way that individuals differ with respect to their relative competence in each kind. Based on this theory, modern educationists insist on prescribing multi-dimensional curriculum and evaluation parameters that enable development and assessment of multiple intelligences. Which of the following statements can be inferred from the given text?
(a) Howard Gardner insists that the teaching curriculum and evaluation needs to be multi-dimensional
(b) Howard Gardner wants to develop and assess the theory of multiple intelligences

- (c) Modern educationists want to develop and assess the theory of multiple intelligences.
- (d) Modern educationists insist that the teaching curriculum and evaluation needs to be multidimensional

7. Five friends P, Q, R, S and T went camping. At night, they had to sleep in a row inside the tent. P, Q and T refused to sleep next to R since he snored loudly. P and S wanted to avoid Q as he usually hugged people in sleep.

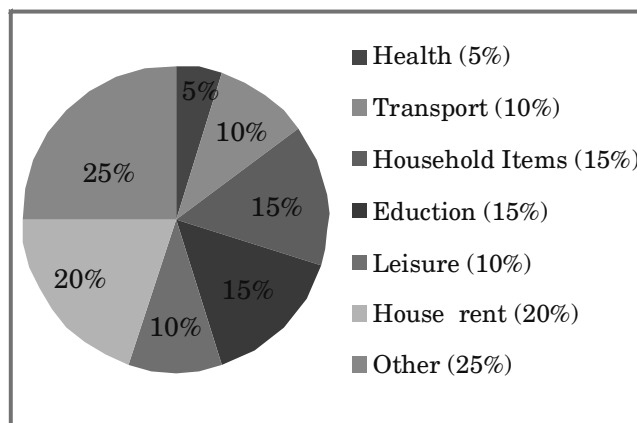
Assuming everyone was satisfied with the sleeping arrangements. What is the order in which they slept?

- (a) RSPTQ
(b) SPRTQ
(c) QRSPT
(d) QTSPR

8. Insert seven numbers between 2 and 34, such that the resulting sequence including 2 and 34 is an arithmetic progression. The sum of these inserted seven numbers is _____.
(a) 120 (b) 124
(c) 126 (d) 130

9. The unit's place in 26591749^{110016} is _____.
(a) 1 (b) 3
(c) 6 (d) 9

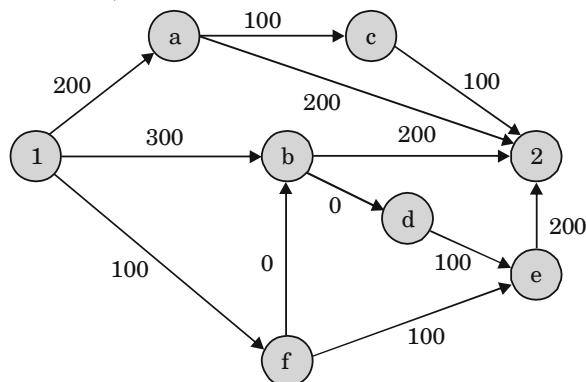
10. The total expenditure of a family, on different activities in a month, is shown in the pie-chart. The extra money spent on education as compared to transport (in percent) is _____.
(a) 5 (b) 33.3
(c) 50 (d) 100



- (a) 5 (b) 33.3
(c) 50 (d) 100

SET-2**Q. 1 to Q. 5 : Carry One Mark Each.**

- Raman is confident of speaking English _____ six months as he has been practising regularly _____ the last three weeks.
(a) during, for (b) for, since
(c) for, in (d) within, for
- His knowledge of the subject was excellent but his classroom performance was _____.
(a) extremely poor (b) good
(c) desirable (d) praiseworthy
- Select the word that fits the analogy:
Cook : Cook :: Fly : _____
(a) Flyer (b) Flying
(c) Flew (d) Flighter
- The dawn of the 21st century witnessed the melting glaciers oscillating between giving too much and too little to billions of people who depend on them for fresh water. The UN climate report estimates that without deep cuts to man-made emissions, at least 30% of the northern hemisphere's surface permafrost could melt by the end of the century. Given this situation of imminent global exodus of billions of people displaced by rising seas, nation-states need to rethink their carbon footprint for political concerns, if not for environmental ones. Which one of the following statements can be inferred from the given passage?
(a) Nation-states do not have environmental concerns.
(b) Nation-states are responsible for providing fresh water to billions of people.
(c) Billions of people are responsible for man-made emissions.
(d) Billions of people are affected by melting glaciers.
- There are multiple routes to reach from node 1 to node 2, as shown in the network.

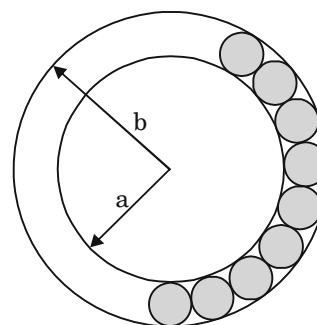


The cost of travel on an edge between two nodes is given in rupees. Nodes 'a', 'b', 'c', 'd', 'e' and 'f' are toll booths. The toll price at toll booths marked 'a' and 'e' is Rs. 200 and is Rs. 100 for the other toll booths. Which is the cheapest route from node 1 to node 2?

- (a) 1-a-c-2 (b) 1-f-b-2
(c) 1-b-2 (d) 1-f-e-2

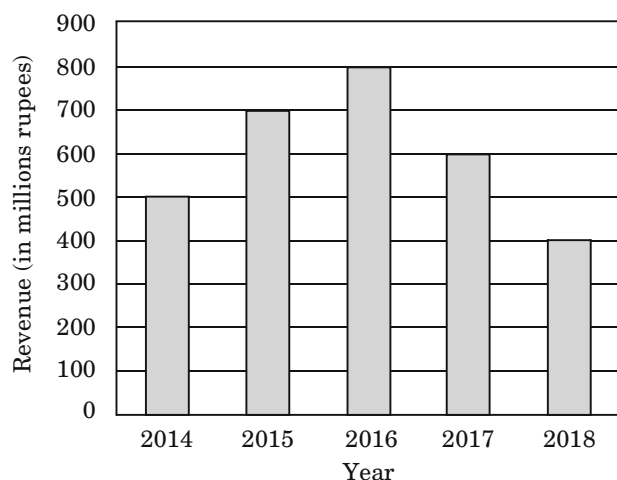
Q. 6 to Q. 10 : Carry Two Marks Each.

- Goods and Services Tax (GST) is an indirect tax introduced in India in 2017 that is imposed on the supply of goods and services, and it subsumes all indirect taxes except few. It is a destination-based tax imposed on goods and services used, and it is not imposed at the point of origin from where goods come. GST also has a few components specific to state governments, central government and Union Territories (UTs). Which one of the following statements can be inferred from the given passage?
(a) GST is imposed on the production of goods and services.
(b) GST includes all indirect taxes.
(c) GST does not have a component specific to UT.
(d) GST is imposed at the point of usage of goods and services.
- If $P = 3$, $R = 27$, $T = 243$, then $Q + S =$ _____.
(a) 40 (b) 80
(c) 90 (d) 110
- The figure below shows an annular ring with outer and inner radii as b and a , respectively. The annular space has been painted in the form of blue colour circles touching the outer and inner periphery of annular space. If maximum n number of circles can be painted, then the unpainted area available in annular space is ____.



- (a) $\pi \left[(b^2 - a^2) - \frac{n}{4} (b - a)^2 \right]$
(b) $\pi [(b^2 - a^2) - n (b - a)^2]$
(c) $\pi \left[(b^2 - a^2) + \frac{n}{4} (b - a)^2 \right]$
(d) $\pi [(b^2 - a^2) + n (b - a)^2]$

9. Two straight lines are drawn perpendicular to each other in X-Y plane. If α and β are the acute angles the straight lines make with the X-axis, then $\alpha + \beta$ is _____.
 (a) 60° (b) 90°
 (c) 120° (d) 180°
10. The total revenue of a company during 2014-2018 is shown in the bar graph. If the total expenditure of the company in each year is 500 million rupees, then the aggregate profit or loss (in percentage) on the total expenditure of the company during 2014-2018 is _____.



- (a) 16.67% profit (b) 16.67% loss
 (c) 20% profit (d) 20% loss

SET - 3**Q 1 – Q 5 Carry one mark each.**

1. This book, including all its chapters, _____ interesting. The students as well as instructor _____ in agreement about it.
 (a) is, was (b) are, are
 (c) is, are (d) were, was
2. People were prohibited _____ their vehicles near the entrance of the main administrative building
 (a) to park (b) from parking
 (c) parking (d) to have parked
3. Select the word that fits the analogy
 Do : Undo :: Trust : _____.
 (a) Entrust (b) Intrust
 (c) Distrust (d) Untrust
4. Stock markets _____ at the news of the coup
 (a) poised
 (b) plunged
 (c) plugged
 (d) probed

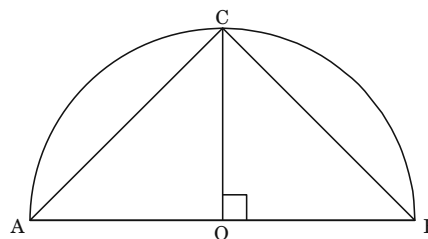
5. If P, Q, R, S are four individuals, how many teams of size exceeding one can be formed, with Q as a member?
 (a) 5 (b) 6
 (c) 7 (d) 8

Q 6 – Q 10 Carry two marks each.

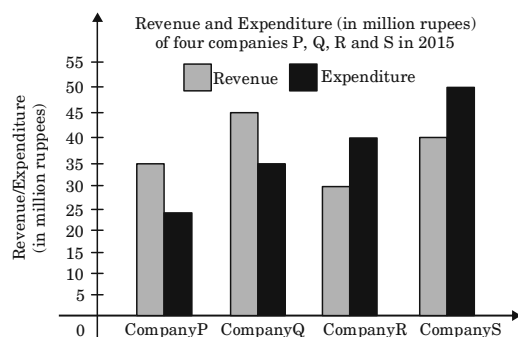
6. Non-performing assets (NPAs) of a bank in India is defined as an asset, which remains unpaid by a borrower for a certain period of time in terms of interest. Principal, or both. Reserve Bank of India (RBI) has changed the definition of NPA thrice during 1993 - 2004 in terms of holding period of loans. The holding period was reduced by one quarter each time. In 1993, the holding period was four quarters (360 days).

Based on the above paragraph, the holding period of loans in 2004 after the third revision was _____ days.

- (a) 45 (b) 90
 (c) 135 (d) 180
7. Select the next element of the series Z, WV, RQP, _____.
 (a) LKJI (b) JIHG
 (c) KJIH (d) NMLK
8. In four-digit integer numbers from 1001 to 9999 the digit group "37" (in the same sequence) appears _____ times.
 (a) 270 (b) 279
 (c) 280 (d) 299
9. Given a semicircle with 'O' as the centre, as shown in the figure, the ratio $\frac{\overline{AC} + \overline{CB}}{\overline{AB}}$ is _____.
 Where $\overline{AC}$, $\overline{CB}$ and $\overline{AB}$ are chords.



- (a) $\sqrt{2}$ (b) $\sqrt{3}$
 (c) 2 (d) 3
10. The revenue and expenditure of four different companies P, Q, R and S in 2015 are shown in the figure. If the revenue of company Q in 2015 was 20% more than that in 2014, and company Q had earned a profit of 10% on expenditure in 2014, then its expenditure (in million rupees) in 2014 was _____.



- (a) 32.7 (b) 33.7
(c) 34.1 (d) 35.1

SET-4

Q. 1 to Q. 5 : Carry One Mark Each.

- The untimely loss of life is a cause of serious global concern as thousands of people get killed _____ accidents every year while many other die _____ diseases like cardio vascular disease, cancer, etc.
(a) in, of (b) from, of
(c) during, from (d) from, from
- He was not only accused of theft _____ of conspiracy.
(a) rather (b) but also
(c) but even (d) rather than
- Select the word that fits the analogy:
Explicit : Implicit :: Express : _____
(a) Impress (b) Repress
(c) Compress (d) Suppress
- The Canadian constitution requires that equal importance be given to English and French. Last year, Air Canada lost a lawsuit, and had to pay a six-figure fine to a French-speaking couple after they filed complaints about formal in-flight announcements in English lasting 15 seconds, as opposed to informal 5 second messages in French.
The French-speaking couple were upset at _____.
(a) the in-flight announcements being made in English.
(b) the English announcements being clearer than the French ones.
(c) the English announcements being longer than the French ones.
(d) equal importance being given to English and French.
- A superadditive function $f(\cdot)$ satisfies the following property

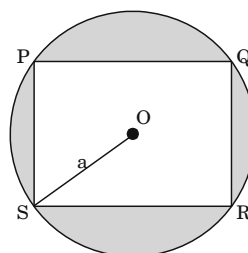
$$f(x_1 + x_2) \geq f(x_1) + f(x_2)$$

Which of the following functions is a superadditive function for $x > 1$?

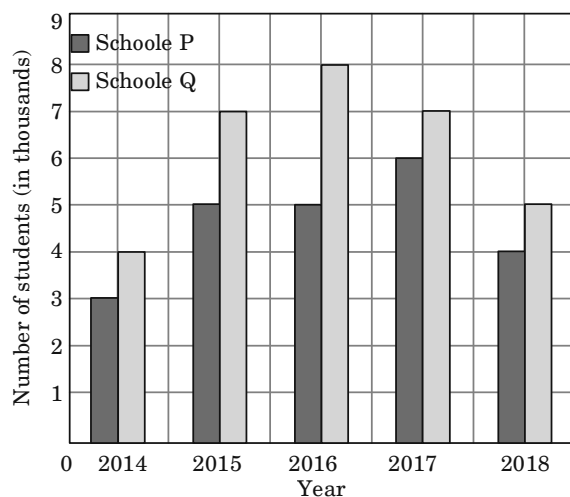
- (a) e^x (b) $\sqrt{x}$
(c) $\frac{1}{x}$ (d) e^{-x}

Q. 6 to Q. 10 : Carry Two Marks Each.

- The global financial crisis in 2008 is considered to be the most serious world-wide financial crisis, which started with the sub-prime lending crisis in USA in 2007. The sub-prime lending crisis led to the banking crisis in 2008 with the collapse of Lehman Brothers in 2008. The sub-prime lending refers to the provision of loans to those borrowers who may have difficulties in repaying loans, and it arises because of excess liquidity following the East Asian crisis. Which one of the following sequences shows the correct precedence as per the given passage?
(a) East Asian crisis — Sub-prime lending crisis — Banking crisis — Global financial crisis
(b) Sub-prime lending crisis — Global financial crisis — Banking crisis — East Asian crisis
(c) Banking crisis — Sub-prime lending crisis — Global financial crisis — East Asian crisis
(d) Global financial crisis — East Asian crisis — Banking crisis — Sub-prime lending crisis
- It is quarter past three in your watch. The angle between the hour hand and the minute hand is ____
(a) 0° (b) 7.5°
(c) 15° (d) 22.5°
- A circle with centre O is shown in the figure. A rectangle PQRS of maximum possible area is inscribed in the circle. If the radius of the circle is a , then the area of the shaded portion is ____.



- (a) $\pi a^2 - a^2$
(b) $\pi a^2 - \sqrt{2}a^2$
(c) $\pi a^2 - 2a^2$
(d) $\pi a^2 - 3a^2$
- a, b, c are real numbers. The quadratic equation $ax^2 - bx + c = 0$ has equal roots, which is β , then
(a) $\beta = b/a$ (b) $\beta^2 = ac$
(c) $\beta^3 = bc/(2a^2)$ (d) $b^2 \neq 4ac$
 - The following figure shows the data of students enrolled in 5 years (2014 to 2018) for two schools P and Q. During this period, the ratio of the average number of the students enrolled in school P to the average of the difference of the number of students enrolled in schools P and Q is ____



- (a) 8 : 23 (b) 23 : 8
(c) 23 : 31 (d) 31 : 23

SET - 5**Q. 1 to Q. 5 : Carry One Mark Each.**

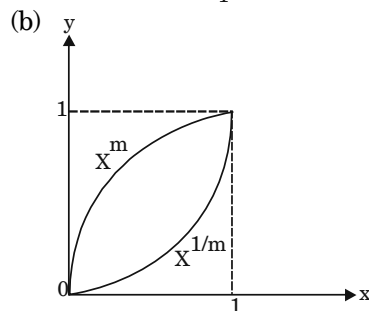
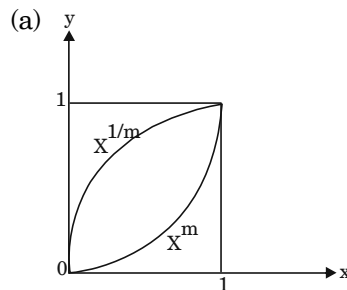
- He is known for his unscrupulous ways. He always sheds _____ tears to deceive people.
(a) fox's (b) crocodile's
(c) crocodile (d) fox
- Jofra Archer, the England fast bowler, is _____ than accurate.
(a) more fast (b) faster
(c) less fast (d) more faster
- Select the word that fits the analogy:
Build : Building :: Grow :
(a) Grown
(b) Grew
(c) Growth
(d) Growed
- I do not think you know the case well enough to have opinions. Having said that, I agree with your other point.
What does the phrase "having said that" mean in the given text?
(a) as opposed to what I have said
(b) despite what I have said
(c) in addition to what I have said
(d) contrary to what I have said
- Define $[x]$ as the greatest integer less than or equal to x , for each $x \in (-\infty, \infty)$. If $y = [x]$, then area under y for $x \in [1, 4]$ is _____.
(a) 1 (b) 3
(c) 4 (d) 6

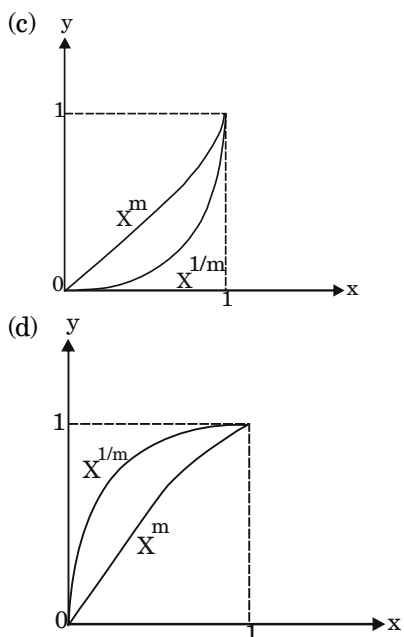
Q. 6 to Q. 10 : Carry Two Marks Each.

- Crowd funding deals with mobilisation of funds for a project from a large number of people, who would be willing to invest smaller amounts through web-based platforms in the project.

Based on the above paragraph, which of the following is correct about crowd funding?

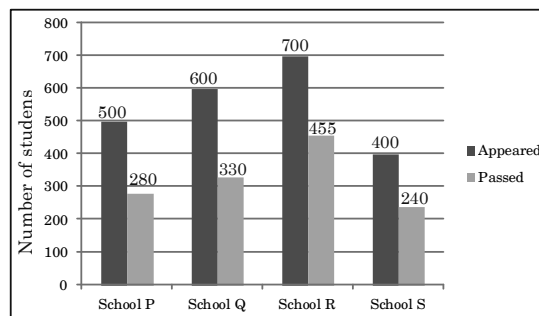
- Funds raised through unwilling contributions on web-based platforms.
 - Funds raised through large contributions on web-based platforms.
 - Funds raised through coerced contributions on web-based platforms.
 - Funds raised through voluntary contributions on web-based platforms.
- P, Q, R and S are to be uniquely coded using α and β . If P is coded as $\alpha\alpha$ and Q as $\alpha\beta$, then R and S, respectively, can be coded as
(a) $\beta\alpha$ and $\alpha\beta$ (b) $\beta\beta$ and $\alpha\alpha$
(c) $\alpha\beta$ and $\beta\beta$ (d) $\beta\alpha$ and $\beta\beta$
 - The sum of the first n terms in the sequence 8, 88, 888, 8888, ... is _____.
(a) $\frac{81}{80}(10^n - 1) + \frac{9}{8}n$
(b) $\frac{81}{80}(10^n - 1) - \frac{9}{8}n$
(c) $\frac{80}{81}(10^n - 1) + \frac{8}{9}n$
(d) $\frac{80}{81}(10^n - 1) - \frac{8}{9}n$
 - Select the graph that schematically represents BOTH $y = x^m$ and $y = x^{1/m}$ properly in the interval $0 \leq x \leq 1$, for integer values of m , where $m > 1$.





10. The bar graph shows the data of the students who appeared and passed in an examination for four schools P, Q, R and S. The average of success rates (in percentage) of these four schools is ____.

Performance of schools P, Q, R and S



- (a) 58.5% (b) 58.8%
(c) 59.0% (d) 59.3%

ANSWERS AND EXPLANATIONS

SET-1

ANSWERS

1. (b) 2. (c) 3. (a) 4. (b) 5. (d)
6. (d) 7. (a) 8. (c) 9. (a) 10. (c)

EXPLANATIONS

- Ivory towers means lack of concern with or interest in practical matters.
- Voracious means hungry for knowledge.
- Verb and noun analogy.
- O – 0, P – 1, Q – 2, R – 4, T – 5, U – 6, V – 7, W – 8, X – 9
So, ST = 45
- As Given that :**
 $a + b = 100$
and $(a - 5)(b - 5) = 0$
So, either $a = 5$ or $b = 5$
Since 5 is not in any of the option.
If $a = 5$, then $b = 95$ or if $b = 5$, then $a = 95$
Hence of the original number is 95.
- The last two lines support the answer.
- P, Q & T refused to sleep next to R
So option (c, b, d) eliminated
Only option (a) is satisfies the given conditions in the paragraph.

8. $2, -, -, -, -, -, -, -, 34$

↓

$$t_1 = a = 2$$

↓

$$t_n = a + (n - 1)d = 34$$

$$= 2 + 8d = 34$$

$$d = 4$$

So, 2, 6, 10, 14, 18, 22, 26, 30, 34

Sum of inserted 7 AP numbers

$$= \frac{n}{2} (\text{1st terms} + \text{last term}) = \frac{n}{2} [a + (a + 6d)]$$

$$= \frac{7}{2} (6 + 30) = 18 \times 7 = 126$$

9. Units place in $26591749^{110016} = 9^{110016}$

$$= 9^{\text{even}} = 1 \quad (\because 9^2 = 81)$$

Hence, answer will be 1.

10. Money spent on Education = $\frac{15}{100} \times 100$
 $= 15\%$

Money spent on transport = $\frac{10}{100} \times 100$
 $= 10\%$

So, required % of extra money spent on education compared to transport :

$$= \frac{15 - 10}{10} \times 100$$

$$= 50\%$$

SET-2

ANSWERS

1. (d) 2. (a) 3. (a) 4. (d)
 5. (b) 6. (d) 7. (c) 8. (a)
 9. (b) 10. (c)

EXPLANATIONS

1. "Within" Preposition that is used to express something that occurs inside a particular period of time.

Here, because Sentence is in 'present perfect continuous tense' "For is" used when we talk about a period of time.

2. Here, we have to note that the conjunction used is 'but'. 'But' is used for introducing an idea which contrasts with the statement that has been already said.

3. (i) Flyer : Relation is Verb : Noun

(ii) One who cooks is a cook, and one who flies any aircraft is a flyer.

5. • Cost of 1-a-C-2 : $200 + 200 + 100 + 100 + 100 = 700$
 • Cost of 1-f-e-2 : $100 + 100 + 100 + 200 + 200 = 700$
 • Cost of 1-b-2 : $300 + 100 + 200 = 600$
 • Cost of 1-f-b-2 : $100 + 100 + 100 + 200 = 500$

Hence, 1-f-b-2 is having minimum cost.

6. According to given passage GST is imposed on the usage or supply of goods and services hence, option (d) is correct.

7. As given that

$$P = 3^1, Q = 9 = 3^2, R = 27 = 3^3, S = 81 = 3^4, \\ T = 243 = 3^5$$

$$\text{Hence, } Q + S = 9 + 81 = 90.$$

8. Unpainted area = Area of annular ring – Area of n blue color circles

$$\text{Area of annular ring} = \pi b^2 - \pi a^2$$

$$\text{i.e. } \pi(b^2 - a^2)$$

$$\text{Area of 1 blue circle} = \pi \left(\frac{b-a}{2} \right)^2$$

$$\text{Hence, area of n blue circles} = n\pi \left(\frac{b-a}{2} \right)^2$$

$$\Rightarrow \text{Unpainted area} = \pi \left[(b^2 - a^2) - \frac{\pi}{4} (b-a)^2 \right]$$

9. Since, α and β are acute angles hence it will form the interior angles of straight lines $\alpha + \beta = 180^\circ - 90^\circ = 90^\circ$

10. As given that,

$$\text{Total expenditure} = 2500 \text{ million}$$

$$\text{Total revenue} = 3000 \text{ million}$$

$$\text{Total profit} = 3000 - 2500 = 500$$

$$\text{So, profit \%} = \frac{500}{2500} \times 100 = 20\%$$

SET-3

ANSWERS

1. (c) 2. (b) 3. (c) 4. (b) 5. (c)
 6. (b) 7. (c) 8. (b) 9. (a) 10. (c)

EXPLANATIONS

1. Subject + Verb + agreement

$\downarrow$ $\downarrow$
 is are

2. Prohibited from parking means "to officially refuse allow something"

3. Do : Undo : Trust : Distrust

The obvious analogical relationship is do and undo are antonyms just as trust and distrust are antonyms.

4. Plunged means plummet especially (prices, temperature, etc.) to decrease suddenly and quickly.

5. When team size is more than 1 then

Case 1 : When team size = 2, then Q is fixed, only 41 member to be selected among P, R & S

$${}^3C_1 = 3 \text{ ways}$$

Case 2 : When team size = 3, then Q is fixed, only 2 members to be selected among P, R & S

$${}^3C_2 = 3 \text{ ways}$$

Case 3 : When team size = 4, then Q is fixed, only 3 members to be selected among P, R & S

$${}^3C_3 = 1 \text{ ways}$$

Case 4 : When team size ≥ 5 ; That Q Not possible.

So, total possible ways = $3 + 3 + 1 = 7$ ways

Hence, possible combinations are :

PQ, RQ, SQ, PRQ, PSQ, RSQ, PQRS.

6. Based on the given paragraph the holding period of loans was reduced by one quarter in 2004. Therefore after the third revision equal to one quarter days of the year.

$$= \frac{1}{4} \times 360 = 90 \text{ days.}$$

7. $\begin{matrix} 26 & 23,22 & 18,17,16 & 11,10,9,8 \\ Z, & WV, & RQP, & KJIH \\ \downarrow & \downarrow & \downarrow & \downarrow \\ YX & UTS & ONML & GFED \end{matrix}$

8. Four digit integer numbers from 1001 to 9999, with 37 as sequence appears as :

$$(1) \begin{array}{cc} 3 & 7 & - & - \\ \downarrow & \downarrow & & \end{array}$$

$$10 \times 10 = 100 \text{ integers}$$

$$(2) \begin{array}{ccc} - & 3 & 7 & - \\ \downarrow & & \downarrow & \end{array}$$

$$9 \times 10 = 90 \text{ Integers}$$

$$(3) \begin{array}{ccc} - & - & 3 & 7 \\ \downarrow & \downarrow & & \end{array}$$

$$9 \times 10 \text{ Integers}$$

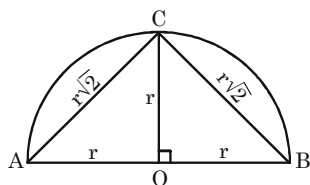
This question asking how many times 37 appears not how many number.

$$\text{So, total ways} = 100 + 90 + 90 = 280 \text{ ways}$$

9. AB = Diameter = 2r

$$\text{OC} = \text{radius} = r = \text{AO} = \text{OB}$$

consider is $\triangle AOC$



using pythagorous theorem

$$AC^2 = r^2 + r^2 \Rightarrow AC^2 = 2r^2$$

$$\Rightarrow AC = r\sqrt{2}$$

$$\text{similarly } \triangle COB \Rightarrow BC = r\sqrt{2}$$

$$\text{so, the ratio : } \frac{\overline{AC} + \overline{CB}}{\overline{AB}} = \frac{r\sqrt{2} + r\sqrt{2}}{2r} = \sqrt{2}$$

10. Revenue of company Q in 2015 = 45 revenue of company Q in 2014 was 20% more than that in 2014.

$$\text{Let, Revenue in 2014} = x,$$

$$R_{2015} = 120\% (R_x)$$

$$45 = \frac{120}{100} x$$

$$x = 37.5 \text{ in million rupees.}$$

$$\text{Revenue of company Q in 2014} = 37.5$$

10% profit in 2014 for company Q.

$$\Rightarrow \text{Revenue in } (R_{2014})x = 110\% \text{ Expenditure of company in 2014.}$$

$$37.5 = \frac{110}{100} \times \text{Expenditure } Q_{2014}$$

$$\text{Expenditure of company Q in 2014} = \frac{375}{11} = 34.1$$

(Note : profit & loss is calculated on Expenditure)

SET-4

ANSWERS

1. (a) 2. (b) 3. (b) 4. (c) 5. (a)
6. (a) 7. (b) 8. (c) 9. (c) 10. (b)

EXPLANATIONS

5. x^m is super additive function, if $x > 1$.

$$\text{example: } 2^3 > 2^1 + 2^2$$

$$f(x_1 + x_2) > f(x_1) + f(x_2) \text{ Satisfying}$$

Hence, e^x is super additive function.

7. As given that, Time: 03 : 15

$$\text{So, } h = 3, m = 15$$

$$\text{Then, the angle between hands } (\theta) = \left| 30h - \frac{11m}{2} \right|$$

$$\text{Hence, } \theta = \left| 30 \times 3 - \frac{165}{2} \right| = 75^\circ$$

8. From given figure:

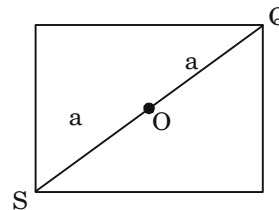
Area of shaded portion

$$= \text{Area of circle} - \text{Area of rectangle}$$

As we know that,

$$\text{Area of the circle} = \pi a^2 \text{ And}$$

The maximum possible, area of rectangle inscribed in the circle, when it becomes square.



$$\text{Diagonal} = 2a$$

$$\text{Area of square} = \frac{1}{2} (\text{diagonal})^2 = 2a^2$$

So, required shaded Area.

$$= \text{Area of circle} - \text{Area of square} = \pi a^2 - 2a^2$$

9. As given that, the quadratic equation:

$$ax^2 - bx + c = 0$$

According to question it has equal roots (β).

$$\text{So, sum of roots} = \beta + \beta = \frac{b}{a} \Rightarrow 2\beta = \frac{b}{a} \quad \dots(1)$$

$$\text{And, product of roots} = \beta * \beta = \frac{c}{a} \Rightarrow \beta^2 = \frac{c}{a} \quad \dots(2)$$

By multiplying (1) & (2) equation:

$$\Rightarrow (2\beta)(\beta^2) = (b/a)(c/a)$$

$$(2b^3) = bc/a^2$$

$$b^3 = bc/(2a^2)$$

10. Average number of students in school (P) in 5 years

$$= \frac{3000 + 5000 + 5000 + 6000 + 4000}{5} = 4600$$

Average number of students in school (Q) in 5 years

$$= \frac{4000 + 7000 + 8000 + 7000 + 5000}{5} = 6200$$

The difference of the number of students enrolled in school P and Q = $(6200 - 4600) = 1600$

The ratio of the average number of the student enrolled in school P to the average of the difference of the number of student enrolled in school P and Q then,

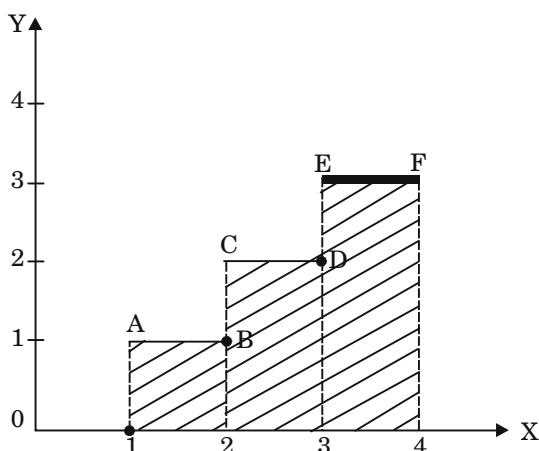
$$\text{Required ratio} = \frac{4600}{1600} = \frac{23}{8} = 23 : 8$$

SET-5**ANSWERS**

- | | | | |
|--------|---------|--------|--------|
| 1. (c) | 2. (a) | 3. (c) | 4. (b) |
| 5. (d) | 6. (d) | 7. (d) | 8. (d) |
| 9. (a) | 10. (c) | | |

EXPLANATIONS

1. Crocodile tears: A false, insincere display of emotion.
2. Jofra Archer, the England fast bowler, is more fast than accurate.
3. Build : Building (Noun)
so, Grow : Growth (Noun)
4. having said that : despite what one just said.
5. Graph of $y = [x]$ for $x \in [1, 4]$



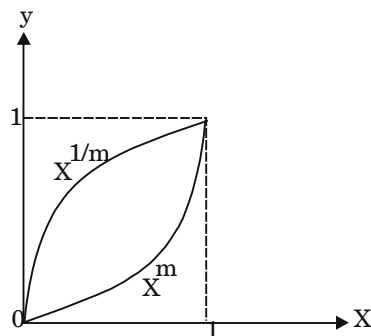
The total area, $A = (1 \times 1) + (2 \times 1) + (3 \times 1)$
 $= 1 + 2 + 3 = 6$

6. The correct statement about crowd funding is to fund raised through voluntary contributions on web-based platforms.

7. P, Q, R, S
 $\alpha\alpha$ $\alpha\beta$ $\boxed{\beta\alpha}$ $\boxed{\beta\beta}$

8. $S = 8 + 88 + 888 + \dots n$
 $= \frac{8}{9}(9 + 99 + 999 + \dots n)$
 $= \frac{8}{9}[(10^1 - 1) + (10^2 - 1) + \dots + (10^n - 1)]$
 $= \frac{8}{9}[(10^1 + 10^2 + \dots 10^n) - n]$
 $= \frac{8}{9}\left[\frac{10 \times (10^n - 1)}{10 - 1}\right] - \frac{8}{9}n$
 $= \frac{80}{81}(10^n - 1) - \frac{8}{9}n$

9. Graph of $y = x^m$ and $y = x^{1/m}$ for $m > 1$



Correct option is (a).

10. Success rate = $\frac{\text{Number of candidates who passed the exam}}{\text{Number of appeared candidates}}$

So, Average success rate,

$$= \frac{\frac{280}{500} + \frac{330}{600} + \frac{455}{700} + \frac{240}{400}}{4} = 0.59 \text{ or } 59\%$$

Correct option is (c).

SOLVED PAPER – 2021

GENERAL APTITUDE

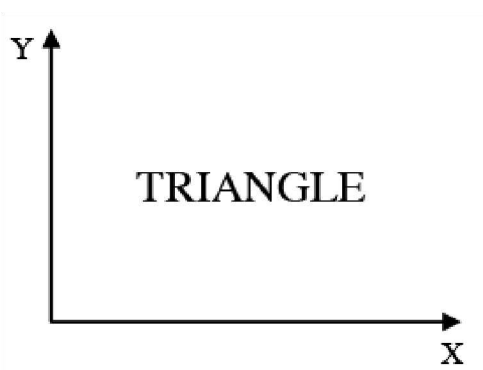
SET 1

Q. 1 to Q. 5 : Carry One Mark Each.

1. Getting to the top is _____ than staying on top.

(a) more easy (b) much easy
(c) easiest (d) easier

2.



The mirror image of the above text about the x-axis is

(a) $\overline{\text{TRIANGLE}}$ (b) $\overline{\text{TRIANGLE}}$
(c) $\overline{\text{TRIANGLE}}$ (d) $\overline{\text{TRIANGLE}}$

3. In a company, 35% of the employees drink coffee, 40% of the employees drink tea and 10% of the employees drink both tea and coffee. What % of employees drink neither tea nor coffee?

(a) 15 (b) 25
(c) 35 (d) 40

4. $\oplus$ and $\odot$ are two operators on numbers p and q such that

$$p \oplus q = \frac{p^2 + q^2}{pq} \text{ and } p \odot q = \frac{p^2}{q};$$

If $x \oplus y = 2 \odot 2$, then $x =$

(a) $\frac{y}{2}$ (b) y
(c) $\frac{3y}{2}$ (d) 2y

5. Four persons P, Q, R and S are to be seated in a row, all facing the same direction, but not necessarily in the same order. P and R cannot sit adjacent to each other. S should be seated to the right of Q. The number of distinct seating arrangements possible is:

(a) 2 (b) 4
(c) 6 (d) 8

Q. 6 to Q. 10 : Carry Two Marks Each.

6. Statement: Either P marries Q or X marries Y. Among the options below, the logical NEGATION of the above statement is:

(a) P does not marry Q and X marries Y.
(b) Neither P marries Q nor X marries Y.
(c) X does not marry Y and P marries Q.
(d) P marries Q and X marries Y.

7. Consider two rectangular sheets, Sheet M and Sheet N of dimensions 6 cm x 4 cm each.

Folding operation 1: The sheet is folded into half by joining the short edges of the current shape.

Folding operation 2: The sheet is folded into half by joining the long edges of the current shape.

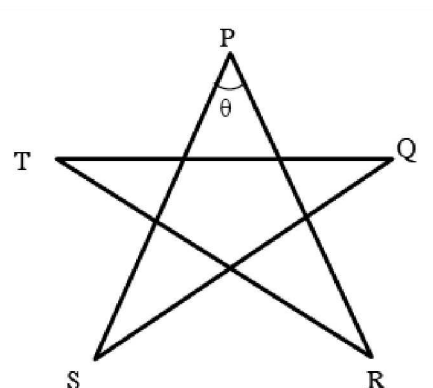
Folding operation 1 is carried out on Sheet M three times.

Folding operation 2 is carried out on Sheet N three times.

The ratio of perimeters of the final folded shape of Sheet N to the final folded shape of Sheet M is _____.

(a) 13 : 7 (b) 3 : 2
(c) 7 : 5 (d) 5 : 13

8.



Five line segments of equal lengths, PR, PS, QS, QT and RT are used to form a star as shown in the figure above.

The value of θ , in degrees, is

(a) 36 (b) 45
(c) 72 (d) 108

9. A function, λ , is defined by

$$\lambda(p, q) = \begin{cases} (p - q)^2, & \text{if } p \geq q, \\ p + q, & \text{if } p < q. \end{cases}$$

The value of the expression

$$\frac{\lambda(-(-3+2), (-2+3))}{(-(-2+1))} \text{ is :}$$

- (a) -1 (b) 0
(c) $\frac{16}{3}$ (d) 16

10. Humans have the ability to construct worlds entirely in their minds, which don't exist in the physical world. So far as we know, no other species possesses this ability. This skill is so important that we have different words to refer to its different flavors, such as imagination, invention and innovation.

Based on the above passage, which one of the following is TRUE?

- (a) No species possess the ability to construct worlds in their minds.
(b) The terms imagination, invention and innovation refer to unrelated skills.
(c) We do not know of any species other than humans who possess the ability to construct mental worlds.
(d) Imagination, invention and innovation are unrelated to the ability to construct mental worlds.

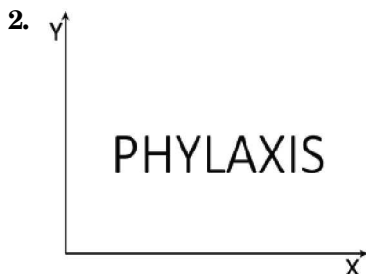
SET 2

Q. 1 – Q. 5 carry one mark each

1. (i) Arun and Aparna are here.
(ii) Arun and Aparna is here.
(iii) Arun's families is here.
(iv) Arun's family is here.

Which of the above sentences are grammatically CORRECT?

- (a) (i) and (ii) (b) (i) and (iv)
(c) (ii) and (iv) (d) (iii) and (iv)



The mirror image of the above text about the x-axis is

- (a) PHYLAXIS (b) PHYLAXIS
(c) PHYLAXIS (d) PHYLAXIS

3. Two identical cube shaped dice each with faces numbered 1 to 6 are rolled simultaneously. The probability that an even number is rolled out on each dice is:

- (a) $\frac{1}{36}$ (b) $\frac{1}{12}$
(c) $\frac{1}{8}$ (d) $\frac{1}{4}$

4. $\oplus$ and $\odot$ are two operators on numbers p and q such that $p \odot q = p - q$, and $p \oplus q = p \times q$

Then, $(9 \odot (6 \oplus 7)) \odot (7 \oplus (6 \odot 5)) =$

- (a) 40 (b) -26
(c) -33 (d) -40

5. Four persons P, Q, R and S are to be seated in a row. R should not be seated at the second position from the left end of the row. The number of distinct seating arrangements possible is:

- (a) 6 (b) 9
(c) 18 (d) 24

Q. 6 – Q. 10 carry two marks each

6. On a planar field, you travelled 3 units East from a point O. Next you travelled 4 units South to arrive at point P. Then you travelled from P in the North-East direction such that you arrive at a point that is 6 units East of point O. Next, you travelled in the North-West direction, so that you arrive at point Q that is 8 units North of point P. The distance of point Q to point O, in the same units, should be _____

- (a) 3 (b) 4
(c) 5 (d) 6

7. The author said, "Musicians rehearse before their concerts. Actors rehearse their roles before the opening of a new play. On the other hand, I find it strange that many public speakers think they can just walk on to the stage and start speaking. In my opinion, it is no less important for public speakers to rehearse their talks."

Based on the above passage, which one of the following is TRUE?

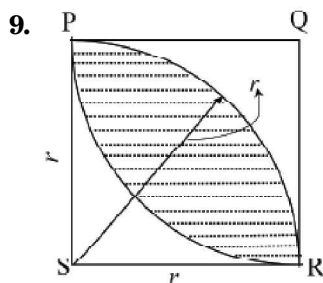
- (a) The author is of the opinion that rehearsing is important for musicians, actors and public speakers.
(b) The author is of the opinion that rehearsing is less important for public speakers than for musicians and actors.

- (c) The author is of the opinion that rehearsing is more important only for musicians than public speakers.
- (d) The author is of the opinion that rehearsal is more important for actors than musicians.

8. 1. Some football players play cricket.
2. All cricket players play hockey.

Among the options given below, the statement that logically follows from the two statements 1 and 2 above, is:

- (a) No football player plays hockey.
(b) Some football players play hockey.
(c) All football players play hockey.
(d) All hockey players play football.



In the figure shown above, PQRS is a square. The shaded portion is formed by the intersection of sectors of circles with radius equal to the side of the square and centers at S and Q.

The probability that any point picked randomly within the square falls in the shaded area is _____

- (a) $4 - \frac{\pi}{2}$ (b) $1/2$
(c) $\frac{\pi}{2} - 1$ (d) $\frac{\pi}{4}$
10. In an equilateral triangle PQR, side PQ is divided into four equal parts, side QR is divided into six equal parts and side PR is divided into eight equal parts. The length of each subdivided part in cm is an integer.

The minimum area of the triangle PQR possible, in cm^2 , is

- (a) 18 (b) 24
(c) $48\sqrt{3}$ (d) $144\sqrt{3}$

SET 3

Q.1 to Q.5: Multiple Choice Questions (MCQ), carry ONE mark each.

1. The current population of a city is 11,02,500. If it has been increasing at the rate of 5% per annum, what was its population 2 years ago?

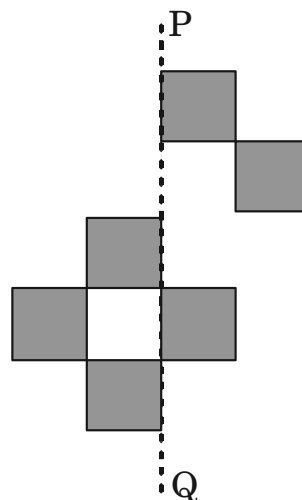
- (a) 9,92,500 (b) 9,95,006
(c) 10,00,000 (d) 12,51,506

2. p and q are positive integers and $\frac{p}{q} + \frac{q}{p} = 3$, then,

$$\frac{p^2}{q^2} + \frac{q^2}{p^2} =$$

- (a) 3 (b) 7
(c) 9 (d) 11

- 3.



The least number of squares that must be added so that the line P-Q becomes the line of symmetry is _____

- (a) 4 (b) 3
(c) 6 (d) 7

4. *Nostalgia* is to *anticipation* as _____ is to _____

Which one of the following options maintains a similar logical relation in the above sentence?

- (a) Present, past (b) Future, past
(c) Past, future (d) Future, present

5. Consider the following sentences:

- (i) I woke up from sleep.
(ii) I woked up from sleep.
(iii) I was woken up from sleep.
(iv) I was wokened up from sleep.

Which of the above sentences are grammatically CORRECT?

- (a) (i) and (ii) (b) (i) and (iii)
(c) (ii) and (iii) (d) (i) and (iv)

Q.6 to Q.10: Multiple Choice Question (MCQ), carry TWO marks each.

6. Given below are two statements and two conclusions.

Statement 1 : All purple are green.

Statement 2 : All black are green.

Conclusion I : Some black are purple.

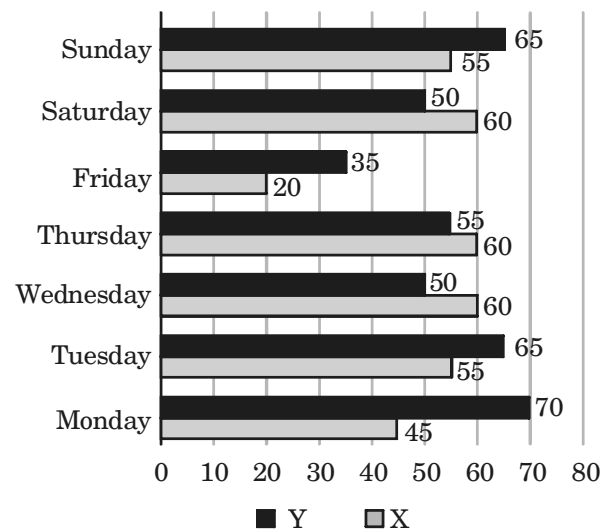
Conclusion II : No black is purple.

Based on the above statements and conclusions, which one of the following options is logically CORRECT?

- (a) Only conclusion I is correct.
 (b) Only conclusion II is correct.
 (c) Either conclusion I or II is correct.
 (d) Both conclusion I and II are correct.
7. Computers are ubiquitous. They are used to improve efficiency in almost all fields from agriculture to space exploration. Artificial intelligence (AI) is currently a hot topic. AI enables computers to learn, given enough training data. For humans, sitting in front of a computer for long hours can lead to health issues. Which of the following can be deduced from the above passage?
- (i) Nowadays, computers are present in almost all places.
 (ii) Computers cannot be used for solving problems in engineering.
 (iii) For humans, there are both positive and negative effects of using computers.
 (iv) Artificial intelligence can be done without data.
- (a) (ii) and (iii)
 (b) (ii) and (iv)
 (c) (i), (iii) and (iv)
 (d) (i) and (iii)
8. Consider a square sheet of side 1 unit. In the first step, it is cut along the main diagonal to get two triangles. In the next step, one of the cut triangles is revolved about its short edge to form a solid cone. The volume of the resulting cone, in cubic units, is _____

- (a) $\frac{\pi}{3}$
 (b) $\frac{2\pi}{3}$
 (c) $\frac{3\pi}{2}$
 (d) 3π

9.

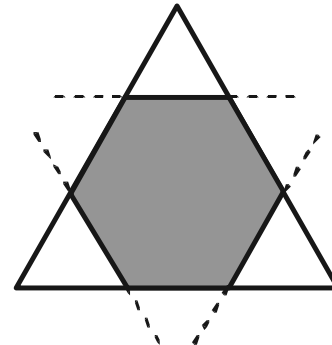


The number of minutes spent by two students, X and Y, exercising every day in a given week are shown in the bar chart above.

The number of days in the given week in which one of the students spent a minimum of 10% more than the other student, on a given day, is

- (a) 4
 (b) 5
 (c) 6
 (d) 7

10.



Corners are cut from an equilateral triangle to produce a regular convex hexagon as shown in the figure above.

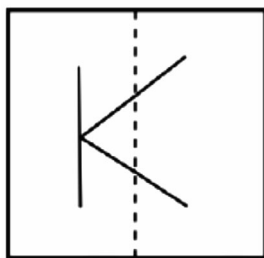
The ratio of the area of the regular convex hexagon to the area of the original equilateral triangle is

- (a) 2 : 3
 (b) 3 : 4
 (c) 4 : 5
 (d) 5 : 6

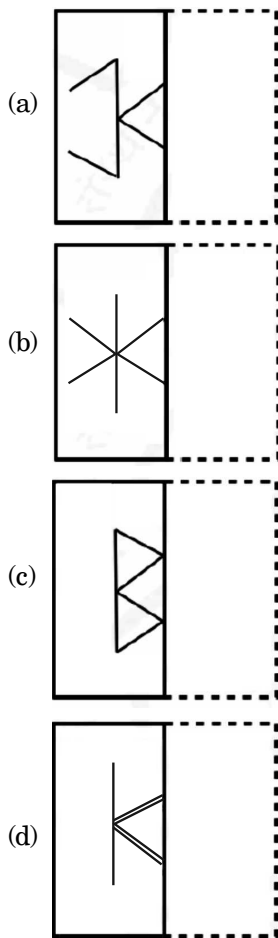
SET 4

1. The people _____ were at the demonstration were from all sections of society.
- (a) whose
 (b) which
 (c) who
 (d) whom

2.



A transparent square sheet shown above is folded along the dotted line. The folded sheet will look like _____.



3. For a regular polygon having 10 sides, the interior angle between the sides of the polygon, in degrees, is :

- (a) 396 (b) 324
(c) 216 (d) 144

4. Which one of the following numbers is exactly divisible by $(11^{13} + 1)$?

- (a) $11^{26} + 1$ (b) $11^{33} + 1$
(c) $11^{39} - 1$ (d) $11^{52} - 1$

5. Oasis is to sand as island is to _____

Which one of the following options maintains a similar logical relation in the above sentence?

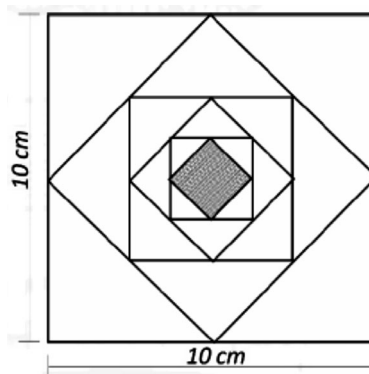
- (a) Stone (b) Land
(c) Water (d) Mountain

Q. 6 to Q. 10 : Multiple Choice Question (MCQ), carry Two marks each.

6. The importance of sleep is often overlooked by students when they are preparing for exams. Research has consistently shown that sleep deprivation greatly reduces the ability to recall the material learnt. Hence, cutting down on sleep to study longer hours can be counter productive. Which one of the following statements is the CORRECT inference from the above passage?

- (a) Sleeping well alone is enough to prepare for an exam. Studying has lesser benefit.
(b) Students are efficient and are not wrong in thinking that sleep is a waste of time.
(c) If a student is extremely well prepared for an exam, he needs little or no sleep.
(d) To do well in an exam, adequate sleep must be part of the preparation.

7.



In the figure shown above, each inside square is formed by joining the midpoints of the sides of the next larger square. The area of the smallest square (shaded) as shown, in cm^2 is :

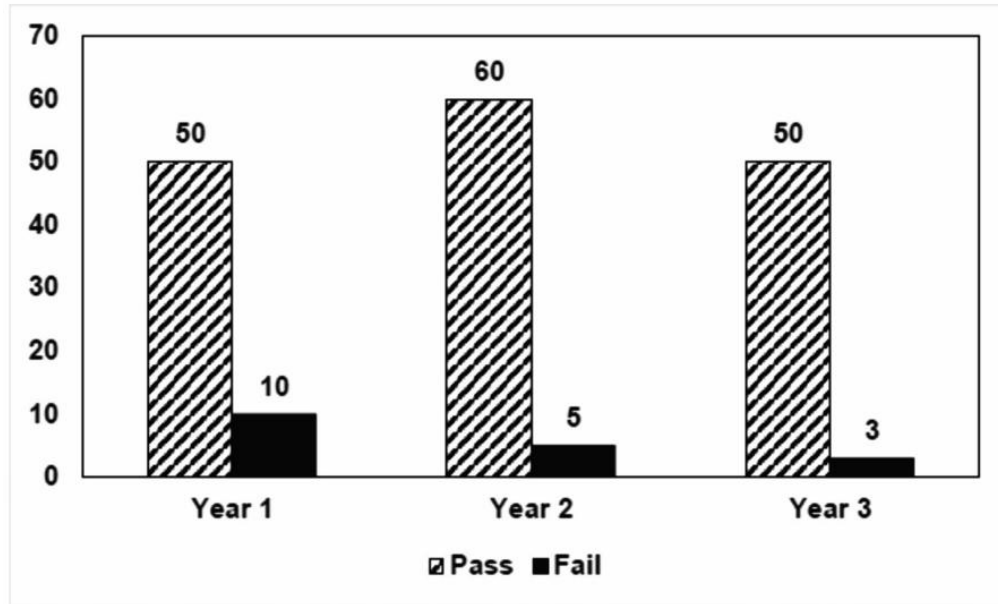
- (a) 12.50
(b) 6.25
(c) 3.125
(d) 1.5625

8. Let X be a continuous random variable denoting the temperature measured. The range of temperature is $[0, 100]$ degree Celsius and let the probability density function of X be $f(x) = 0.01$ for $0 \leq X \leq 100$.

The mean of X is _____

- (a) 2.5 (b) 5.0
(c) 25.0 (d) 50.0

9.



The number of students passing or failing in an exam for a particular subject are presented in the bar chart above. Students who pass the exam cannot appear for the exam again. Students who fail the exam in the first attempt must appear for the exam in the following year. Students always pass the exam in their second attempt.

The number of students who took the exam for the first time in the year 2 and the year 3 respectively, are _____

- (a) 65 and 53 (b) 60 and 50 (c) 55 and 53 (d) 55 and 48

10. Seven cars P, Q, R, S, T, U and V are parked in a row not necessarily in that order. The cars T and U should be parked next to each other. The cars S and V also should be parked next to each other, whereas P and Q cannot be parked next to each other. Q and S must be parked next to each other. R is parked to the immediate right of V. T is parked to the left of U.

Based on the above statements, the only INCORRECT option given below is :

- (a) There are two cars parked in between Q and V.
 (b) Q and R are not parked together.
 (c) V is the only car parked in between S and R.
 (d) Car P is parked at the extreme end.

SET 5

Q. 1 to Q. 5 : Carry One Mark Each.

1. Consider the following sentences:

- (i) After his surgery, Raja hardly could walk.
 (ii) After his surgery, Raja could barely walk.
 (iii) After his surgery, Raja barely could walk.
 (iv) After his surgery, Raja could hardly walk.

Which of the above sentences are grammatically CORRECT?

- (a) (i) and (ii) (b) (i) and (iii)
 (c) (iii) and (iv) (d) (ii) and (iv)

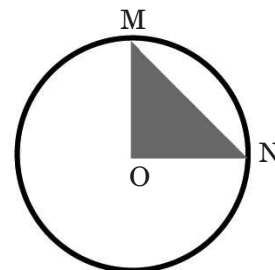
2. Ms. X came out of a building through its front door to find her shadow due to the morning sun falling to her right side with the building to her back. From this, it can be inferred that building is facing _____

- (a) North (b) East
 (c) West (d) South

3. In the above figure, O is the center of the circle and, M and N lie on the circle.

The area of the right triangle MON is 50 cm^2 .

What is the area of the circle in cm^2 ?



- (a) 2π (b) 50π
 (c) 75π (d) 100π

4. If $\left\{ \begin{array}{l} \oplus \text{ "means" } - \\ \otimes \text{ "means" } \div \\ \Delta \text{ "means" } + \\ \nabla \text{ "means" } \times \end{array} \right.$

then, the value of the expression $\Delta 2 \oplus 3 \Delta ((4 \otimes 2) \nabla 4) =$

- (a) -1 (b) -0.5
(c) 6 (d) 7
5. "The increased consumption of leafy vegetables in the recent months is a clear indication that the people in the state have begun to lead a healthy lifestyle"

Which of the following can be logically inferred from the information presented in the above statement?

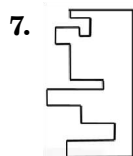
- (a) The people in the state did not consume leafy vegetables earlier.
(b) Consumption of leafy vegetables may not be the only indicator of healthy lifestyle.
(c) Leading a healthy lifestyle is related to a diet with leafy vegetables.
(d) The people in the state have increased awareness of health hazards causing by consumption of junk foods.

Q. 6 to Q. 10: Multiple Choice Question (MCQ), Carry Two Marks Each

6. Oxpeckers and rhinos manifest a symbiotic relationship in the wild. The oxpeckers warn the rhinos about approaching poachers, thus possibly saving the lives of the rhinos. Oxpeckers also feed on the parasitic ticks found on rhinos.

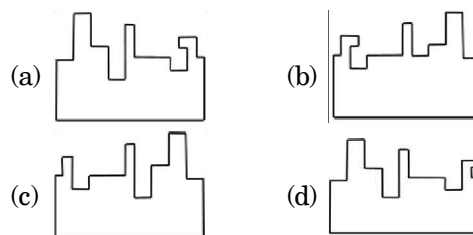
In the symbiotic relationship described above, the primary benefits for oxpeckers and rhinos respectively are,

- (a) Oxpeckers get a food source, rhinos have no benefit.
(b) Oxpeckers save their habitat from poachers while the rhinos have no benefit.
(c) Oxpeckers get a food source, rhinos may be saved from the poachers.
(d) Oxpeckers save the lives of poachers, rhinos save their own lives.



A jigsaw puzzle has 2 pieces. One of the pieces is shown above. Which one of the given options for the missing piece when assembled will form a

rectangle? The piece can be moved, rotated or flipped to assemble with the above piece.

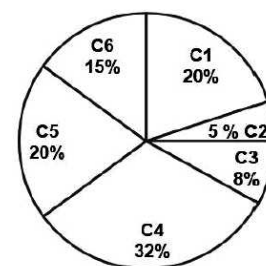


8. The number of hens, ducks and goats in farm P are 65, 91 and 169, respectively. The total number of hens, ducks and goats in a nearby farm Q is 416. The ratio of hens:ducks:goats in farm Q is 5:14:13. All the hens, ducks and goats are sent from farm Q to farm P.

The new ratio of hens:ducks:goats in farm P is

- (a) 5:7:13 (b) 5:14:13
(c) 10:21:26 (d) 21:10:26

9.



Company	Ratio
C1	3:2
C2	1:4
C3	5:3
C4	2:3
C5	9:1
C6	3:4

The distribution of employees at the rank of executives, across different companies C1, C2, ..., C6 is presented in the chart given above. The ratio of executives with a management degree to those without a management degree in each of these companies is provided in the table above. The total number of executives across all companies is 10,000.

The total number of management degree holders among the executives in companies C2 and C5 together is _____.

- (a) 225 (b) 600
(c) 1900 (d) 2500

10. Five persons P, Q, R, S and T are sitting in a row not necessarily in the same order. Q and R are separated by one person, and S should not be seated adjacent to Q.

The number of distinct seating arrangements possible is:

- (a) 4 (b) 8
(c) 10 (d) 16

SET 6

Q. 1 to Q. 5 : Carry One Mark Each.

1. Five persons P, Q, R, S and T are to be seated in a row, all facing the same direction, but not necessarily in the same order. P and T cannot be

seated at either end of the row. P should not be seated adjacent to S. R is to be seated at the second position from the left end of the row. The number of distinct seating arrangements possible is:

- (a) 2 (b) 3
(c) 4 (d) 5

2. Consider the following sentences:

- (i) The number of candidates who appear for the GATE examination is staggering.
(ii) A number of candidates from my class are appearing for the GATE examination.
(iii) The number of candidates who appear for the GATE examination are staggering.
(iv) A number of candidates from my class is appearing for the GATE examination.

Which of the above sentences are grammatically CORRECT?

- (a) (i) and (ii) (b) (i) and (iii)
(c) (ii) and (iii) (d) (ii) and (iv)

3. A digital watch X beeps every 30 seconds while watch Y beeps every 32 seconds. They beeped together at 10 AM.

The immediate next time that they will beep together is _____

- (a) 10.08 AM (b) 10.42 AM
(c) 11.00 AM (d) 10.00 PM

4. If $\oplus \div \odot = 2$; $\oplus \div \Delta = 3$; $\odot + \Delta = 5$; $\Delta \times \otimes = 10$,

Then, the value of $(\otimes - \oplus)^2$, is:

- (a) 0 (b) 1
(c) 4 (d) 16

5. The front door of Mr. X's house faces East. Mr. X leaves the house, walking 50 m straight from the back door that is situated directly opposite to the front door. He then turns to his right, walks for another 50 m and stops. The direction of the point Mr. X is now located at with respect to the starting point is _____

- (a) South-East (b) North-East
(c) West (d) North-West

Q. 6 to Q. 10: Multiple Choice Questions (MCQ), Carry Two Marks Each

6. Given below are two statements 1 and 2, and two conclusions I and II.

Statement 1: All entrepreneurs are wealthy.

Statement 2: All wealthy are risk seekers.

Conclusion I: All risk seekers are wealthy.

Conclusion II: Only some entrepreneurs are risk seekers.

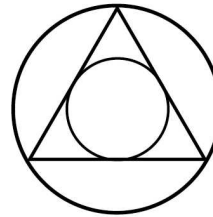
Based on the above statements and conclusions, which one of the following options is CORRECT?

- (a) Only conclusion I is correct
(b) Only conclusion II is correct
(c) Neither conclusion I nor II is correct
(d) Both conclusions I and II are correct

7. A box contains 15 blue balls and 45 black balls. If 2 balls are selected randomly, without replacement, the probability of an outcome in which the first selected is a blue ball and the second selected is a black ball, is _____

- (a) $\frac{3}{16}$ (b) $\frac{45}{236}$
(c) $\frac{1}{4}$ (d) $\frac{3}{4}$

8.



The ratio of the area of the inscribed circle to the area of the circumscribed circle of an equilateral triangle is _____

- (a) $\frac{1}{8}$ (b) $\frac{1}{6}$
(c) $\frac{1}{4}$ (d) $\frac{1}{2}$

9. Consider a square sheet of side 1 unit. The sheet is first folded along the main diagonal. This is followed by a fold along its line of symmetry. The resulting folded shape is again folded along its line of symmetry. The area of each face of the final folded shape, in square units, equal to _____

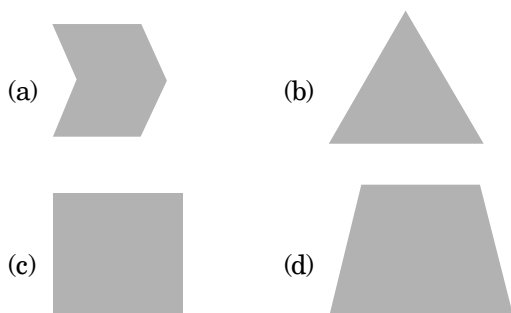
- (a) $\frac{1}{4}$ (b) $\frac{1}{8}$
(c) $\frac{1}{16}$ (d) $\frac{1}{32}$

10. The world is going through the worst pandemic in the past hundred years. The air travel industry is facing a crisis, as the resulting quarantine requirement for travelers led to weak demand. In relation to the first sentence above, what does the second sentence do?

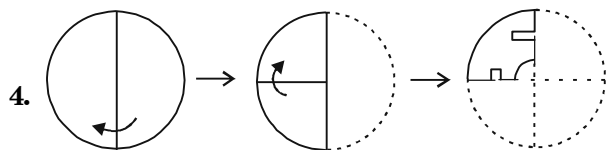
- (a) Restates an idea from the first sentence.
(b) Second sentence entirely contradicts the first sentence.
(c) The two statements are unrelated.
(d) States an effect of the first sentence.

SET 7**Q. 1 to Q. 5 : Carry One Mark Each.**

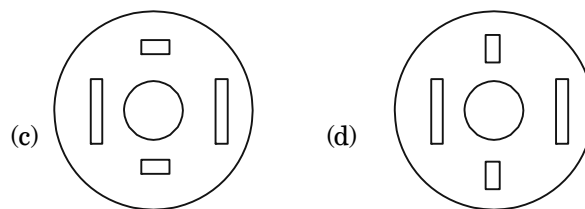
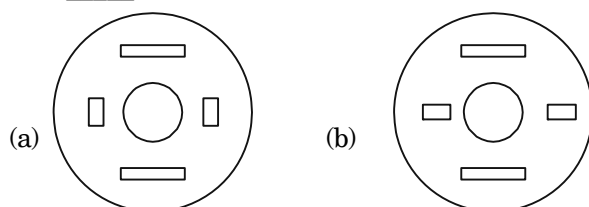
1. The ratio of boys to girls in a class is 7 to 3.
Among the options below, an acceptable value for the total number of students in the class is :
(a) 21 (b) 37
(c) 50 (d) 73
2. A polygon is convex if, for every pair of points, P and Q belonging to the polygon, the line segment PQ lies completely inside or on the polygon.
Which one of the following is NOT a convex polygon ?



3. Consider the following sentences :
(i) Everybody in the class is prepared for the exam.
(ii) Babu invited Danish to his home because he enjoys playing chess.
Which of the following is the CORRECT observation about the above two sentences ?
(a) (i) is grammatically correct and (ii) is unambiguous
(b) (i) is grammatically incorrect and (ii) is unambiguous
(c) (i) is grammatically correct and (ii) is ambiguous
(d) (i) is grammatically incorrect and (ii) is ambiguous



4. A circular sheet of paper is folded along the lines in the directions shown. The paper, after being punched in the final folded state as shown and unfolded in the reverse order of folding, will look like ____.



5. ____ is to *surgery* as *writer* is to ____.
Which one of the following options maintains a similar logical relation in the above sentence ?
(a) Plan, outline (b) Hospital, library
(c) Doctor, book (d) Medicine, grammar

Q. 6 to Q. 10 : Carry Two Marks Each

6. We have 2 rectangular sheets of paper, M and N, of dimensions $6 \text{ cm} \times 1 \text{ cm}$ each. Sheet M is rolled to form an open cylinder by bringing the short edges of the sheet together. Sheet N is cut into equal square patches and assembled to form the largest possible closed cube. Assuming the ends of the cylinder are closed, the ratio of the volume of the cylinder to that of the cube is ____.
- (a) $\frac{\pi}{2}$ (b) $\frac{3}{\pi}$
(c) $\frac{9}{\pi}$ (d) 3π

Items	Cost (₹)	Profit %	Marked Price (₹)
P	5,400	--	5,860
Q	---	25	10,000

Details of prices of two items P and Q are presented in the above table. The ratio of cost of item P to cost of item Q is 3:4. Discount is calculated as the difference between the marked price and the selling price. The profit percentage is calculated as the ratio of the difference between selling price and cost, to the cost

$$\left(\text{Profit \%} = \frac{\text{Selling price} - \text{Cost}}{\text{Cost}} \times 100 \right).$$

The discount on item Q, as a percentage of its marked price, is ____.

- (a) 25 (b) 12.5
(c) 10 (d) 5
8. There are five bags each containing identical sets of ten distinct chocolates. One chocolate is picked from each bag.
The probability that at least two chocolates are identical is ____.
- (a) 0.3024 (b) 0.4235
(c) 0.6976 (d) 0.8125

9. Given below are two statements 1 and 2, and two conclusions I and II.

Statement 1 : All bacteria are microorganisms.

Statement 2 : All pathogens are microorganisms.

Conclusion I : Some pathogens are bacteria.

Conclusion II : All pathogens are not bacteria.

Based on the above statements and conclusions, which one of the following option is logically CORRECT ?

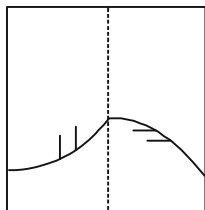
- (a) Only conclusion I is correct
 (b) Only conclusion II is correct
 (c) Either conclusion I or II is correct.
 (d) Neither conclusion I nor II is correct.
10. Some people suggest anti-obesity measures (AOM) such as displaying calorie information in restaurant menus. Such measures sidestep addressing the core problems that cause obesity: poverty and income inequality. Which one of the following statements summarizes the passage ?
- (a) The proposed AOM addresses the core problems that cause obesity.
 (b) If obesity reduces, poverty will naturally reduce, since obesity causes poverty.
 (c) AOM are addressing the core problems and are likely to succeed.
 (d) AOM are addressing the problem superficially.

SET 8

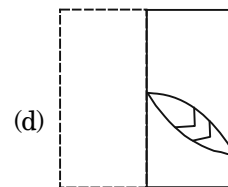
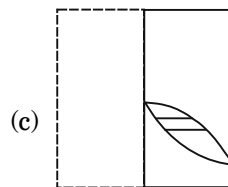
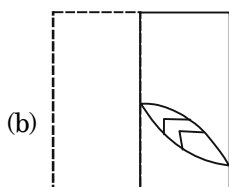
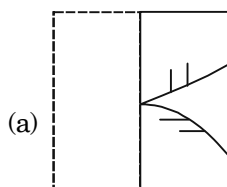
Q. 1 to Q. 5 : Carry One Mark Each.

1. Gauri said that she can play the keyboard ____ her sister.
- (a) as well as (b) as better as
 (c) as nicest as (d) as worse as

2.



As transparent square sheet shown above is folded along the dotted line. The folded sheet will look like ____.



3. If θ is the angle, in degrees, between the longest diagonal of the cube and any one of the edges of the cube, then, $\cos \theta =$

- (a) $\frac{1}{2}$ (b) $\frac{1}{\sqrt{3}}$
 (c) $\frac{1}{\sqrt{2}}$ (d) $\frac{\sqrt{3}}{2}$

4. If $\left(x - \frac{1}{2}\right)^2 - \left(x - \frac{3}{2}\right)^2 = x + 2$, then the value of x is :

- (a) 2 (b) 4
 (c) 6 (d) 8

5. Pen : Write :: Knife : ____.

Which one of the following options maintains a similar logical relation in the above ?

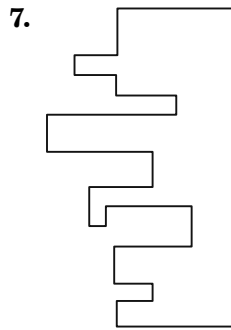
- (a) Vegetables
 (b) Sharp
 (c) Cut
 (d) Blunt

Q. 6 to Q. 10 : Carry Two Marks Each.

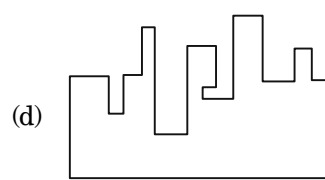
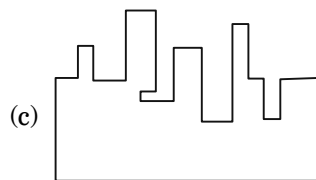
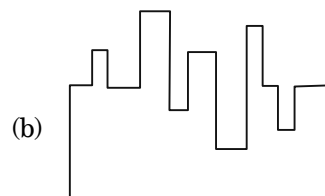
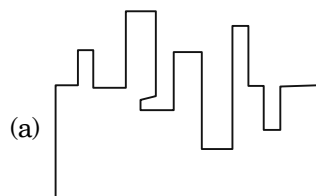
6. Listening to music during exercise improve exercise performance and reduces discomfort. Scientists researched whether listening to music while studying can help students learn better and the results were inconclusive. Students who needed external stimulation for studying fared worse while students who did not need any external stimulation benefited from music.

Which one of the following statements is the CORRECT inference of the above passage ?

- (a) Listening to music has no effect on learning and a positive effect on physical exercise.
 (b) Listening to music has a clear positive effect both on physical exercise and on learning.
 (c) Listening to music has a clear positive effect on physical exercise. Music has a positive effect on learning only in some students.
 (d) Listening to music has a clear positive effect on learning in all students. Music has a positive effect only in some students who exercise.



A jigsaw puzzle has 2 pieces. One of the pieces is shown above. Which one of the given options for the missing piece when assembled will form a rectangle? The piece can be moved, rotated or flipped to assemble with the above piece.

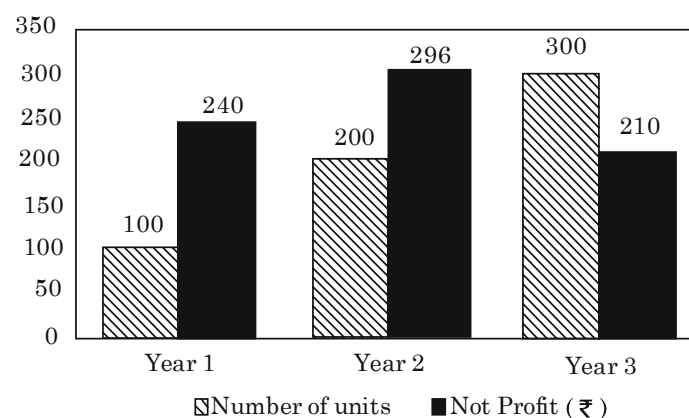


8. The number of students in three classes is in the ratio 3 : 13 : 6. If 18 students are added to each class, the ratio changes to 15 : 35 : 21.

The total number of students in all the three classes in the beginning was :

- (a) 22 (b) 66 (c) 88 (d) 110

9.



The number of units of a product sold in three different years and the respective net profits are presented in the figure above. The cost/unit in Year 3 was ₹1, which was half the cost/unit in Year 2. The cost/unit in Year 3 was one-third of the cost/unit in Year 1. Taxes were paid on the selling price at 10%, 13% and 15% respectively for the three years. Net profit is calculated as the difference between the selling price and the sum of cost and taxes paid in that year.

The ratio of the selling price in Year 2 to the selling price in Year 3 is ____.

- (a) 4 : 3 (b) 1 : 1 (c) 3 : 4 (d) 1 : 2

10. Six students P, Q, R, S, T and U, with distinct heights, compare their heights and make the following observations.

Observation I: S is taller than R.

Observation II: Q is the shortest of all.

Observation III: U is taller than only one student.

Observation IV: T is taller than S but is not the tallest.

The number of students that are taller than R is the same as the number of students shorter than ____.

- (a) T (b) R (c) S (d) P

ANSWERS

Set 1

1. (d) 2. (b) 3. (c) 4. (b) 5. (c) 6. (b) 7. (a) 8. (a) 9. (b) 10. (c)

Set 2

1. (b) 2. (b) 3. (d) 4. (d) 5. (c) 6. (c) 7. (a) 8. (b) 9. (c) 10. (d)

Set 3

1. (c) 2. (b) 3. (c) 4. (c) 5. (b) 6. (c) 7. (d) 8. (a) 9. (c) 10. (a)

Set 4

1. (c) 2. (c) 3. (d) 4. (d) 5. (c) 6. (d) 7. (c) 8. (d) 9. (d) 10. (a)

Set 5

1. (d) 2. (d) 3. (d) 4. (d) 5. (c) 6. (c) 7. (a) 8. (c) 9. (c) 10. (d)

Set 6

1. (b) 2. (a) 3. (a) 4. (b) 5. (d) 6. (c) 7. (b) 8. (c) 9. (b) 10. (d)

Set 7

1. (c) 2. (a) 3. (c) 4. (a) 5. (c) 6. (c) 7. (c) 8. (c) 9. (d) 10. (d)

Set 8

1. (a) 2. (b) 3. (b) 4. (b) 5. (c) 6. (c) 7. (a) 8. (c) 9. (a) 10. (c)

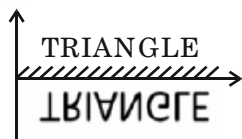
EXPLANATIONS

SET 1

1. When the comparison is between two things we use the second degree of the adjective the degree from of easy are : easy $\rightarrow$ easier $\rightarrow$ easiest.

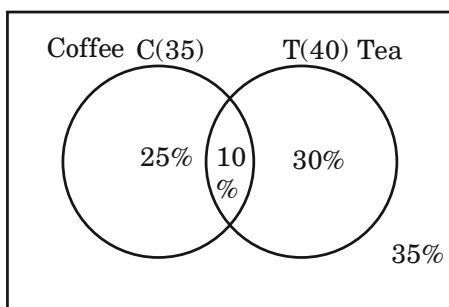
Hence, comparative degree is easier.

2. The mirror image of TRIANGLE about the X-axis is



$\therefore$ Top to Bottom will interchange (or) symmetrical about X-axis.

3.



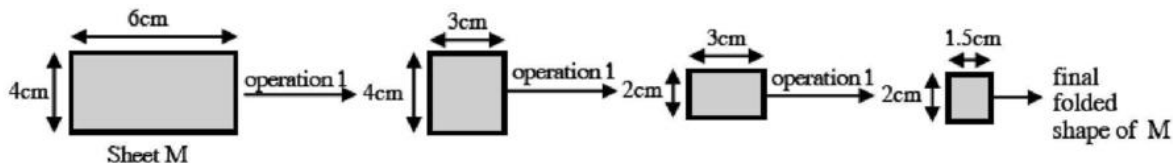
C $\rightarrow$ Coffee $\rightarrow$ Percentage of employees drink Coffee = 35%

T $\rightarrow$ Tea $\rightarrow$ Percentage of employees drink Tea = 40%

$\rightarrow$ Percentage of employees drink both Tea and Coffee = 10%

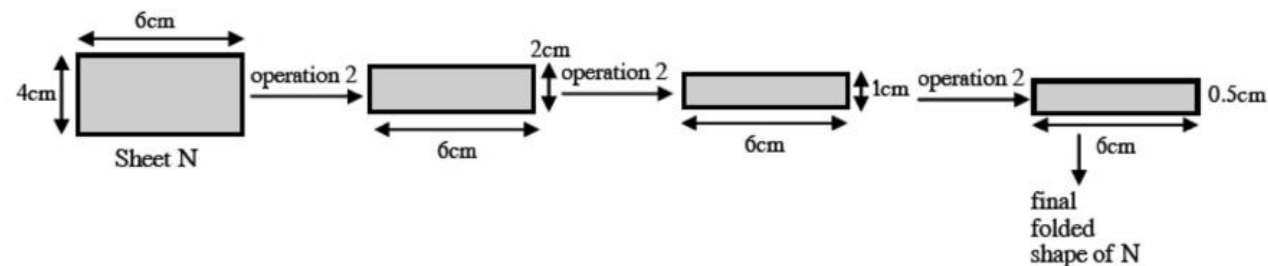
$\rightarrow$ Percentage of employees drink neither Tea and Coffee = $100\% - n(\text{TUE})$
 $= 100\% - (25\% + 10\% + 30\%)$
 $= 35\%$

7. Shortest side of folding:



So, perimeter of final folded shape of M = $2(2 + 1.5) = 7$

Longest side in folded:



4. As given that,

$$\left[\because p \oplus q = \frac{p^2 + q^2}{pq} \text{ and } p \odot q = \frac{p^2}{q} \right]$$

$$\text{Now, L.H.S} \Rightarrow x \oplus y = \frac{x^2 + y^2}{x \cdot y}$$

$$\text{R.H.S} \Rightarrow x \oplus y = 2 \odot 2 = \frac{2^2}{2}$$

So, L.H.S = R.H.S

$$\Rightarrow \frac{x^2 + y^2}{xy} = \frac{2^2}{2}$$

$$\Rightarrow x^2 + y^2 = 2xy \Rightarrow x^2 + y^2 - 2xy = 0$$

$$\Rightarrow (x - y)^2 = 0 \Rightarrow x - y = 0 \Rightarrow \boxed{x = y}$$

5. According to the given information,

Possible way are

R Q S P

P Q S R

R Q P S

P Q R S

Q R S P

Q P S R

$\Rightarrow$ Total number of ways = $2 + 2 + 2 = 6$

$\therefore$ The number of distinct seating arrangements possible is 6.

6. As Given statement:

Either P marries Q or X marries Y.

The logical negation of the Either P marries Q or X marries Y

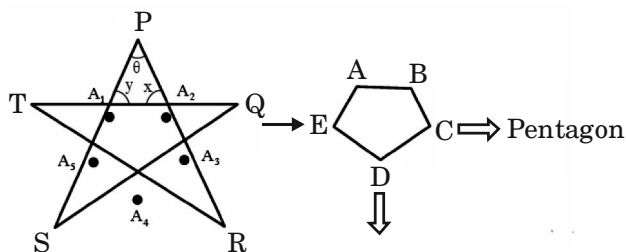
"above statement is Neither P marries Q nor X marries Y."

Hence option (b) is correct.

$\therefore$ Perimeter of final folded shape of N = $2(6 + 0.5)$
= 13

$\therefore$ Required ratio is (N:M) = 13:7

8.



PR = PS = QS = QT = RT (Given)

Then ABCDE will be regular pentagon.

So, Sum of internal angles formed at pentagon
= $(5 - 2) \times 180$

$$\Rightarrow \text{Each internal angle} = \frac{3 \times 180}{5} = 108^\circ$$

So, $\angle \theta = ?$; $\angle x = 180^\circ - 108^\circ = 72^\circ$;
 $\angle y = 180 - 108 = 72$

$\therefore \angle x + \angle y + \theta = 180$

$$\Rightarrow \theta = 180 - 72 - 72 = 36^\circ$$

9. According to the question

$$\frac{\lambda(-(-3+2), (-2+3))}{(-(-2+1))} = \frac{\lambda(-1(-1), 1)}{1}$$

So, 1st definition will be applicable as $p = q$

$[\therefore \lambda(p, q) = (p - q)^2 \text{ if } p = q]$

So, $\lambda(1, 1) = (1 - 1)^2 = 0$

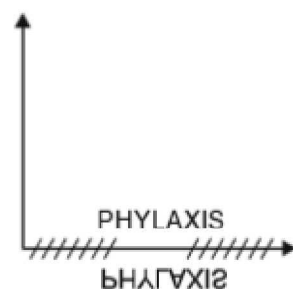
10. Only humans have the ability to construct worlds entirely in their minds and no other species has this.

Hence, answer is option (c) which reflects the information given in the passage.

SET 2

1. When two or more singular nouns joined by 'and' the verb should be plural.

2.



With respect 'x' - axis.

3. Let, $E_1 \rightarrow$ occurrence of even number on first dice

$E_2 \rightarrow$ occurrence of even number on second dice

Probability of even number on a first dice is:

$$P(E_1) = \frac{3}{6} = \frac{1}{2}$$

And probability of even number on a second dice is:

$$P(E_2) = \frac{3}{6} = \frac{1}{2}$$

So, the probability that an even number is rolled out on each dice is:

$$P(E_1 \cap E_2) = P(E_1) \times P(E_2) \\ = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

4. As we know that

$$p \odot q = p - q$$

$$p \oplus q = p \times q$$

$$\Rightarrow 9 \odot (6 \oplus 7) \odot (7 \oplus (6 \odot 5)) \\ = 9 - (6 \times 7) - (7 \times (6 - 5)) \\ = 9 - 42 - (7 \times 1) \\ = -40$$

5. Possible arrangements:

$\therefore$ 'R' should not be seated at second position from the left end:

(-, R, -, -)

So, possible arrangements:

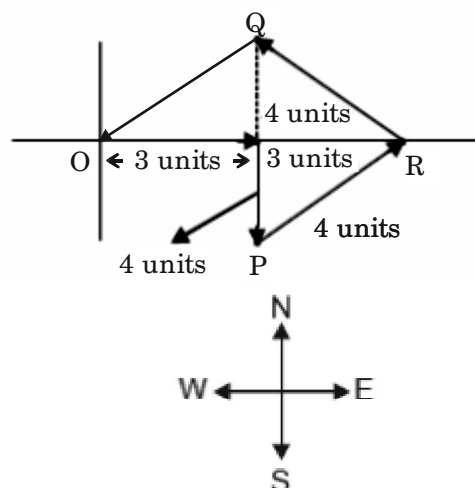
$$\boxed{} \boxed{P} \boxed{} \boxed{} = 3!$$

$$\boxed{} \boxed{Q} \boxed{} \boxed{} = 3!$$

$$\boxed{} \boxed{S} \boxed{} \boxed{} = 3!$$

Hence, total number of ways = $3 \times 3! = 18$ chances.

6.

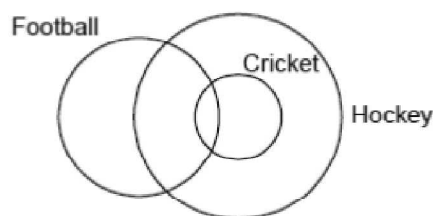


$$\therefore OQ = \sqrt{(3)^2 + (4)^2} \\ = 5 \text{ units}$$

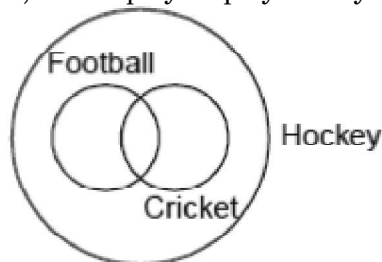
7. The author states musicians rehearse, actors rehearse and like wise he states public speakers

also should rehearse and this is stated in option (a).

8. (i) Some foot ball player play cricket:

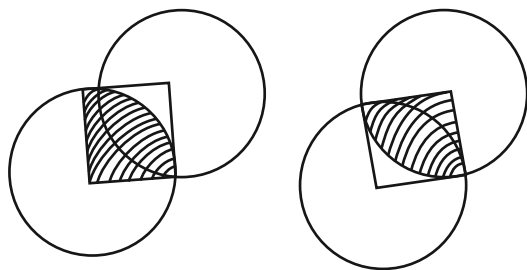


- (ii) All, cricket players play hockey:



Hence, some football players play hockey is satisfied.

9. According to question:



Area of shaded region from the figure is $\frac{\pi r^2}{4}$.

Shaded area from the figure:

$$= 2 \times \left[\frac{\pi r^2}{4} - \frac{1}{2} \times r \times r \right]$$

$$= r^2 \left[\frac{\pi}{2} - 1 \right]$$

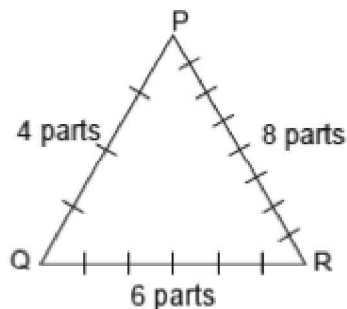
Total area = r^2

As we know that,

$$\therefore \text{Required probability} = \left[\frac{\text{Shaded area}}{\text{Total area}} \right]$$

$$= \frac{r^2 \left[\frac{\pi}{2} - 1 \right]}{r^2} = \left(\frac{\pi}{2} - 1 \right)$$

10.



Let the dimension of equilateral

$\Delta PQR = a$ units

As per question,

$$\left(\frac{a}{4}, \frac{a}{6}, \frac{a}{8} \right) \text{ is integer,}$$

Minimum area of triangle

$$(A) = \text{LCM}(4, 6, 8) = 24 \text{ units}$$

$$\therefore \text{Area of } \Delta PQR = \frac{\sqrt{3}}{4} \times a^2$$

$$= \left[\frac{\sqrt{3}}{4} \times 24 \times 24 \right] = 144\sqrt{3} \text{ sq. units}$$

SET 3

1. As given that,

Current population of a city = 11, 02, 500

Increasing rate = 5%/Annum

The population 2 years ago = ?

Let us Consider, the population 2 years ago = x

Using compound interest formula, we have

$$A = P \left(1 + \frac{R}{100} \right)^n$$

$$\Rightarrow 11,02,500 = x \left(1 + \frac{5}{100} \right)^2$$

$$\Rightarrow 11,02,500 \times \frac{100 \times 100}{105 \times 105} = x$$

$$\Rightarrow \boxed{x = 10,00,000}$$

Hence, the population 2 years ago = 10,00,000.

2. As Given that, $\frac{p}{q} + \frac{q}{p} = 3$

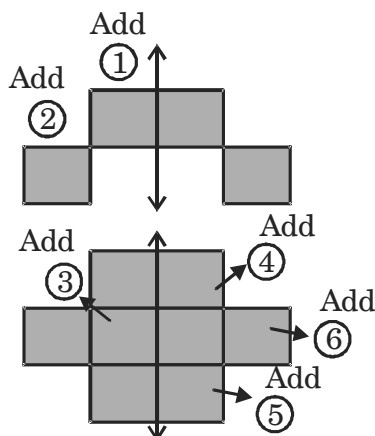
By Squaring on both sides;

$$\left(\frac{p}{q} + \frac{q}{p}\right)^2 = 9$$

$$\Rightarrow \frac{p^2}{q^2} + \frac{q^2}{p^2} + 2 = 9$$

Hence, $\boxed{\frac{p^2}{q^2} + \frac{q^2}{p^2} = 7}$

3.



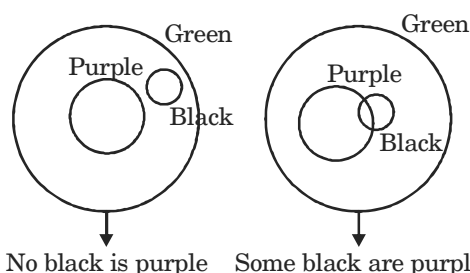
Hence, the least number of squares added = 6.

4. Nostalgia (means excessively sentimental yearning for return to or of some past period) is to anticipation. (means visualization of a future event or state)

5. Wake pastense is woke and third form is woken so, the verb forms are:

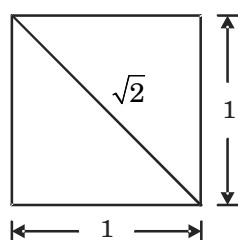
V1	V2	V3
Wake	Woke	Woken

6.

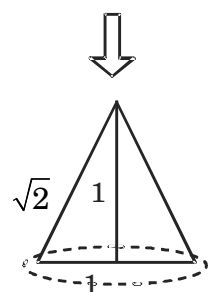
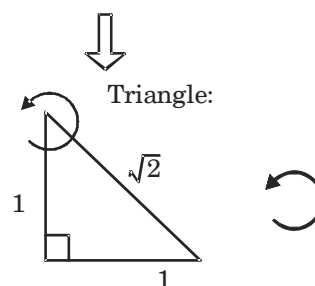


7. The passage deduces that computers are present every where and they have both positive and negative effects on humans.

8. Square:



Solid core formed by revolving tringle about its short edge.



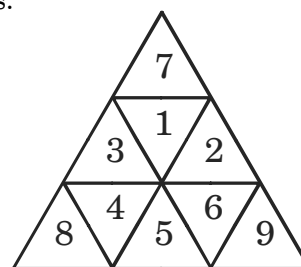
Radius of base circle(r) = 1 unit

Height of cone(h) = 1 unit

$$\text{Volume of cone} = \frac{\pi}{3} \times r^2 h = \frac{\pi}{3} \times 1 \times 1 = \frac{\pi}{3} \text{ unit}^3$$

9. From the bar graph, it is clear that except Thursday, total 6 days one of the students spent a minimum of 10% more than the other student.

10. In regular hexagon, each interior angle = 120° ,
So, if we cut corners of an equilateral triangle, then the removed triangles are also equilateral triangles.



$\therefore$ Let, side of regular hexagon = x units

Then, side of original equilateral triangle = $3x$

$$\therefore \text{Area of regular hexagon} = \frac{3\sqrt{3}}{2} x^2$$

$$\text{Area of equilateral triangle} = \frac{\sqrt{3}}{4} (3x)^2$$

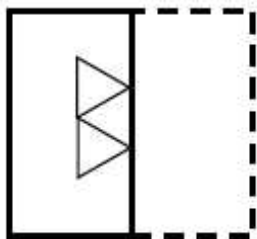
$$\therefore \text{Required ratio} = \frac{3\sqrt{3}}{2} x^2 : \frac{\sqrt{3}}{4} (3x)^2$$

$$= \frac{3\sqrt{3}}{2} x^2 : \frac{9\sqrt{3}}{4} x^2$$

$$= 1 : \frac{3}{2} = 2 : 3$$

SET 4

- Here the subject of the verb is 'who' because we can not use 'Which' for people, 'Who' is the only word which acts as a subject for the verb.
- If the square sheet is folded along the dotted line by upon rotation, then the folded sheet will look like as follows.



- As given that,
 $n = 10$
 As we know that
 $\therefore$ The sum of interior angles for a regular polygon $= (n - 2) \times 180^\circ$
 $= (10 - 2) \times 180 = 8 \times 180^\circ = 1440$

$$\text{So, Each interior side angle} = \frac{8 \times 180^\circ}{10} = 144^\circ.$$

- $(a^n - b^n)$ is divisible by $(a + b)$, only when 'n' is even

So, we take the option (d).

$$\begin{aligned} 11^{52} - 1 &= (11^{26})^2 - 1^2 \\ &= (11^{26} + 1)(11^{26} - 1) \\ &= (11^{26} + 1)[(11^{13})^2 - (1^2)] \\ &= (11^{26} + 1)[(11^{13}) - 1](11^{13} + 1) \end{aligned}$$

$\therefore 11^{13} + 1$ is divisor of $11^{52} - 1$.

- Oasis is a water pool amidst sand just as island is a piece of land amidst water.
- Deprivation of sleep during exams will have negative effect on exams. So, considerable sleep is necessary to well in exams.
- According to the question.

$$\therefore \text{Total number of squares} = 6$$

Now, by using Pythagoras theorem, in each square, we can find side length.

$$\therefore \text{Side of smallest square becomes} = \frac{5}{\sqrt{8}}$$

$$\text{So, Area of smallest square} = \frac{5}{\sqrt{8}} \times \frac{5}{\sqrt{8}} = \frac{25}{8} = 3.125$$

- As we know that,

$$E(X) = \text{Mean of } X = \int_{-\infty}^{\infty} x \cdot f_x(x) dx$$

$$\begin{aligned} E[x] &= \int_0^{100} x \cdot f(x) dx \quad (\because 0 \leq x \leq 100) \\ &= \int_0^{100} x \cdot (0.01) dx = 0.01 \left(\frac{x^2}{2} \right)_0^{100} \\ &= \frac{1}{200} \times 100 \times 100 = 50 \end{aligned}$$

- From the given bar graph:

In year 2 : Total number of students who took the exam $= 60 + 5 = 65$

Failed students $= 0$

But who appeared the exam for 1st time $= 65 - 10 = 55$.

In year 3 : Total number of students who took the exam $= 50 + 3 = 53$

Failed students $= 5$

But who took the exam for 1st exam $= 53 - 5 = 48$

Hence option (d) is correct.

- According to the given information, possible arrange:

$$\begin{bmatrix} T & U & P & Q & S & V & R \\ U & Q & S & V & R & P & I \\ I & U & O & S & V & R & P \end{bmatrix}$$

Statements : (b), (c), (d) are true and

Statements (A) is incorrect.

Hence, there is only one car between Q and V.

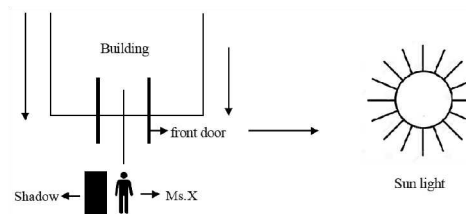
SET 5

- When you say you could hardly do some thing you are emphasizing that it is very difficult for you to do it.

$\therefore$ Could + Adverb + Verb (I).

Hence, option (d) is the correct answer.

- According to the questions:



$\therefore$ From the above figure, it is clear that building is facing towards **South**.

Hence, option (d) is the correct answer.

3. Given that,

Area of right triangle MON = 50

According to given figure:

$$\Rightarrow \frac{1}{2} \times ON \times OM = 50$$

[$\because ON = OM = \text{radius of circle} = r(\text{say})$]

$$\Rightarrow \frac{1}{2} \times r \times r = 50$$

$$\Rightarrow r^2 = 100$$

$\therefore$ Area of circle with radius $r = \pi r^2$

$$= \pi(100) = 100\pi.$$

Hence option (d) is the correct answer.

4. As given expression =

$$\Delta 2 \oplus 3\Delta((4 \otimes 2) \nabla 4)$$

$$= +2 - 3 + ((4 \div 2) \times 4)$$

$$= 2 - 3 + (2 \times 4) = 2 - 3 + 8 = 7$$

Hence option (d) is the correct answer.

5. One can clearly infer that eating leafy vegetables undoubtedly lead a healthy style.

7. From the options, the figure in option (b) is the missing piece. If the figure is rotated clockwise then it will form a rectangular with the given piece in the picture.

8. **For farm P:**

Number of hens = 65

Number of ducks = 91

Number of goats = 169

For farm Q:

$$\text{Number of hens} = 416 \times \frac{5}{32} = 65$$

$$\text{Number of ducks} = 416 \times \frac{14}{32} = 182$$

$$\text{Number of goats} = 416 \times \frac{13}{32} = 169$$

$$\therefore \text{Total number of hens} = 65 + 65 = 130$$

$$\text{Total number of ducks} = 91 + 182 = 273$$

$$\text{Total number of goats} = 169 + 169 = 338$$

$$\therefore \text{The new ratio of hens: ducks: goats} = 130:273:338 = 10:21:26.$$

9. As given that,

Total number of executives, across all companies = 10,000

$$\therefore \text{Number of executives in } C_2 = 10,000 \times \frac{5}{100} = 500$$

$$\text{Number of executives in } C_5 = 10,000 \times \frac{20}{100} = 2000$$

$\therefore$ Number of management degree holders among

$$\text{the executives in } C_2 = 500 \times \frac{1}{5} = 100$$

And, Number of management degree holders

$$\text{among the executives in } C_5 = 2000 \times \frac{9}{10} = 1800$$

So, the total number of management degree holders among the executives in companies C_5 and C_2 together = $100 + 1800 = 1900$.

Hence, option (c) is the correct answer.

10. As given that

Total block is (Q_R)

Total number of ways,

$$\frac{1}{Q} \times \frac{2}{R} \times \frac{1}{S} \times \frac{2}{S} \rightarrow 4 \text{ choices}$$

(OR)

$$\frac{2}{Q} \times \frac{1}{R} \times \frac{1}{S} \times \frac{2}{S} \rightarrow 2 \text{ choices}$$

$$\frac{S}{Q} \times \frac{2}{R} \times \frac{1}{S} \rightarrow 2 \text{ choices}$$

(OR)

$$\frac{2}{R} \times \frac{1}{Q} \times \frac{1}{S} \rightarrow 2 \text{ choices}$$

(OR)

$$\frac{S}{R} \times \frac{R}{Q} \times \frac{2}{S} \times \frac{Q}{S} \times \frac{1}{S} \rightarrow 2 \text{ choices}$$

(OR)

$$\frac{2}{S} \times \frac{2}{R} \times \frac{1}{Q} \times \frac{1}{S} \rightarrow 4 \text{ choices}$$

Hence, total number of distinct seating arrangement possible = $4 + 2 + 2 + 2 + 2 + 4 = 16$.

SET 6

1. Positive arrangements are

$$\underline{S} \underline{R} \underline{P} \underline{T} \underline{Q} \text{ (OR) } \underline{Q} \underline{R} \underline{P} \underline{T} \underline{Q} \text{ (OR) } \underline{S} \underline{R} \underline{T} \underline{P} \underline{Q}$$

Total number of combinations possible is 3.

2. (i) The first sentence informs us about the number with is singular.

(ii) A number of means several, some.

The second sentence takes about the students which is plural word.

3. As given that,

Watch x beeps every 30 sec

Watch y beeps every 32 sec

L.C.M of 30, 32 = 480 seconds = 8 minutes

They beeps together every 8 min

$\therefore$ i.e., 10 AM, 10.08 AM; 10.16 AM, 10.24 AM,

Hence option (a) is the correct answer.

4. (i) $\oplus \div \ominus \Rightarrow \frac{\oplus}{\ominus} = 2$

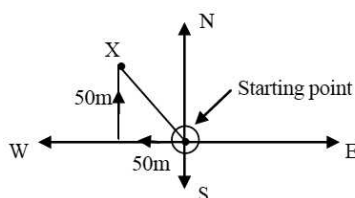
(ii) $\oplus \div \Delta = 3 \Rightarrow \frac{\oplus}{\Delta} = 3$

(iii) $\ominus + \Delta = 5 \Rightarrow \frac{\oplus}{2} + \frac{\oplus}{3} = 5$, $\oplus = 6$ and $\Delta = \frac{6}{3} = 2$.

(iv) $\Delta \times \otimes = 10 \Rightarrow 2 \times \otimes = 10$ $\otimes = 5$

So, $(\otimes - \oplus)^2 = (5 - 6)^2 = 1$

5. From the figure, it is clear that, the direction of X is North West

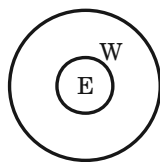


6. E $\rightarrow$ Entrepreneurs

W $\rightarrow$ Wealthy

R $\rightarrow$ Risk seekers

Statement-1



All entrepreneurs are wealthy

$\therefore$ All risk sectors are not wealthy $\Rightarrow$ Conclusion I is incorrect.

All entrepreneurs are risk seekers $\Rightarrow$ Conclusion II is incorrect

Hence, neither conclusion I nor II is correct.

7. Box

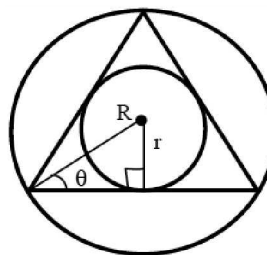
Total balls =	15 Blue	45 Black
---------------	------------	-------------

If the two balls are drawn without replacement the required probability of getting first ball blue and second ball black will be

$\therefore$ Required probability = $\frac{15}{60} \times \frac{45}{59} = \frac{45}{236}$

8. $\therefore$ From the figure,

$$\sin 30^\circ = \frac{r}{R} \Rightarrow \frac{1}{2} = \frac{r}{R} \Rightarrow R = 2r$$



$r \rightarrow$ radius of inscribed circle

$R \rightarrow$ radius of circumscribed circle

$\therefore$ Area of inscribed circle = πr^2

Area of circumscribed circle = πR^2

$$= \pi(4r^2)$$

$\therefore$ Required ratio = $\pi r^2 : 4\pi r^2$

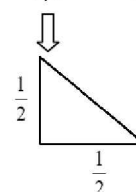
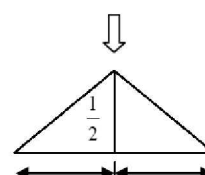
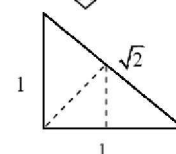
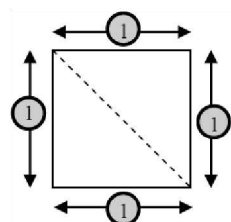
$$= 1 : 4 = \frac{1}{4}$$

9. $\therefore$ Area of each face of the final folded shape.

(Right angle triangle)

$$= \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \left(\frac{1}{2} \times \text{Box} \times \text{weight} \right)$$

$$= \frac{1}{8} \text{ square units}$$



Hence, option (b) is the correct answer.

10. Due to worst pandemic the air travel industry is facing a crisis is nothing but restarting the idea from first sentence.

SET 7

1. Let us consider the number of boys = B

And total number of students = S

So, we have

$$\frac{B}{(S - B)} = \frac{7}{3}$$

$$B = \frac{7S}{10}$$

Thus, S must be multiple of 10. Hence, answer is 50.

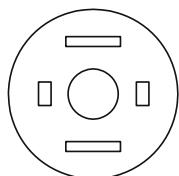
2. If we choose two points P, Q such that P is the top-left corner and Q is the bottom left corner then the line joining P, Q will not lie on the polygon. Hence, option (a) is the correct answer.

3. Every body is prepared means they are ready for the exam, the second sentence does not clearly state whether babu will play with Danish or not. Babu loves playing chess the does not means danish knows how to play chess. So statement (i) is grammatically correct and (ii) is ambiguous.

Statement (ii) is ambiguous because we do not know who enjoys playing chess, Babu or Danish. So, statement (i) is grammatically correct.

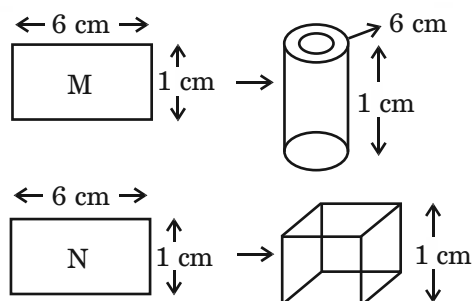
Hence, option (c) is the correct answer.

4. Unforhed in reverse order of folding, it's look like option (a)



5. A doctor performs surgery just as a writer writes a book. It is the relation of person and skill.

6.



The circumference of lower circle $2\pi r = 6$

$$\Rightarrow r = \frac{3}{\pi}$$

$$\text{Volume of a cylinder} = \pi r^2 h = \pi \times \frac{3}{\pi} \times \frac{3}{\pi} \times 1 = \frac{9}{\pi}$$

Volume of a cube = $(1)^3$

So, request ratio

$$= \frac{\left(\frac{9}{\pi}\right)}{1} = \frac{9}{\pi}$$

Hence, option (c) is the correct answer.

7. As given that

$$CP_p = 5400$$

$$MP_p = 5860$$

$$\text{Cost of Q}(CP_q) = \left(\frac{4}{3}\right) * CP_p = 7200$$

Let, selling price (S.P) of Q = S

$$25 = (SP_q - 7200) * \frac{100}{7200}$$

$$SP_q = 9000$$

$$\text{Discount on Q} = MP_q - SP_q = 10000 - 9000 = 1000$$

$$\text{Discount\%} = \frac{1000}{10000} \times 100 = 10\%$$

Discount on SP_q i.e. 1000 is 10% of marked price (MP_q) of Q i.e. 10,000.

So, answer is 10%.

Hence, option (c) is the correct answer.

8. P (exactly 1 =) Probability that all the picked chocolates are different:

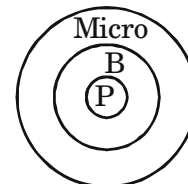
$$= \frac{(10 * 9 * 8 * 7 * 6)}{10^5} = 0.3024$$

Probability that at least two of the picked chocolates are identical:

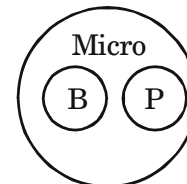
$\Rightarrow 1 - (\text{Probability that all the picked chocolates are different})$

$$\Rightarrow 1 - 0.3024 = 0.6976$$

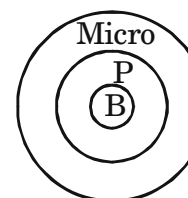
9.



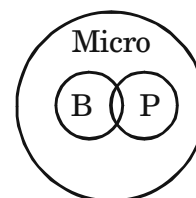
Case I



Case II



Case III



Case IV

So, none of the two conclusions will satisfy all the 4 case. Hence, option (d) is the correct answer.

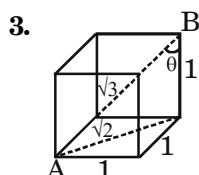
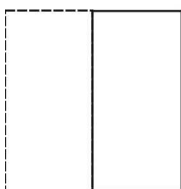
10. Superficially is the deciding key word which means apparently or seemingly.

As AOM are not addressing the core problems, they are superficial.

Superficial means shallow, cursory mean lacking in depth or solidity. Superficial implies a concern only with surface aspects or obvious features. A superficial analysis of the problem shallow is more generally derogatory in implying lack of depth in knowledge, reasoning, emotions, or character. Hence, option (d) is the correct answer.

SET 8

1. 'As well as' is used in positive degree.
2. The folded sheet is look like option (b).



$$\cos \theta = \frac{BX}{AB} = \frac{1}{\sqrt{3}}$$

4. As we know that
 $(a - b)^2 = a^2 + b^2 - 2ab$
 As given expression

$$\left(x - \frac{1}{2}\right)^2 - \left(x - \frac{3}{2}\right)^2 = (x + 2)$$

So,

$$\Rightarrow \left[x^2 + \left(\frac{1}{2}\right)^2 - 2(x) \left(\frac{1}{2}\right) \right] - \left[x^2 + \left(\frac{9}{4}\right) - 2(x) \left(\frac{3}{2}\right) \right]$$

$$= [x + 2]$$

$$\Rightarrow \left[x^2 + \frac{1}{4} - x - x^2 - \frac{9}{4} + 3x \right] = [x + 2]$$

$$\Rightarrow [-2 + 2x] = [x + 2]$$

$$x = 4$$

5. Pen is to write just as knife is to cut.

6. Listening to music improves exercise performance and reduces discomfort. Music as positive effort only on few students during learning.

7. When the piece can be moved, rotated or flipped to assemble with the given piece is option (a)

8. The total number of students in three classes in the beginning.

Let us consider:

$$3a + 13a + 6a = 22a = n$$

after addition of 18 student total number of students.

$$15b + 35b + 21b = 71b$$

$$3a + 18 + 13a + 18 + 6a + 18 = 22a + 54$$

$$22a + 54 = 71b$$

$$n + 54 = 71b$$

$$n = 71b - 54$$

$$\text{Let } (b = 2)$$

$$\text{So, } n = 71 \times 2 - 54$$

$$= 88$$

9. Let us consider, the selling price in year '2' be 'x'.

$$\therefore \text{Net profit} = \text{Selling price} - (\text{Cost} + \text{Taxes})$$

$$296 = x - (400 + 0.13x)$$

$$296 = 0.87x - 400$$

$$x = \frac{696}{0.87} = 800$$

So, selling price in year '2' = Rs.800

In year 3, total cost = $300 \times 1 = \text{Rs.}300$

Taxes paid = 0.15% of selling price.

Now, let us consider, the selling price in year 3 be 'y'

$$\therefore 210 = y - (300 + 0.15y)$$

$$\Rightarrow 210 = (0.85y - 300)$$

$$y = \frac{510}{0.85} = 600$$

$$\therefore \text{Selling price in year '3'} = 600$$

$$\text{So, } \frac{x}{y} = \frac{800}{600} = \frac{4}{3}$$

10. I: $S > R$

II: $Q < P, R, S, T, U$

From III & IV observation

$$P > T > S > R > U > Q$$

The number of students that are taller than 'R' is same as the number of students shorter than S.